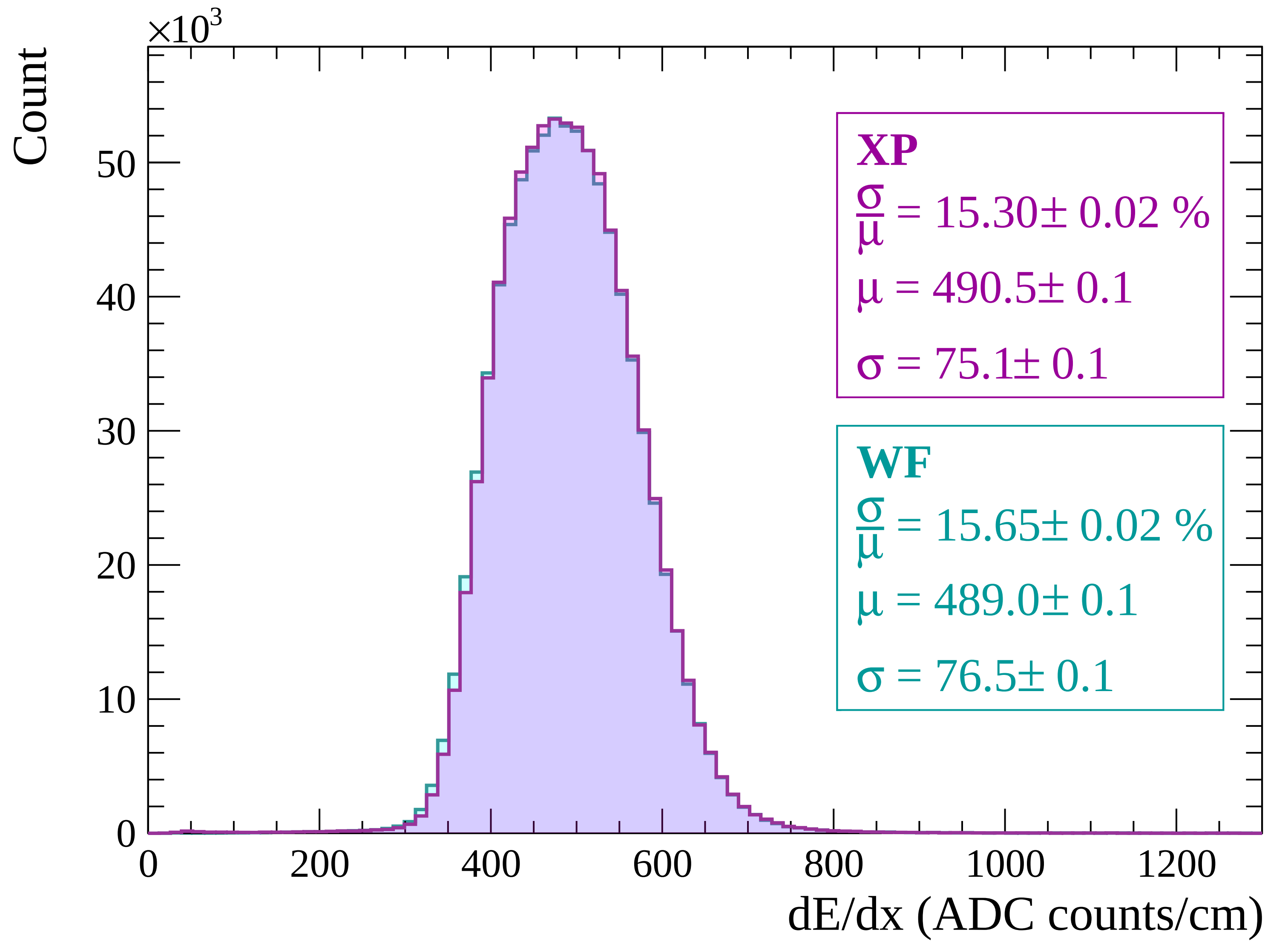
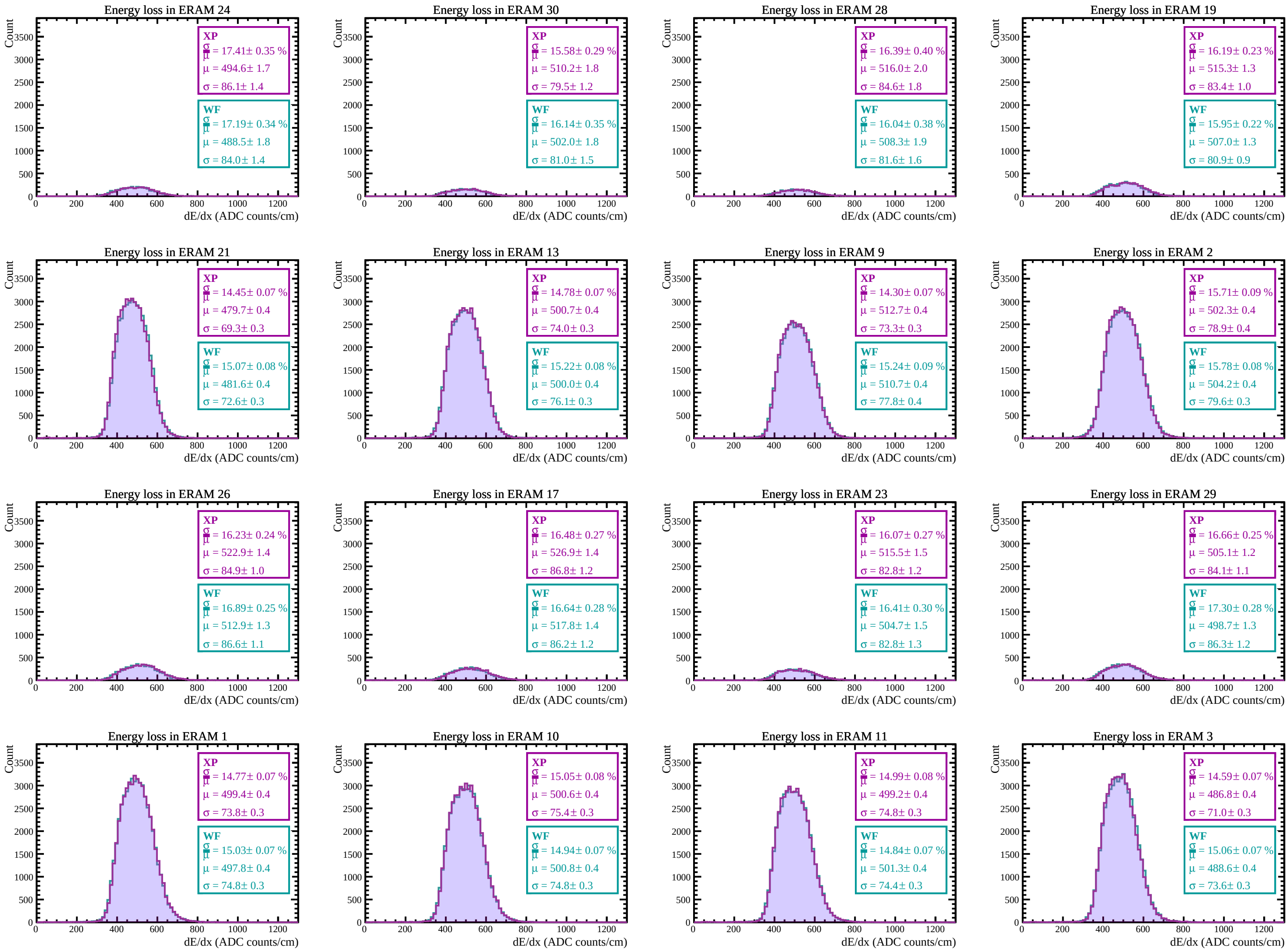
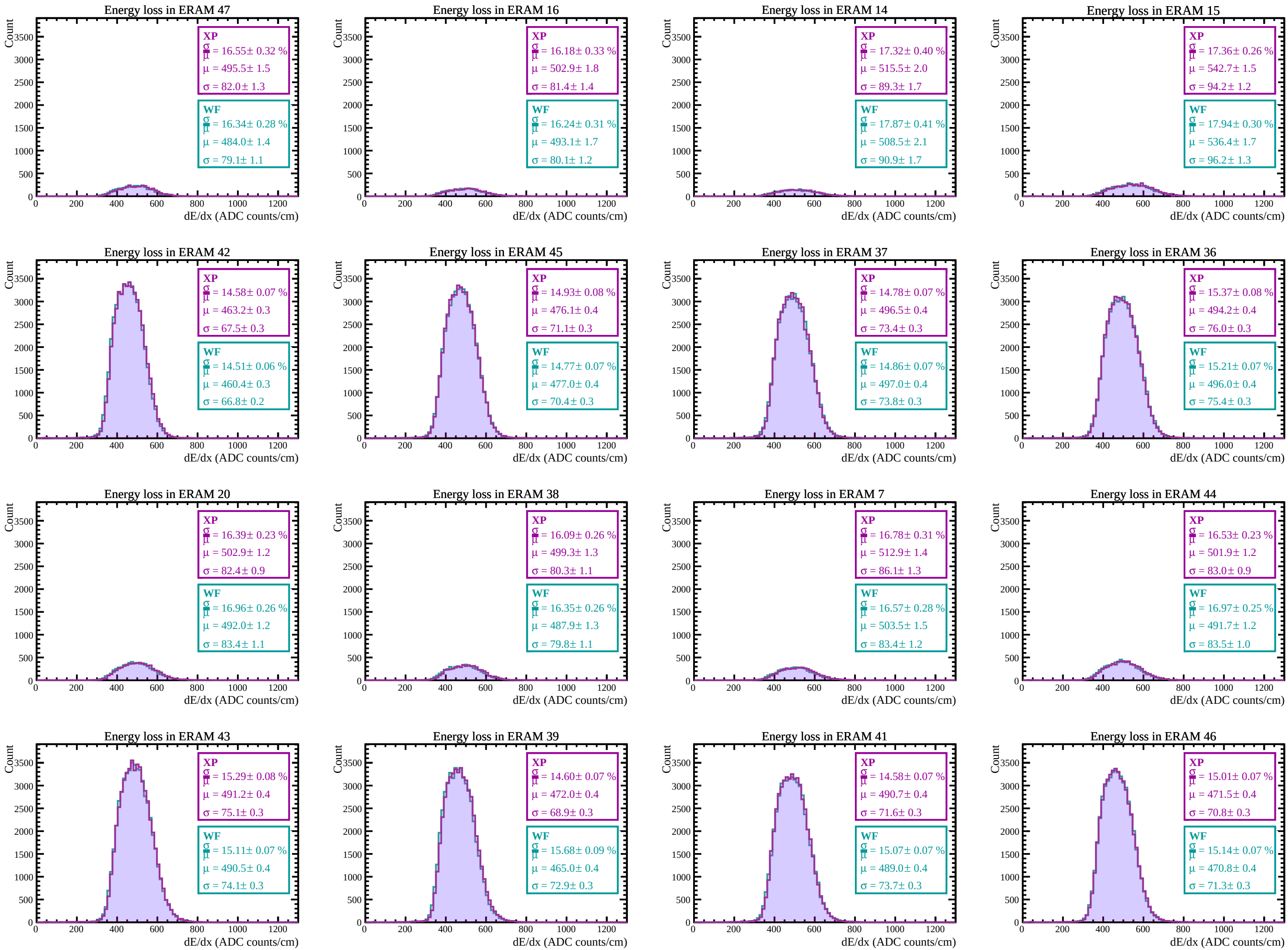


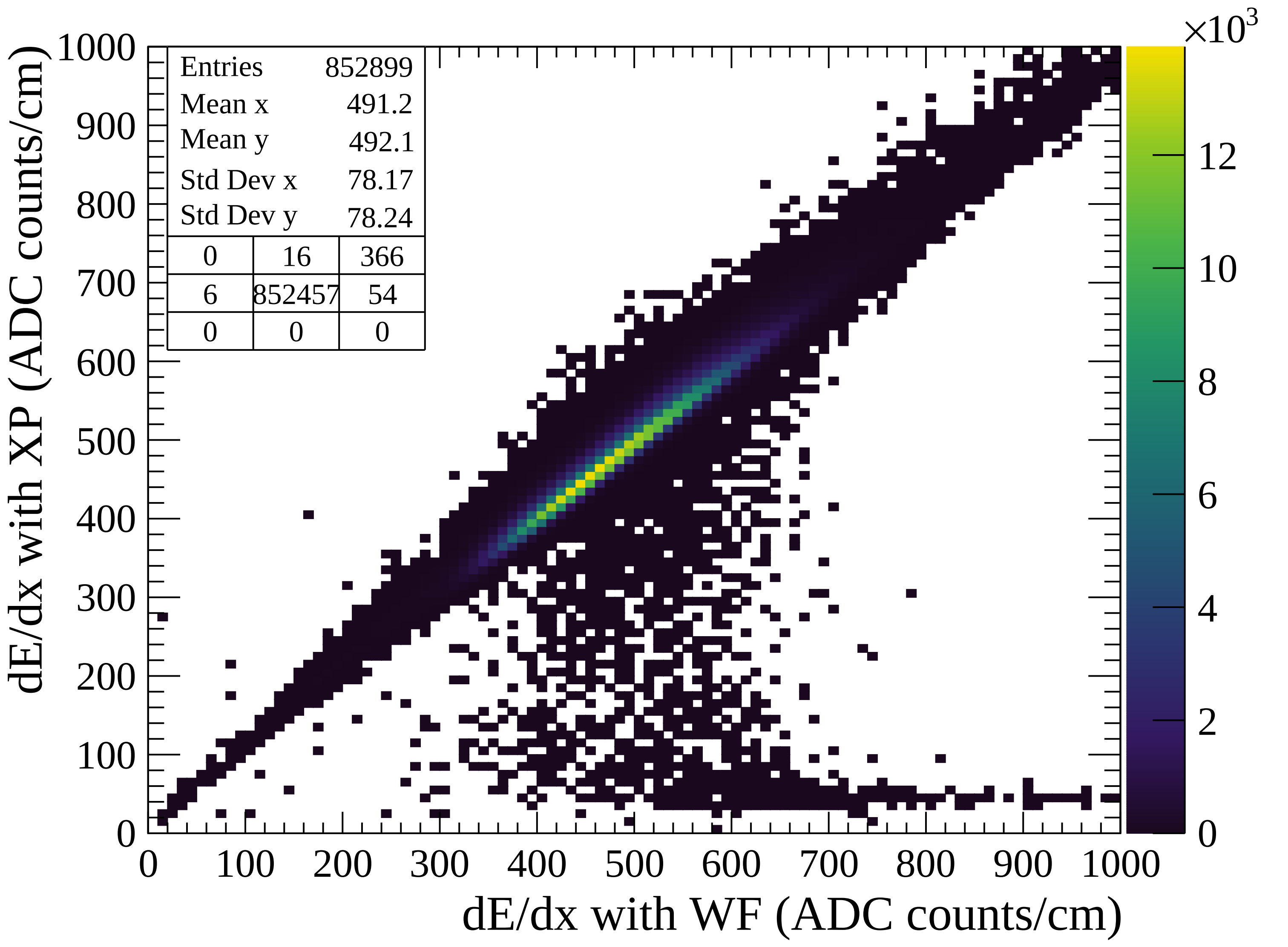
Count
×10^3
0
10
20
30
40
50
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.30± 0.02 %
μ = 490.5± 0.1
σ = 75.1± 0.1
WF
σ/μ = 15.65± 0.02 %
μ = 489.0± 0.1
σ = 76.5± 0.1



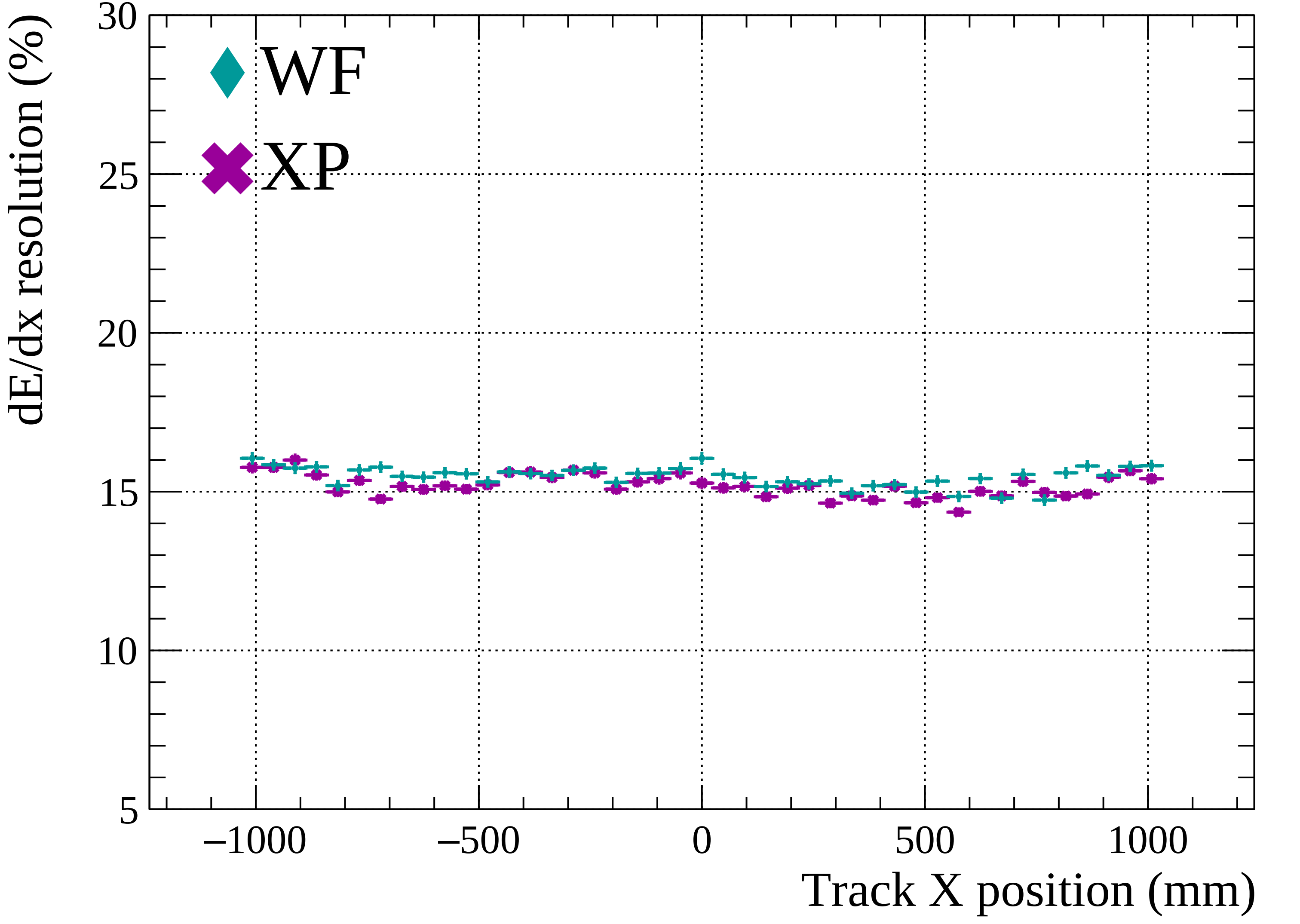
Energy loss in ERAM 24
XP
σ/μ = 17.41± 0.35 %
μ = 494.6± 1.7
σ = 86.1± 1.4
WF
σ/μ = 17.19± 0.34 %
μ = 488.5± 1.8
σ = 84.0± 1.4
Energy loss in ERAM 30
XP
σ/μ = 15.58± 0.29 %
μ = 510.2± 1.8
σ = 79.5± 1.2
WF
σ/μ = 16.14± 0.35 %
μ = 502.0± 1.8
σ = 81.0± 1.5
Energy loss in ERAM 28
XP
σ/μ = 16.39± 0.40 %
μ = 516.0± 2.0
σ = 84.6± 1.8
WF
σ/μ = 16.04± 0.38 %
μ = 508.3± 1.9
σ = 81.6± 1.6
Energy loss in ERAM 19
XP
σ/μ = 16.19± 0.23 %
μ = 515.3± 1.3
σ = 83.4± 1.0
WF
σ/μ = 15.95± 0.22 %
μ = 507.0± 1.3
σ = 80.9± 0.9
Energy loss in ERAM 21
XP
σ/μ = 14.45± 0.07 %
μ = 479.7± 0.4
σ = 69.3± 0.3
WF
σ/μ = 15.07± 0.08 %
μ = 481.6± 0.4
σ = 72.6± 0.3
Energy loss in ERAM 13
XP
σ/μ = 14.78± 0.07 %
μ = 500.7± 0.4
σ = 74.0± 0.3
WF
σ/μ = 15.22± 0.08 %
μ = 500.0± 0.4
σ = 76.1± 0.3
Energy loss in ERAM 9
XP
σ/μ = 14.30± 0.07 %
μ = 512.7± 0.4
σ = 73.3± 0.3
WF
σ/μ = 15.24± 0.09 %
μ = 510.7± 0.4
σ = 77.8± 0.4
Energy loss in ERAM 2
XP
σ/μ = 15.71± 0.09 %
μ = 502.3± 0.4
σ = 78.9± 0.4
WF
σ/μ = 15.78± 0.08 %
μ = 504.2± 0.4
σ = 79.6± 0.3
Energy loss in ERAM 26
XP
σ/μ = 16.23± 0.24 %
μ = 522.9± 1.4
σ = 84.9± 1.0
WF
σ/μ = 16.89± 0.25 %
μ = 512.9± 1.3
σ = 86.6± 1.1
Energy loss in ERAM 17
XP
σ/μ = 16.48± 0.27 %
μ = 526.9± 1.4
σ = 86.8± 1.2
WF
σ/μ = 16.64± 0.28 %
μ = 517.8± 1.4
σ = 86.2± 1.2
Energy loss in ERAM 23
XP
σ/μ = 16.07± 0.27 %
μ = 515.5± 1.5
σ = 82.8± 1.2
WF
σ/μ = 16.41± 0.30 %
μ = 504.7± 1.5
σ = 82.8± 1.3
Energy loss in ERAM 29
XP
σ/μ = 16.66± 0.25 %
μ = 505.1± 1.2
σ = 84.1± 1.1
WF
σ/μ = 17.30± 0.28 %
μ = 498.7± 1.3
σ = 86.3± 1.2
Energy loss in ERAM 1
XP
σ/μ = 14.77± 0.07 %
μ = 499.4± 0.4
σ = 73.8± 0.3
WF
σ/μ = 15.03± 0.07 %
μ = 497.8± 0.4
σ = 74.8± 0.3
Energy loss in ERAM 10
XP
σ/μ = 15.05± 0.08 %
μ = 500.6± 0.4
σ = 75.4± 0.3
WF
σ/μ = 14.94± 0.07 %
μ = 500.8± 0.4
σ = 74.8± 0.3
Energy loss in ERAM 11
XP
σ/μ = 14.99± 0.08 %
μ = 499.2± 0.4
σ = 74.8± 0.3
WF
σ/μ = 14.84± 0.07 %
μ = 501.3± 0.4
σ = 74.4± 0.3
Energy loss in ERAM 3
XP
σ/μ = 14.59± 0.07 %
μ = 486.8± 0.4
σ = 71.0± 0.3
WF
σ/μ = 15.06± 0.07 %
μ = 488.6± 0.4
σ = 73.6± 0.3
Count
dE/dx (ADC counts/cm)



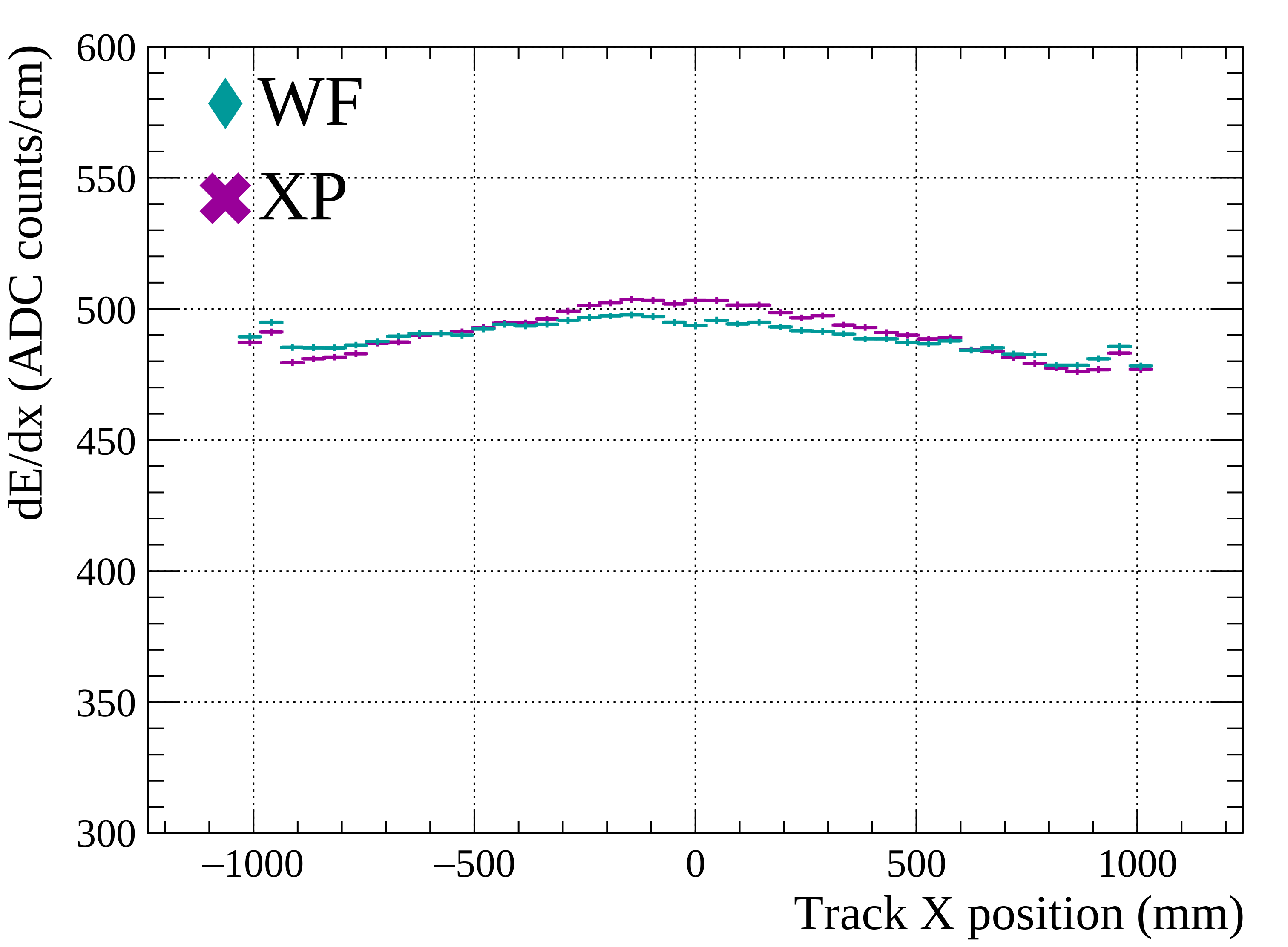
Energy loss in ERAM 47
XP
σ/μ = 16.55± 0.32 %
μ = 495.5± 1.5
σ = 82.0± 1.3
WF
σ/μ = 16.34± 0.28 %
μ = 484.0± 1.4
σ = 79.1± 1.1
Energy loss in ERAM 16
XP
σ/μ = 16.18± 0.33 %
μ = 502.9± 1.8
σ = 81.4± 1.4
WF
σ/μ = 16.24± 0.31 %
μ = 493.1± 1.7
σ = 80.1± 1.2
Energy loss in ERAM 14
XP
σ/μ = 17.32± 0.40 %
μ = 515.5± 2.0
σ = 89.3± 1.7
WF
σ/μ = 17.87± 0.41 %
μ = 508.5± 2.1
σ = 90.9± 1.7
Energy loss in ERAM 15
XP
σ/μ = 17.36± 0.26 %
μ = 542.7± 1.5
σ = 94.2± 1.2
WF
σ/μ = 17.94± 0.30 %
μ = 536.4± 1.7
σ = 96.2± 1.3
Energy loss in ERAM 42
XP
σ/μ = 14.58± 0.07 %
μ = 463.2± 0.3
σ = 67.5± 0.3
WF
σ/μ = 14.51± 0.06 %
μ = 460.4± 0.3
σ = 66.8± 0.2
Energy loss in ERAM 45
XP
σ/μ = 14.93± 0.08 %
μ = 476.1± 0.4
σ = 71.1± 0.3
WF
σ/μ = 14.77± 0.07 %
μ = 477.0± 0.4
σ = 70.4± 0.3
Energy loss in ERAM 37
XP
σ/μ = 14.78± 0.07 %
μ = 496.5± 0.4
σ = 73.4± 0.3
WF
σ/μ = 14.86± 0.07 %
μ = 497.0± 0.4
σ = 73.8± 0.3
Energy loss in ERAM 36
XP
σ/μ = 15.37± 0.08 %
μ = 494.2± 0.4
σ = 76.0± 0.3
WF
σ/μ = 15.21± 0.07 %
μ = 496.0± 0.4
σ = 75.4± 0.3
Energy loss in ERAM 20
XP
σ/μ = 16.39± 0.23 %
μ = 502.9± 1.2
σ = 82.4± 0.9
WF
σ/μ = 16.96± 0.26 %
μ = 492.0± 1.2
σ = 83.4± 1.1
Energy loss in ERAM 38
XP
σ/μ = 16.09± 0.26 %
μ = 499.3± 1.3
σ = 80.3± 1.1
WF
σ/μ = 16.35± 0.26 %
μ = 487.9± 1.3
σ = 79.8± 1.1
Energy loss in ERAM 7
XP
σ/μ = 16.78± 0.31 %
μ = 512.9± 1.4
σ = 86.1± 1.3
WF
σ/μ = 16.57± 0.28 %
μ = 503.5± 1.5
σ = 83.4± 1.2
Energy loss in ERAM 44
XP
σ/μ = 16.53± 0.23 %
μ = 501.9± 1.2
σ = 83.0± 0.9
WF
σ/μ = 16.97± 0.25 %
μ = 491.7± 1.2
σ = 83.5± 1.0
Energy loss in ERAM 43
XP
σ/μ = 15.29± 0.08 %
μ = 491.2± 0.4
σ = 75.1± 0.3
WF
σ/μ = 15.11± 0.07 %
μ = 490.5± 0.4
σ = 74.1± 0.3
Energy loss in ERAM 39
XP
σ/μ = 14.60± 0.07 %
μ = 472.0± 0.4
σ = 68.9± 0.3
WF
σ/μ = 15.68± 0.09 %
μ = 465.0± 0.4
σ = 72.9± 0.3
Energy loss in ERAM 41
XP
σ/μ = 14.58± 0.07 %
μ = 490.7± 0.4
σ = 71.6± 0.3
WF
σ/μ = 15.07± 0.07 %
μ = 489.0± 0.4
σ = 73.7± 0.3
Energy loss in ERAM 46
XP
σ/μ = 15.01± 0.07 %
μ = 471.5± 0.4
σ = 70.8± 0.3
WF
σ/μ = 15.14± 0.07 %
μ = 470.8± 0.4
σ = 71.3± 0.3
Count
dE/dx (ADC counts/cm)



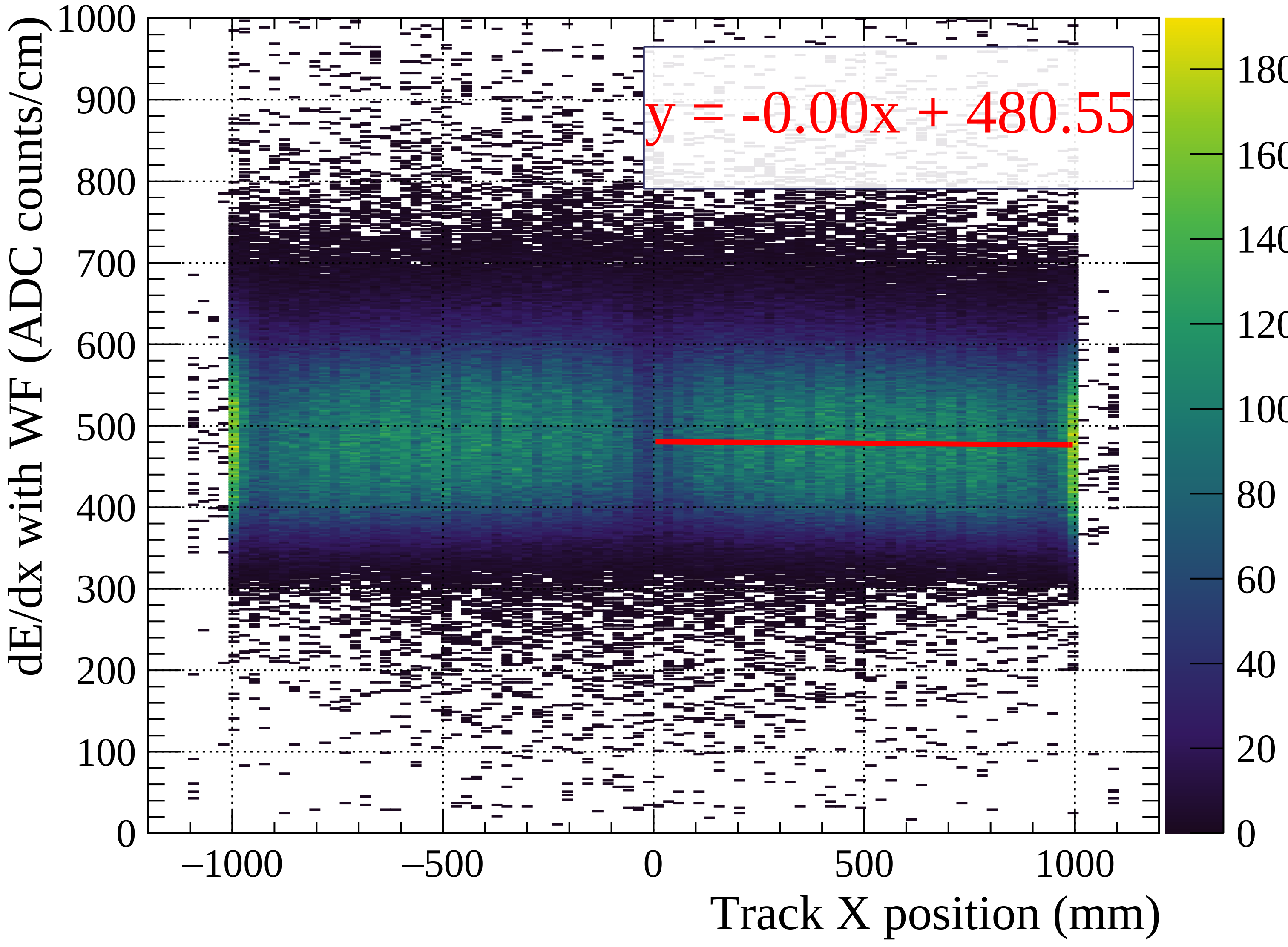
Entries 852899
Mean x 491.2
Mean y 492.1
Std Dev x 78.17
Std Dev y 78.24
0 16 366
6 852457 54
0 0 0
dE/dx with XP (ADC counts/cm)
dE/dx with WF (ADC counts/cm)
×10^3
0
100
200
300
400
500
600
700
800
900
1000
2
4
6
8
10
12



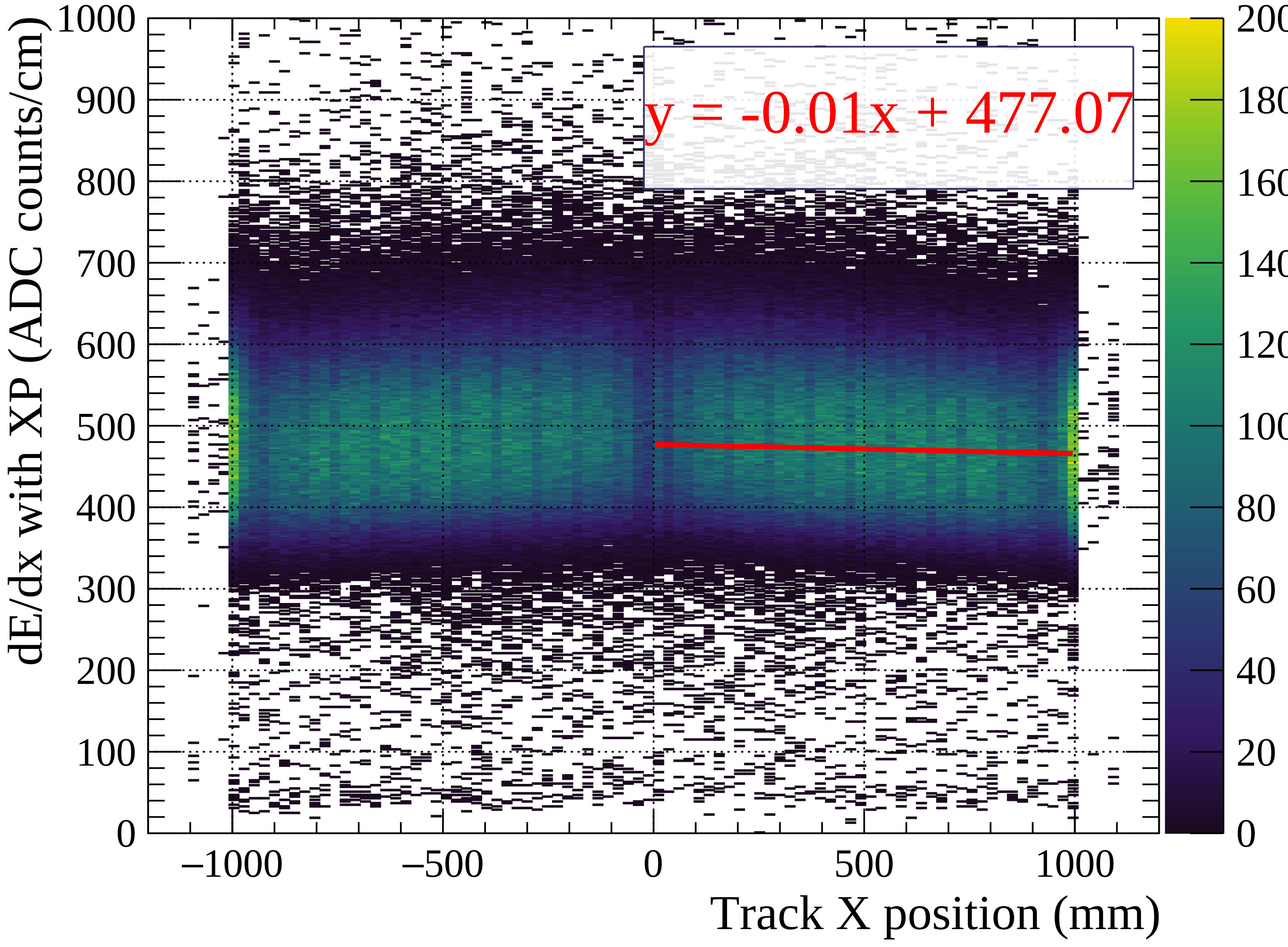
dE/dx resolution (%)
WF
XP
30
25
20
15
10
5
−1000
−500
0
500
1000
Track X position (mm)



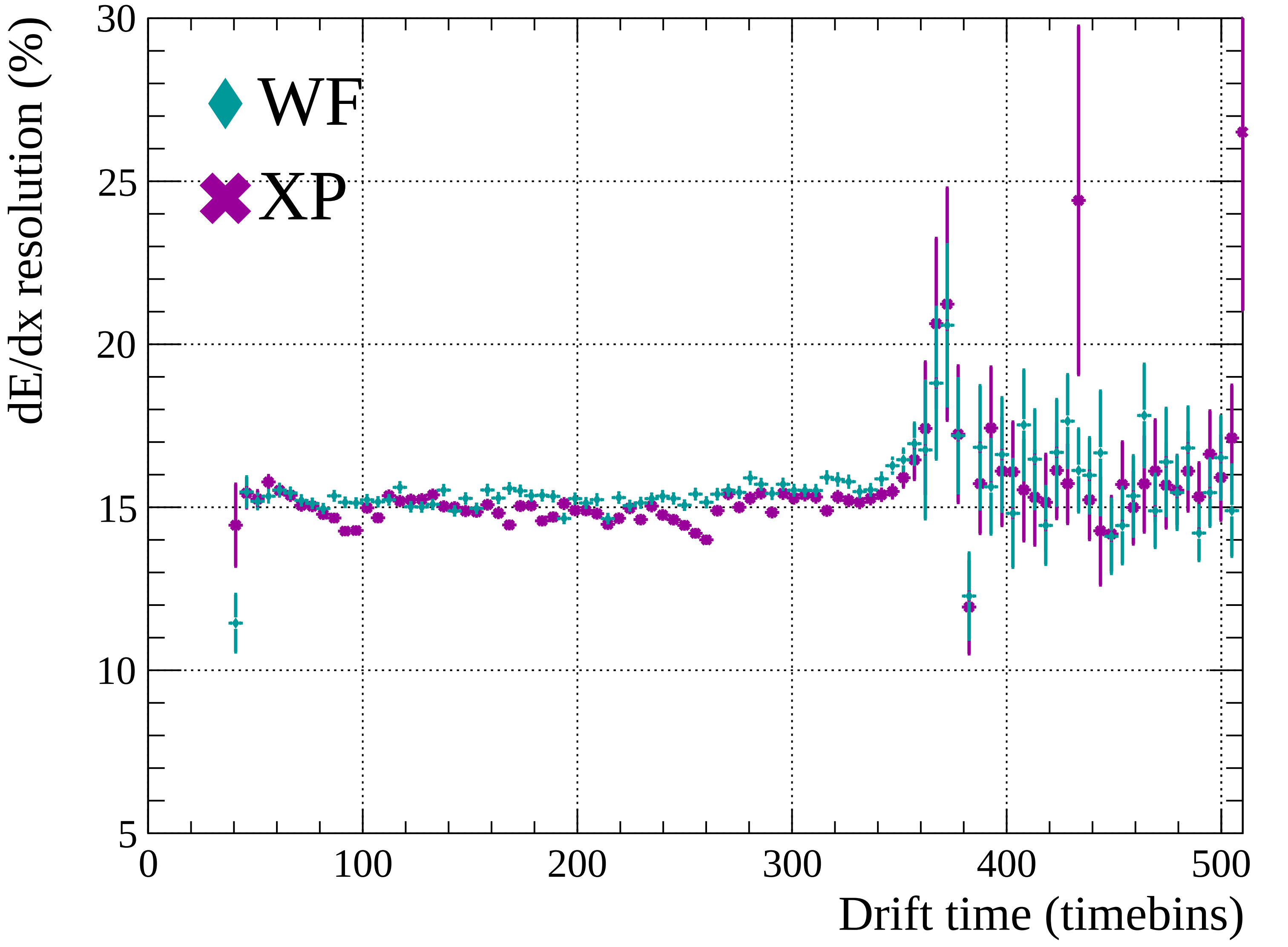
dE/dx (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
−1000
−500
0
500
1000
Track X position (mm)



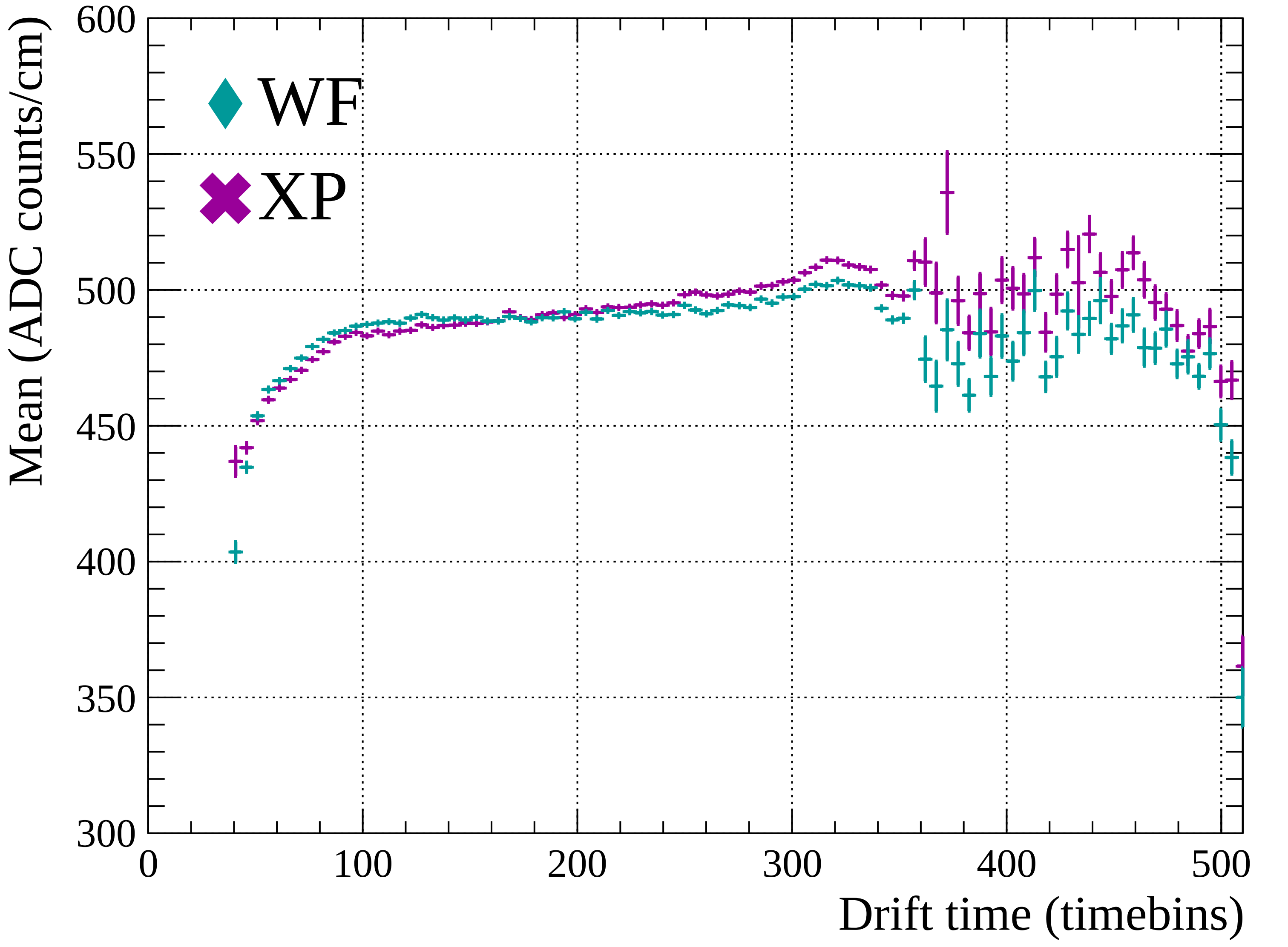
y = -0.00x + 480.55
dE/dx with WF (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000
0
20
40
60
80
100
120
140
160
180



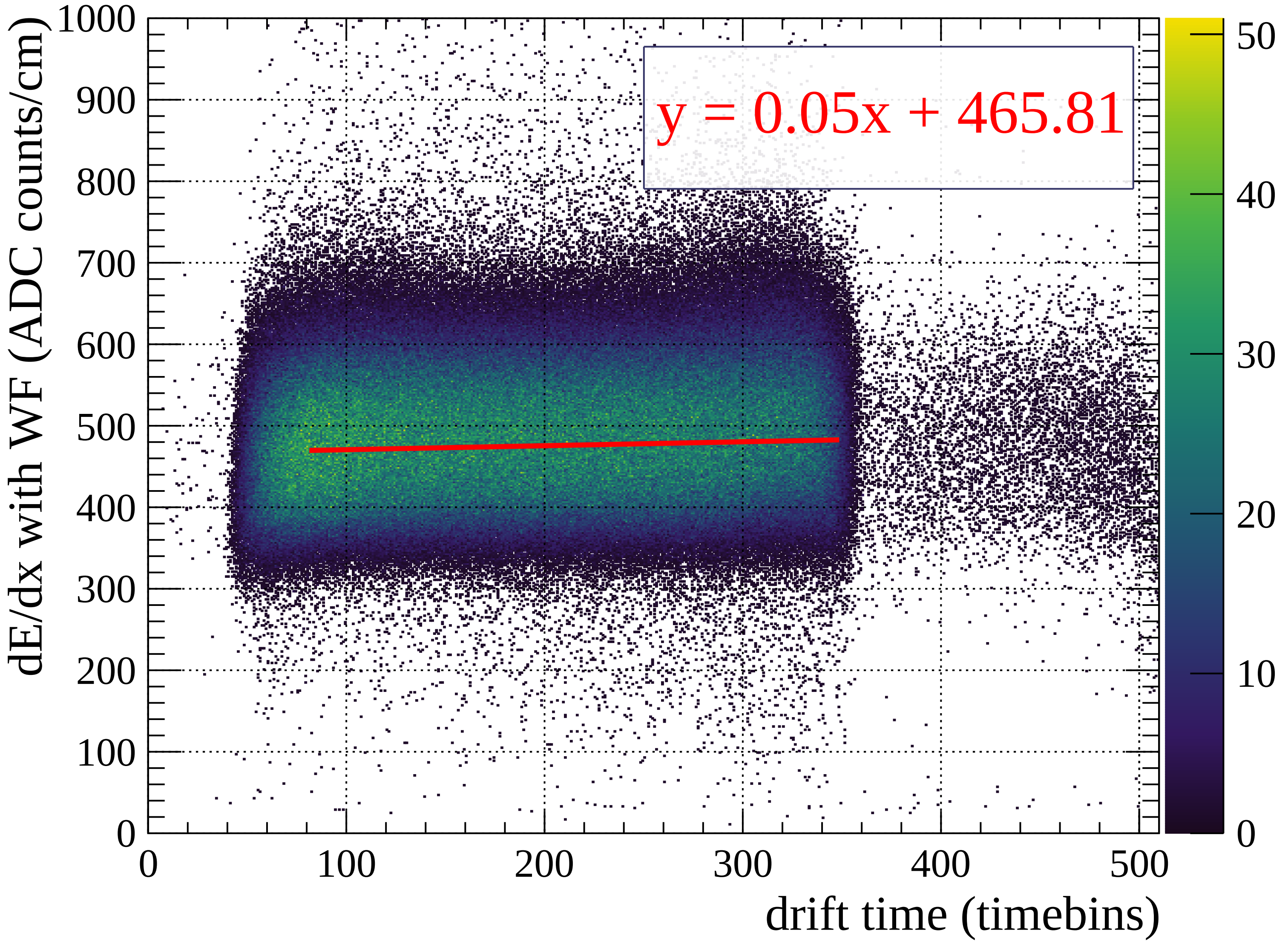
y = -0.01x + 477.07
dE/dx with XP (ADC counts/cm)
Track X position (mm)
0
100
200
300
400
500
600
700
800
900
1000
−1000
−500
0
500
1000



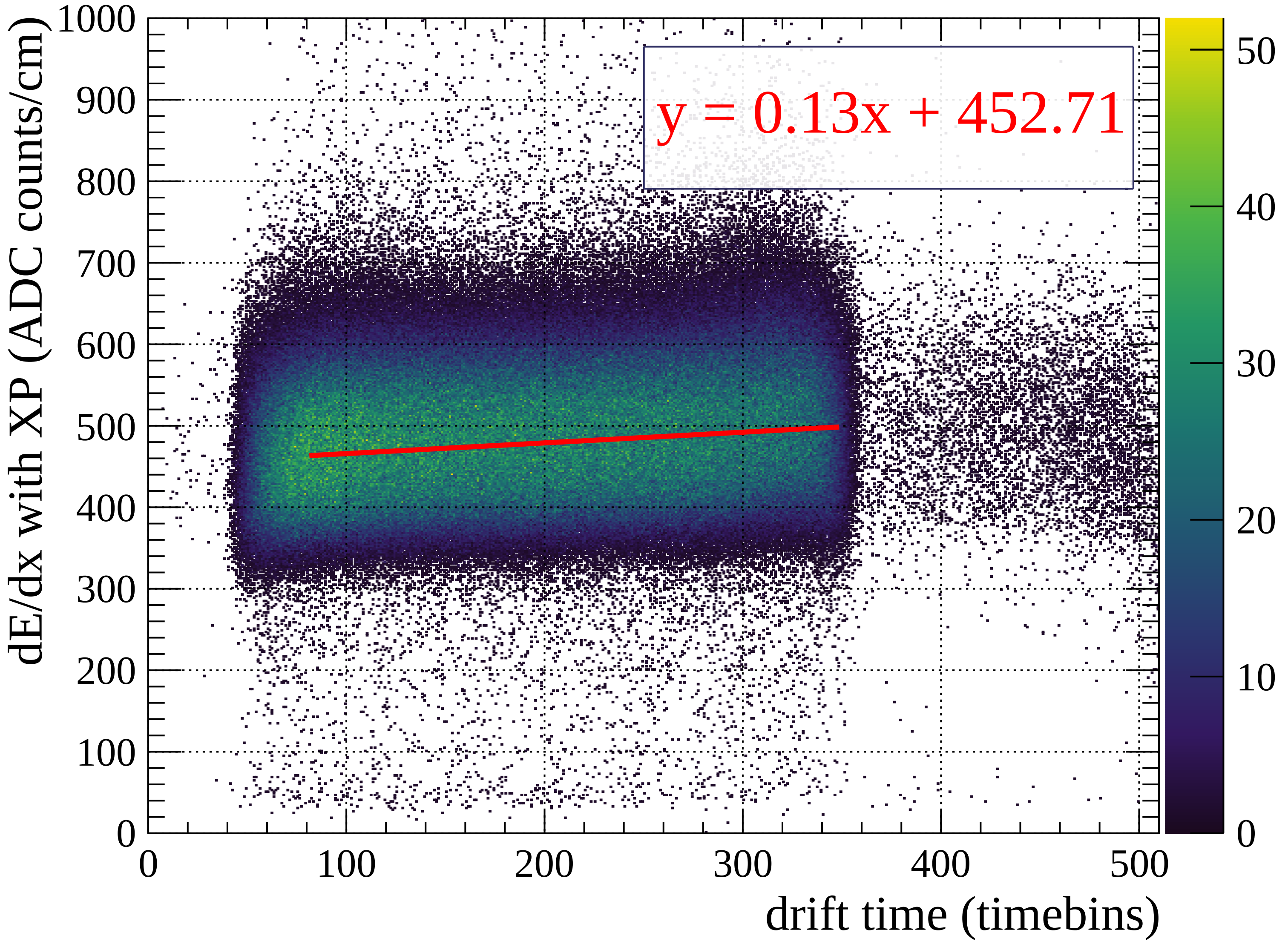
dE/dx resolution (%)
Drift time (timebins)
WF
XP
30
25
20
15
10
5
0
100
200
300
400
500



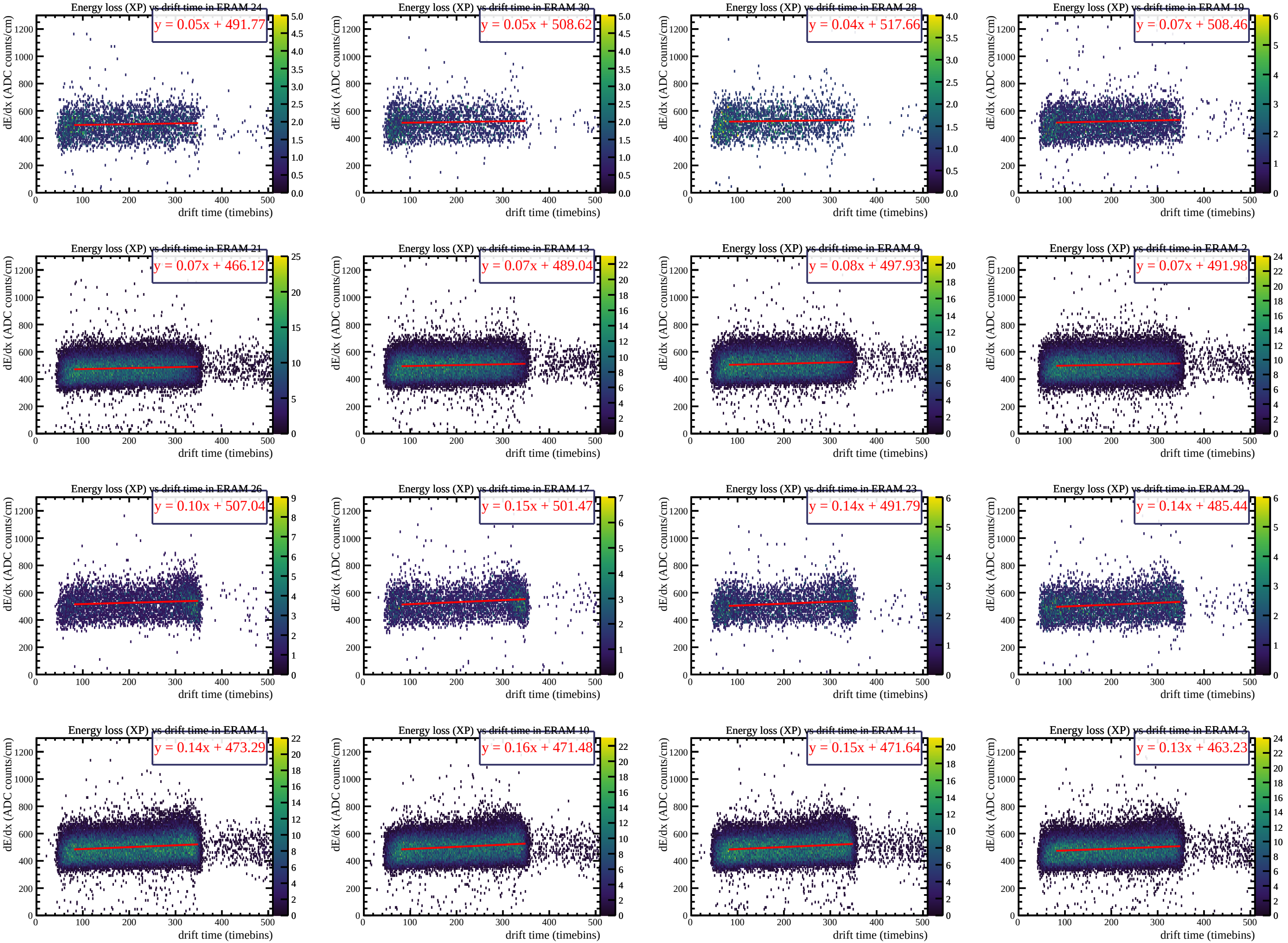
WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
100
200
300
400
500
Drift time (timebins)



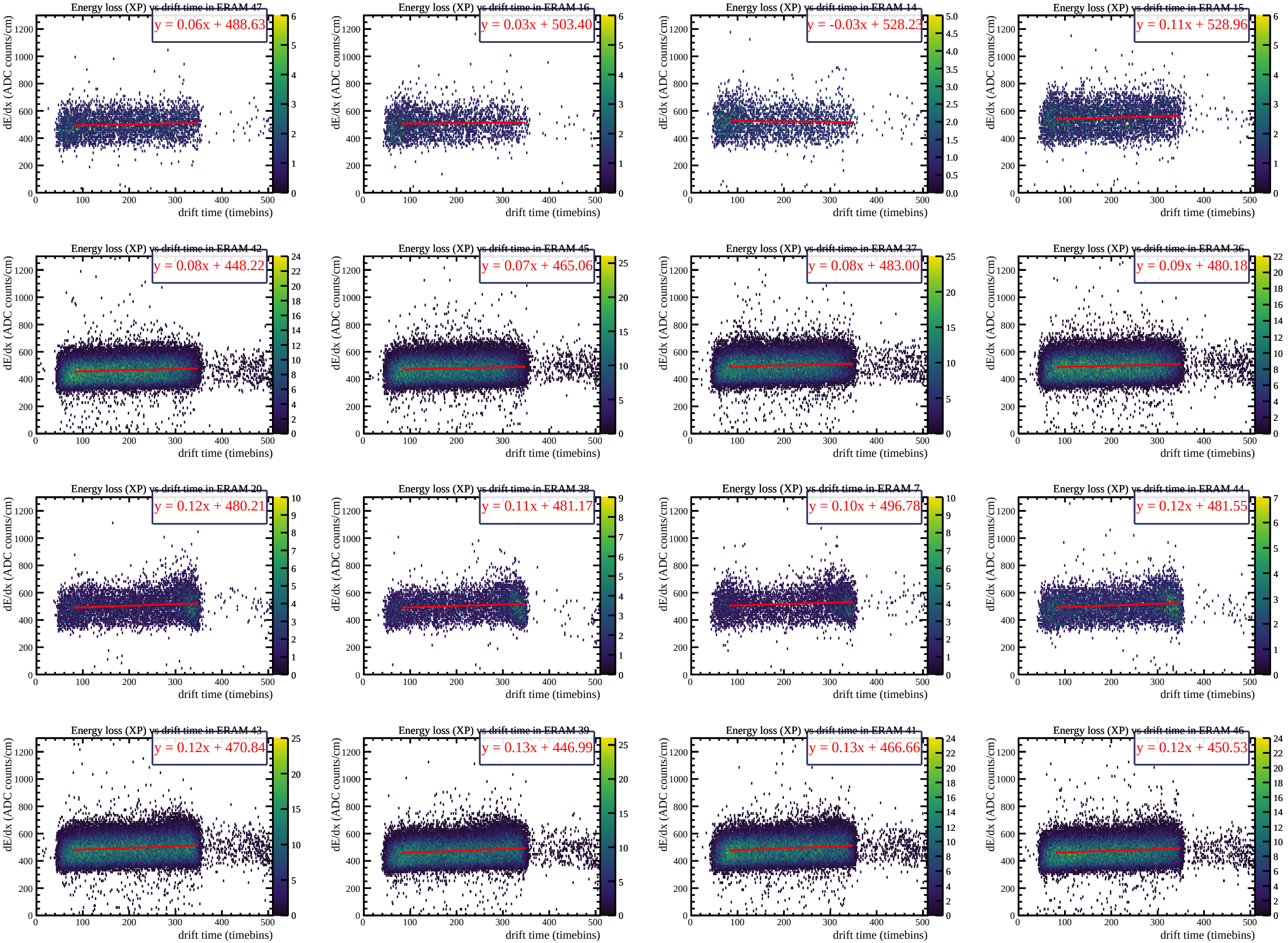
y = 0.05x + 465.81
dE/dx with WF (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
10
20
30
40
50

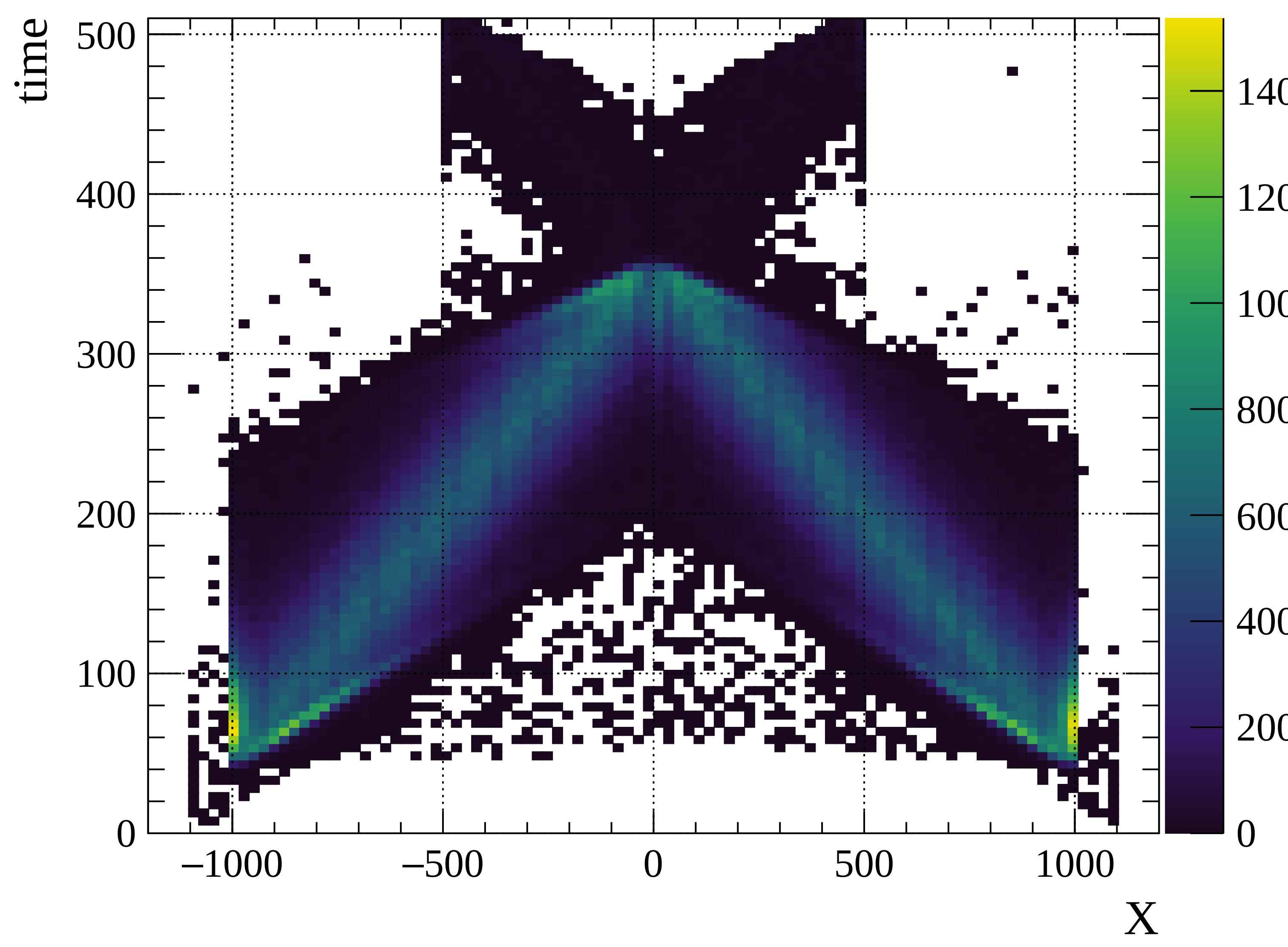


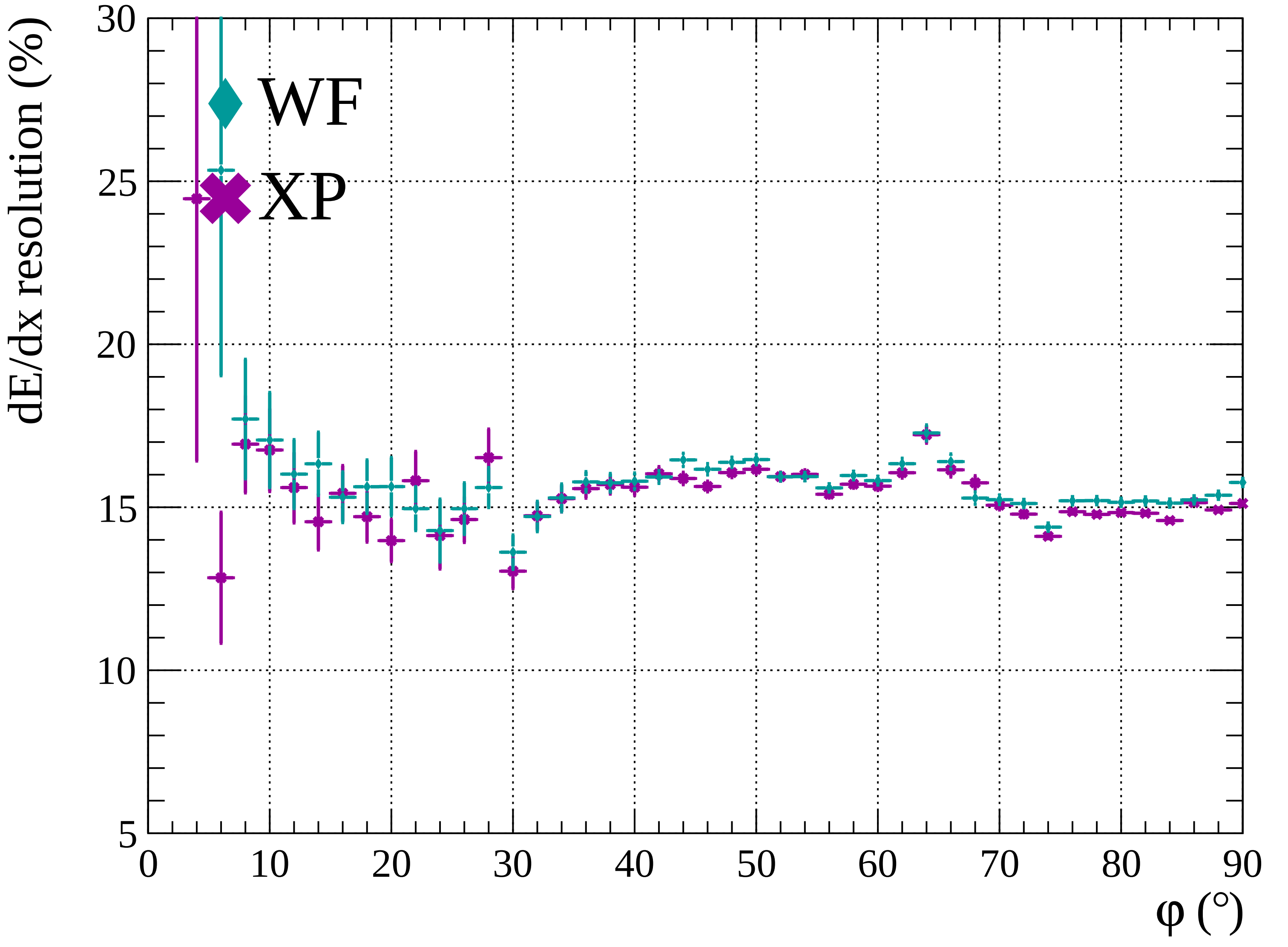
y = 0.13x + 452.71
dE/dx with XP (ADC counts/cm)
drift time (timebins)
0
100
200
300
400
500
600
700
800
900
1000
0
100
200
300
400
500
0
10
20
30
40
50



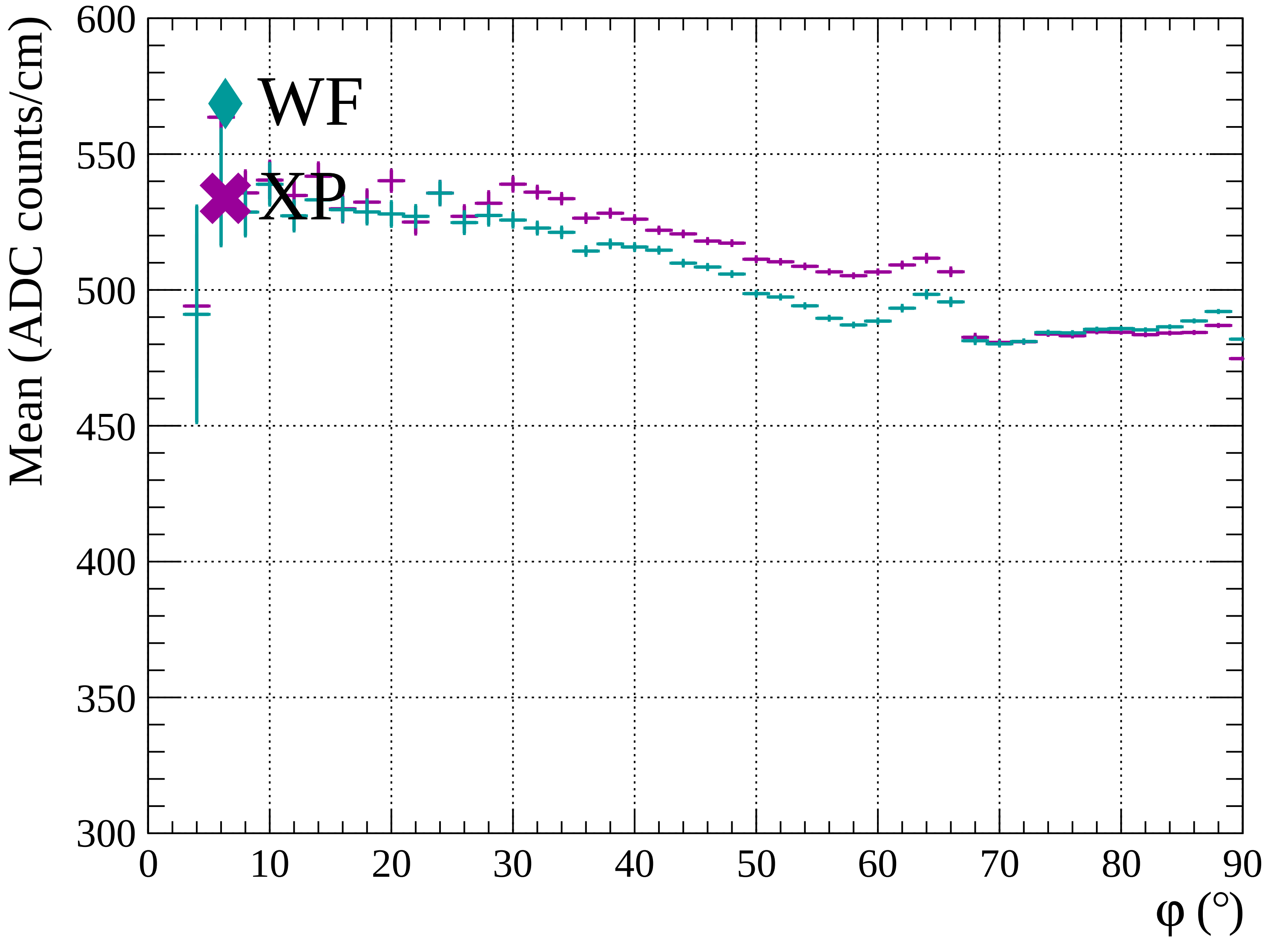
Energy loss (XP) vs drift time in ERAM 24
y = 0.05x + 491.77
Energy loss (XP) vs drift time in ERAM 30
y = 0.05x + 508.62
Energy loss (XP) vs drift time in ERAM 28
y = 0.04x + 517.66
Energy loss (XP) vs drift time in ERAM 19
y = 0.07x + 508.46
Energy loss (XP) vs drift time in ERAM 21
y = 0.07x + 466.12
Energy loss (XP) vs drift time in ERAM 13
y = 0.07x + 489.04
Energy loss (XP) vs drift time in ERAM 9
y = 0.08x + 497.93
Energy loss (XP) vs drift time in ERAM 2
y = 0.07x + 491.98
Energy loss (XP) vs drift time in ERAM 26
y = 0.10x + 507.04
Energy loss (XP) vs drift time in ERAM 17
y = 0.15x + 501.47
Energy loss (XP) vs drift time in ERAM 23
y = 0.14x + 491.79
Energy loss (XP) vs drift time in ERAM 29
y = 0.14x + 485.44
Energy loss (XP) vs drift time in ERAM 1
y = 0.14x + 473.29
Energy loss (XP) vs drift time in ERAM 10
y = 0.16x + 471.48
Energy loss (XP) vs drift time in ERAM 11
y = 0.15x + 471.64
Energy loss (XP) vs drift time in ERAM 3
y = 0.13x + 463.23
dE/dx (ADC counts/cm)
drift time (timebins)


Energy loss (XP) vs drift time in ERAM 47
y = 0.06x + 488.63
Energy loss (XP) vs drift time in ERAM 16
y = 0.03x + 503.40
Energy loss (XP) vs drift time in ERAM 14
y = -0.03x + 528.23
Energy loss (XP) vs drift time in ERAM 15
y = 0.11x + 528.96
Energy loss (XP) vs drift time in ERAM 42
y = 0.08x + 448.22
Energy loss (XP) vs drift time in ERAM 45
y = 0.07x + 465.06
Energy loss (XP) vs drift time in ERAM 37
y = 0.08x + 483.00
Energy loss (XP) vs drift time in ERAM 36
y = 0.09x + 480.18
Energy loss (XP) vs drift time in ERAM 20
y = 0.12x + 480.21
Energy loss (XP) vs drift time in ERAM 38
y = 0.11x + 481.17
Energy loss (XP) vs drift time in ERAM 7
y = 0.10x + 496.78
Energy loss (XP) vs drift time in ERAM 44
y = 0.12x + 481.55
Energy loss (XP) vs drift time in ERAM 43
y = 0.12x + 470.84
Energy loss (XP) vs drift time in ERAM 39
y = 0.13x + 446.99
Energy loss (XP) vs drift time in ERAM 41
y = 0.13x + 466.66
Energy loss (XP) vs drift time in ERAM 46
y = 0.12x + 450.53
dE/dx (ADC counts/cm)
drift time (timebins)

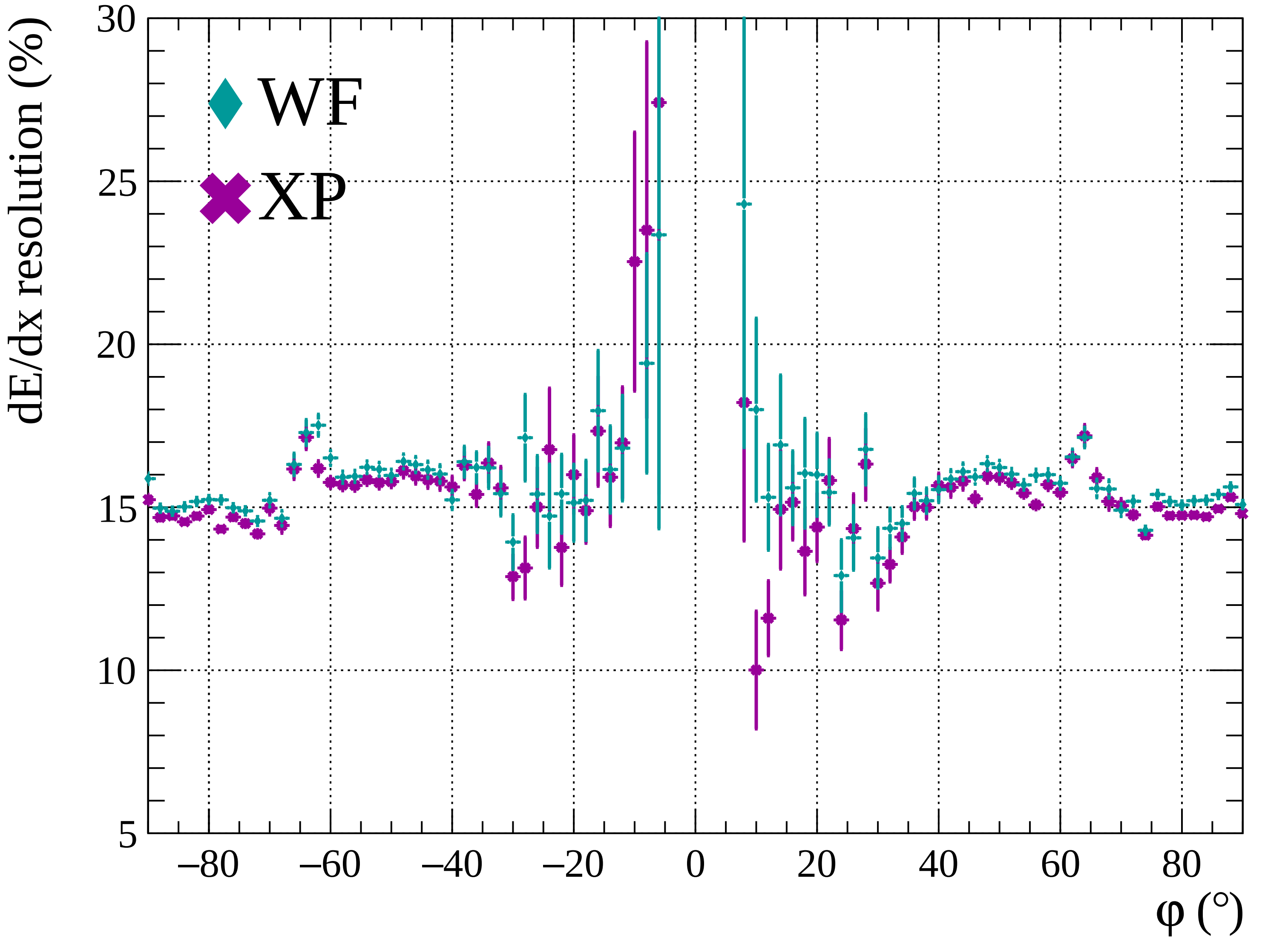

time
500
400
300
200
100
0
−1000
−500
0
500
1000
X
140
120
100
800
600
400
200
0



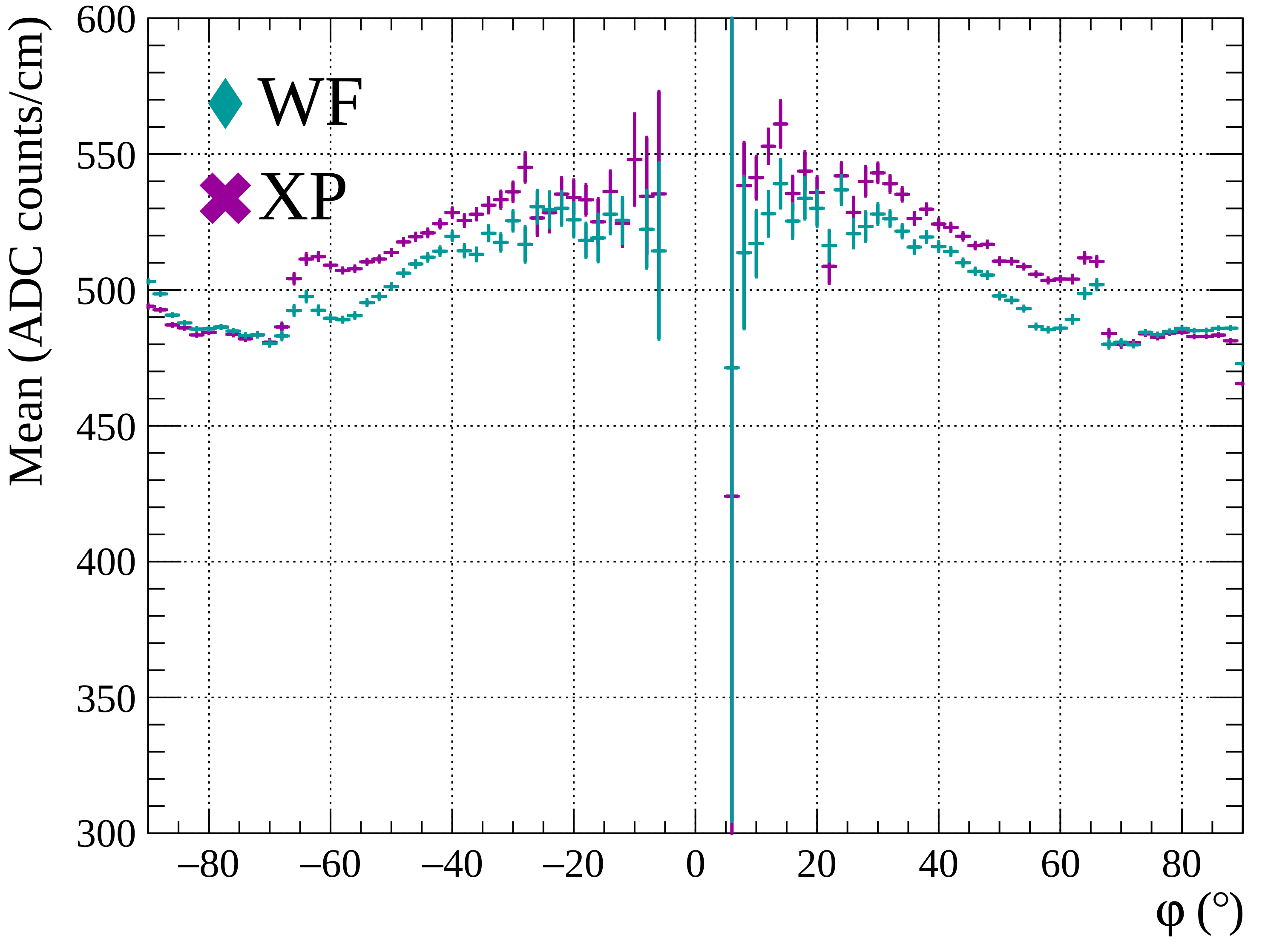
dE/dx resolution (%)
WF
XP
φ (°)
30
25
20
15
10
5
0
10
20
30
40
50
60
70
80
90



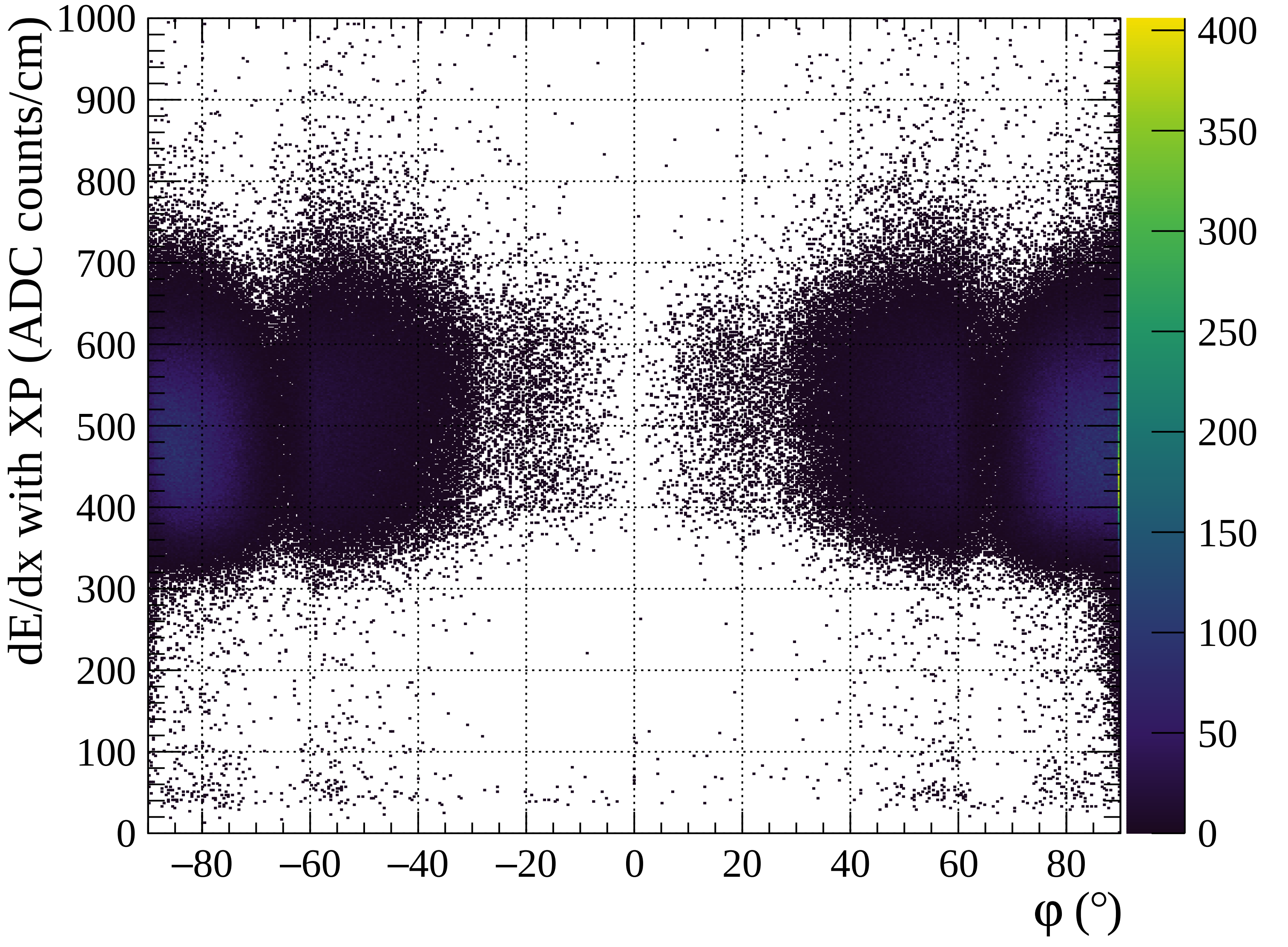
WF
XP
Mean (ADC counts/cm)
φ (°)
300
350
400
450
500
550
600
0
10
20
30
40
50
60
70
80
90



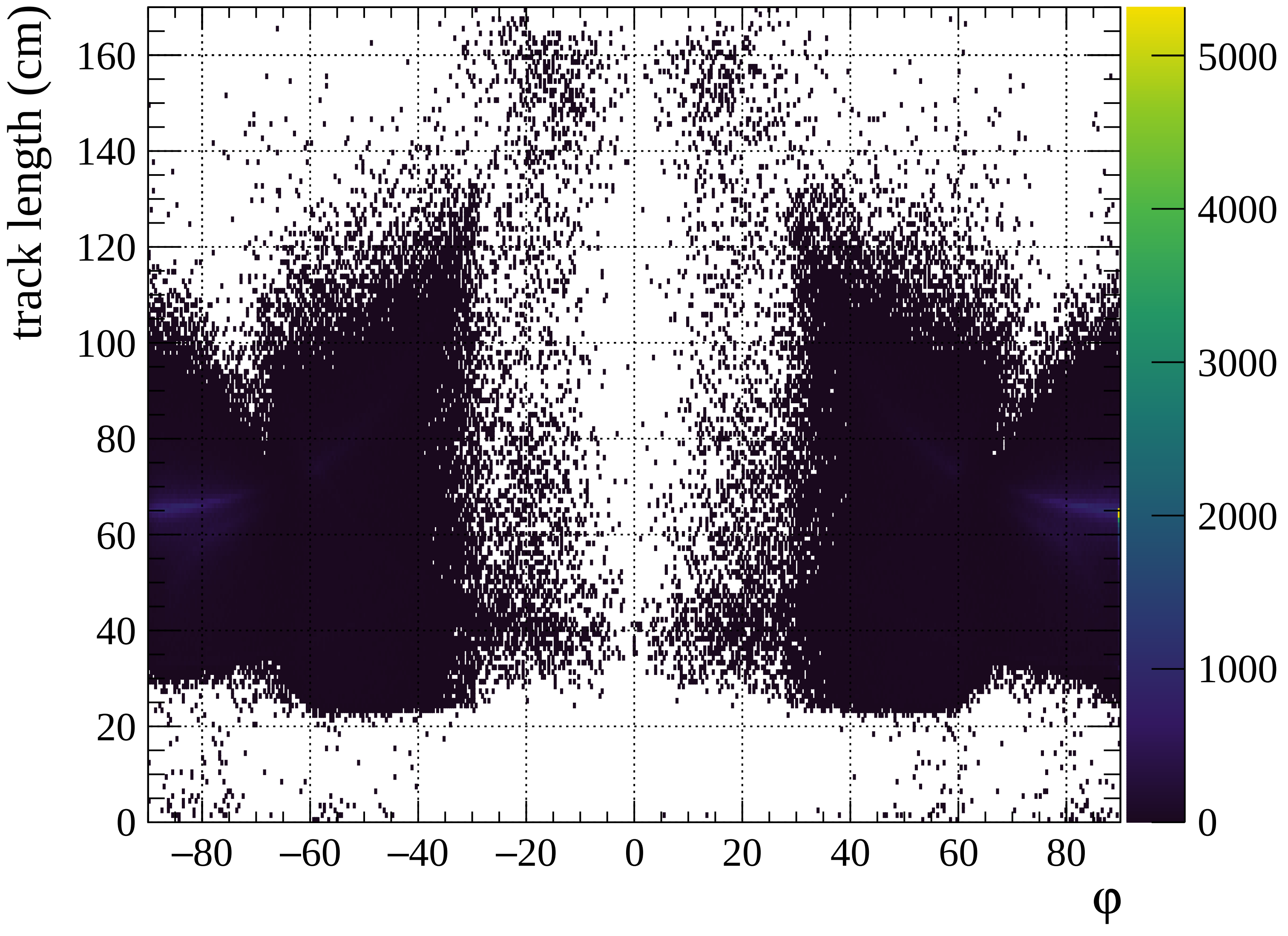
dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−80
−60
−40
−20
0
20
40
60
80
φ (°)



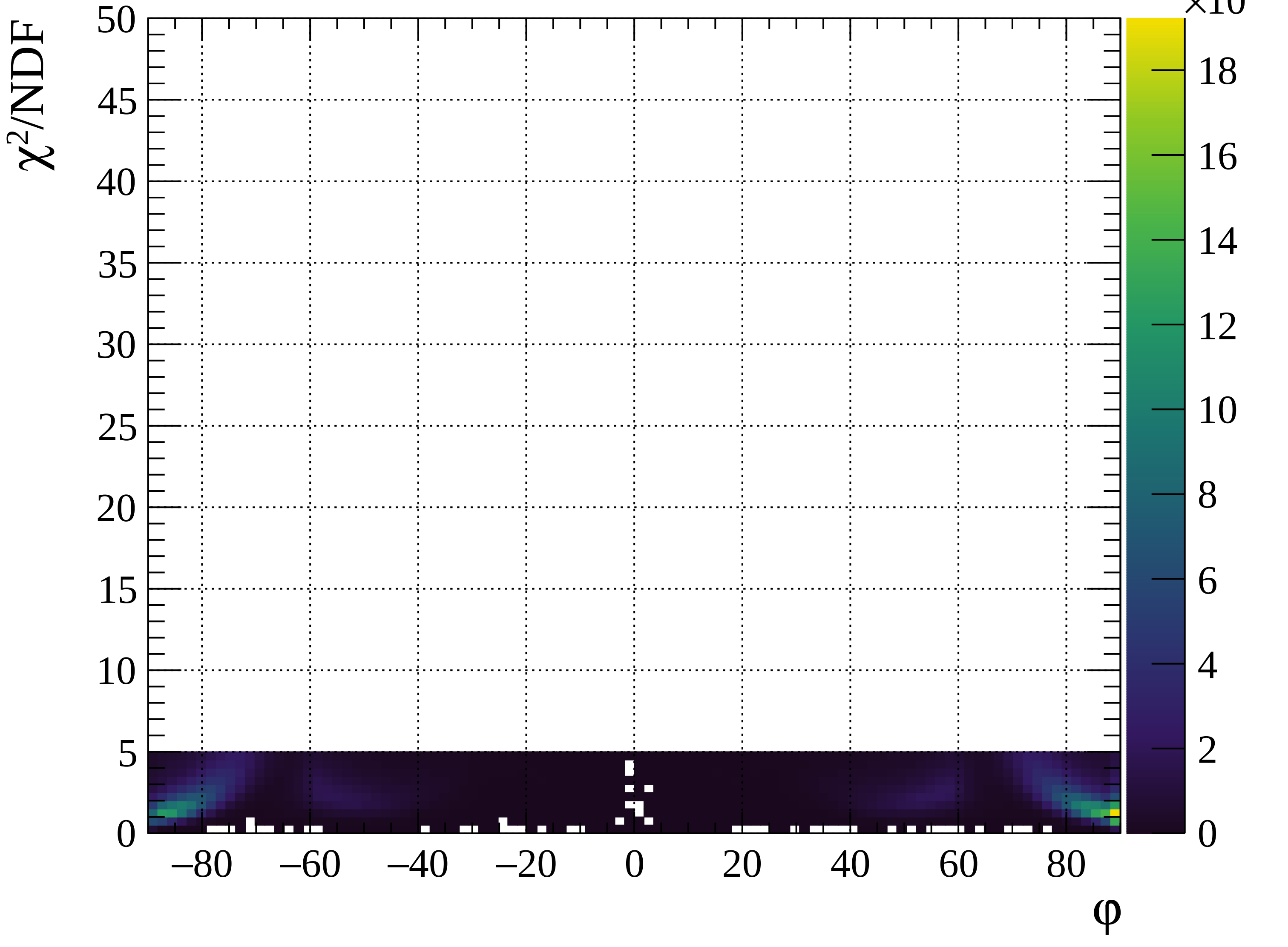
Mean (ADC counts/cm)
600
550
500
450
400
350
300
WF
XP
−80
−60
−40
−20
0
20
40
60
80
φ (°)



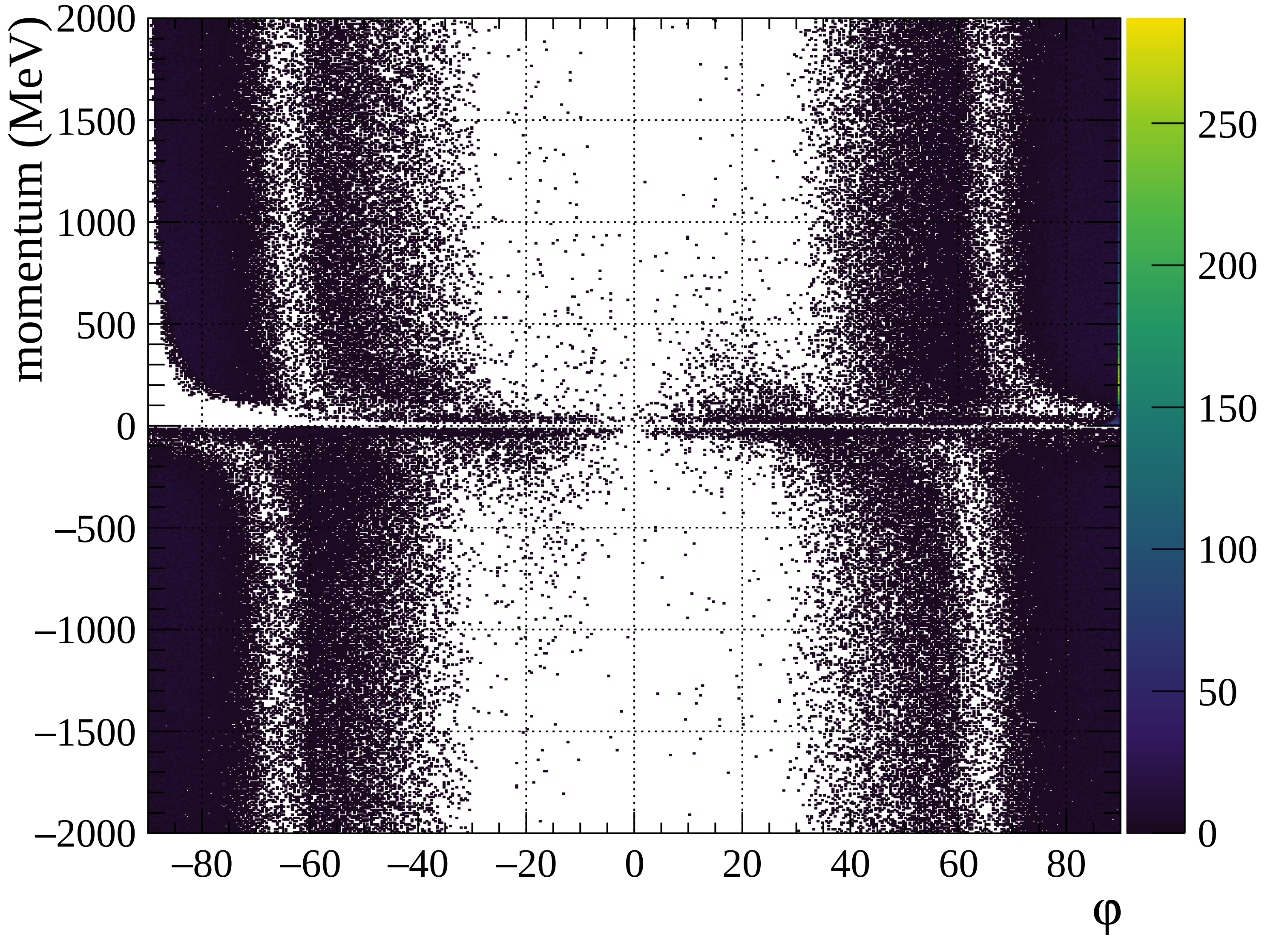
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
φ (°)
400
350
300
250
200
150
100
50
0

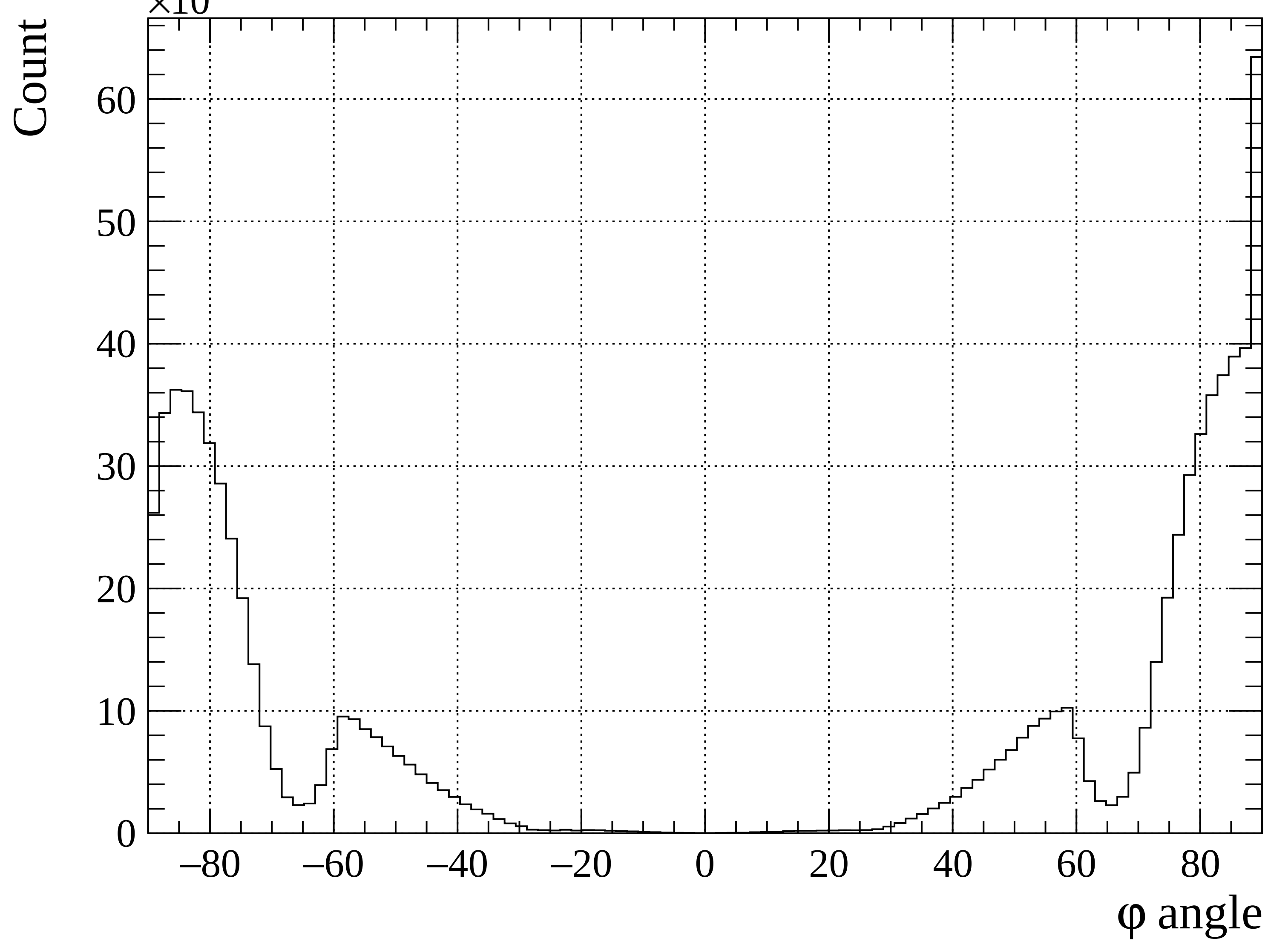


track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
φ
5000
4000
3000
2000
1000
0

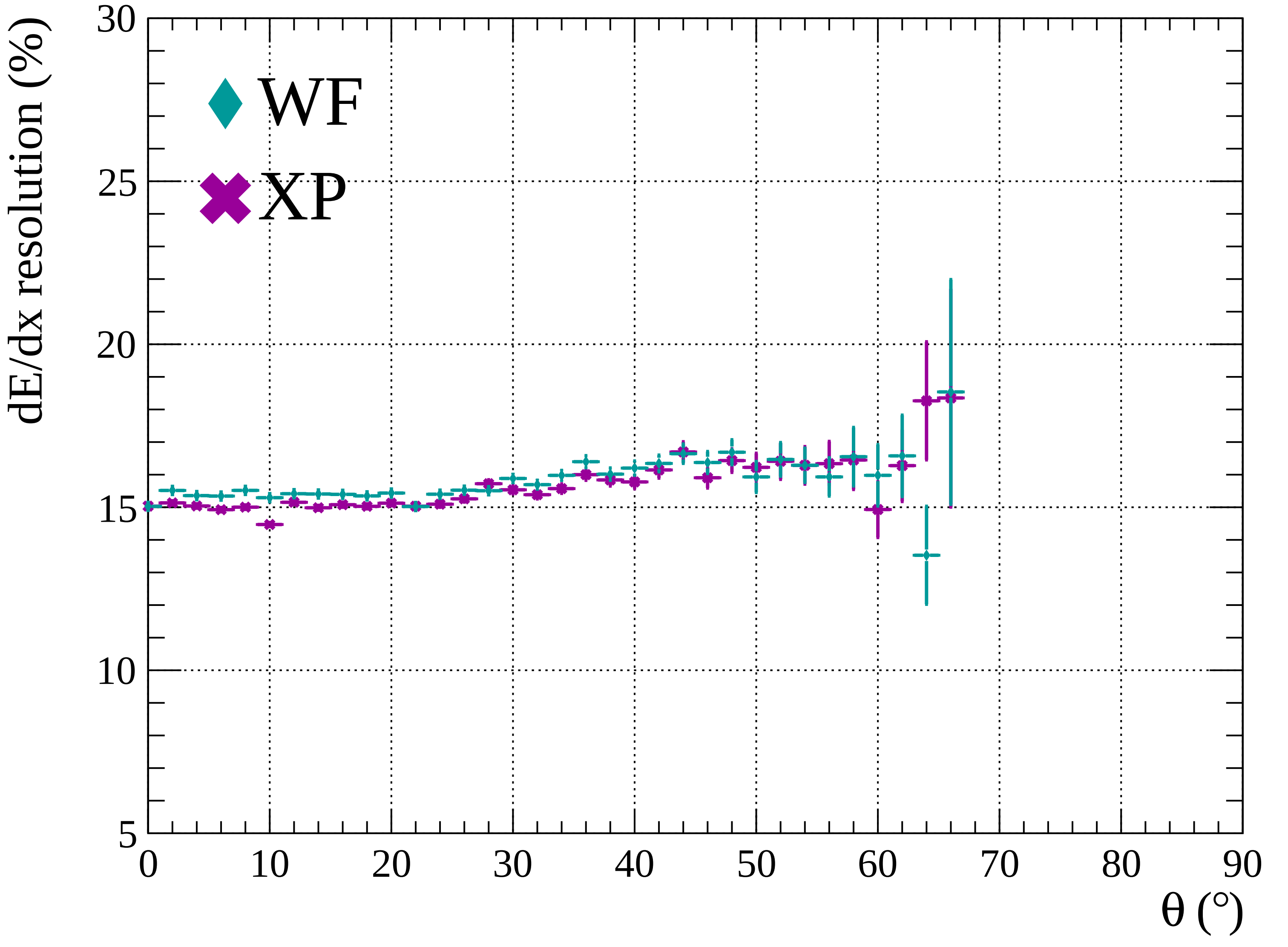


χ^2/NDF
φ
×10
0
2
4
6
8
10
12
14
16
18

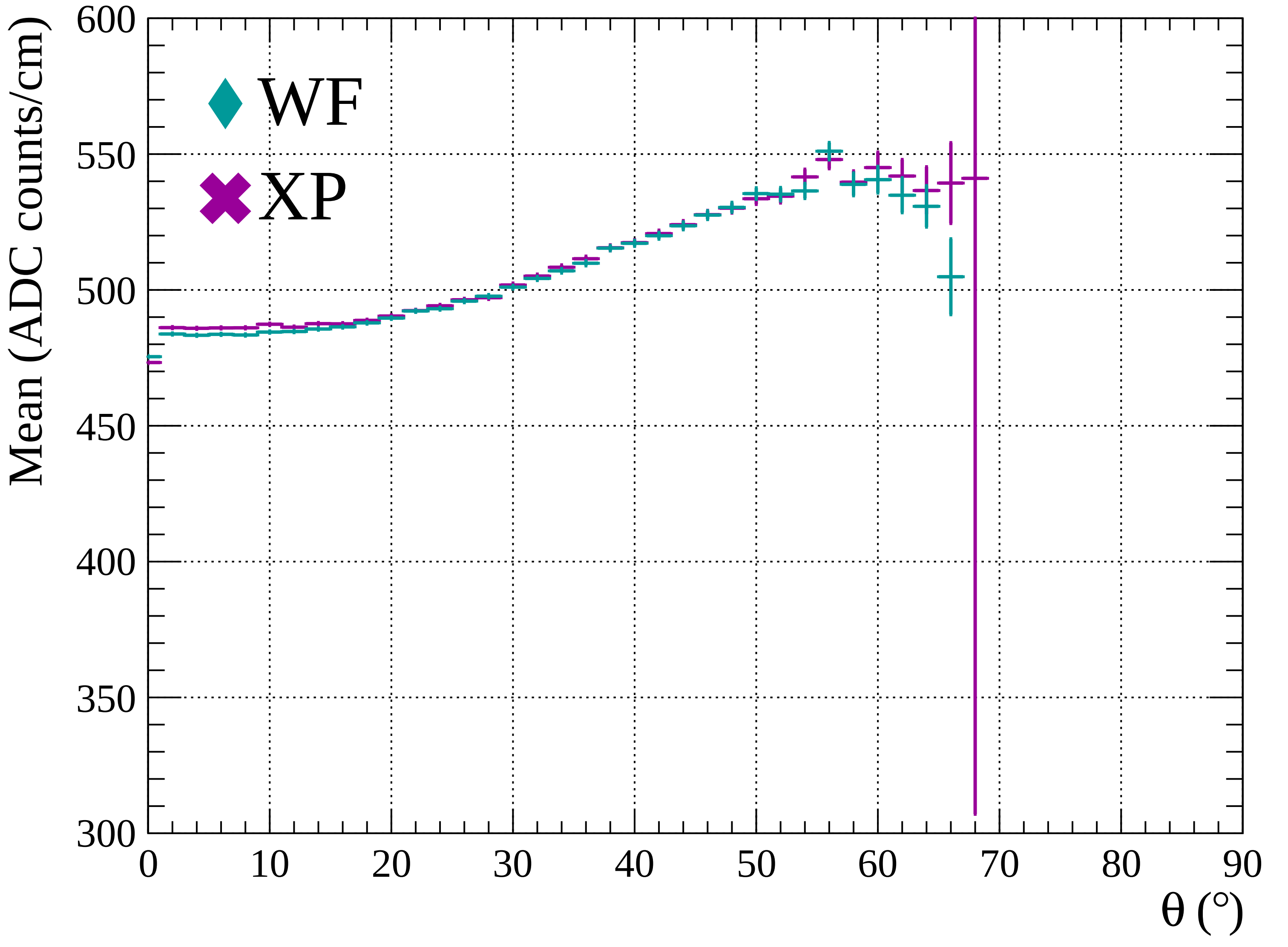

momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
φ
250
200
150
100
50
0



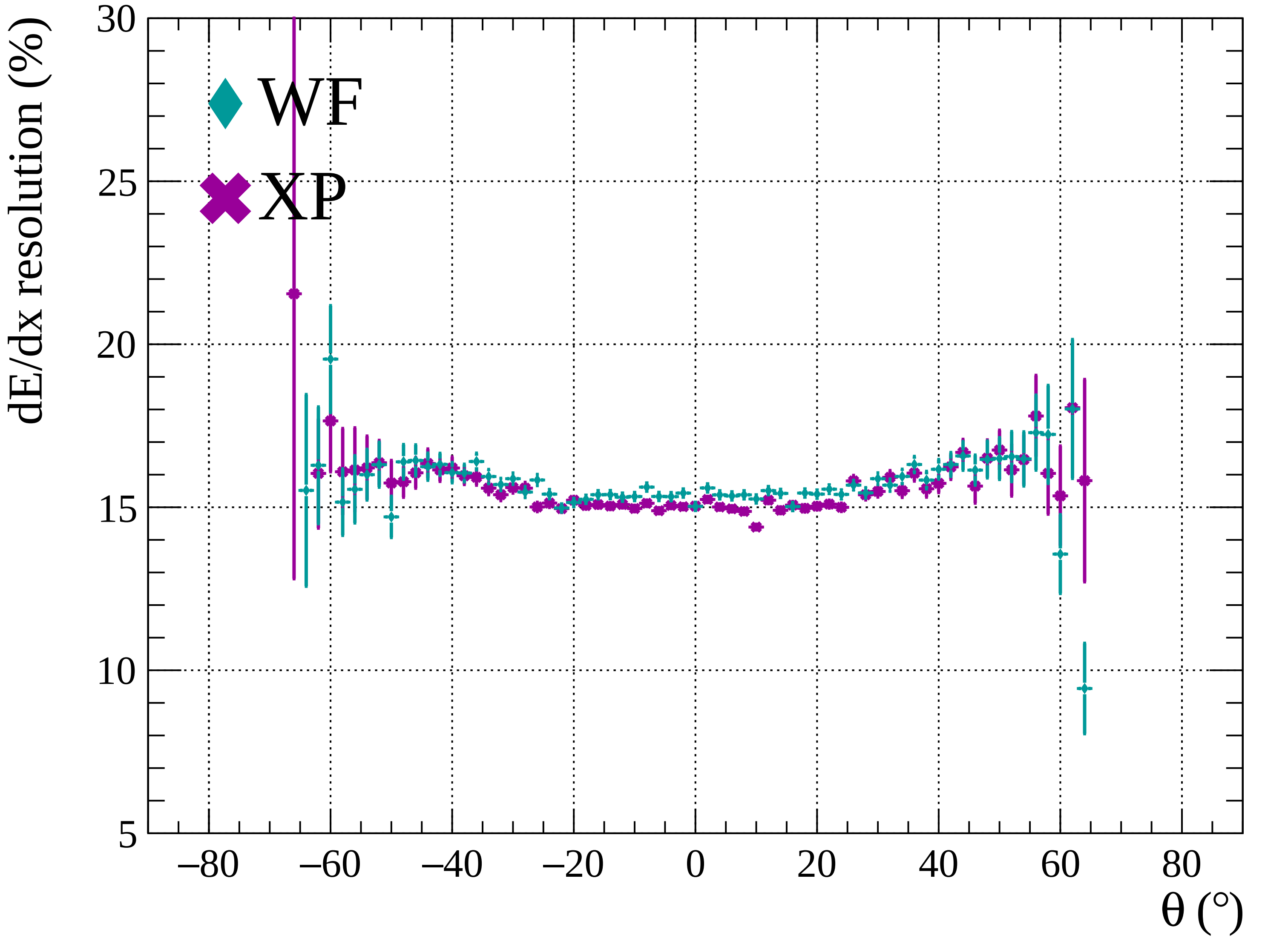
×10
Count
0
10
20
30
40
50
60
−80
−60
−40
−20
0
20
40
60
80
φ angle



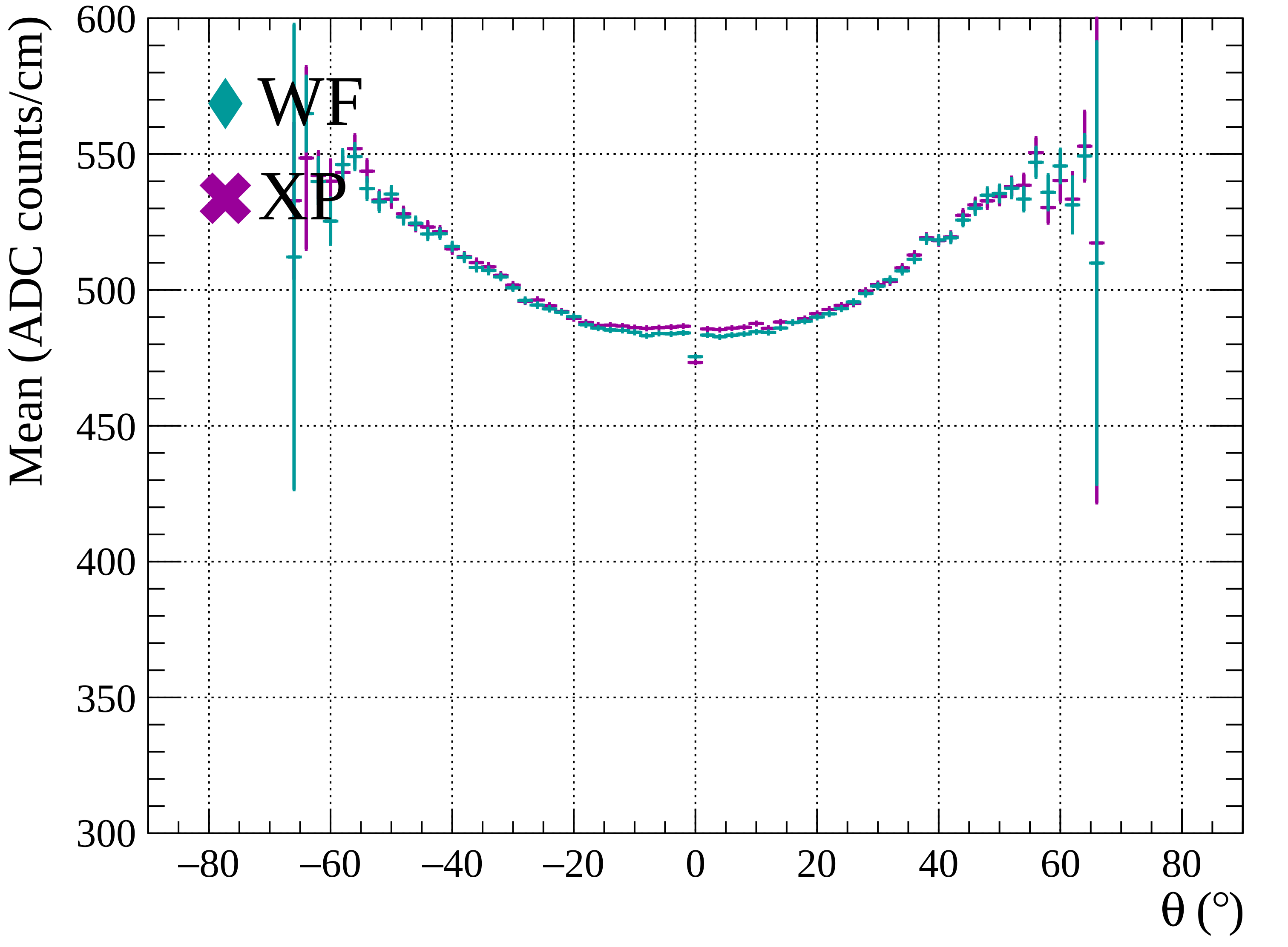
dE/dx resolution (%)
30
25
20
15
10
5
0
10
20
30
40
50
60
70
80
90
θ (°)
WF
XP



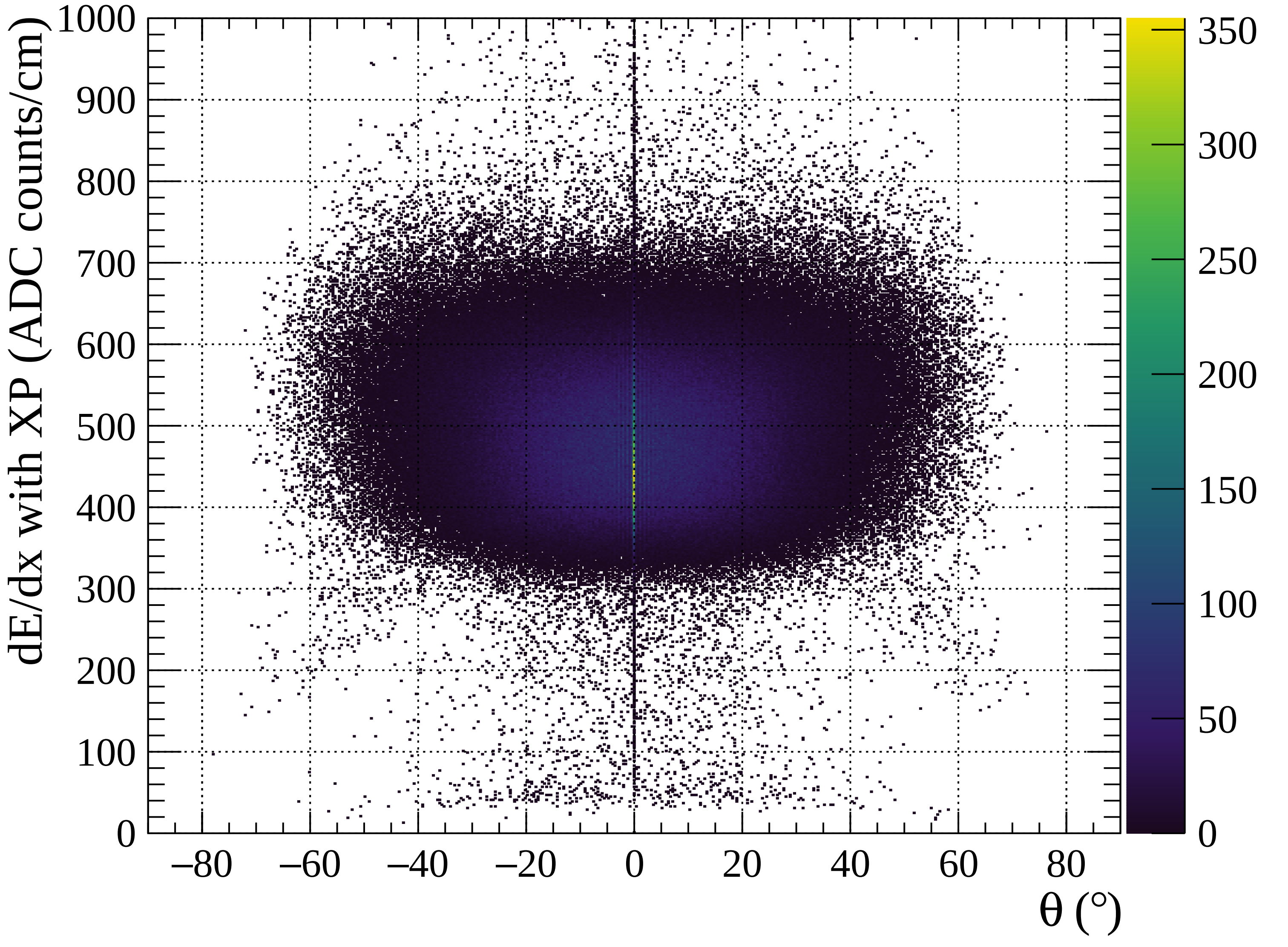
WF
XP
Mean (ADC counts/cm)
600
550
500
450
400
350
300
0
10
20
30
40
50
60
70
80
90
θ (°)



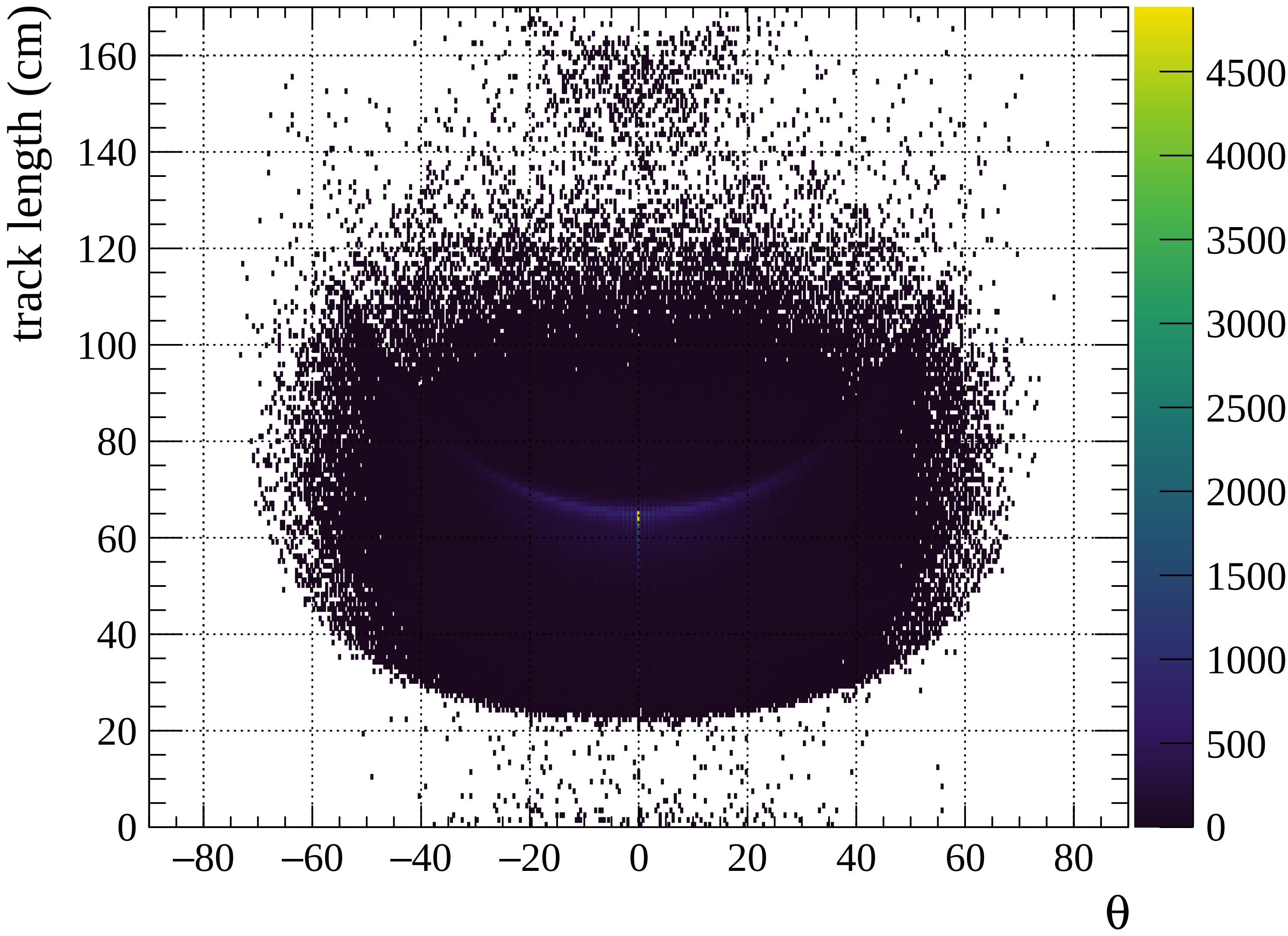
dE/dx resolution (%)
30
25
20
15
10
5
WF
XP
−80
−60
−40
−20
0
20
40
60
80
θ (°)



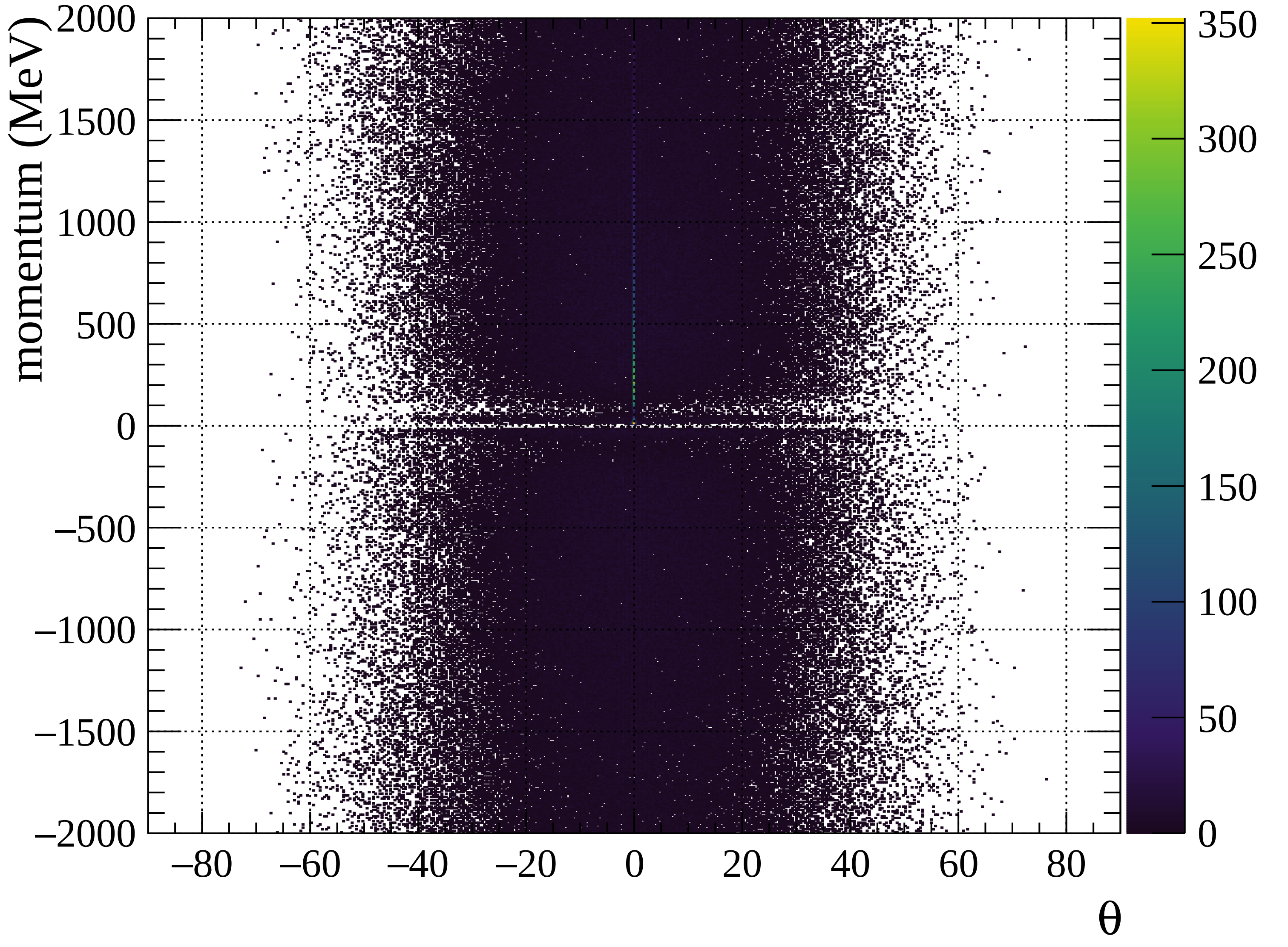
Mean (ADC counts/cm)
WF
XP
600
550
500
450
400
350
300
−80
−60
−40
−20
0
20
40
60
80
θ (°)



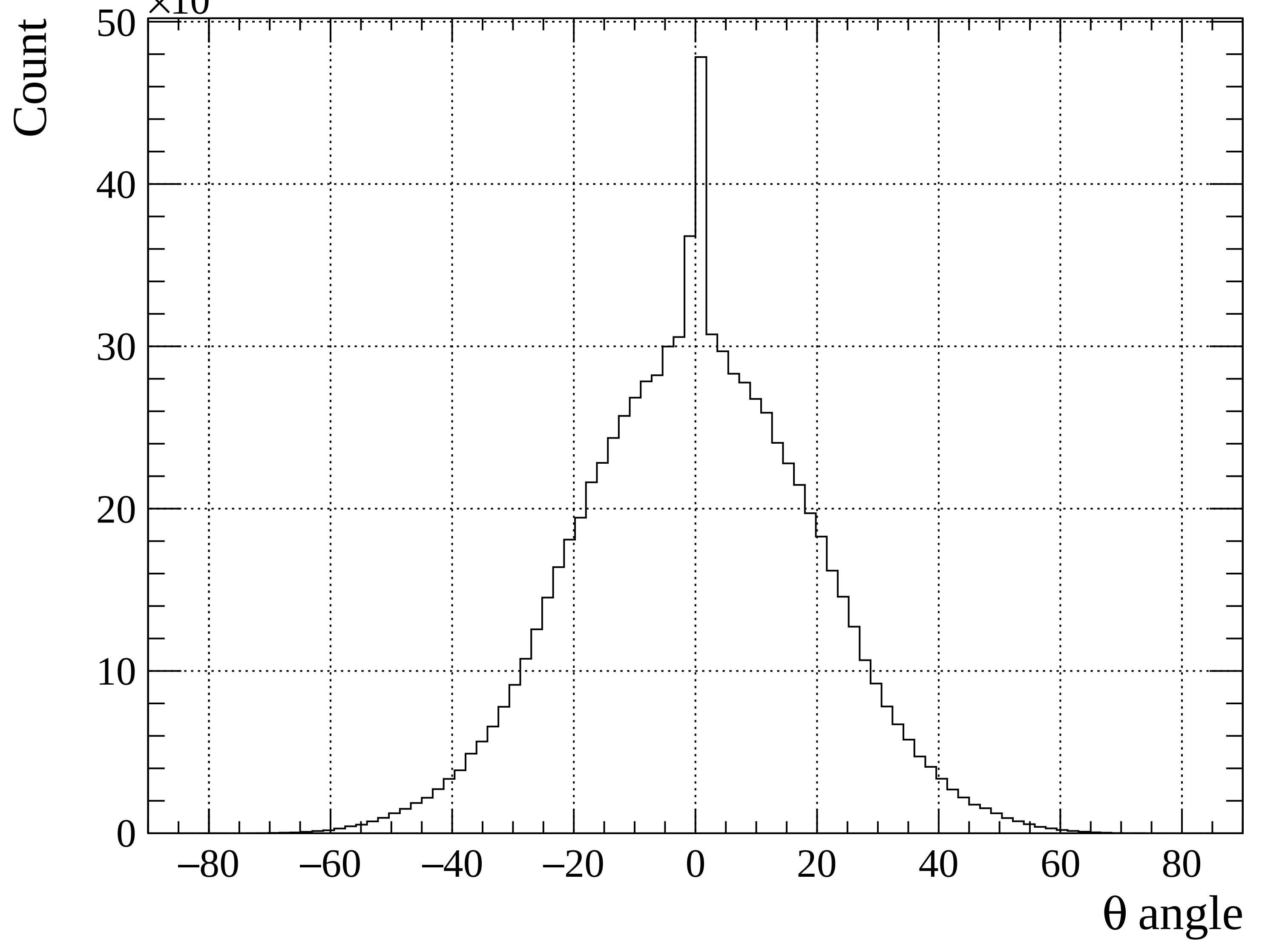
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−80
−60
−40
−20
0
20
40
60
80
θ (°)
350
300
250
200
150
100
50
0



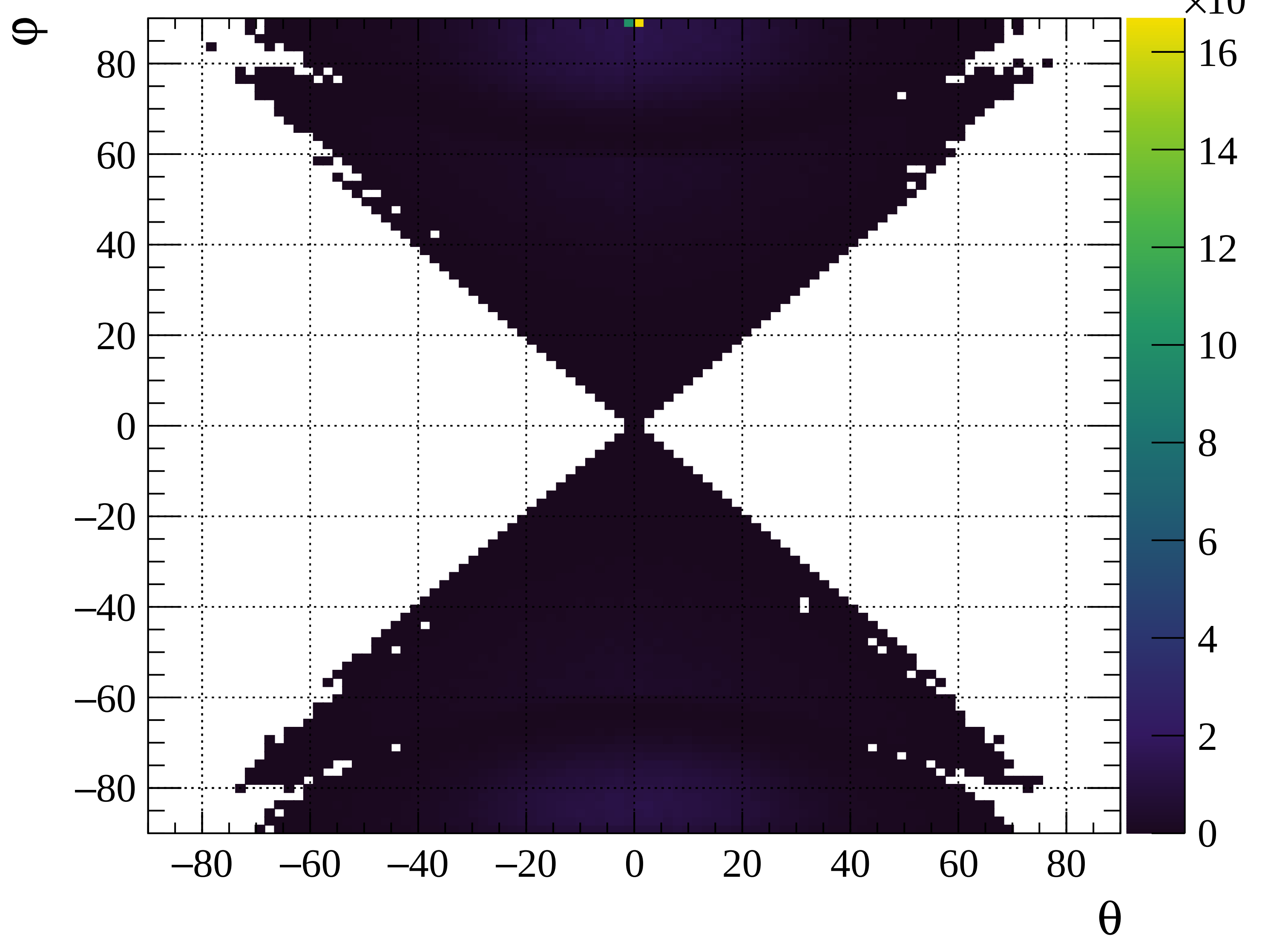
track length (cm)
160
140
120
100
80
60
40
20
0
−80
−60
−40
−20
0
20
40
60
80
θ
4500
4000
3500
3000
2500
2000
1500
1000
500
0



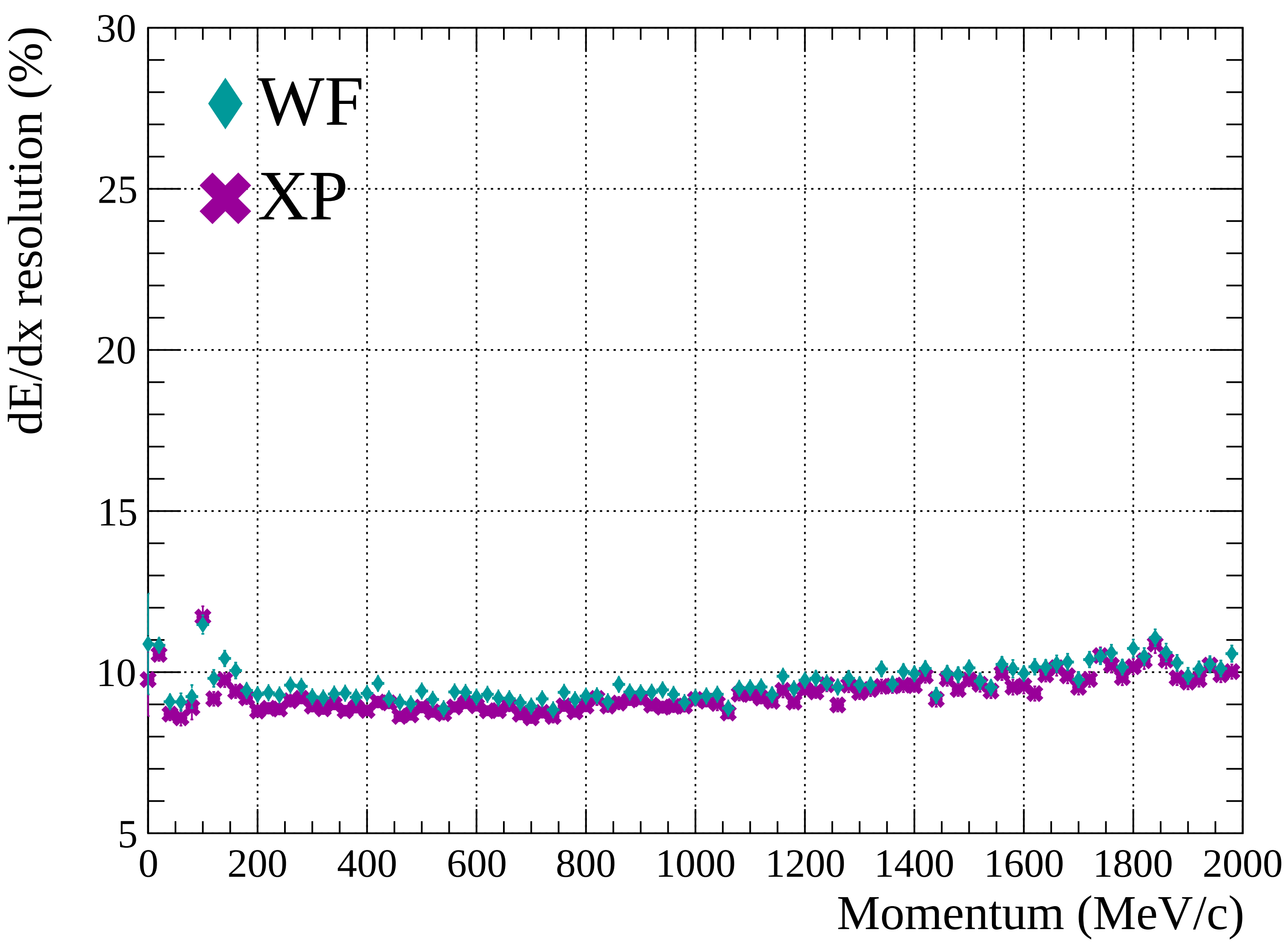
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−80
−60
−40
−20
0
20
40
60
80
θ
350
300
250
200
150
100
50
0

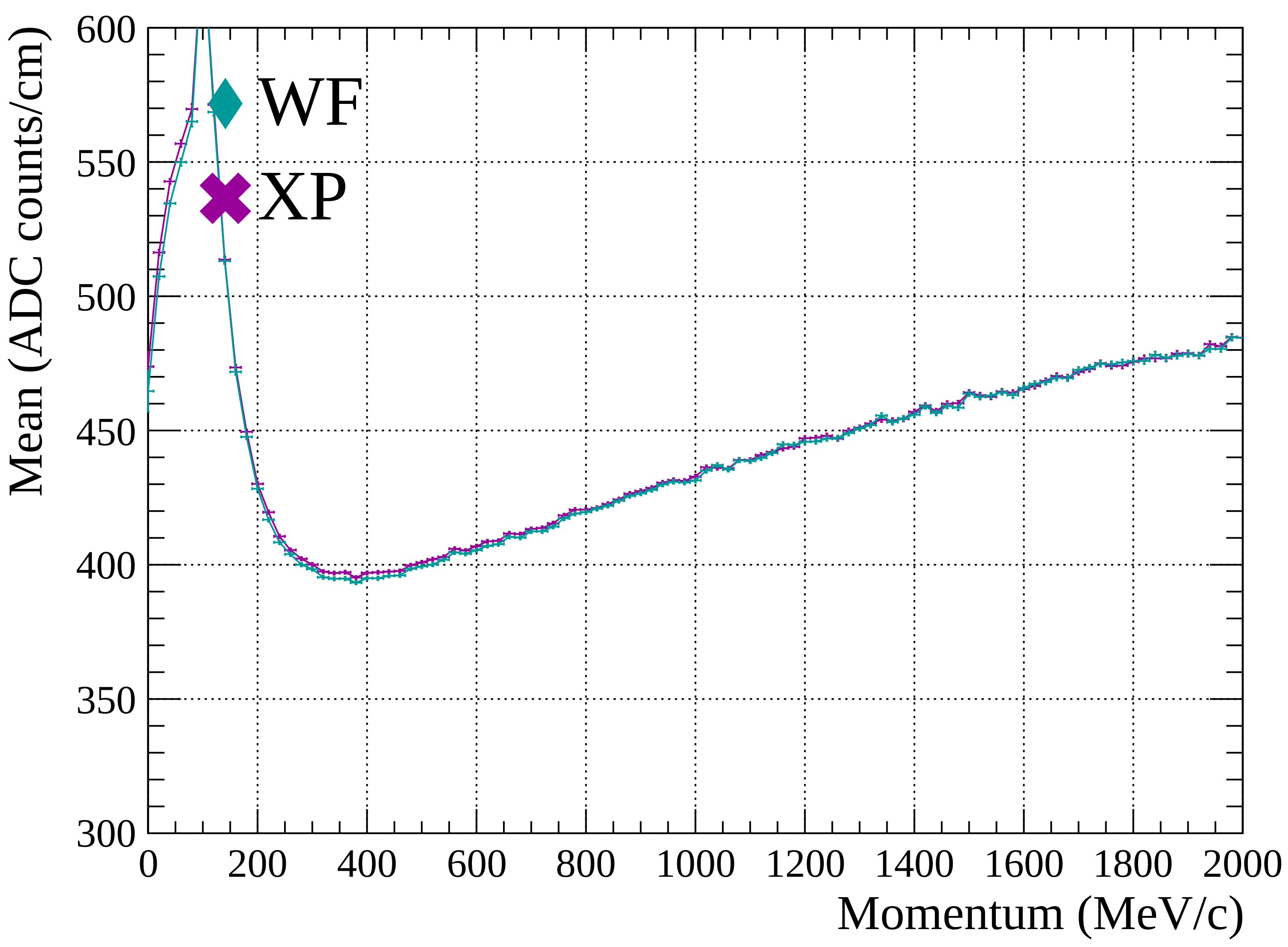


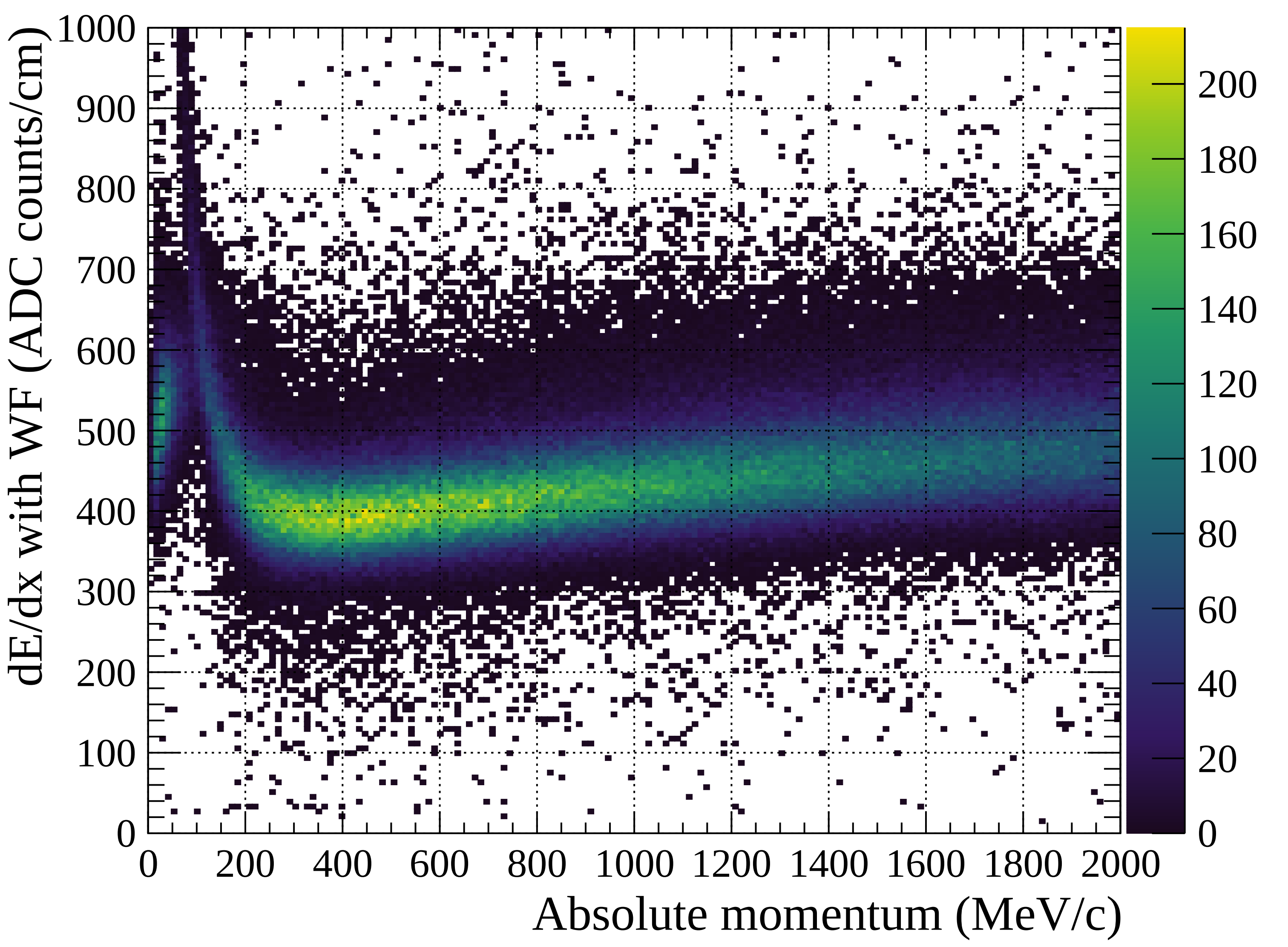
Count
×10
50
40
30
20
10
0
−80
−60
−40
−20
0
20
40
60
80
θ angle

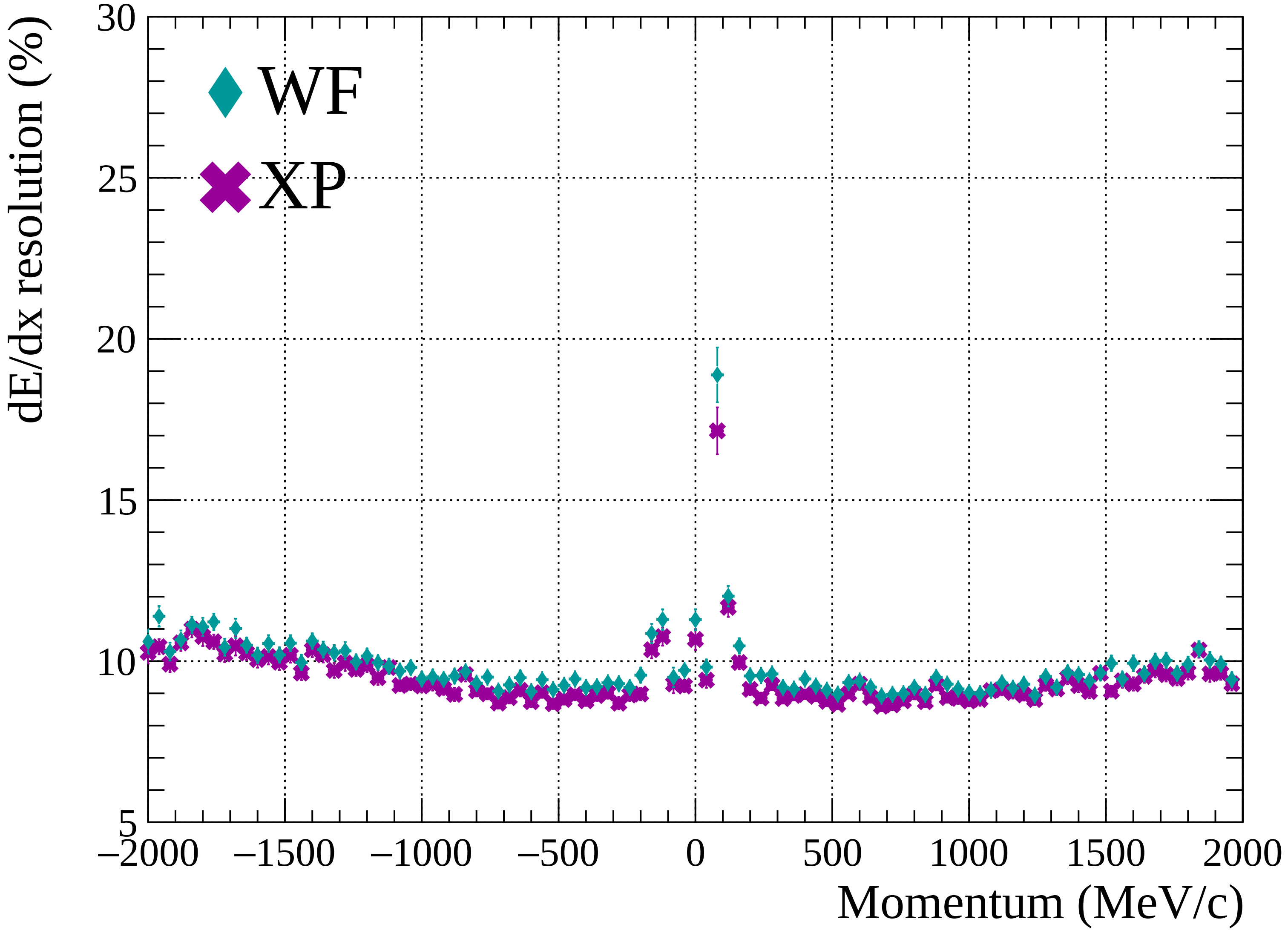


φ
80
60
40
20
0
−20
−40
−60
−80
−80
−60
−40
−20
0
20
40
60
80
θ
×10
16
14
12
10
8
6
4
2
0

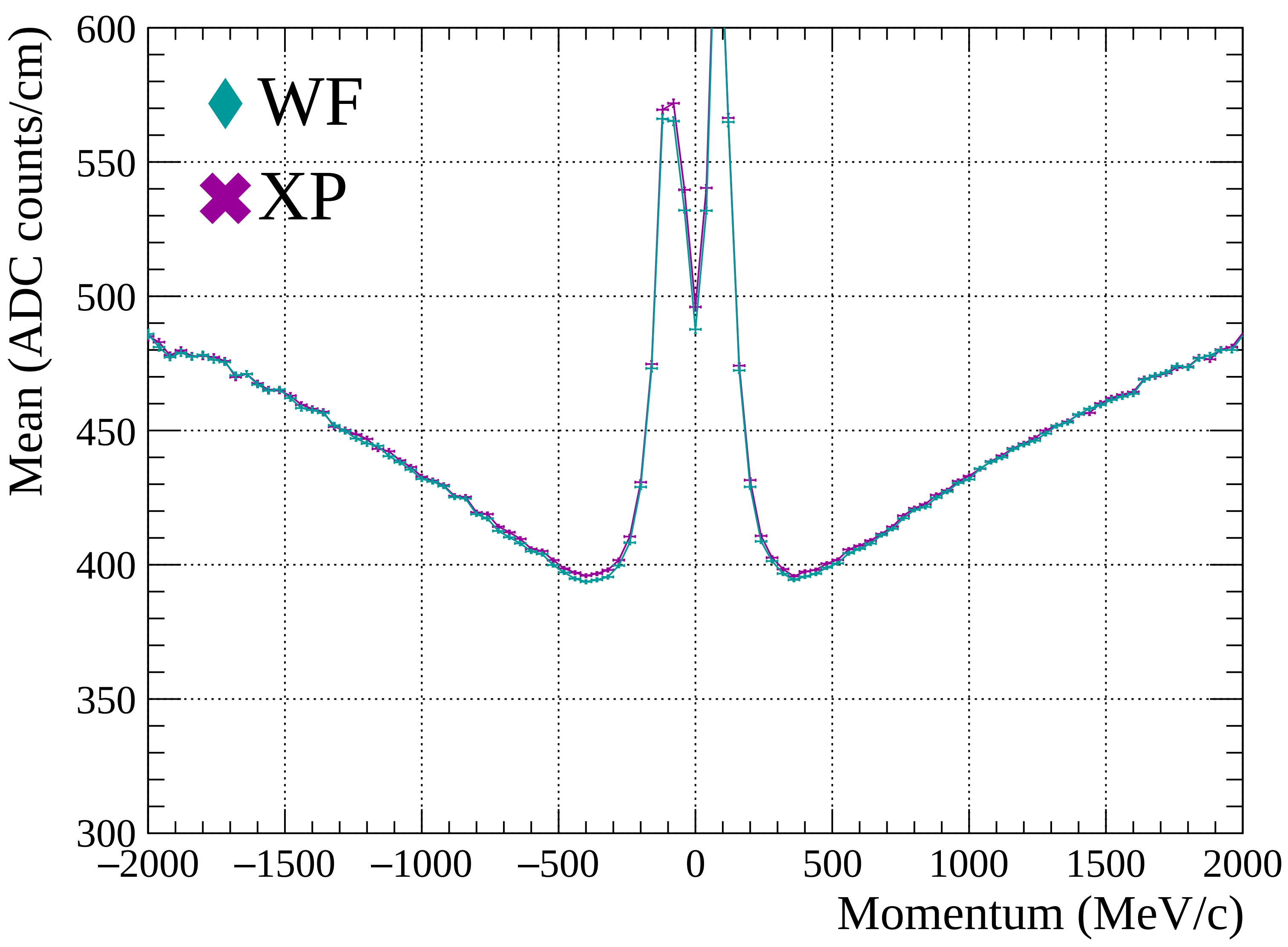

WF
XP
dE/dx resolution (%)
Momentum (MeV/c)
30
25
20
15
10
5
0
200
400
600
800
1000
1200
1400
1600
1800
2000


WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
0
200
400
600
800
1000
1200
1400
1600
1800
2000

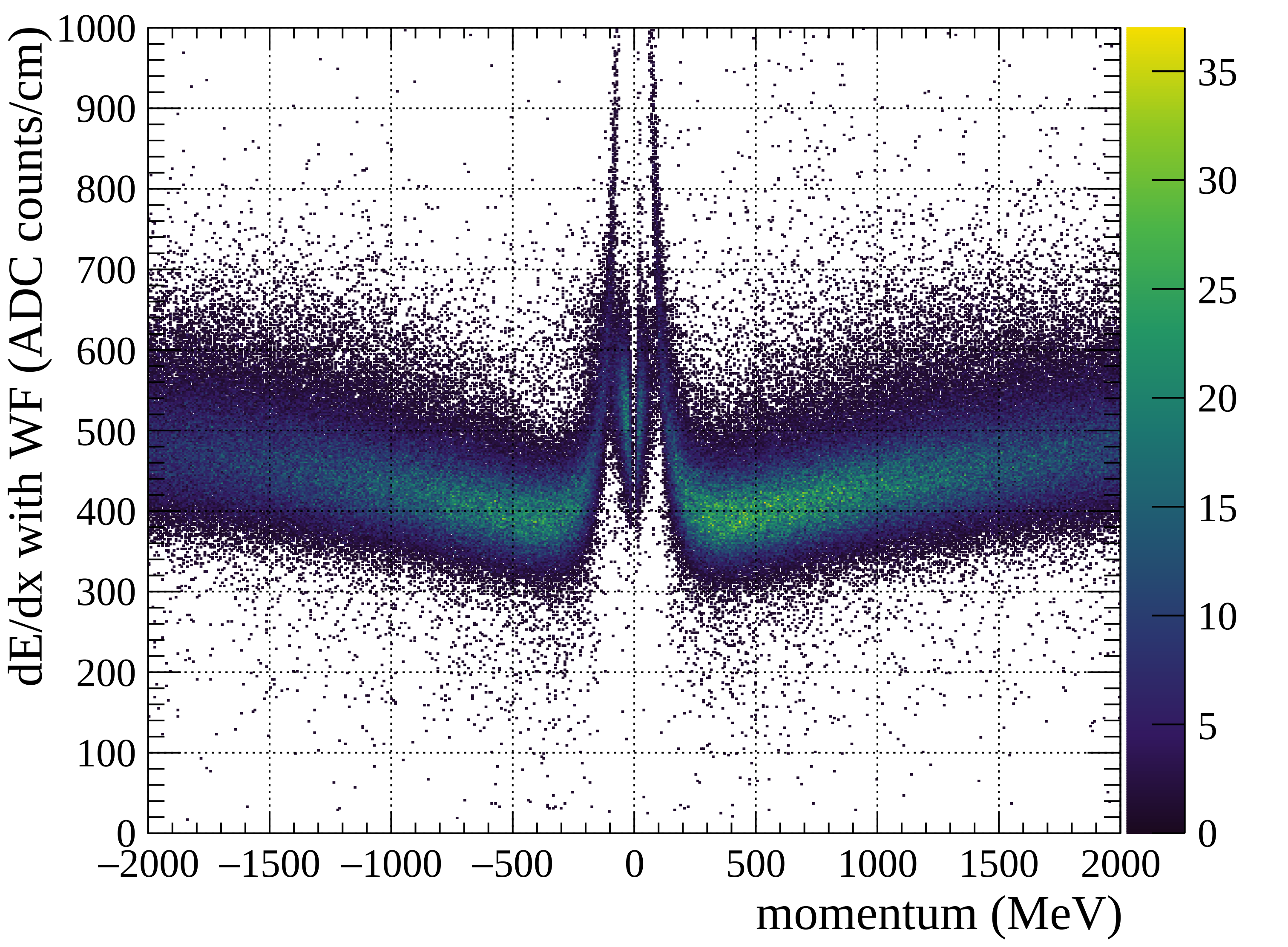

dE/dx with WF (ADC counts/cm)
Absolute momentum (MeV/c)
0
100
200
300
400
500
600
700
800
900
1000
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
20
40
60
80
100
120
140
160
180
200

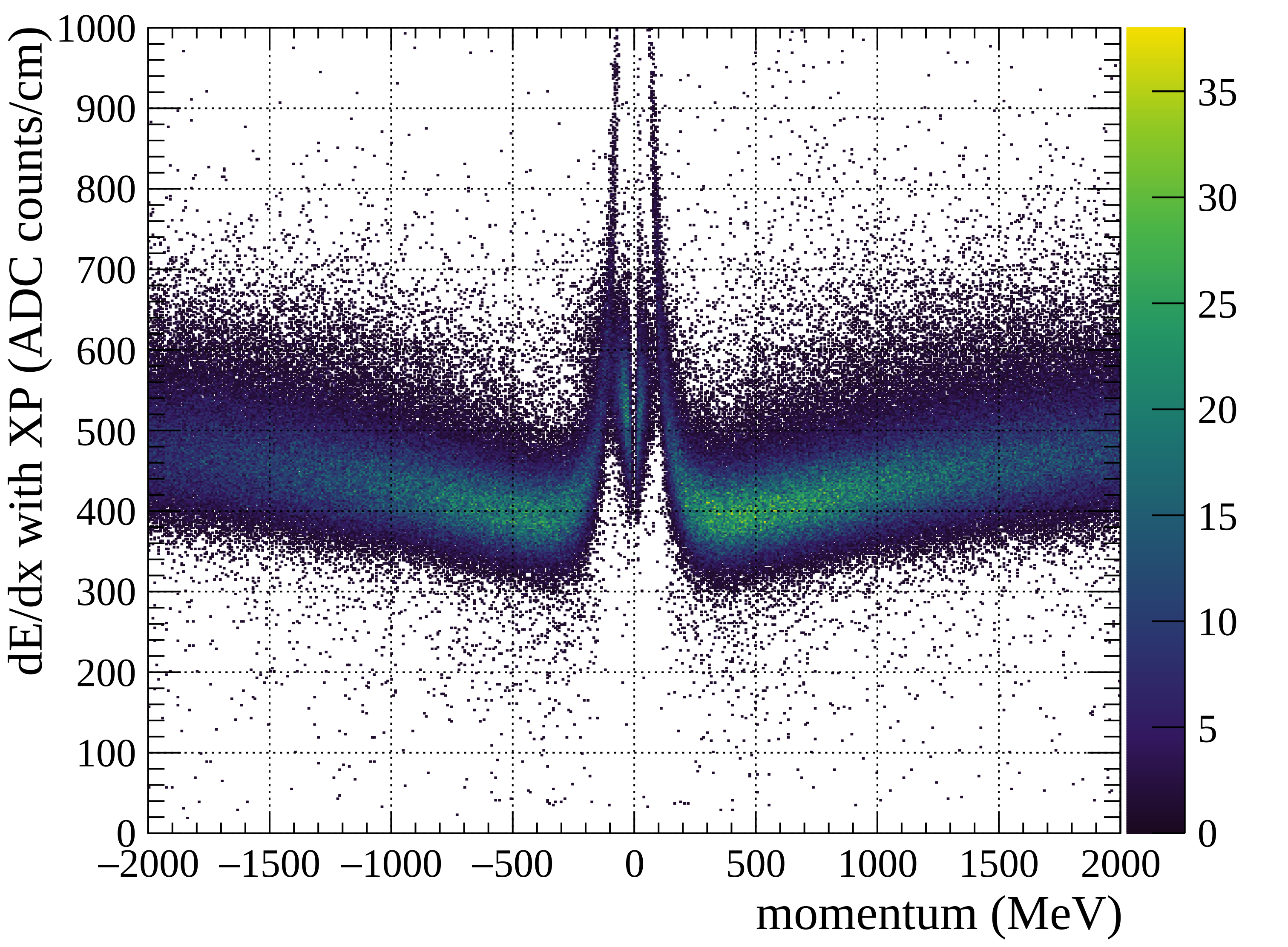


WF
XP
dE/dx resolution (%)
Momentum (MeV/c)
30
25
20
15
10
5
−2000
−1500
−1000
−500
0
500
1000
1500
2000

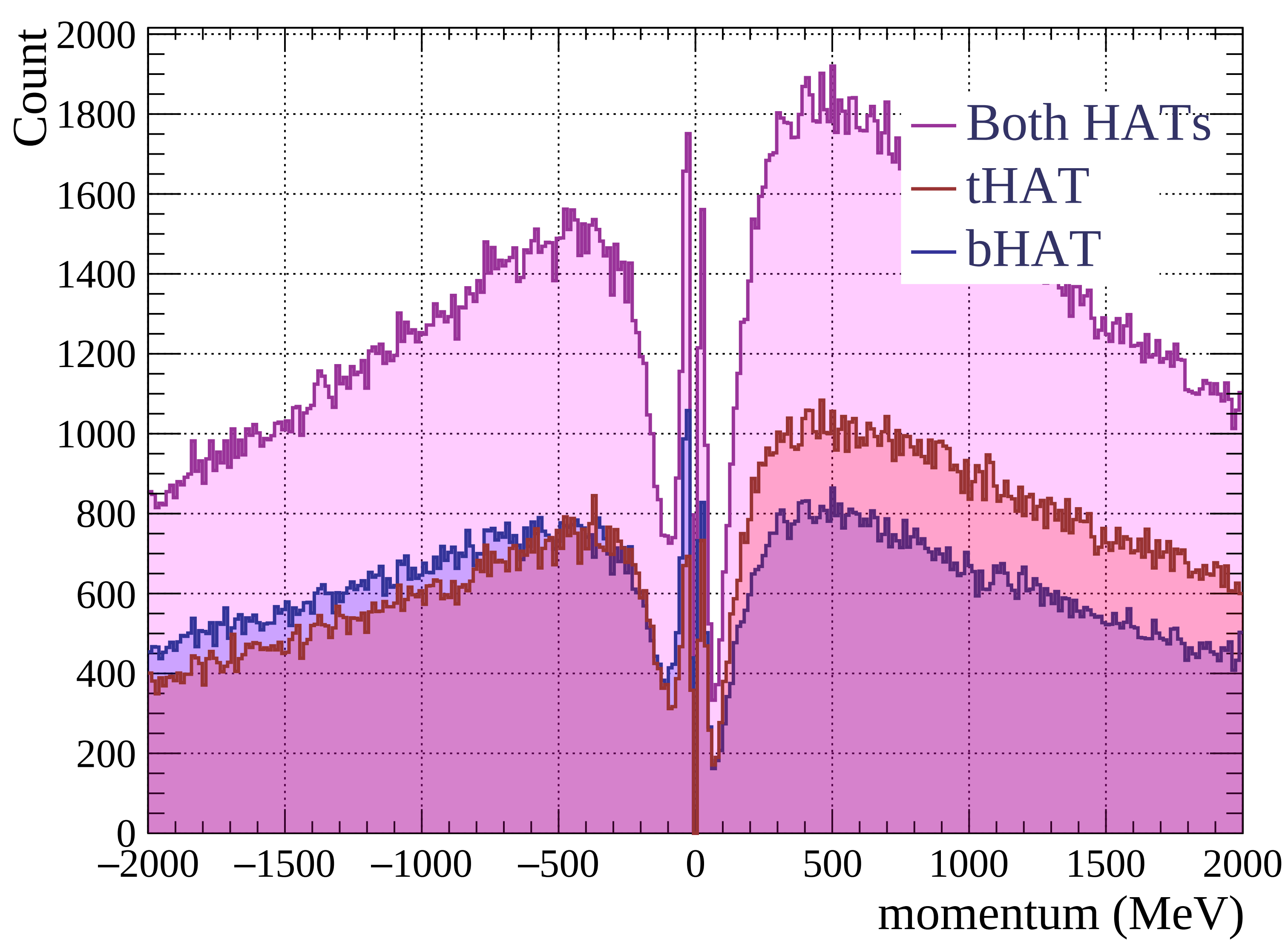


WF
XP
Mean (ADC counts/cm)
Momentum (MeV/c)
600
550
500
450
400
350
300
−2000
−1500
−1000
−500
0
500
1000
1500
2000

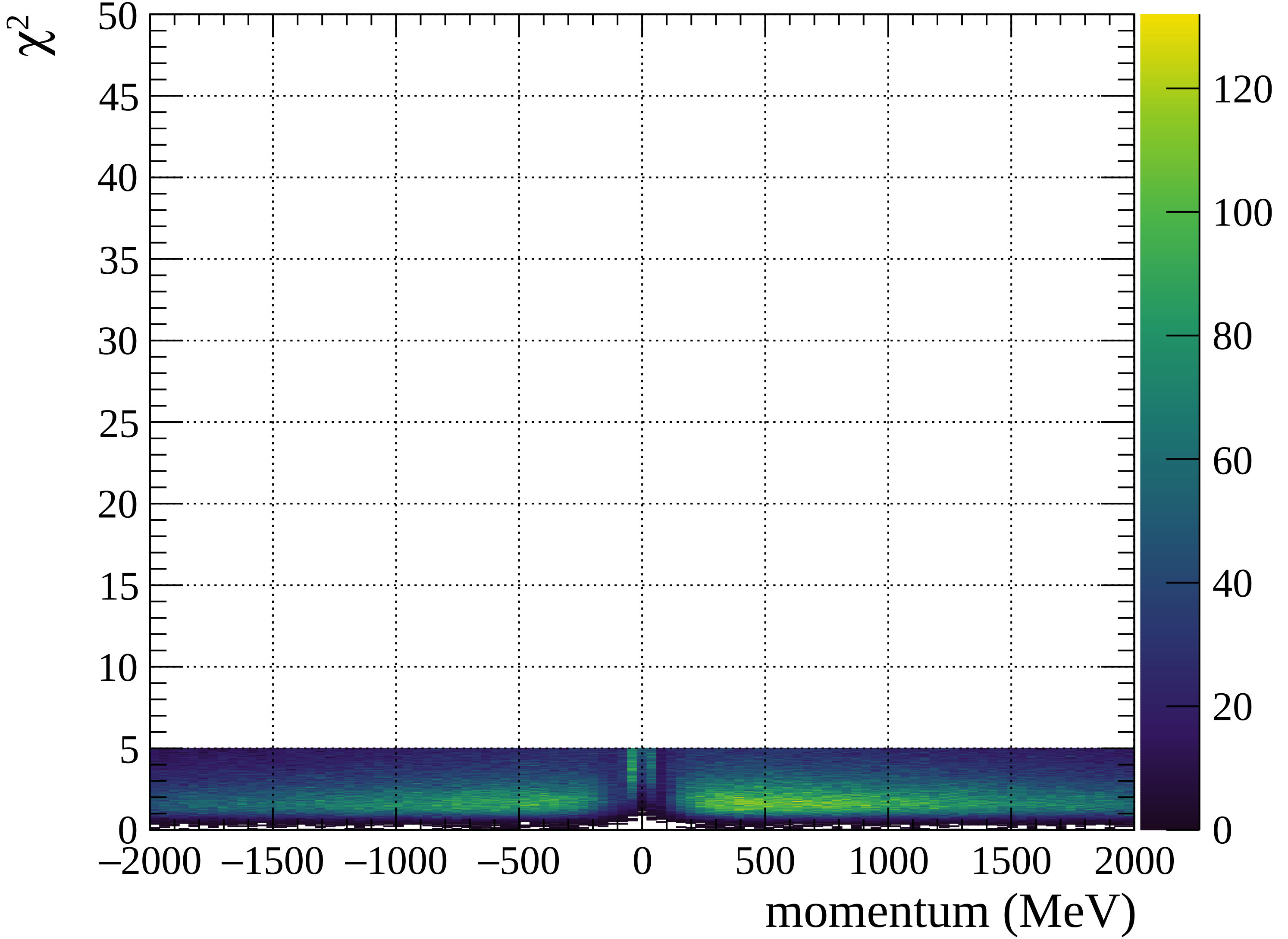

dE/dx with WF (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
1000
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
0
5
10
15
20
25
30
35



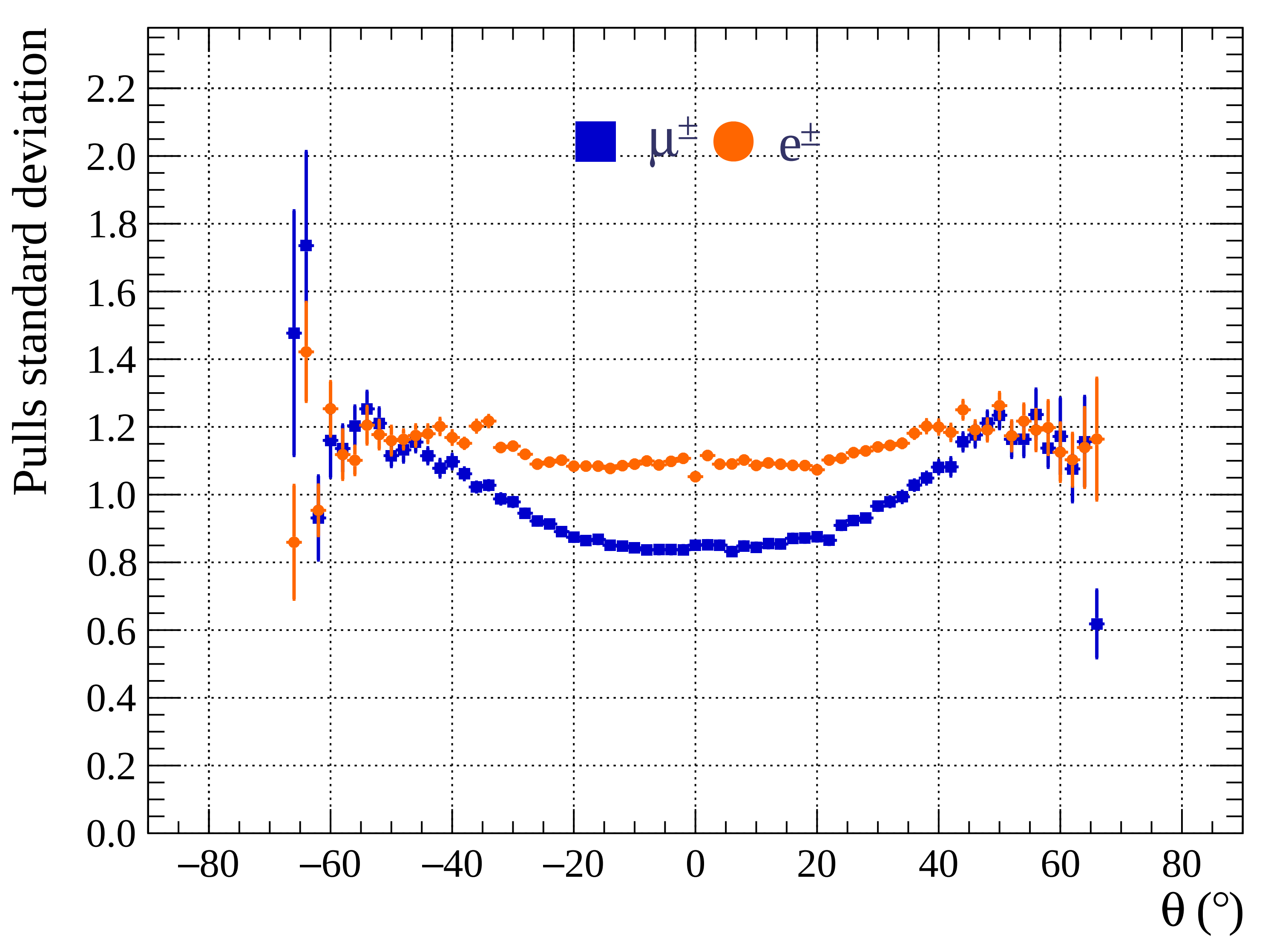
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
35
30
25
20
15
10
5
0



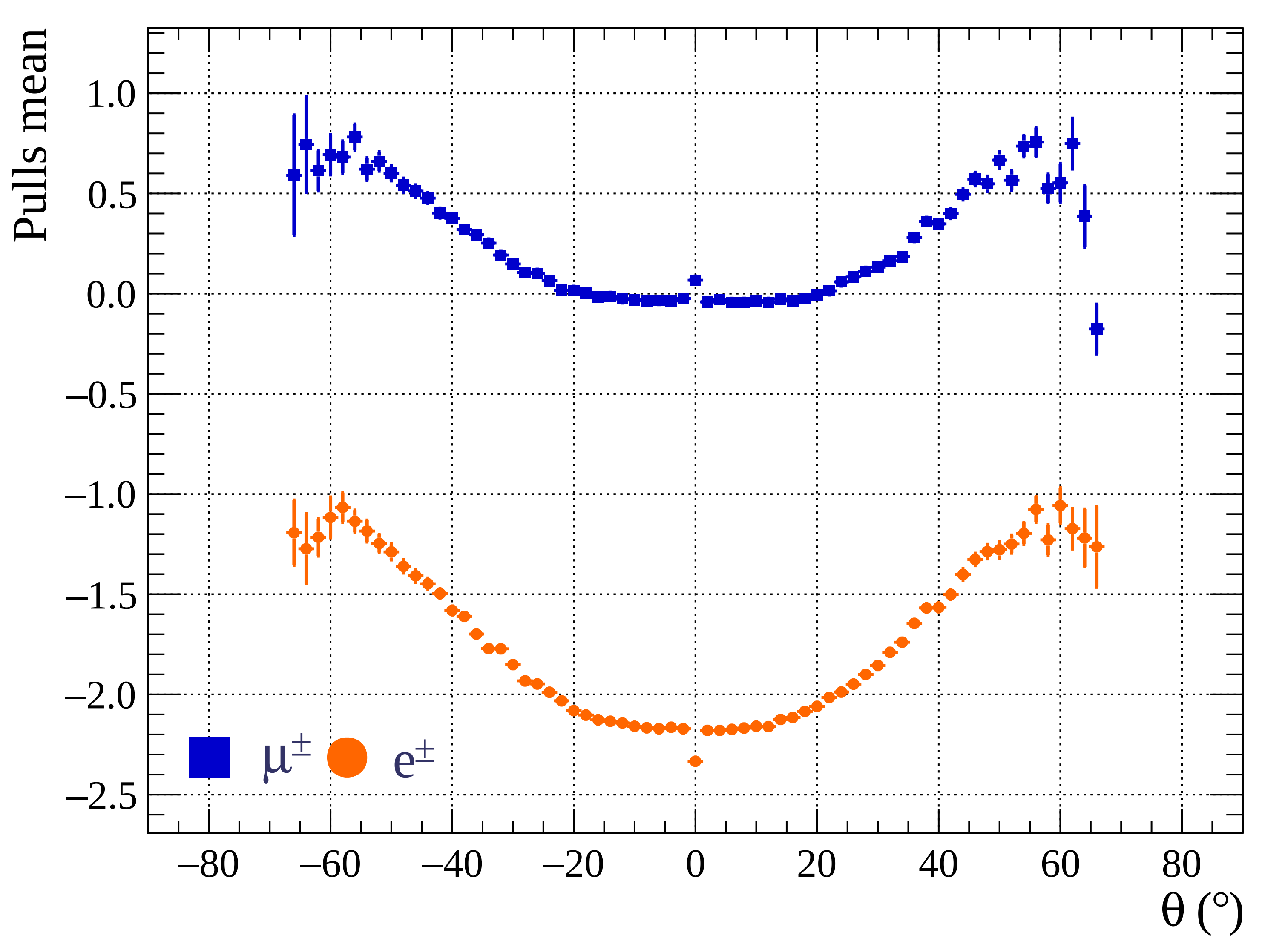
Count
2000
1800
1600
1400
1200
1000
800
600
400
200
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
Both HATs
tHAT
bHAT



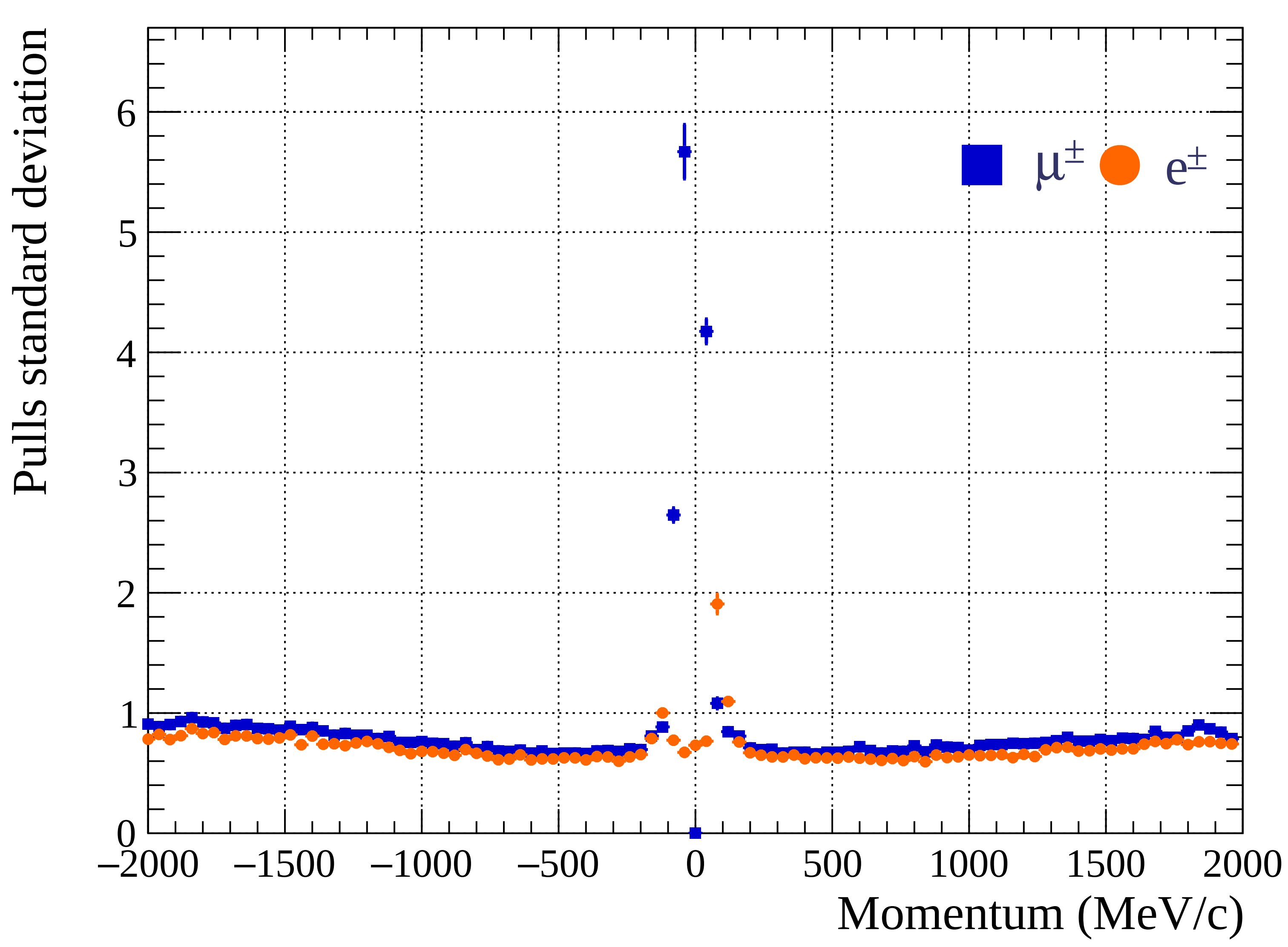
χ^2
50
45
40
35
30
25
20
15
10
5
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
momentum (MeV)
120
100
80
60
40
20
0



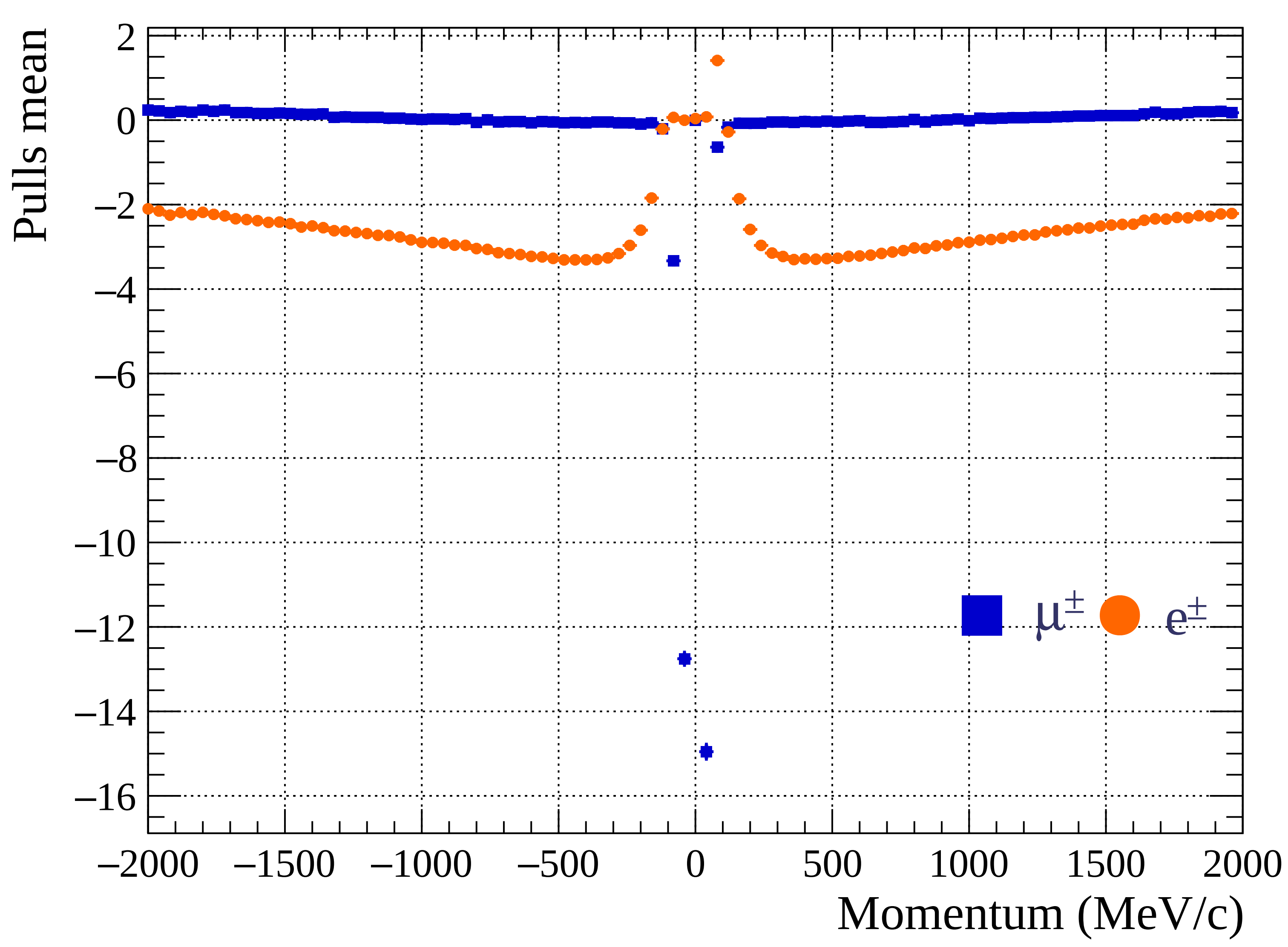
Pulls standard deviation
μ±
e±
θ (°)



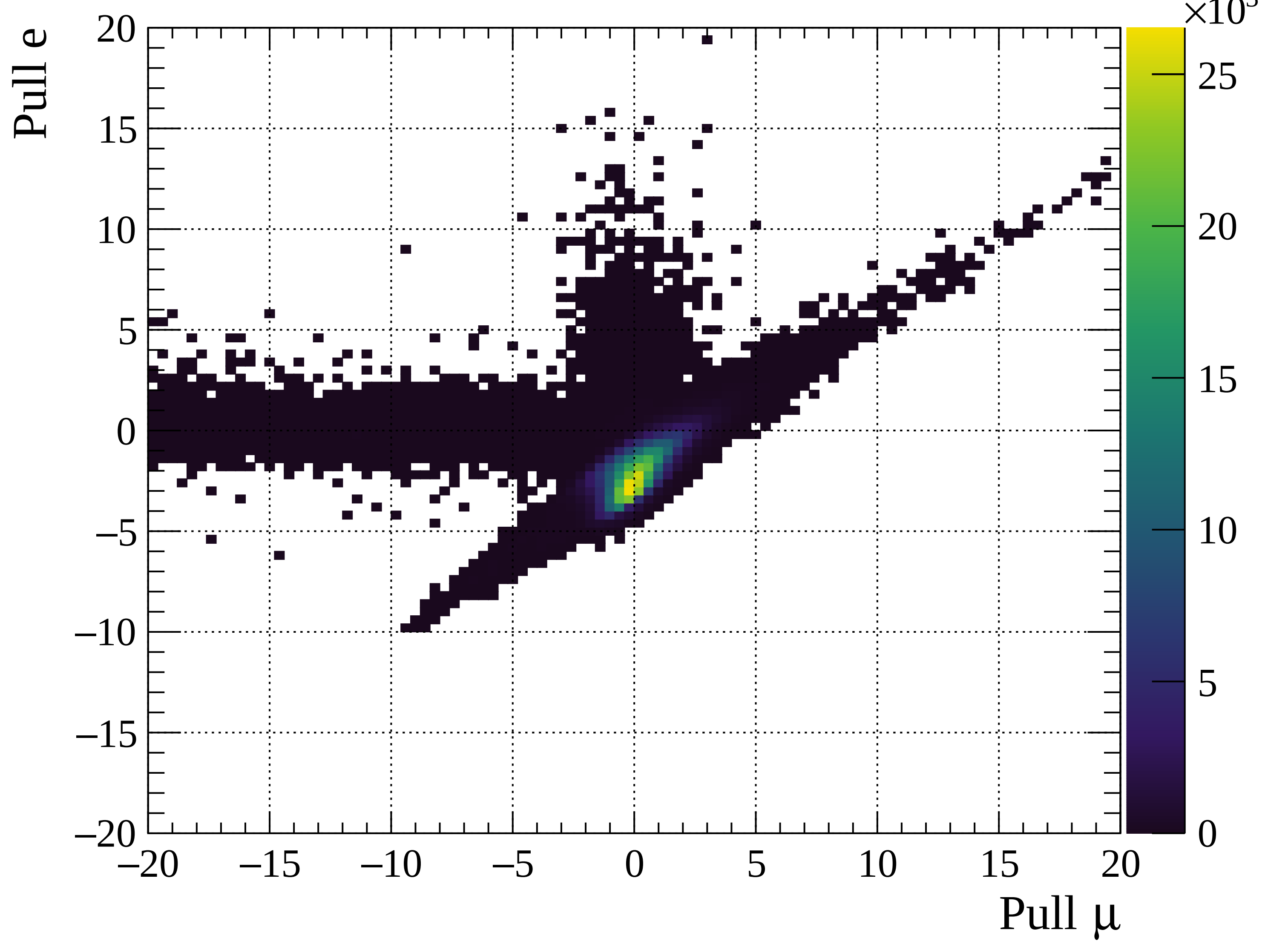
Pulls mean
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
−2.5
−80
−60
−40
−20
0
20
40
60
80
θ (°)
μ±
e±



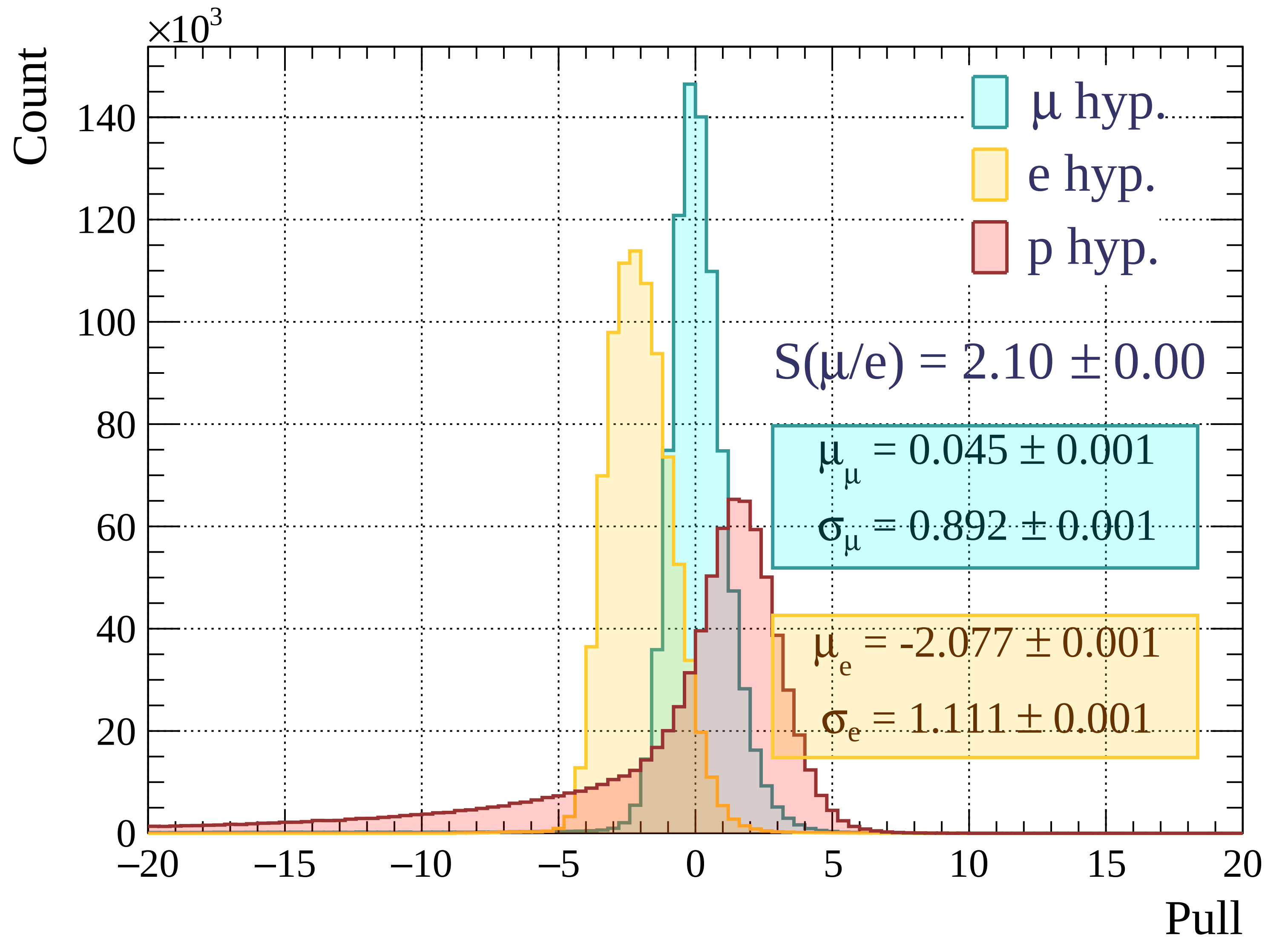
Pulls standard deviation
6
5
4
3
2
1
0
−2000
−1500
−1000
−500
0
500
1000
1500
2000
Momentum (MeV/c)
μ±
e±



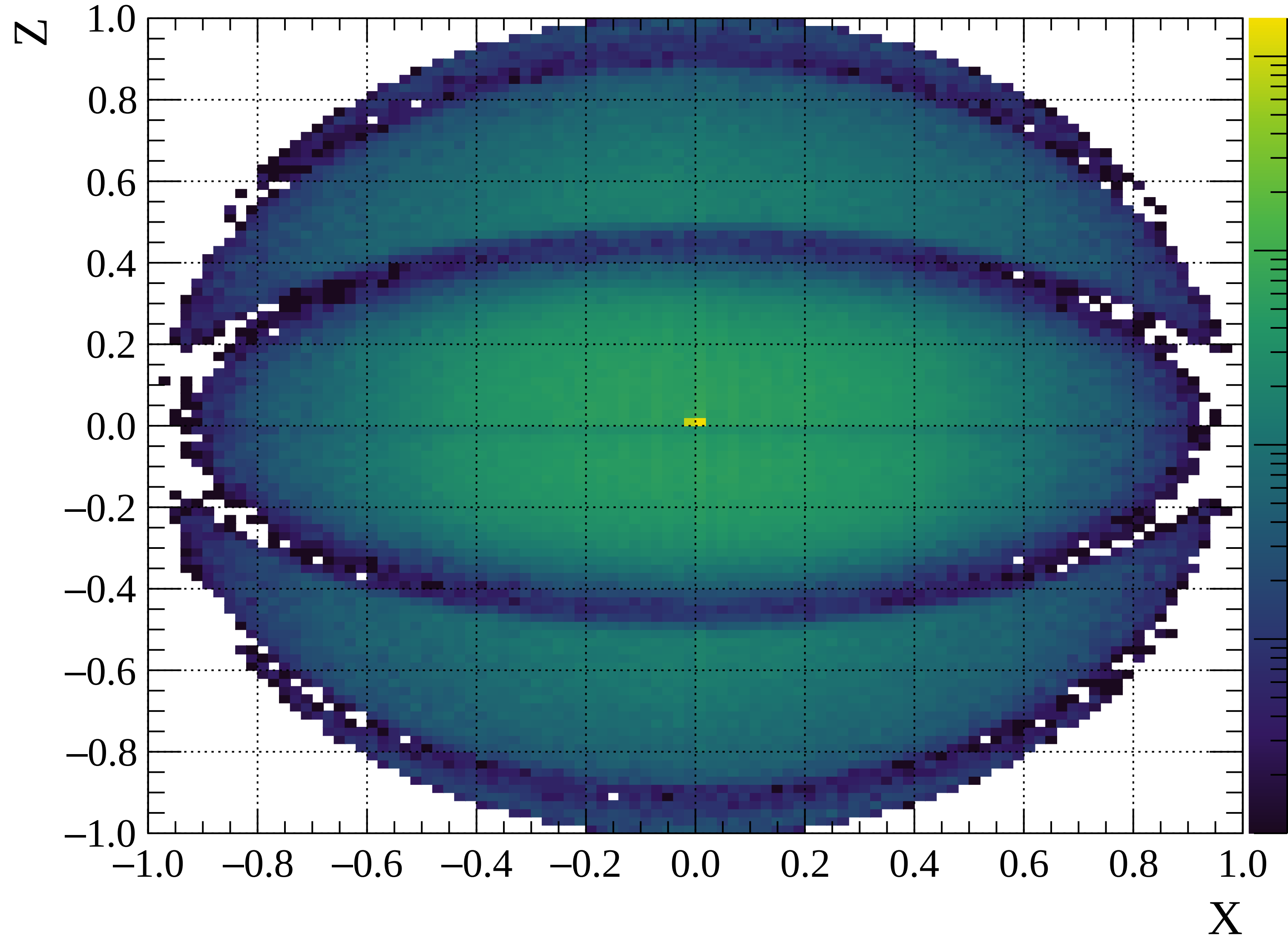
Pulls mean
Momentum (MeV/c)
μ±
e±
2
0
−2
−4
−6
−8
−10
−12
−14
−16
−2000
−1500
−1000
−500
0
500
1000
1500
2000



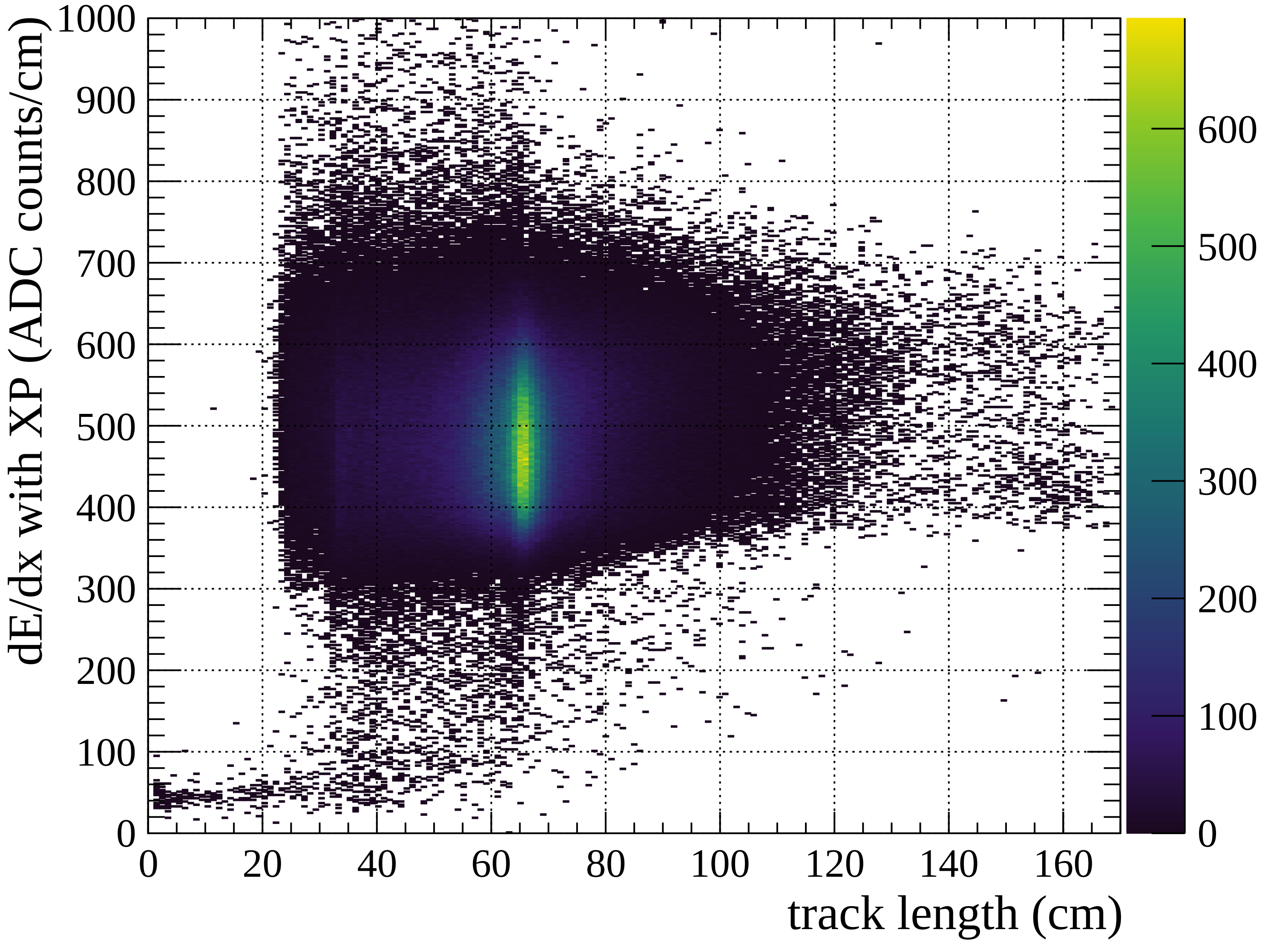
Pull e
Pull μ
×10³
20
15
10
5
0
−5
−10
−15
−20
25



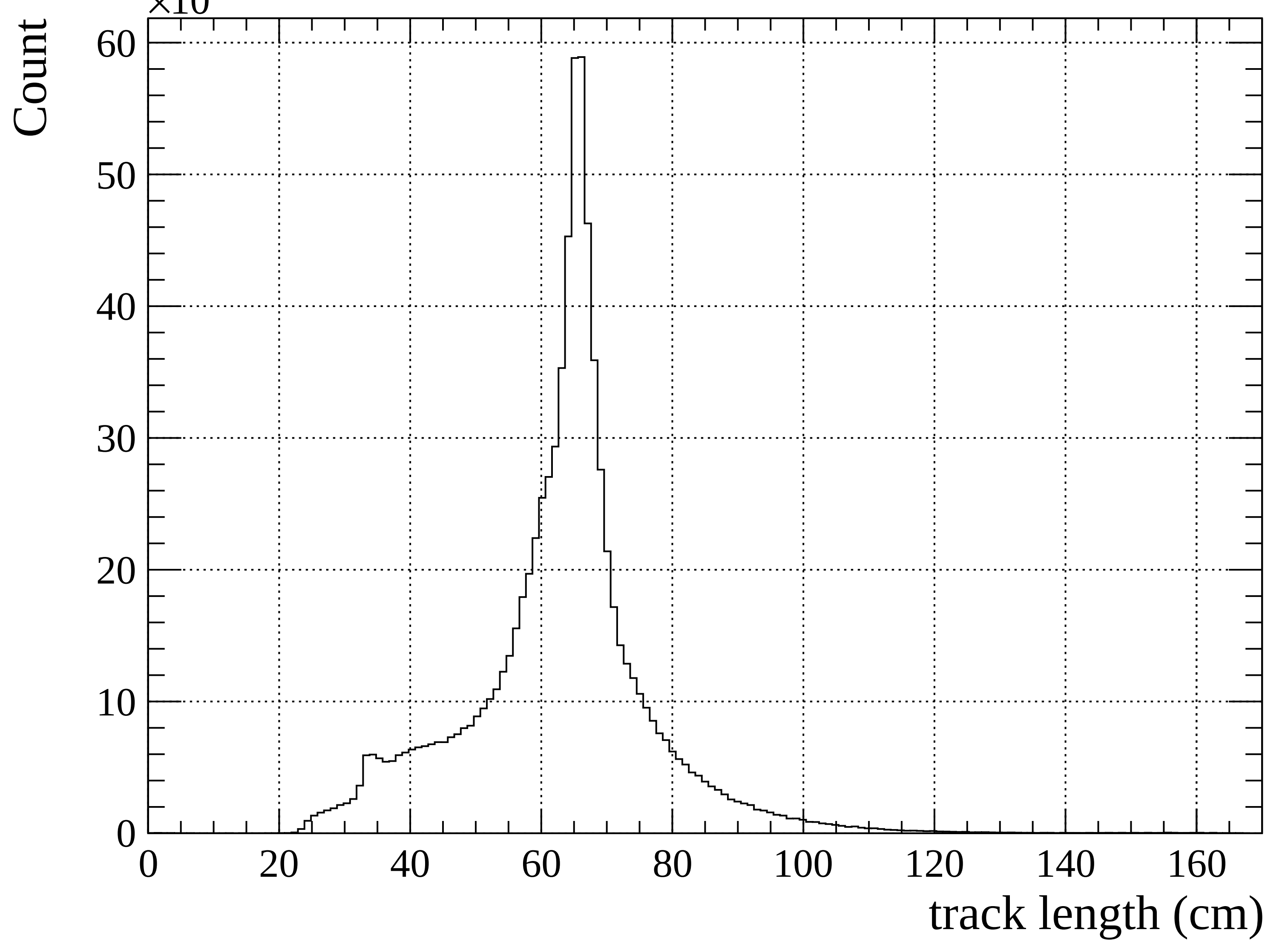
$\times 10^3$
Count
140
120
100
80
60
40
20
0
−20
−15
−10
−5
0
5
10
15
20
Pull
μ hyp.
e hyp.
p hyp.
S(μ/e) = 2.10 ± 0.00
μ_μ = 0.045 ± 0.001
σ_μ = 0.892 ± 0.001
μ_e = -2.077 ± 0.001
σ_e = 1.111 ± 0.001

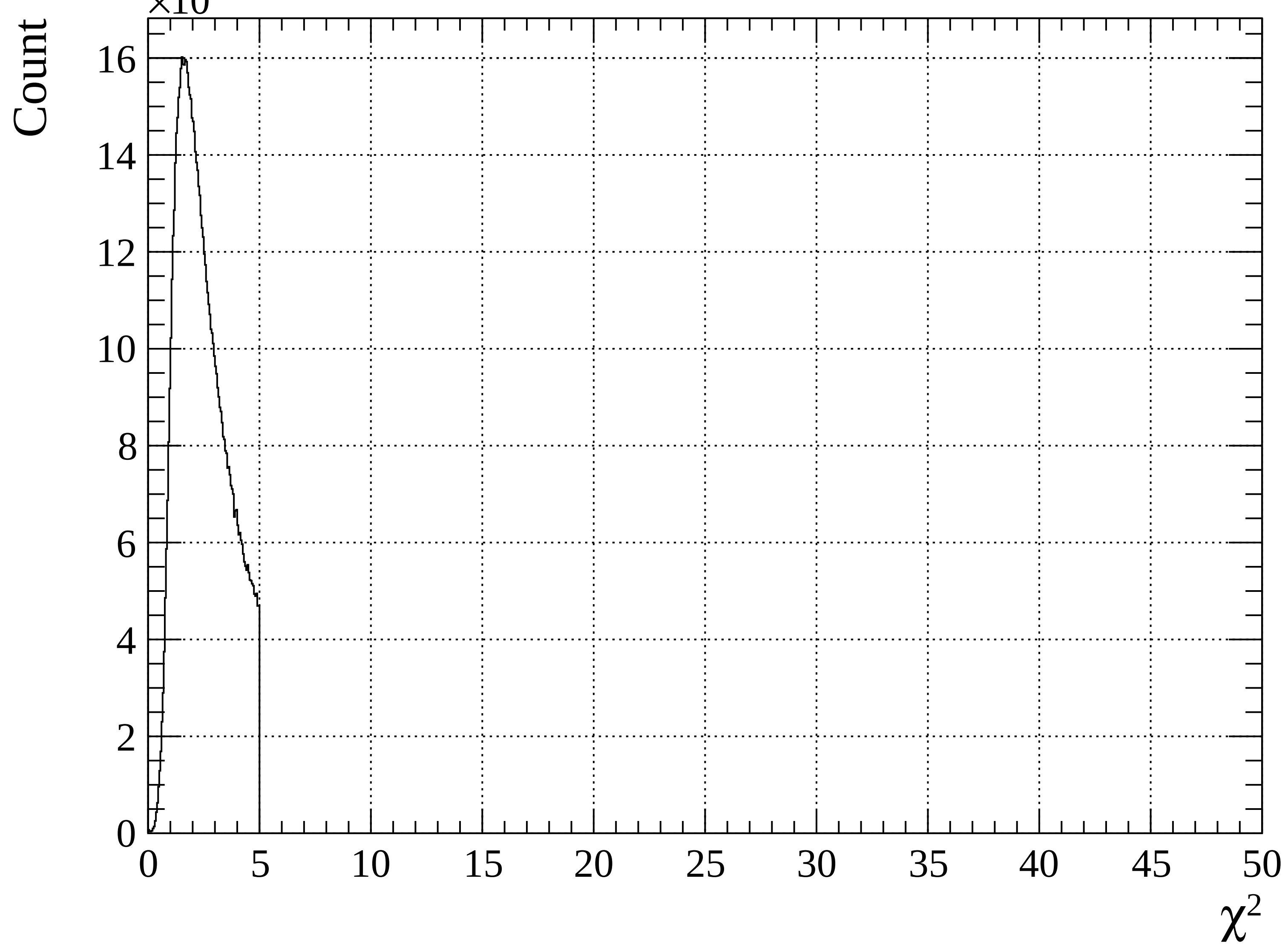


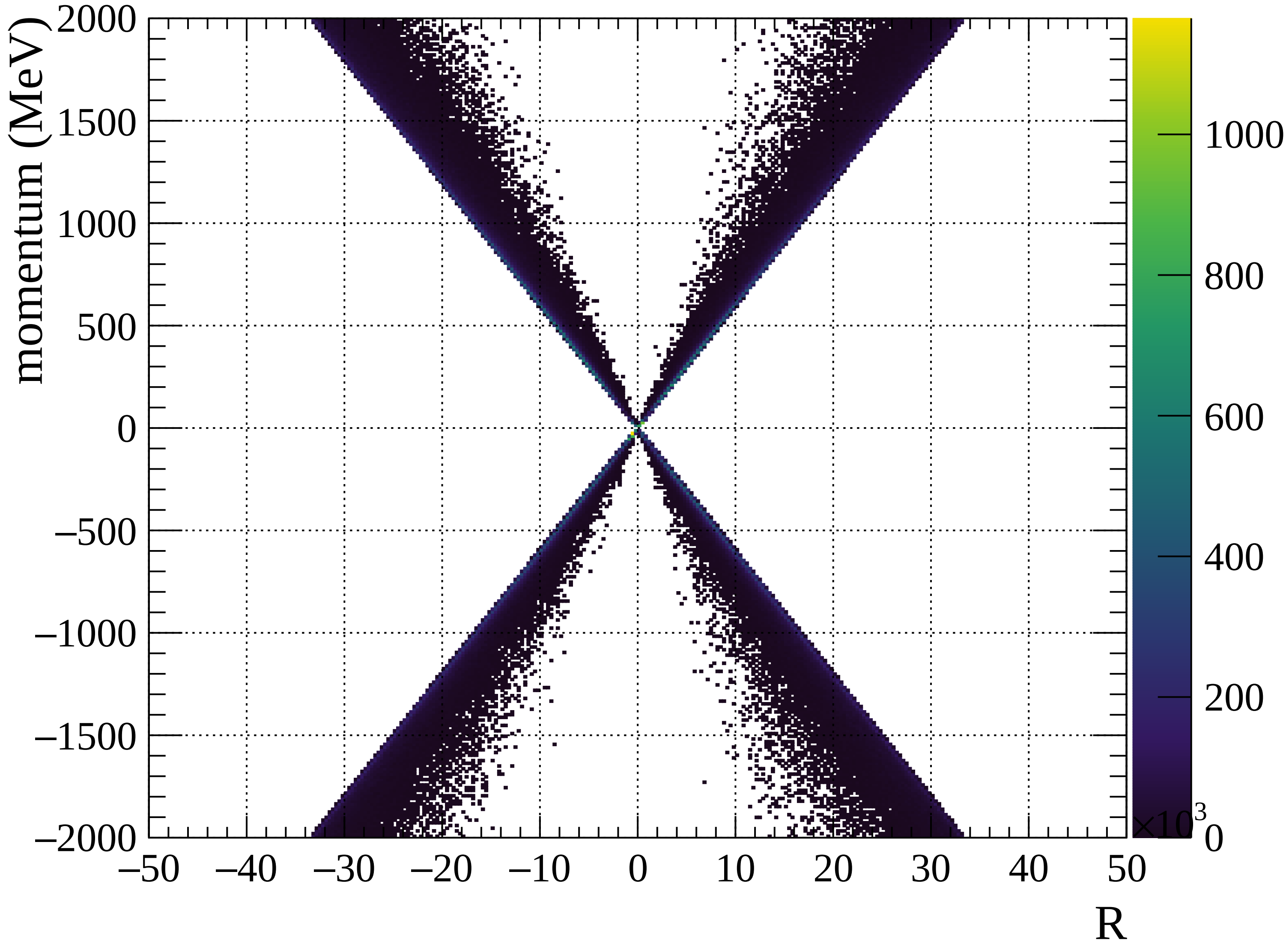
Z
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
X



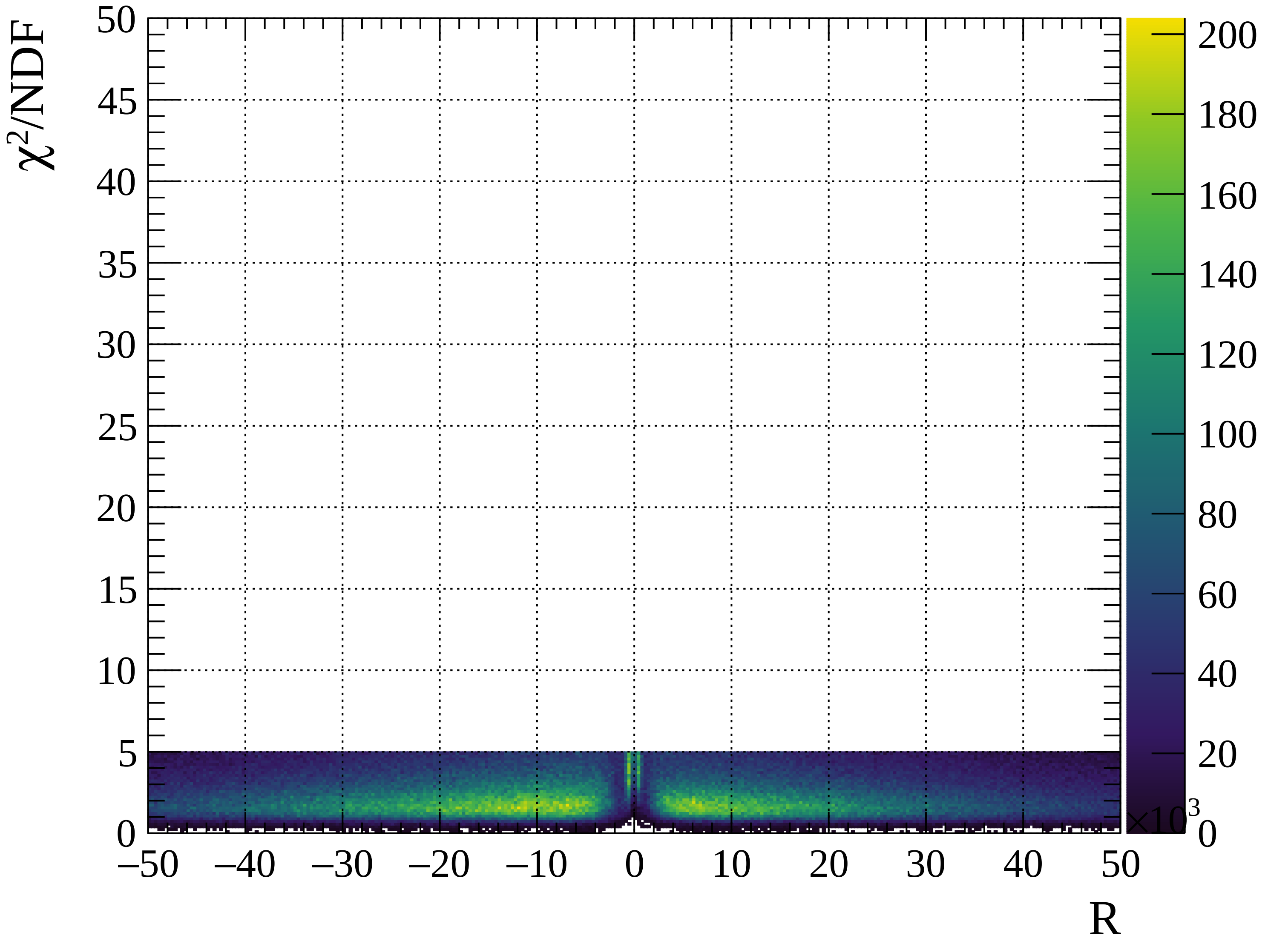
dE/dx with XP (ADC counts/cm)
1000
900
800
700
600
500
400
300
200
100
0
0
20
40
60
80
100
120
140
160
track length (cm)
600
500
400
300
200
100
0


×10
Count
60
50
40
30
20
10
0
0
20
40
60
80
100
120
140
160
track length (cm)

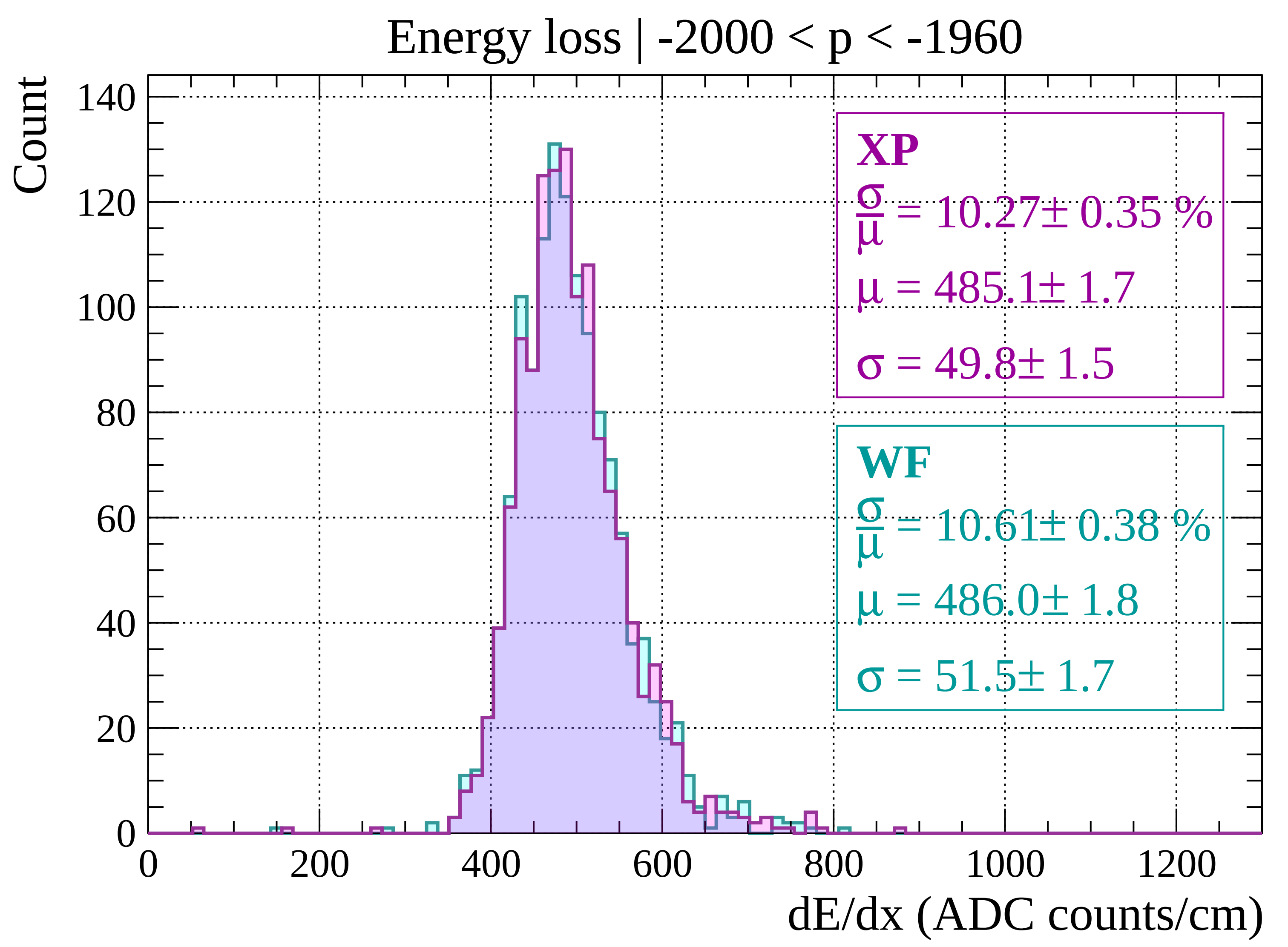

×10
Count
16
14
12
10
8
6
4
2
0
0
5
10
15
20
25
30
35
40
45
50
χ^2



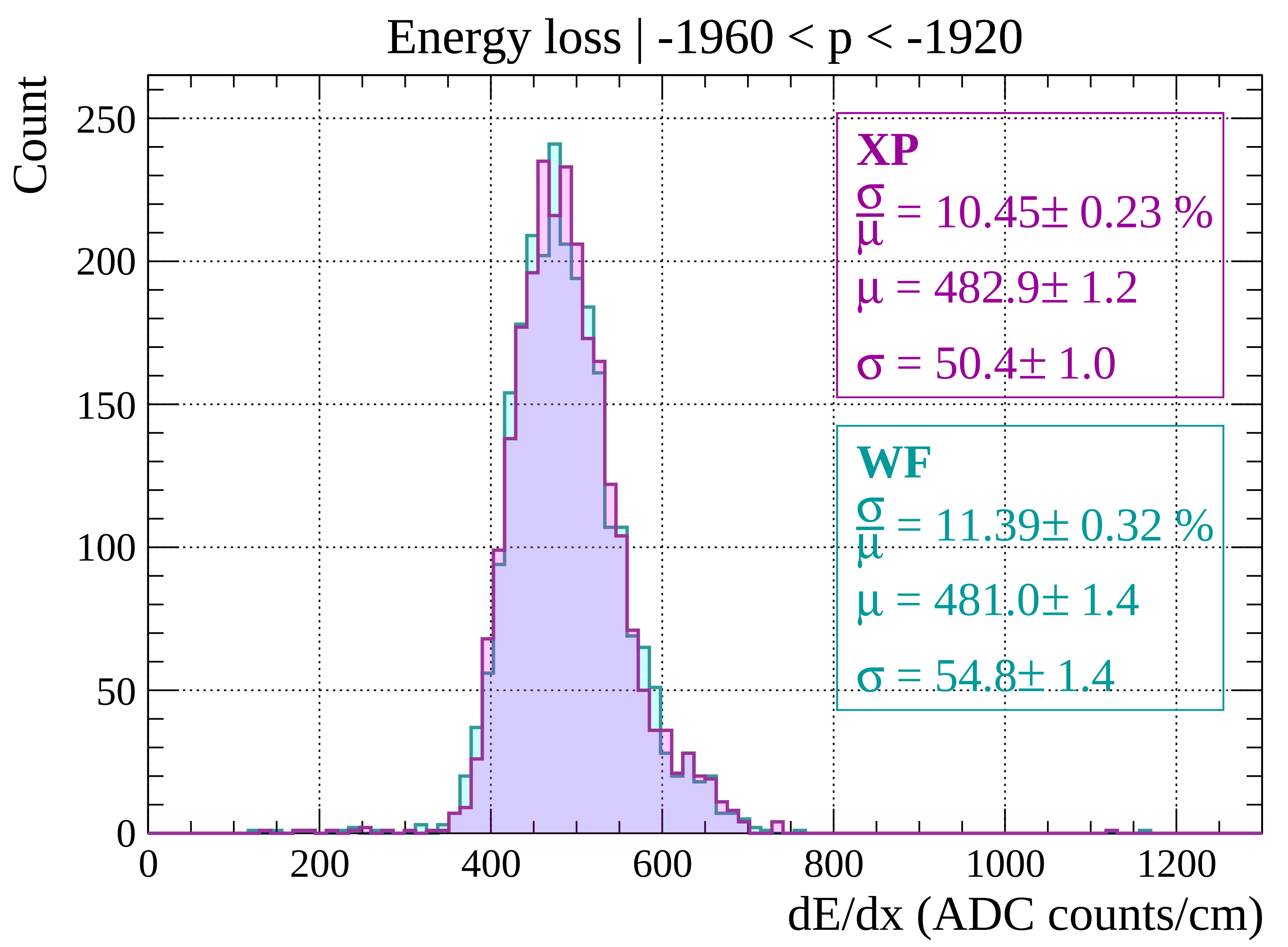
momentum (MeV)
2000
1500
1000
500
0
−500
−1000
−1500
−2000
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
1000
800
600
400
200
0
×10³



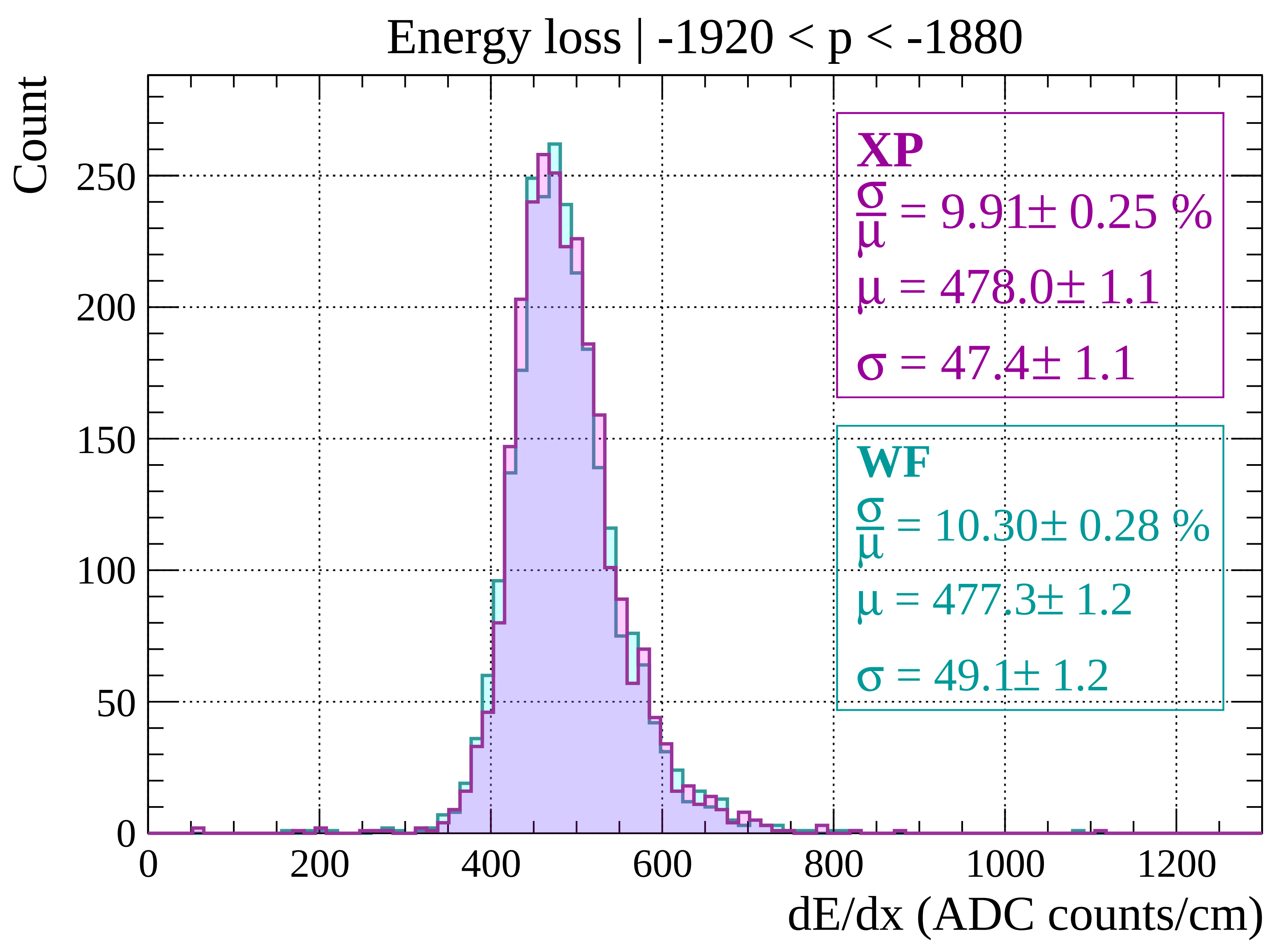
χ^2/NDF
50
45
40
35
30
25
20
15
10
5
0
−50
−40
−30
−20
−10
0
10
20
30
40
50
R
200
180
160
140
120
100
80
60
40
20
0
$\times 10^3$



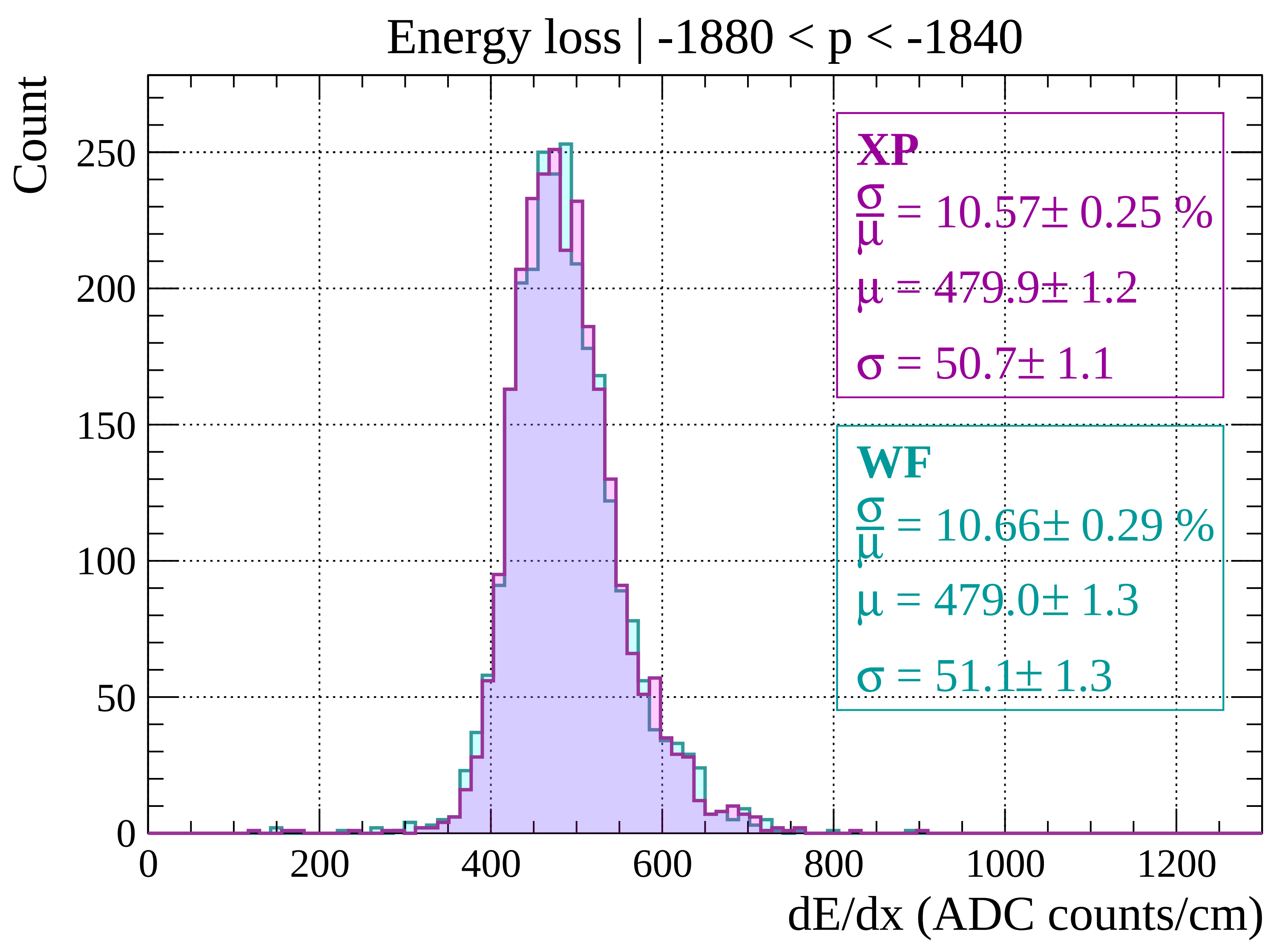
Energy loss | -2000 < p < -1960
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200
XP
σ/μ = 10.27± 0.35 %
μ = 485.1± 1.7
σ = 49.8± 1.5
WF
σ/μ = 10.61± 0.38 %
μ = 486.0± 1.8
σ = 51.5± 1.7



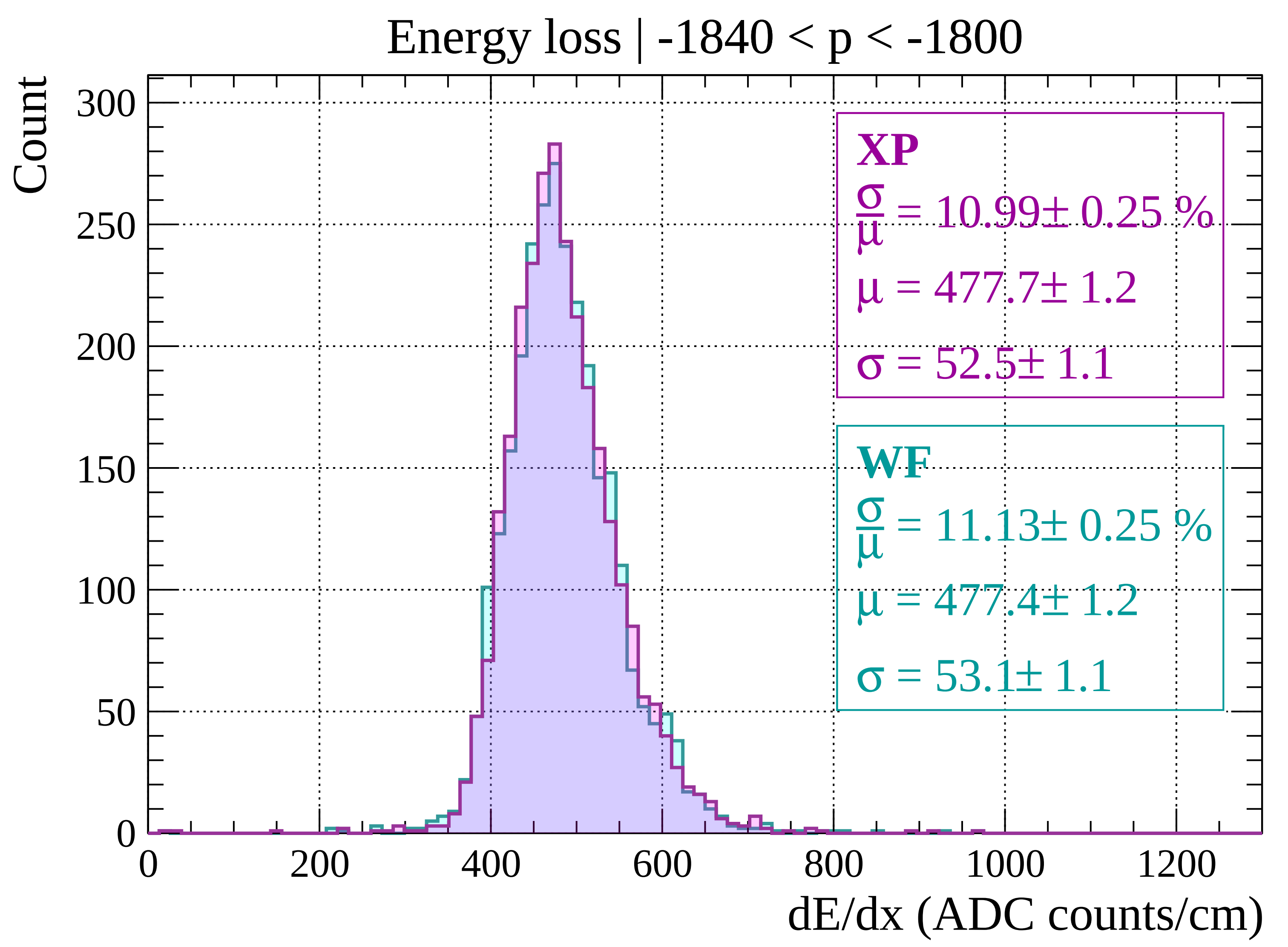
Energy loss | -1960 < p < -1920
Count
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.45± 0.23 %
μ = 482.9± 1.2
σ = 50.4± 1.0
WF
σ/μ = 11.39± 0.32 %
μ = 481.0± 1.4
σ = 54.8± 1.4

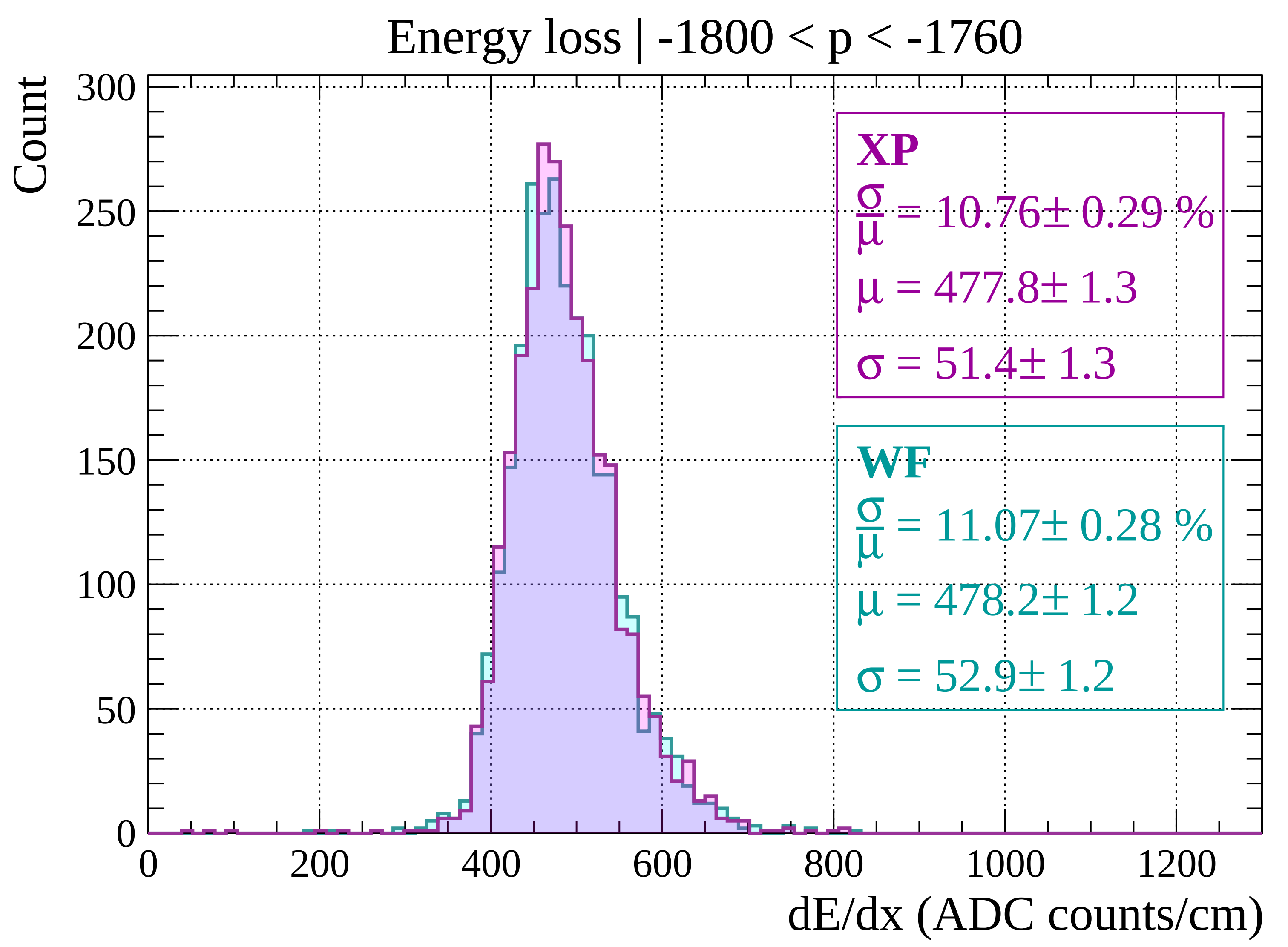


Energy loss | -1920 < p < -1880
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 9.91± 0.25 %
μ = 478.0± 1.1
σ = 47.4± 1.1
WF
σ/μ = 10.30± 0.28 %
μ = 477.3± 1.2
σ = 49.1± 1.2

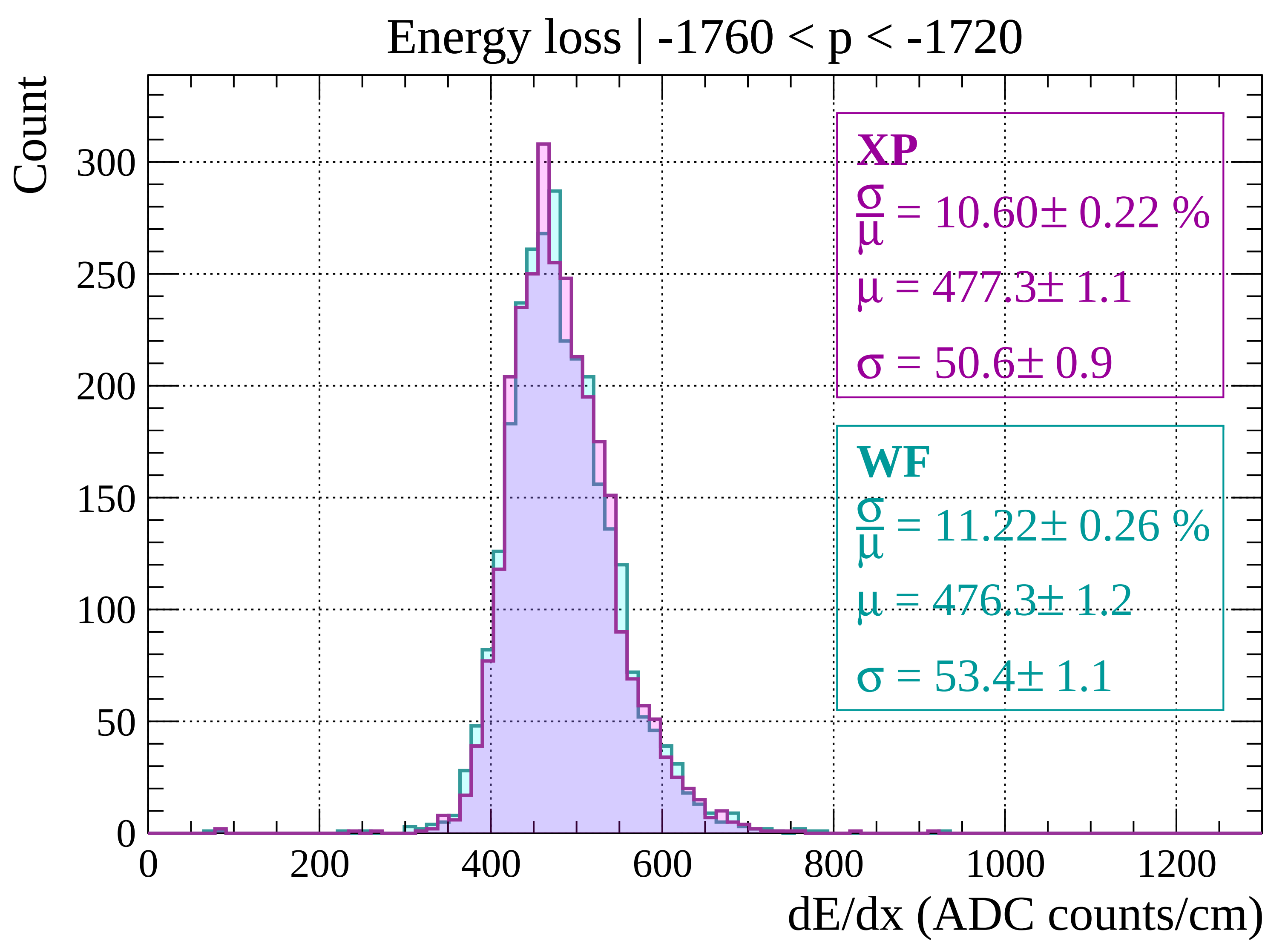


Energy loss | -1880 < p < -1840
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 10.57± 0.25 %
μ = 479.9± 1.2
σ = 50.7± 1.1
WF
σ/μ = 10.66± 0.29 %
μ = 479.0± 1.3
σ = 51.1± 1.3

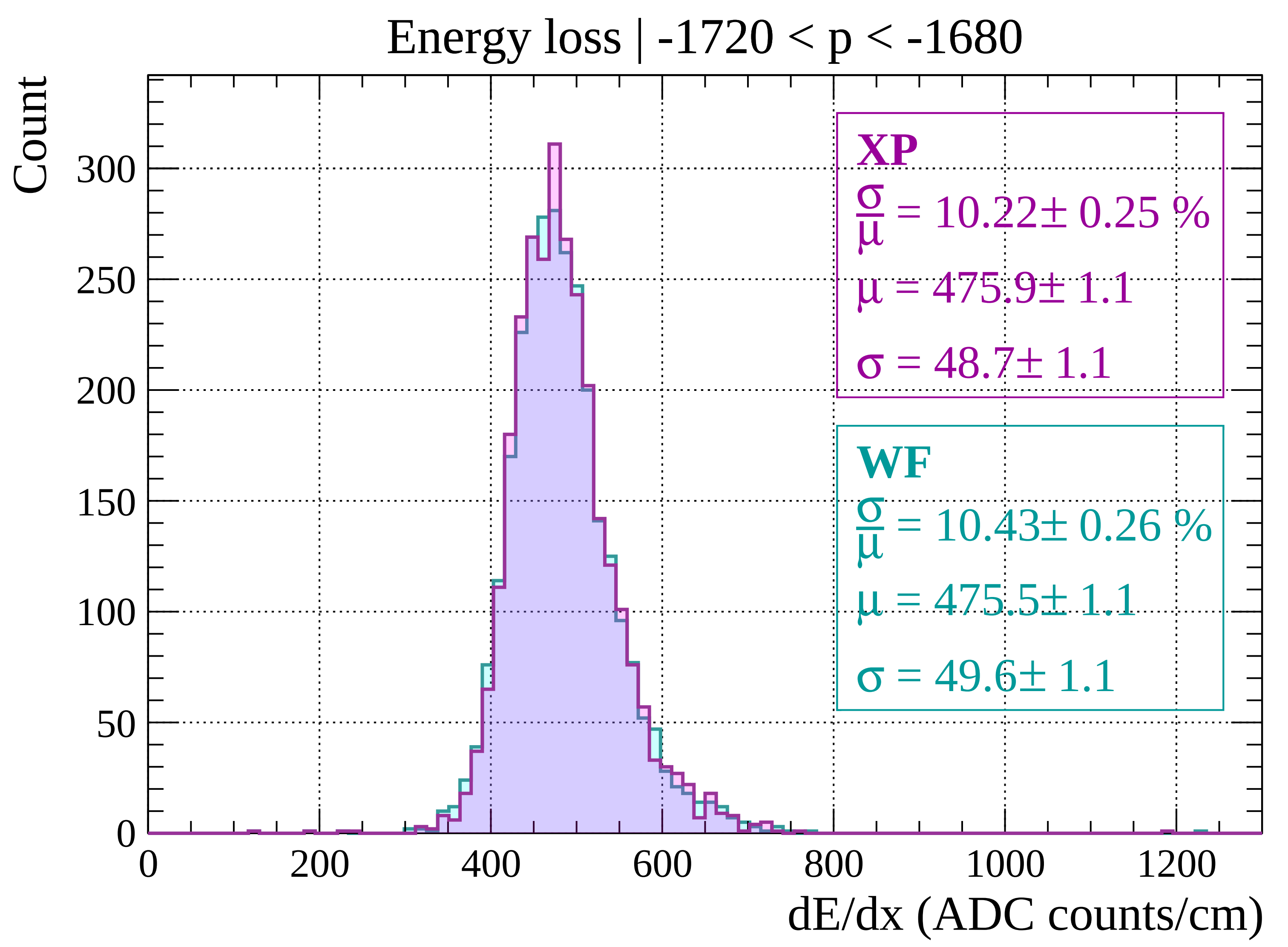

Energy loss | -1840 < p < -1800
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.99± 0.25 %
μ = 477.7± 1.2
σ = 52.5± 1.1
WF
σ/μ = 11.13± 0.25 %
μ = 477.4± 1.2
σ = 53.1± 1.1
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200



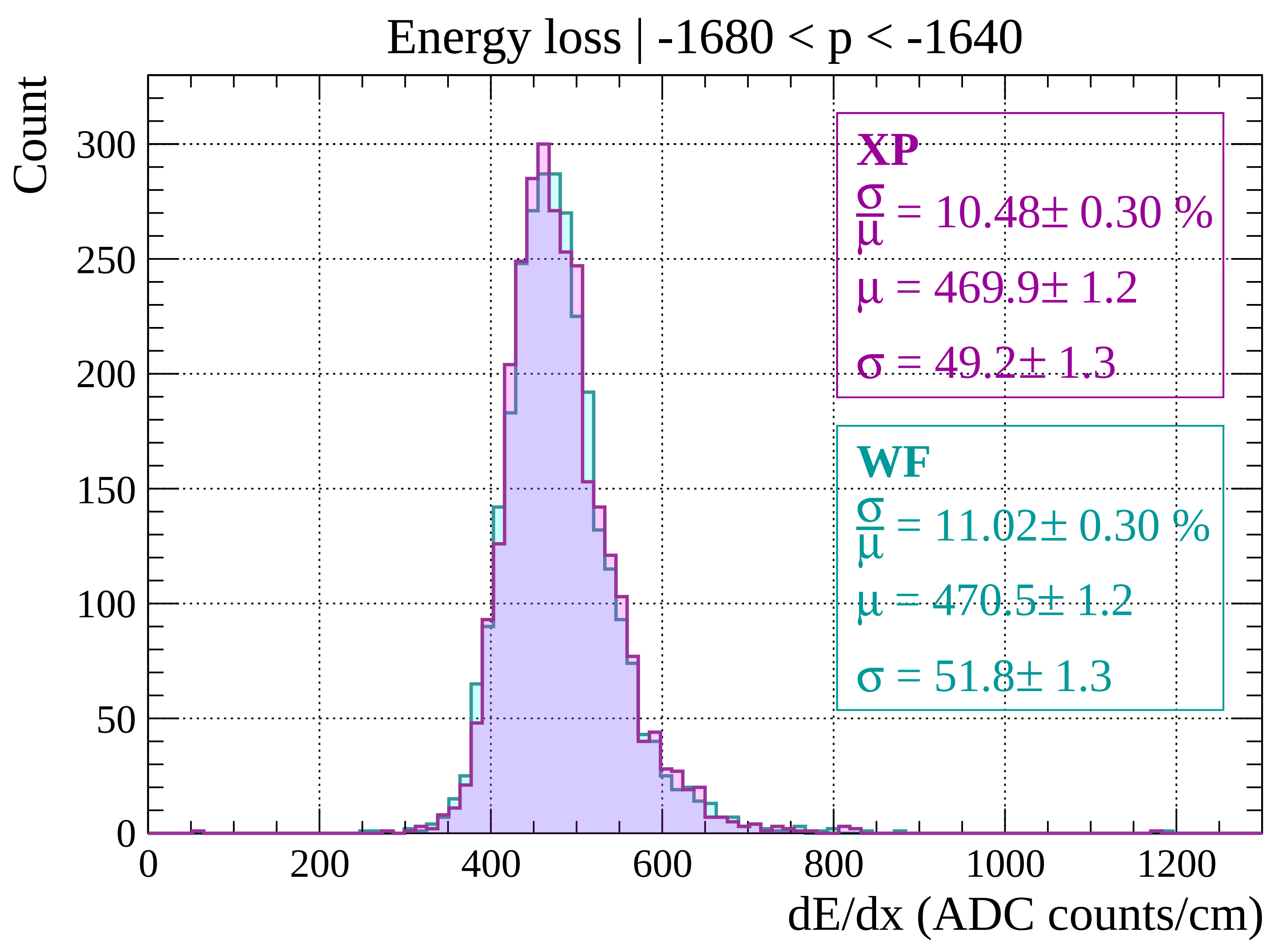
Energy loss | -1800 < p < -1760
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.76± 0.29 %
μ = 477.8± 1.3
σ = 51.4± 1.3
WF
σ/μ = 11.07± 0.28 %
μ = 478.2± 1.2
σ = 52.9± 1.2



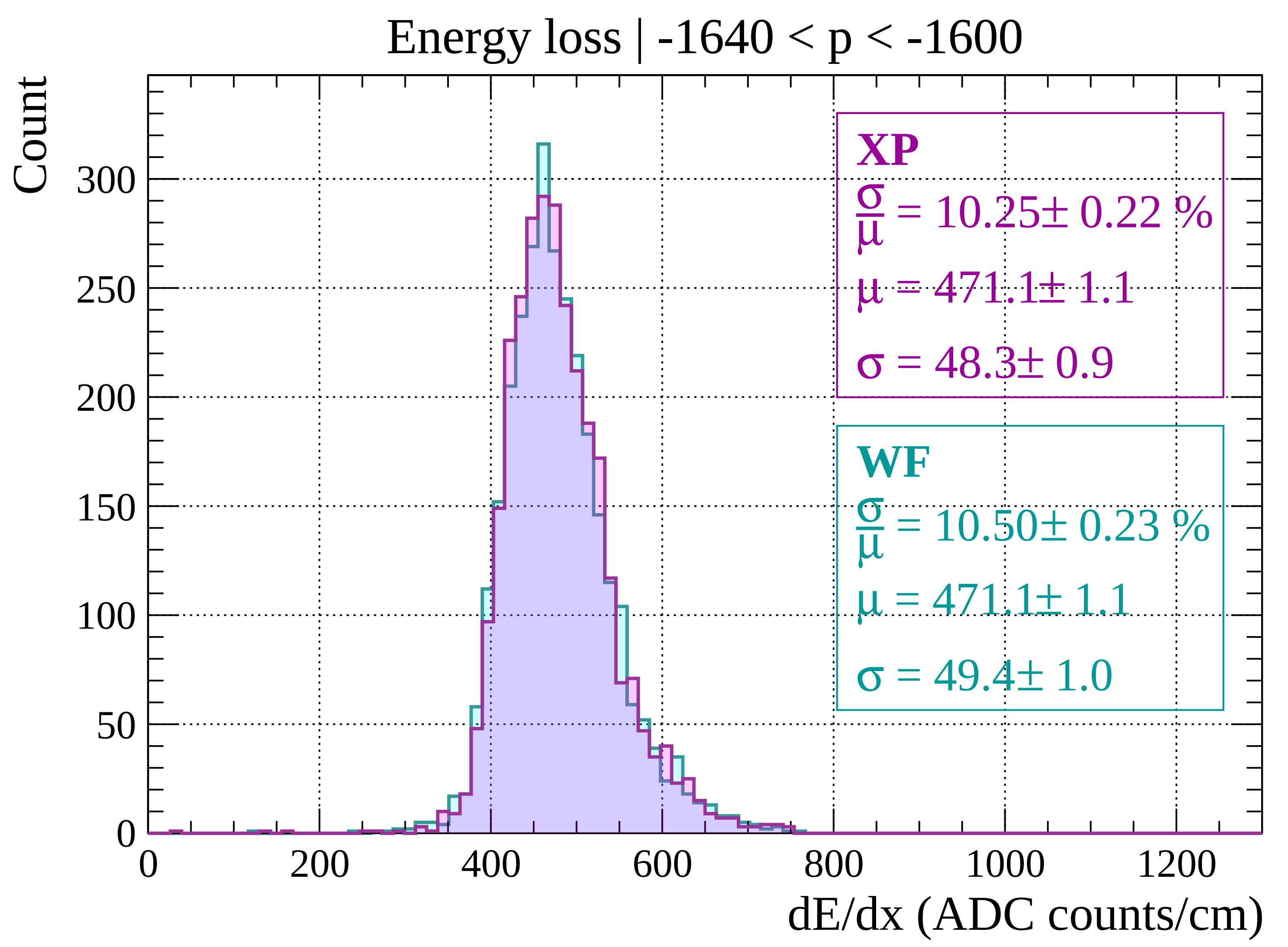
Energy loss | -1760 < p < -1720
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.60± 0.22 %
μ = 477.3± 1.1
σ = 50.6± 0.9
WF
$\frac{\sigma}{\mu}$ = 11.22± 0.26 %
μ = 476.3± 1.2
σ = 53.4± 1.1
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200



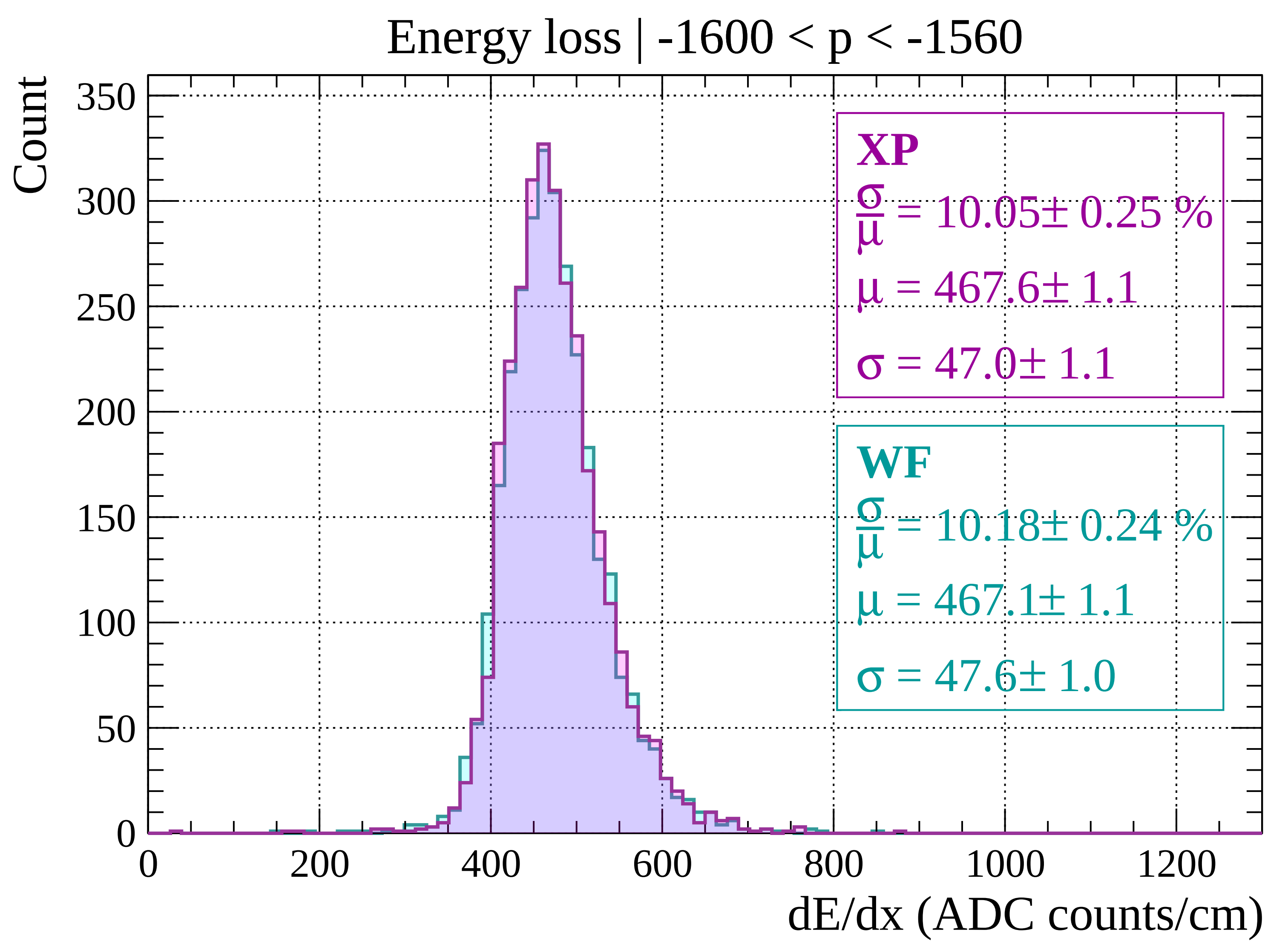
Energy loss | -1720 < p < -1680
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.22± 0.25 %
μ = 475.9± 1.1
σ = 48.7± 1.1
WF
$\frac{\sigma}{\mu}$ = 10.43± 0.26 %
μ = 475.5± 1.1
σ = 49.6± 1.1



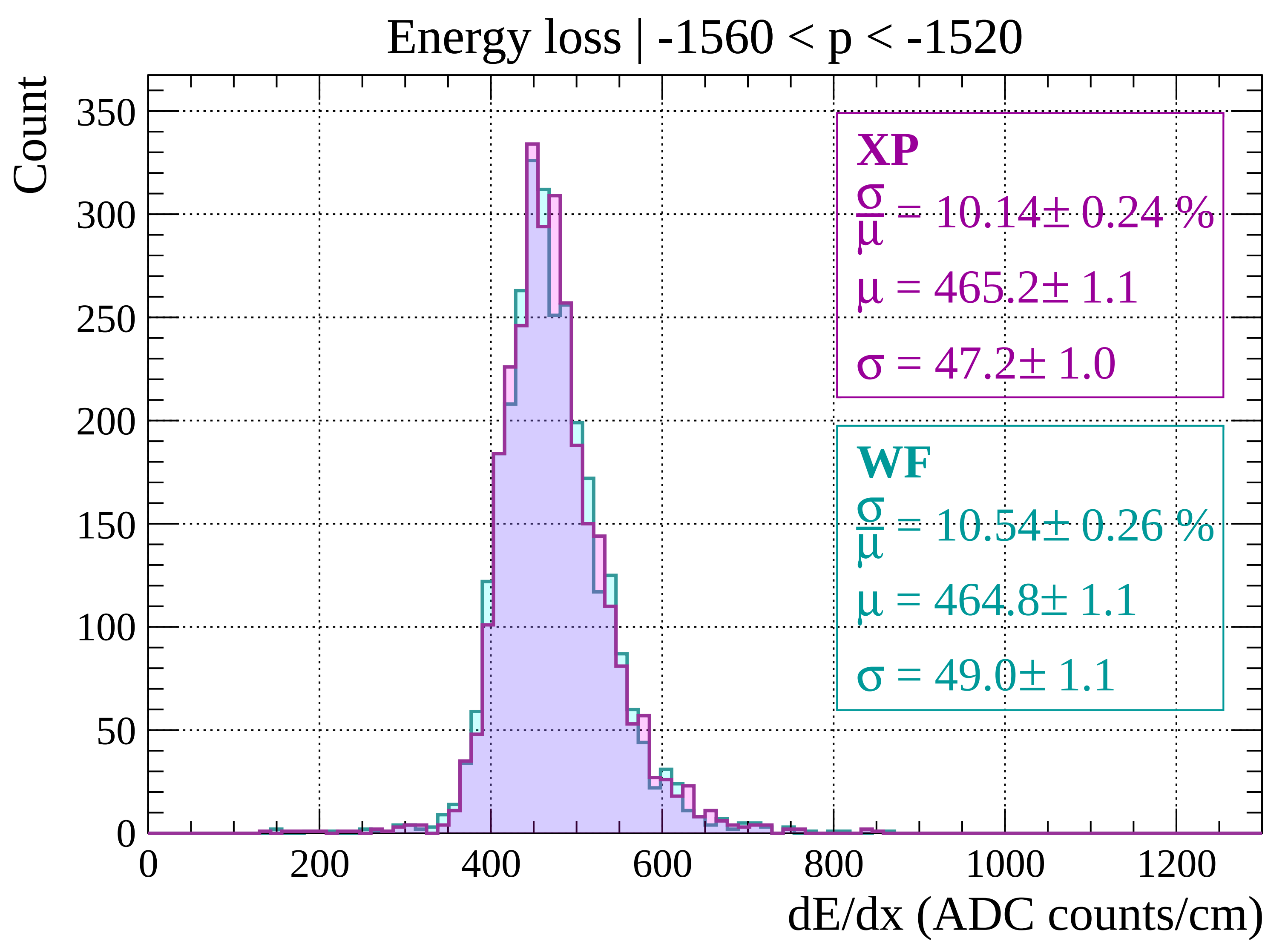
Energy loss | -1680 < p < -1640
Count
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.48± 0.30 %
μ = 469.9± 1.2
σ = 49.2± 1.3
WF
σ/μ = 11.02± 0.30 %
μ = 470.5± 1.2
σ = 51.8± 1.3



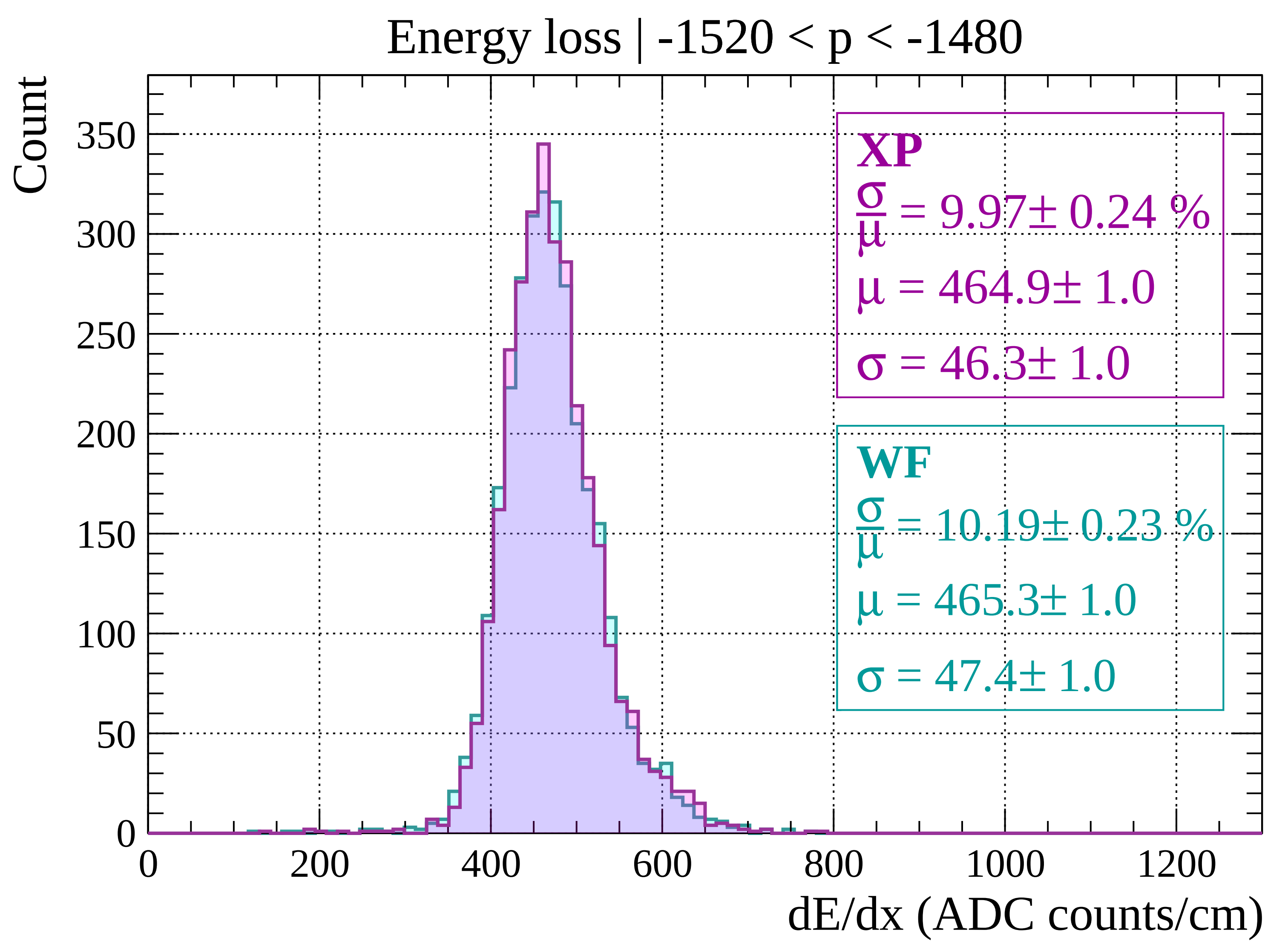
Energy loss | -1640 < p < -1600
Count
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.25± 0.22 %
μ = 471.1± 1.1
σ = 48.3± 0.9
WF
σ/μ = 10.50± 0.23 %
μ = 471.1± 1.1
σ = 49.4± 1.0

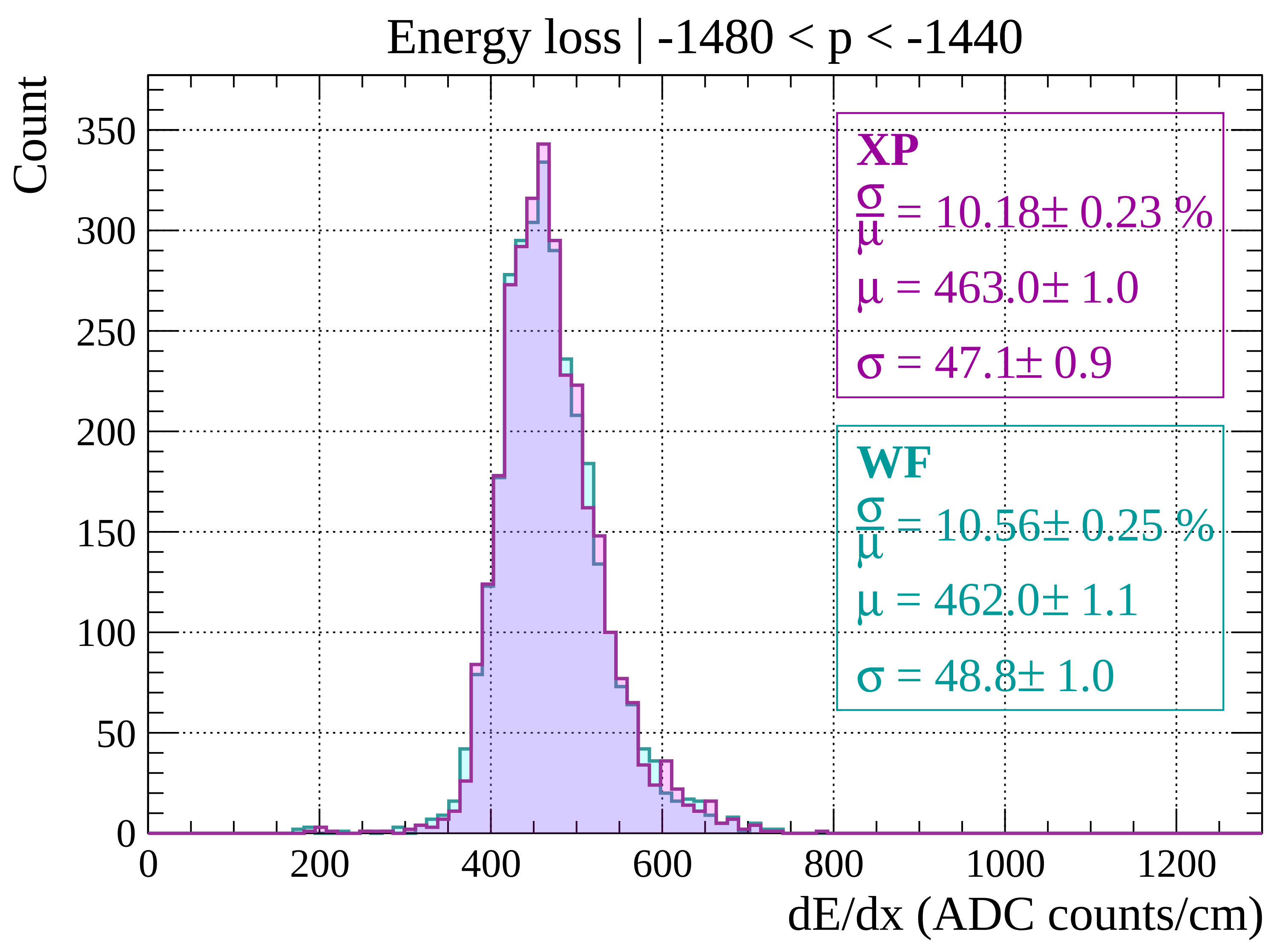


Energy loss | -1600 < p < -1560
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 10.05± 0.25 %
μ = 467.6± 1.1
σ = 47.0± 1.1
WF
σ/μ = 10.18± 0.24 %
μ = 467.1± 1.1
σ = 47.6± 1.0

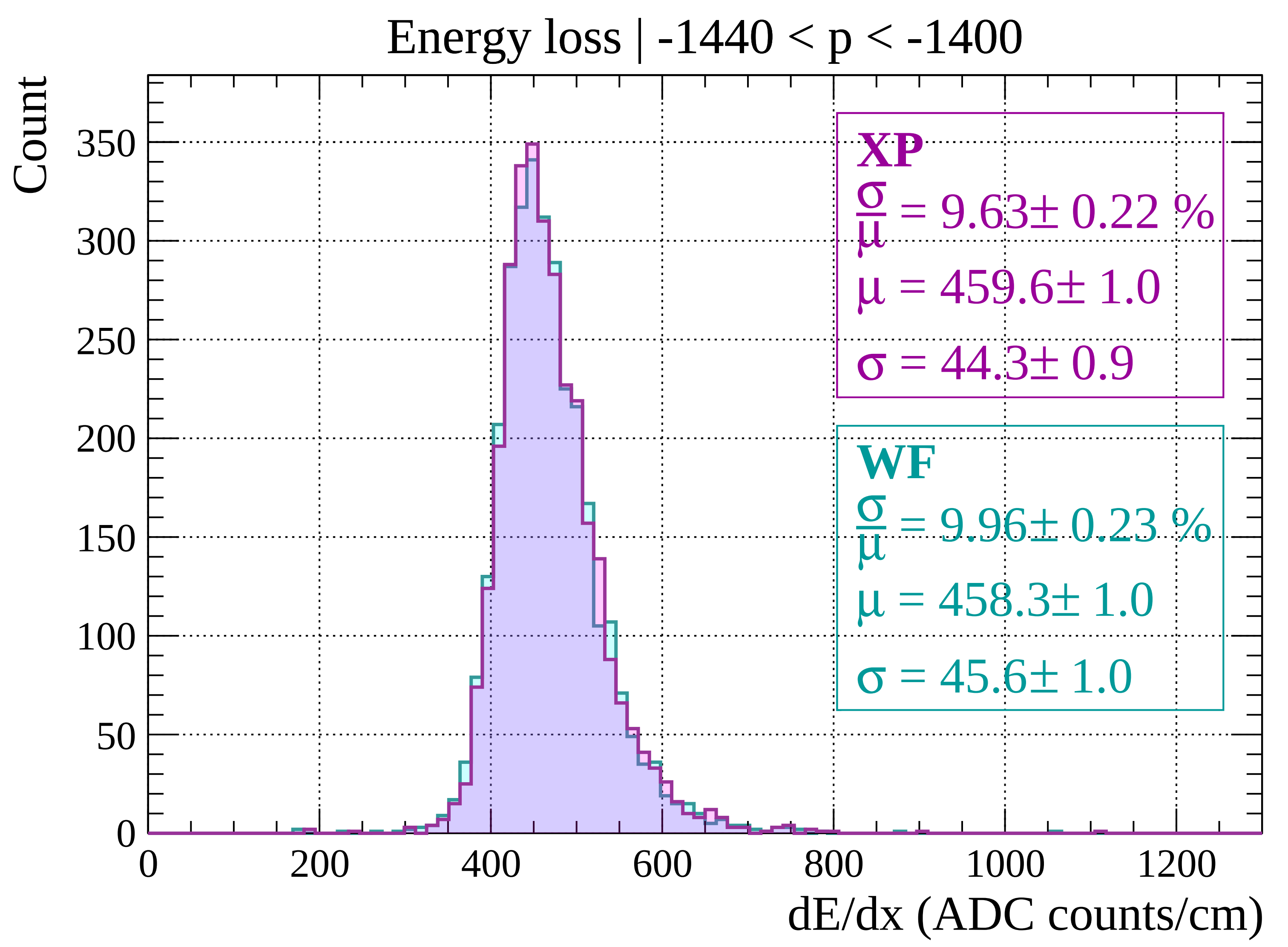


Energy loss | -1560 < p < -1520
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.14 ± 0.24 %
μ = 465.2 ± 1.1
σ = 47.2 ± 1.0
WF
σ/μ = 10.54 ± 0.26 %
μ = 464.8 ± 1.1
σ = 49.0 ± 1.1

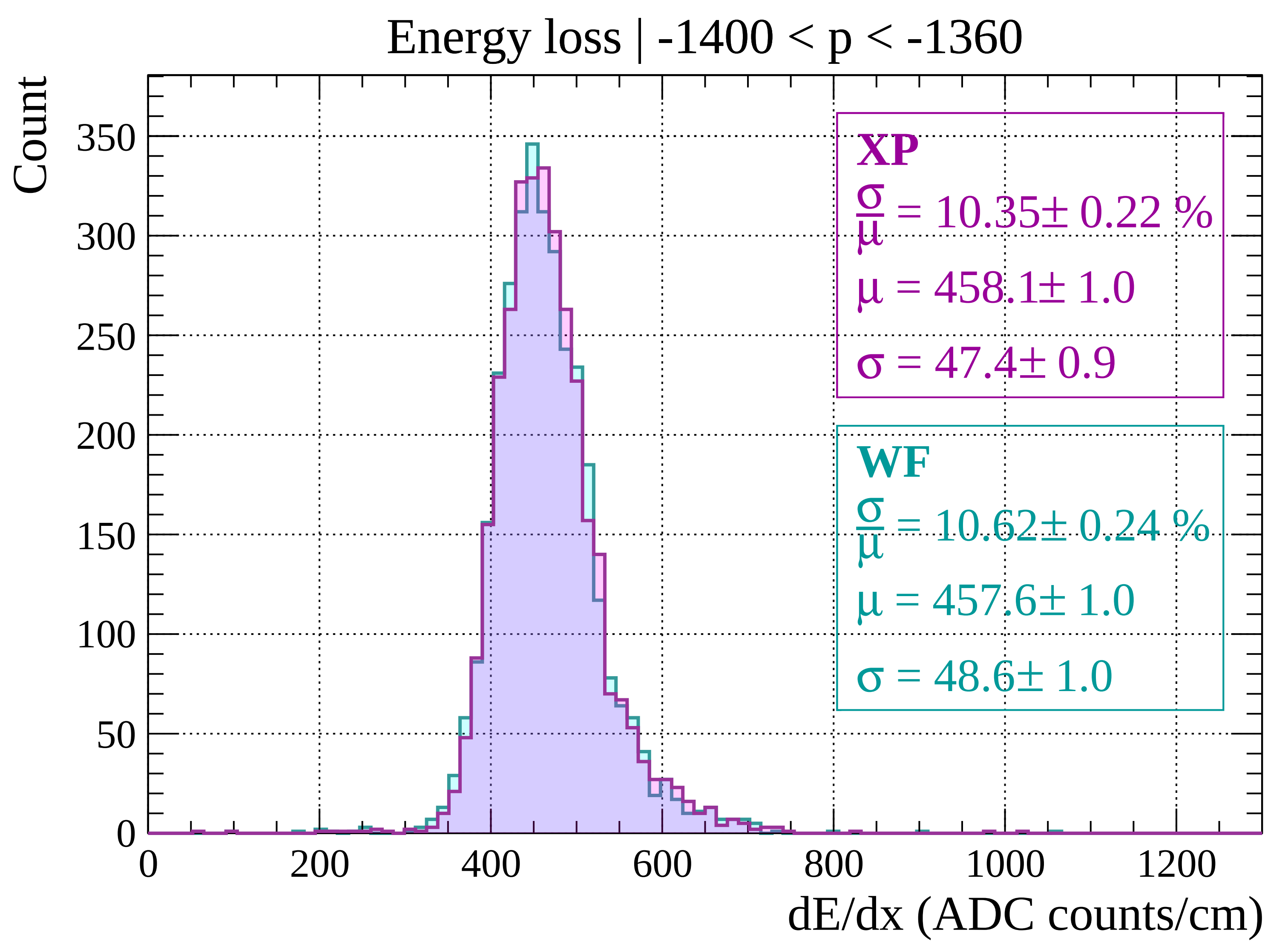

Energy loss | -1520 < p < -1480
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 9.97± 0.24 %
μ = 464.9± 1.0
σ = 46.3± 1.0
WF
σ/μ = 10.19± 0.23 %
μ = 465.3± 1.0
σ = 47.4± 1.0



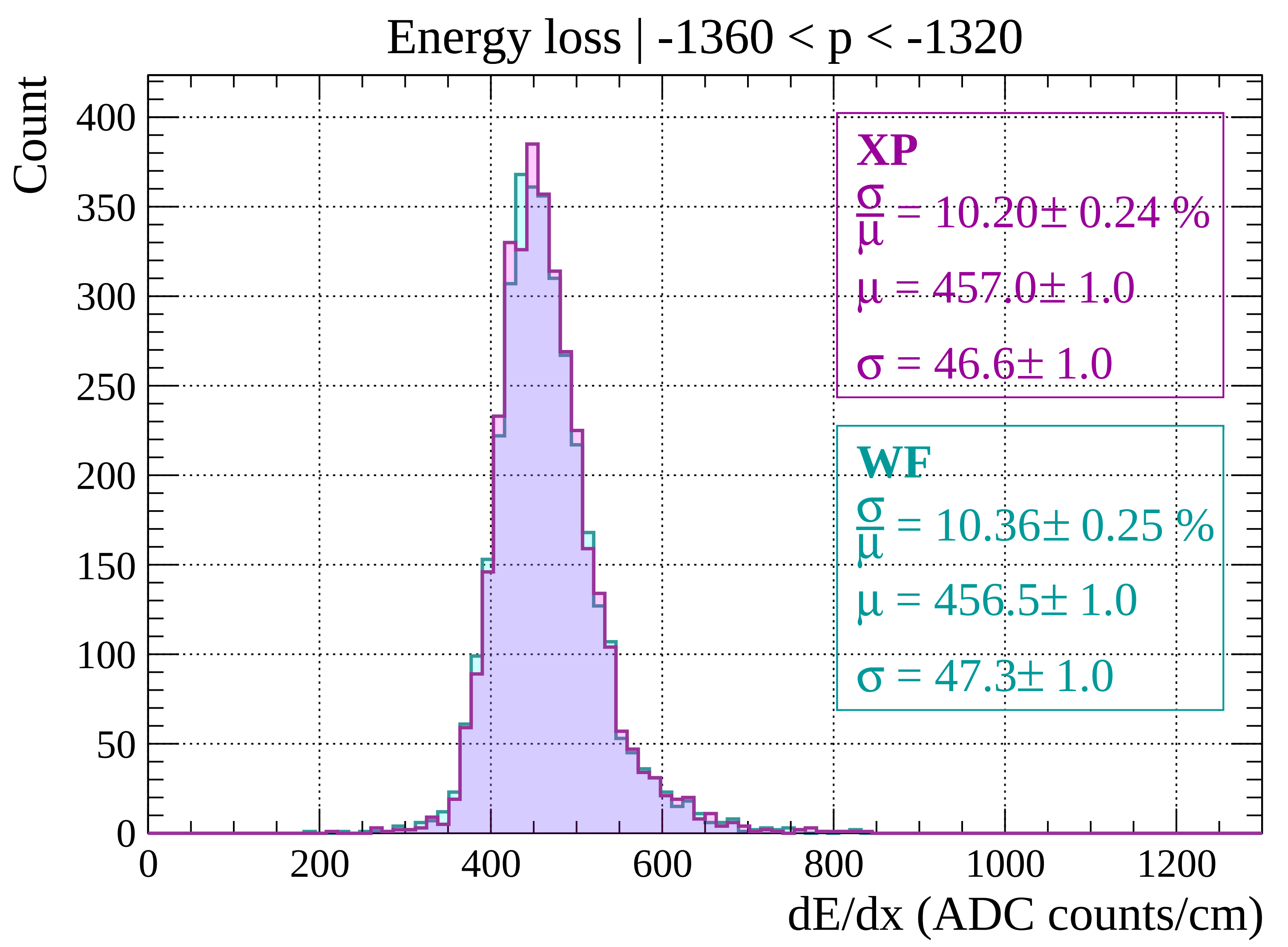
Energy loss | -1480 < p < -1440
Count
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 10.18± 0.23 %
μ = 463.0± 1.0
σ = 47.1± 0.9
WF
σ/μ = 10.56± 0.25 %
μ = 462.0± 1.1
σ = 48.8± 1.0



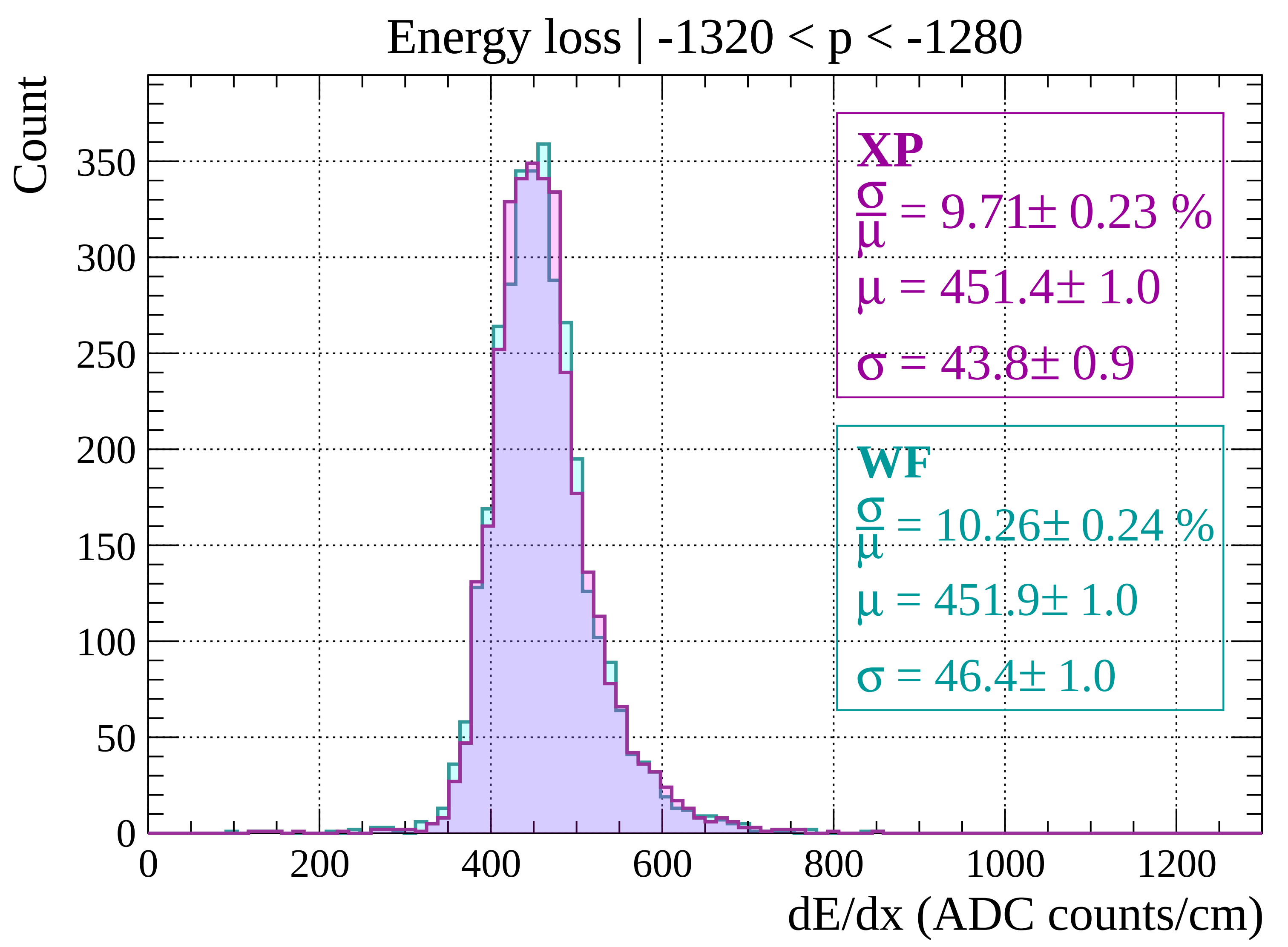
Energy loss | -1440 < p < -1400
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 9.63± 0.22 %
μ = 459.6± 1.0
σ = 44.3± 0.9
WF
σ/μ = 9.96± 0.23 %
μ = 458.3± 1.0
σ = 45.6± 1.0



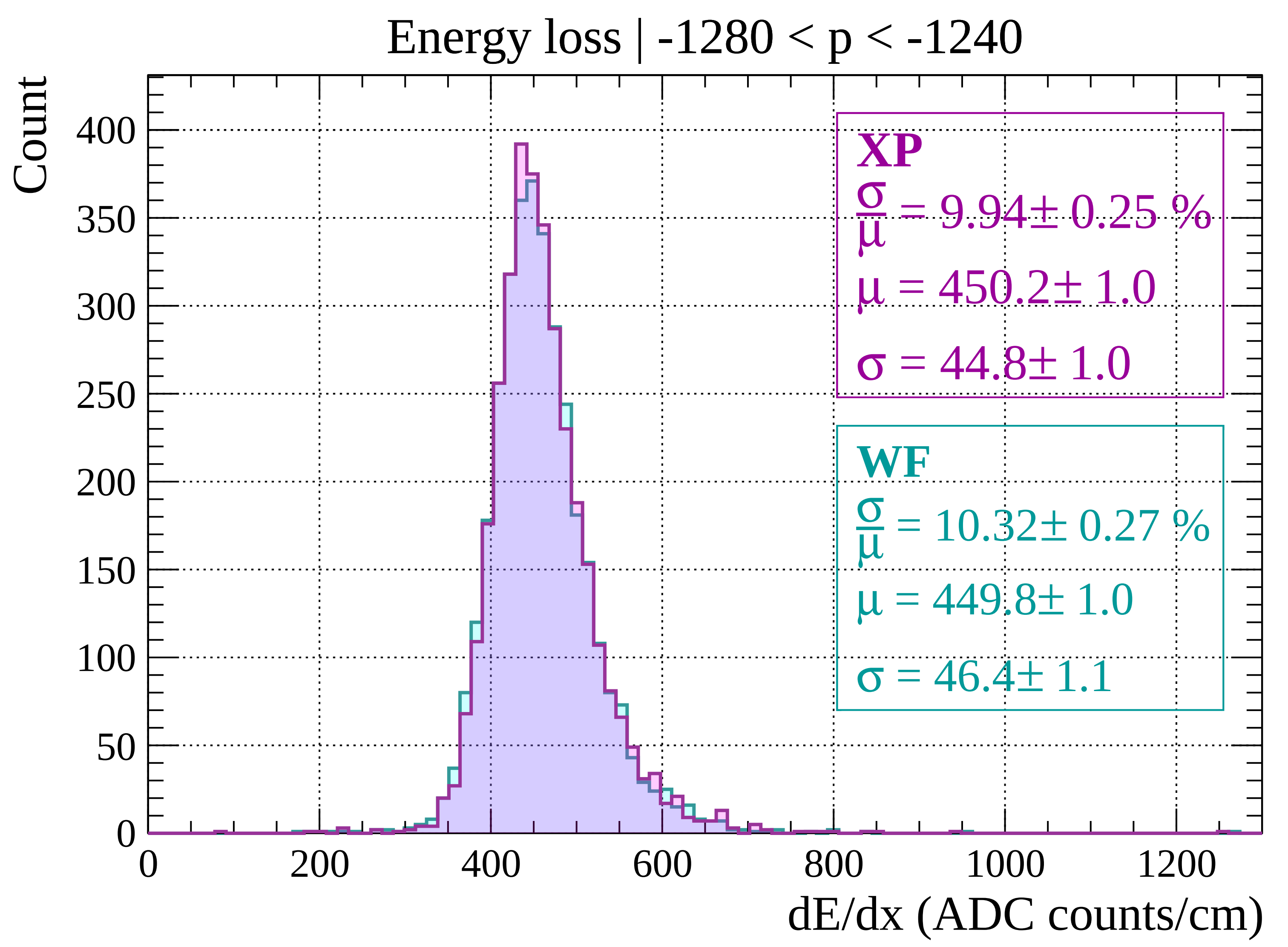
Energy loss | -1400 < p < -1360
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
XP
σ/μ = 10.35± 0.22 %
μ = 458.1± 1.0
σ = 47.4± 0.9
WF
σ/μ = 10.62± 0.24 %
μ = 457.6± 1.0
σ = 48.6± 1.0



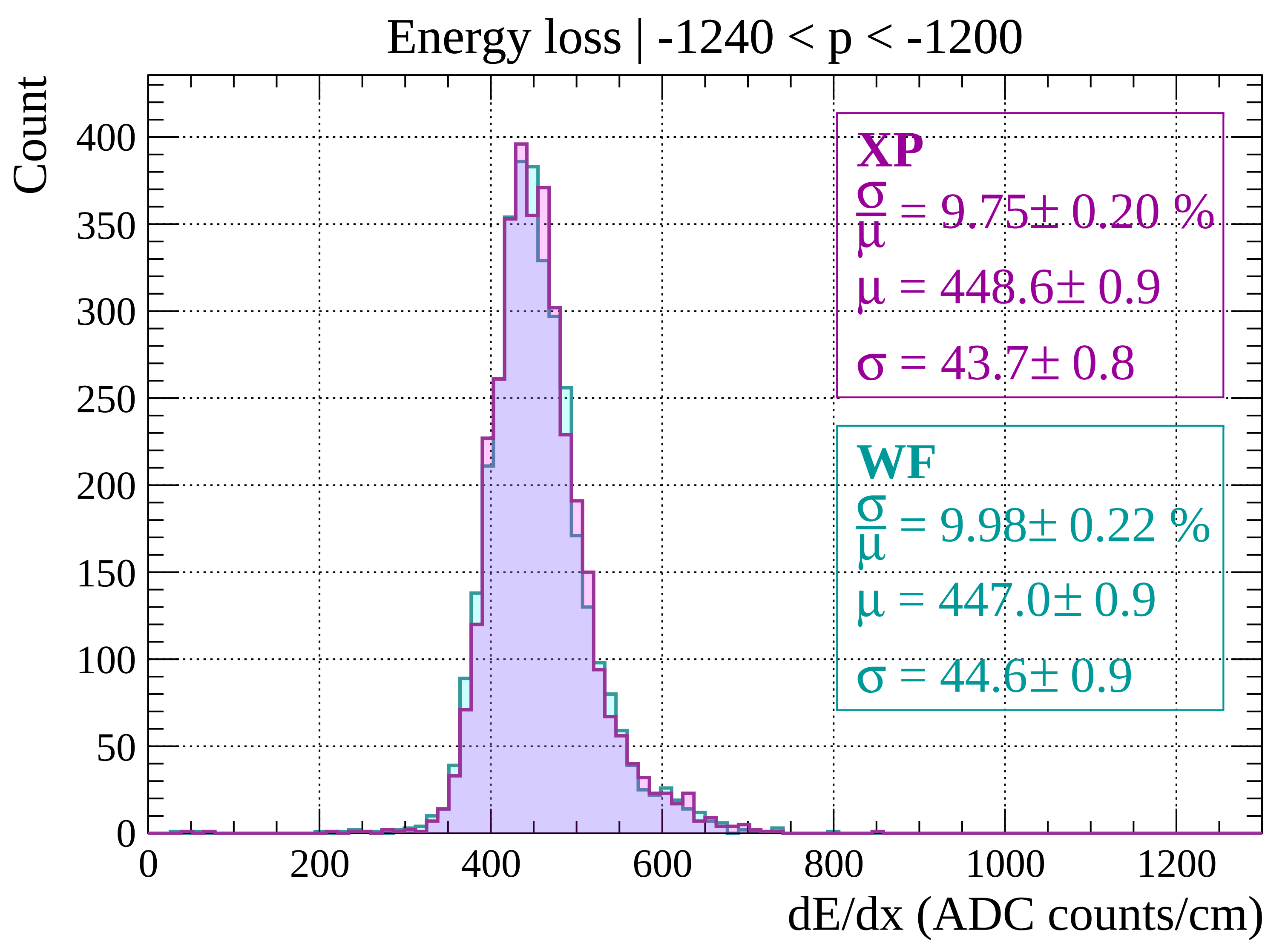
Energy loss | -1360 < p < -1320
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 10.20 \pm 0.24$ %
$\mu = 457.0 \pm 1.0$
$\sigma = 46.6 \pm 1.0$
WF
$\frac{\sigma}{\mu} = 10.36 \pm 0.25$ %
$\mu = 456.5 \pm 1.0$
$\sigma = 47.3 \pm 1.0$



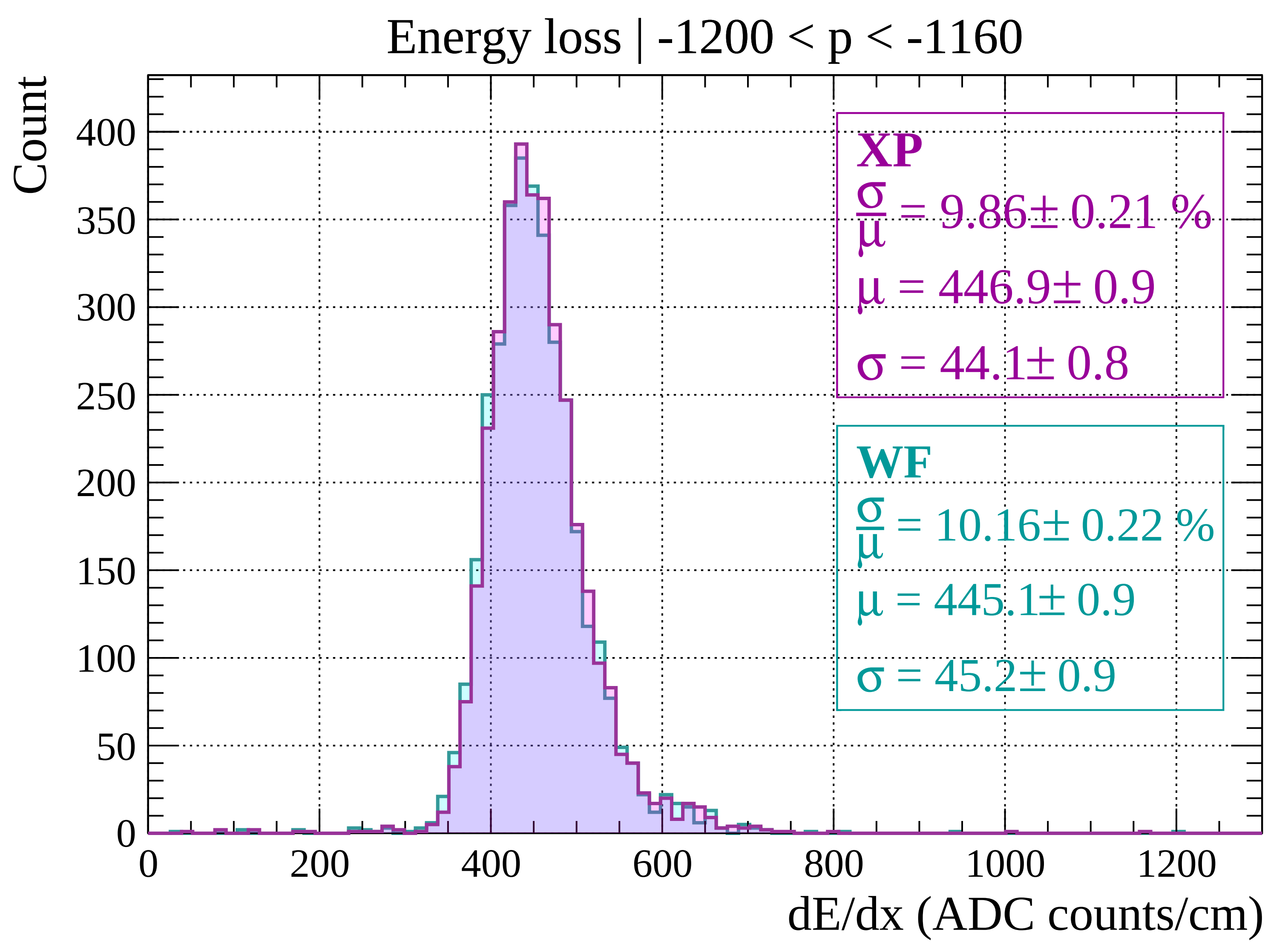
Energy loss | -1320 < p < -1280
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
0
200
400
600
800
1000
1200
XP
σ/μ = 9.71± 0.23 %
μ = 451.4± 1.0
σ = 43.8± 0.9
WF
σ/μ = 10.26± 0.24 %
μ = 451.9± 1.0
σ = 46.4± 1.0



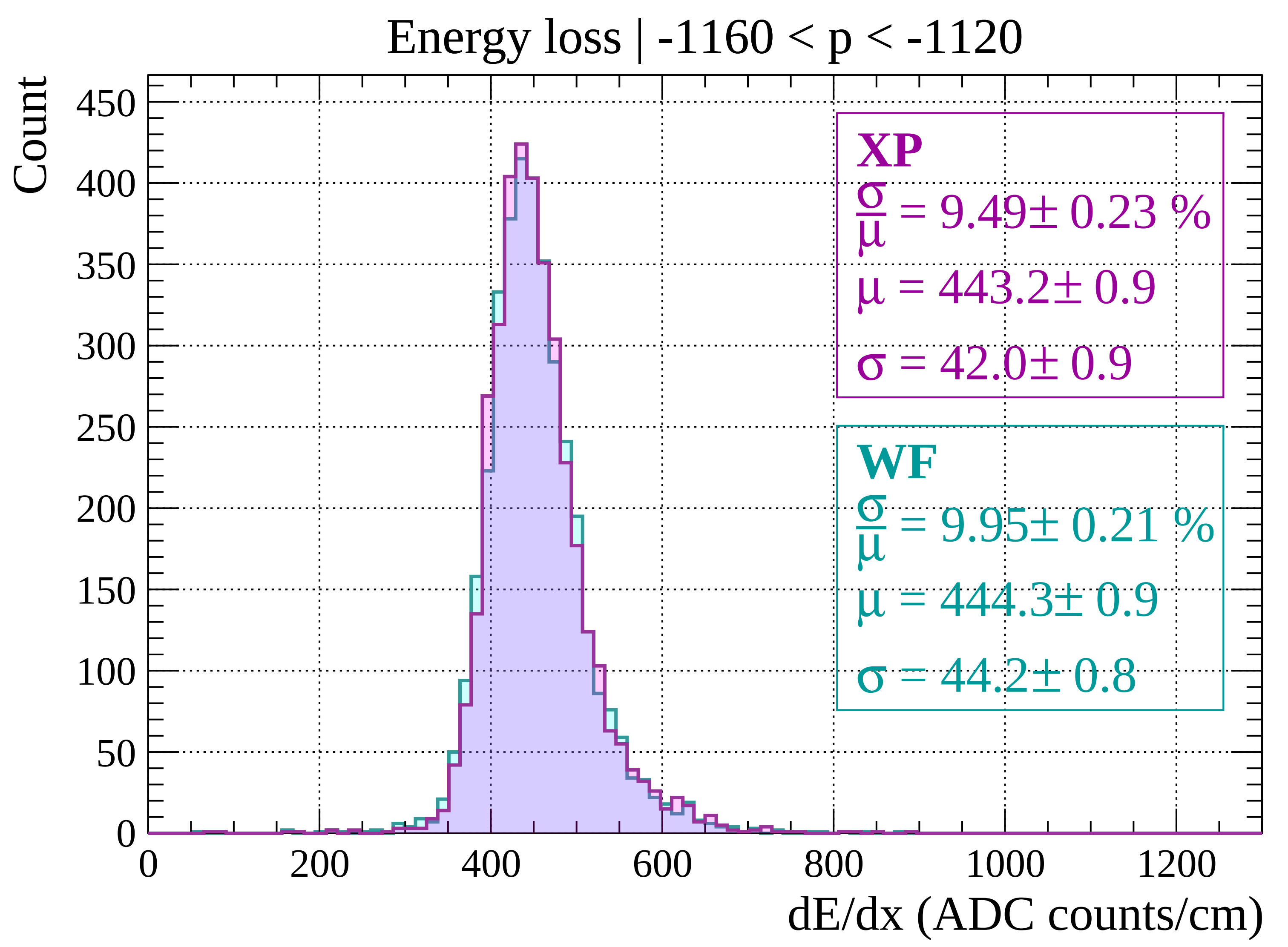
Energy loss | -1280 < p < -1240
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
XP
σ/μ = 9.94 ± 0.25 %
μ = 450.2 ± 1.0
σ = 44.8 ± 1.0
WF
σ/μ = 10.32 ± 0.27 %
μ = 449.8 ± 1.0
σ = 46.4 ± 1.1

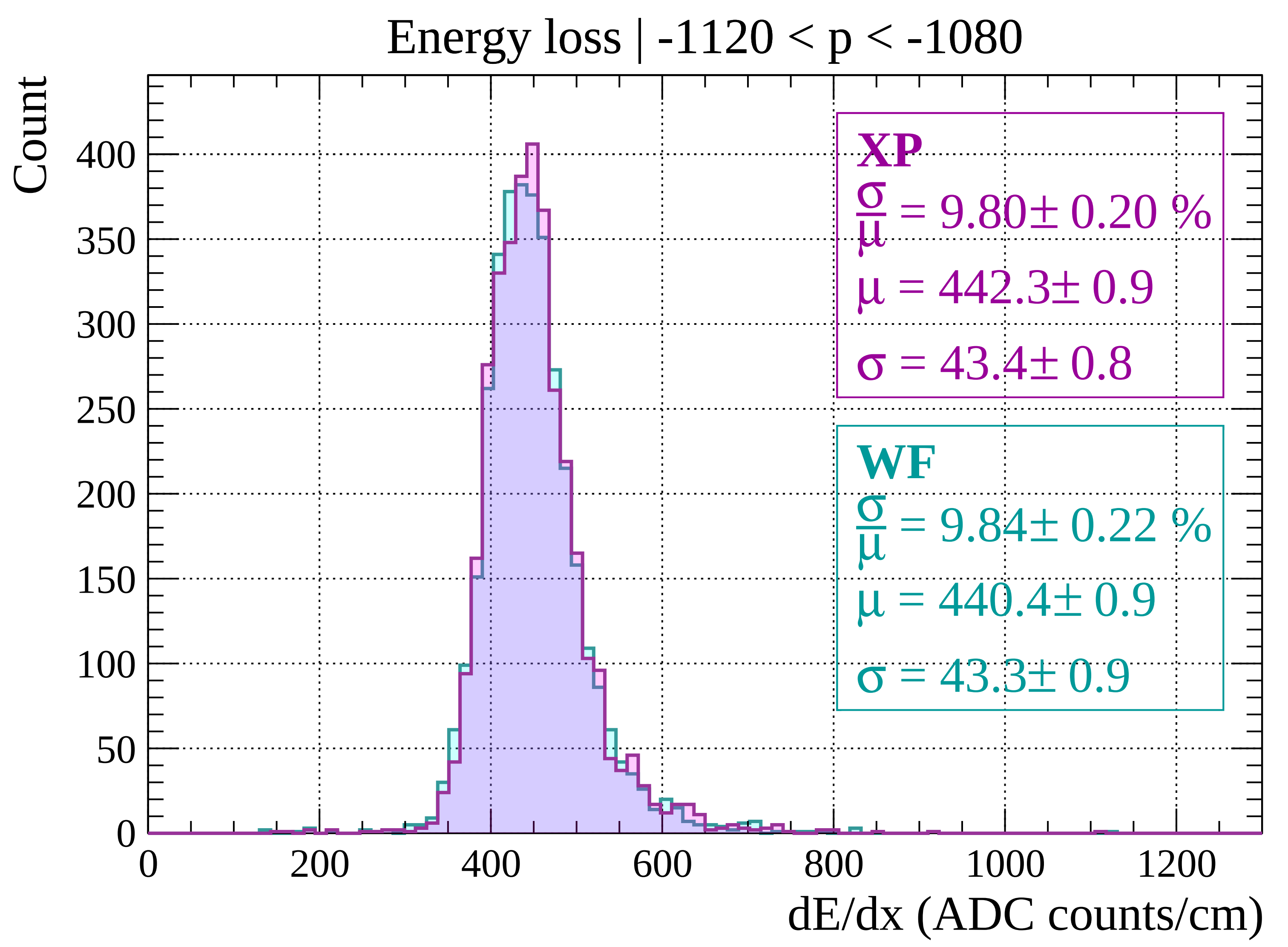


Energy loss | -1240 < p < -1200
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 9.75± 0.20 %
μ = 448.6± 0.9
σ = 43.7± 0.8
WF
σ/μ = 9.98± 0.22 %
μ = 447.0± 0.9
σ = 44.6± 0.9

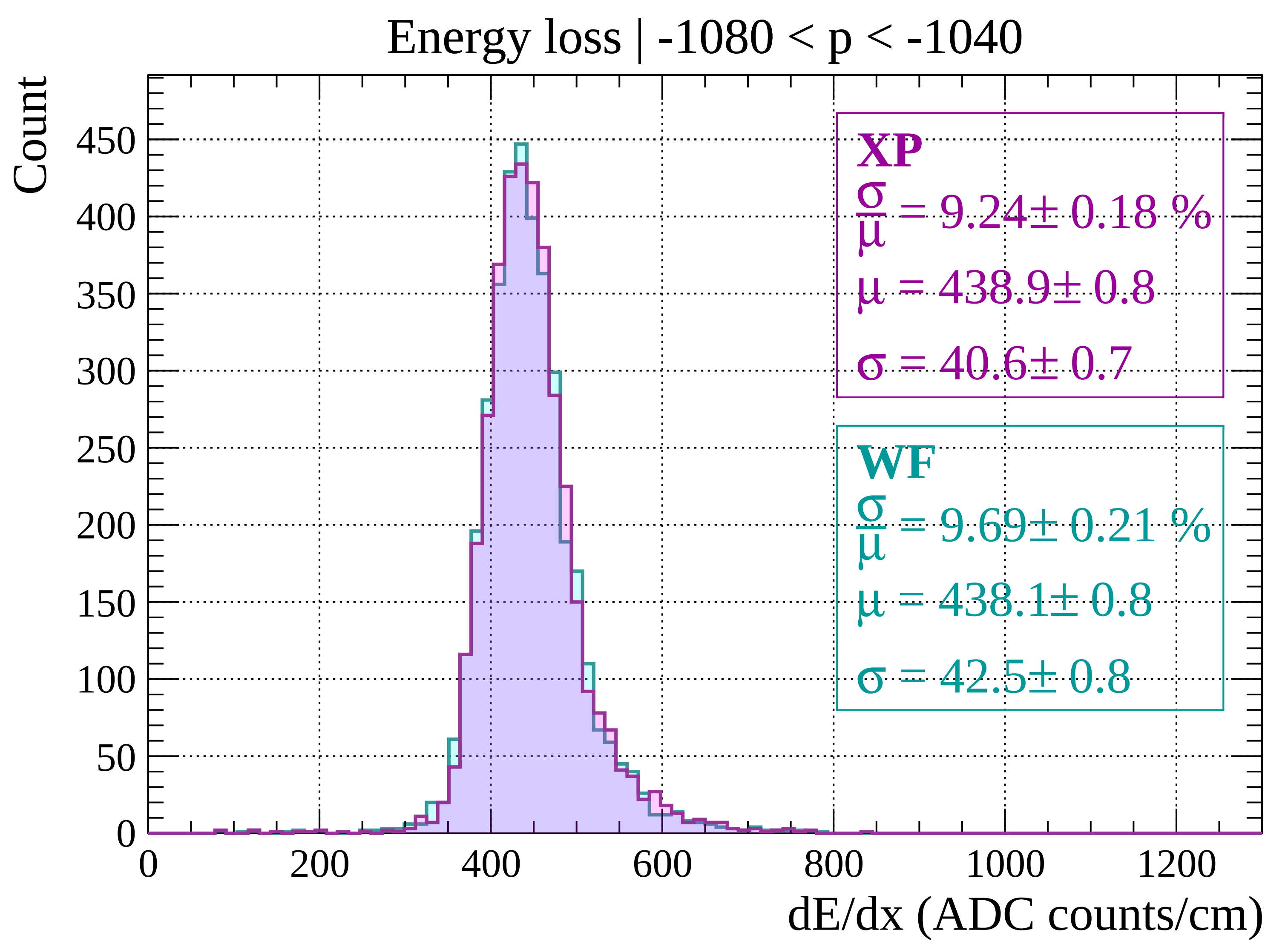


Energy loss | -1200 < p < -1160
Count
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.86± 0.21 %
μ = 446.9± 0.9
σ = 44.1± 0.8
WF
$\frac{\sigma}{\mu}$ = 10.16± 0.22 %
μ = 445.1± 0.9
σ = 45.2± 0.9

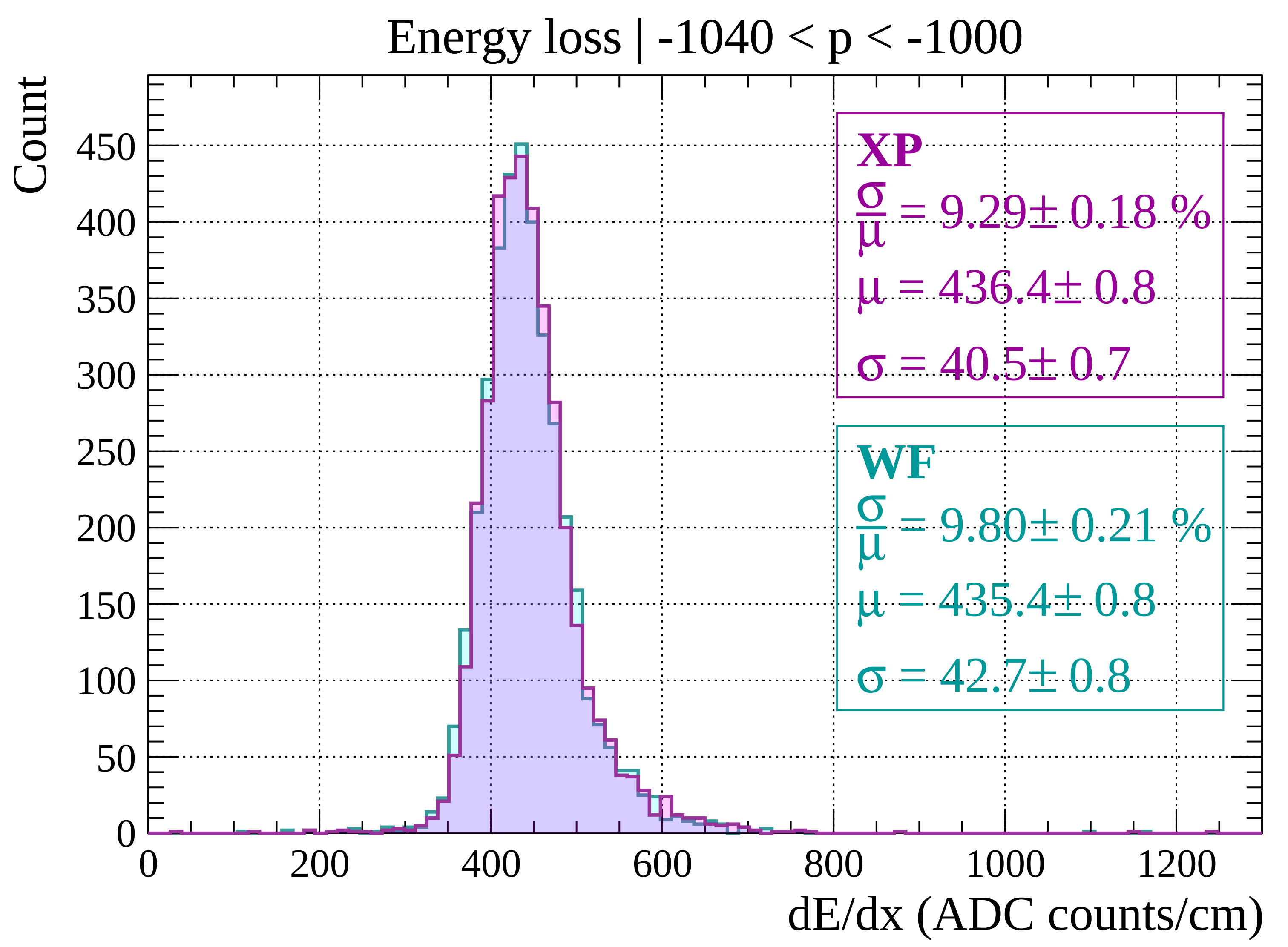

Energy loss | -1160 < p < -1120
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.49± 0.23 %
μ = 443.2± 0.9
σ = 42.0± 0.9
WF
$\frac{\sigma}{\mu}$ = 9.95± 0.21 %
μ = 444.3± 0.9
σ = 44.2± 0.8



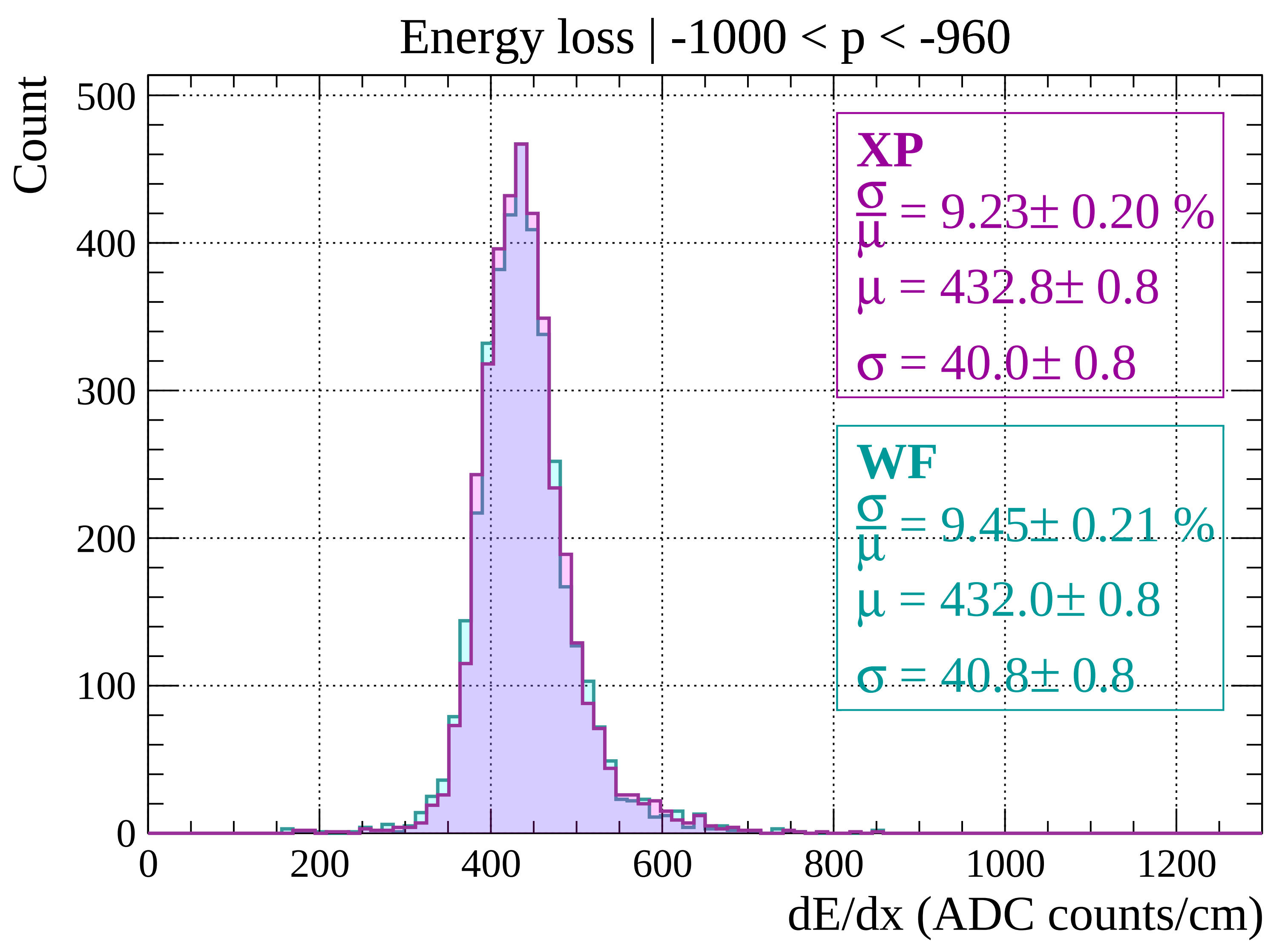
Energy loss | -1120 < p < -1080
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
XP
σ/μ = 9.80± 0.20 %
μ = 442.3± 0.9
σ = 43.4± 0.8
WF
σ/μ = 9.84± 0.22 %
μ = 440.4± 0.9
σ = 43.3± 0.9



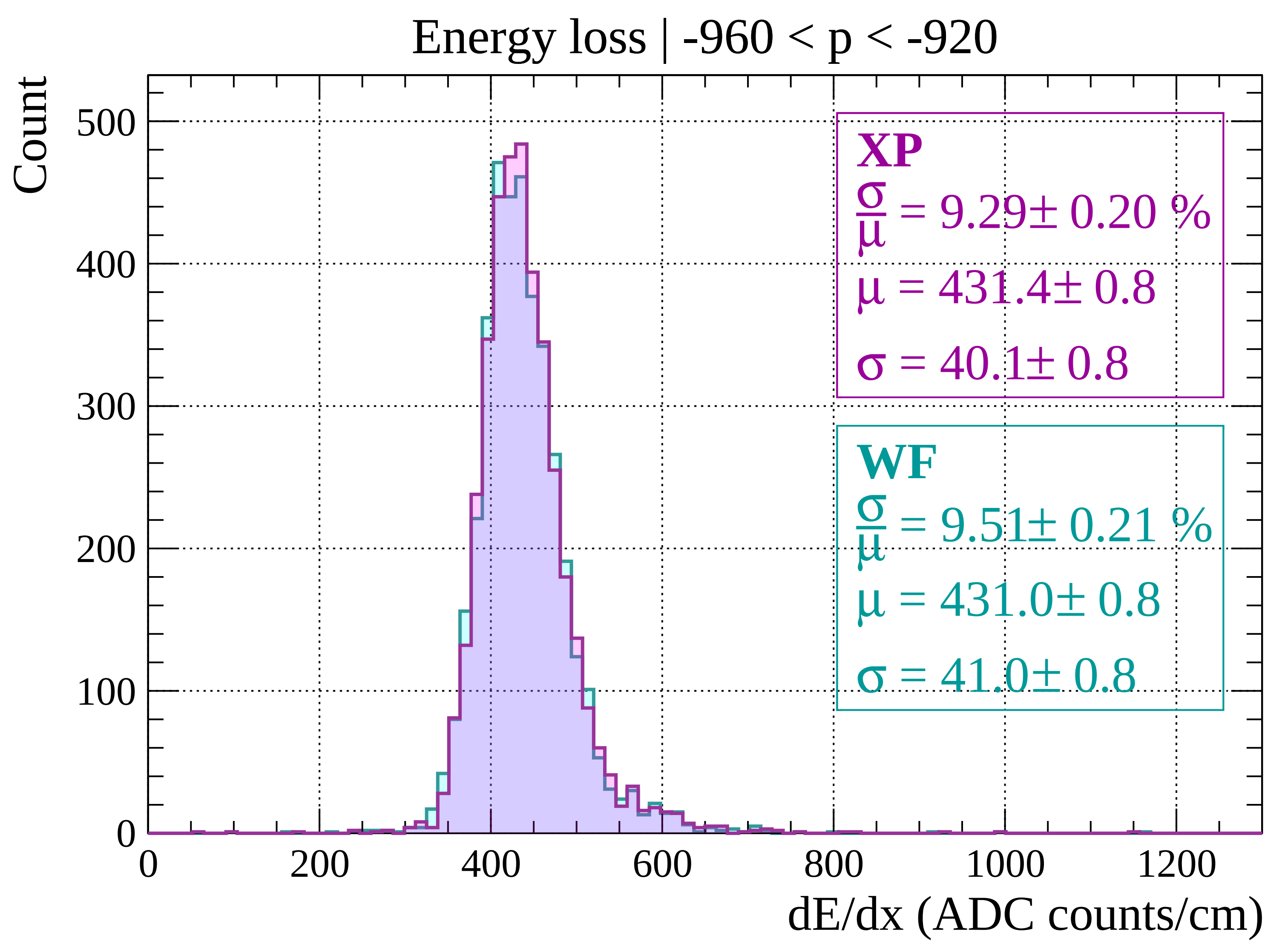
Energy loss | -1080 < p < -1040
Count
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.24 ± 0.18 %
μ = 438.9 ± 0.8
σ = 40.6 ± 0.7
WF
σ/μ = 9.69 ± 0.21 %
μ = 438.1 ± 0.8
σ = 42.5 ± 0.8



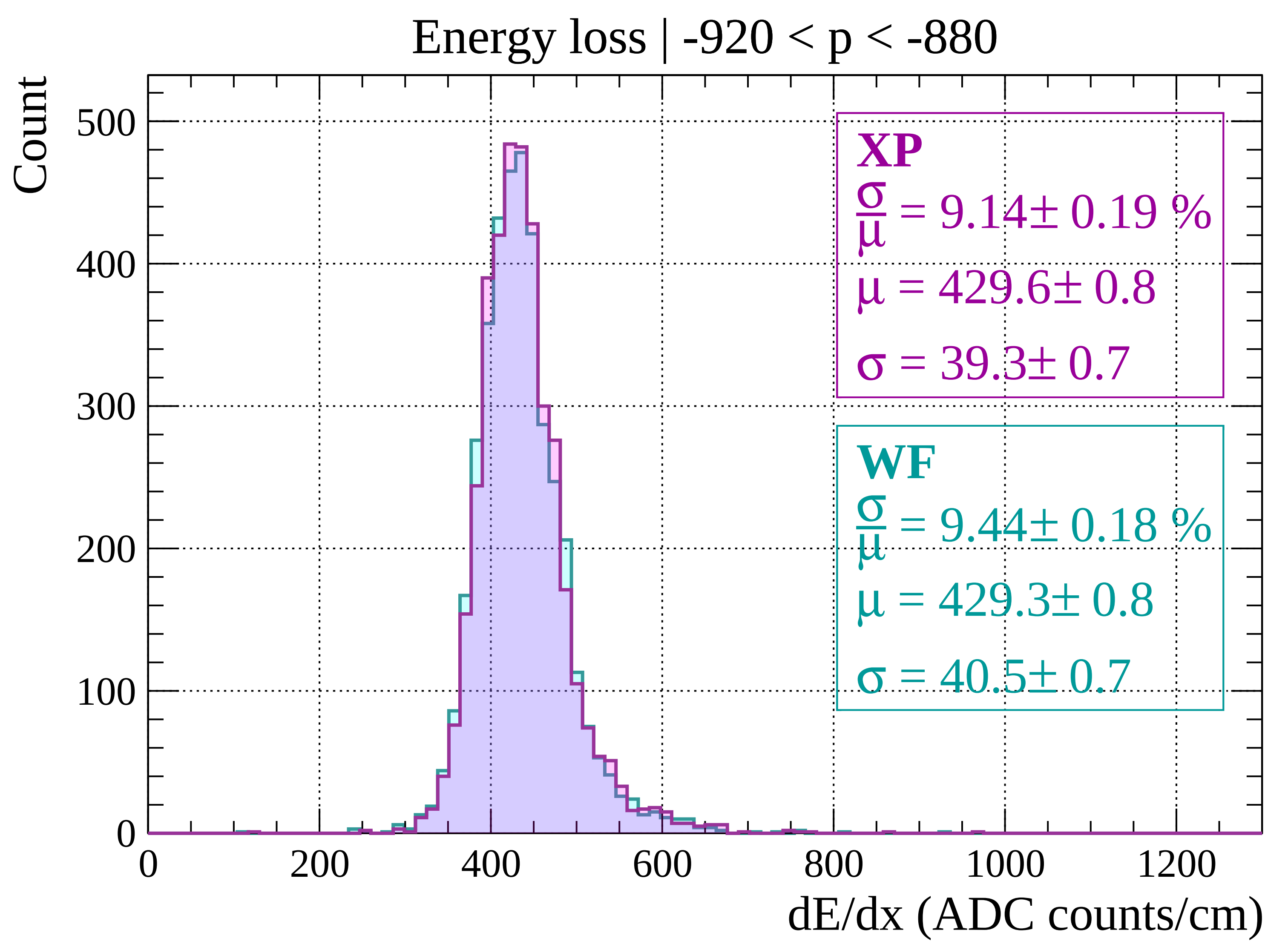
Energy loss | -1040 < p < -1000
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
σ/μ = 9.29± 0.18 %
μ = 436.4± 0.8
σ = 40.5± 0.7
WF
σ/μ = 9.80± 0.21 %
μ = 435.4± 0.8
σ = 42.7± 0.8



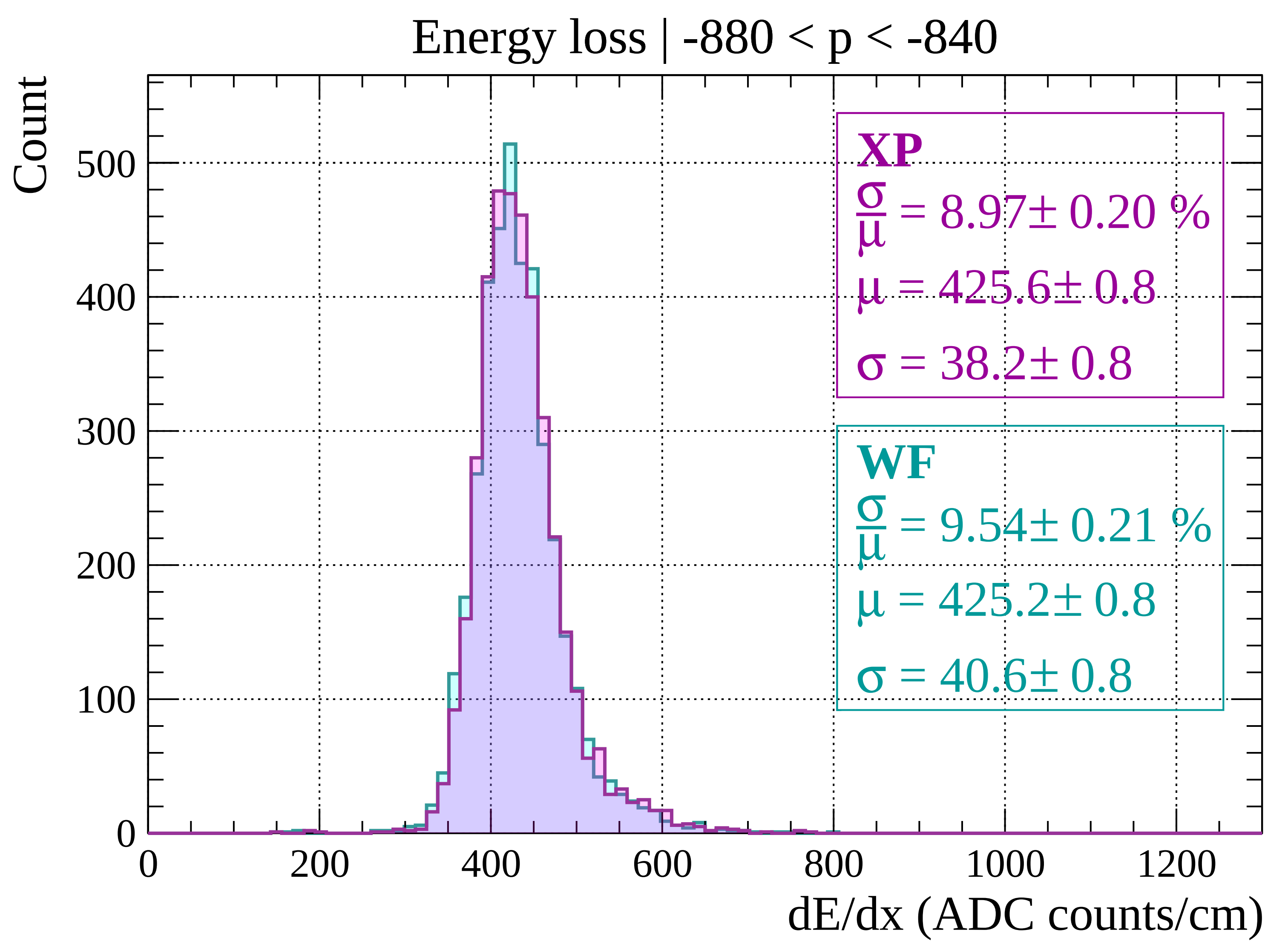
Energy loss | -1000 < p < -960
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 9.23± 0.20 %
μ = 432.8± 0.8
σ = 40.0± 0.8
WF
σ/μ = 9.45± 0.21 %
μ = 432.0± 0.8
σ = 40.8± 0.8



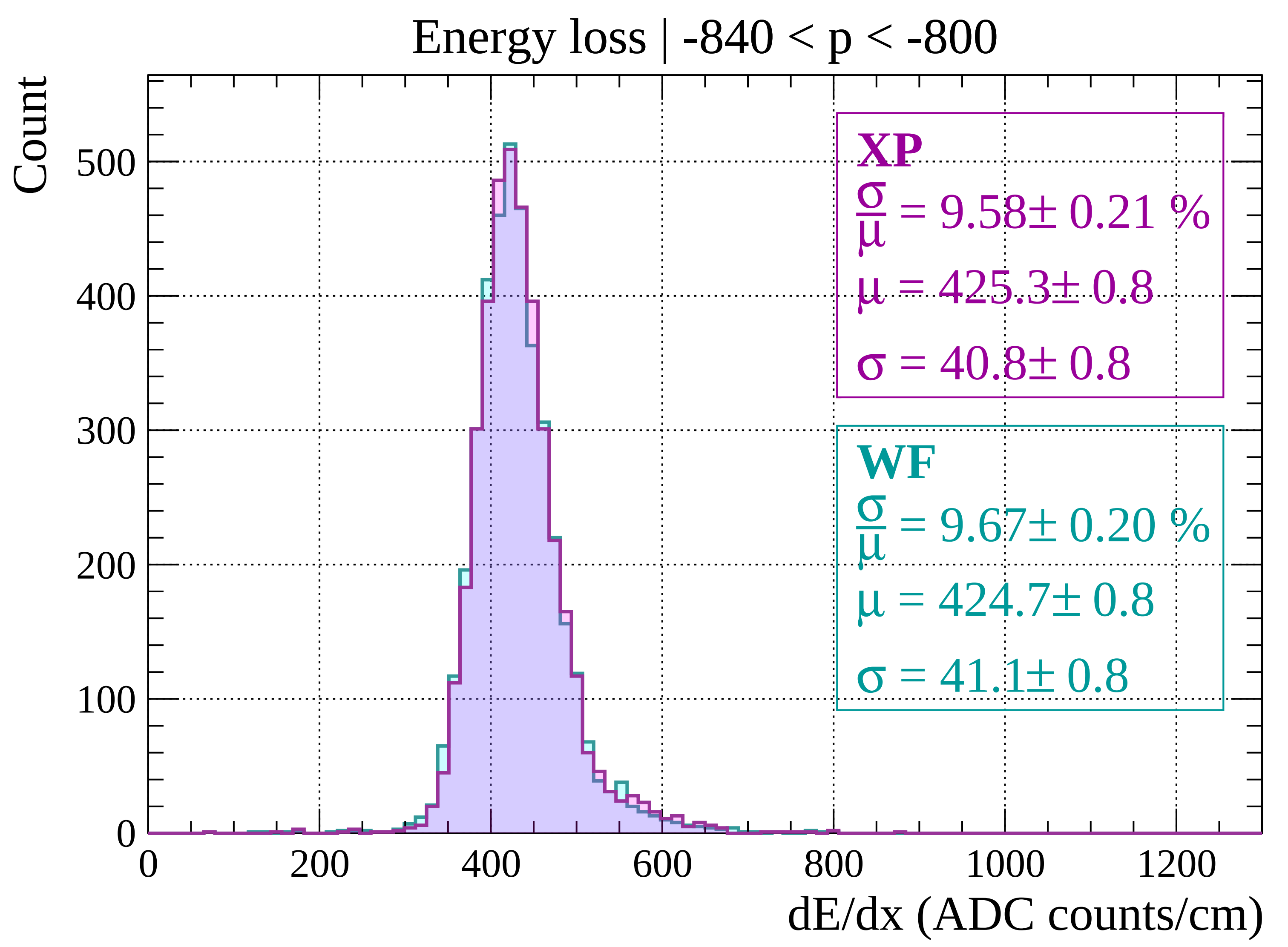
Energy loss | -960 < p < -920
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 9.29± 0.20 %
μ = 431.4± 0.8
σ = 40.1± 0.8
WF
σ/μ = 9.51± 0.21 %
μ = 431.0± 0.8
σ = 41.0± 0.8



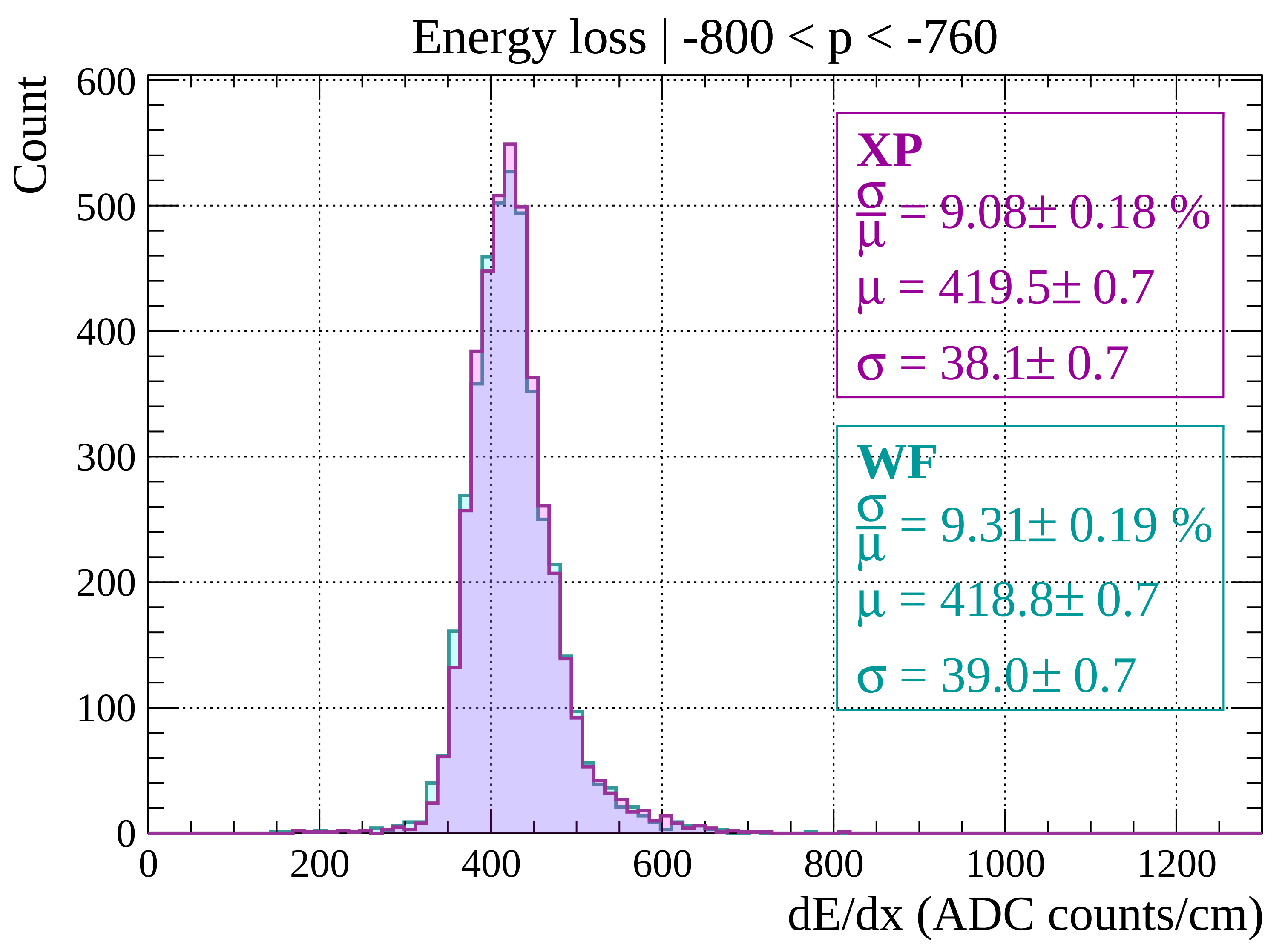
Energy loss | -920 < p < -880
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 9.14± 0.19 %
μ = 429.6± 0.8
σ = 39.3± 0.7
WF
σ/μ = 9.44± 0.18 %
μ = 429.3± 0.8
σ = 40.5± 0.7



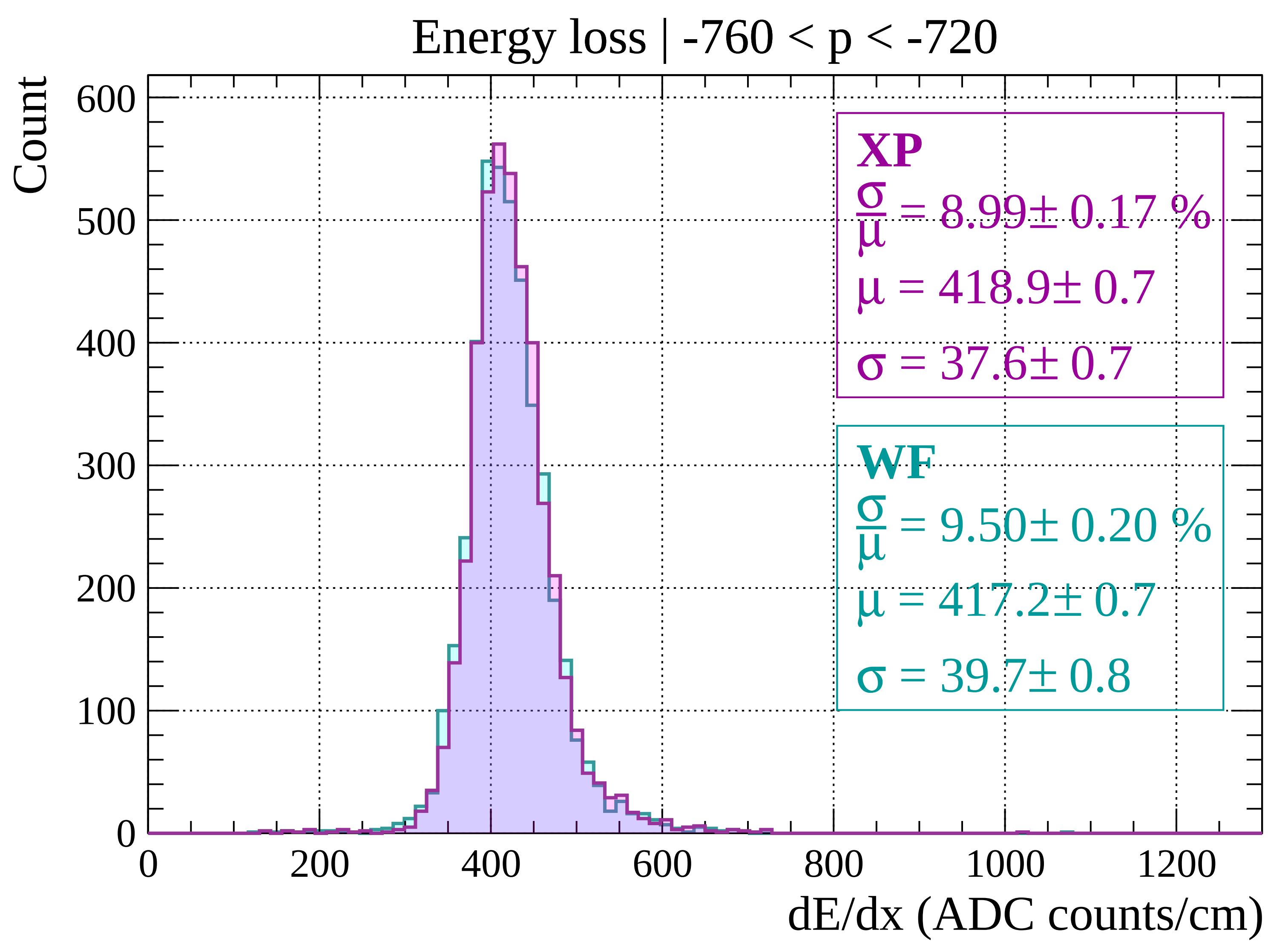
Energy loss | -880 < p < -840
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 8.97± 0.20 %
μ = 425.6± 0.8
σ = 38.2± 0.8
WF
σ/μ = 9.54± 0.21 %
μ = 425.2± 0.8
σ = 40.6± 0.8



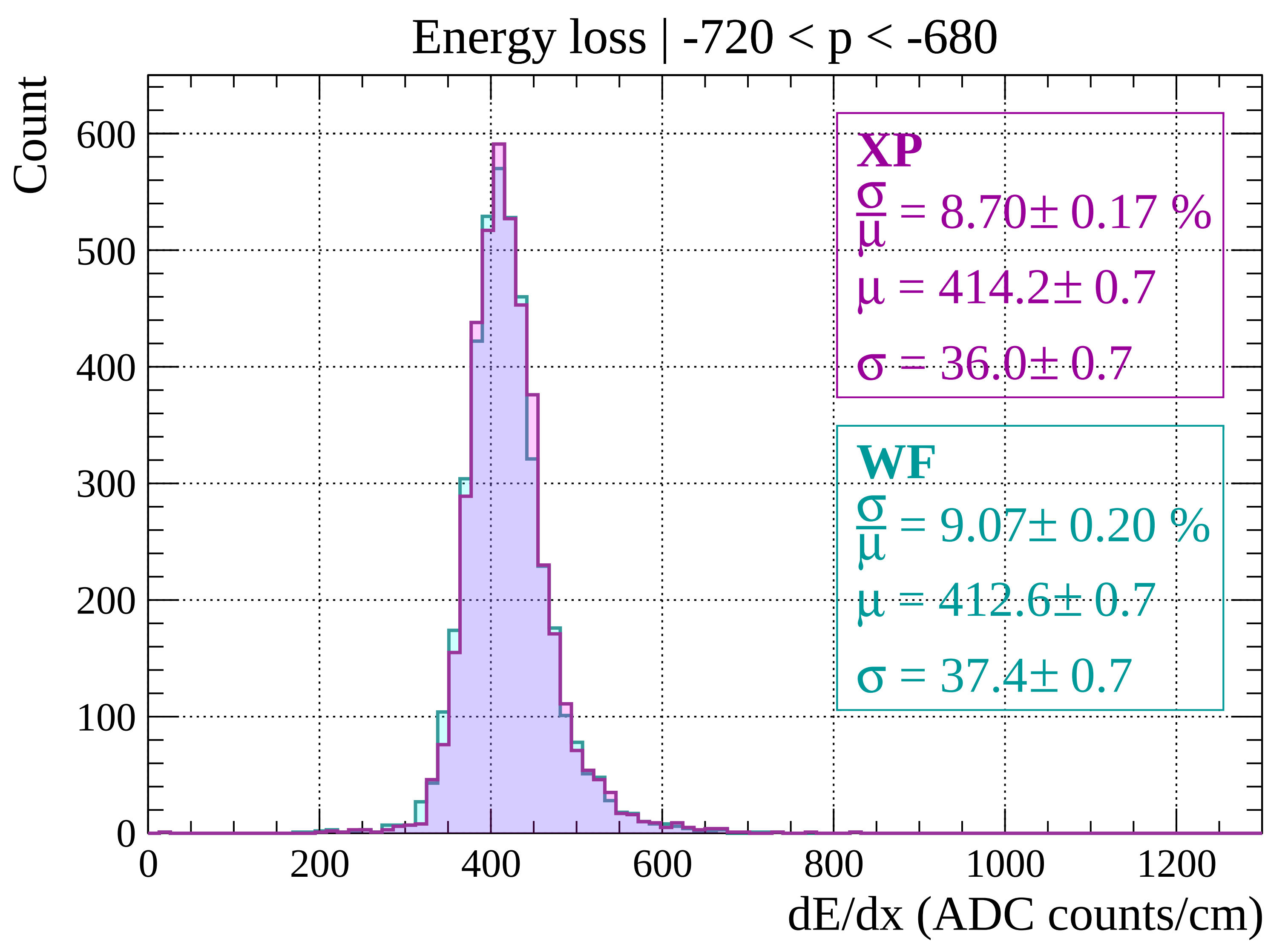
Energy loss | -840 < p < -800
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.58± 0.21 %
μ = 425.3± 0.8
σ = 40.8± 0.8
WF
$\frac{\sigma}{\mu}$ = 9.67± 0.20 %
μ = 424.7± 0.8
σ = 41.1± 0.8



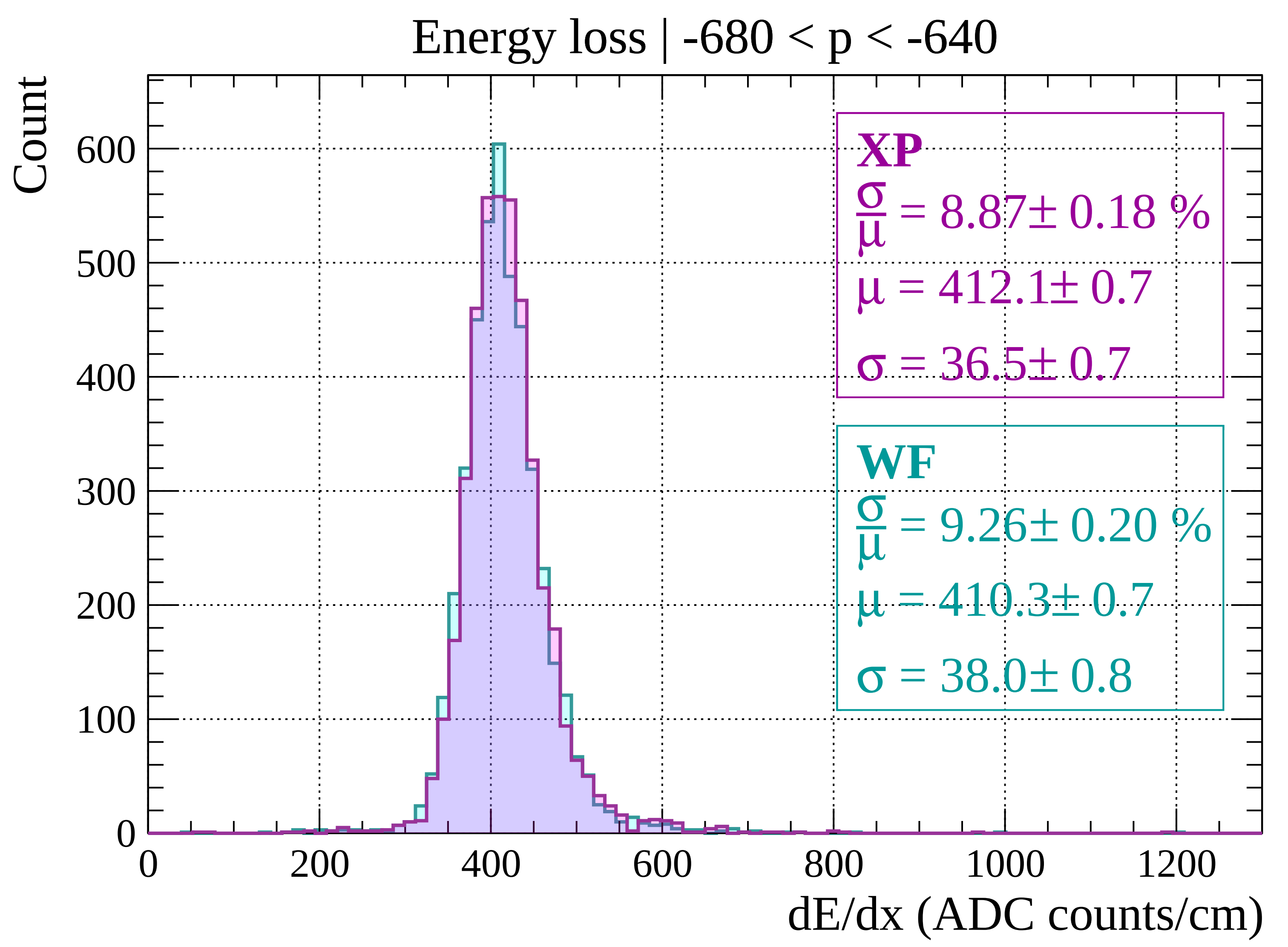
Energy loss | -800 < p < -760
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 9.08 \pm 0.18$ %
$\mu = 419.5 \pm 0.7$
$\sigma = 38.1 \pm 0.7$
WF
$\frac{\sigma}{\mu} = 9.31 \pm 0.19$ %
$\mu = 418.8 \pm 0.7$
$\sigma = 39.0 \pm 0.7$



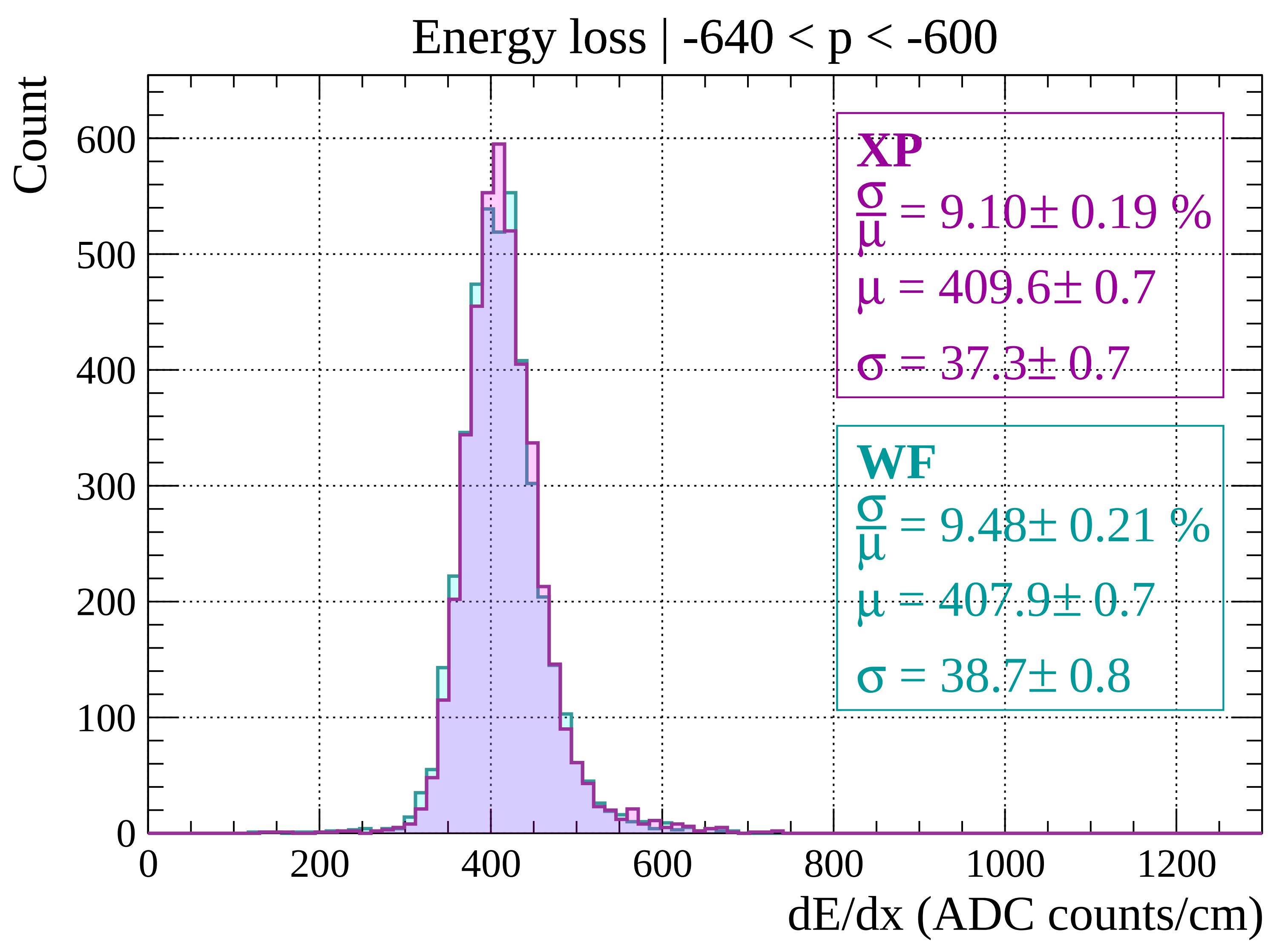
Energy loss | -760 < p < -720
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 8.99 ± 0.17 %
μ = 418.9 ± 0.7
σ = 37.6 ± 0.7
WF
σ/μ = 9.50 ± 0.20 %
μ = 417.2 ± 0.7
σ = 39.7 ± 0.8



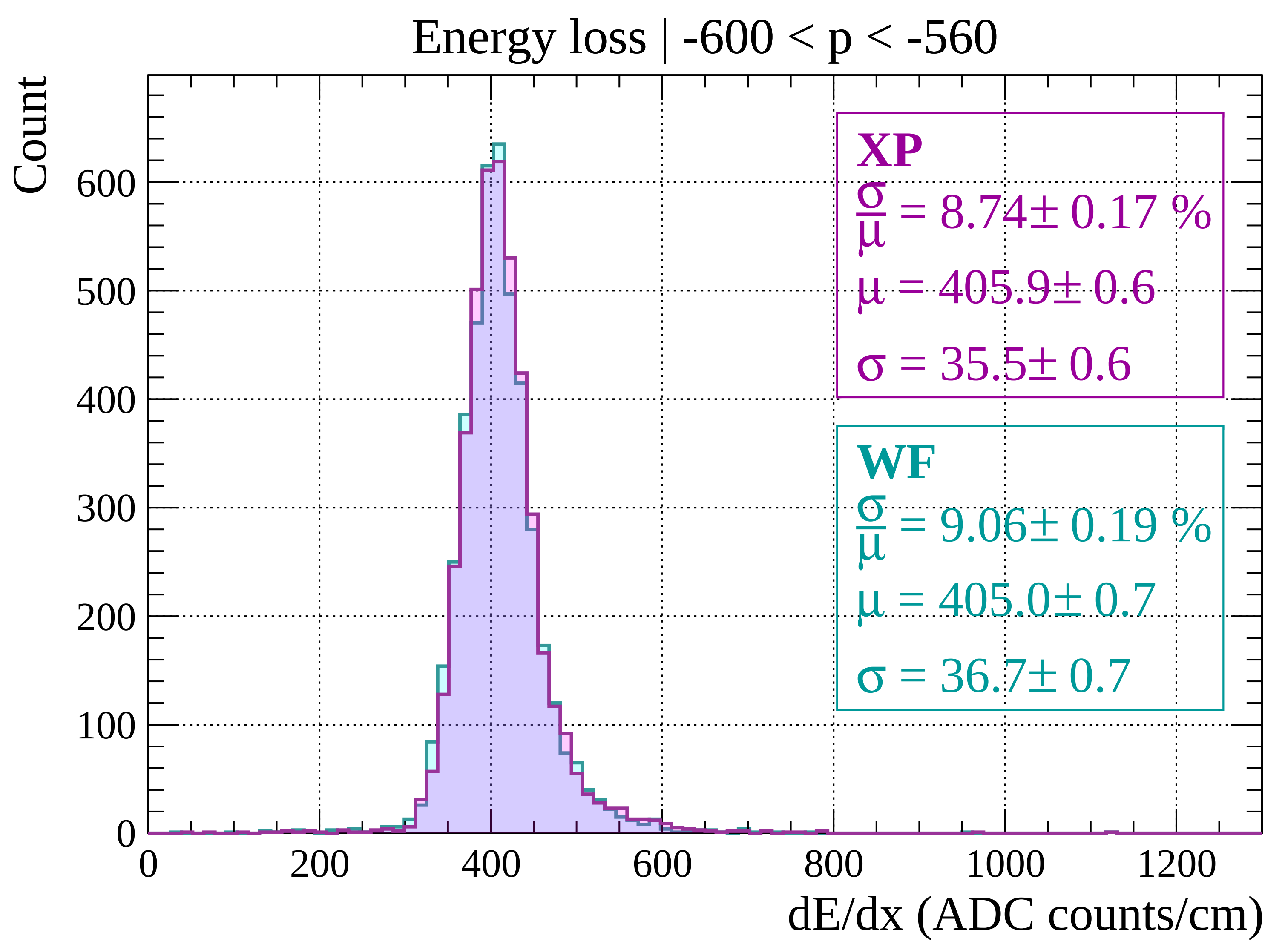
Energy loss | -720 < p < -680
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
XP
$\frac{\sigma}{\mu}$ = 8.70± 0.17 %
μ = 414.2± 0.7
σ = 36.0± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.07± 0.20 %
μ = 412.6± 0.7
σ = 37.4± 0.7



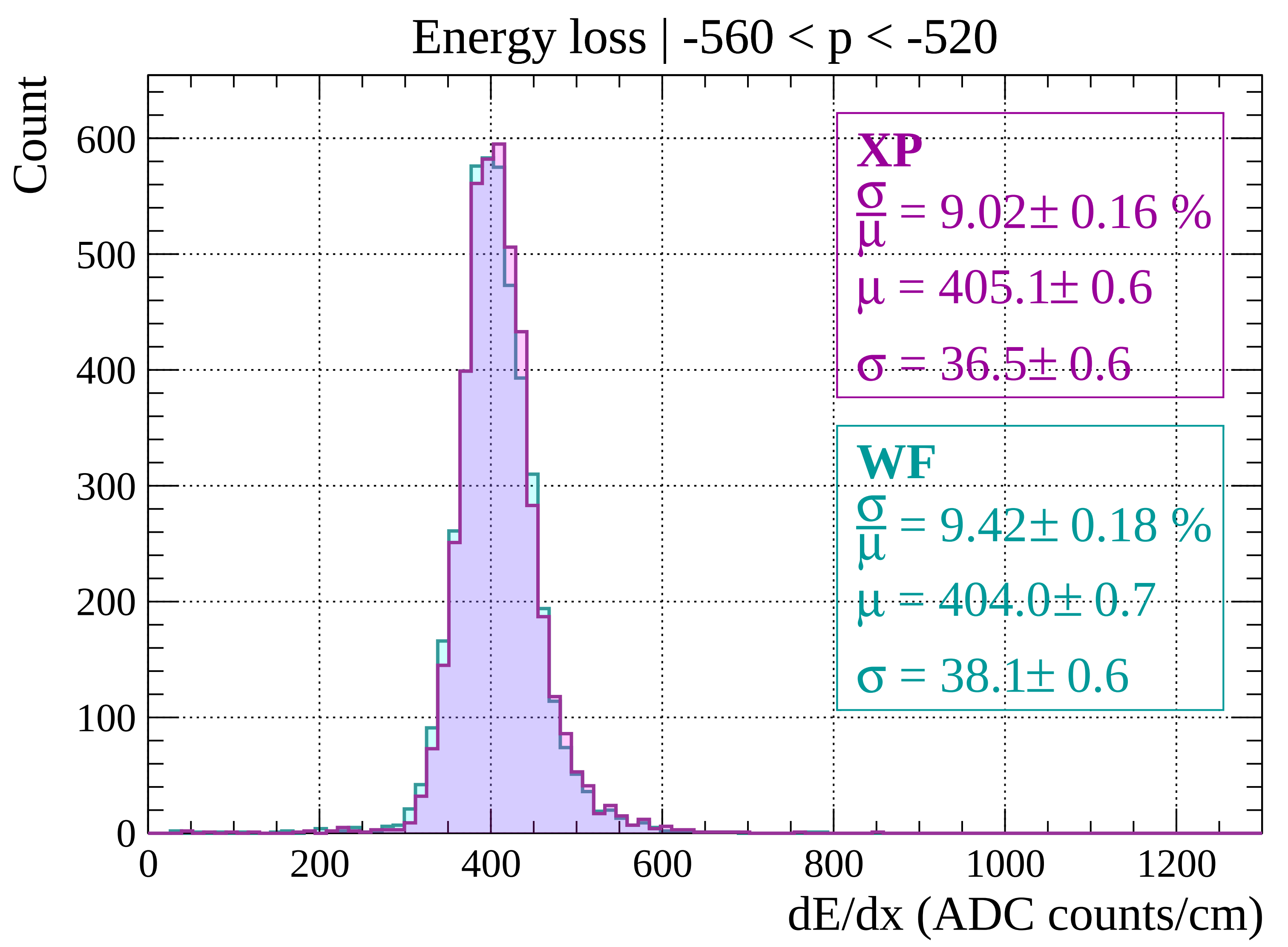
Energy loss | -680 < p < -640
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 8.87± 0.18 %
μ = 412.1± 0.7
σ = 36.5± 0.7
WF
σ/μ = 9.26± 0.20 %
μ = 410.3± 0.7
σ = 38.0± 0.8



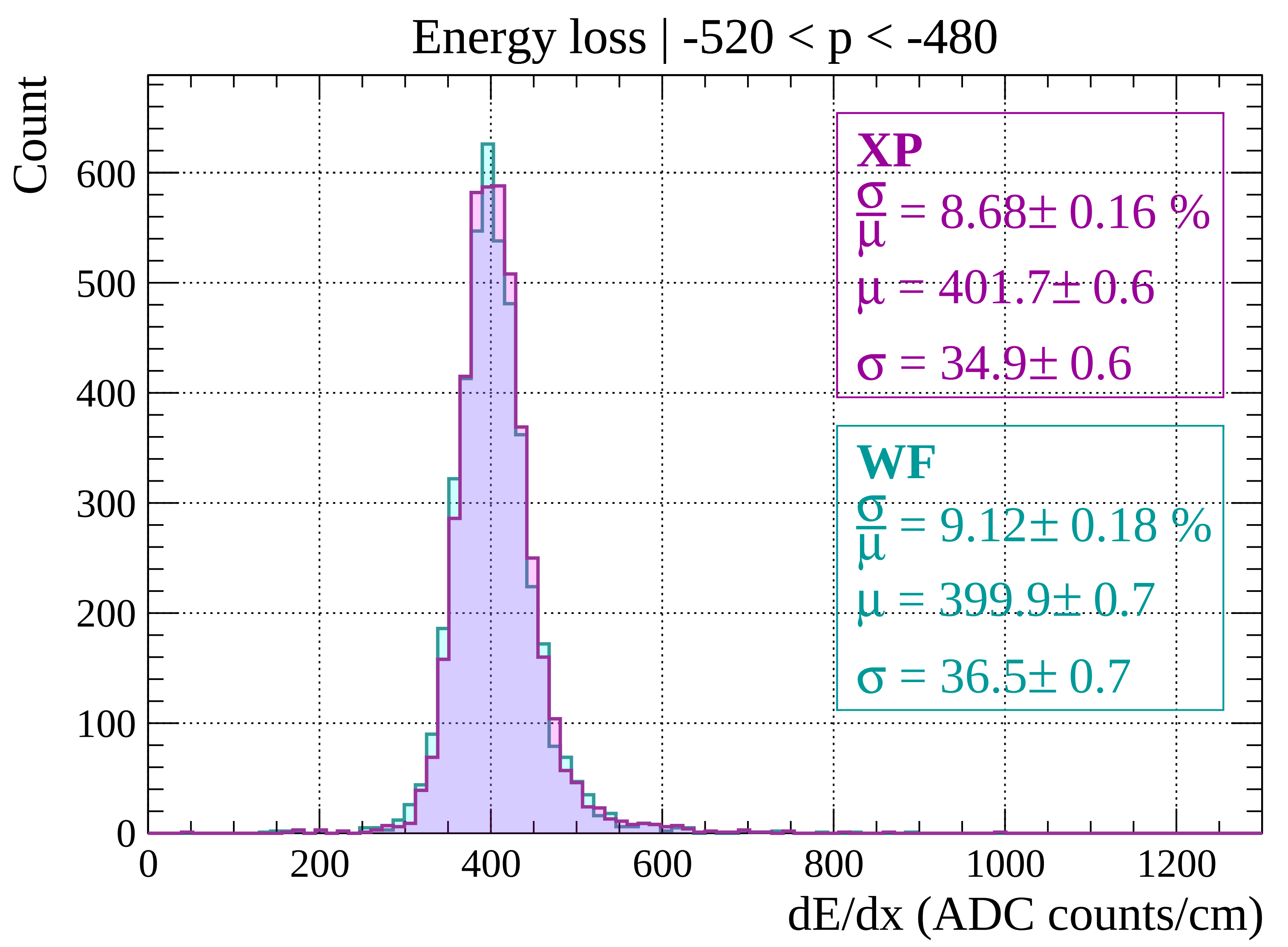
Energy loss | -640 < p < -600
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.10± 0.19 %
μ = 409.6± 0.7
σ = 37.3± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.48± 0.21 %
μ = 407.9± 0.7
σ = 38.7± 0.8



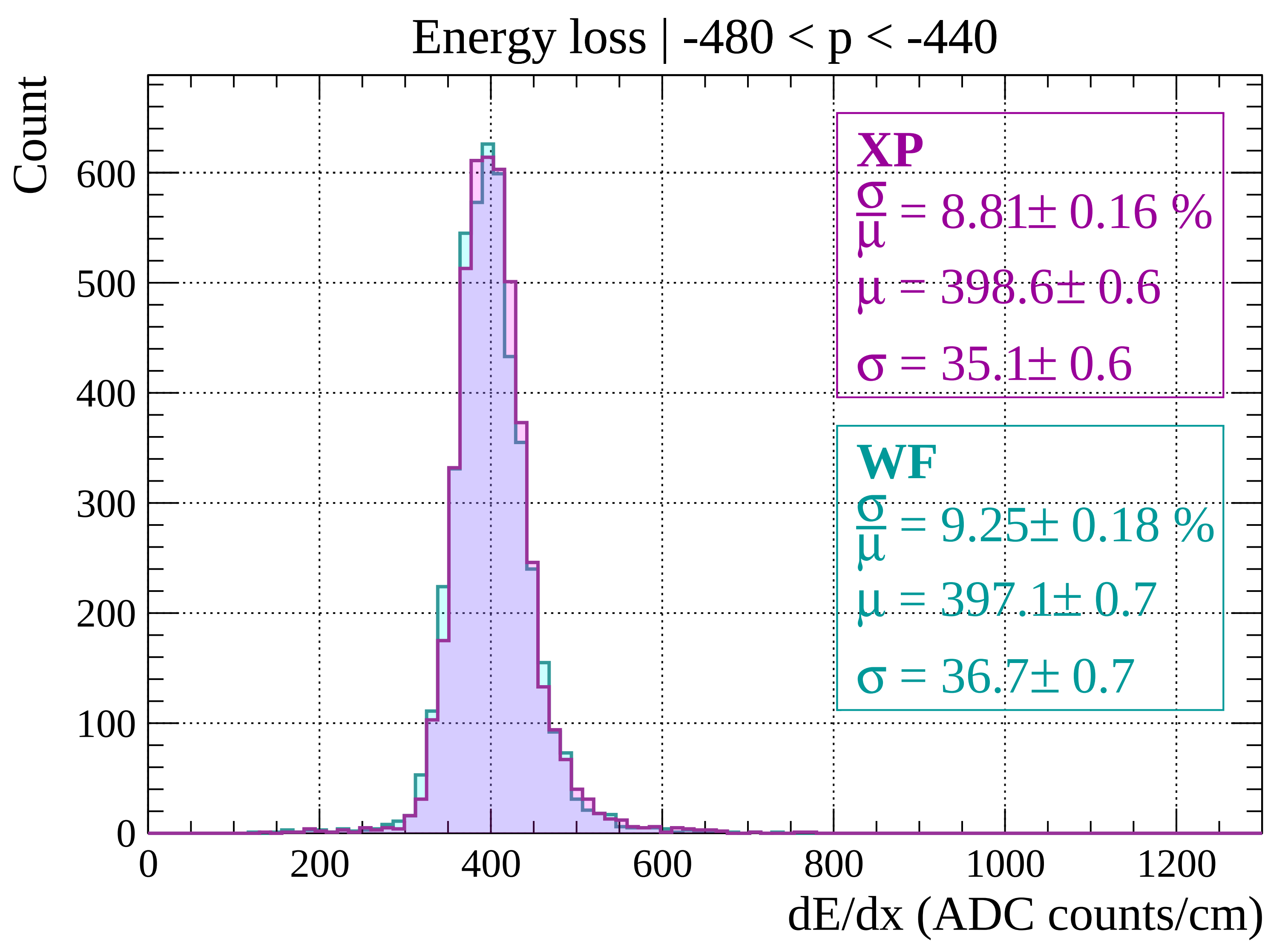
Energy loss | -600 < p < -560
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.74± 0.17 %
μ = 405.9± 0.6
σ = 35.5± 0.6
WF
$\frac{\sigma}{\mu}$ = 9.06± 0.19 %
μ = 405.0± 0.7
σ = 36.7± 0.7



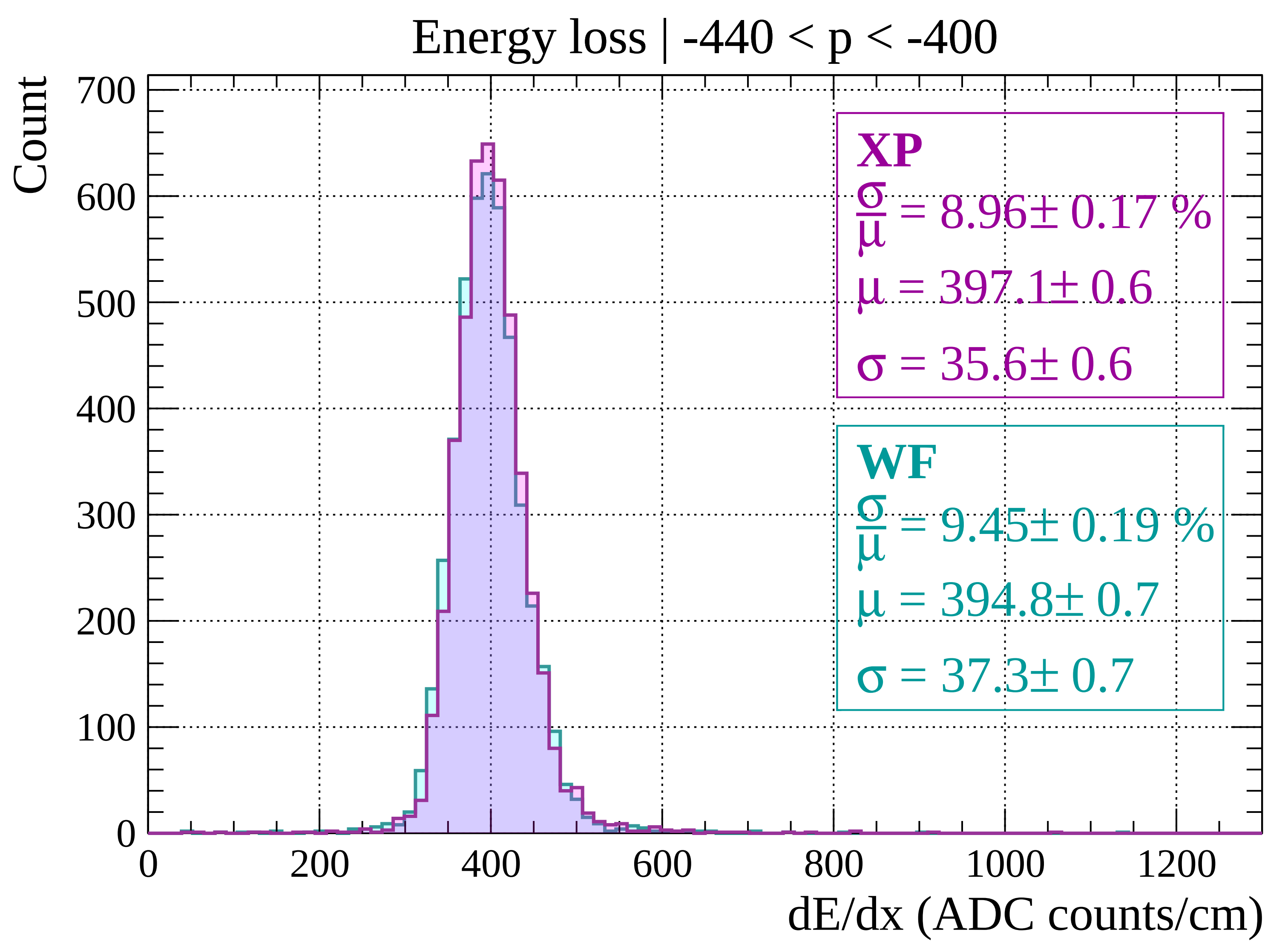
Energy loss | -560 < p < -520
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 9.02± 0.16 %
μ = 405.1± 0.6
σ = 36.5± 0.6
WF
σ/μ = 9.42± 0.18 %
μ = 404.0± 0.7
σ = 38.1± 0.6



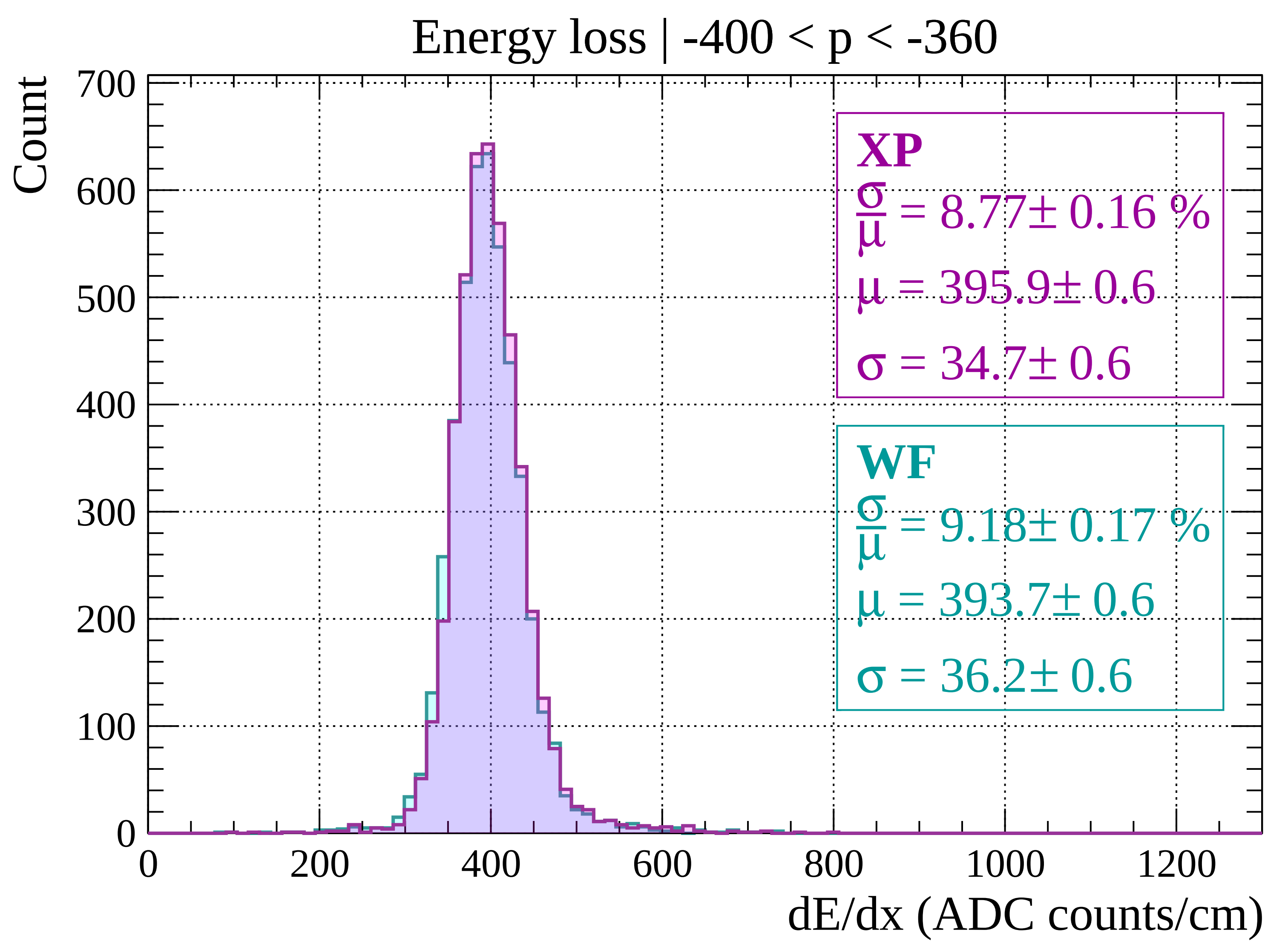
Energy loss | -520 < p < -480
Count
600
500
400
300
200
100
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.68± 0.16 %
μ = 401.7± 0.6
σ = 34.9± 0.6
WF
σ/μ = 9.12± 0.18 %
μ = 399.9± 0.7
σ = 36.5± 0.7



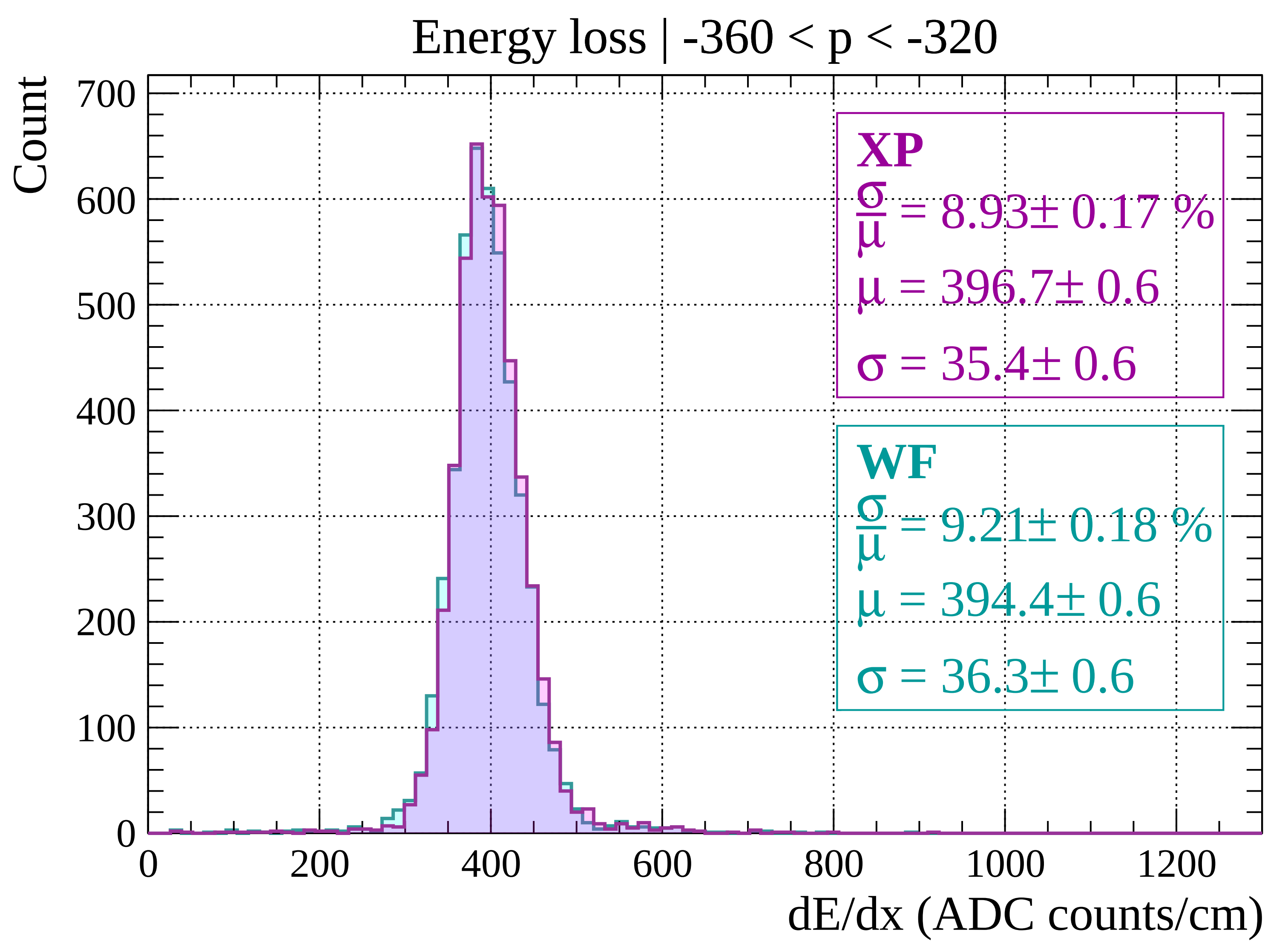
Energy loss | -480 < p < -440
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
100
300
500
XP
σ/μ = 8.81± 0.16 %
μ = 398.6± 0.6
σ = 35.1± 0.6
WF
σ/μ = 9.25± 0.18 %
μ = 397.1± 0.7
σ = 36.7± 0.7

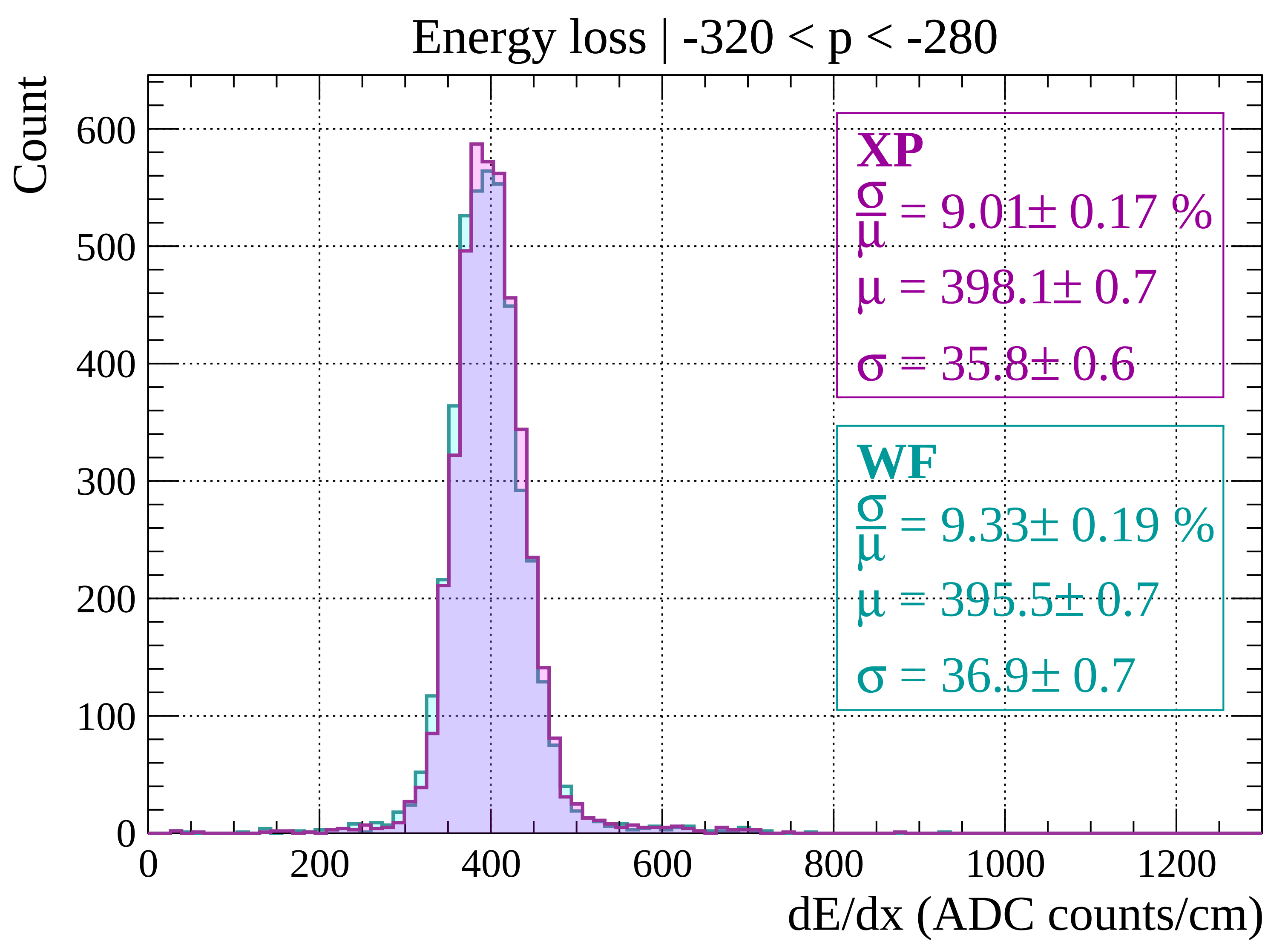


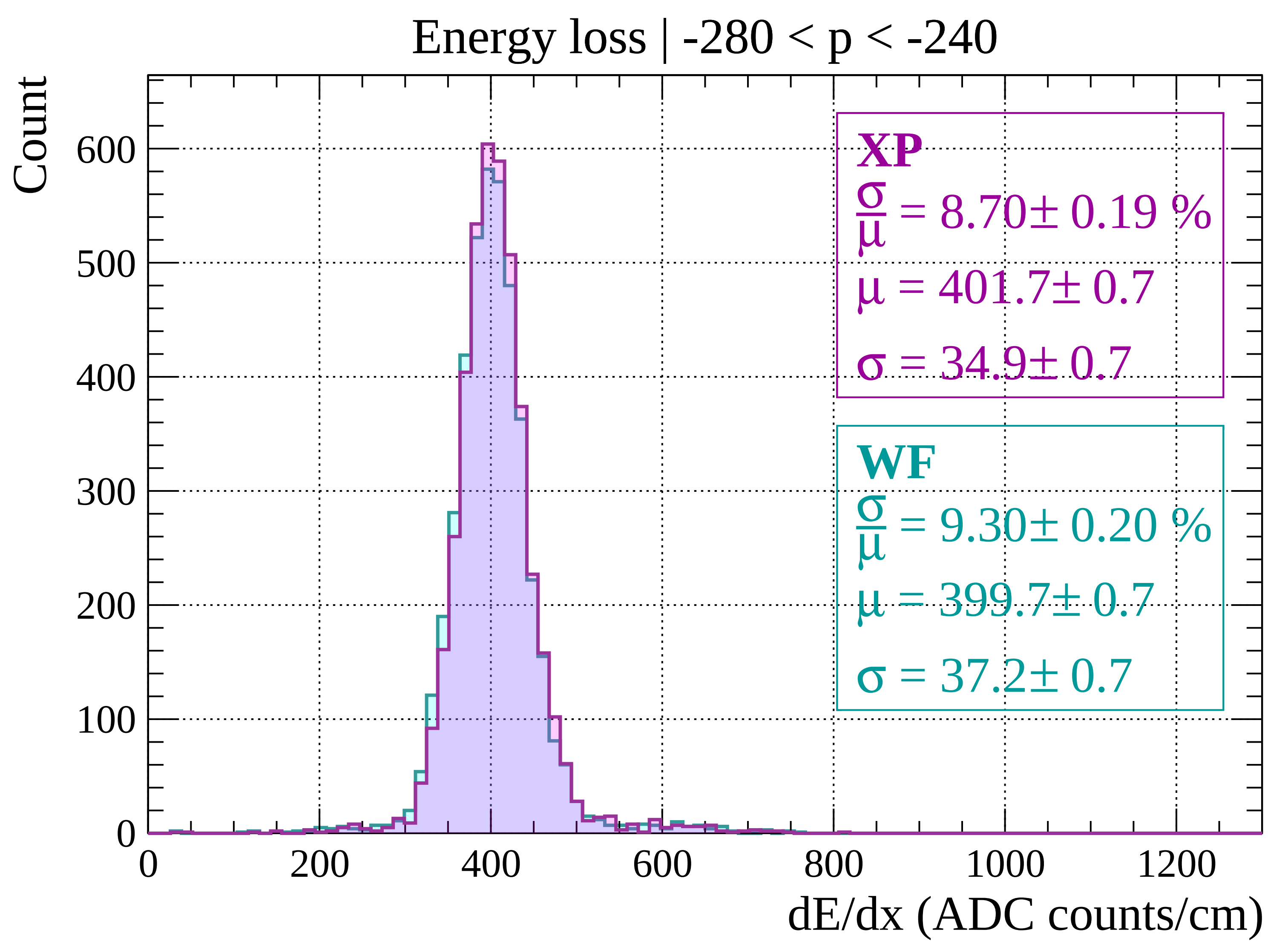
Energy loss | -440 < p < -400
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 8.96± 0.17 %
μ = 397.1± 0.6
σ = 35.6± 0.6
WF
σ/μ = 9.45± 0.19 %
μ = 394.8± 0.7
σ = 37.3± 0.7



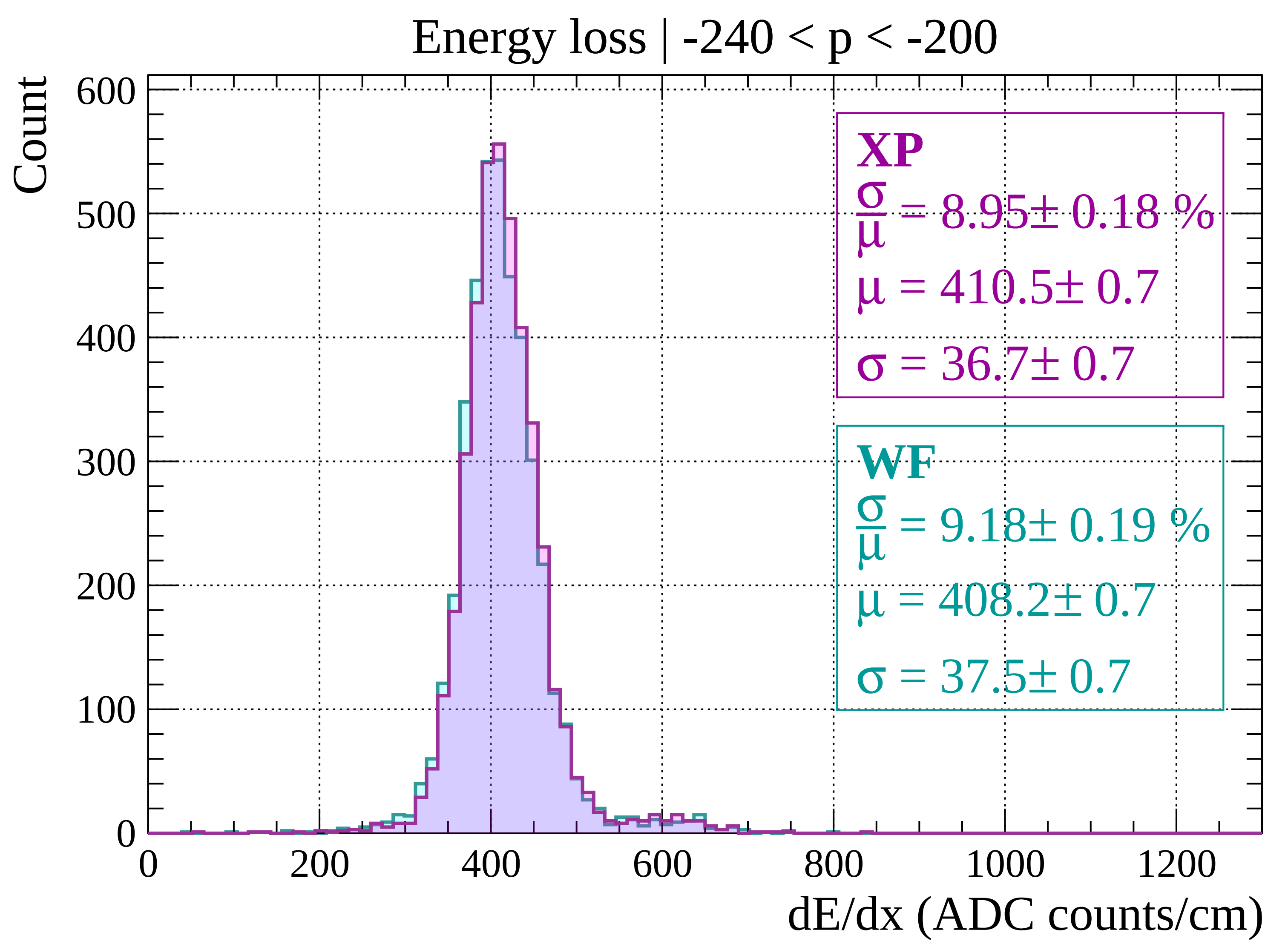
Energy loss | -400 < p < -360
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 8.77± 0.16 %
μ = 395.9± 0.6
σ = 34.7± 0.6
WF
σ/μ = 9.18± 0.17 %
μ = 393.7± 0.6
σ = 36.2± 0.6


Energy loss | -360 < p < -320
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 8.93± 0.17 %
μ = 396.7± 0.6
σ = 35.4± 0.6
WF
σ/μ = 9.21± 0.18 %
μ = 394.4± 0.6
σ = 36.3± 0.6

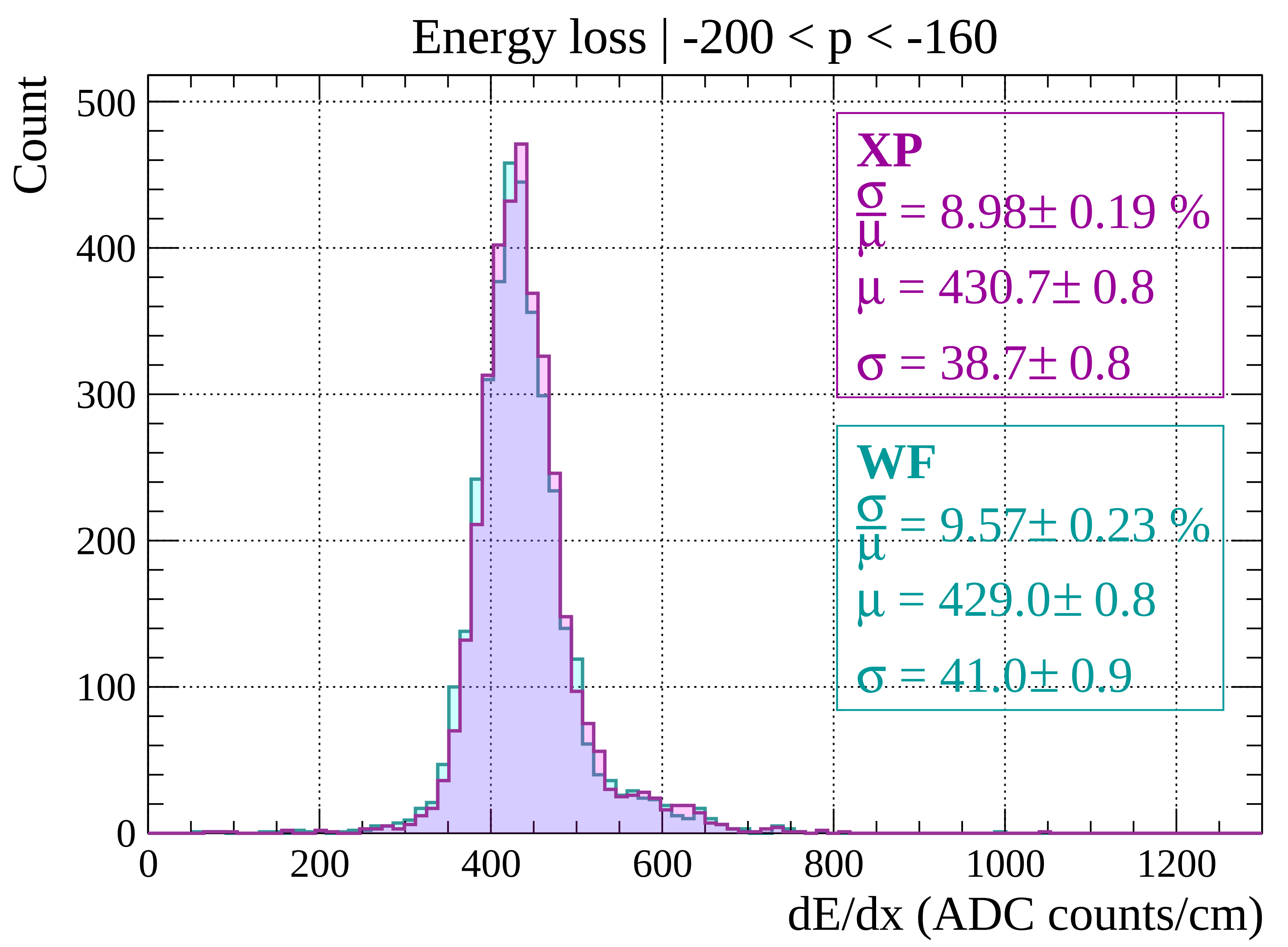

Energy loss | -320 < p < -280
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.01± 0.17 %
μ = 398.1± 0.7
σ = 35.8± 0.6
WF
σ/μ = 9.33± 0.19 %
μ = 395.5± 0.7
σ = 36.9± 0.7



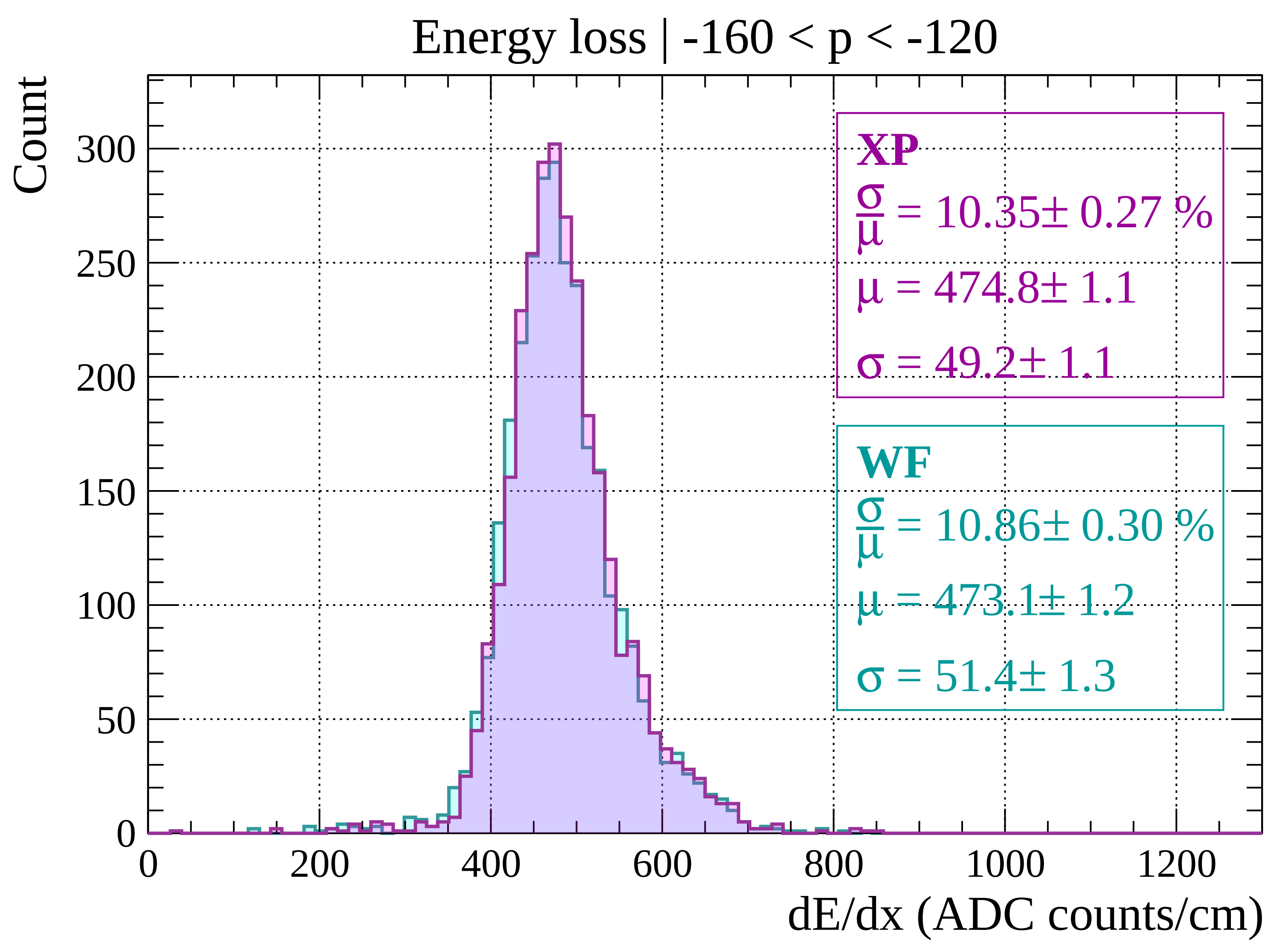
Energy loss | -280 < p < -240
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 8.70± 0.19 %
μ = 401.7± 0.7
σ = 34.9± 0.7
WF
σ/μ = 9.30± 0.20 %
μ = 399.7± 0.7
σ = 37.2± 0.7

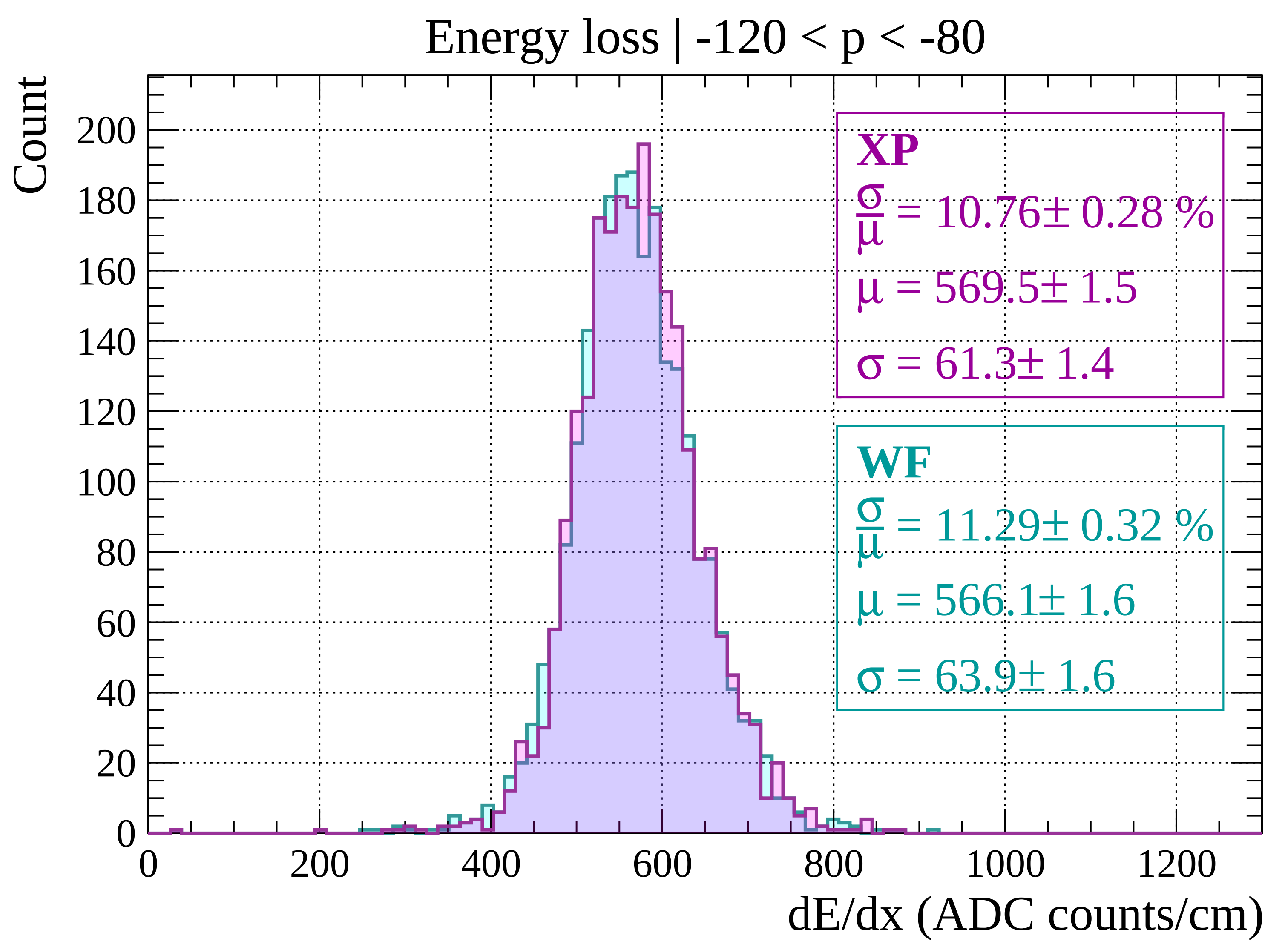


Energy loss | -240 < p < -200
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 8.95 \pm 0.18$ %
$\mu = 410.5 \pm 0.7$
$\sigma = 36.7 \pm 0.7$
WF
$\frac{\sigma}{\mu} = 9.18 \pm 0.19$ %
$\mu = 408.2 \pm 0.7$
$\sigma = 37.5 \pm 0.7$

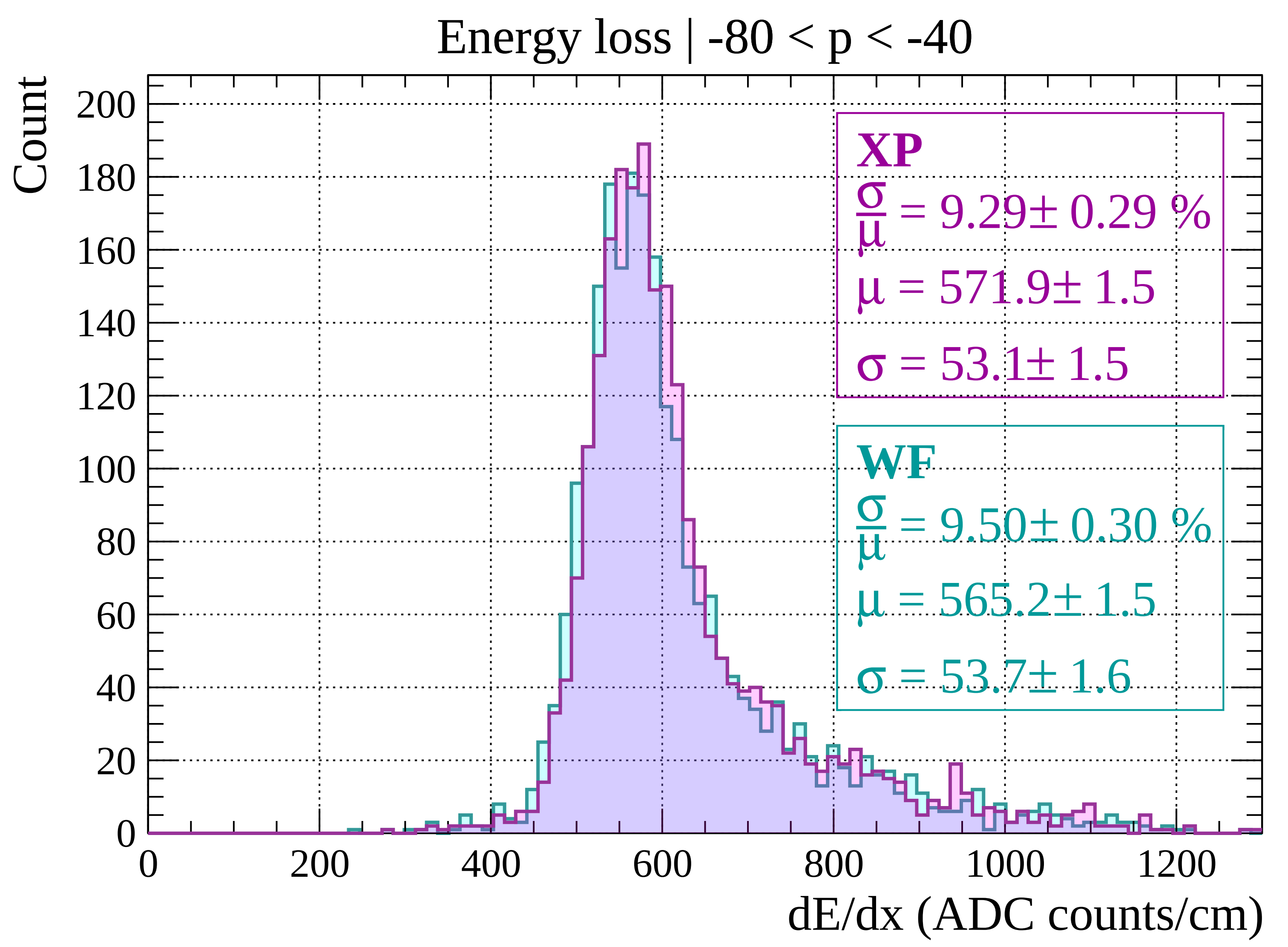


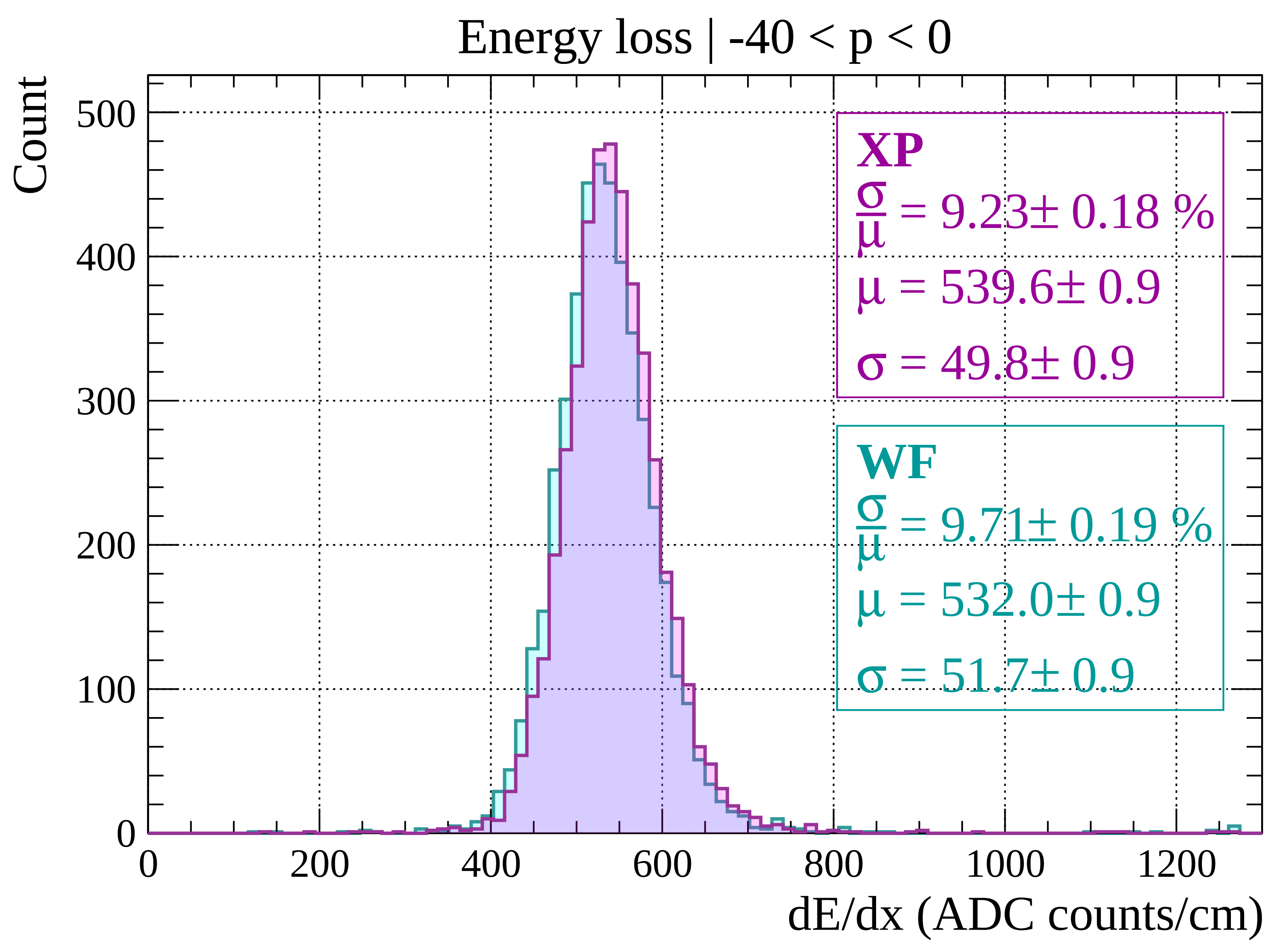
Energy loss | -200 < p < -160
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.98± 0.19 %
μ = 430.7± 0.8
σ = 38.7± 0.8
WF
σ/μ = 9.57± 0.23 %
μ = 429.0± 0.8
σ = 41.0± 0.9


Energy loss | -160 < p < -120
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 10.35± 0.27 %
μ = 474.8± 1.1
σ = 49.2± 1.1
WF
σ/μ = 10.86± 0.30 %
μ = 473.1± 1.2
σ = 51.4± 1.3
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200

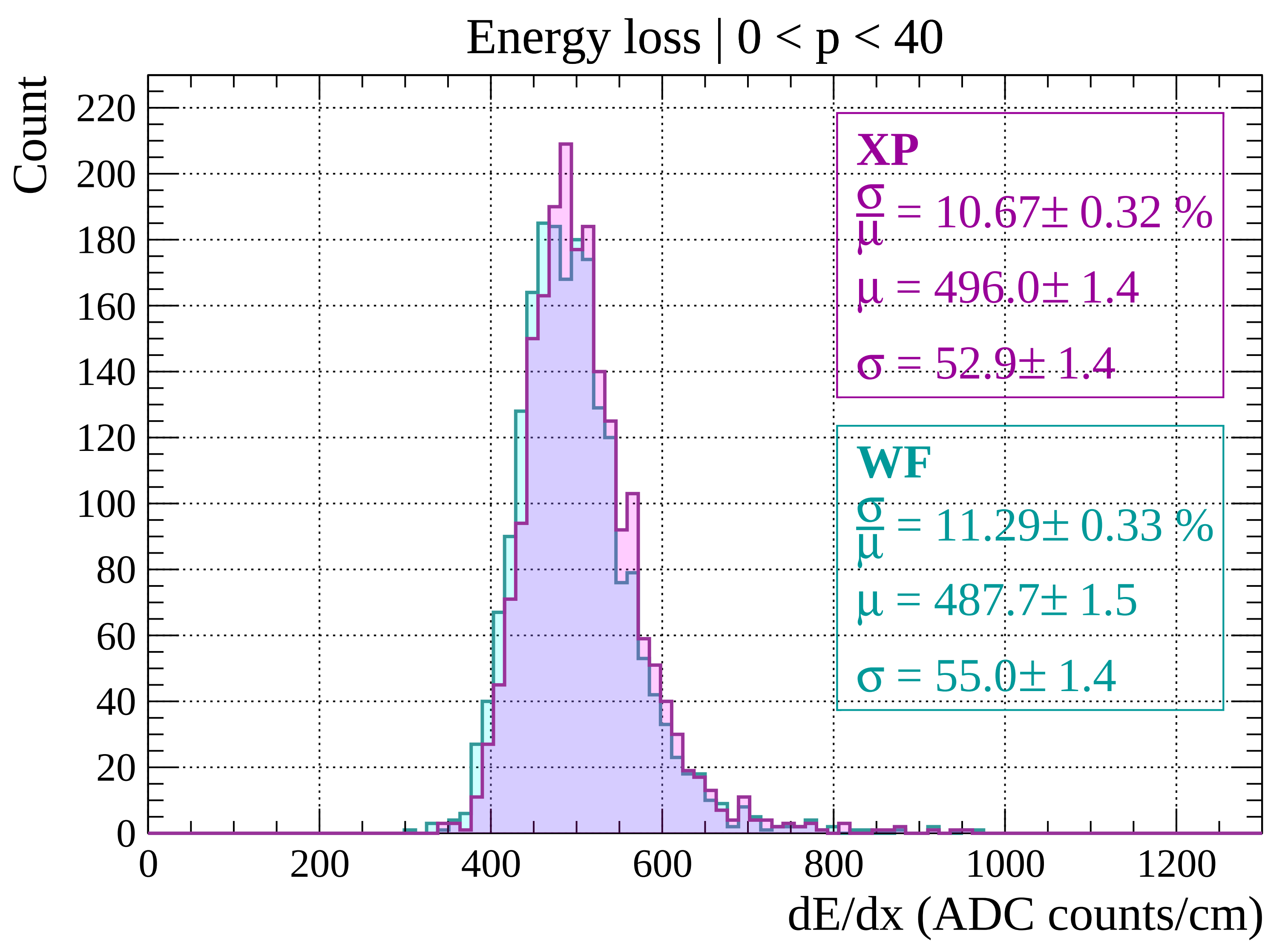


Energy loss | -120 < p < -80
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200
XP
$\frac{\sigma}{\mu}$ = 10.76± 0.28 %
μ = 569.5± 1.5
σ = 61.3± 1.4
WF
$\frac{\sigma}{\mu}$ = 11.29± 0.32 %
μ = 566.1± 1.6
σ = 63.9± 1.6

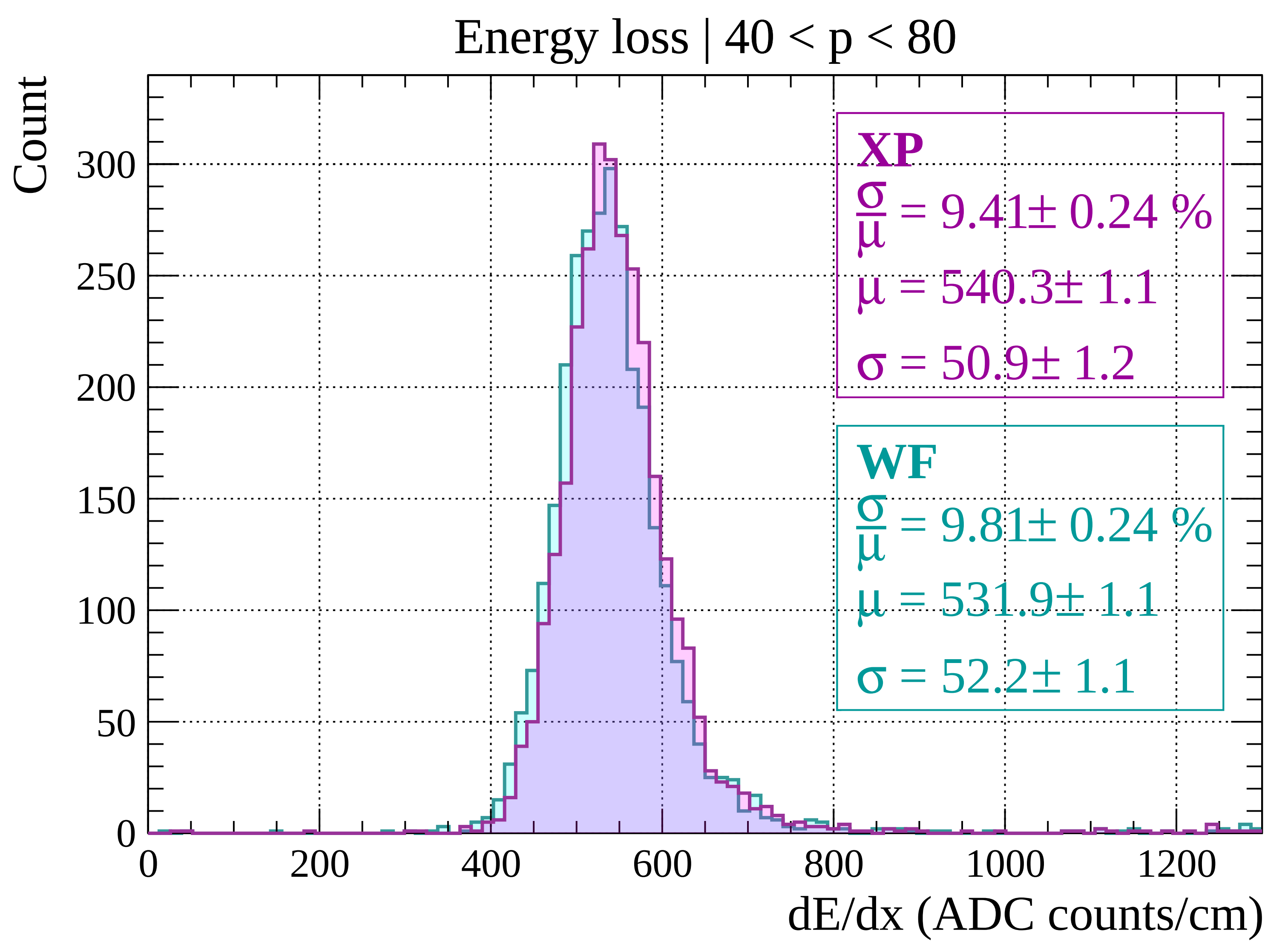

Energy loss | -80 < p < -40
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.29± 0.29 %
μ = 571.9± 1.5
σ = 53.1± 1.5
WF
σ/μ = 9.50± 0.30 %
μ = 565.2± 1.5
σ = 53.7± 1.6
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
200



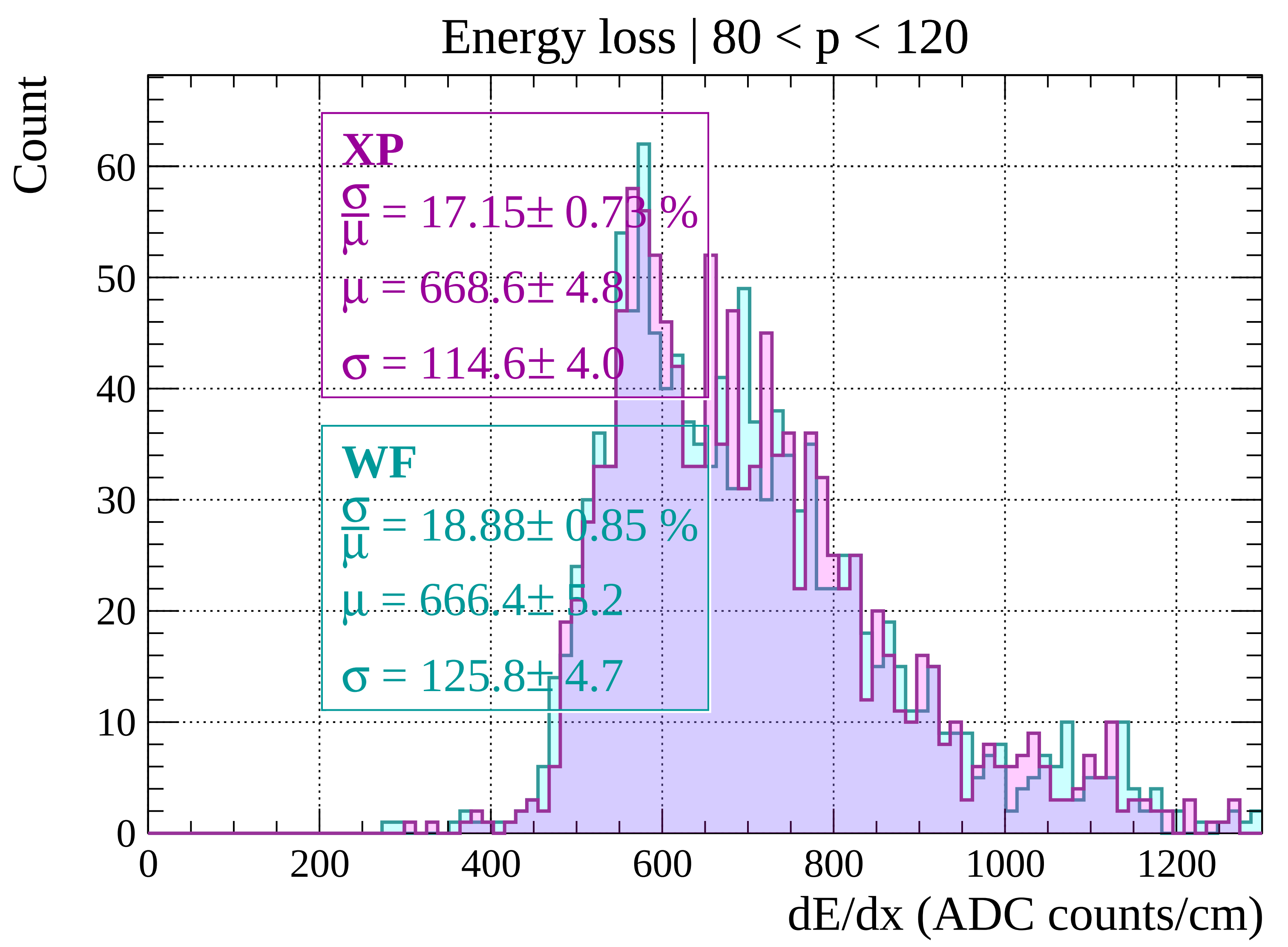
Energy loss | -40 < p < 0
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 9.23± 0.18 %
μ = 539.6± 0.9
σ = 49.8± 0.9
WF
σ/μ = 9.71± 0.19 %
μ = 532.0± 0.9
σ = 51.7± 0.9



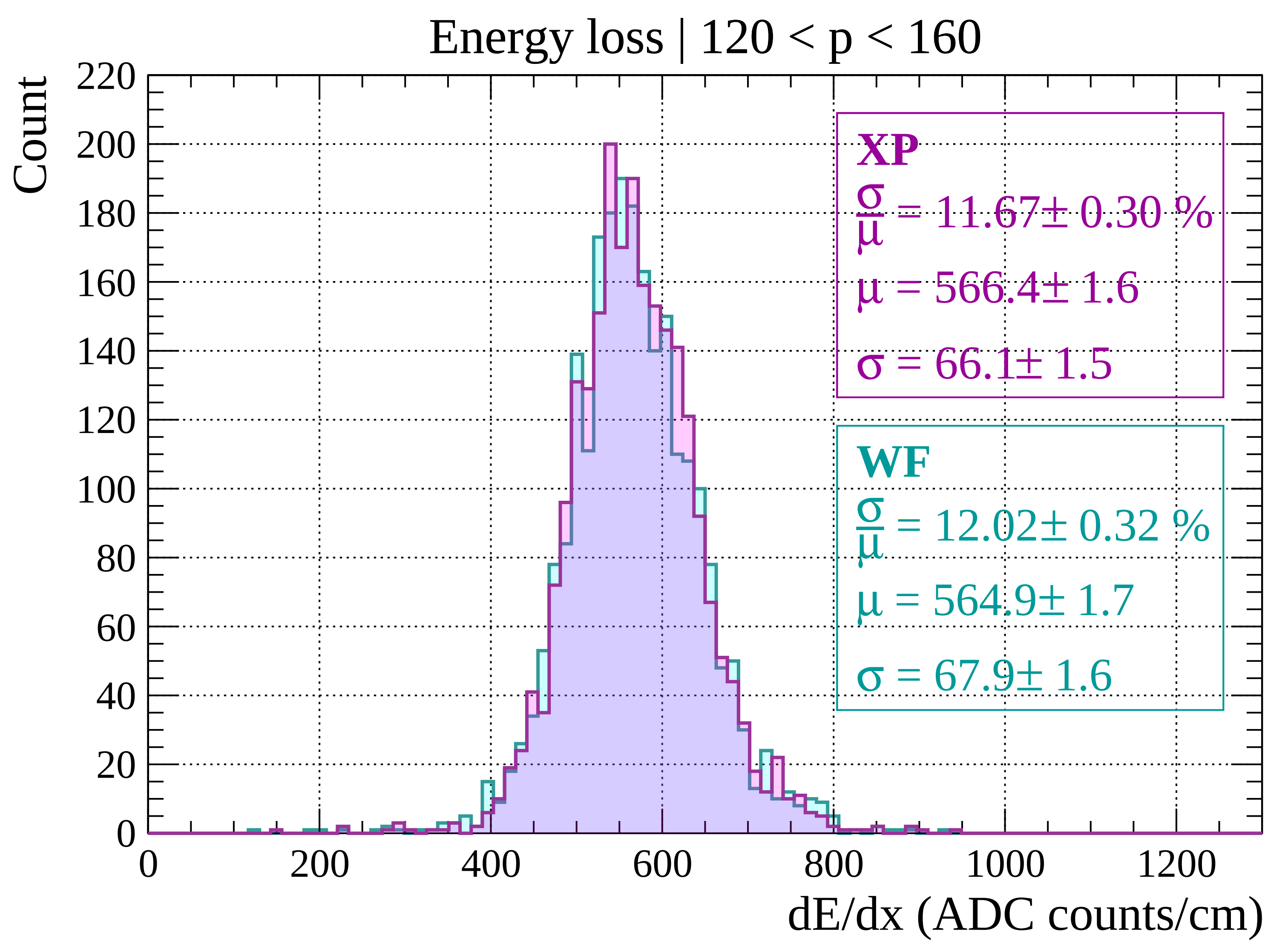
Energy loss | 0 < p < 40
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 10.67± 0.32 %
μ = 496.0± 1.4
σ = 52.9± 1.4
WF
$\frac{\sigma}{\mu}$ = 11.29± 0.33 %
μ = 487.7± 1.5
σ = 55.0± 1.4

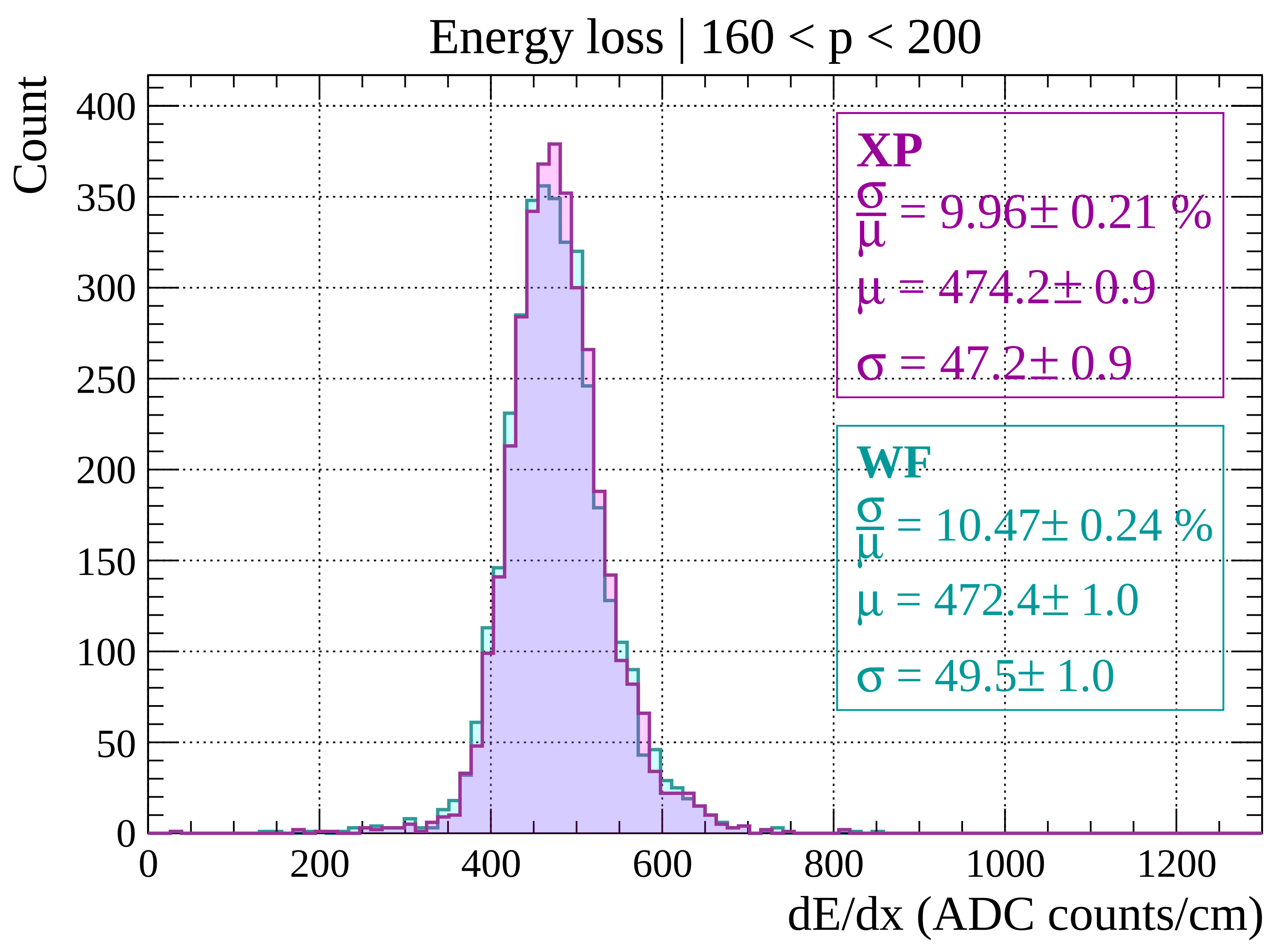


Energy loss | 40 < p < 80
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
XP
σ/μ = 9.41± 0.24 %
μ = 540.3± 1.1
σ = 50.9± 1.2
WF
σ/μ = 9.81± 0.24 %
μ = 531.9± 1.1
σ = 52.2± 1.1

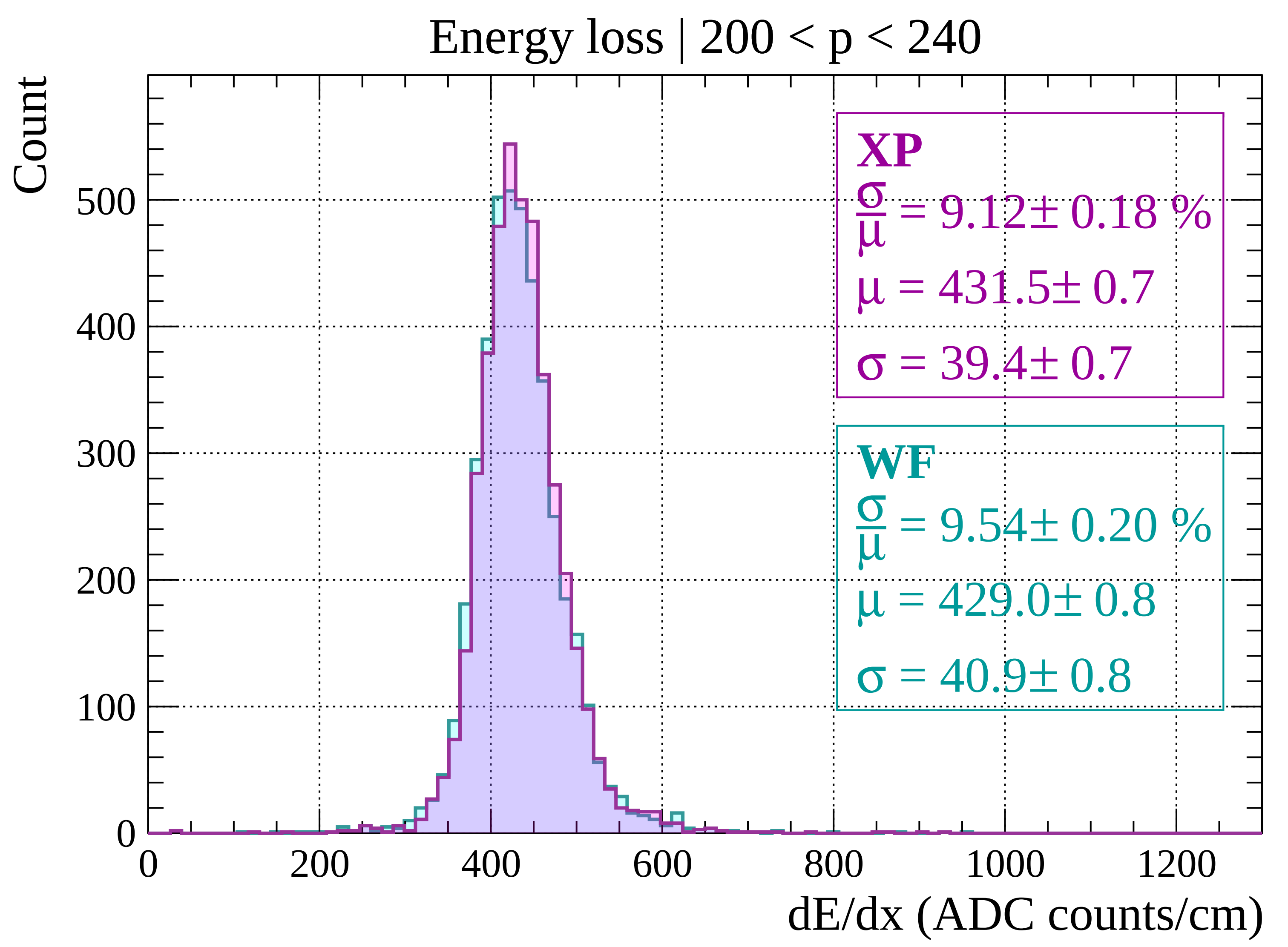


Energy loss | 80 < p < 120
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 17.15± 0.73 %
μ = 668.6± 4.8
σ = 114.6± 4.0
WF
σ/μ = 18.88± 0.85 %
μ = 666.4± 5.2
σ = 125.8± 4.7
0
10
20
30
40
50
60
0
200
400
600
800
1000
1200

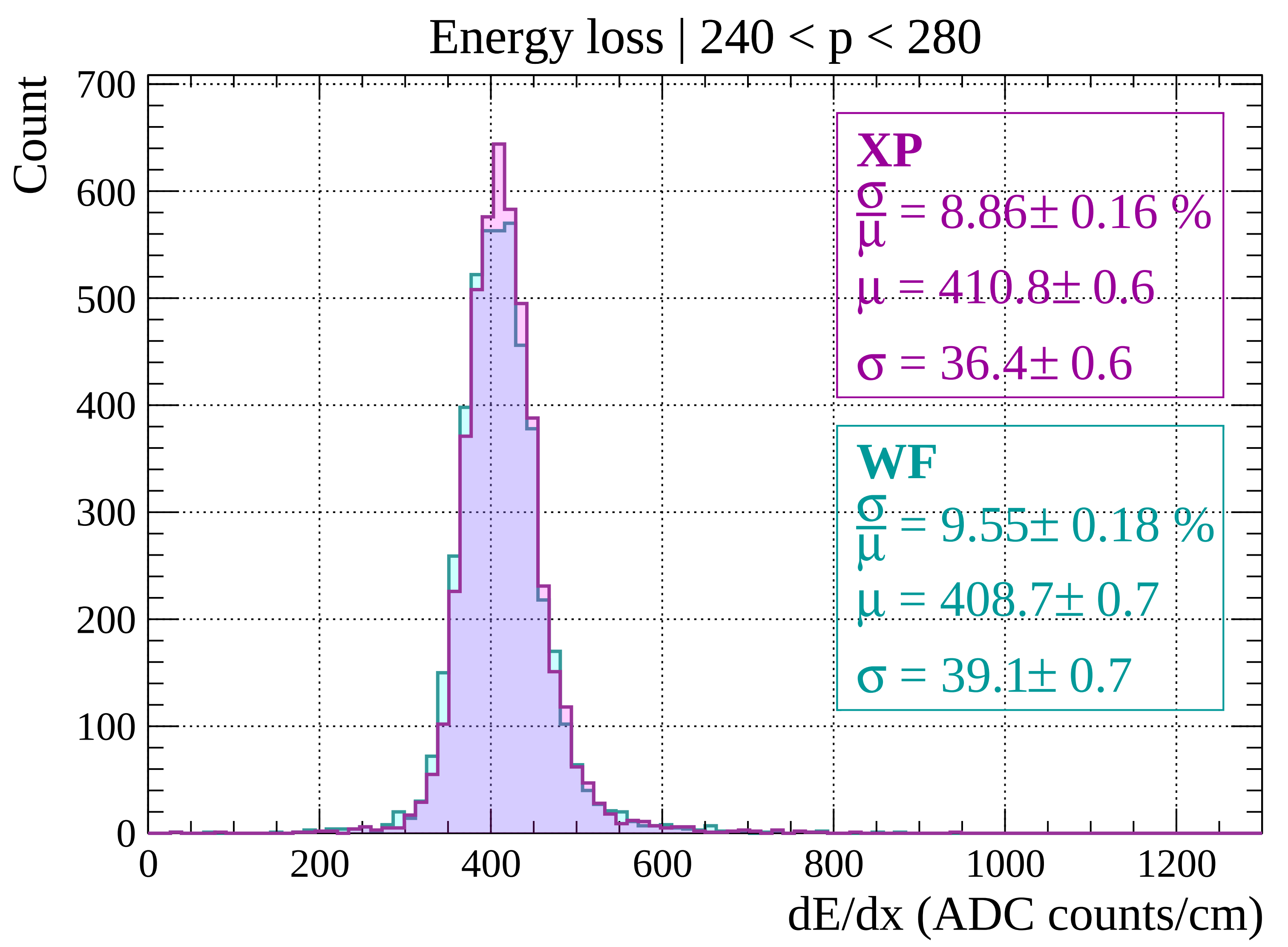

Energy loss | 120 < p < 160
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 11.67± 0.30 %
μ = 566.4± 1.6
σ = 66.1± 1.5
WF
$\frac{\sigma}{\mu}$ = 12.02± 0.32 %
μ = 564.9± 1.7
σ = 67.9± 1.6



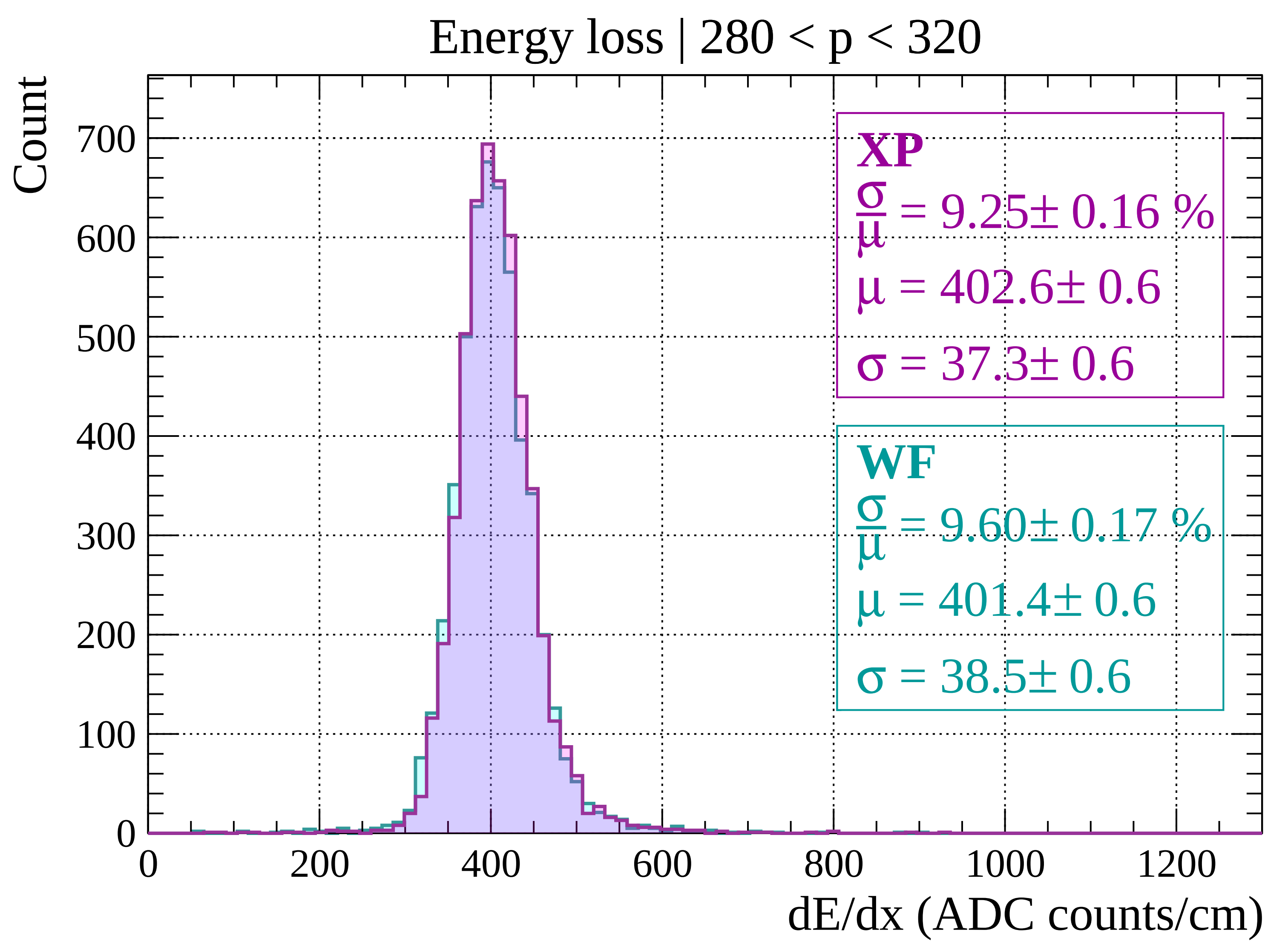
Energy loss | 160 < p < 200
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.96± 0.21 %
μ = 474.2± 0.9
σ = 47.2± 0.9
WF
σ/μ = 10.47± 0.24 %
μ = 472.4± 1.0
σ = 49.5± 1.0
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200



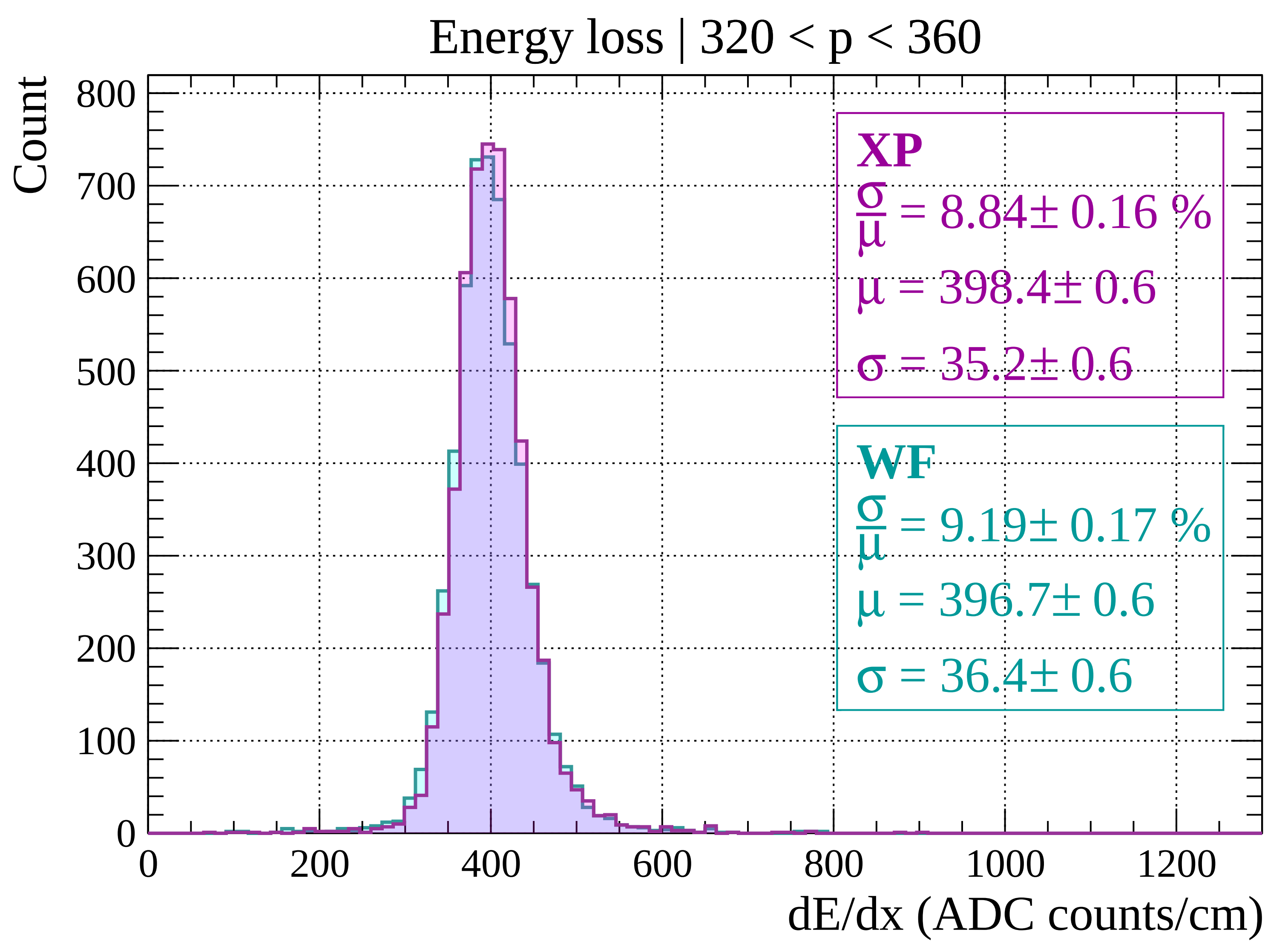
Energy loss | 200 < p < 240
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 9.12± 0.18 %
μ = 431.5± 0.7
σ = 39.4± 0.7
WF
σ/μ = 9.54± 0.20 %
μ = 429.0± 0.8
σ = 40.9± 0.8



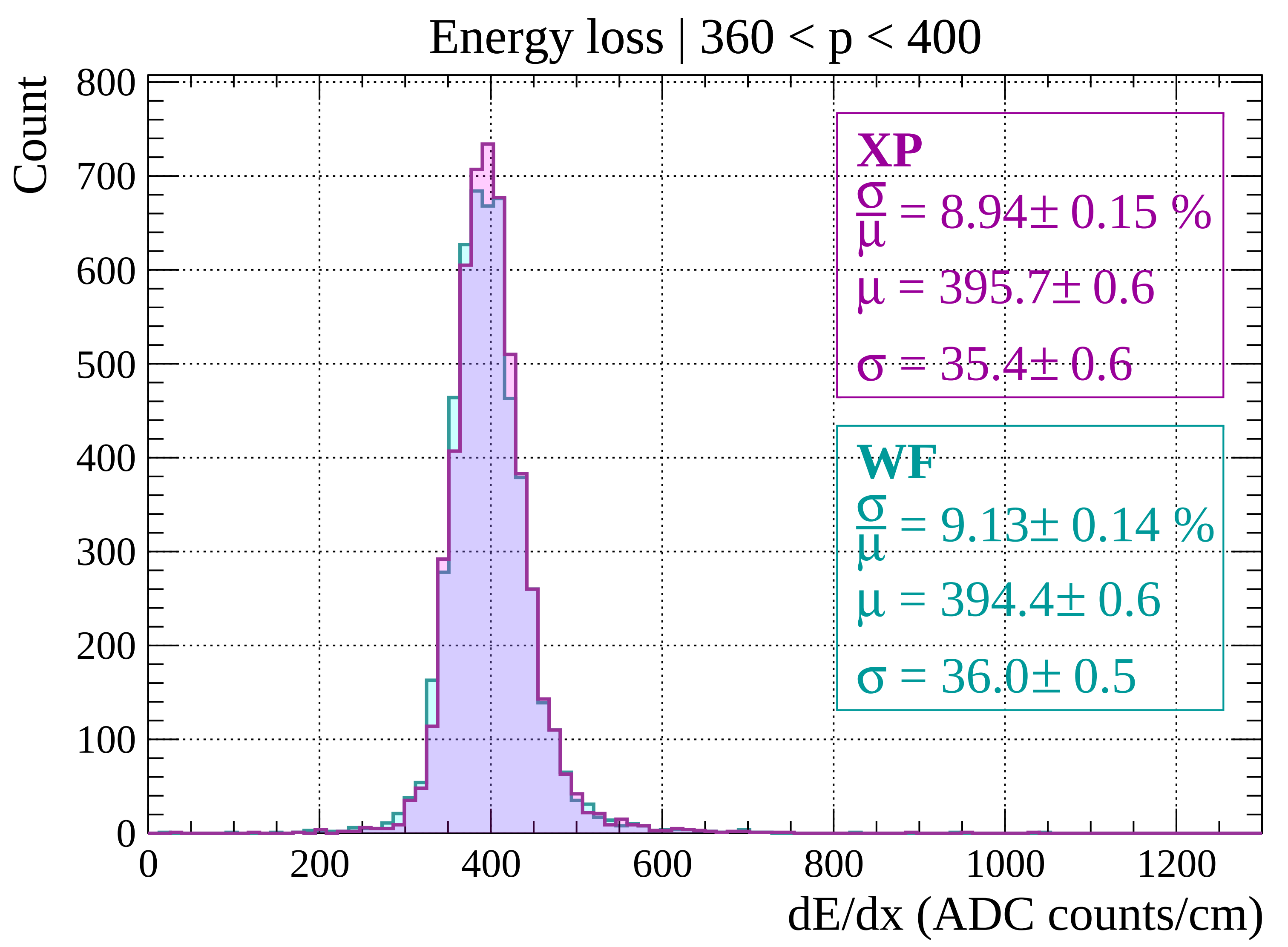
Energy loss | 240 < p < 280
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 8.86± 0.16 %
μ = 410.8± 0.6
σ = 36.4± 0.6
WF
σ/μ = 9.55± 0.18 %
μ = 408.7± 0.7
σ = 39.1± 0.7



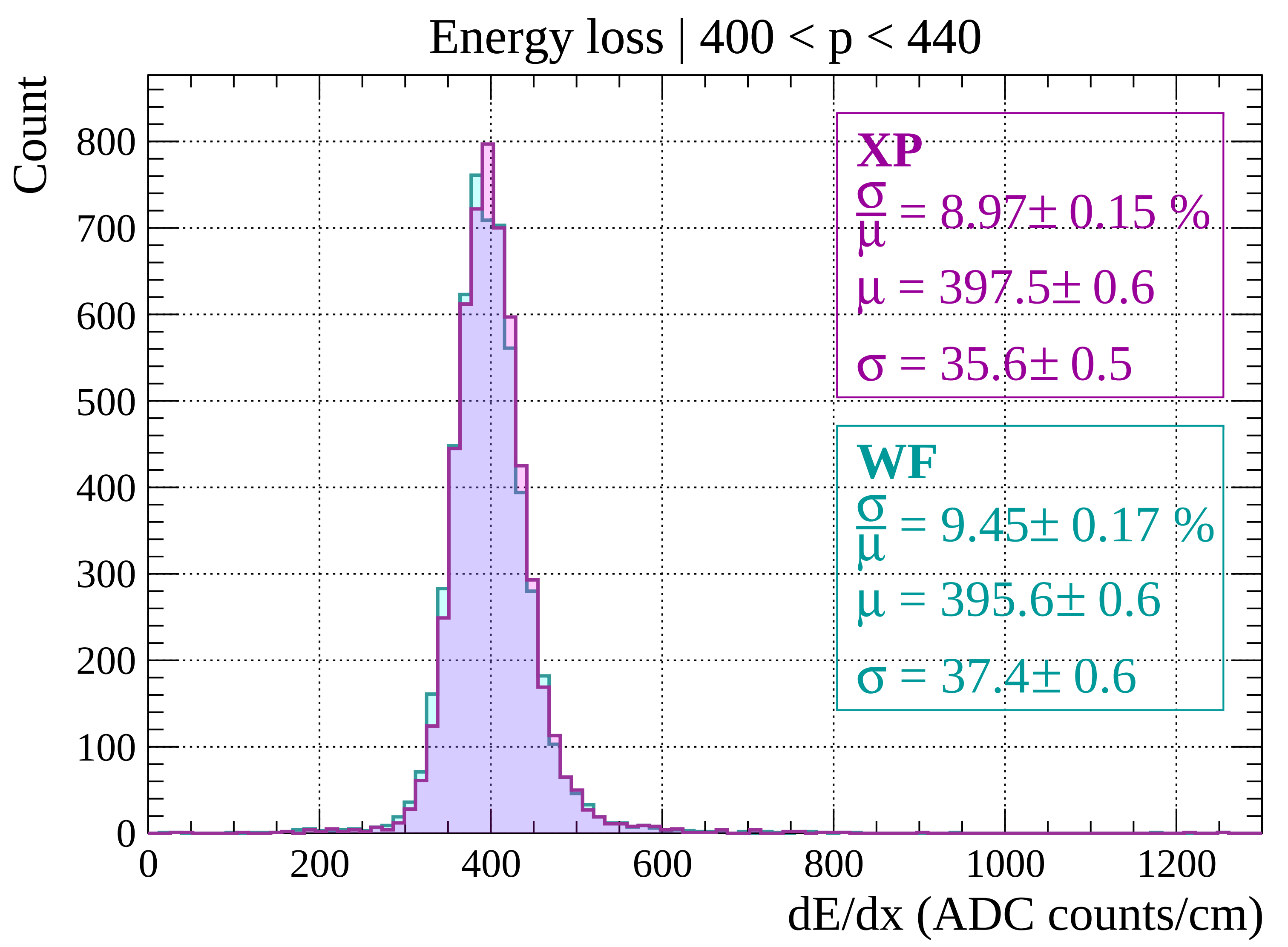
Energy loss | 280 < p < 320
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 9.25± 0.16 %
μ = 402.6± 0.6
σ = 37.3± 0.6
WF
σ/μ = 9.60± 0.17 %
μ = 401.4± 0.6
σ = 38.5± 0.6

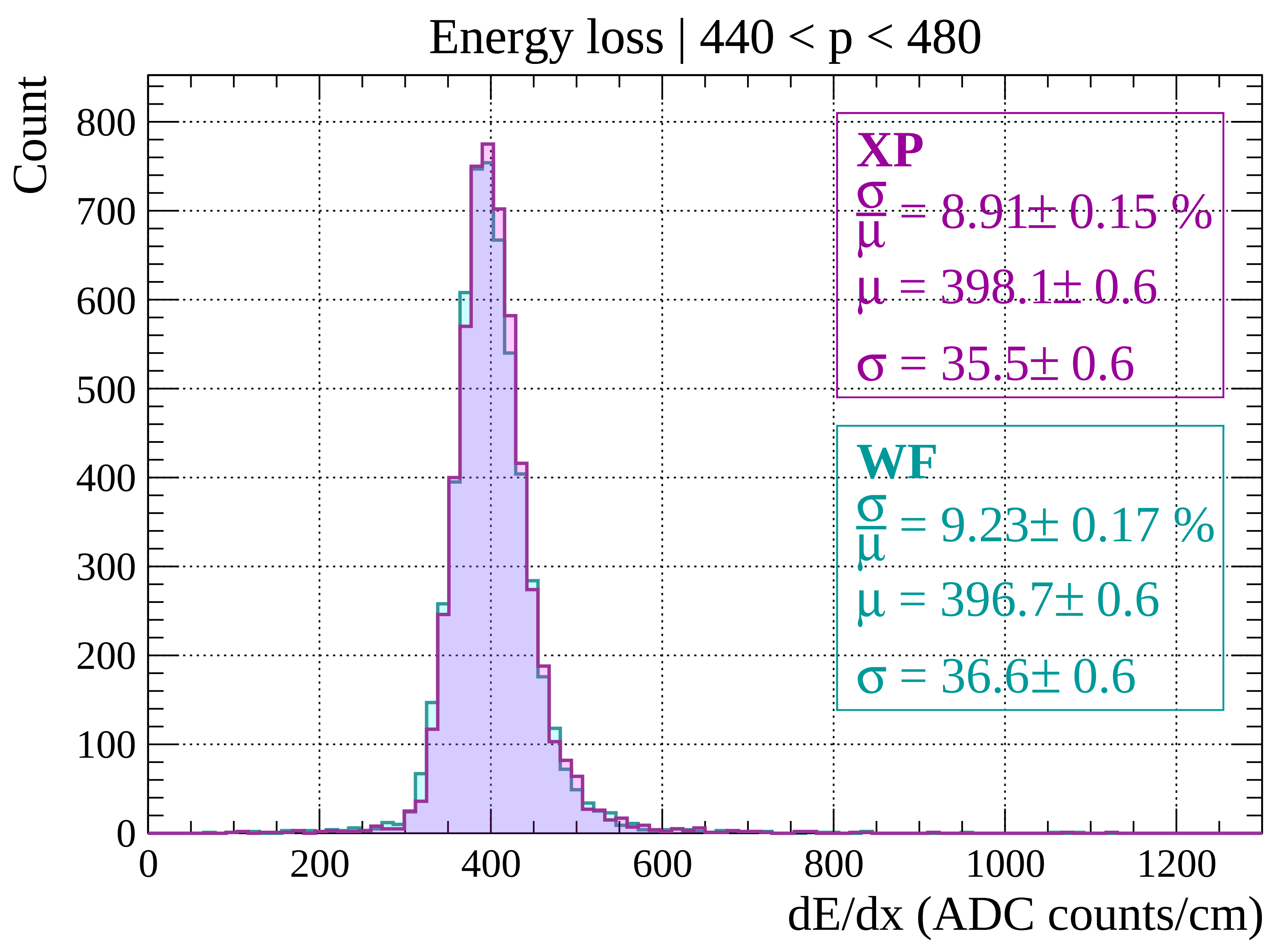


Energy loss | 320 < p < 360
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 8.84 \pm 0.16$ %
$\mu = 398.4 \pm 0.6$
$\sigma = 35.2 \pm 0.6$
WF
$\frac{\sigma}{\mu} = 9.19 \pm 0.17$ %
$\mu = 396.7 \pm 0.6$
$\sigma = 36.4 \pm 0.6$

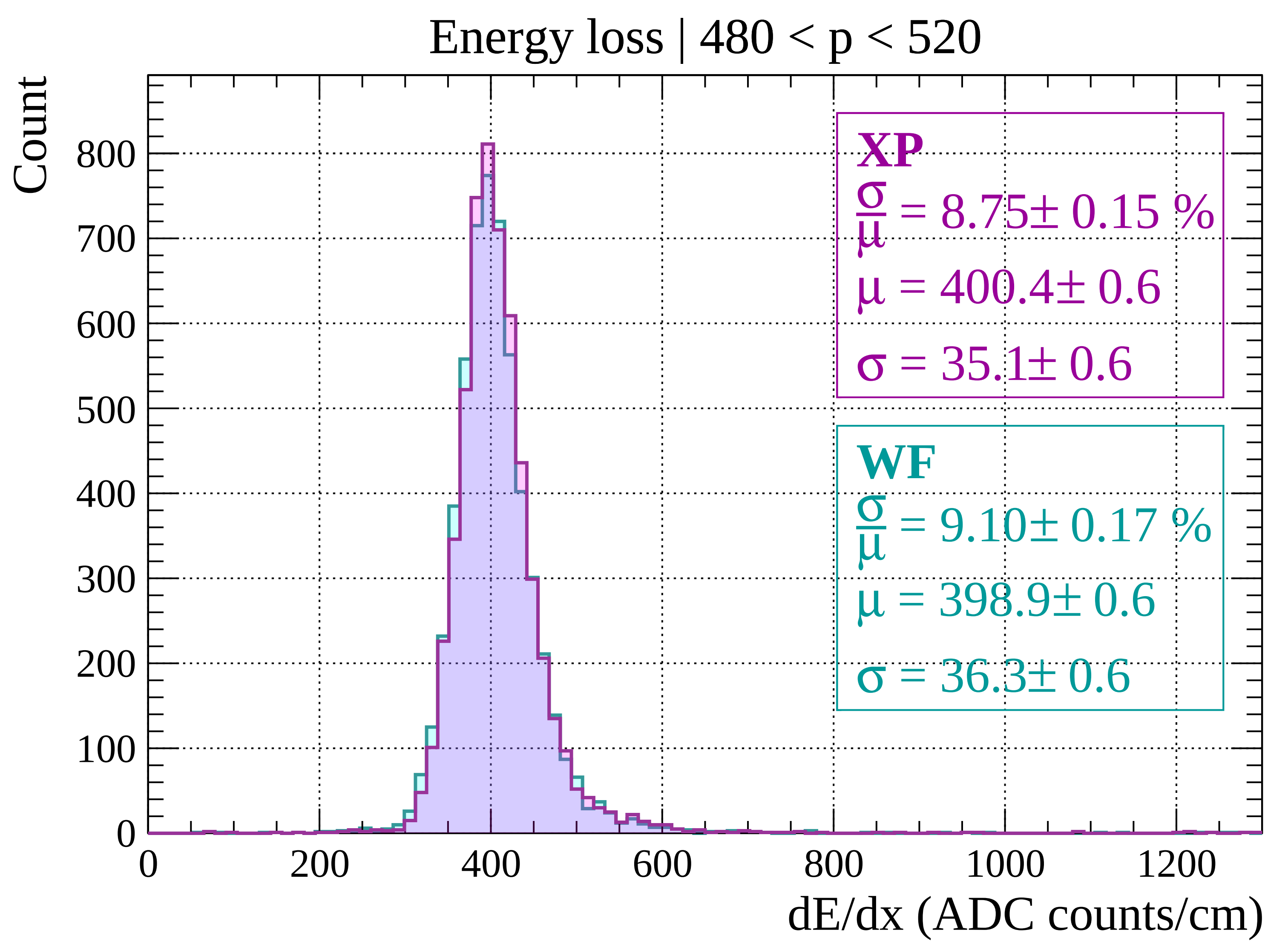


Energy loss | 360 < p < 400
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 8.94± 0.15 %
μ = 395.7± 0.6
σ = 35.4± 0.6
WF
$\frac{\sigma}{\mu}$ = 9.13± 0.14 %
μ = 394.4± 0.6
σ = 36.0± 0.5

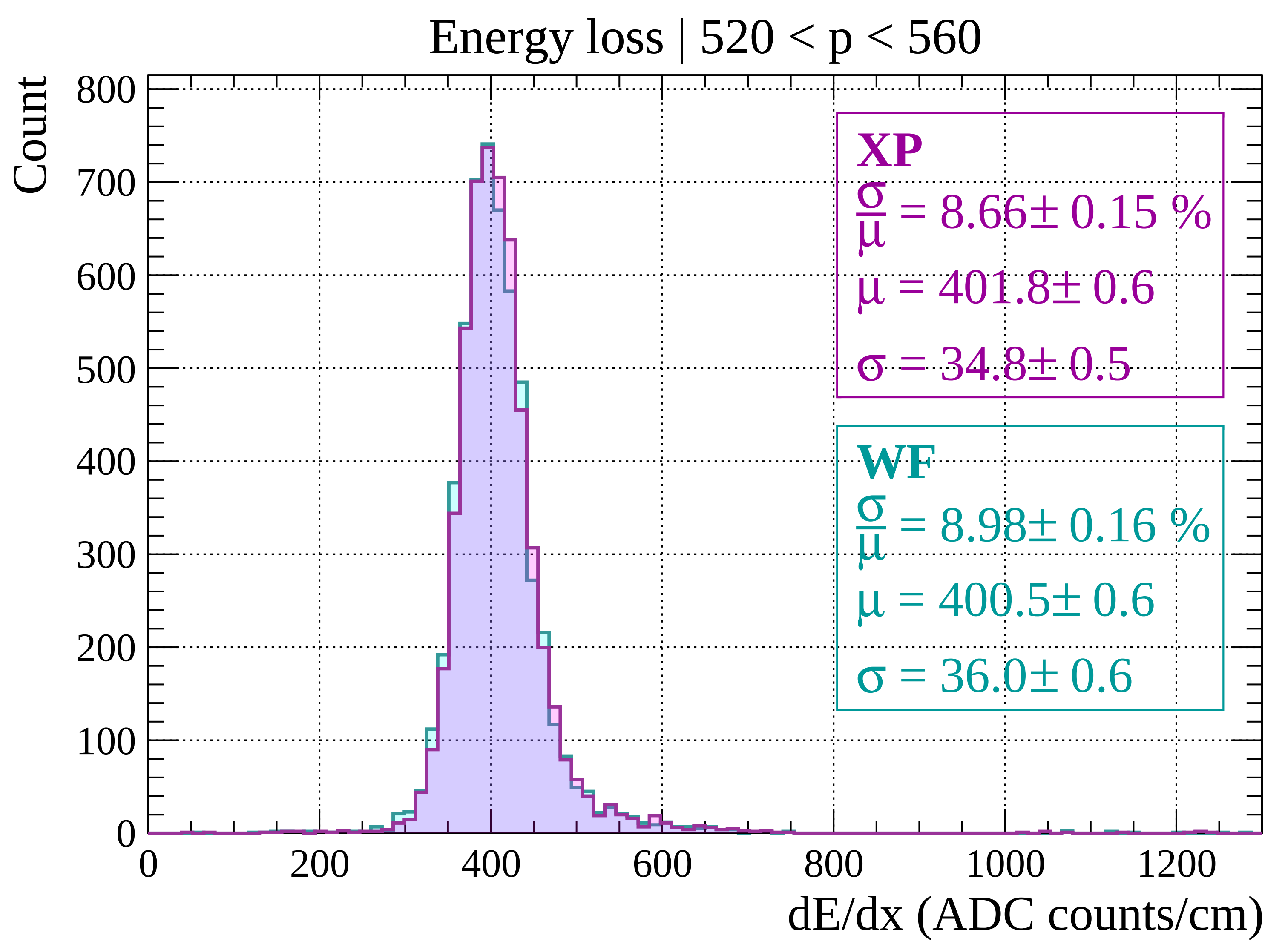

Energy loss | 400 < p < 440
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.97± 0.15 %
μ = 397.5± 0.6
σ = 35.6± 0.5
WF
$\frac{\sigma}{\mu}$ = 9.45± 0.17 %
μ = 395.6± 0.6
σ = 37.4± 0.6



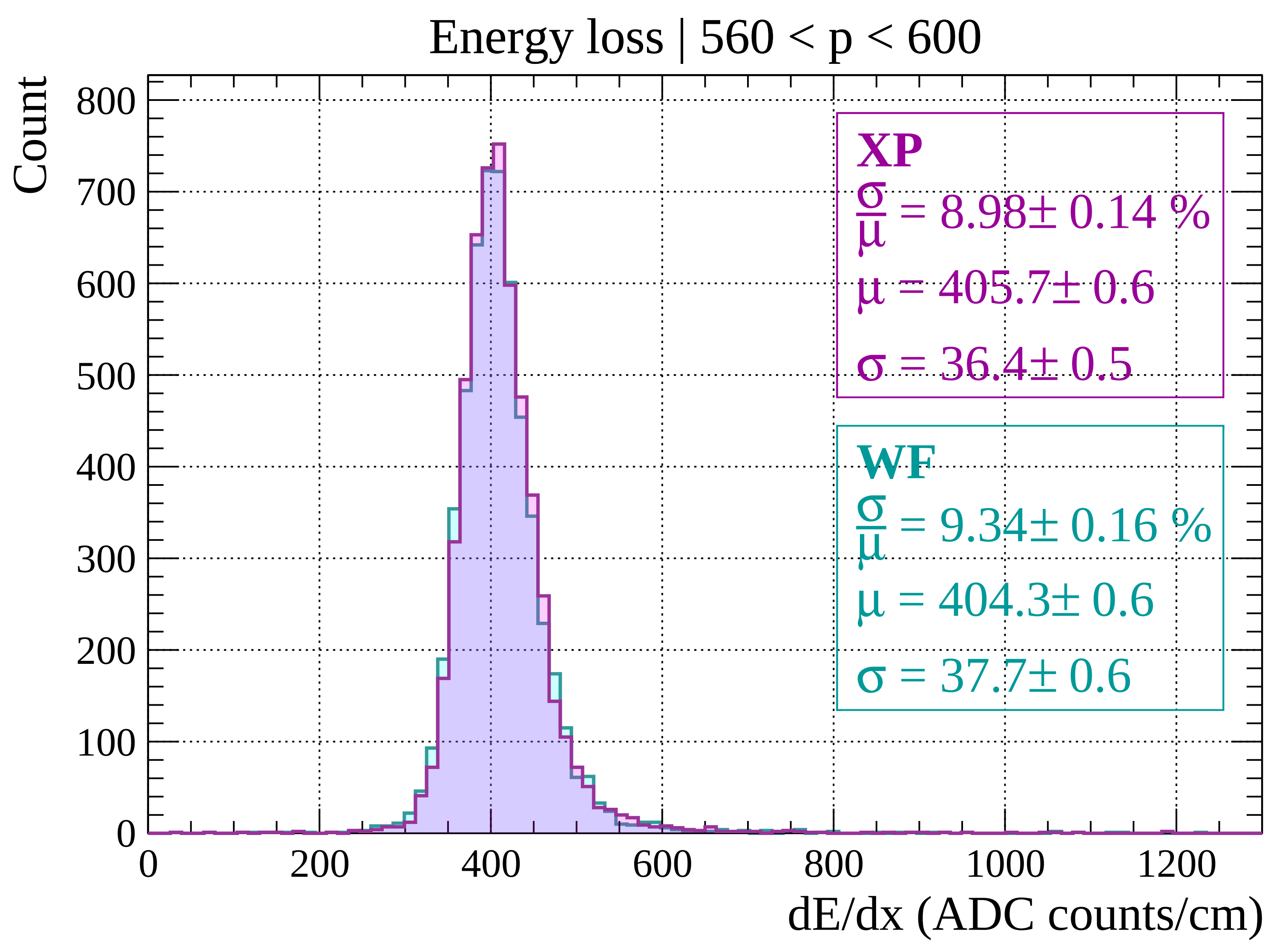
Energy loss | 440 < p < 480
Count
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.91± 0.15 %
μ = 398.1± 0.6
σ = 35.5± 0.6
WF
σ/μ = 9.23± 0.17 %
μ = 396.7± 0.6
σ = 36.6± 0.6



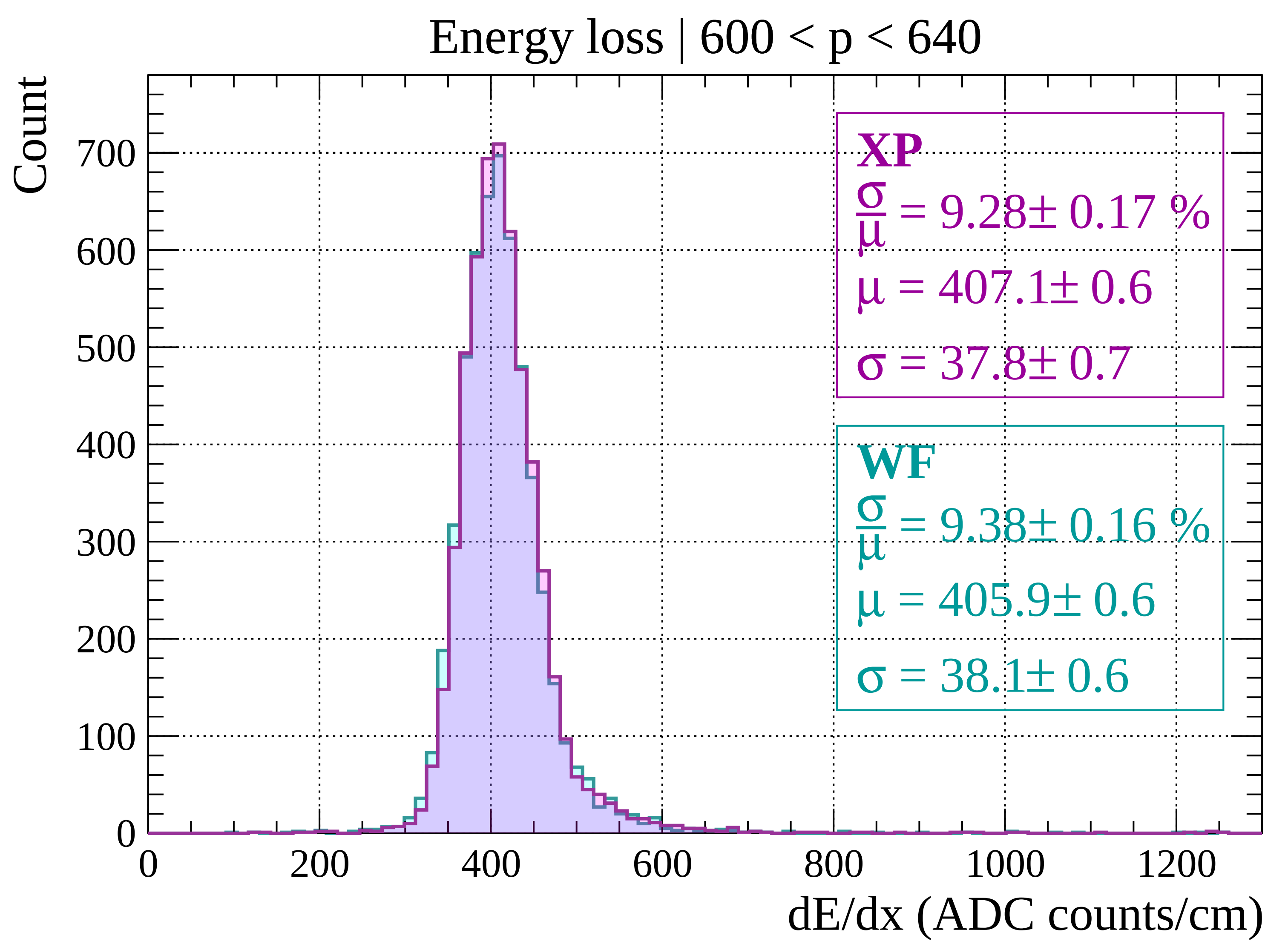
Energy loss | 480 < p < 520
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 8.75± 0.15 %
μ = 400.4± 0.6
σ = 35.1± 0.6
WF
σ/μ = 9.10± 0.17 %
μ = 398.9± 0.6
σ = 36.3± 0.6

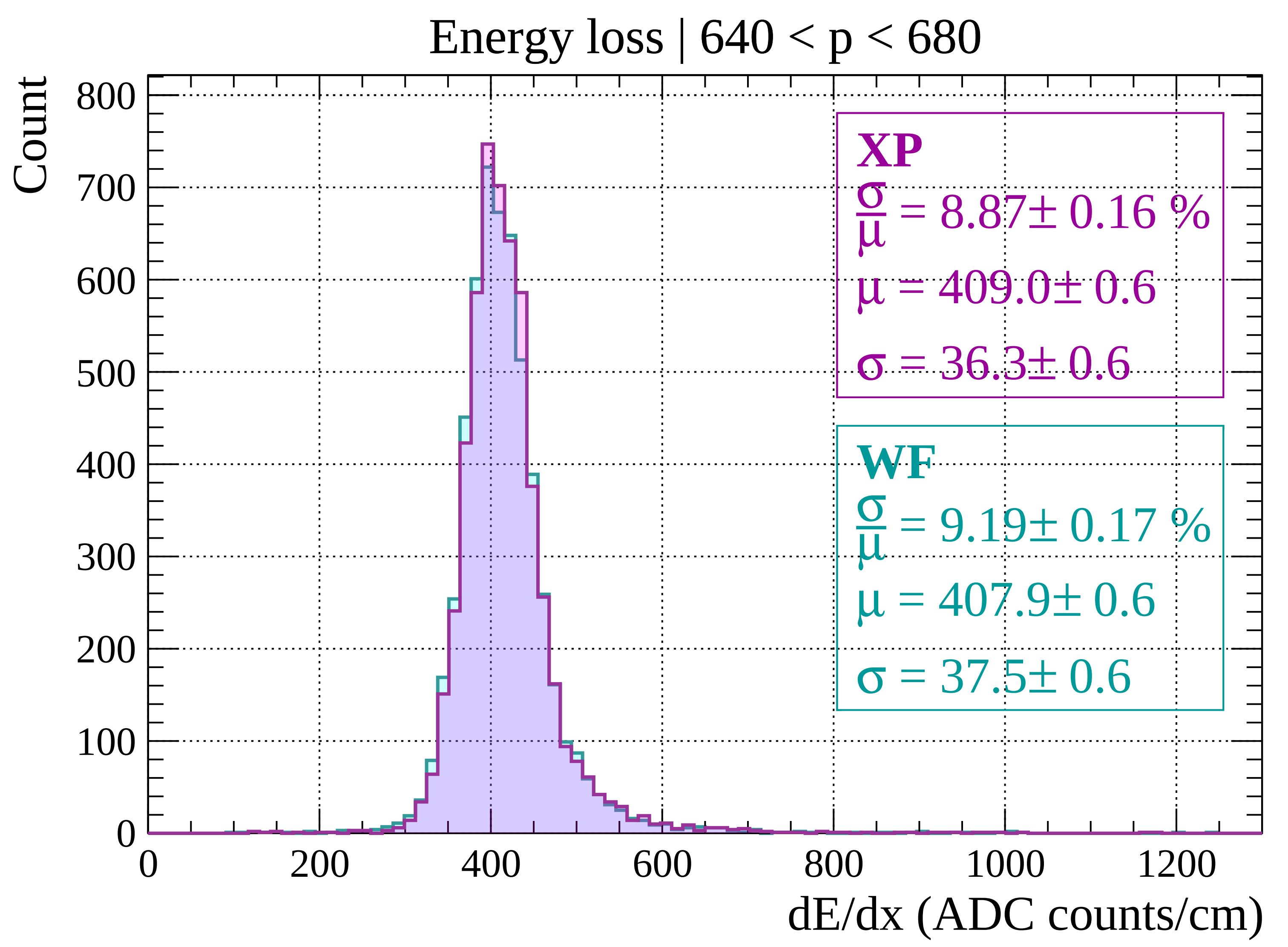


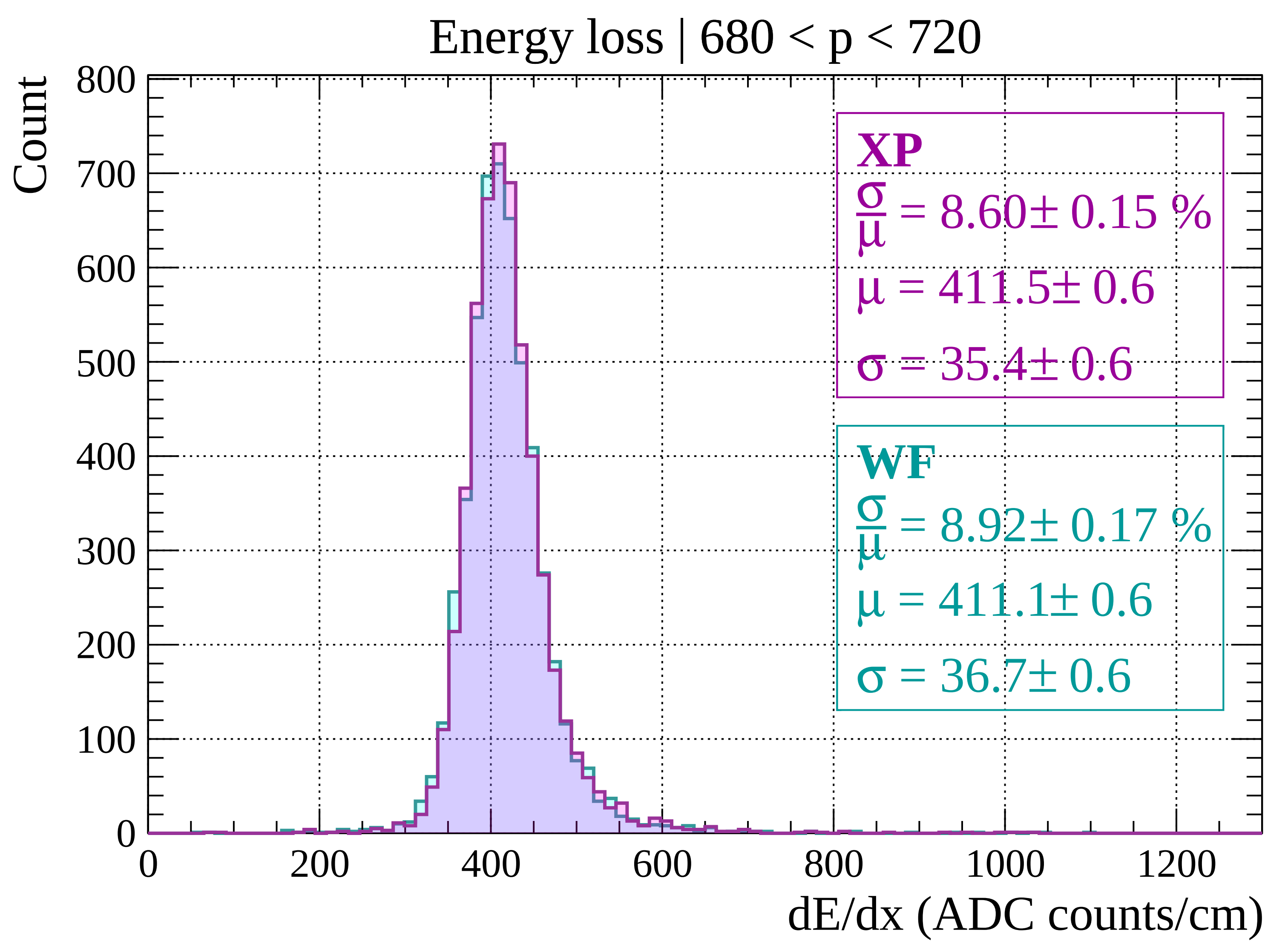
Energy loss | 520 < p < 560
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.66± 0.15 %
μ = 401.8± 0.6
σ = 34.8± 0.5
WF
σ/μ = 8.98± 0.16 %
μ = 400.5± 0.6
σ = 36.0± 0.6
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200



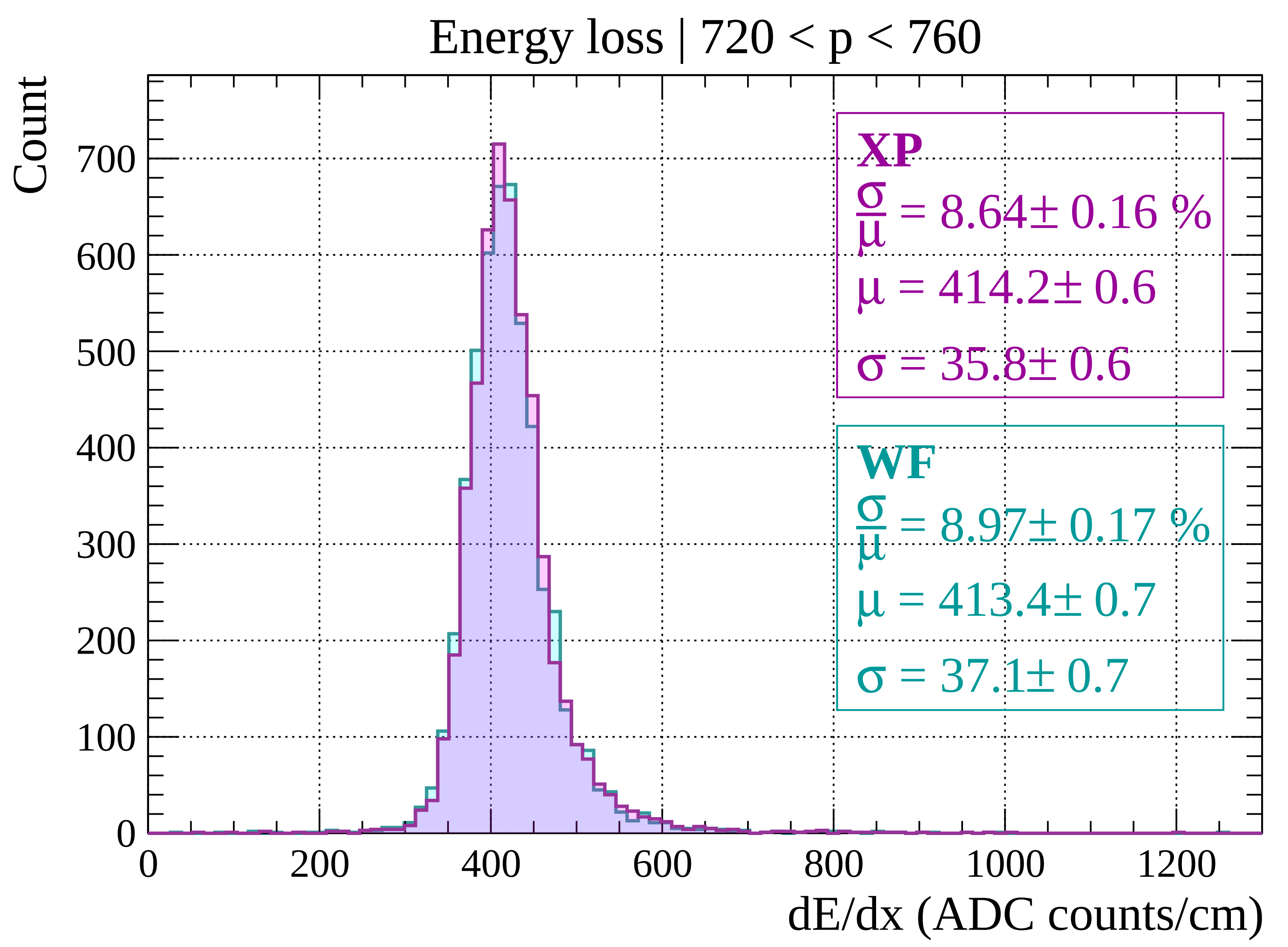
Energy loss | 560 < p < 600
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 8.98± 0.14 %
μ = 405.7± 0.6
σ = 36.4± 0.5
WF
σ/μ = 9.34± 0.16 %
μ = 404.3± 0.6
σ = 37.7± 0.6


Energy loss | 600 < p < 640
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 9.28± 0.17 %
μ = 407.1± 0.6
σ = 37.8± 0.7
WF
σ/μ = 9.38± 0.16 %
μ = 405.9± 0.6
σ = 38.1± 0.6

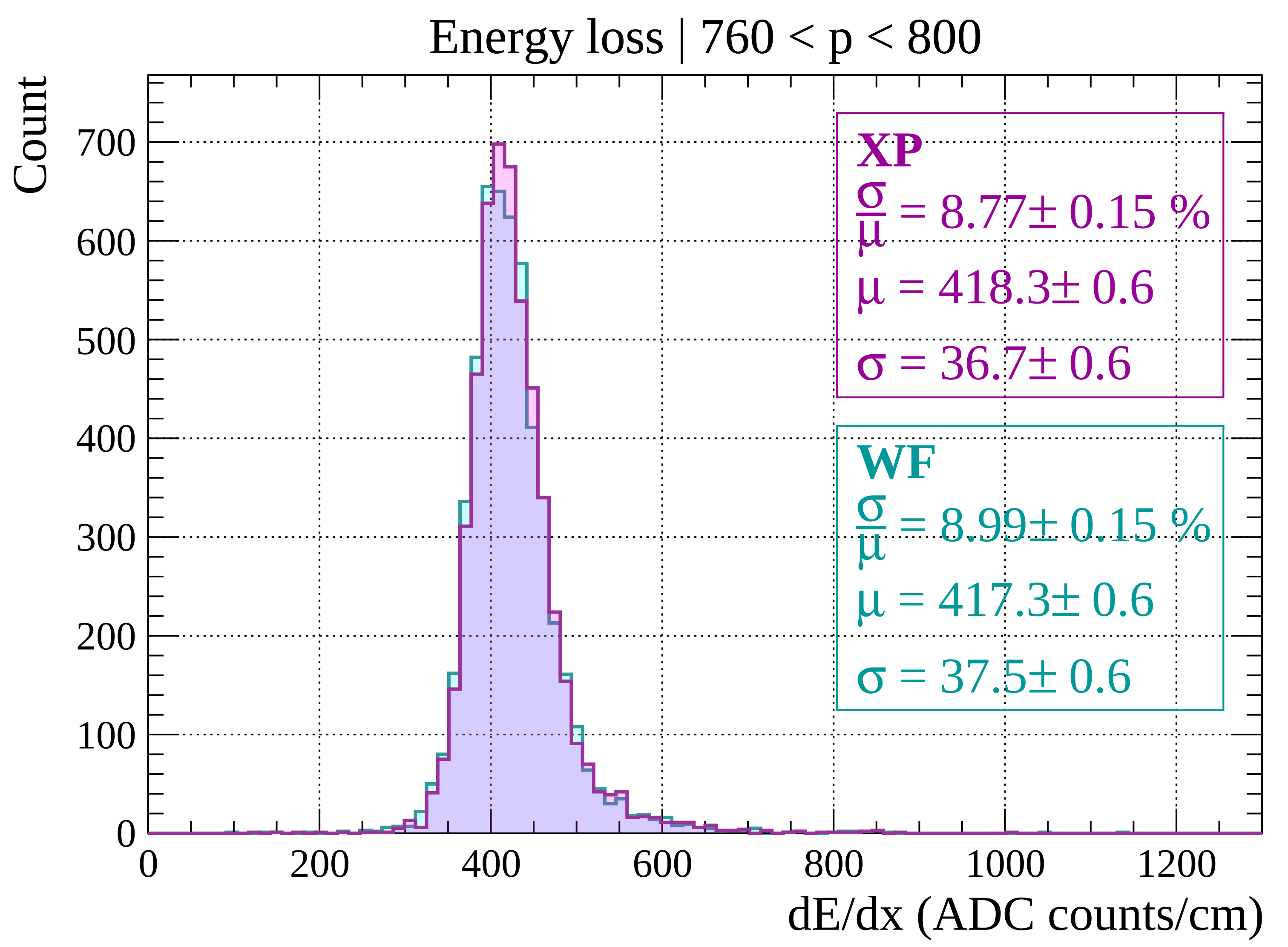

Energy loss | 640 < p < 680
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
XP
σ/μ = 8.87 ± 0.16 %
μ = 409.0 ± 0.6
σ = 36.3 ± 0.6
WF
σ/μ = 9.19 ± 0.17 %
μ = 407.9 ± 0.6
σ = 37.5 ± 0.6



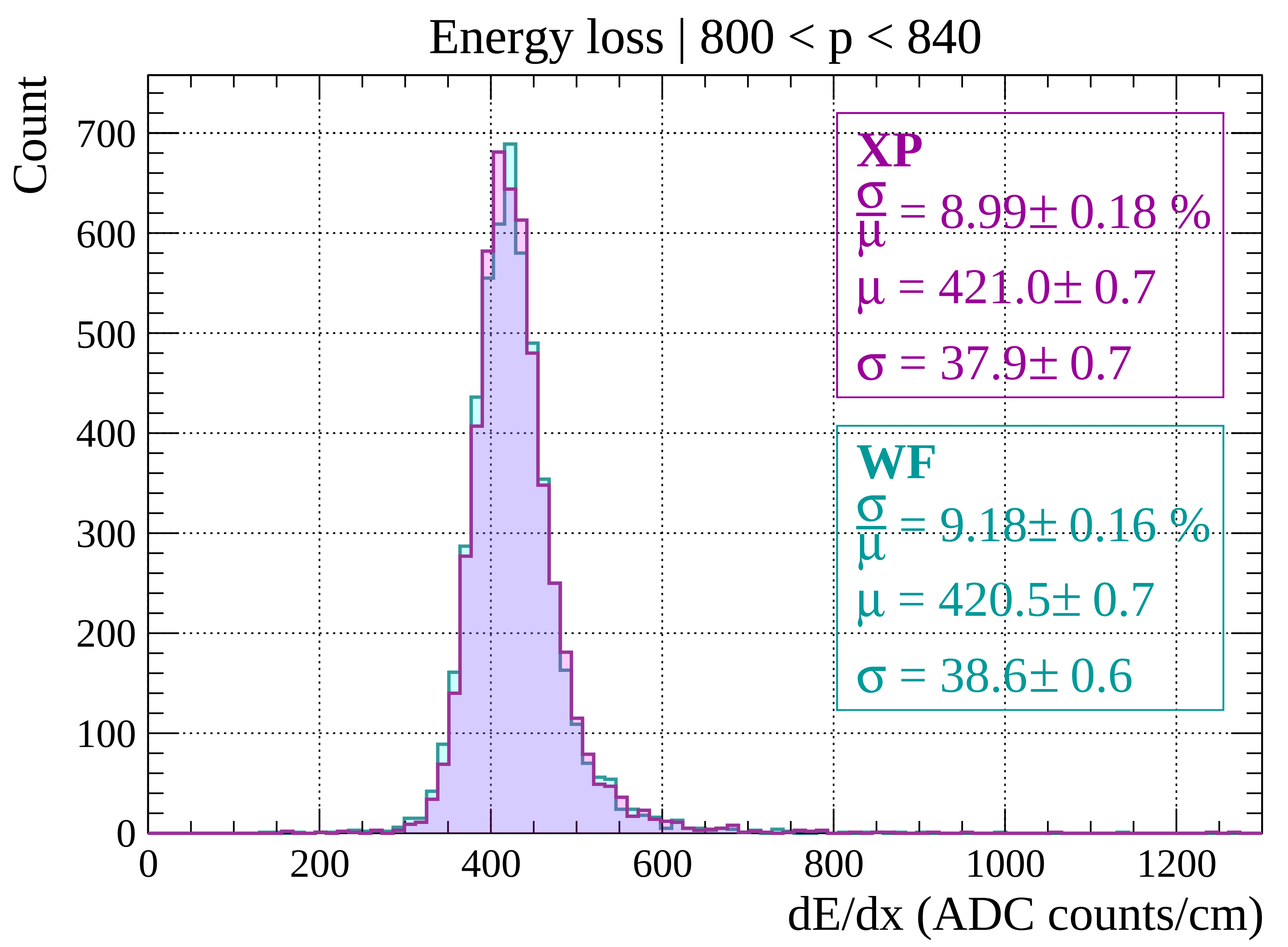
Energy loss | 680 < p < 720
Count
0
100
200
300
400
500
600
700
800
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.60± 0.15 %
μ = 411.5± 0.6
σ = 35.4± 0.6
WF
σ/μ = 8.92± 0.17 %
μ = 411.1± 0.6
σ = 36.7± 0.6



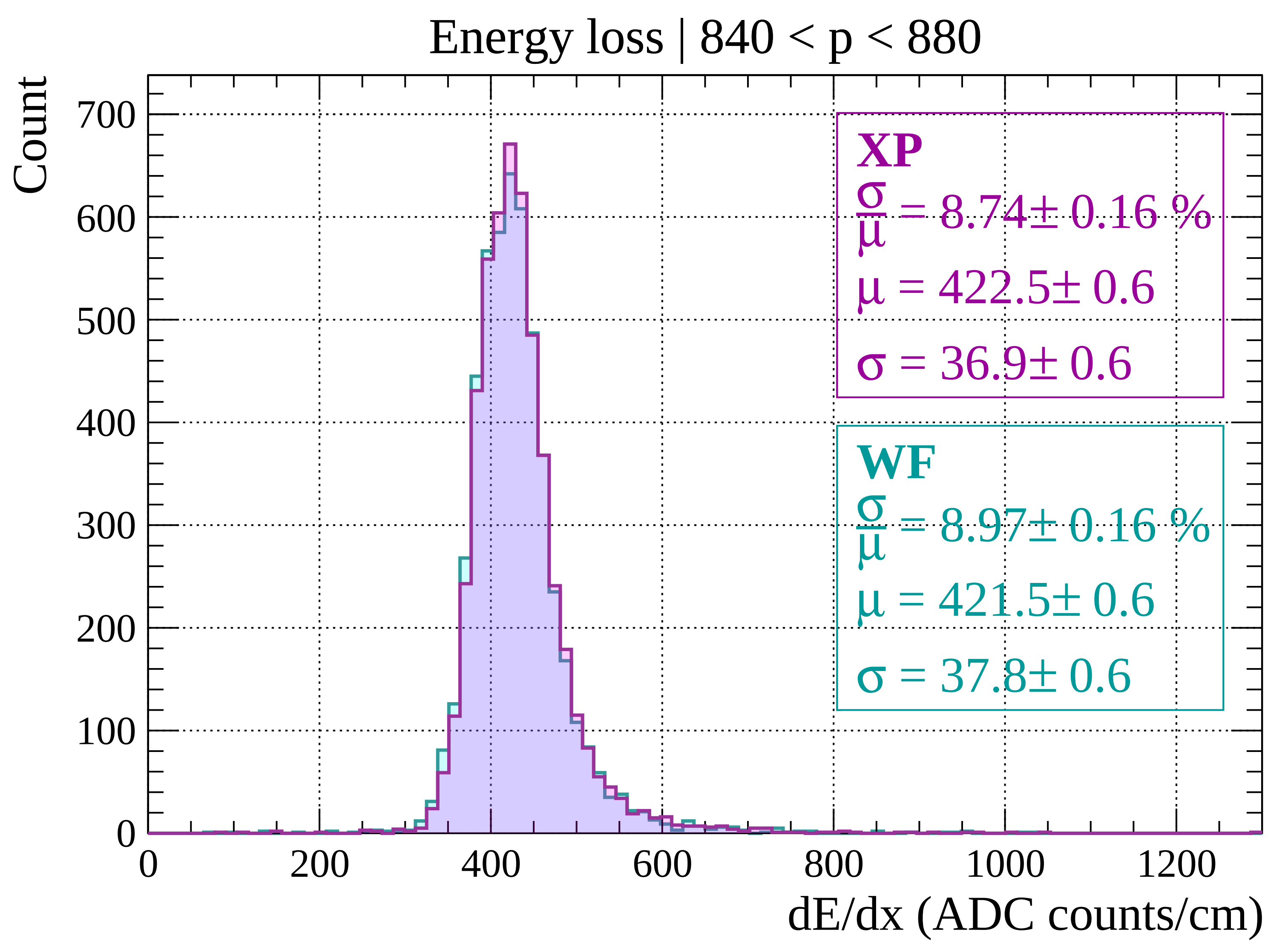
Energy loss | 720 < p < 760
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
σ/μ = 8.64± 0.16 %
μ = 414.2± 0.6
σ = 35.8± 0.6
WF
σ/μ = 8.97± 0.17 %
μ = 413.4± 0.7
σ = 37.1± 0.7



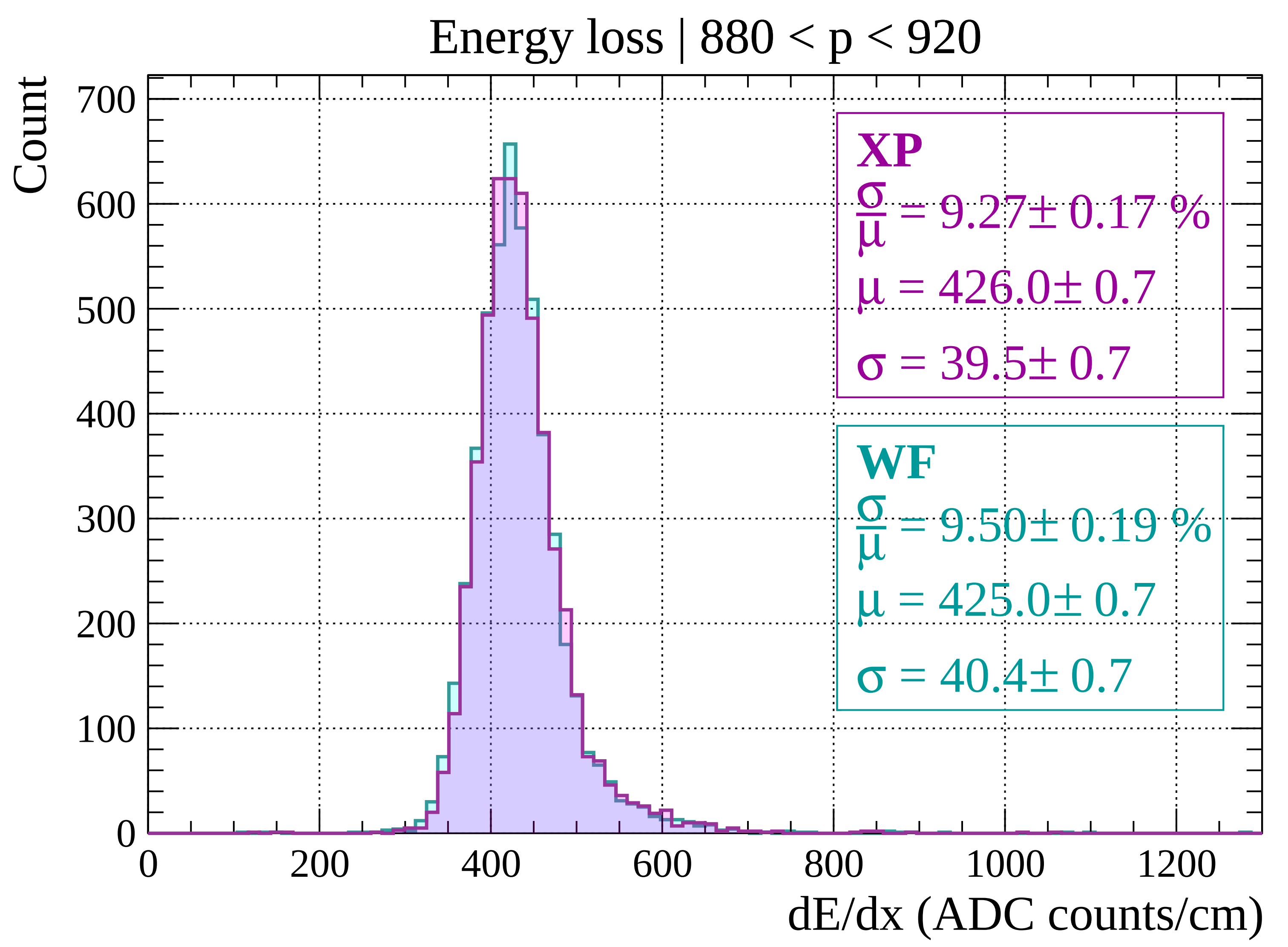
Energy loss | 760 < p < 800
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 8.77± 0.15 %
μ = 418.3± 0.6
σ = 36.7± 0.6
WF
σ/μ = 8.99± 0.15 %
μ = 417.3± 0.6
σ = 37.5± 0.6
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200



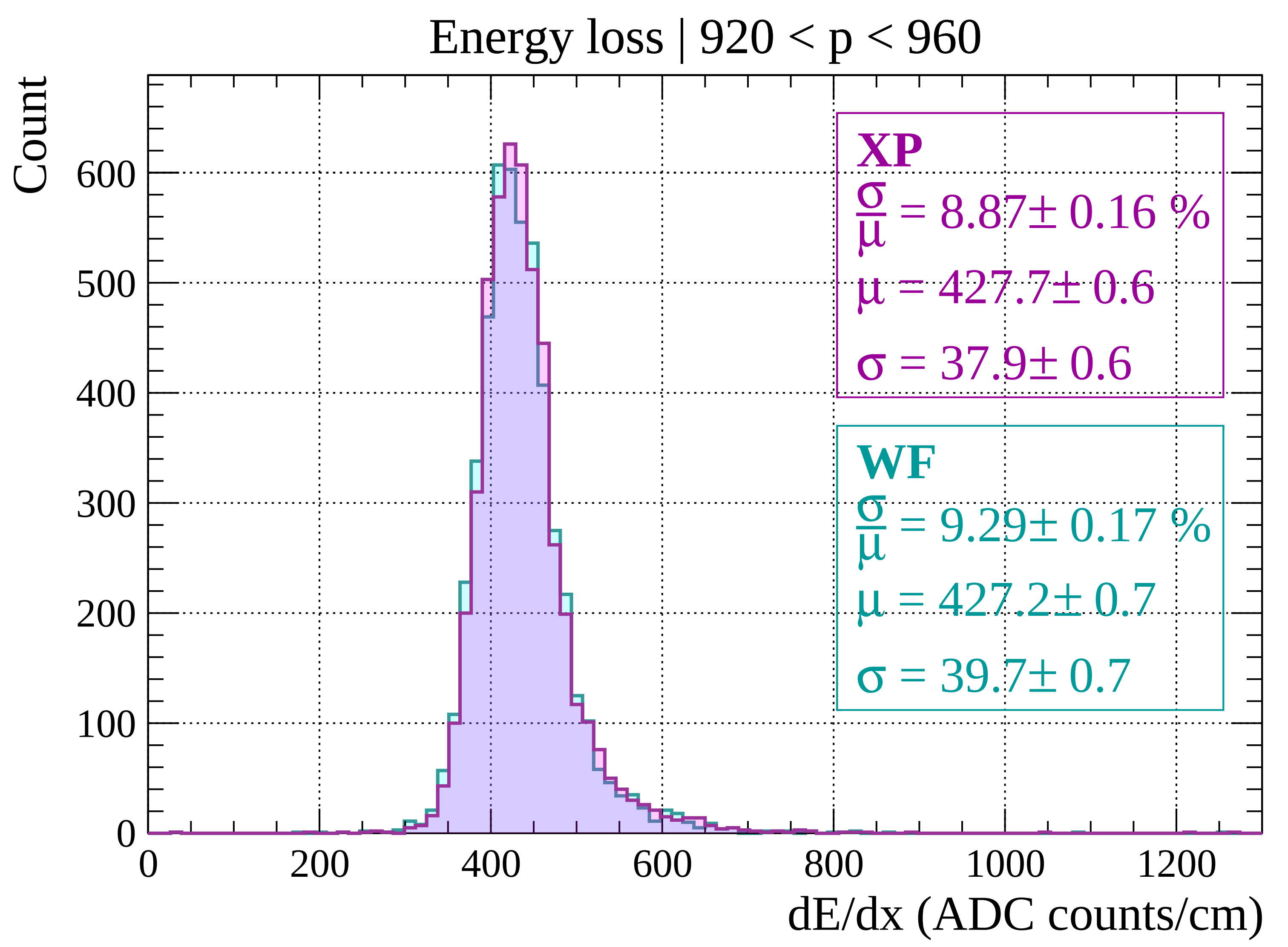
Energy loss | 800 < p < 840
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 8.99 \pm 0.18$ %
$\mu = 421.0 \pm 0.7$
$\sigma = 37.9 \pm 0.7$
WF
$\frac{\sigma}{\mu} = 9.18 \pm 0.16$ %
$\mu = 420.5 \pm 0.7$
$\sigma = 38.6 \pm 0.6$



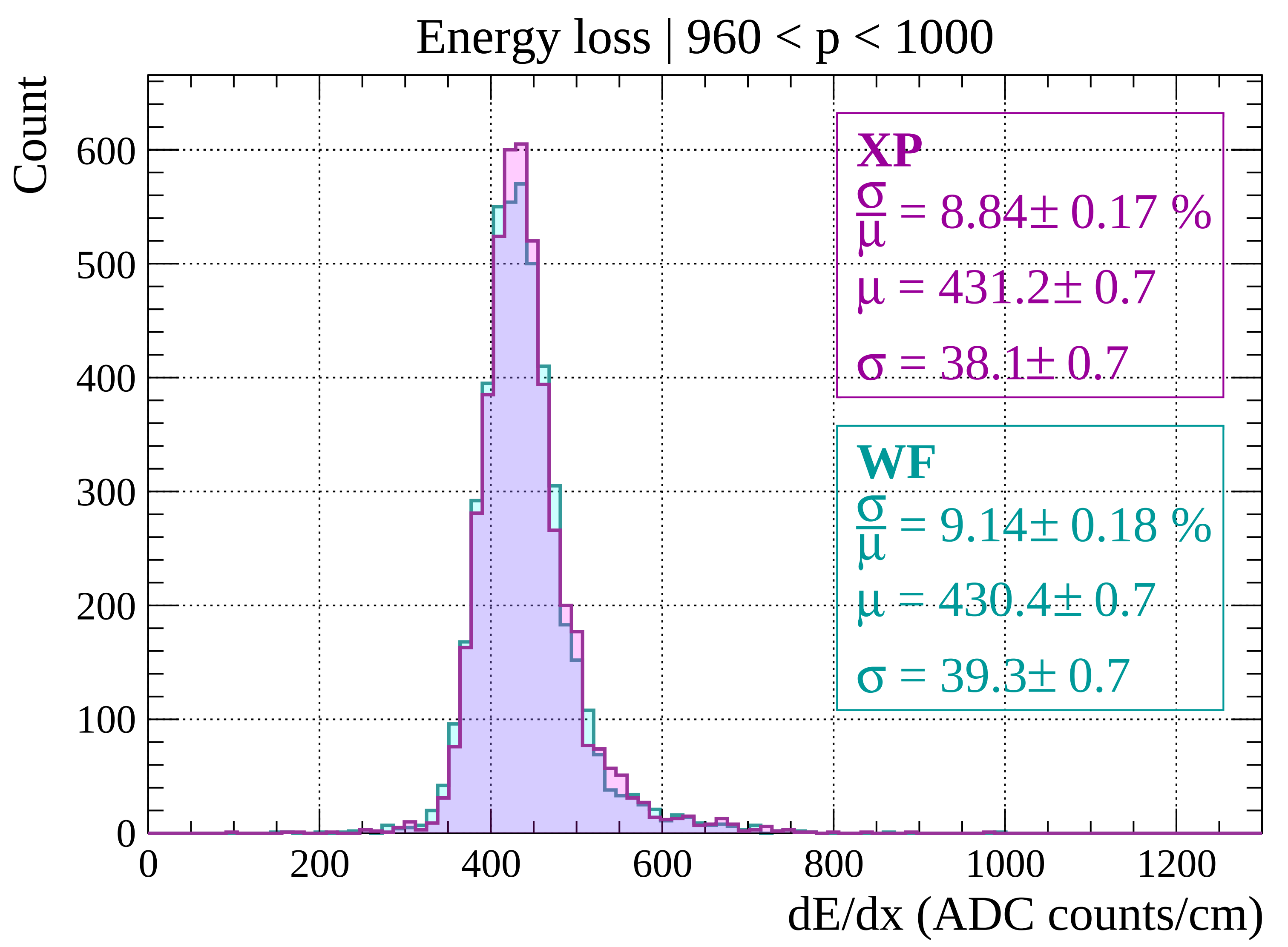
Energy loss | 840 < p < 880
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.74± 0.16 %
μ = 422.5± 0.6
σ = 36.9± 0.6
WF
$\frac{\sigma}{\mu}$ = 8.97± 0.16 %
μ = 421.5± 0.6
σ = 37.8± 0.6



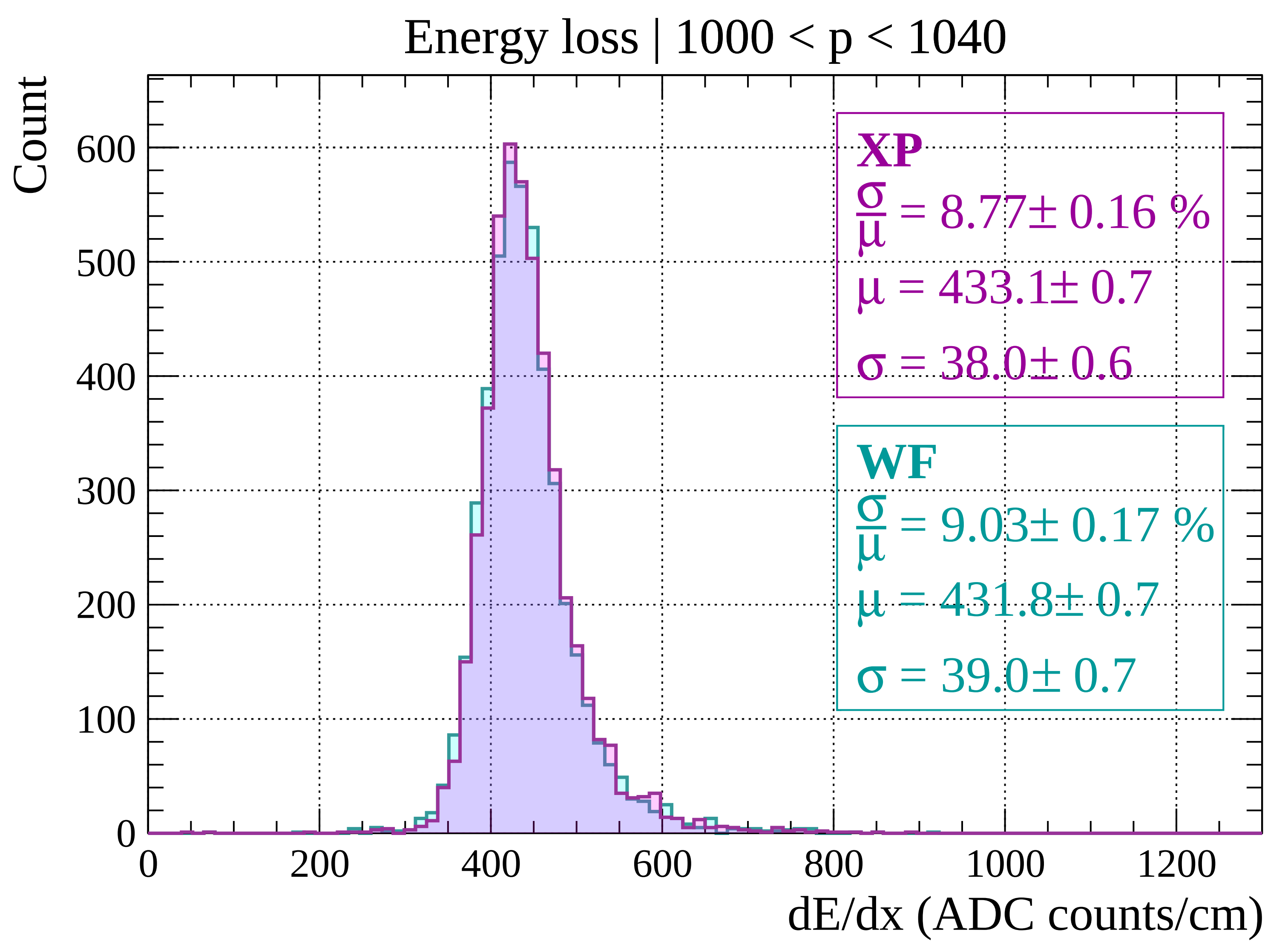
Energy loss | 880 < p < 920
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 9.27 \pm 0.17$ %
$\mu = 426.0 \pm 0.7$
$\sigma = 39.5 \pm 0.7$
WF
$\frac{\sigma}{\mu} = 9.50 \pm 0.19$ %
$\mu = 425.0 \pm 0.7$
$\sigma = 40.4 \pm 0.7$



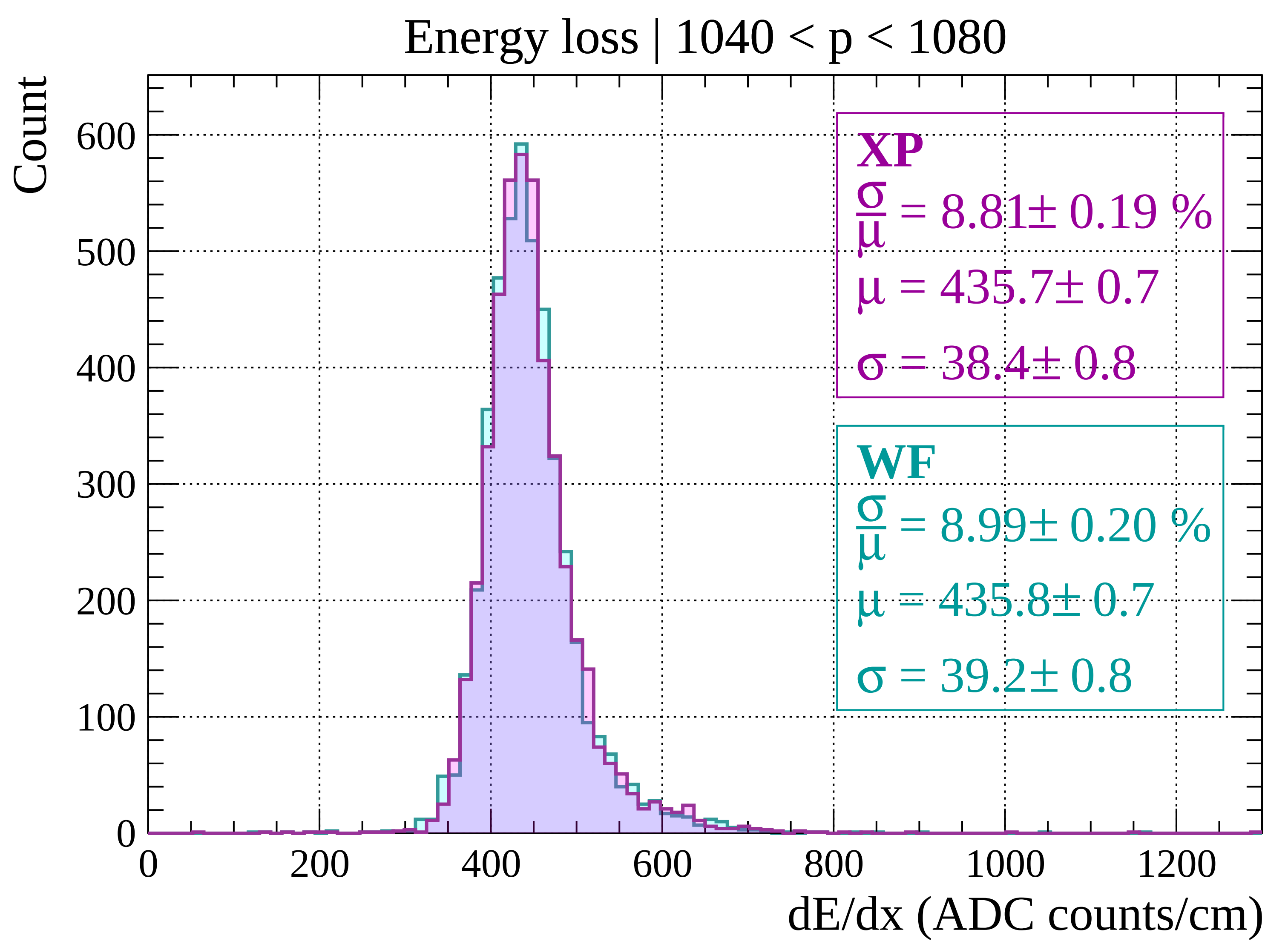
Energy loss | 920 < p < 960
Count
XP
σ/μ = 8.87± 0.16 %
μ = 427.7± 0.6
σ = 37.9± 0.6
WF
σ/μ = 9.29± 0.17 %
μ = 427.2± 0.7
σ = 39.7± 0.7
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200



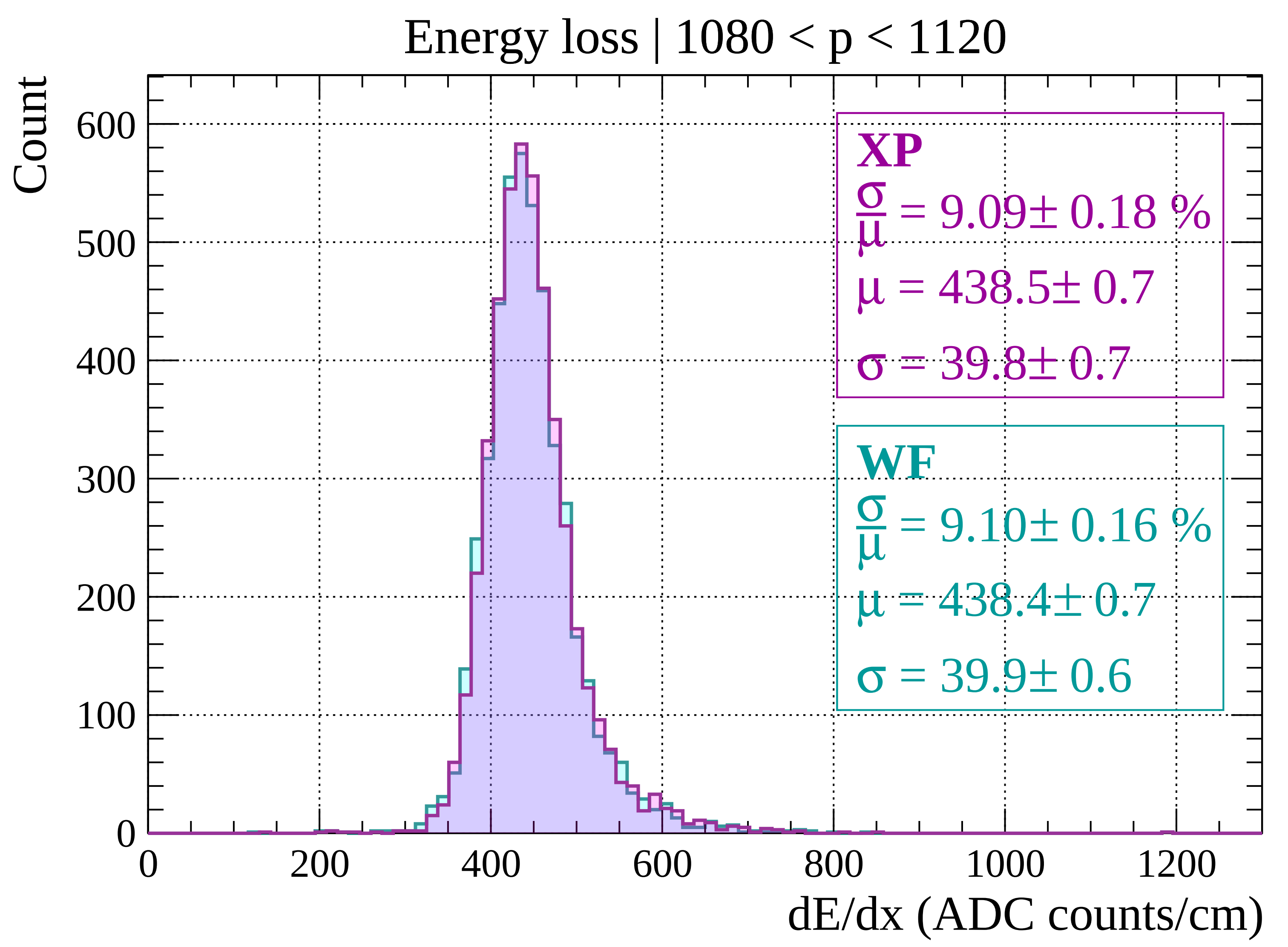
Energy loss | 960 < p < 1000
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 8.84± 0.17 %
μ = 431.2± 0.7
σ = 38.1± 0.7
WF
σ/μ = 9.14± 0.18 %
μ = 430.4± 0.7
σ = 39.3± 0.7



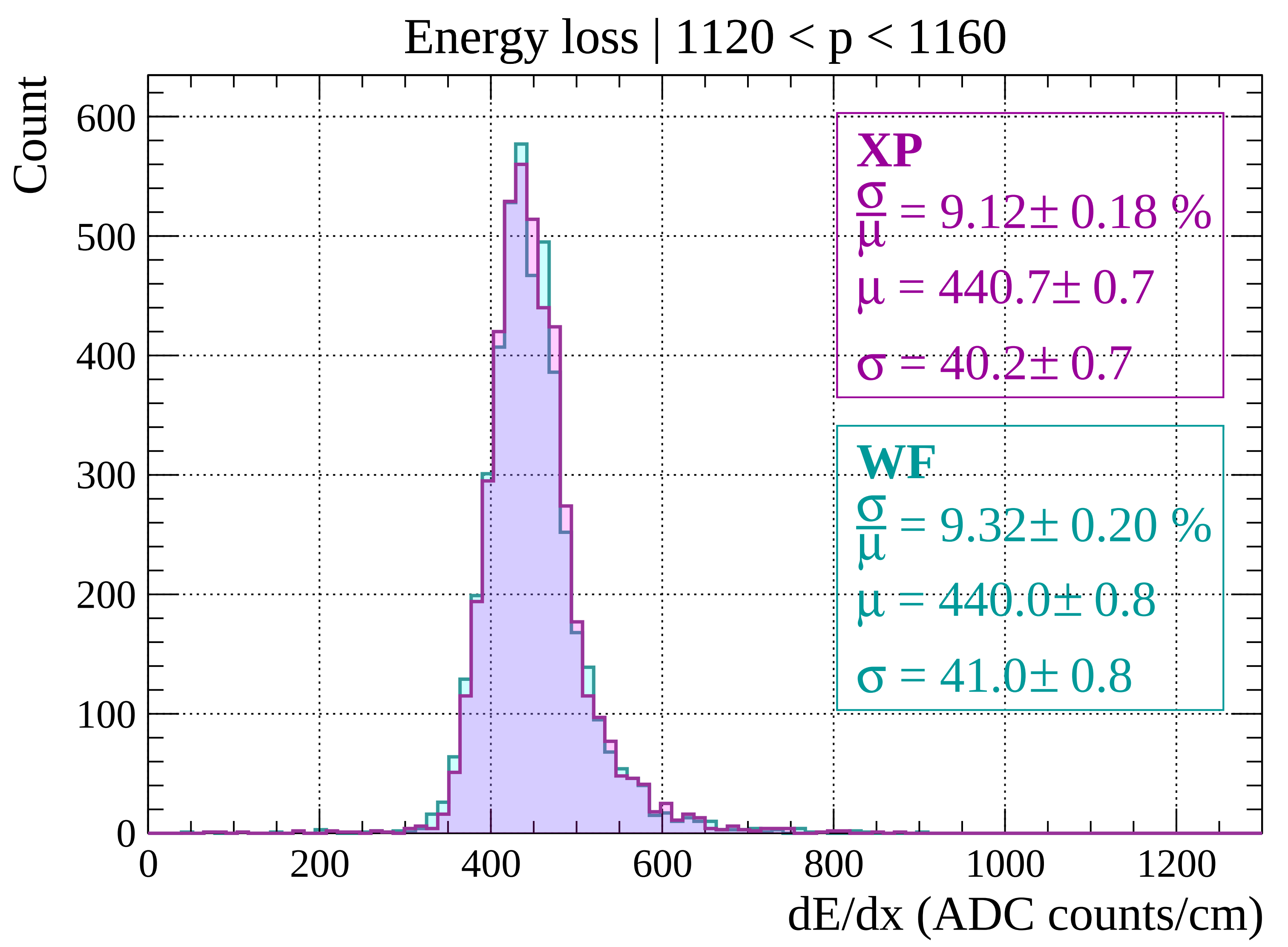
Energy loss | 1000 < p < 1040
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 8.77± 0.16 %
μ = 433.1± 0.7
σ = 38.0± 0.6
WF
$\frac{\sigma}{\mu}$ = 9.03± 0.17 %
μ = 431.8± 0.7
σ = 39.0± 0.7



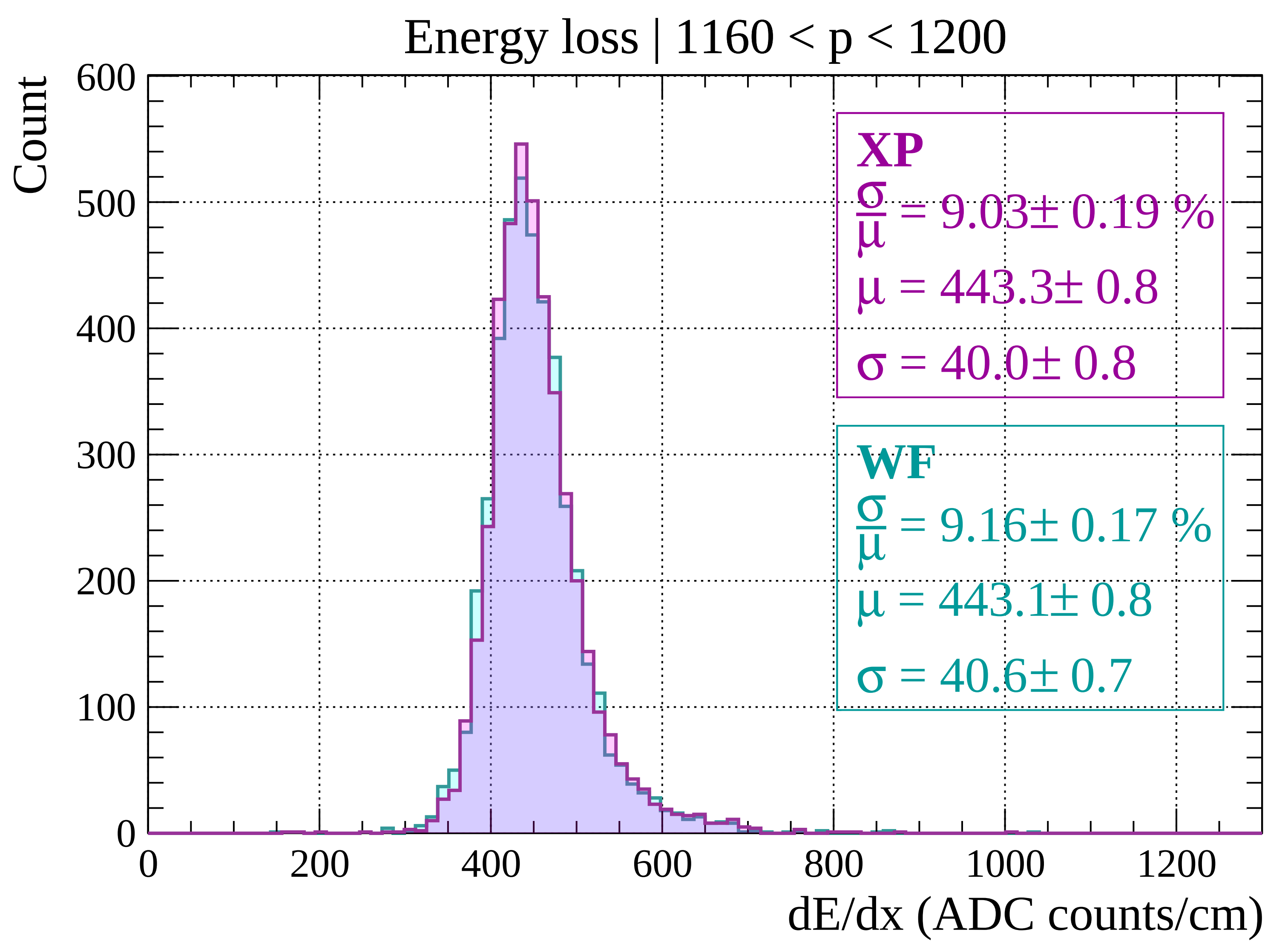
Energy loss | 1040 < p < 1080
Count
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 8.81± 0.19 %
μ = 435.7± 0.7
σ = 38.4± 0.8
WF
σ/μ = 8.99± 0.20 %
μ = 435.8± 0.7
σ = 39.2± 0.8



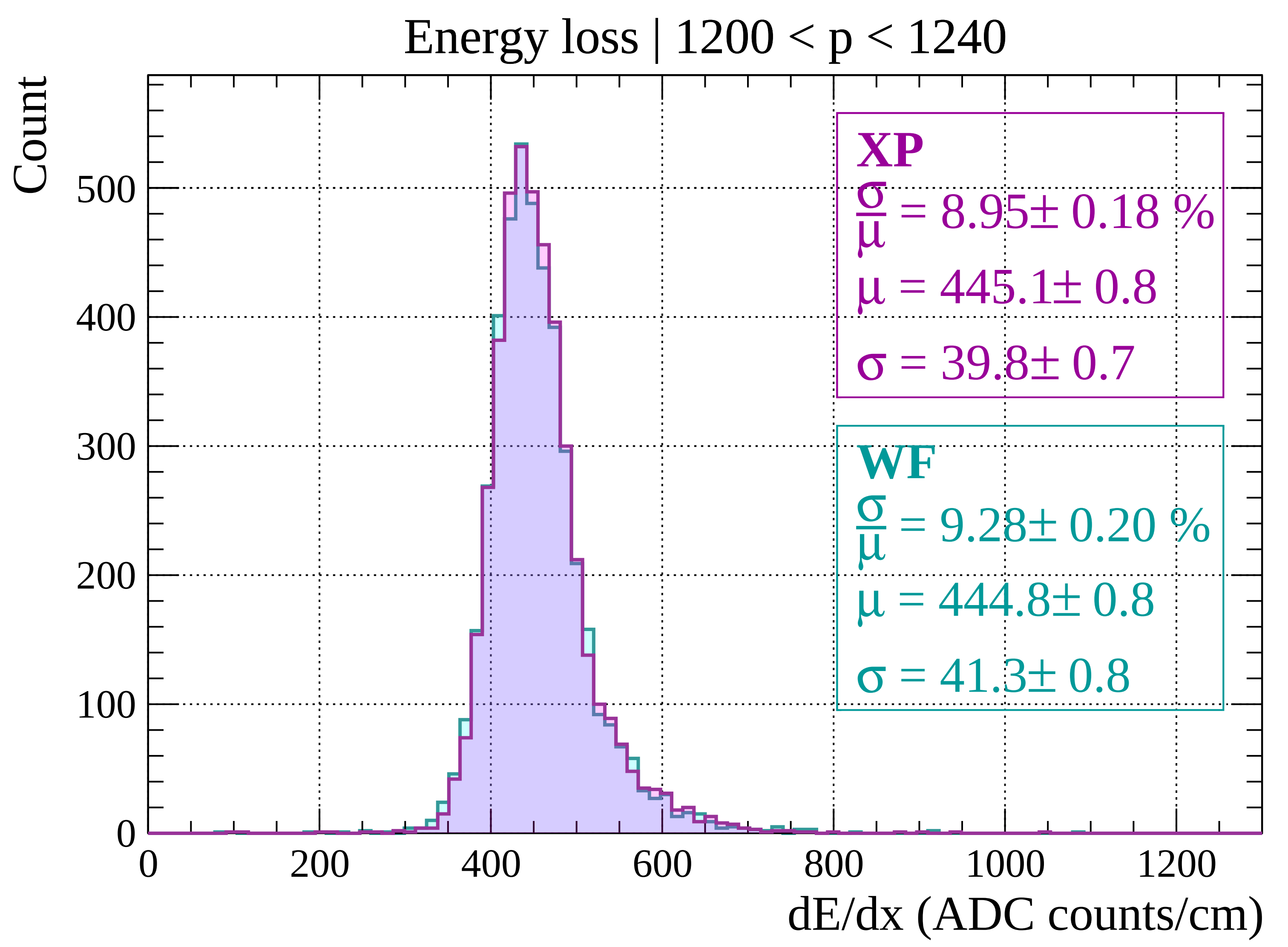
Energy loss | 1080 < p < 1120
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 9.09± 0.18 %
μ = 438.5± 0.7
σ = 39.8± 0.7
WF
σ/μ = 9.10± 0.16 %
μ = 438.4± 0.7
σ = 39.9± 0.6

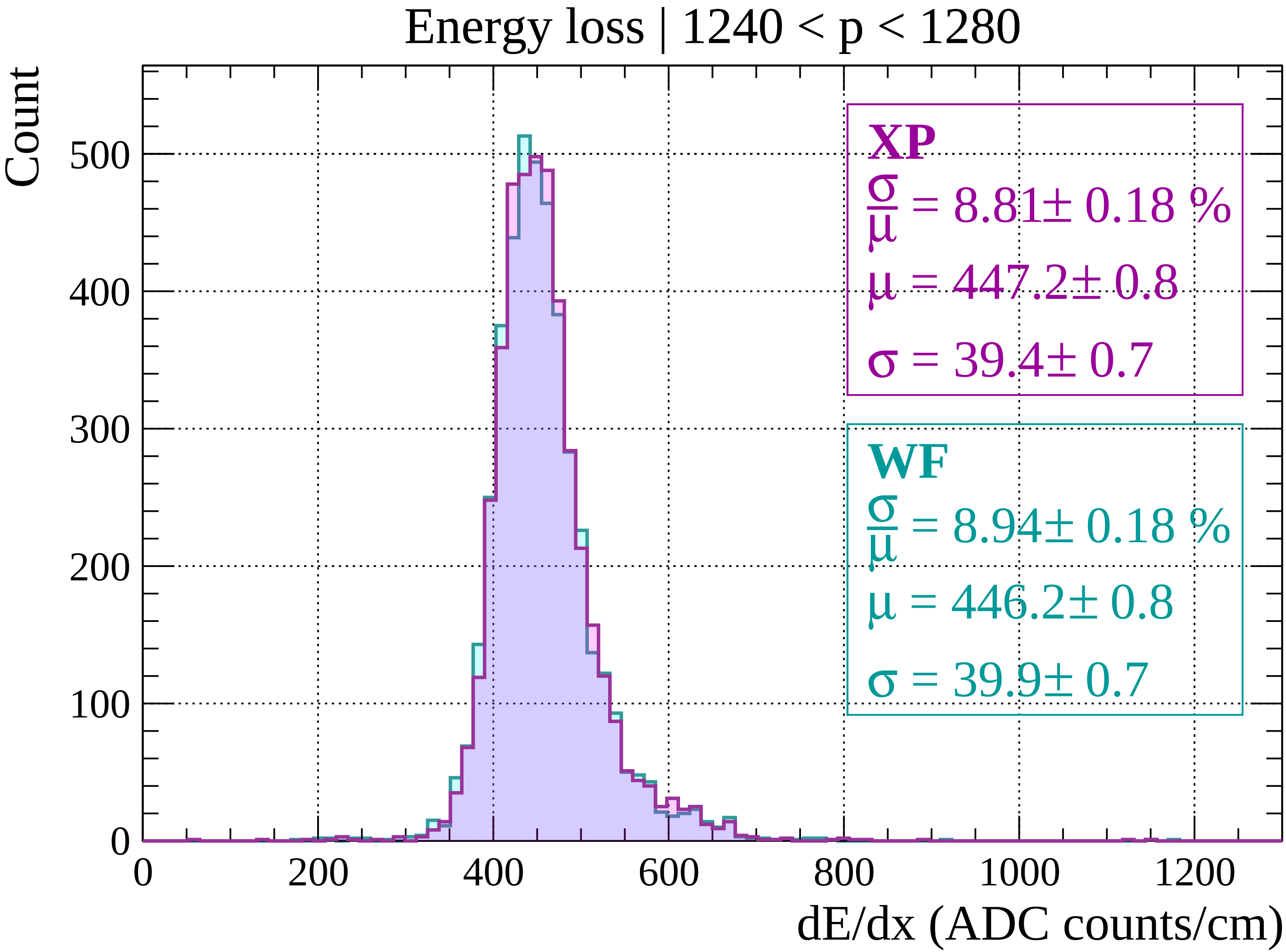


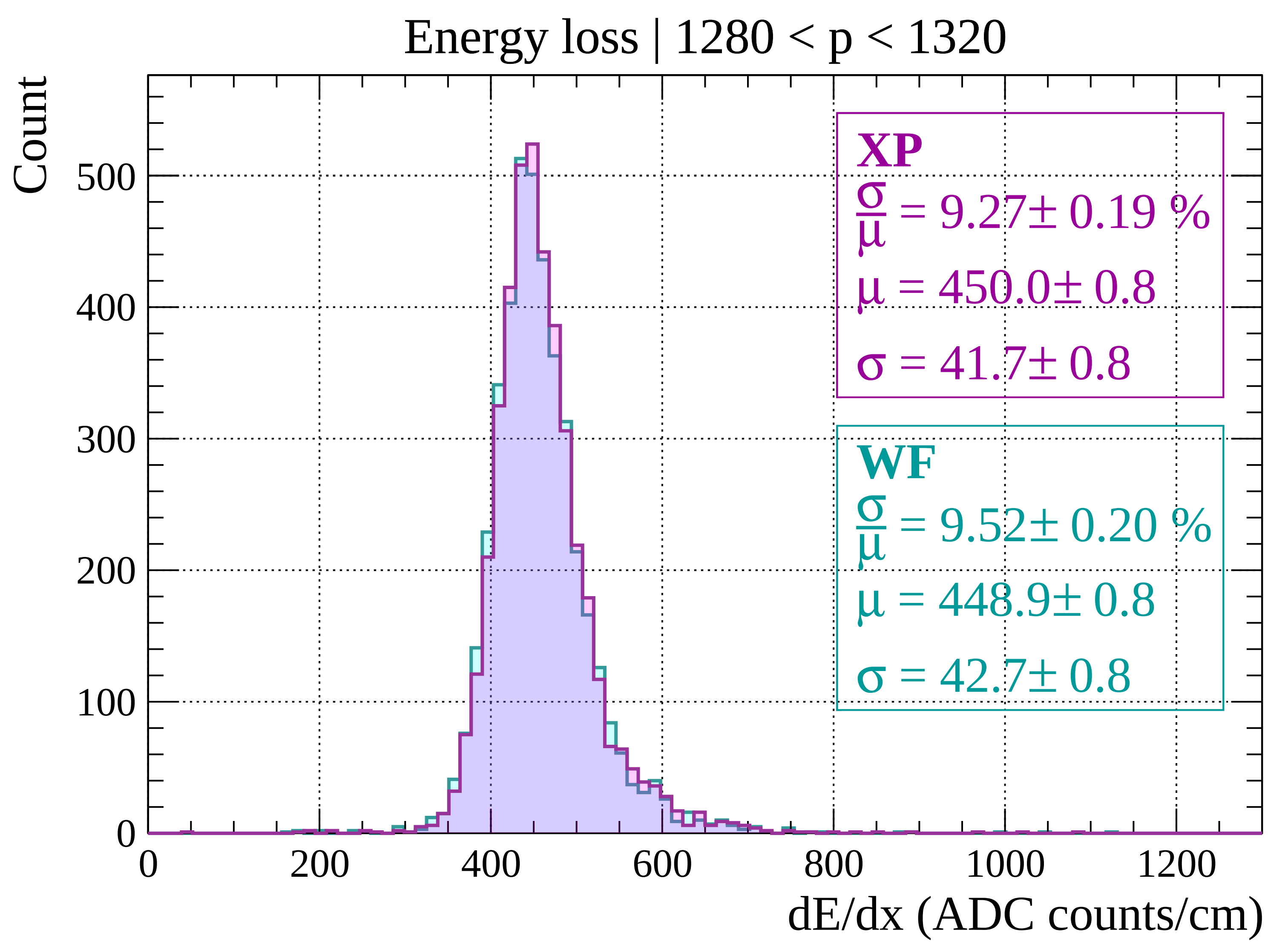
Energy loss | 1120 < p < 1160
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 9.12 \pm 0.18$ %
$\mu = 440.7 \pm 0.7$
$\sigma = 40.2 \pm 0.7$
WF
$\frac{\sigma}{\mu} = 9.32 \pm 0.20$ %
$\mu = 440.0 \pm 0.8$
$\sigma = 41.0 \pm 0.8$



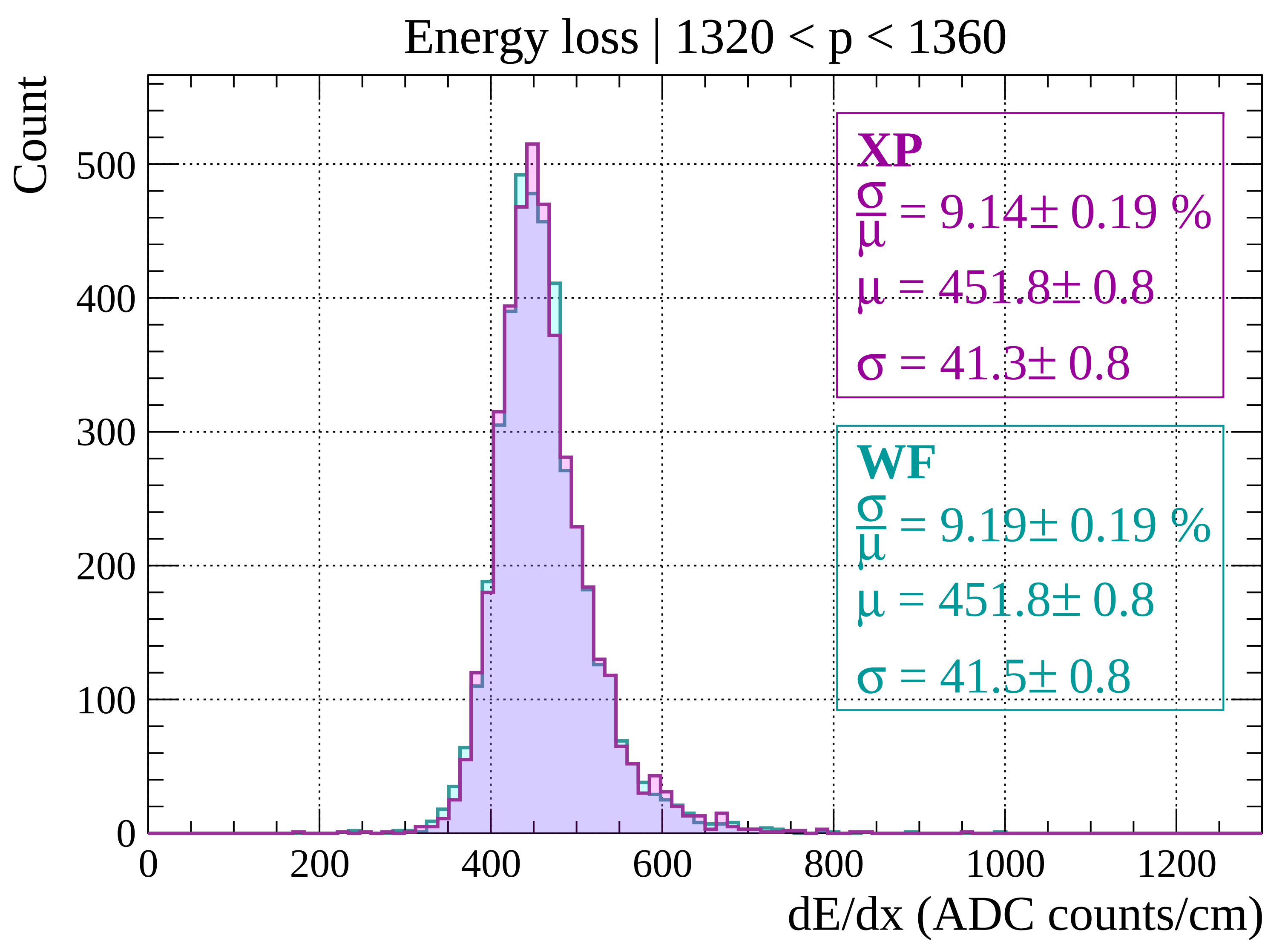
Energy loss | 1160 < p < 1200
Count
600
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.03± 0.19 %
μ = 443.3± 0.8
σ = 40.0± 0.8
WF
σ/μ = 9.16± 0.17 %
μ = 443.1± 0.8
σ = 40.6± 0.7


Energy loss | 1200 < p < 1240
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 8.95± 0.18 %
μ = 445.1± 0.8
σ = 39.8± 0.7
WF
σ/μ = 9.28± 0.20 %
μ = 444.8± 0.8
σ = 41.3± 0.8

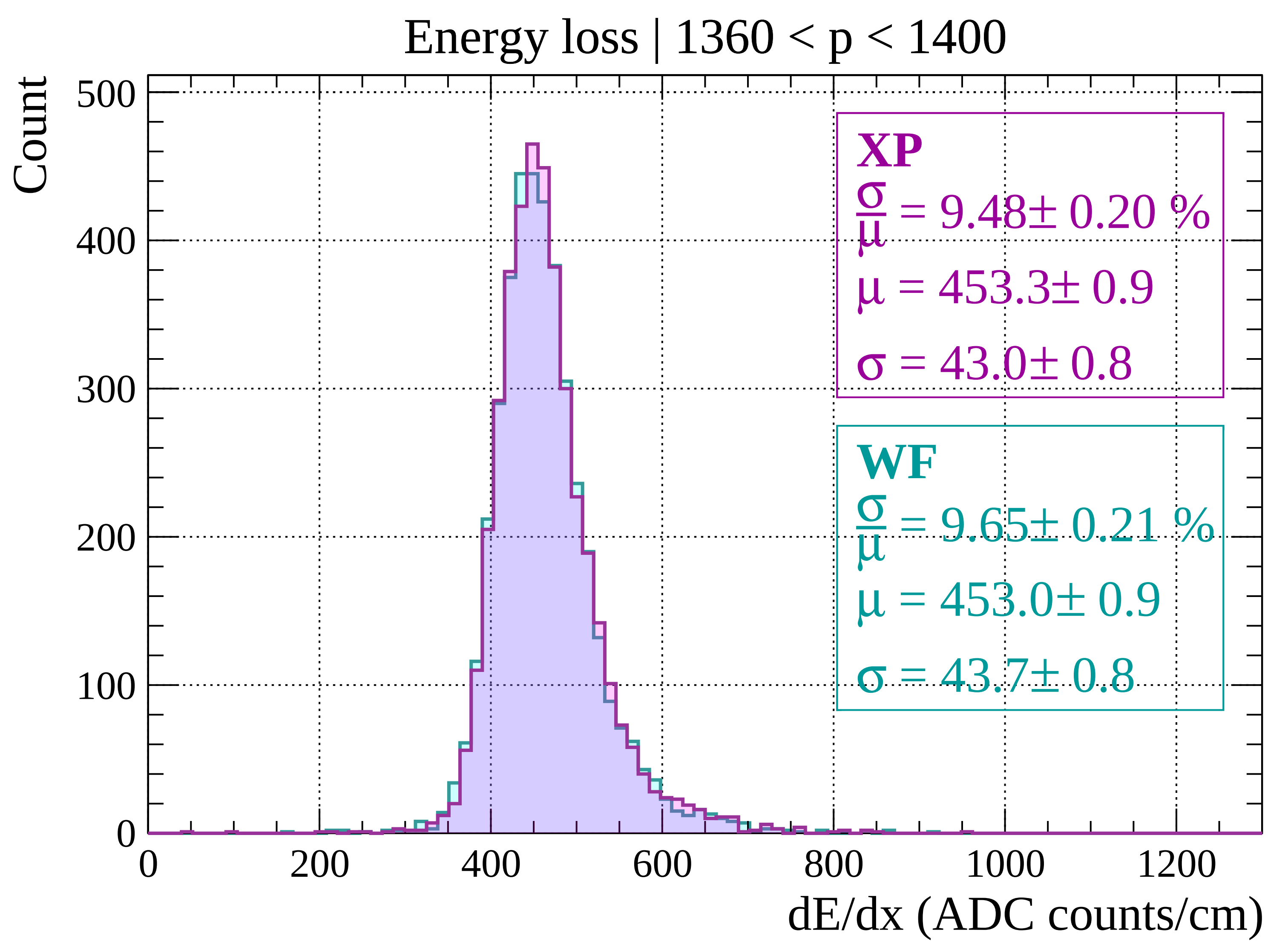

Energy loss | 1240 < p < 1280
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 8.81± 0.18 %
μ = 447.2± 0.8
σ = 39.4± 0.7
WF
σ/μ = 8.94± 0.18 %
μ = 446.2± 0.8
σ = 39.9± 0.7



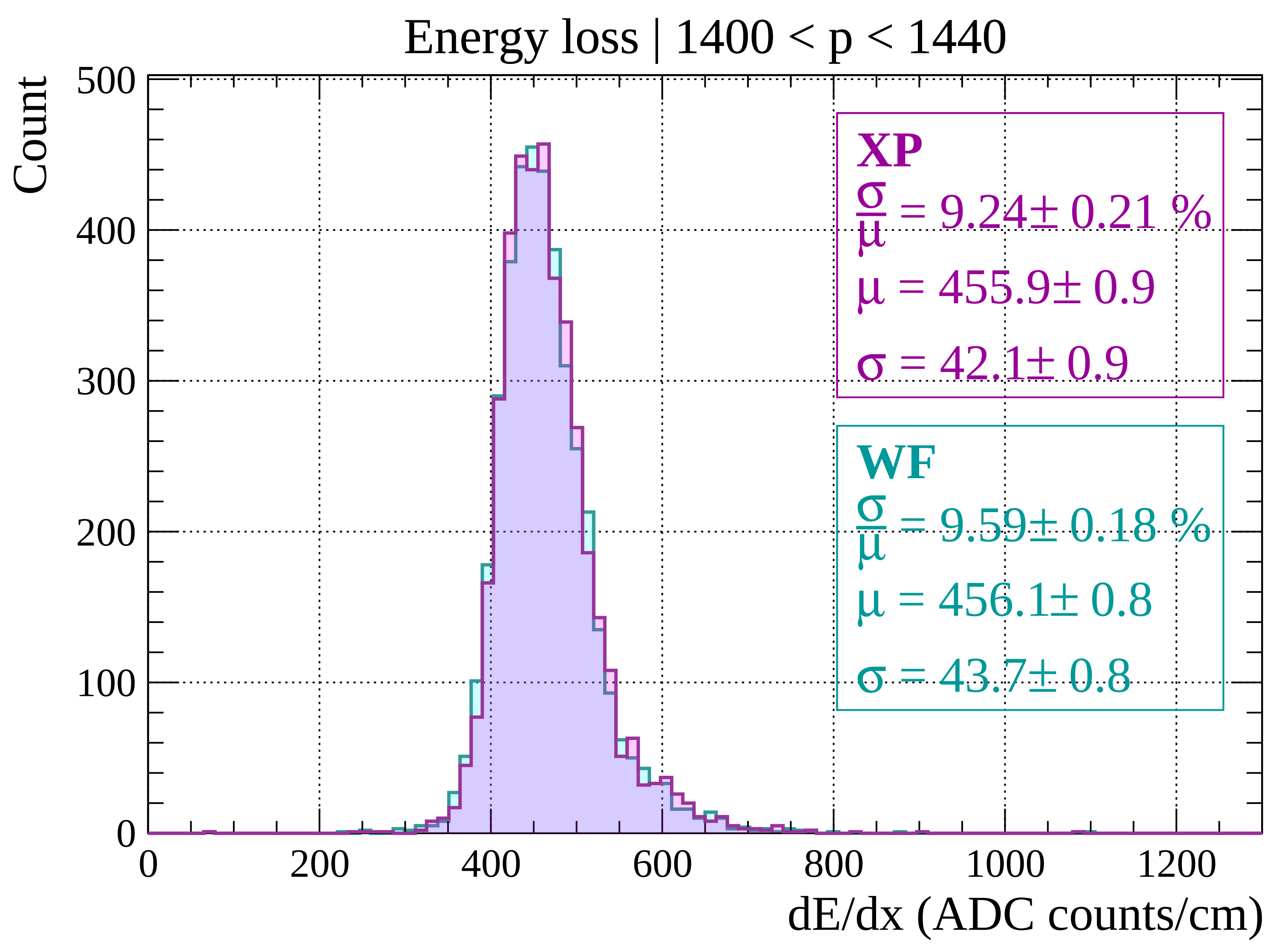
Energy loss | 1280 < p < 1320
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 9.27± 0.19 %
μ = 450.0± 0.8
σ = 41.7± 0.8
WF
σ/μ = 9.52± 0.20 %
μ = 448.9± 0.8
σ = 42.7± 0.8

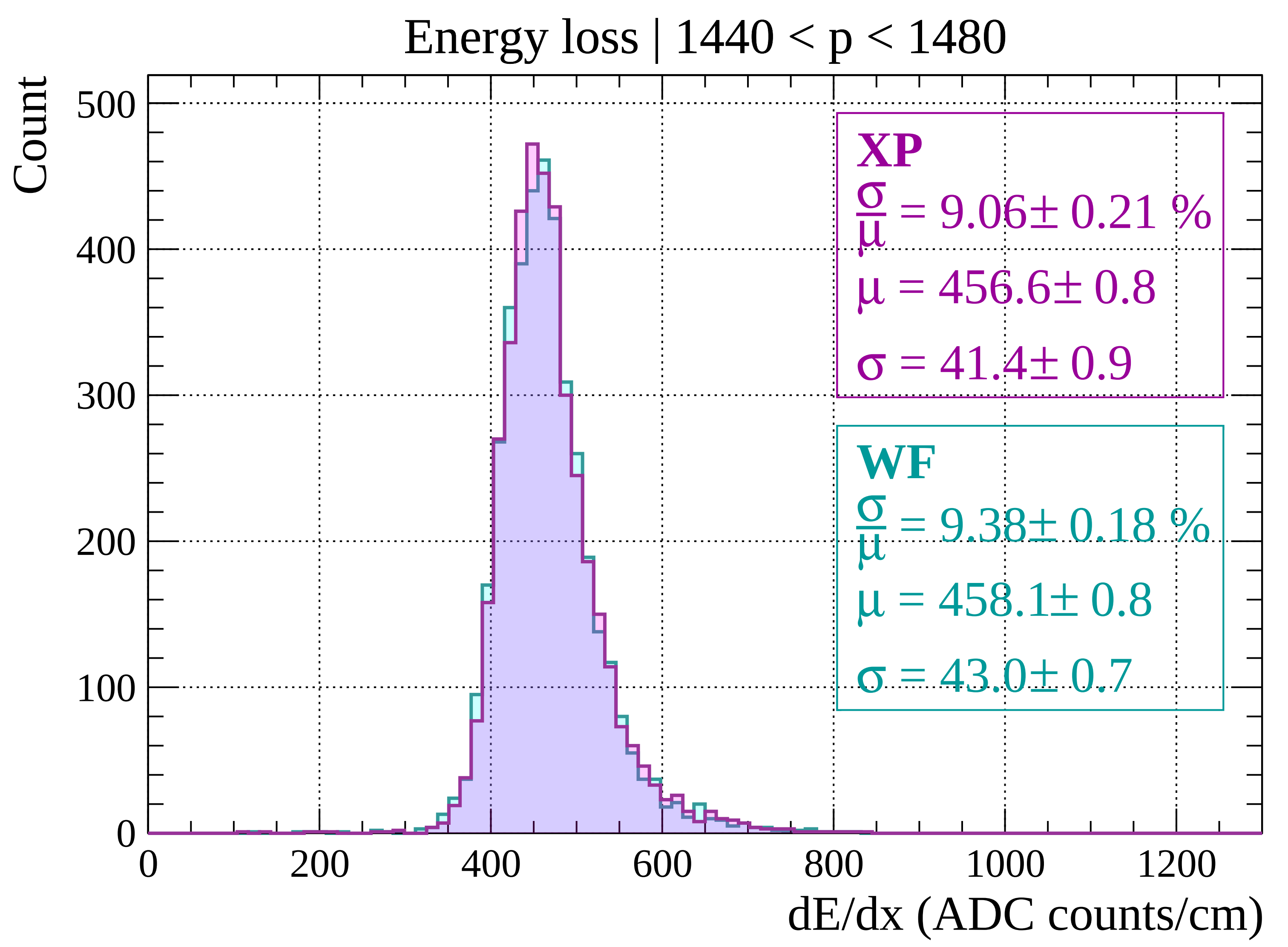


Energy loss | 1320 < p < 1360
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.14± 0.19 %
μ = 451.8± 0.8
σ = 41.3± 0.8
WF
$\frac{\sigma}{\mu}$ = 9.19± 0.19 %
μ = 451.8± 0.8
σ = 41.5± 0.8
0
100
200
300
400
500
0
200
400
600
800
1000
1200

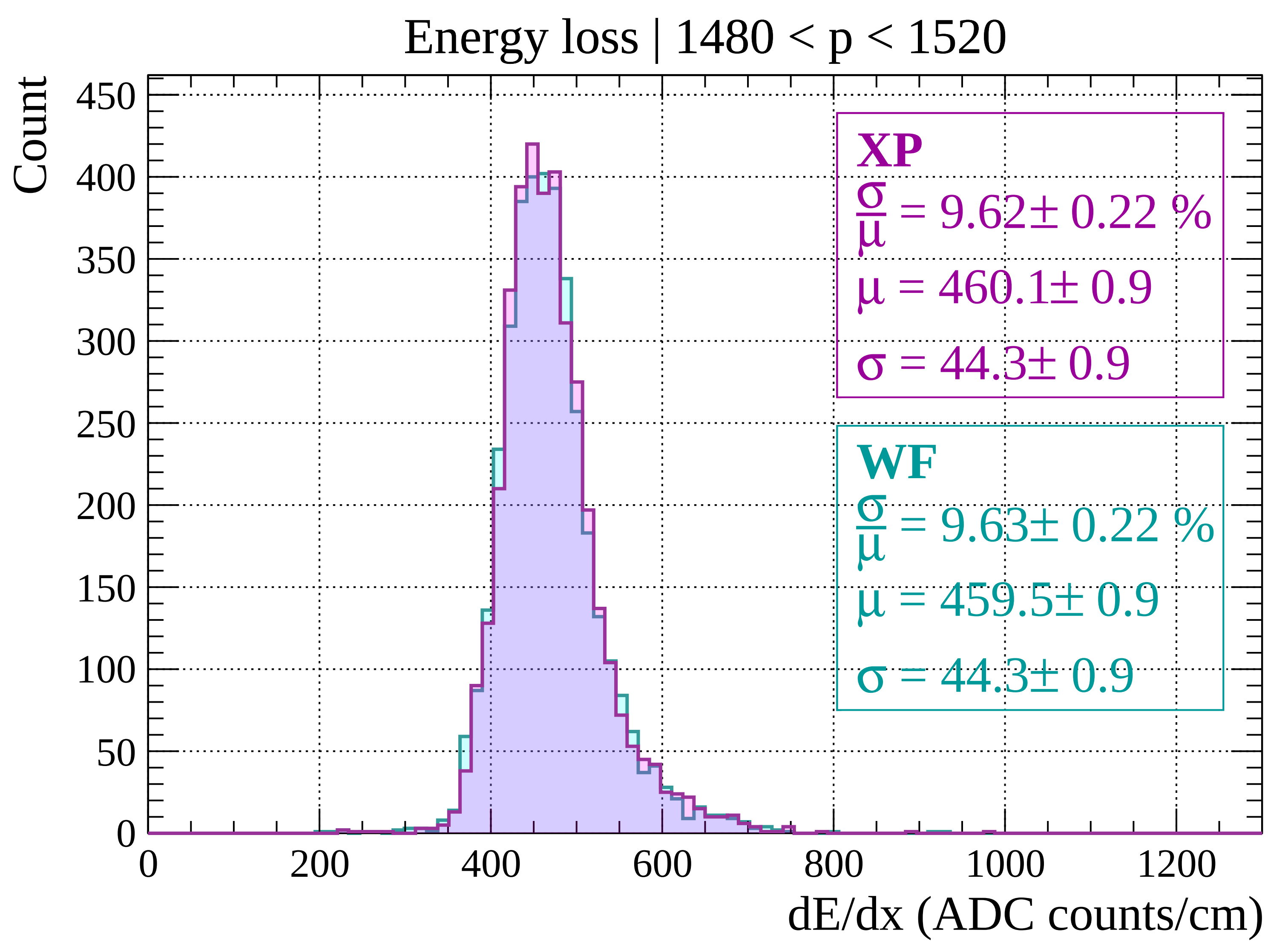


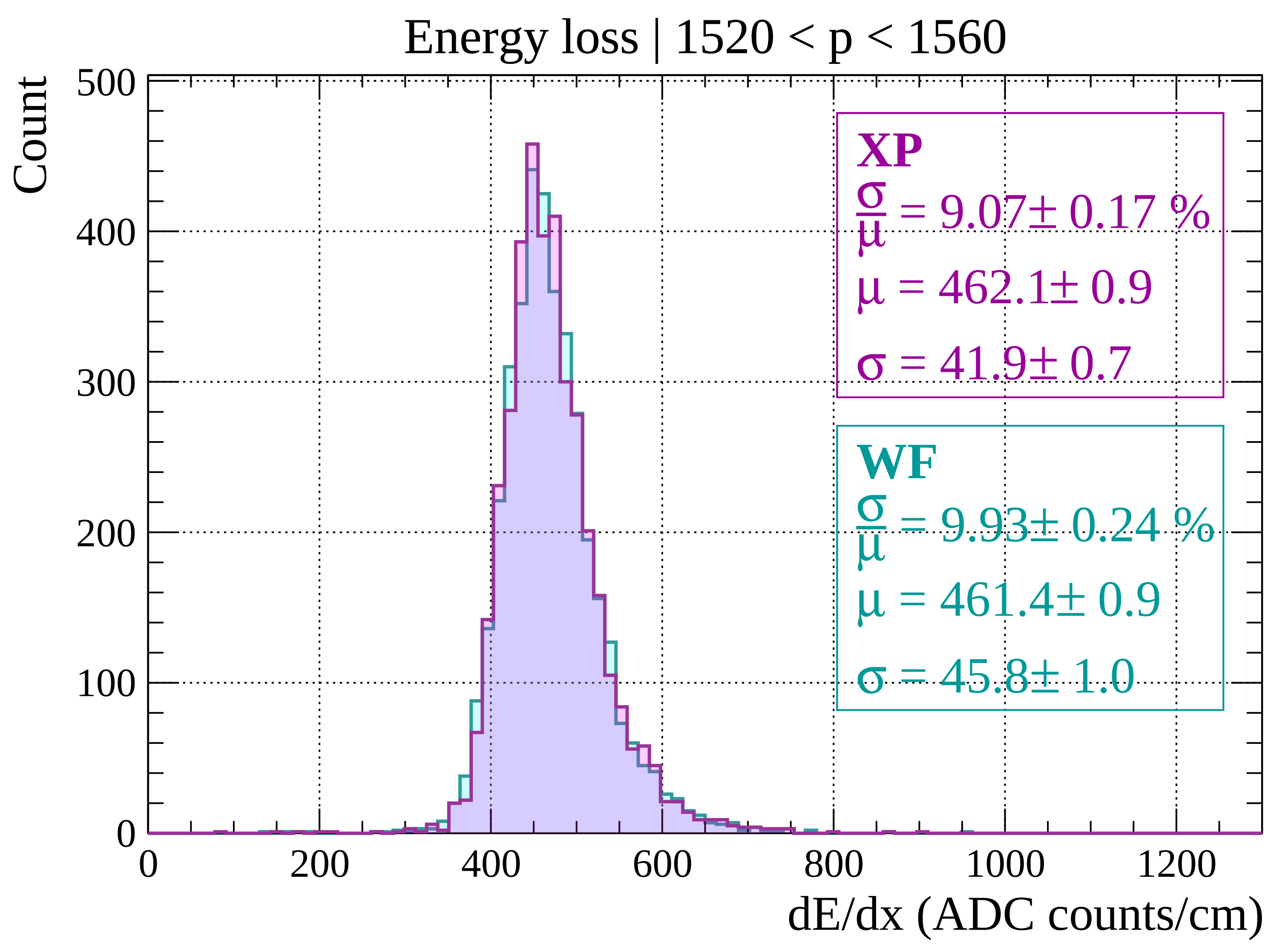
Energy loss | 1360 < p < 1400
Count
0
100
200
300
400
500
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.48± 0.20 %
μ = 453.3± 0.9
σ = 43.0± 0.8
WF
σ/μ = 9.65± 0.21 %
μ = 453.0± 0.9
σ = 43.7± 0.8


Energy loss | 1400 < p < 1440
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 9.24± 0.21 %
μ = 455.9± 0.9
σ = 42.1± 0.9
WF
σ/μ = 9.59± 0.18 %
μ = 456.1± 0.8
σ = 43.7± 0.8

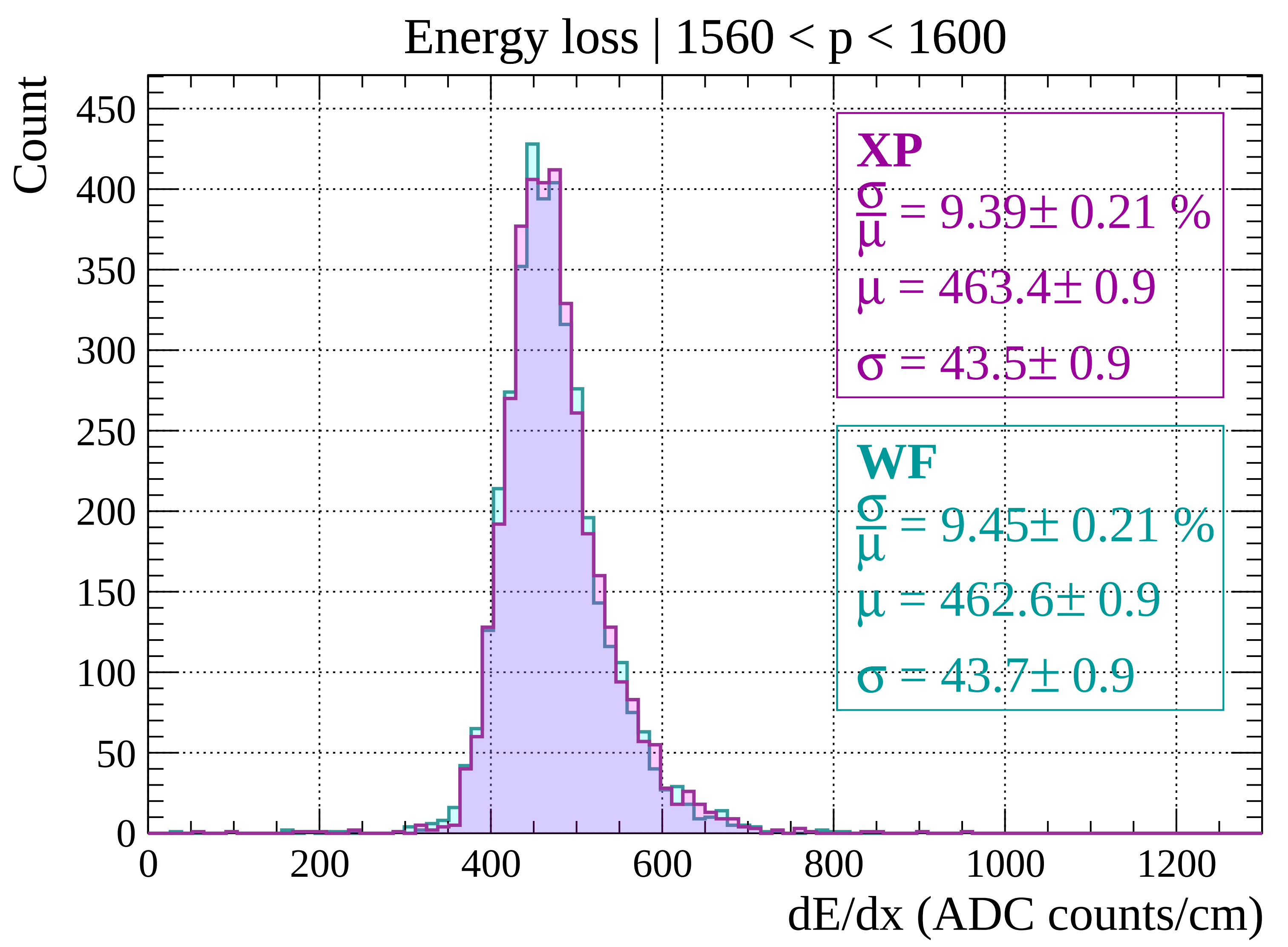


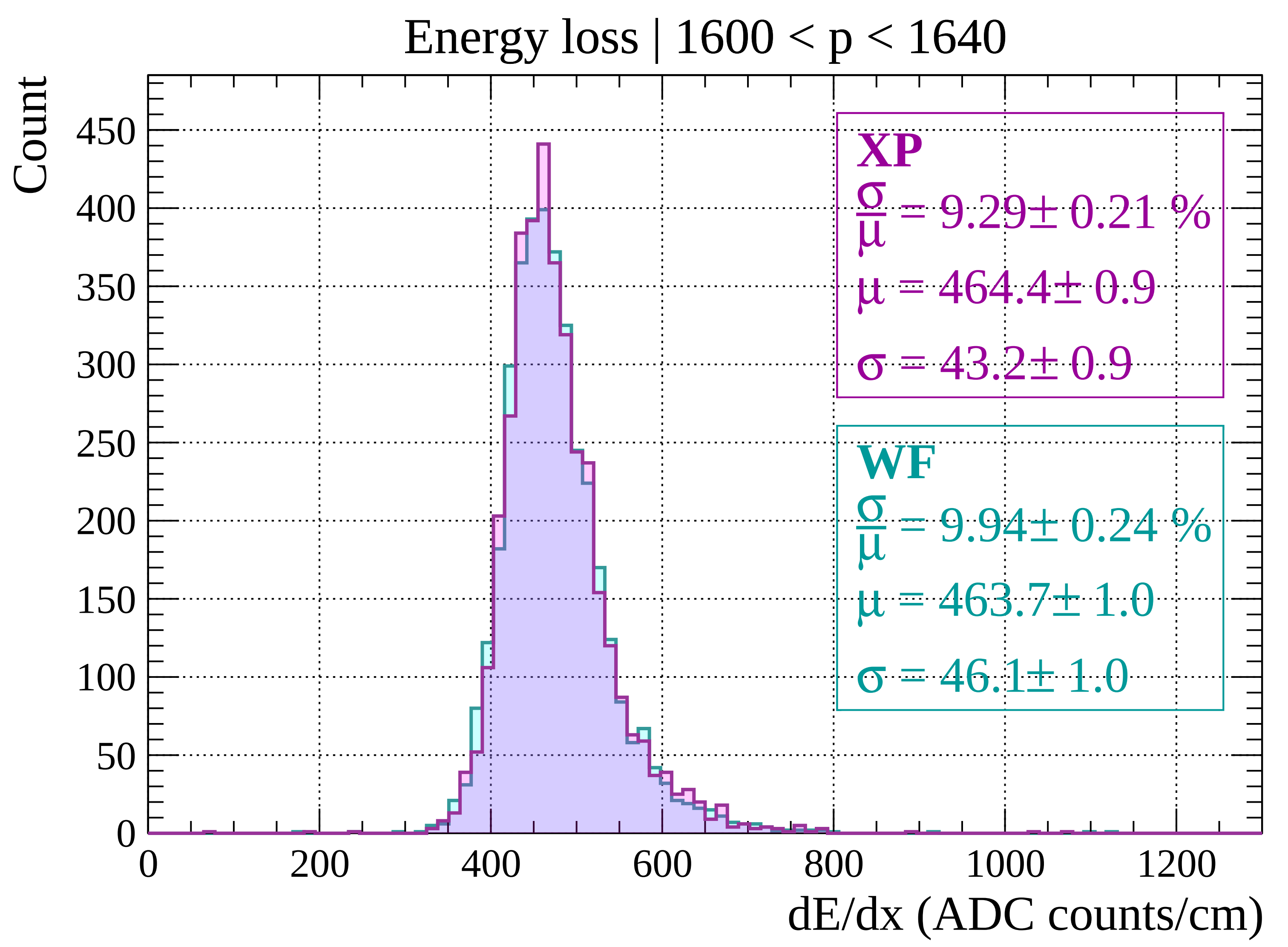
Energy loss | 1440 < p < 1480
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 9.06± 0.21 %
μ = 456.6± 0.8
σ = 41.4± 0.9
WF
σ/μ = 9.38± 0.18 %
μ = 458.1± 0.8
σ = 43.0± 0.7


Energy loss | 1480 < p < 1520
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
σ/μ = 9.62± 0.22 %
μ = 460.1± 0.9
σ = 44.3± 0.9
WF
σ/μ = 9.63± 0.22 %
μ = 459.5± 0.9
σ = 44.3± 0.9

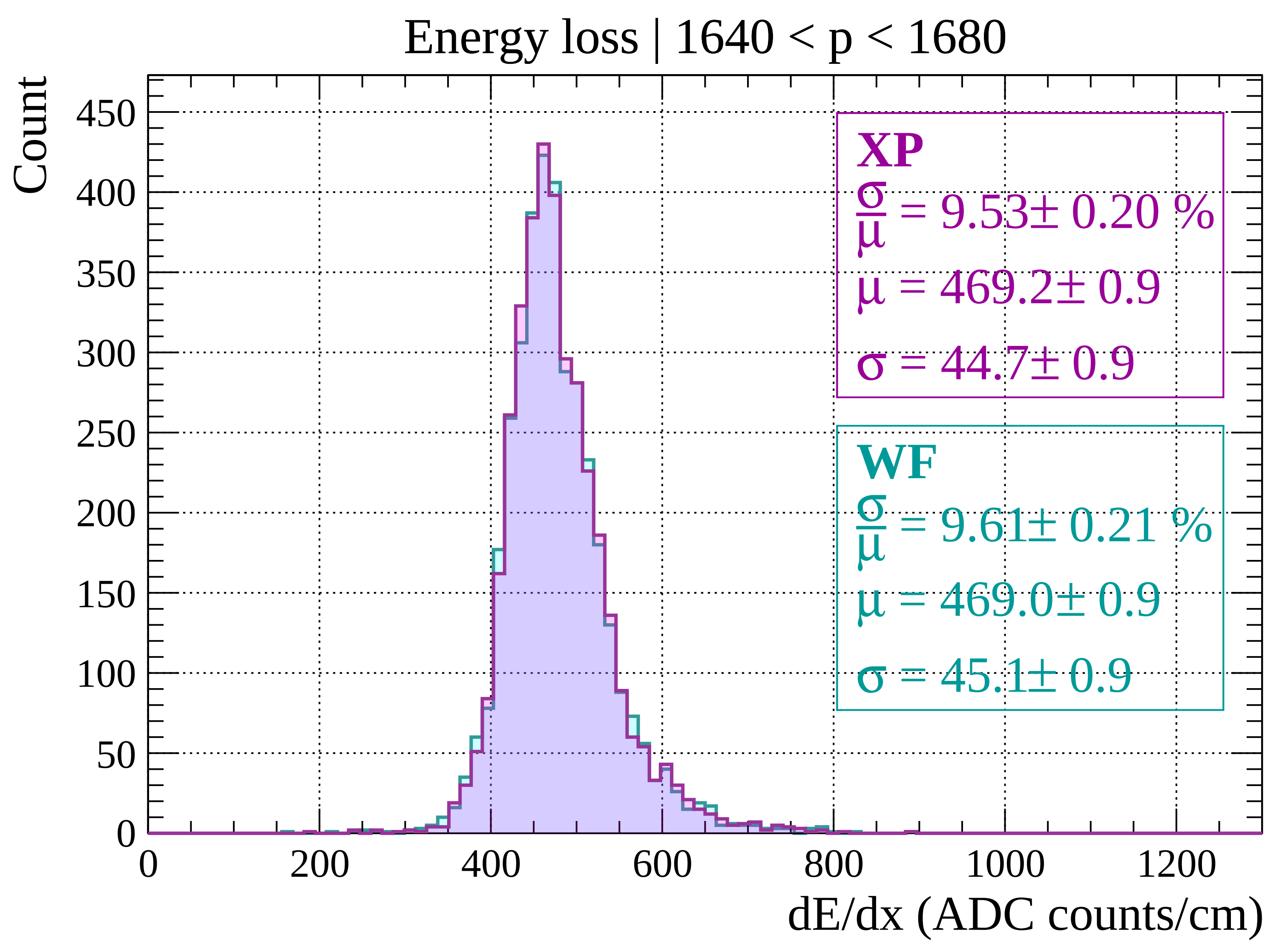


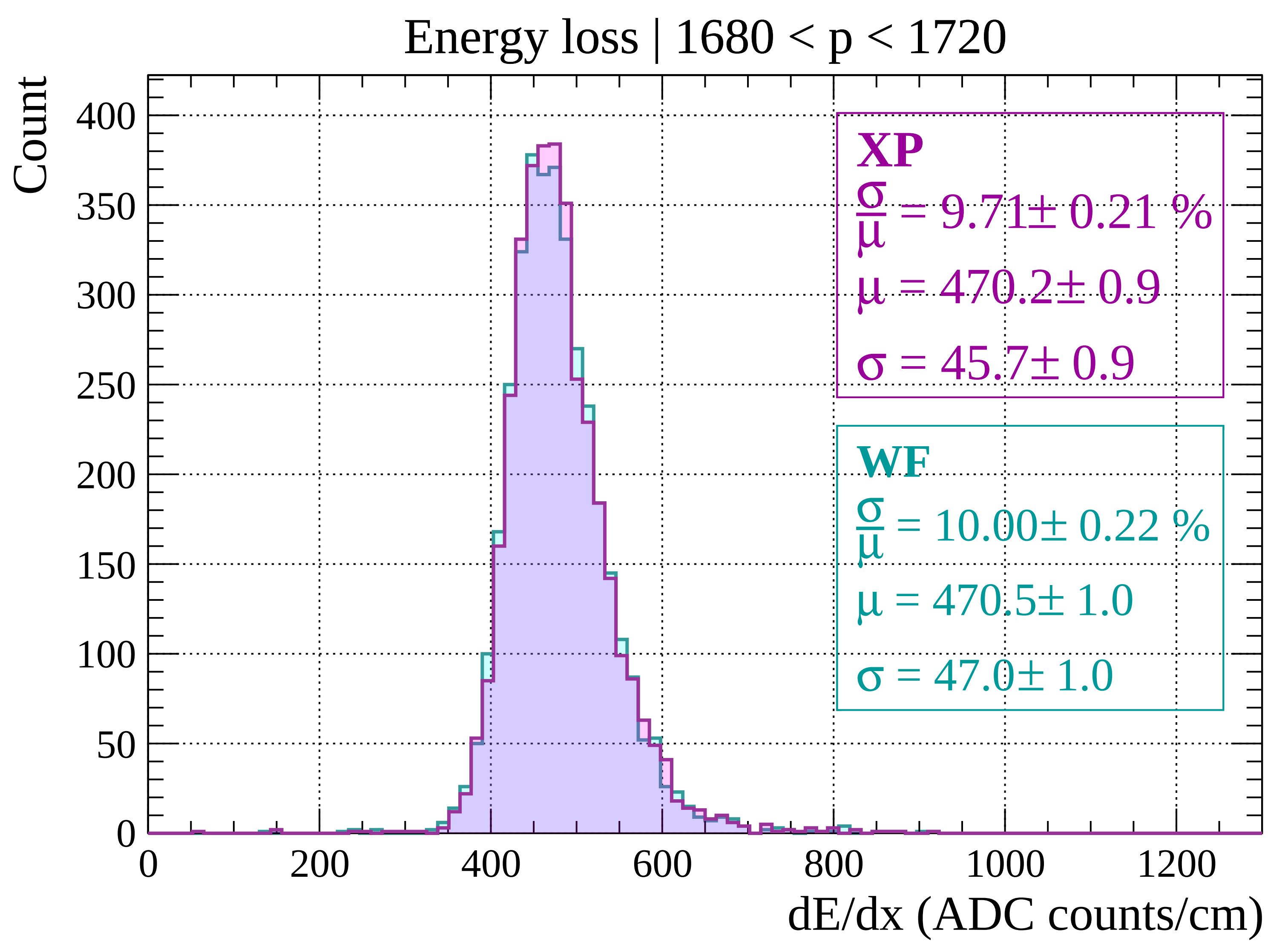
Energy loss | 1520 < p < 1560
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 9.07± 0.17 %
μ = 462.1± 0.9
σ = 41.9± 0.7
WF
$\frac{\sigma}{\mu}$ = 9.93± 0.24 %
μ = 461.4± 0.9
σ = 45.8± 1.0


Energy loss | 1560 < p < 1600
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 9.39± 0.21 %
μ = 463.4± 0.9
σ = 43.5± 0.9
WF
$\frac{\sigma}{\mu}$ = 9.45± 0.21 %
μ = 462.6± 0.9
σ = 43.7± 0.9

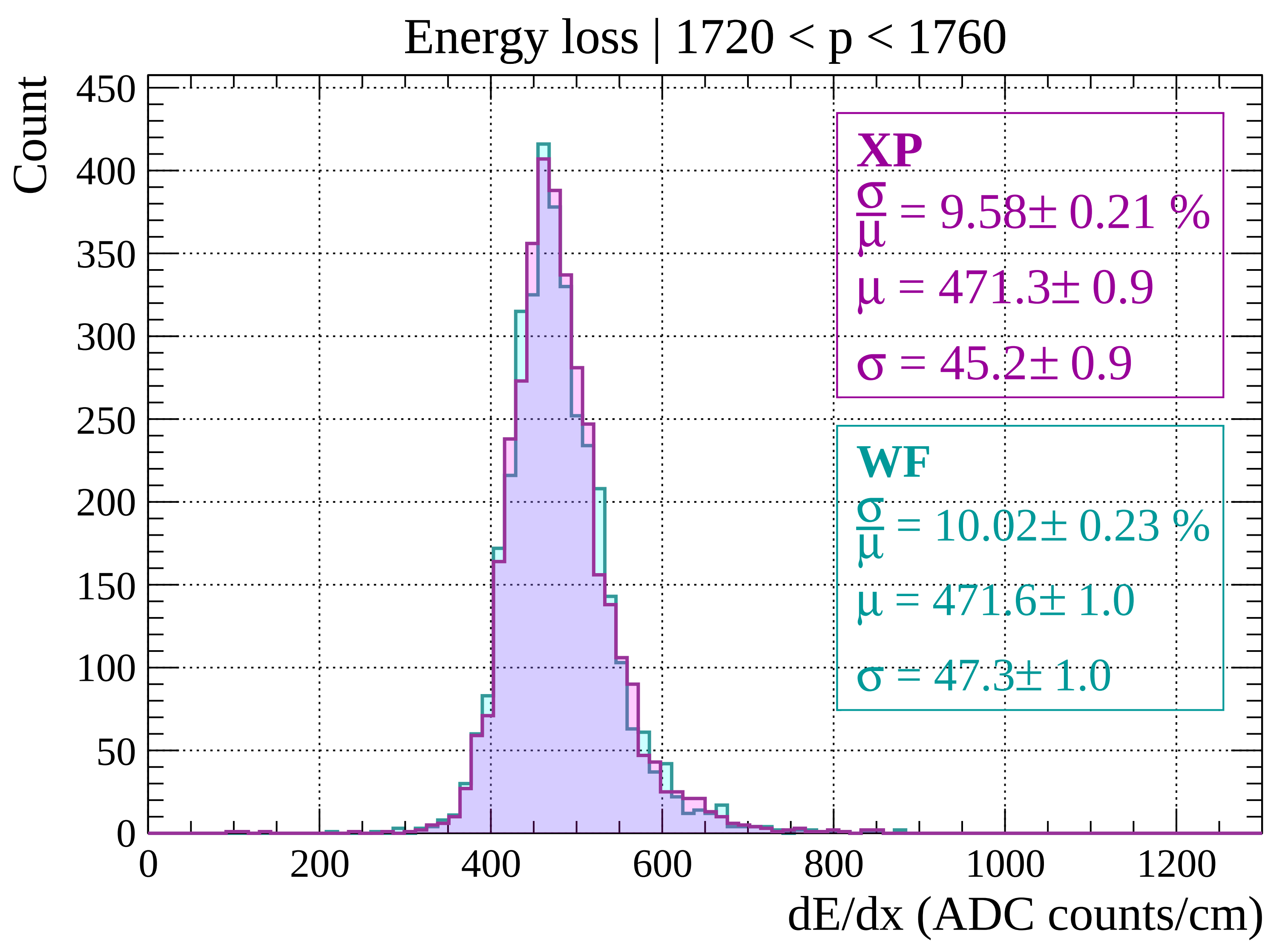


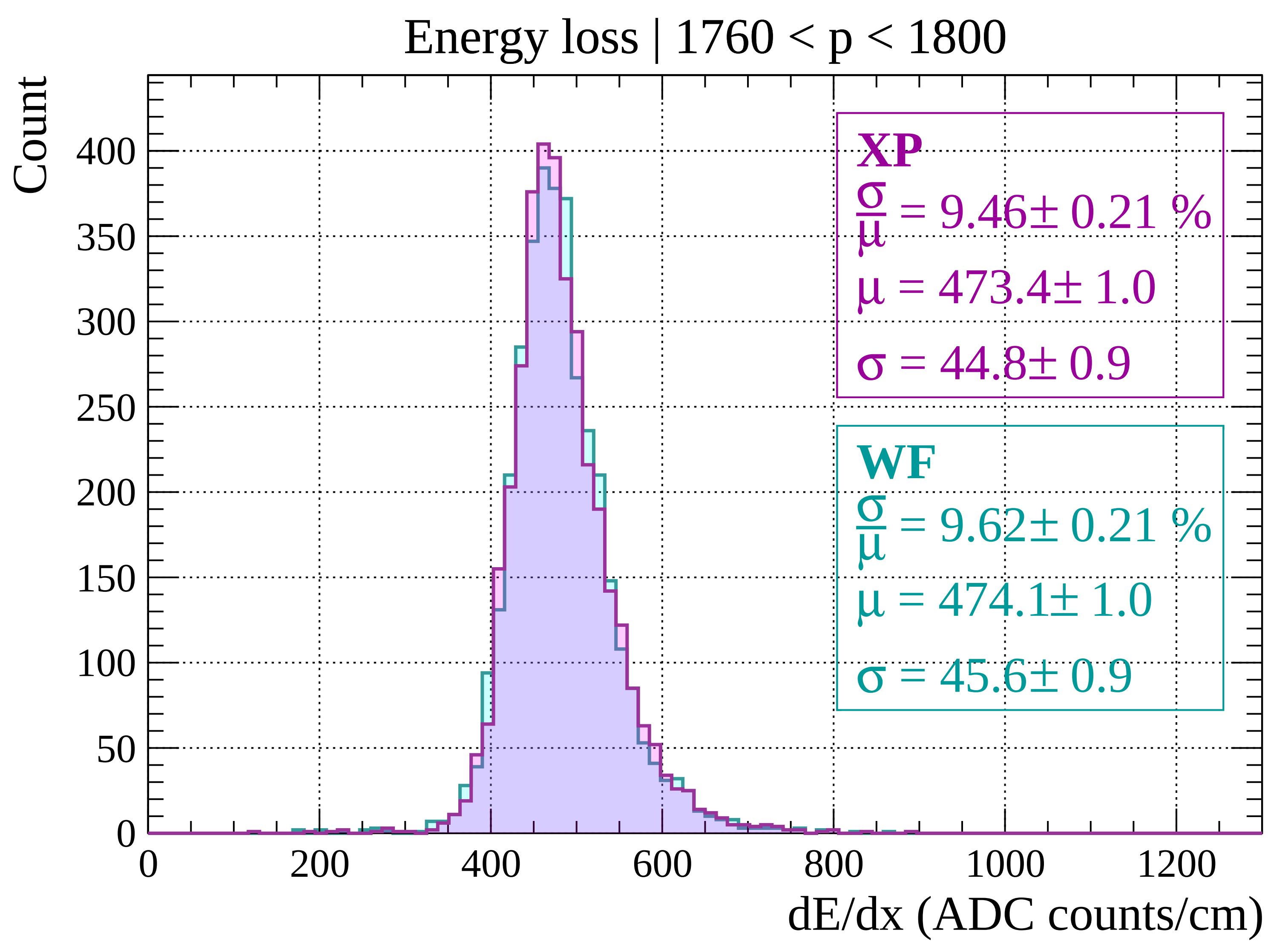
Energy loss | 1600 < p < 1640
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu} = 9.29 \pm 0.21$ %
$\mu = 464.4 \pm 0.9$
$\sigma = 43.2 \pm 0.9$
WF
$\frac{\sigma}{\mu} = 9.94 \pm 0.24$ %
$\mu = 463.7 \pm 1.0$
$\sigma = 46.1 \pm 1.0$
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200


Energy loss | 1640 < p < 1680
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
σ/μ = 9.53± 0.20 %
μ = 469.2± 0.9
σ = 44.7± 0.9
WF
σ/μ = 9.61± 0.21 %
μ = 469.0± 0.9
σ = 45.1± 0.9

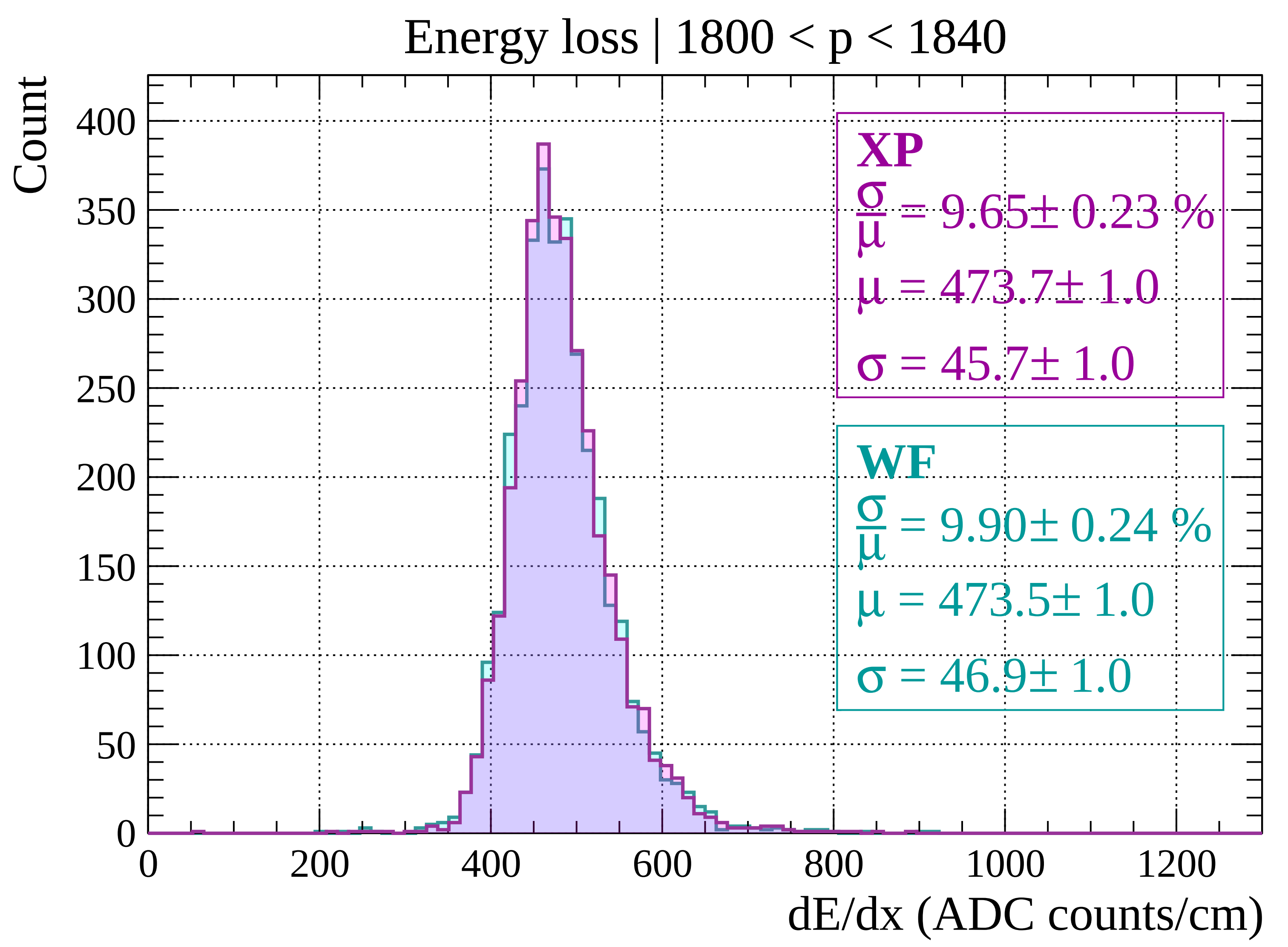


Energy loss | 1680 < p < 1720
Count
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.71± 0.21 %
μ = 470.2± 0.9
σ = 45.7± 0.9
WF
σ/μ = 10.00± 0.22 %
μ = 470.5± 1.0
σ = 47.0± 1.0

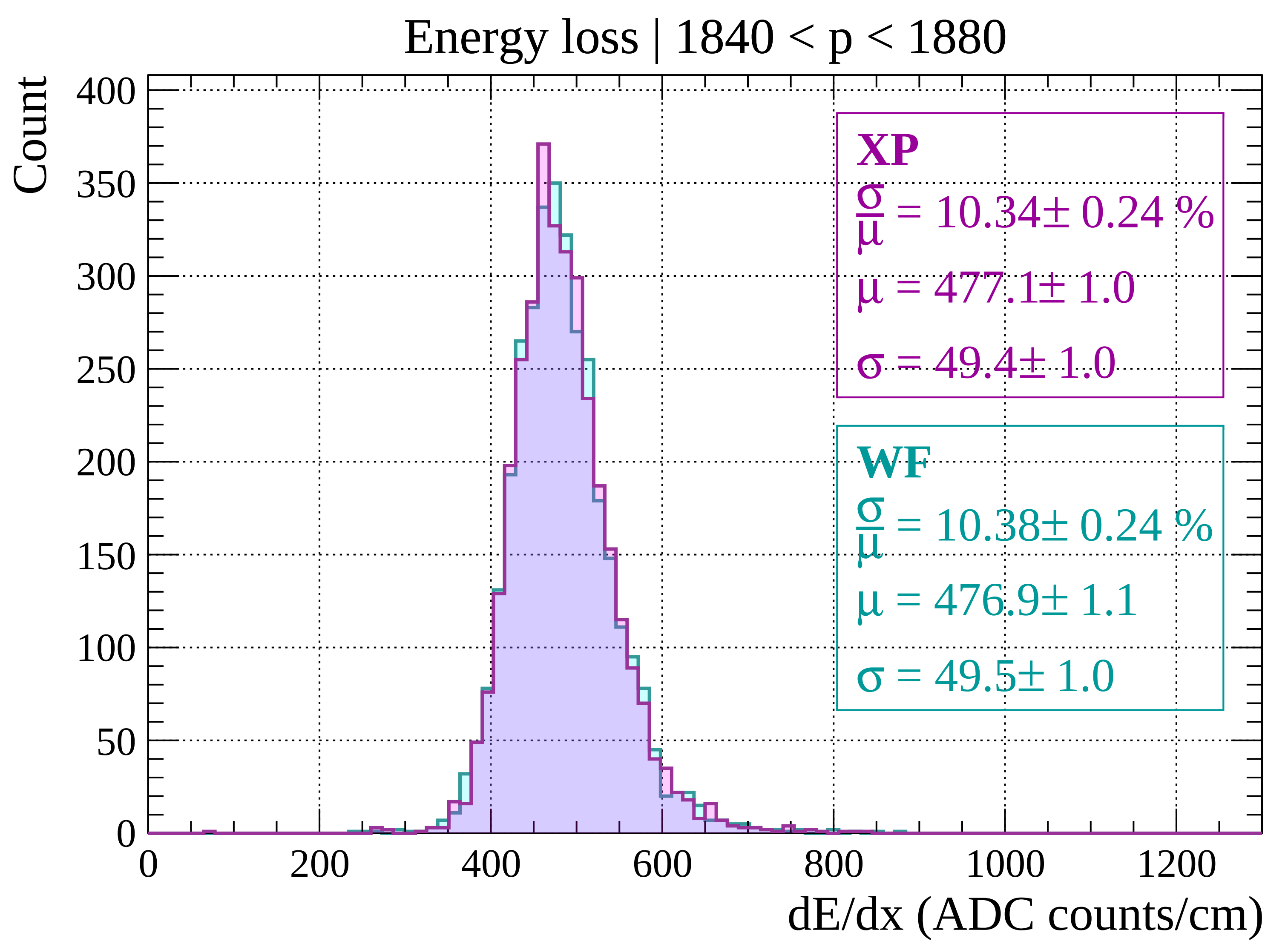

Energy loss | 1720 < p < 1760
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
350
400
450
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu} = 9.58 \pm 0.21\ \%$
$\mu = 471.3 \pm 0.9$
$\sigma = 45.2 \pm 0.9$
WF
$\frac{\sigma}{\mu} = 10.02 \pm 0.23\ \%$
$\mu = 471.6 \pm 1.0$
$\sigma = 47.3 \pm 1.0$



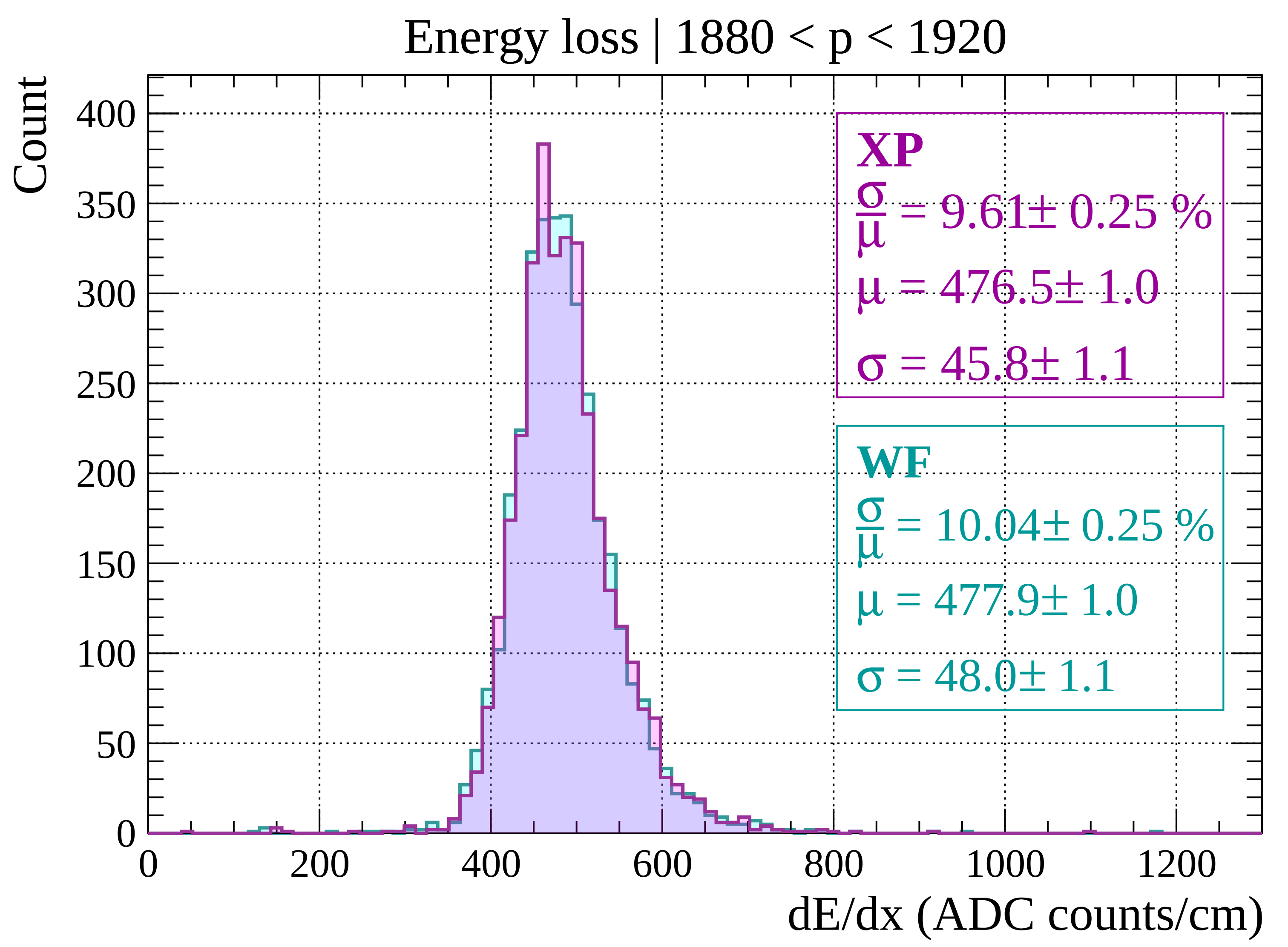
Energy loss | 1760 < p < 1800
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
XP
$\frac{\sigma}{\mu}$ = 9.46± 0.21 %
μ = 473.4± 1.0
σ = 44.8± 0.9
WF
$\frac{\sigma}{\mu}$ = 9.62± 0.21 %
μ = 474.1± 1.0
σ = 45.6± 0.9



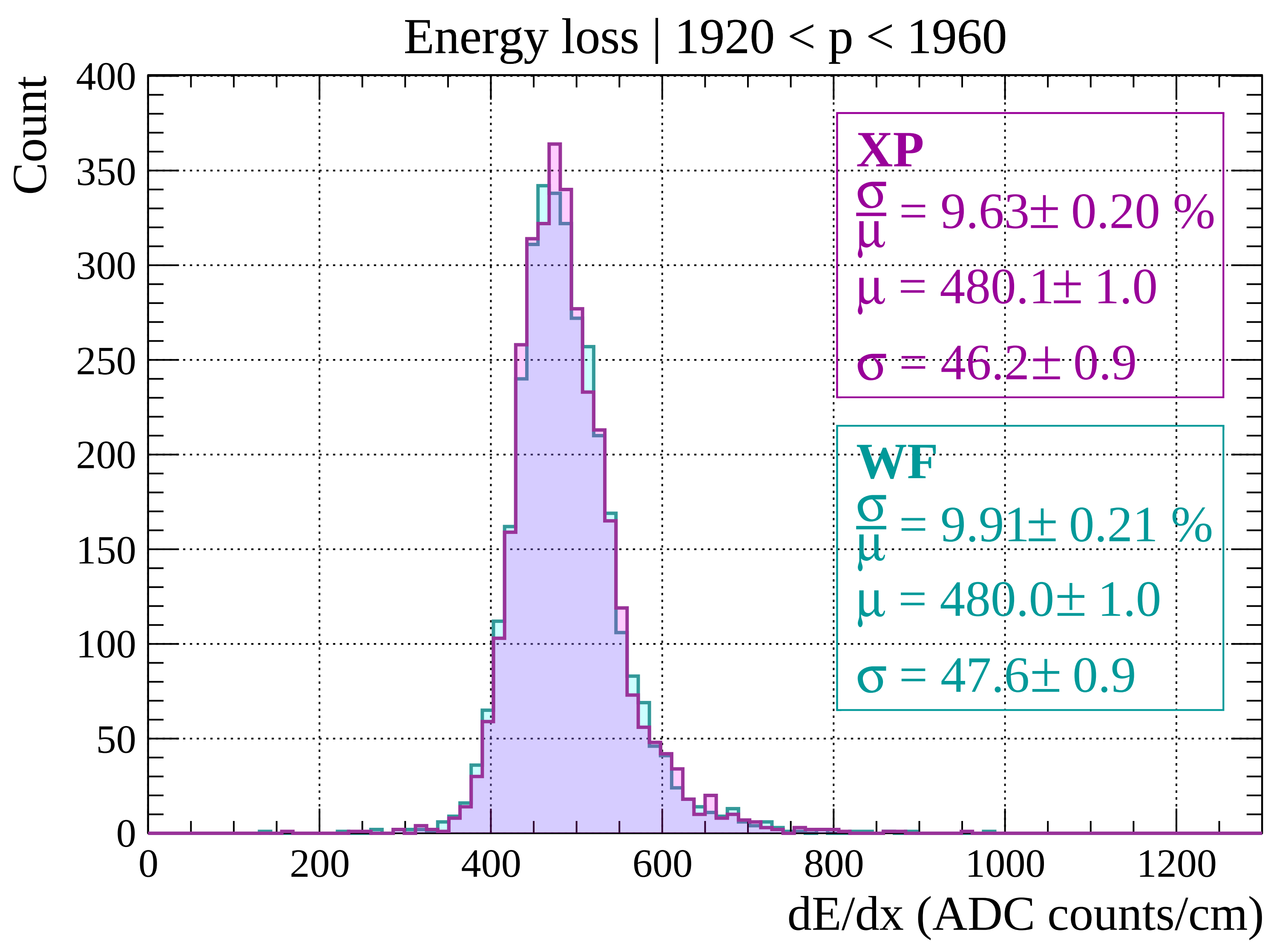
Energy loss | 1800 < p < 1840
Count
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.65± 0.23 %
μ = 473.7± 1.0
σ = 45.7± 1.0
WF
σ/μ = 9.90± 0.24 %
μ = 473.5± 1.0
σ = 46.9± 1.0



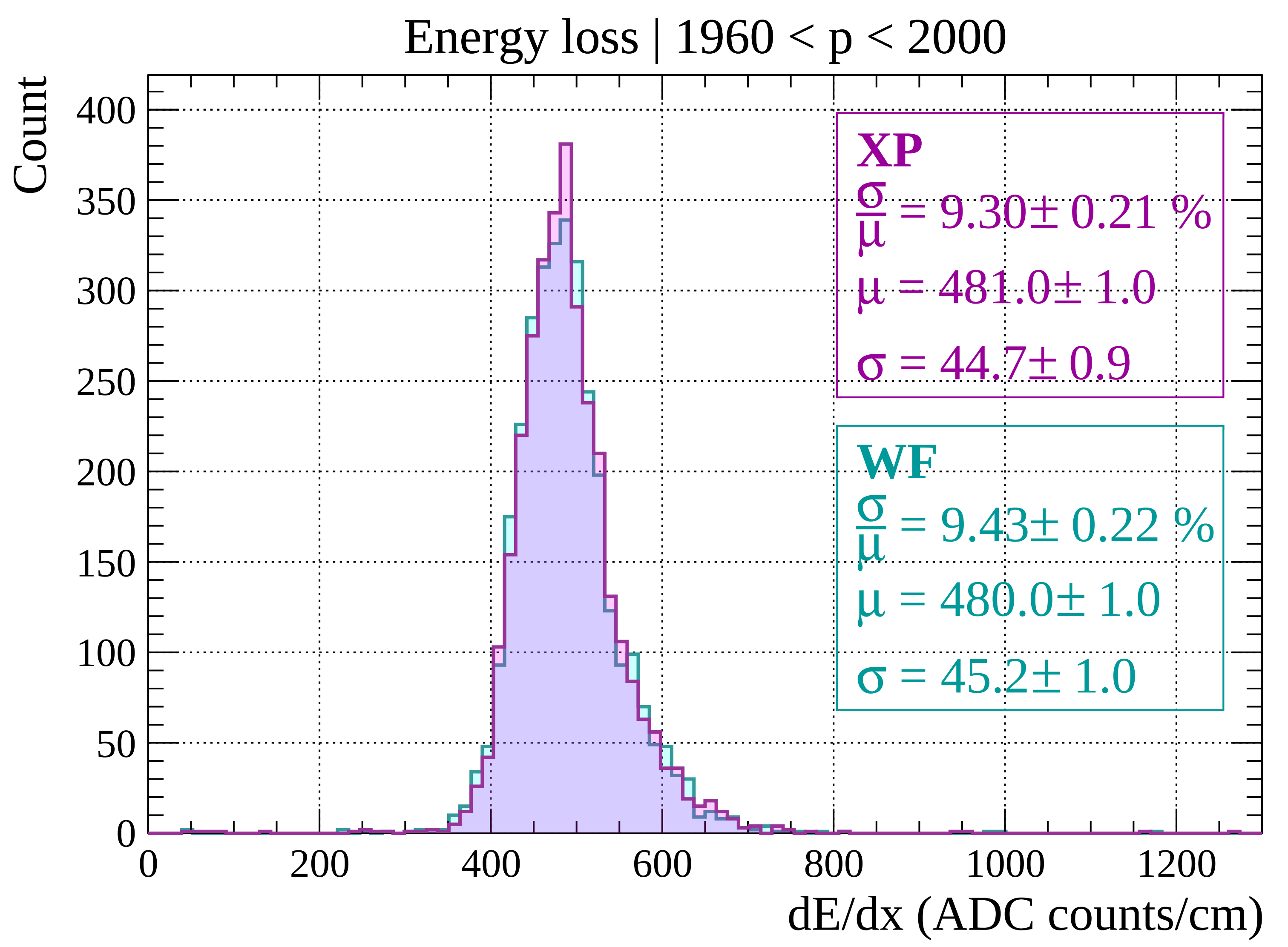
Energy loss | 1840 < p < 1880
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
50
100
150
200
250
300
350
400
XP
σ/μ = 10.34± 0.24 %
μ = 477.1± 1.0
σ = 49.4± 1.0
WF
σ/μ = 10.38± 0.24 %
μ = 476.9± 1.1
σ = 49.5± 1.0



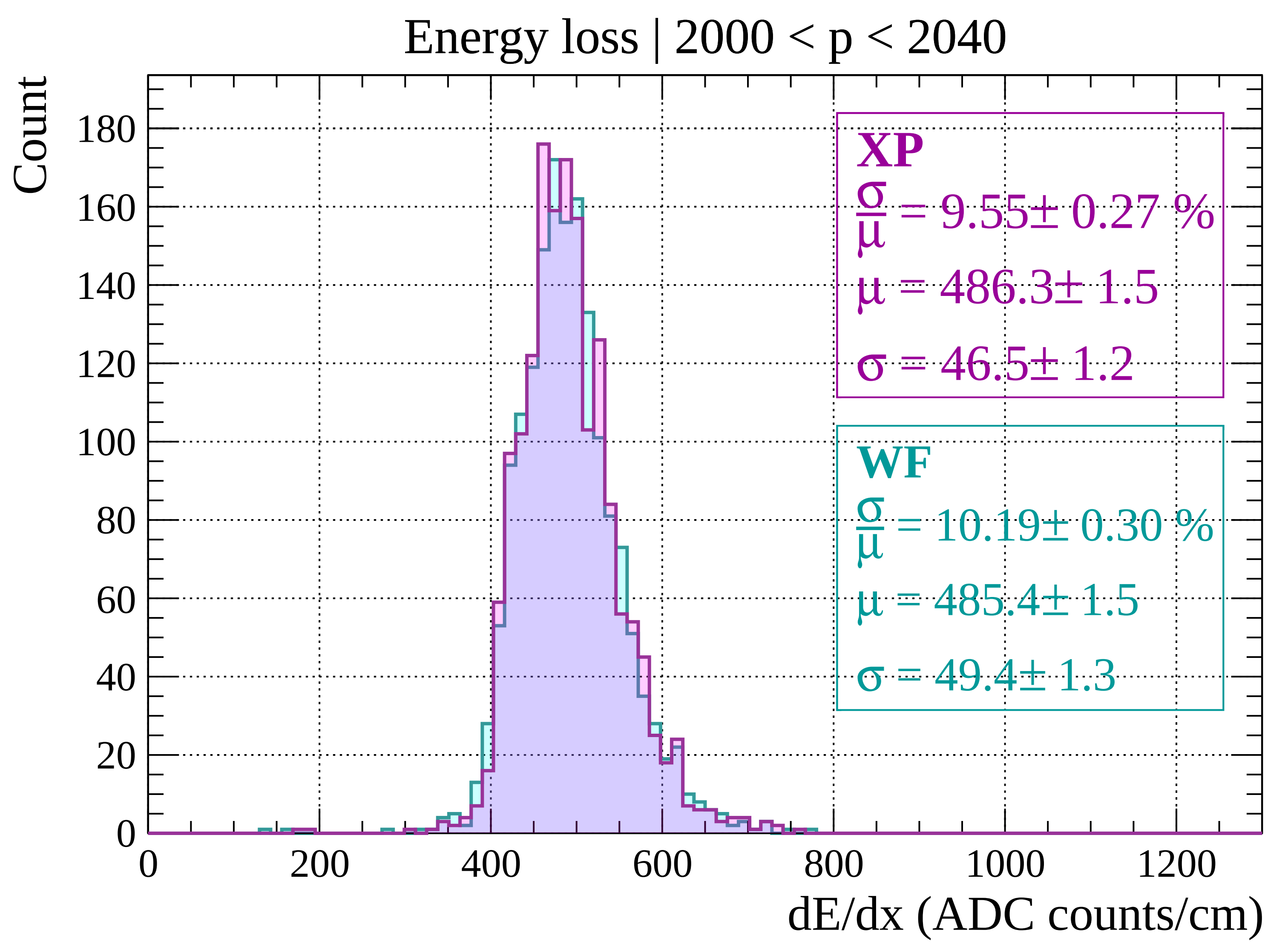
Energy loss | 1880 < p < 1920
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.61± 0.25 %
μ = 476.5± 1.0
σ = 45.8± 1.1
WF
σ/μ = 10.04± 0.25 %
μ = 477.9± 1.0
σ = 48.0± 1.1



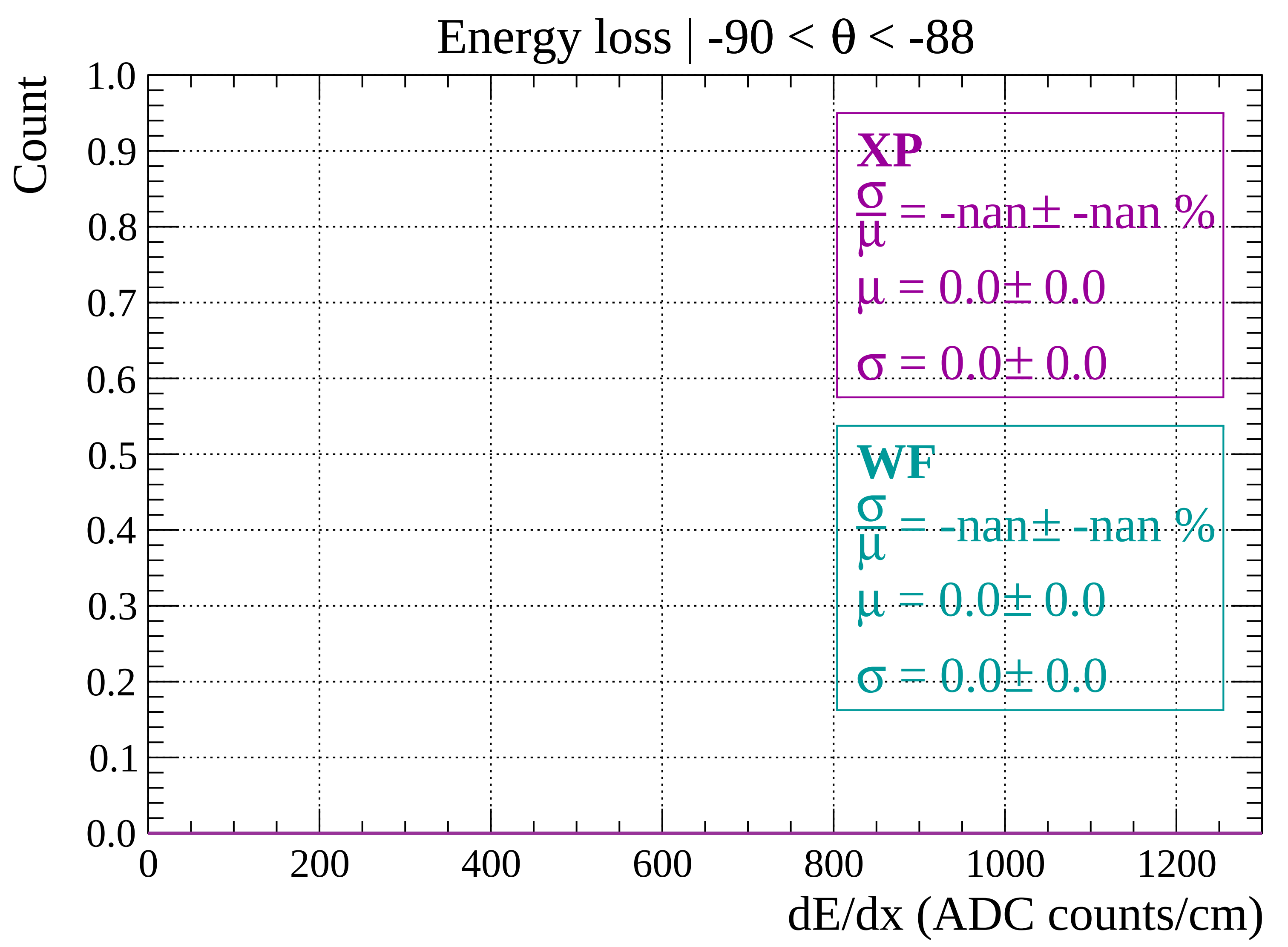
Energy loss | 1920 < p < 1960
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 9.63± 0.20 %
μ = 480.1± 1.0
σ = 46.2± 0.9
WF
σ/μ = 9.91± 0.21 %
μ = 480.0± 1.0
σ = 47.6± 0.9
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200

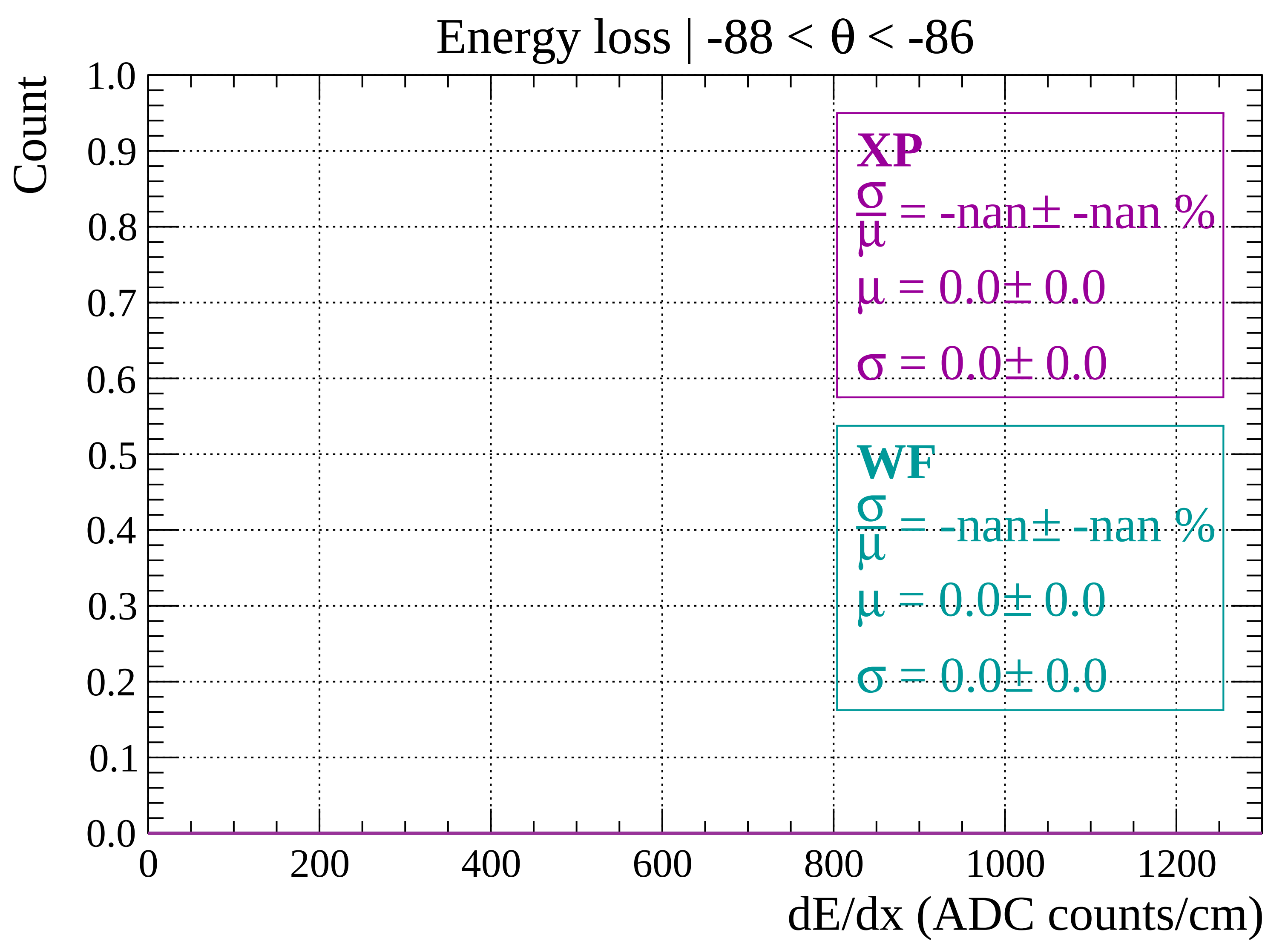


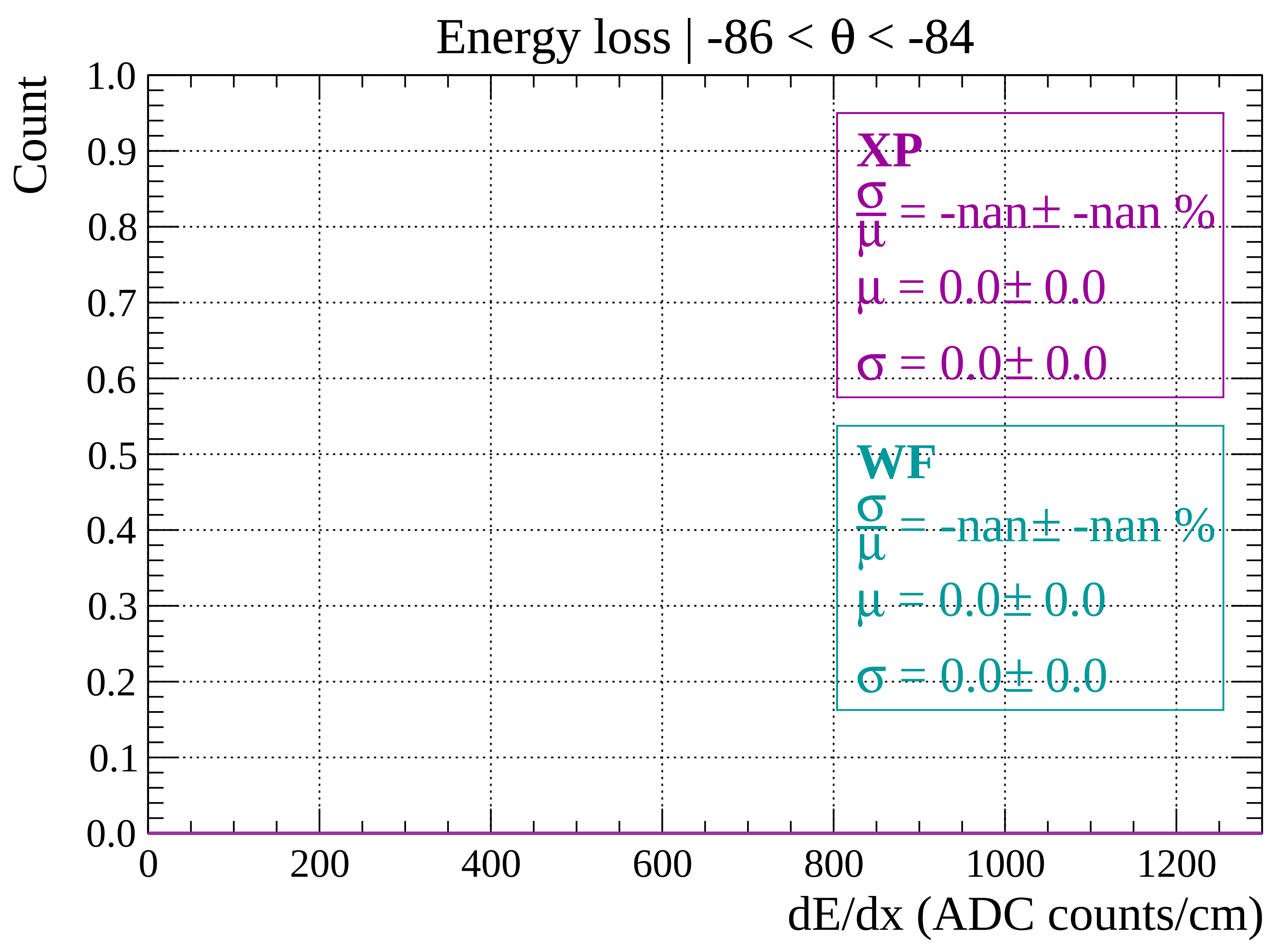
Energy loss | 1960 < p < 2000
Count
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 9.30± 0.21 %
μ = 481.0± 1.0
σ = 44.7± 0.9
WF
σ/μ = 9.43± 0.22 %
μ = 480.0± 1.0
σ = 45.2± 1.0



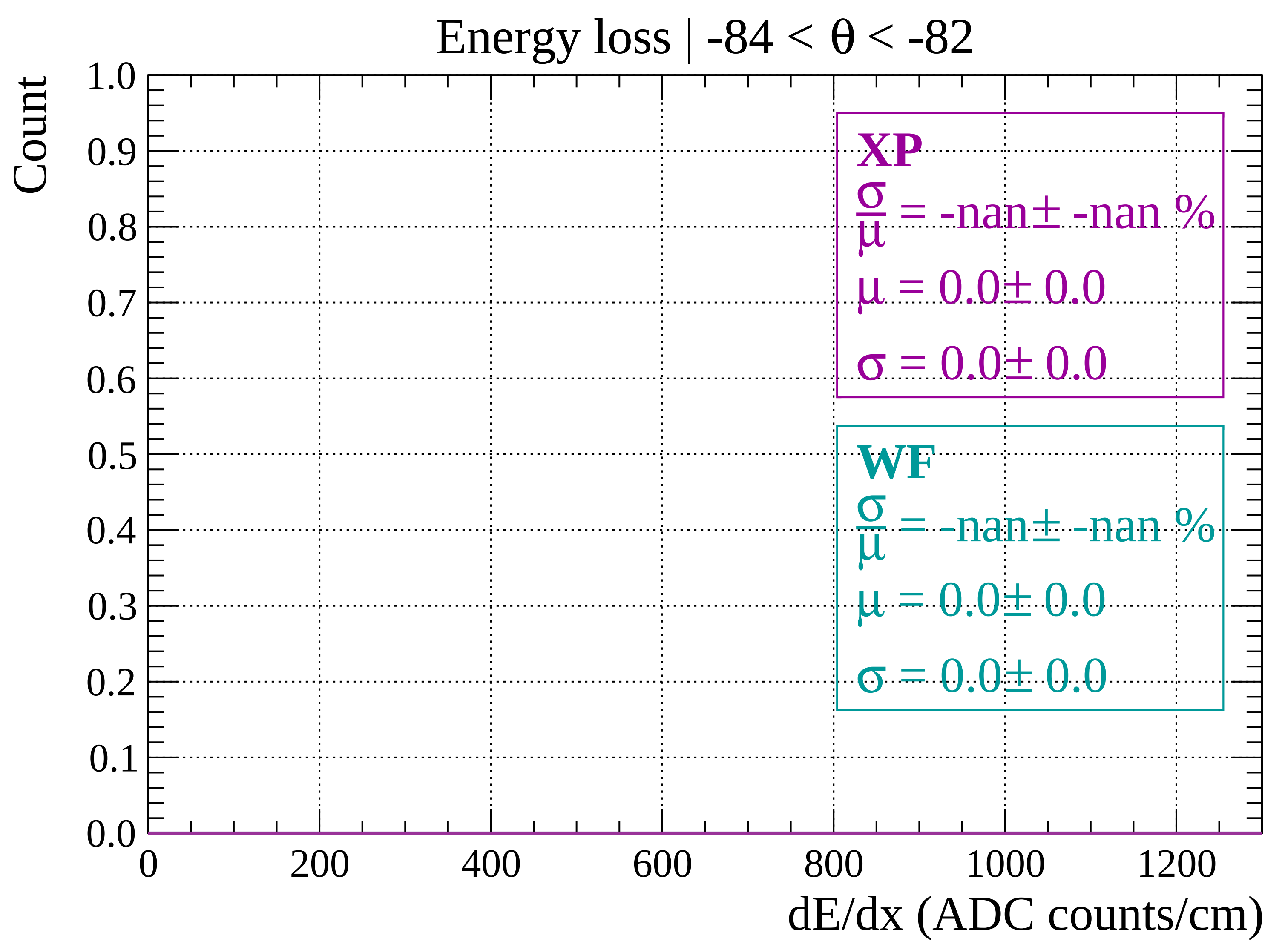
Energy loss | 2000 < p < 2040
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
$\frac{\sigma}{\mu}$ = 9.55± 0.27 %
μ = 486.3± 1.5
σ = 46.5± 1.2
WF
$\frac{\sigma}{\mu}$ = 10.19± 0.30 %
μ = 485.4± 1.5
σ = 49.4± 1.3


Energy loss | -90 < θ < -88
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

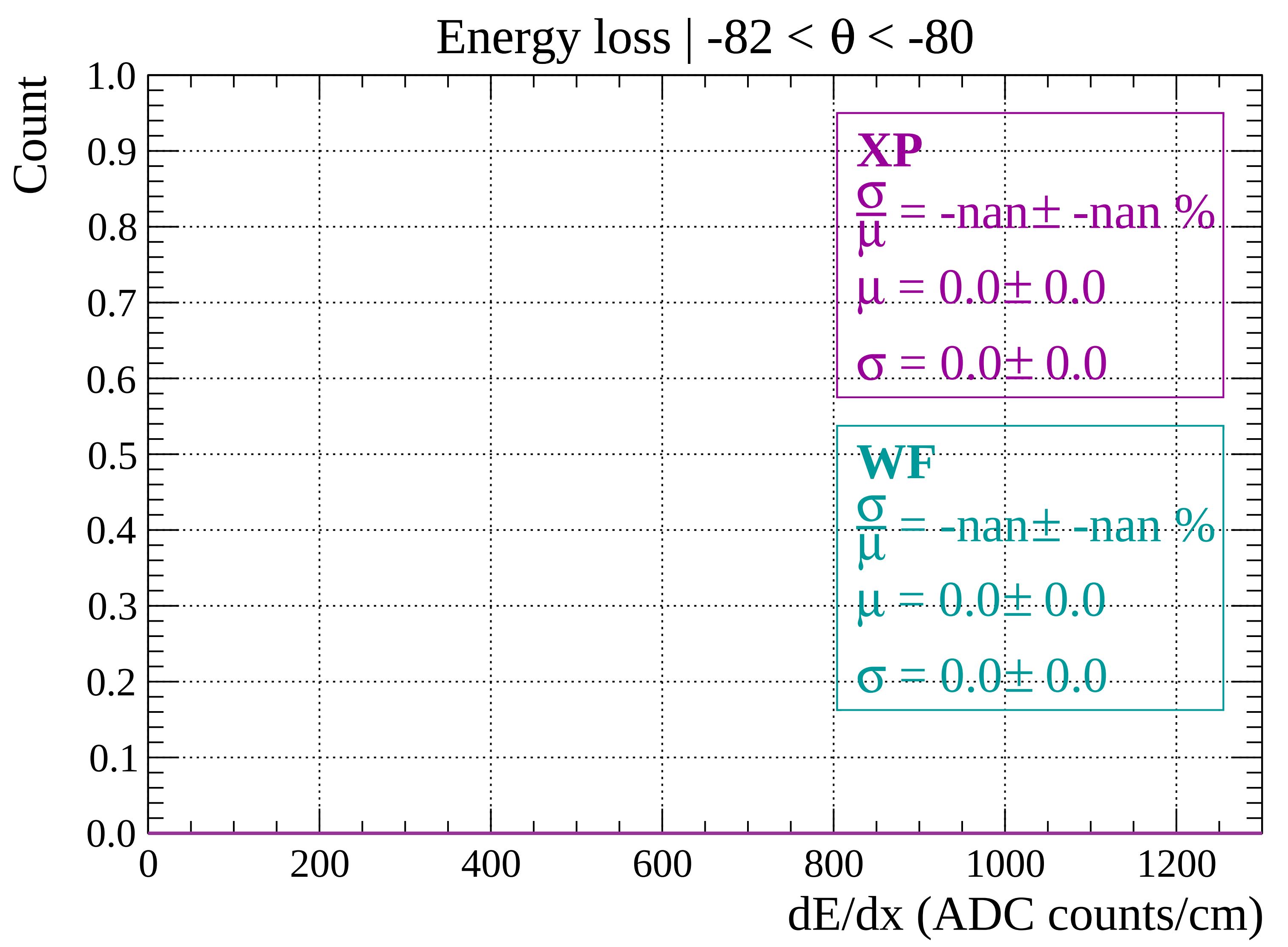

Energy loss | -88 < θ < -86
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

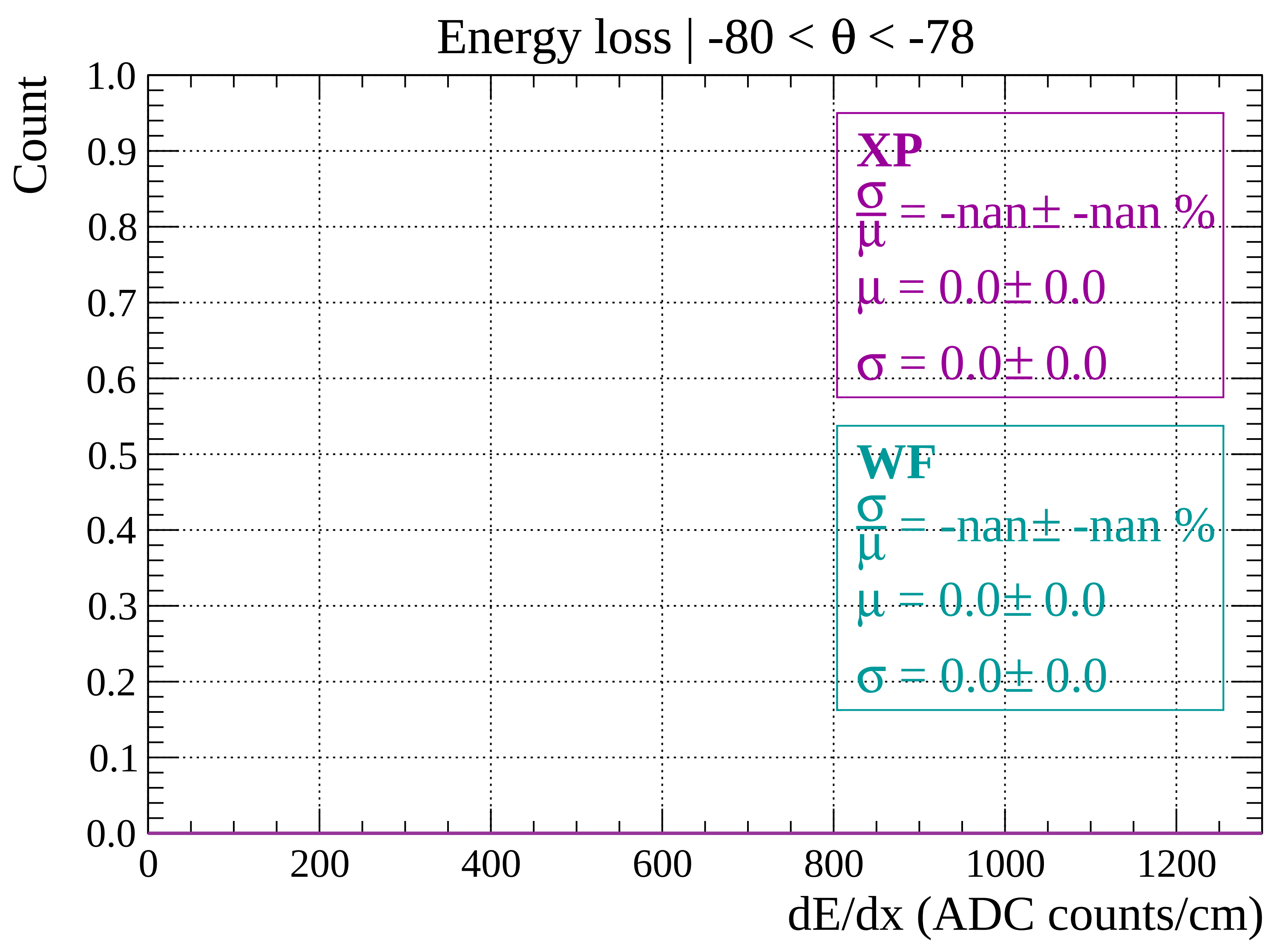


Energy loss | -86 < θ < -84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

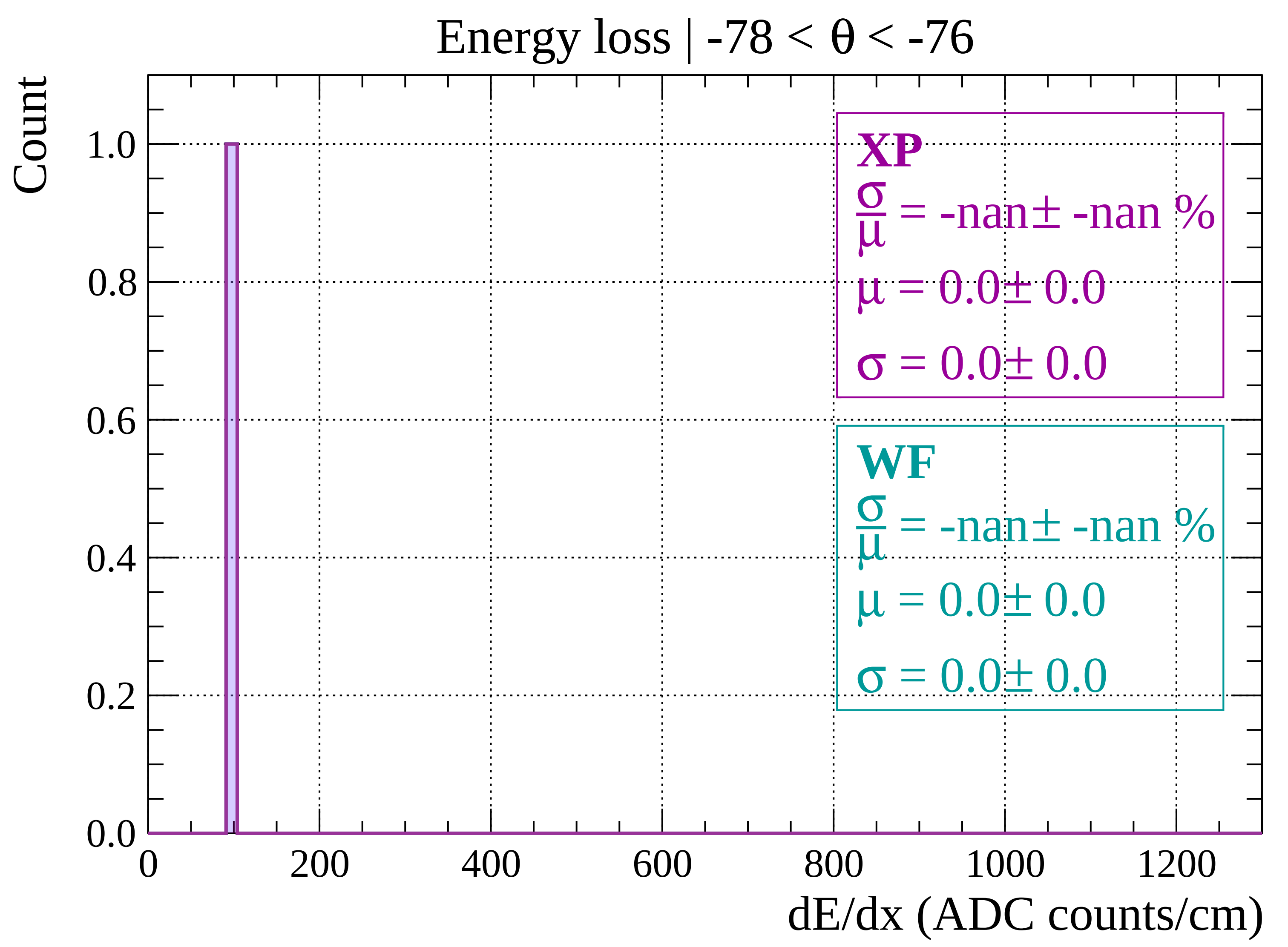


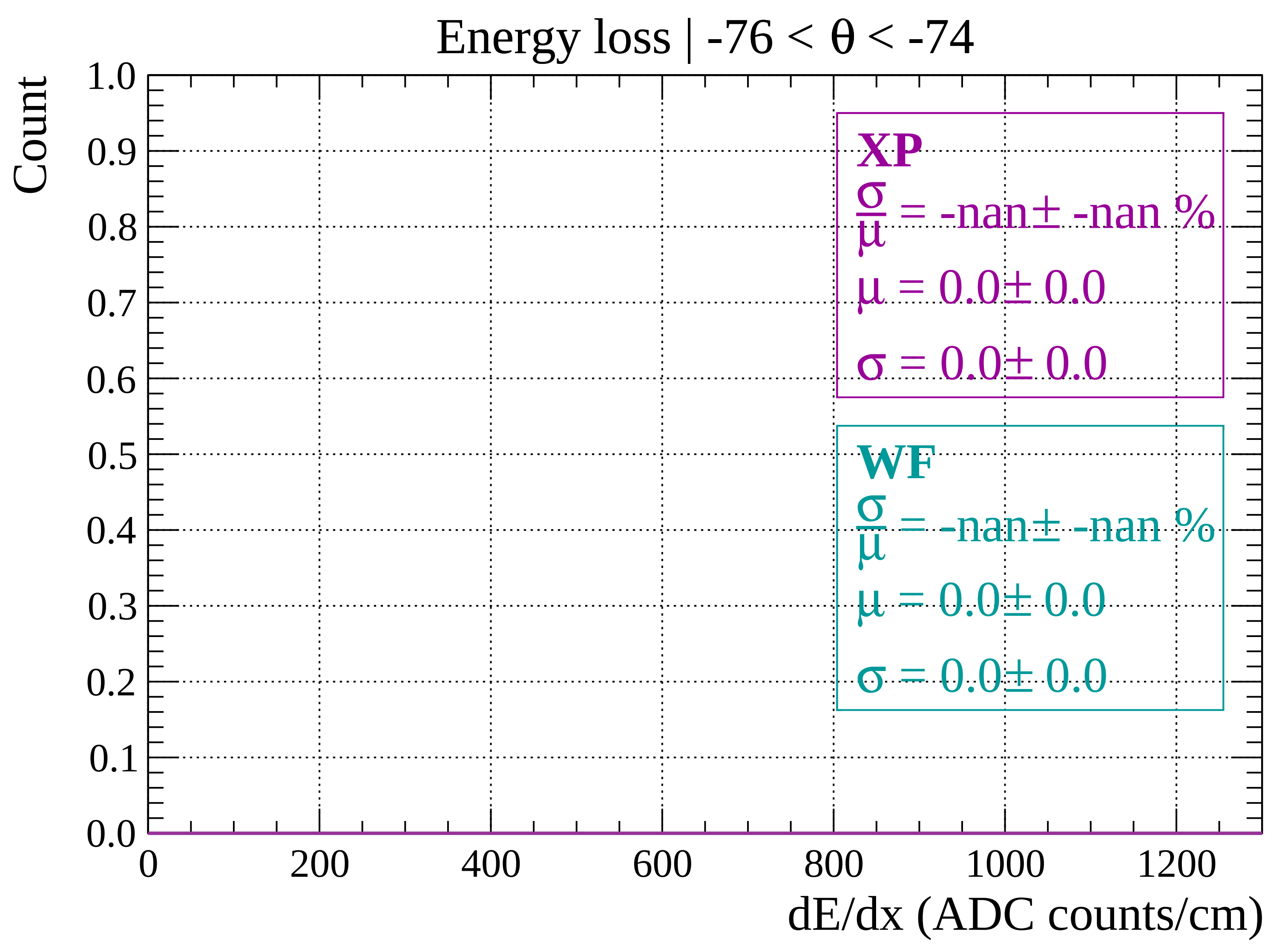
Energy loss | -84 < θ < -82
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -82 < θ < -80
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

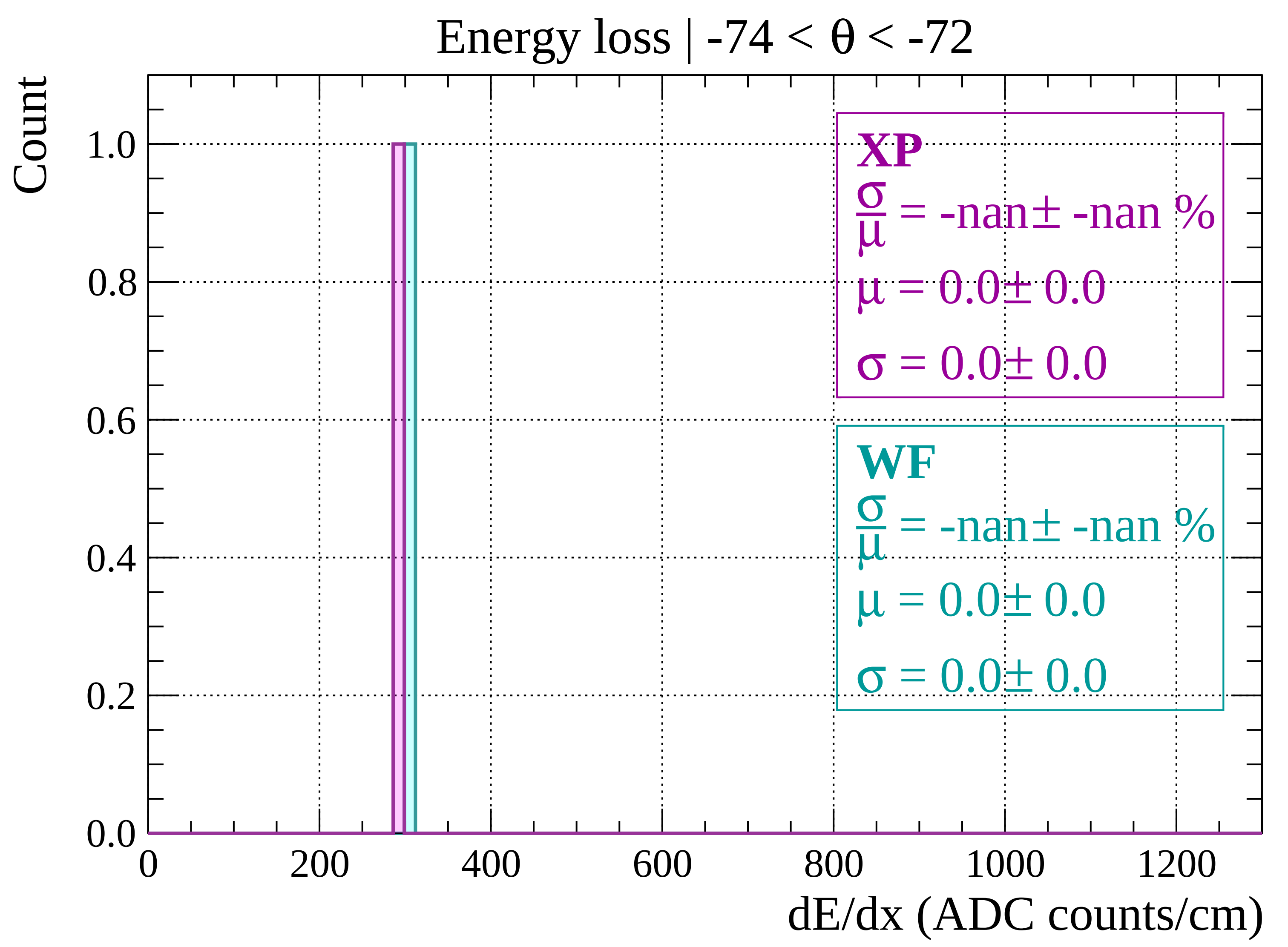


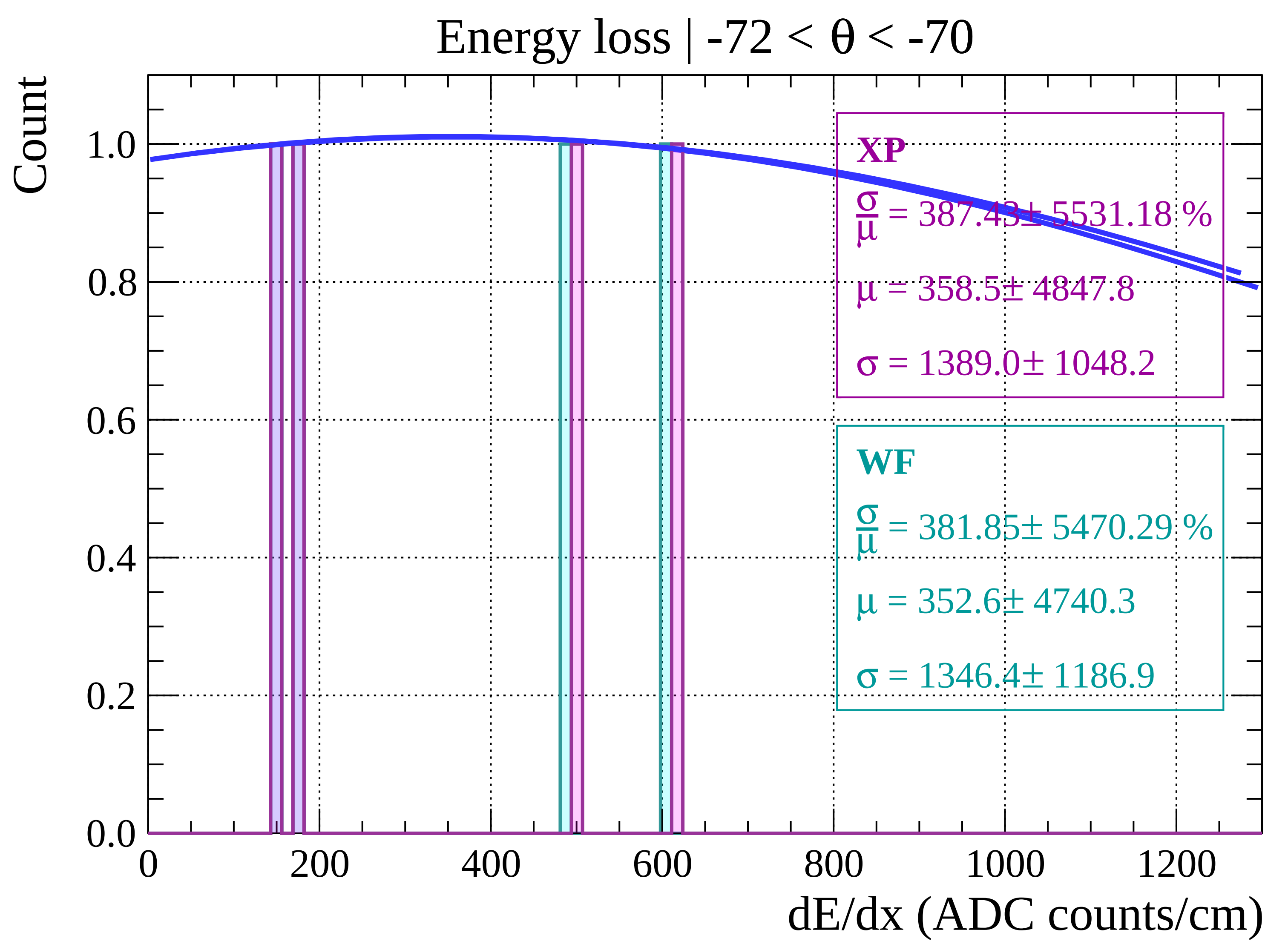
Energy loss | -80 < θ < -78
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | -78 < θ < -76
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

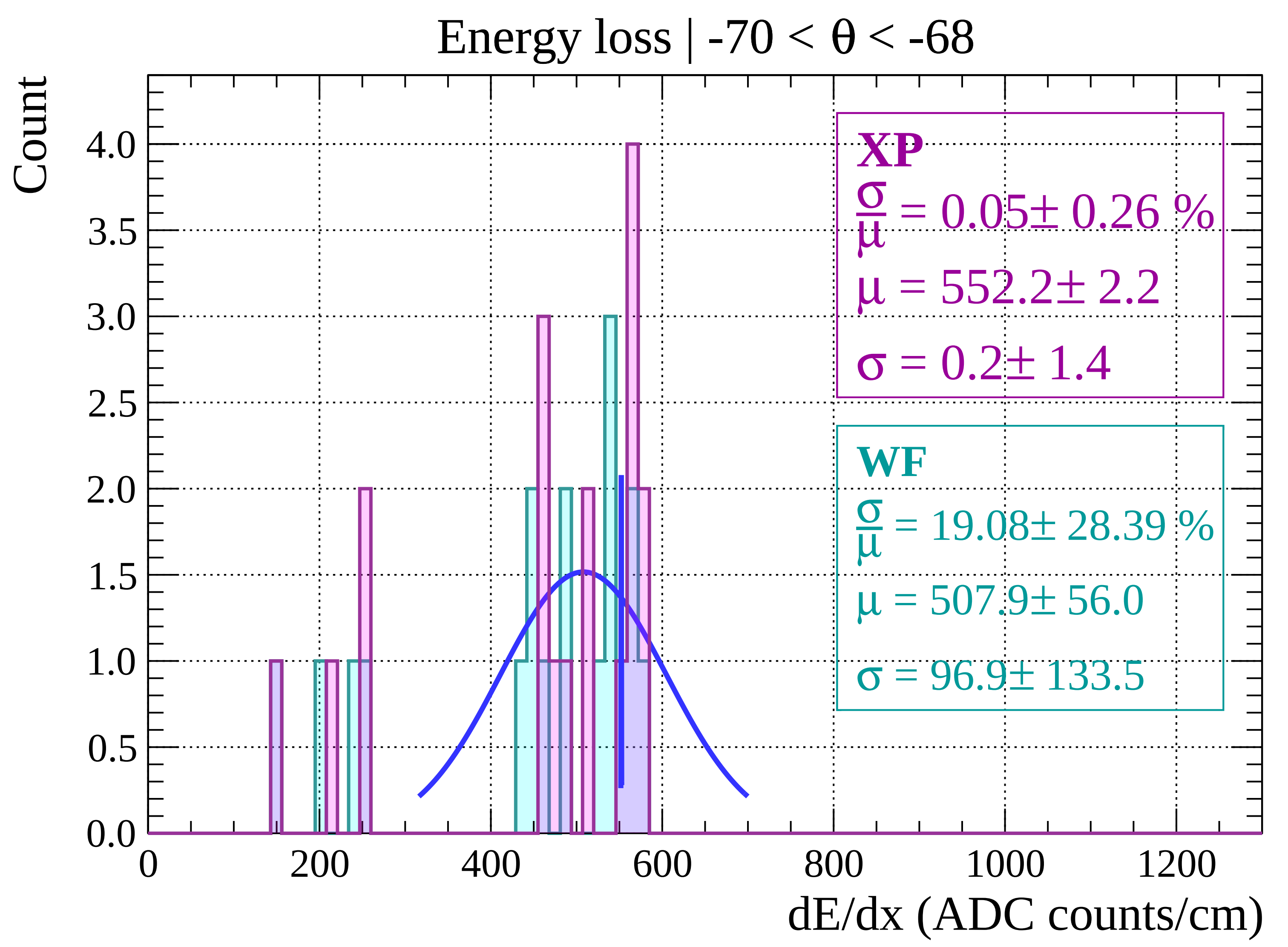


Energy loss | -76 < θ < -74
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

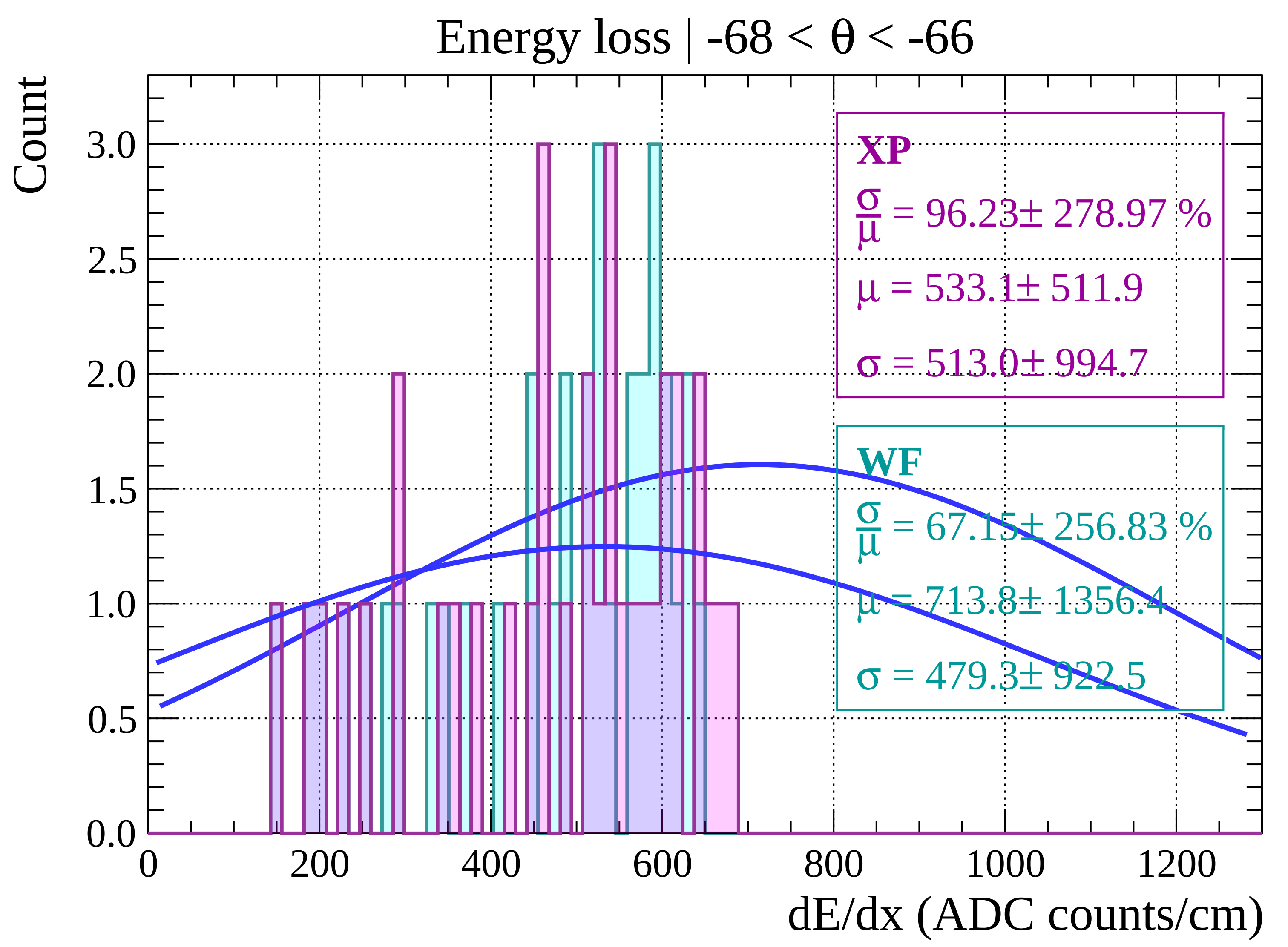

Energy loss | -74 < θ < -72
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

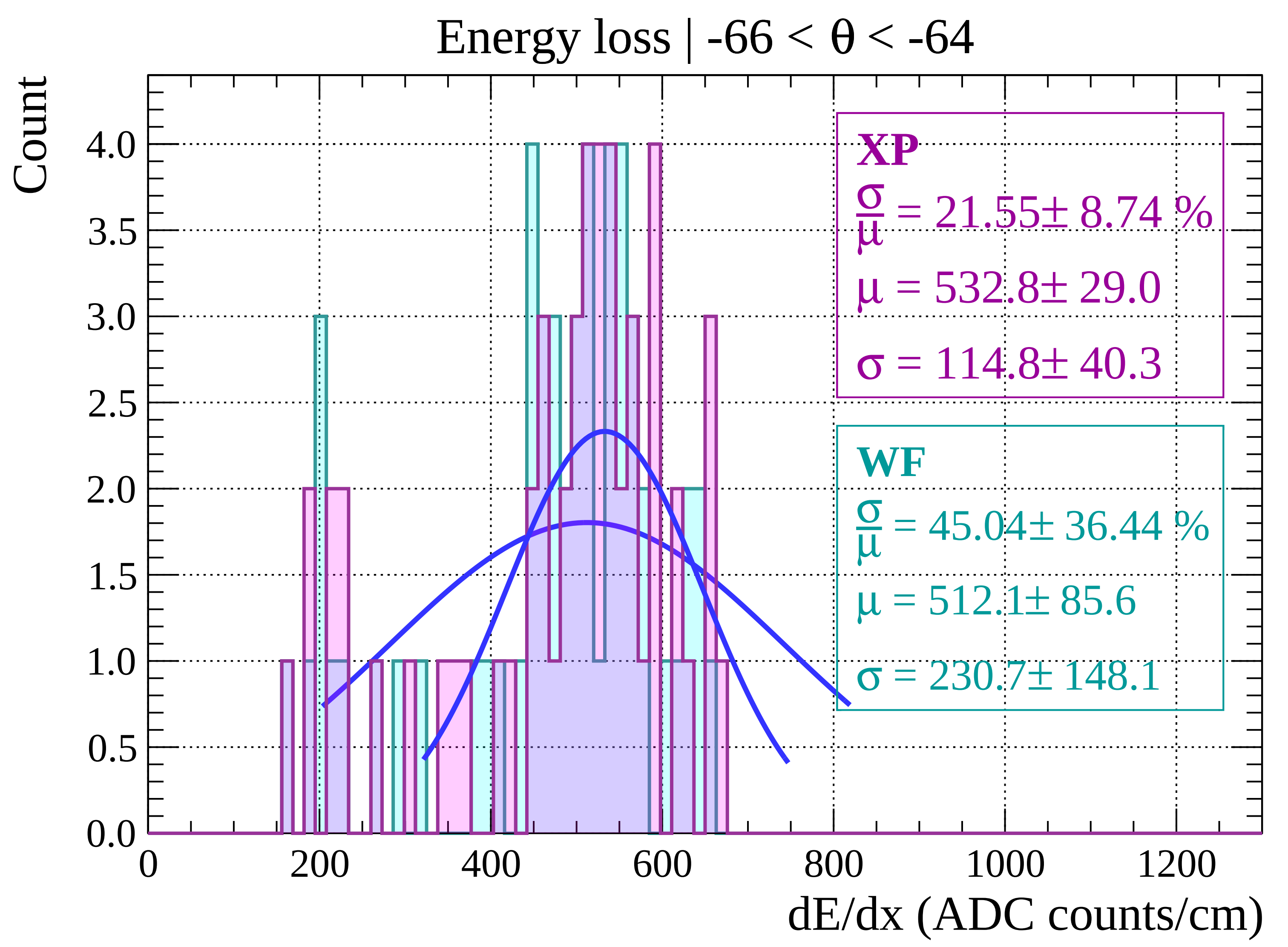


Energy loss | -72 < θ < -70
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 387.43± 5531.18 %
μ = 358.5± 4847.8
σ = 1389.0± 1048.2
WF
σ/μ = 381.85± 5470.29 %
μ = 352.6± 4740.3
σ = 1346.4± 1186.9

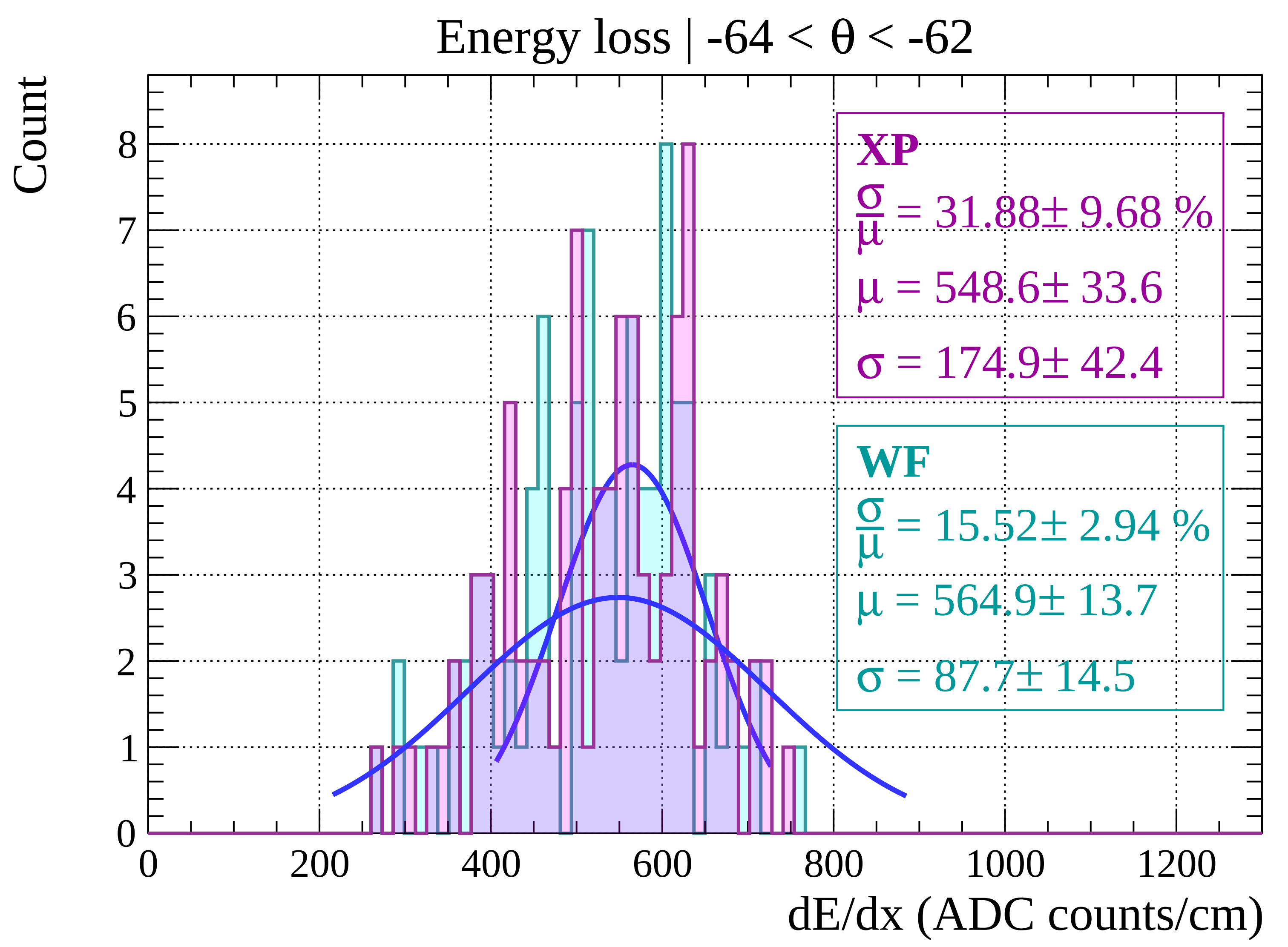


Energy loss | -70 < θ < -68
Count
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.05± 0.26 %
μ = 552.2± 2.2
σ = 0.2± 1.4
WF
σ/μ = 19.08± 28.39 %
μ = 507.9± 56.0
σ = 96.9± 133.5

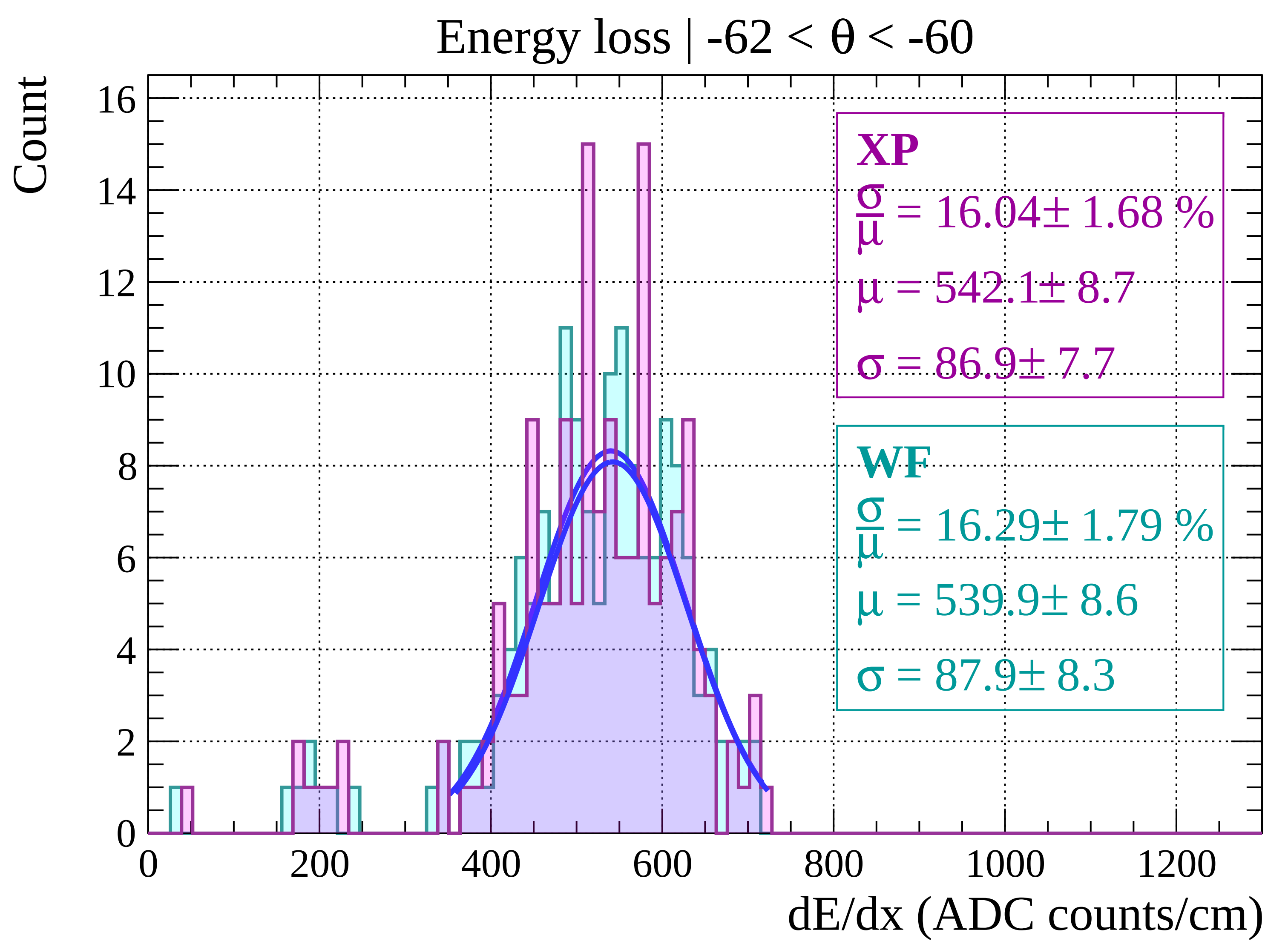

Energy loss | -68 < θ < -66
Count
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 96.23± 278.97 %
μ = 533.1± 511.9
σ = 513.0± 994.7
WF
$\frac{\sigma}{\mu}$ = 67.15± 256.83 %
μ = 713.8± 1356.4
σ = 479.3± 922.5



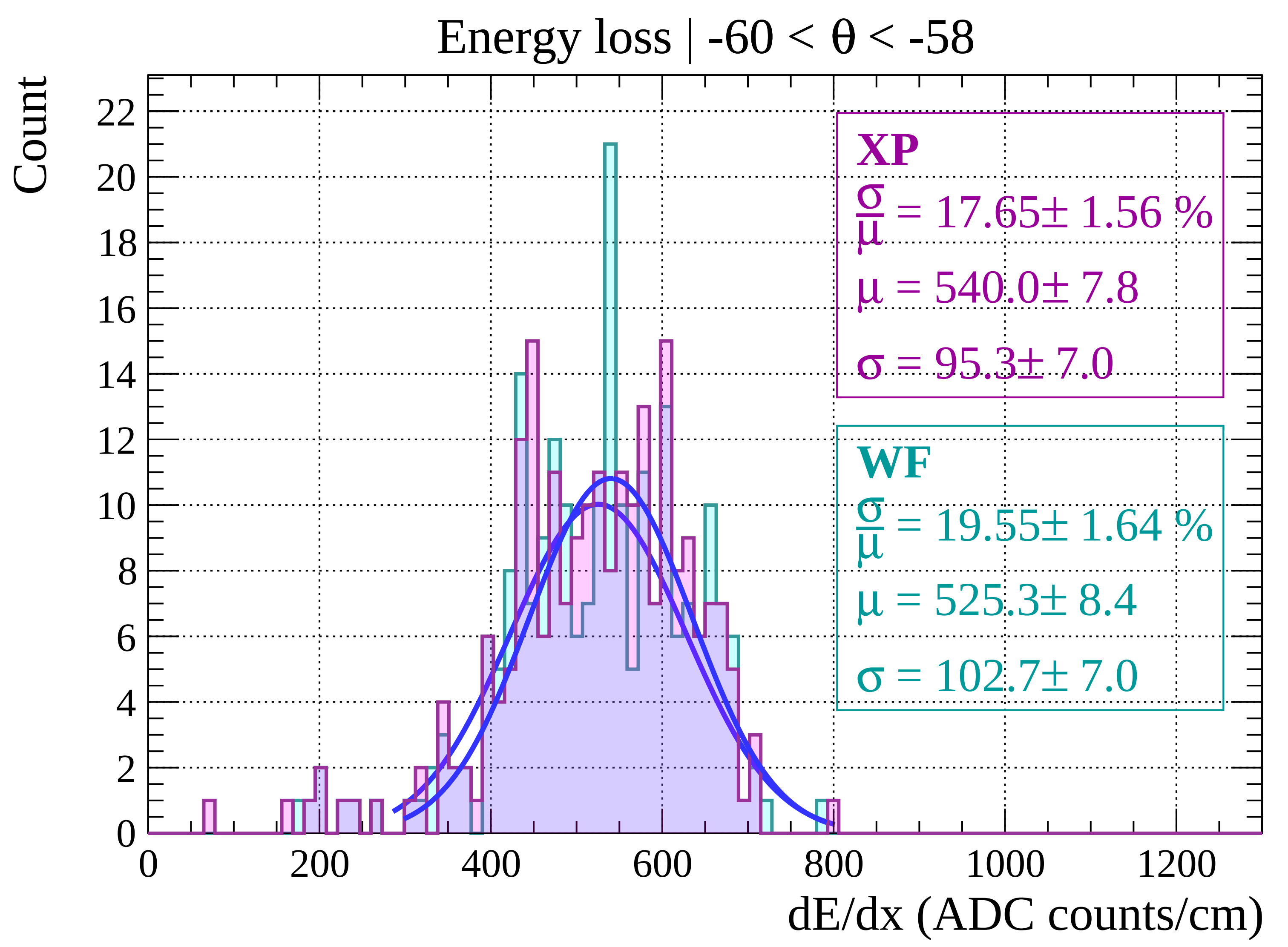
Energy loss | -66 < θ < -64
Count
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 21.55± 8.74 %
μ = 532.8± 29.0
σ = 114.8± 40.3
WF
σ/μ = 45.04± 36.44 %
μ = 512.1± 85.6
σ = 230.7± 148.1



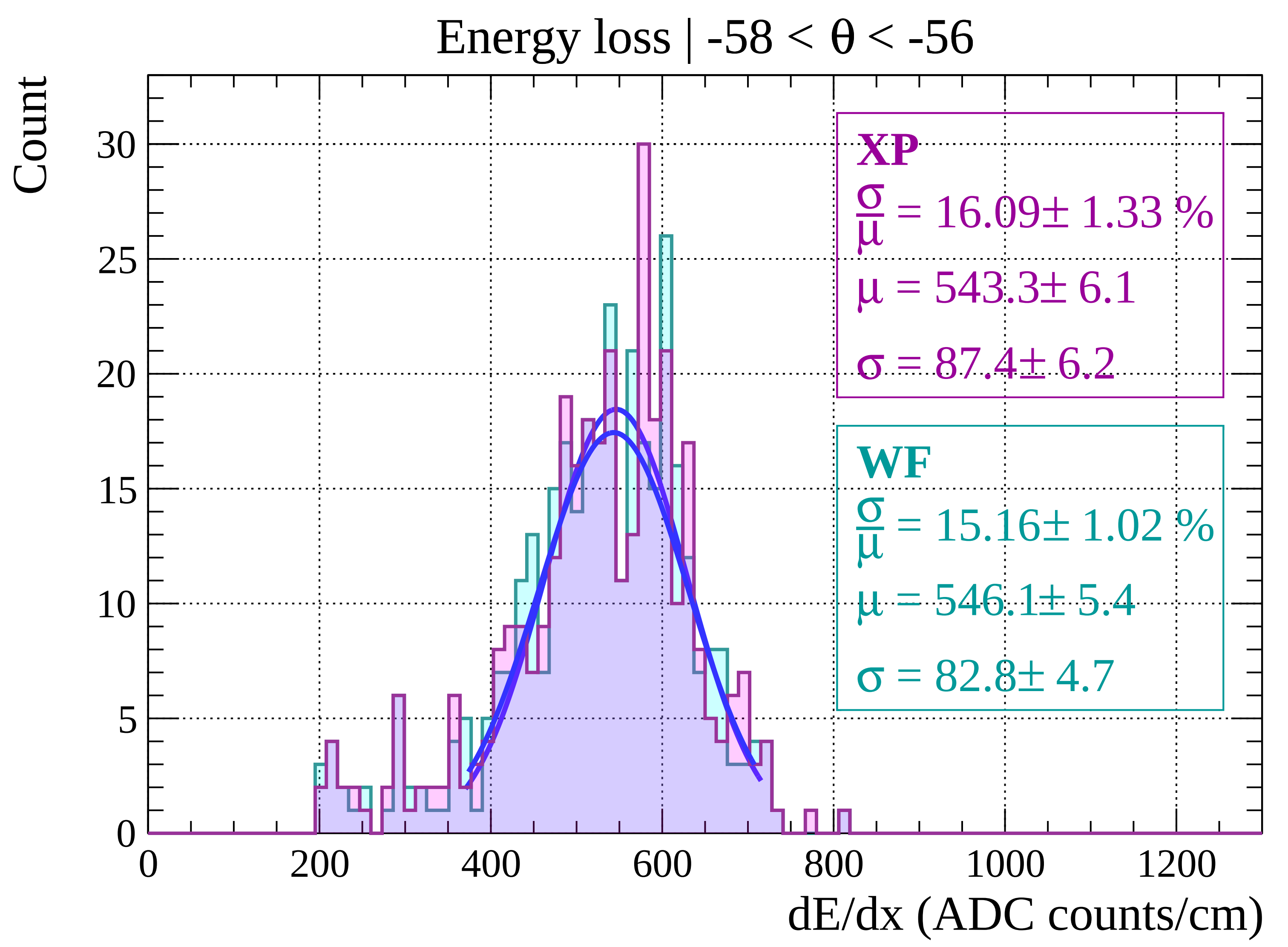
Energy loss | -64 < θ < -62
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
6
7
8
XP
σ/μ = 31.88± 9.68 %
μ = 548.6± 33.6
σ = 174.9± 42.4
WF
σ/μ = 15.52± 2.94 %
μ = 564.9± 13.7
σ = 87.7± 14.5



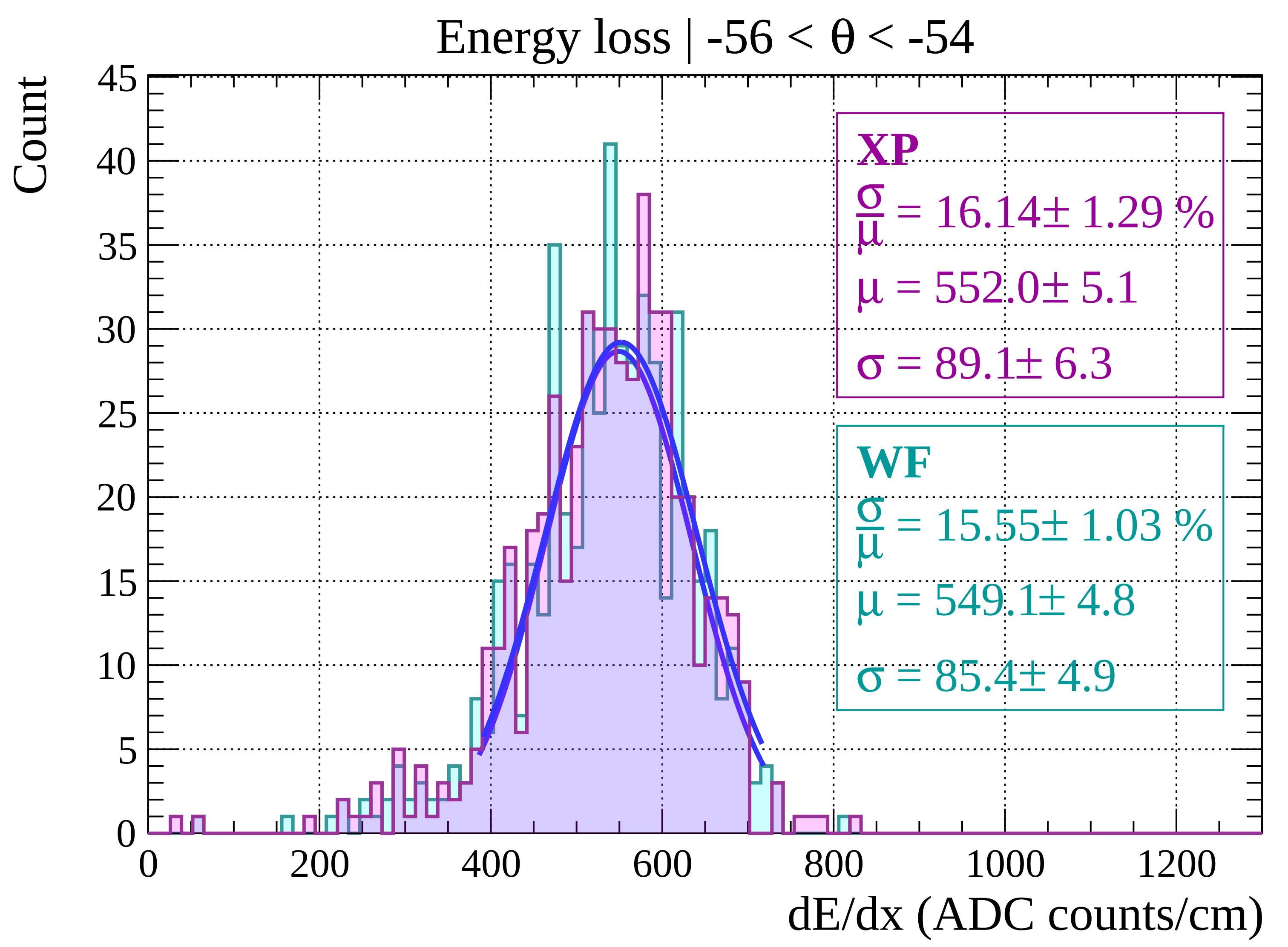
Energy loss | -62 < θ < -60
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
2
4
6
8
10
12
14
16
XP
σ/μ = 16.04± 1.68 %
μ = 542.1± 8.7
σ = 86.9± 7.7
WF
σ/μ = 16.29± 1.79 %
μ = 539.9± 8.6
σ = 87.9± 8.3

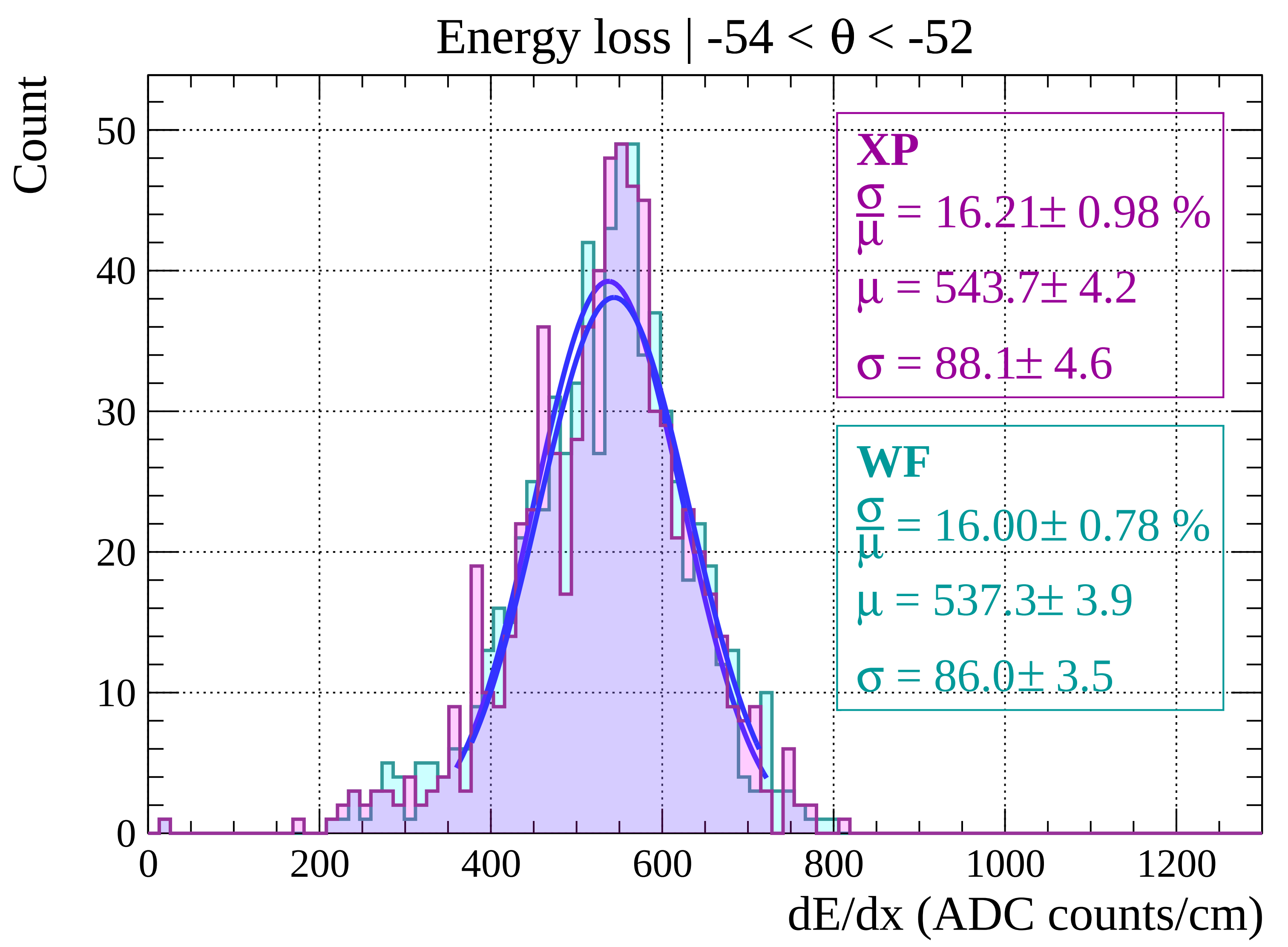


Energy loss | -60 < θ < -58
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 17.65± 1.56 %
μ = 540.0± 7.8
σ = 95.3± 7.0
WF
σ/μ = 19.55± 1.64 %
μ = 525.3± 8.4
σ = 102.7± 7.0
0
2
4
6
8
10
12
14
16
18
20
22
0
200
400
600
800
1000
1200

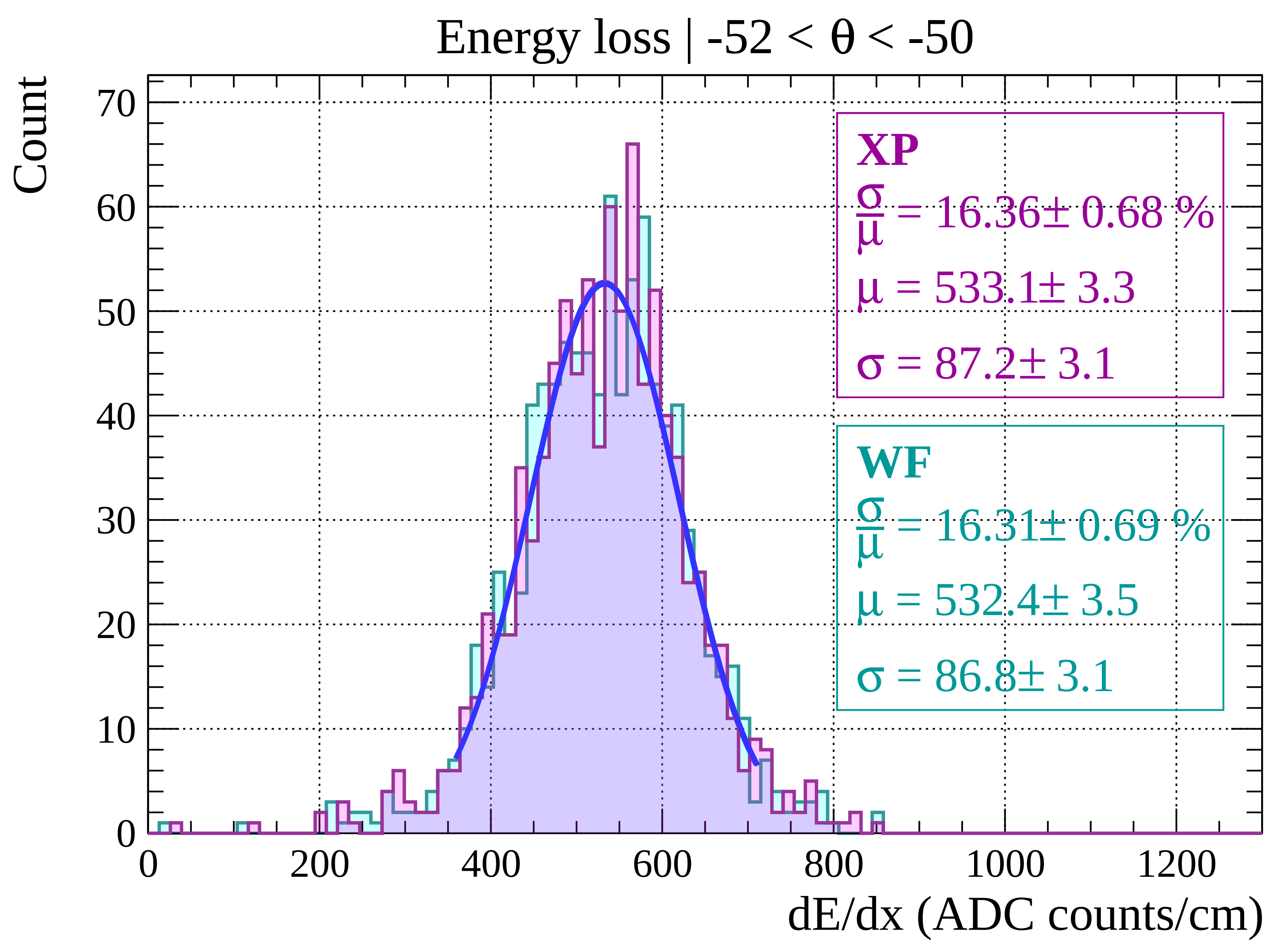


Energy loss | -58 < θ < -56
Count
dE/dx (ADC counts/cm)
0
5
10
15
20
25
30
0
200
400
600
800
1000
1200
XP
σ/μ = 16.09± 1.33 %
μ = 543.3± 6.1
σ = 87.4± 6.2
WF
σ/μ = 15.16± 1.02 %
μ = 546.1± 5.4
σ = 82.8± 4.7

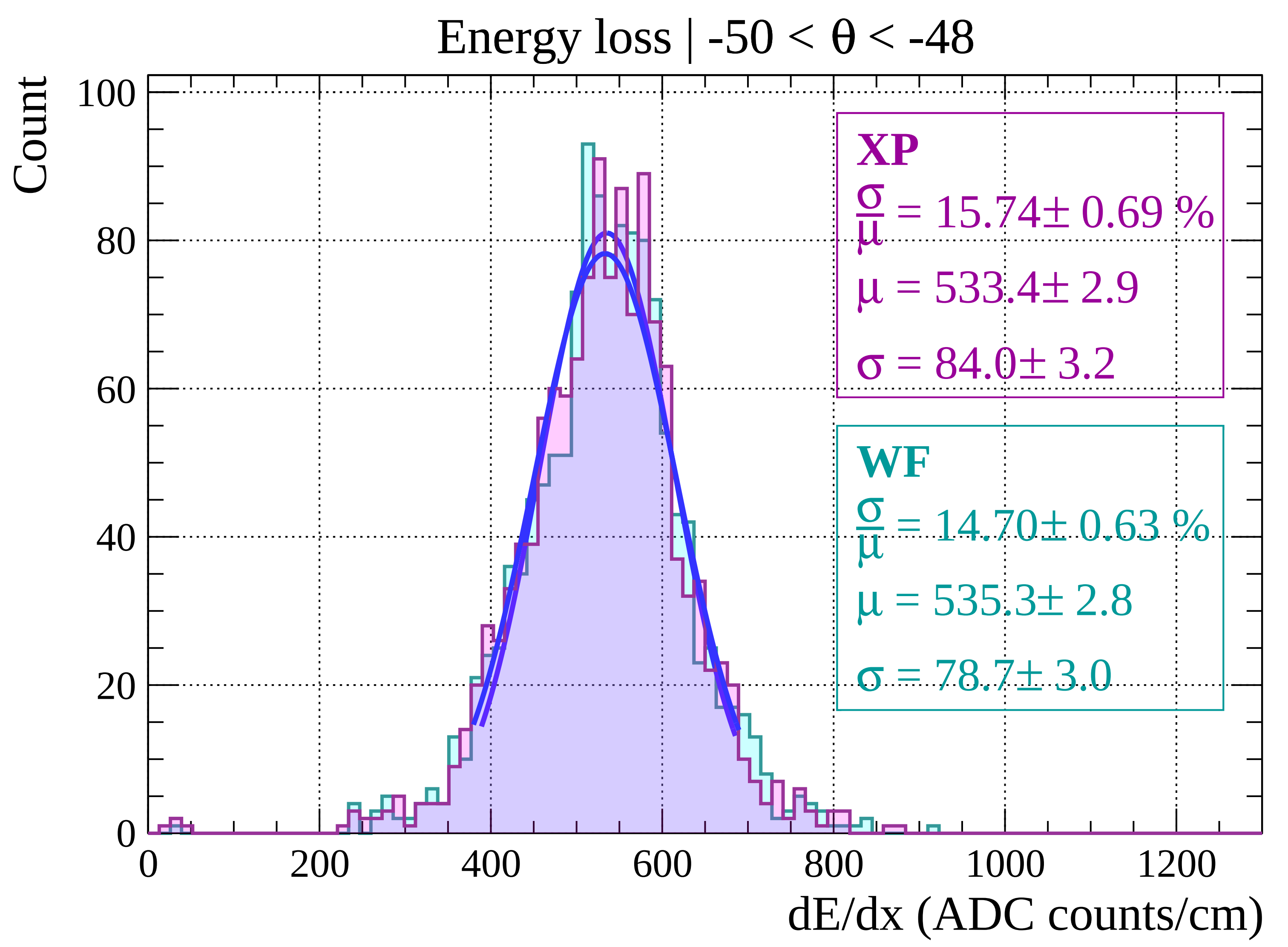

Energy loss | -56 < θ < -54
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.14± 1.29 %
μ = 552.0± 5.1
σ = 89.1± 6.3
WF
σ/μ = 15.55± 1.03 %
μ = 549.1± 4.8
σ = 85.4± 4.9
0
5
10
15
20
25
30
35
40
45
0
200
400
600
800
1000
1200



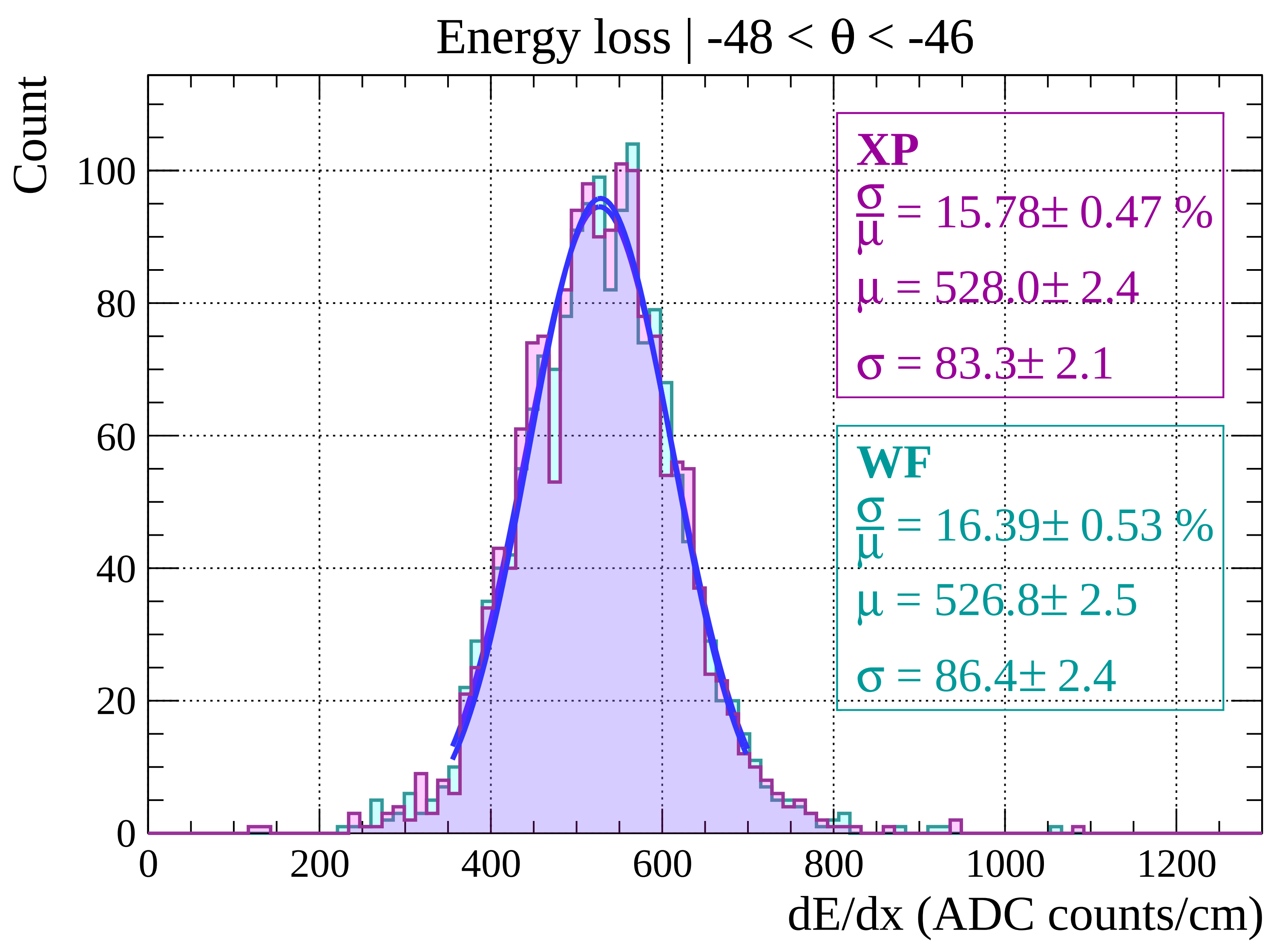
Energy loss | -54 < θ < -52
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.21± 0.98 %
μ = 543.7± 4.2
σ = 88.1± 4.6
WF
σ/μ = 16.00± 0.78 %
μ = 537.3± 3.9
σ = 86.0± 3.5
0
10
20
30
40
50
0
200
400
600
800
1000
1200

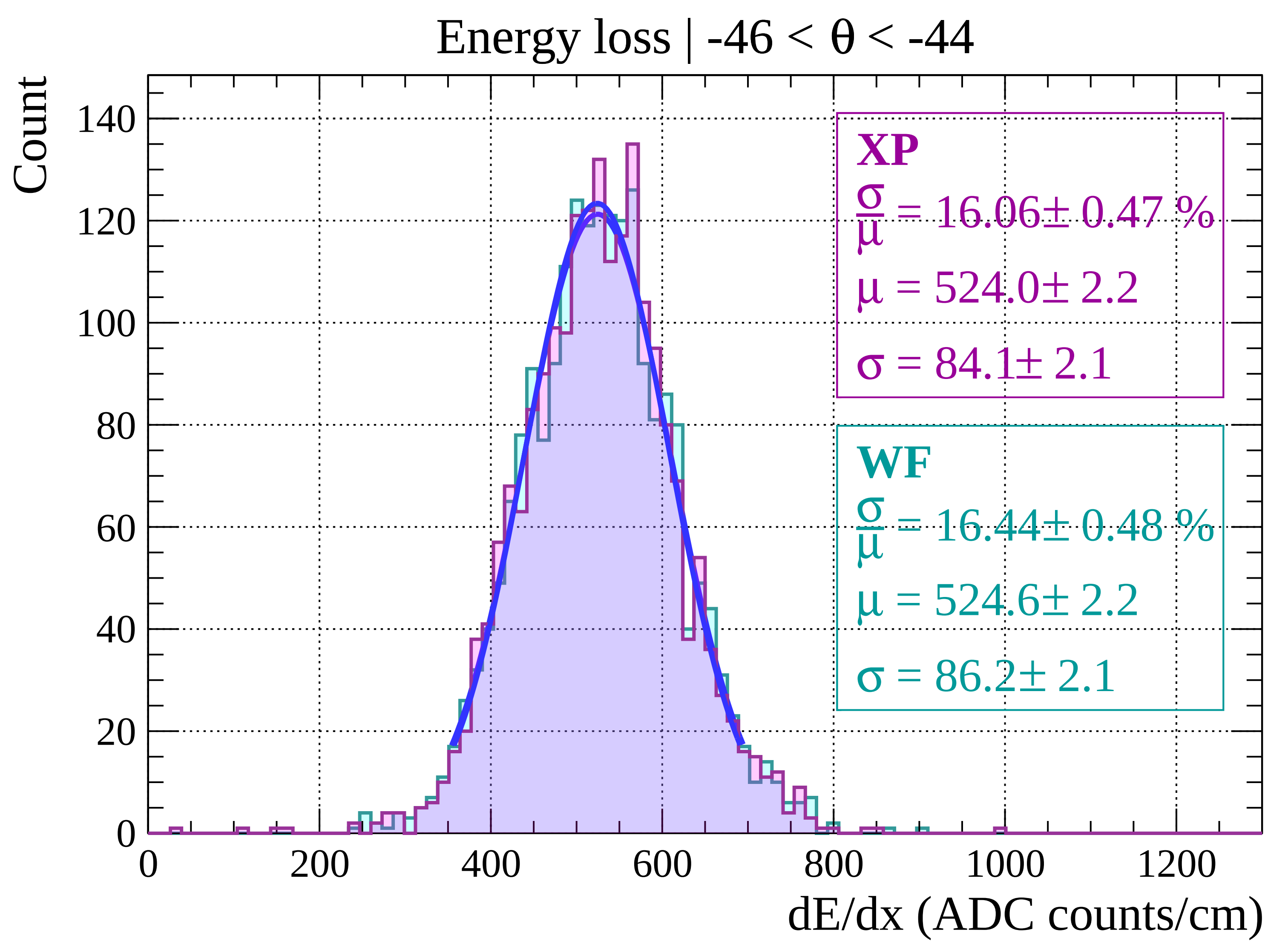


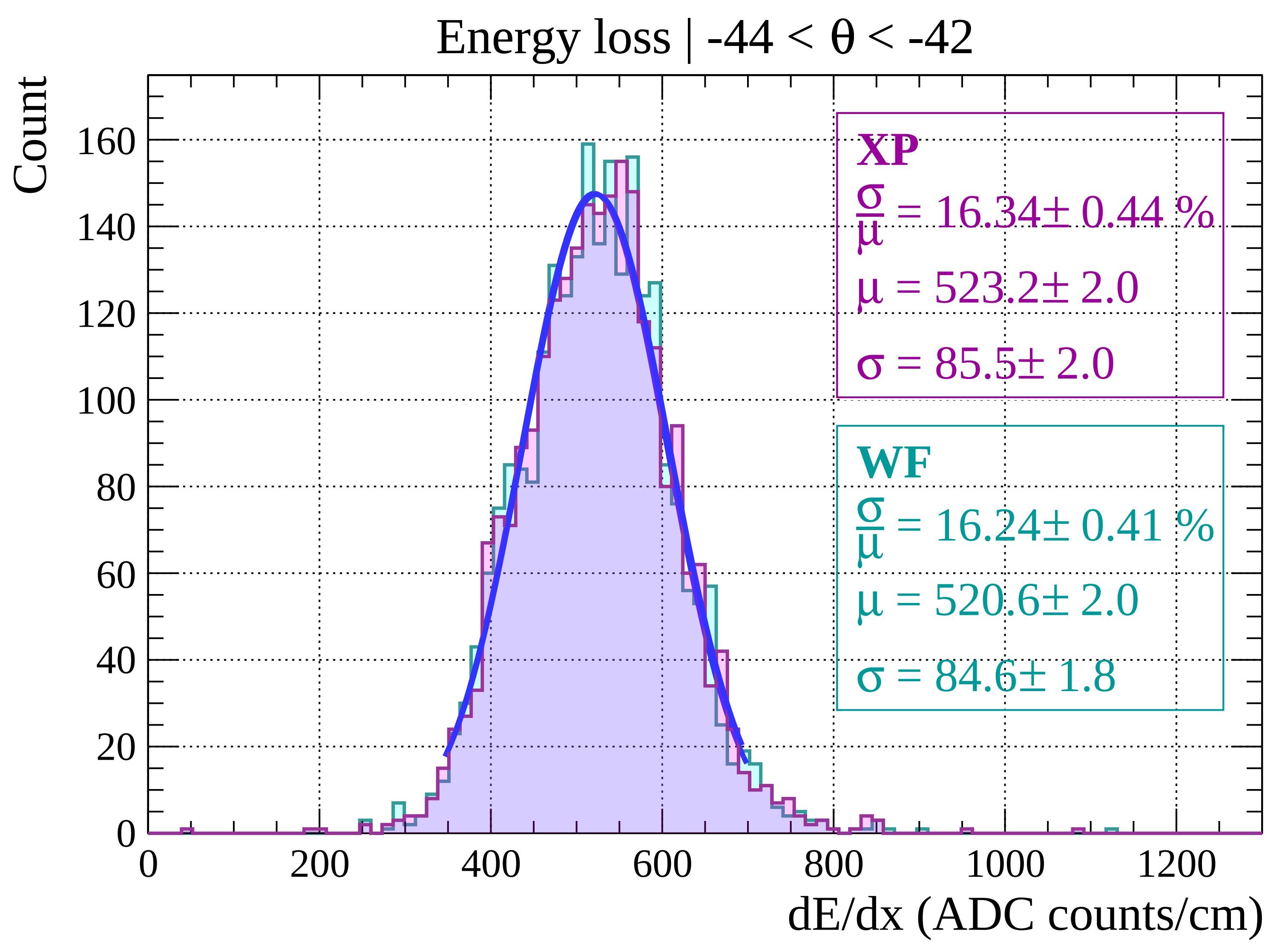
Energy loss | -52 < θ < -50
Count
70
60
50
40
30
20
10
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 16.36± 0.68 %
μ = 533.1± 3.3
σ = 87.2± 3.1
WF
σ/μ = 16.31± 0.69 %
μ = 532.4± 3.5
σ = 86.8± 3.1

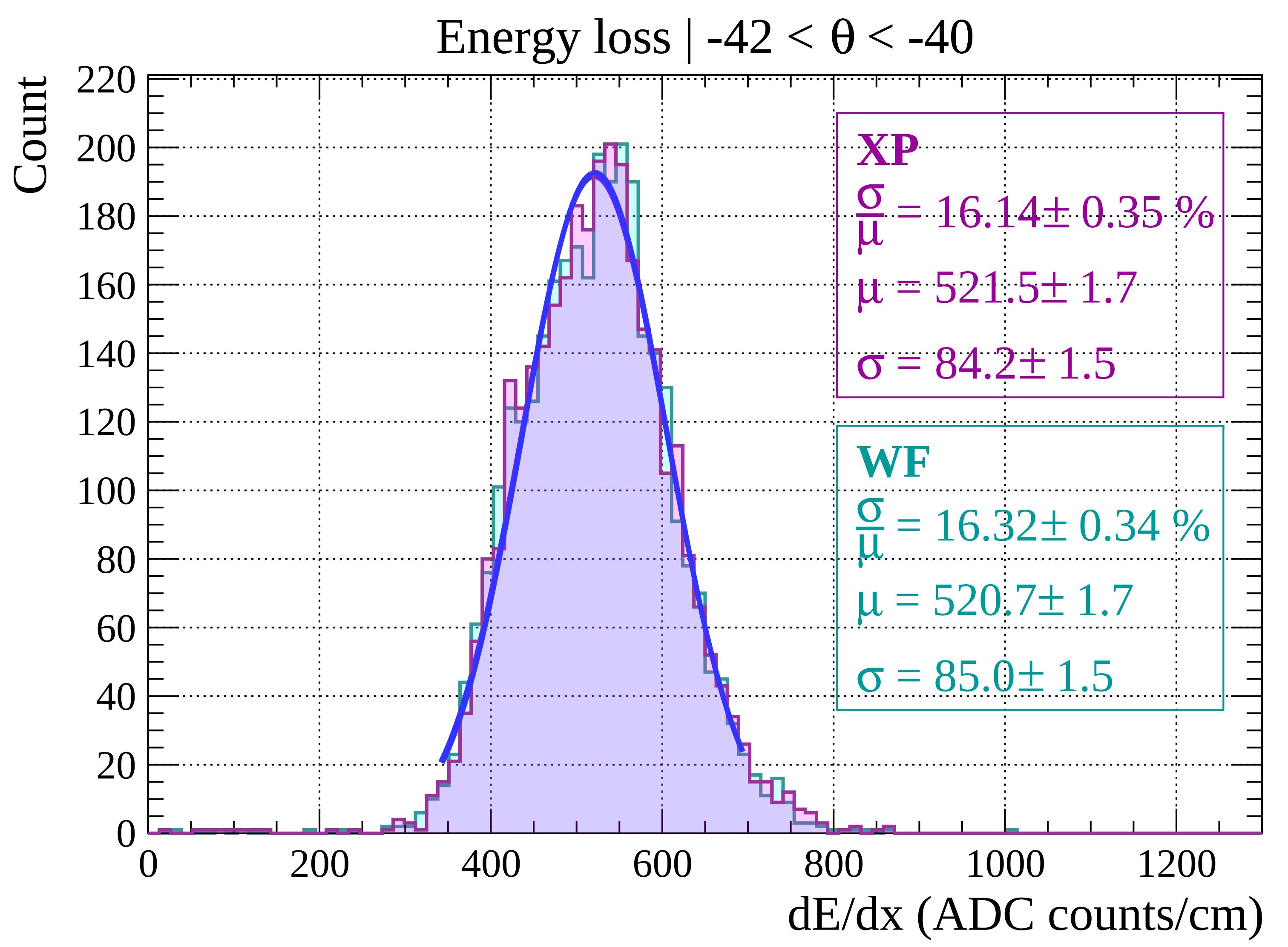


Energy loss | -50 < θ < -48
Count
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 15.74± 0.69 %
μ = 533.4± 2.9
σ = 84.0± 3.2
WF
$\frac{\sigma}{\mu}$ = 14.70± 0.63 %
μ = 535.3± 2.8
σ = 78.7± 3.0

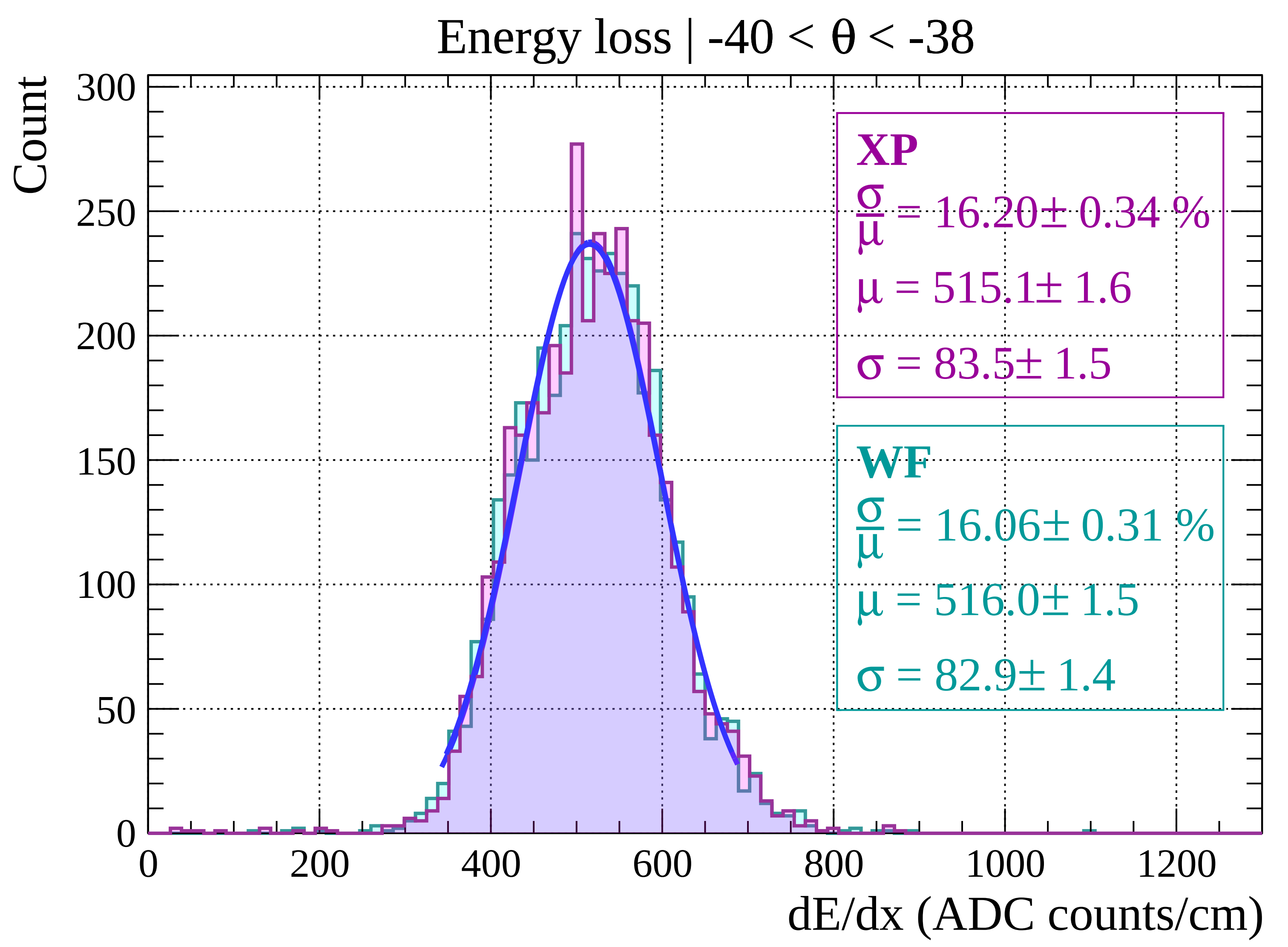

Energy loss | -48 < θ < -46
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.78± 0.47 %
μ = 528.0± 2.4
σ = 83.3± 2.1
WF
σ/μ = 16.39± 0.53 %
μ = 526.8± 2.5
σ = 86.4± 2.4
0
20
40
60
80
100
0
200
400
600
800
1000
1200


Energy loss | -46 < θ < -44
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.06± 0.47 %
μ = 524.0± 2.2
σ = 84.1± 2.1
WF
σ/μ = 16.44± 0.48 %
μ = 524.6± 2.2
σ = 86.2± 2.1
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200

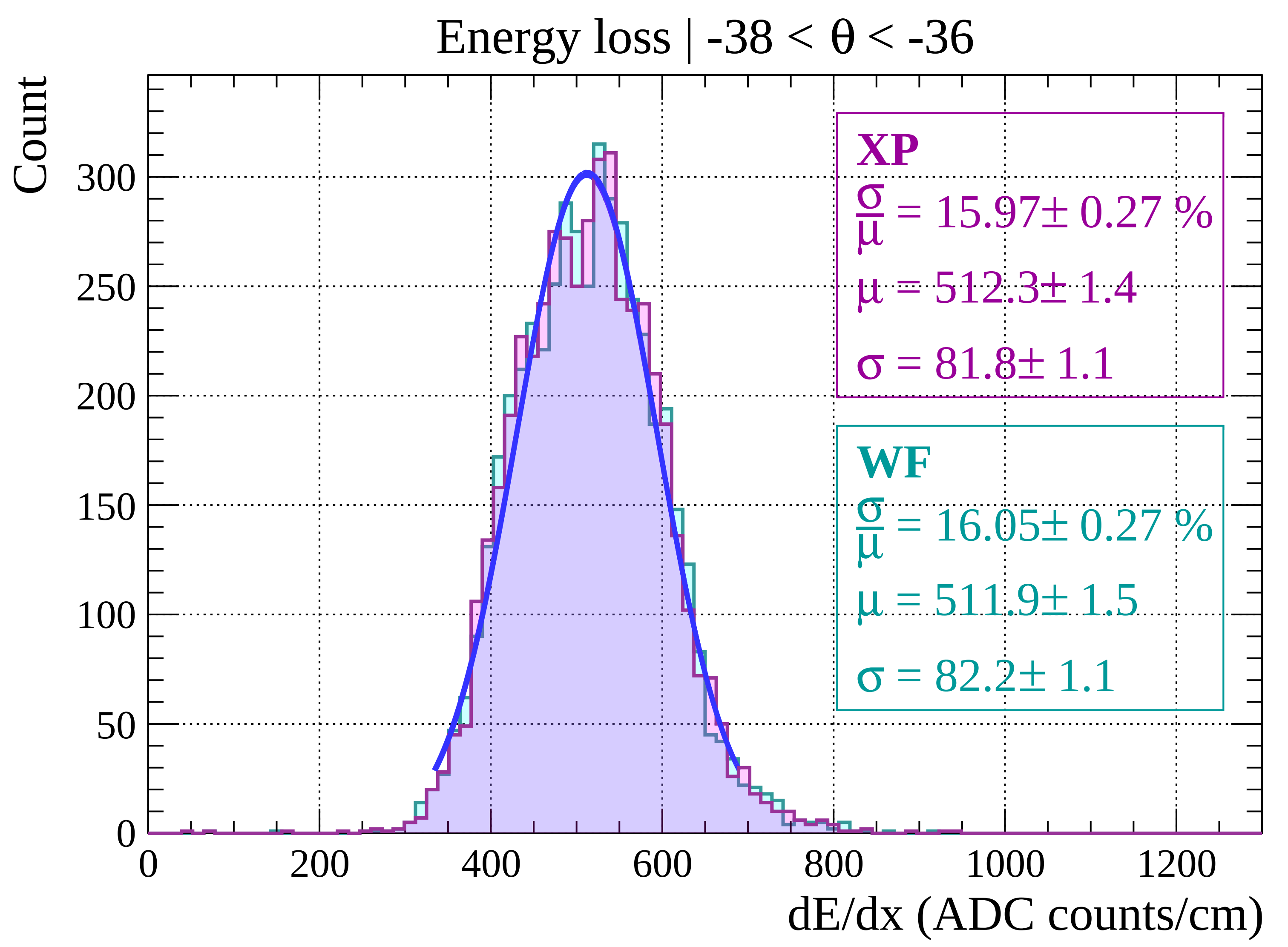

Energy loss | -44 < θ < -42
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.34± 0.44 %
μ = 523.2± 2.0
σ = 85.5± 2.0
WF
σ/μ = 16.24± 0.41 %
μ = 520.6± 2.0
σ = 84.6± 1.8
0
20
40
60
80
100
120
140
160
0
200
400
600
800
1000
1200



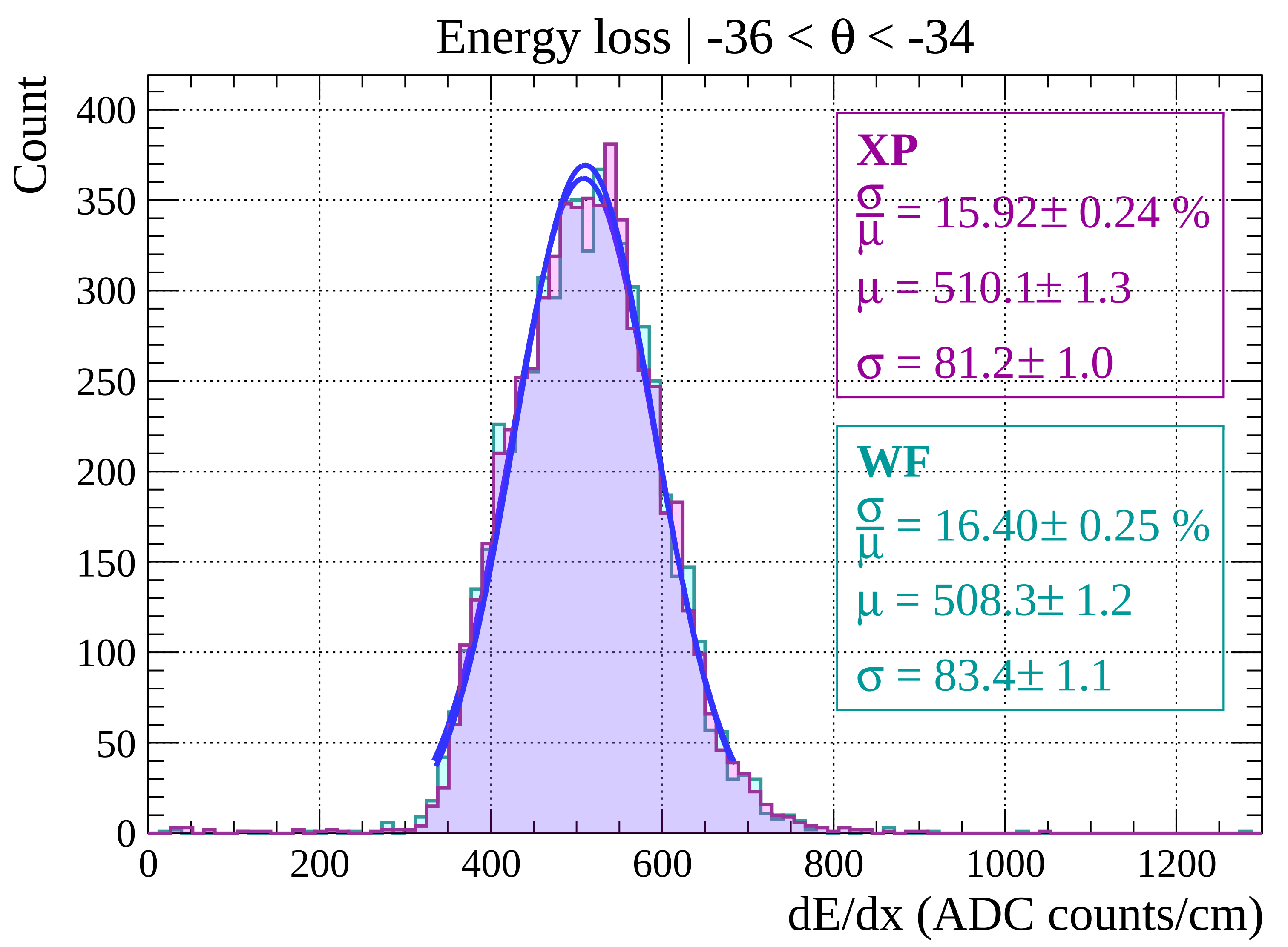
Energy loss | -42 < θ < -40
Count
220
200
180
160
140
120
100
80
60
40
20
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 16.14± 0.35 %
μ = 521.5± 1.7
σ = 84.2± 1.5
WF
σ/μ = 16.32± 0.34 %
μ = 520.7± 1.7
σ = 85.0± 1.5



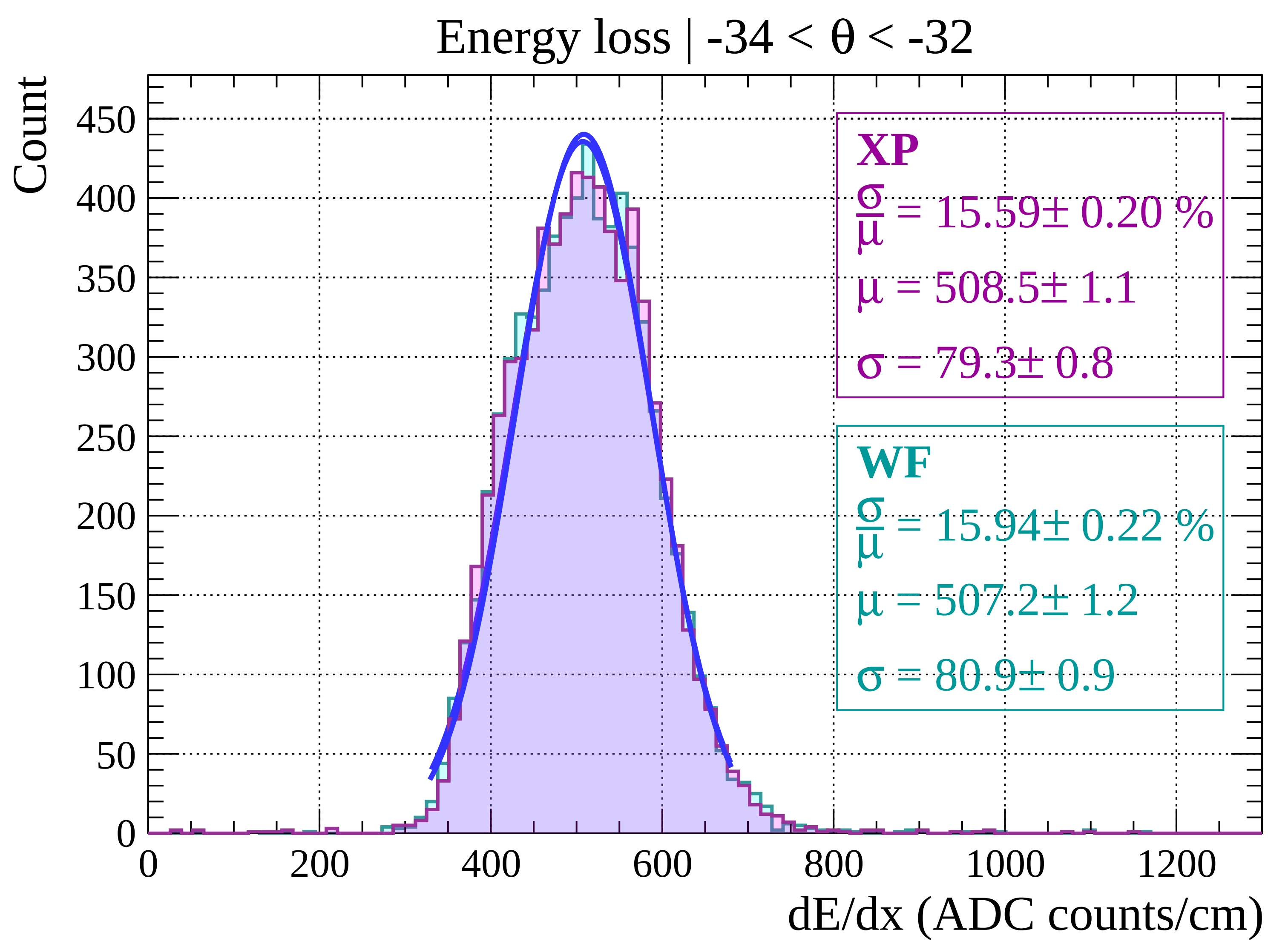
Energy loss | -40 < θ < -38
Count
XP
$\frac{\sigma}{\mu}$ = 16.20± 0.34 %
μ = 515.1± 1.6
σ = 83.5± 1.5
WF
$\frac{\sigma}{\mu}$ = 16.06± 0.31 %
μ = 516.0± 1.5
σ = 82.9± 1.4
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)



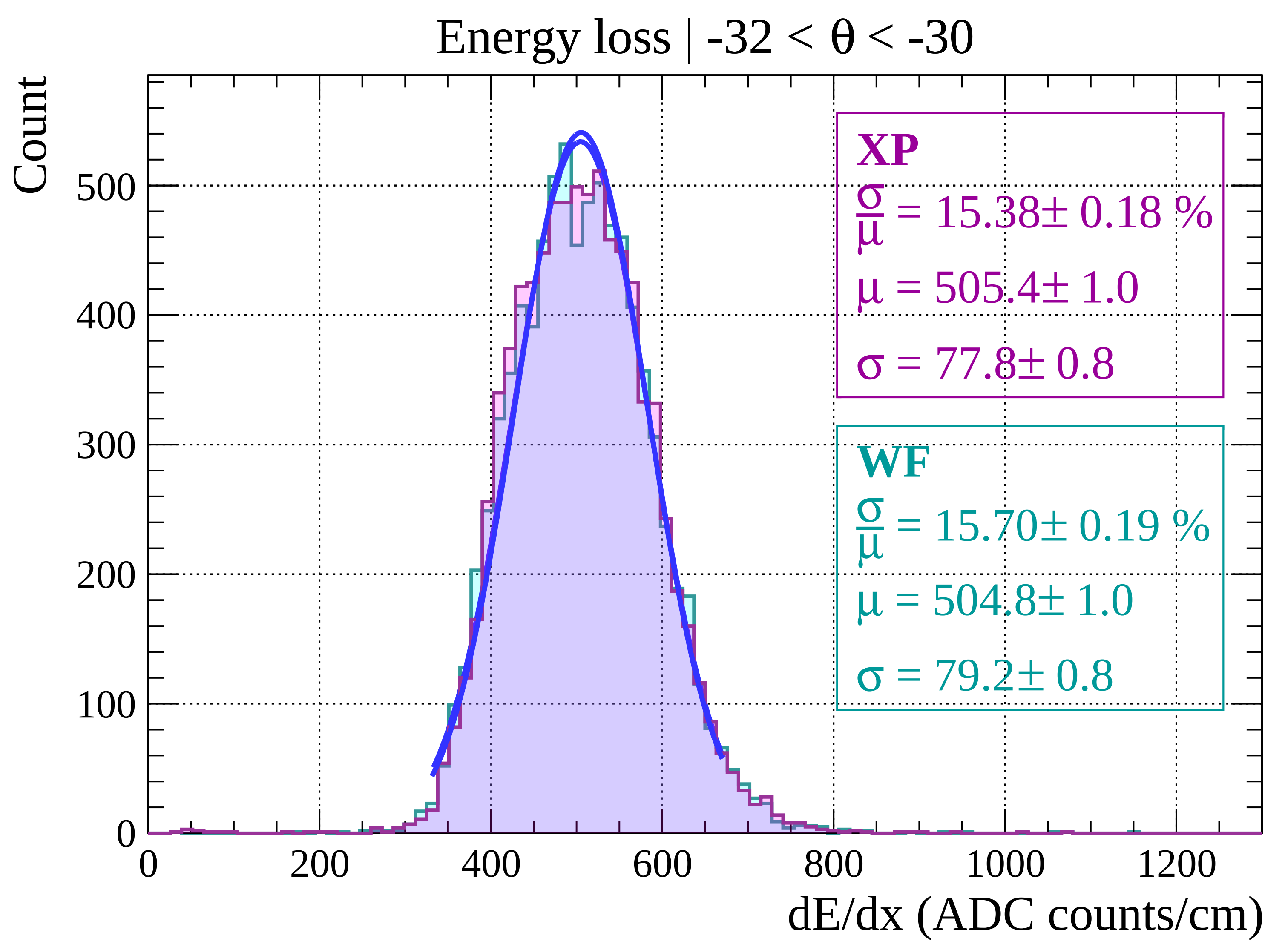
Energy loss | -38 < θ < -36
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 15.97± 0.27 %
μ = 512.3± 1.4
σ = 81.8± 1.1
WF
σ/μ = 16.05± 0.27 %
μ = 511.9± 1.5
σ = 82.2± 1.1



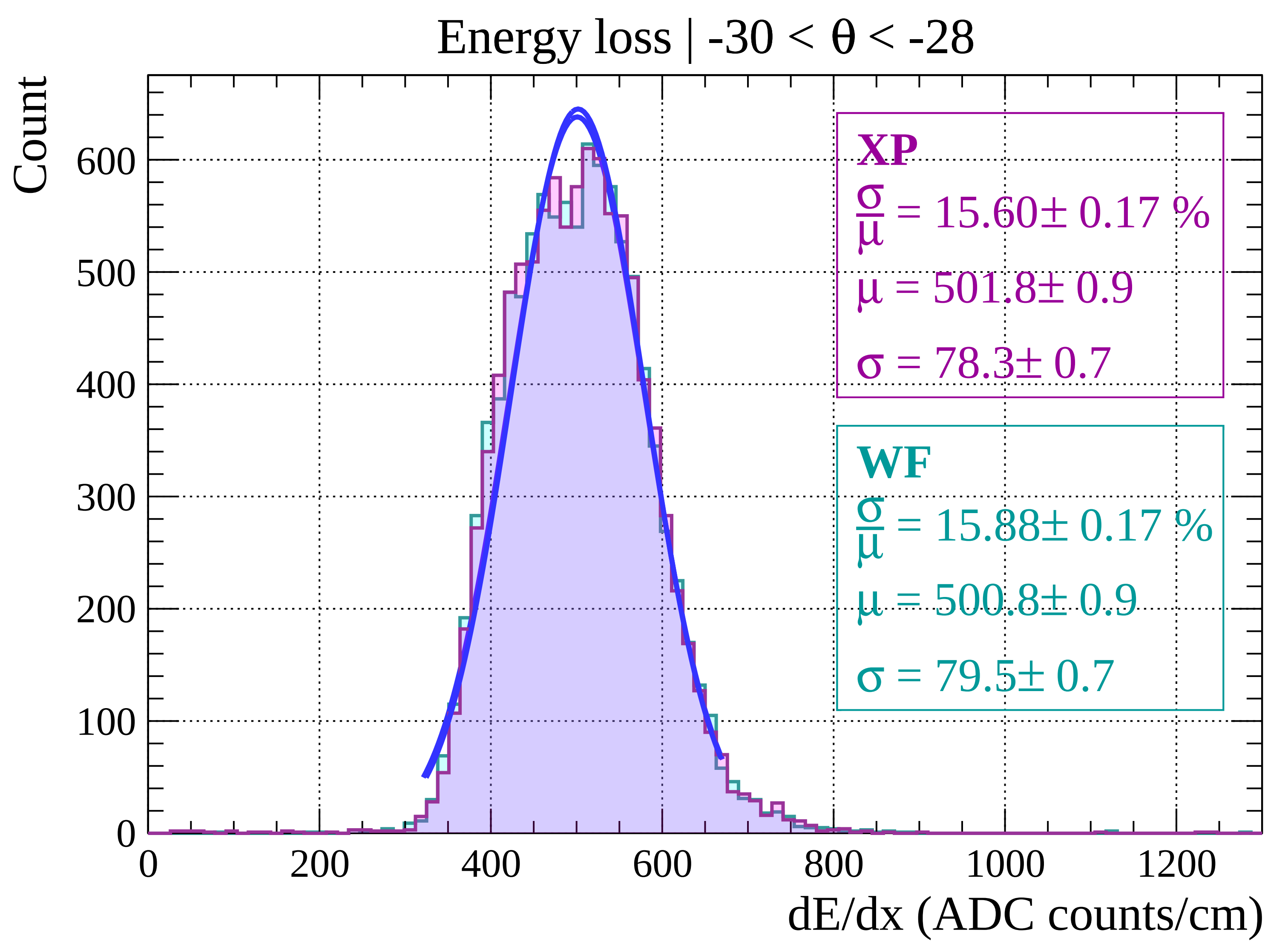
Energy loss | -36 < θ < -34
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.92± 0.24 %
μ = 510.1± 1.3
σ = 81.2± 1.0
WF
σ/μ = 16.40± 0.25 %
μ = 508.3± 1.2
σ = 83.4± 1.1
0
50
100
150
200
250
300
350
400
0
200
400
600
800
1000
1200



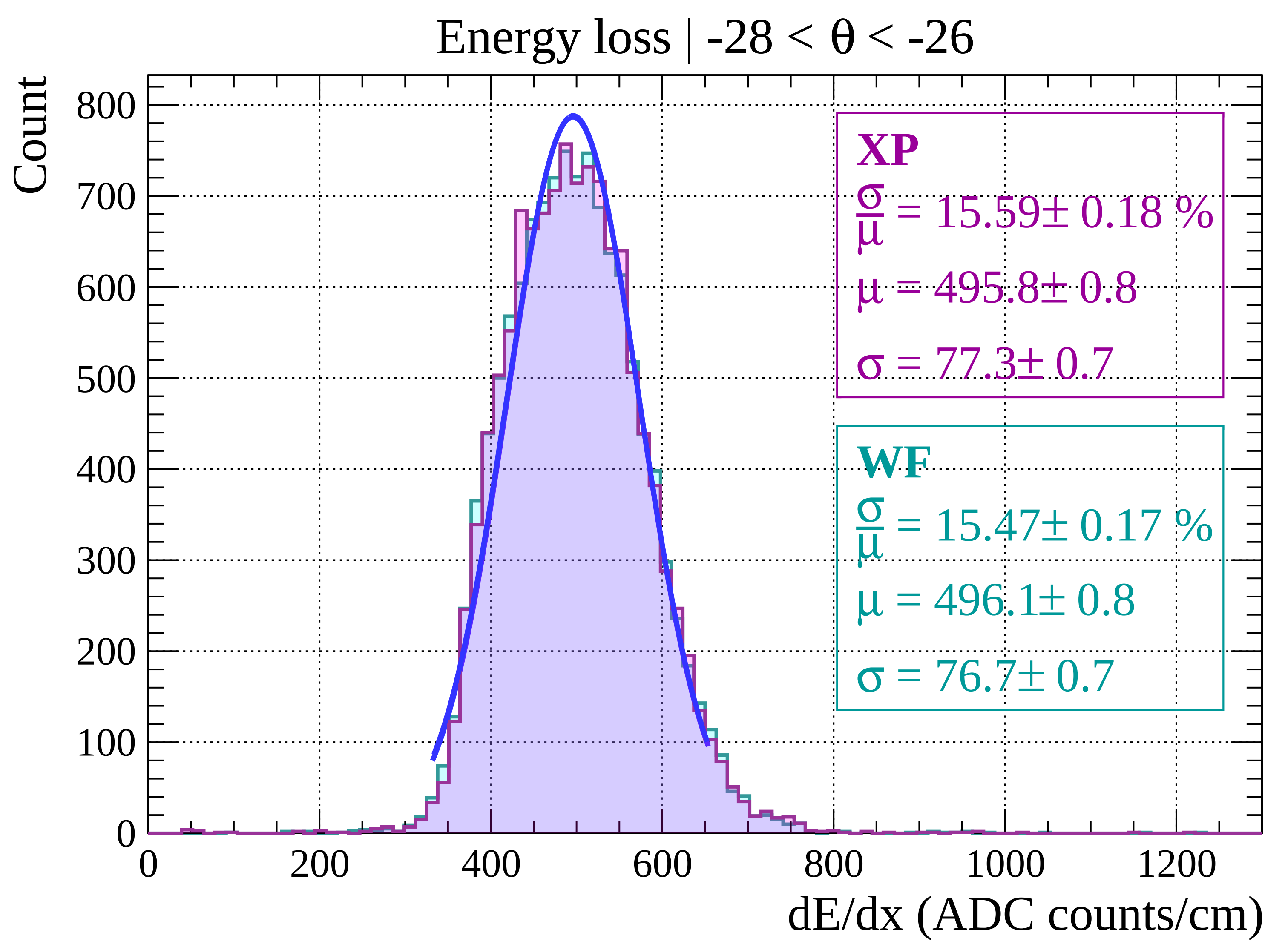
Energy loss | -34 < θ < -32
Count
450
400
350
300
250
200
150
100
50
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.59± 0.20 %
μ = 508.5± 1.1
σ = 79.3± 0.8
WF
σ/μ = 15.94± 0.22 %
μ = 507.2± 1.2
σ = 80.9± 0.9



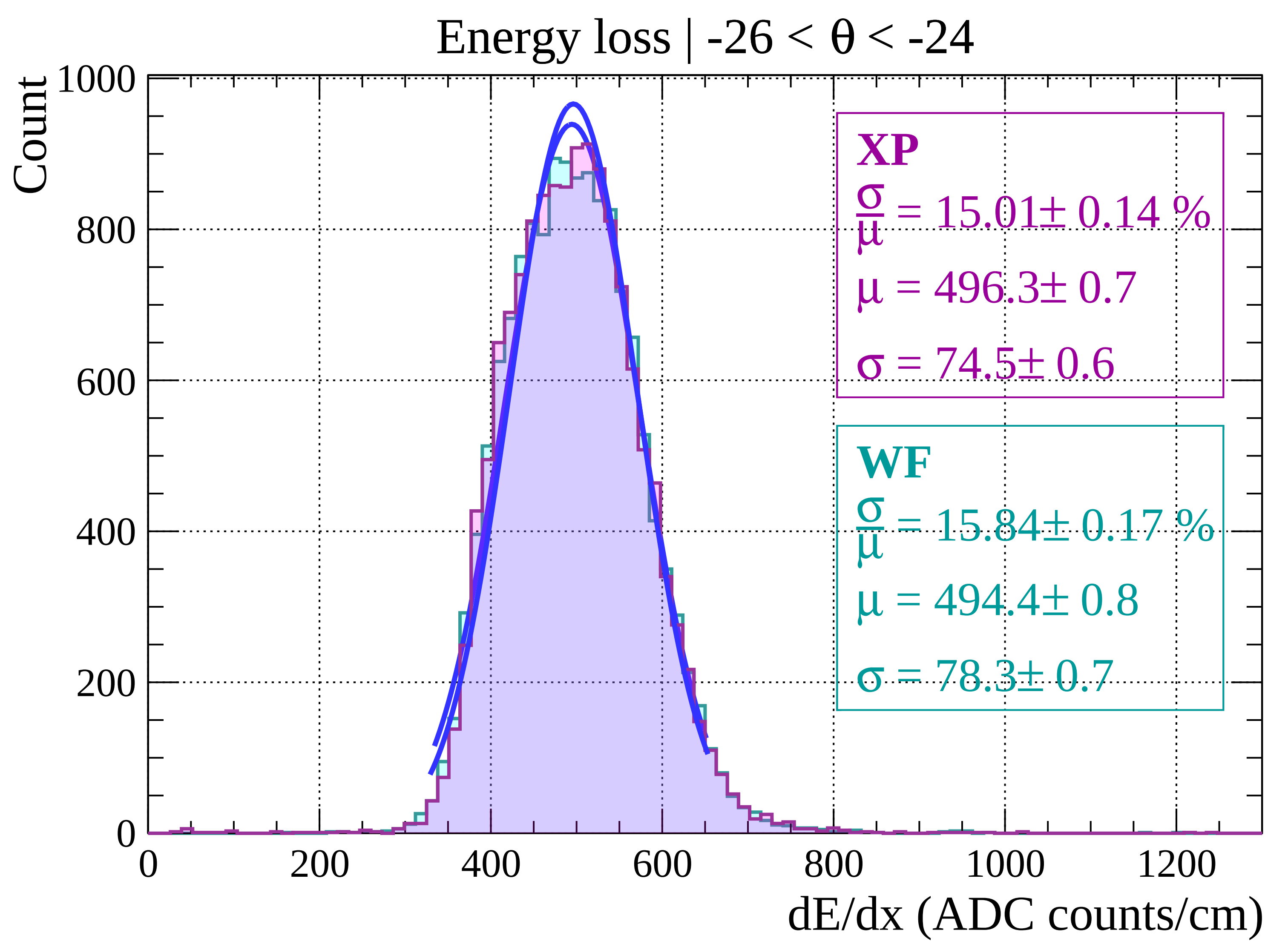
Energy loss | -32 < θ < -30
Count
500
400
300
200
100
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.38± 0.18 %
μ = 505.4± 1.0
σ = 77.8± 0.8
WF
σ/μ = 15.70± 0.19 %
μ = 504.8± 1.0
σ = 79.2± 0.8



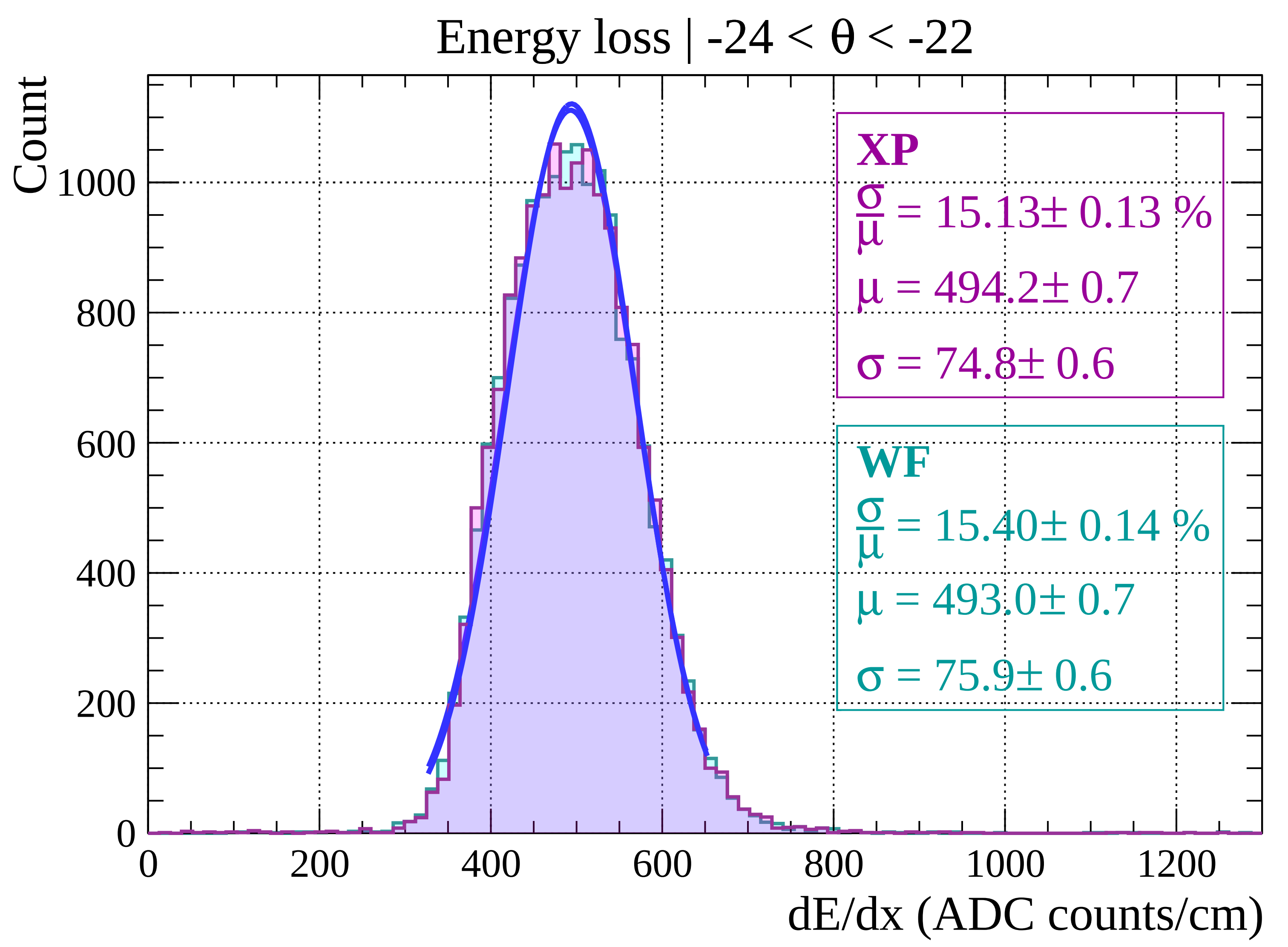
Energy loss | -30 < θ < -28
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 15.60± 0.17 %
μ = 501.8± 0.9
σ = 78.3± 0.7
WF
σ/μ = 15.88± 0.17 %
μ = 500.8± 0.9
σ = 79.5± 0.7



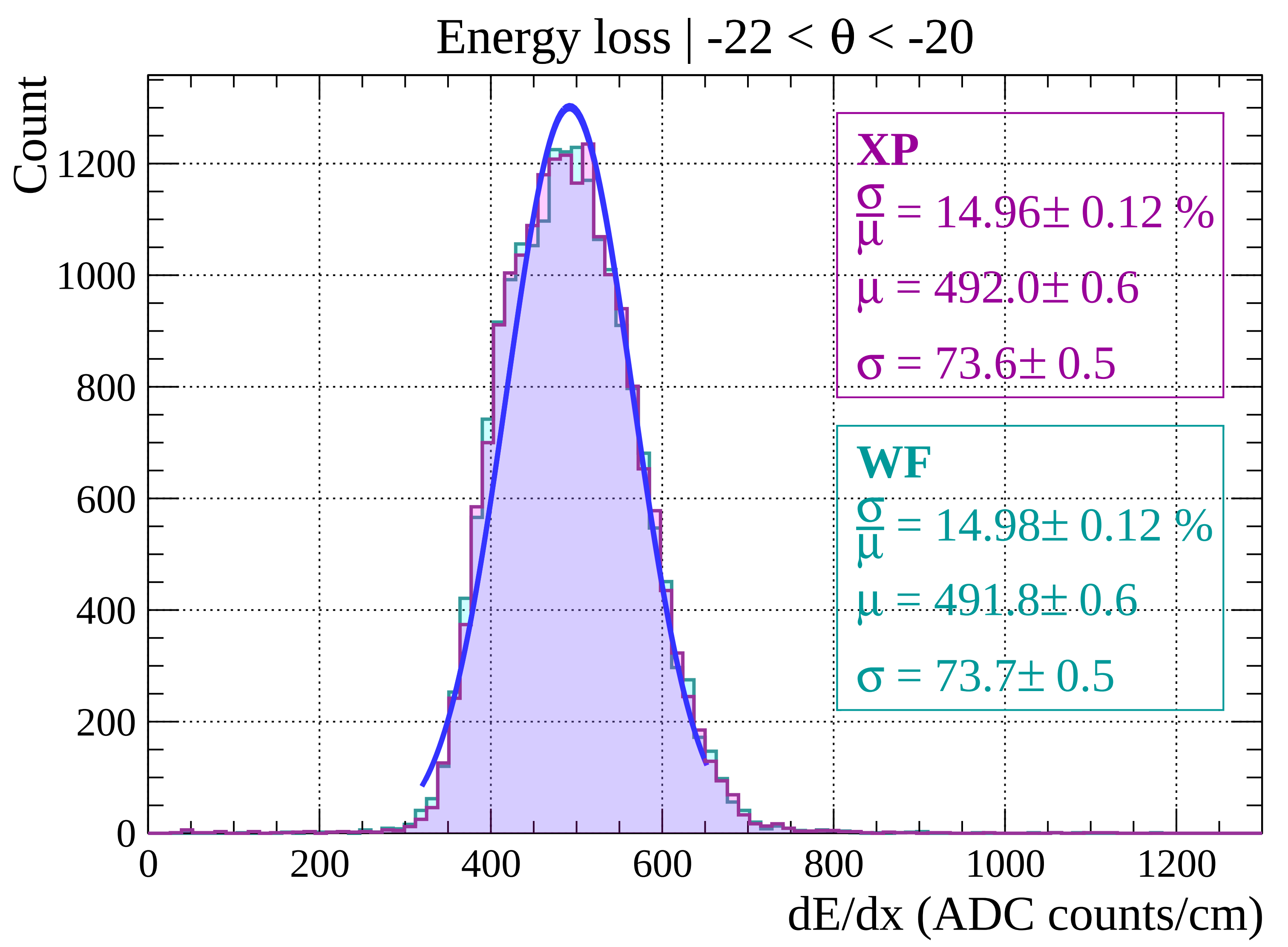
Energy loss | -28 < θ < -26
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.59± 0.18 %
μ = 495.8± 0.8
σ = 77.3± 0.7
WF
σ/μ = 15.47± 0.17 %
μ = 496.1± 0.8
σ = 76.7± 0.7



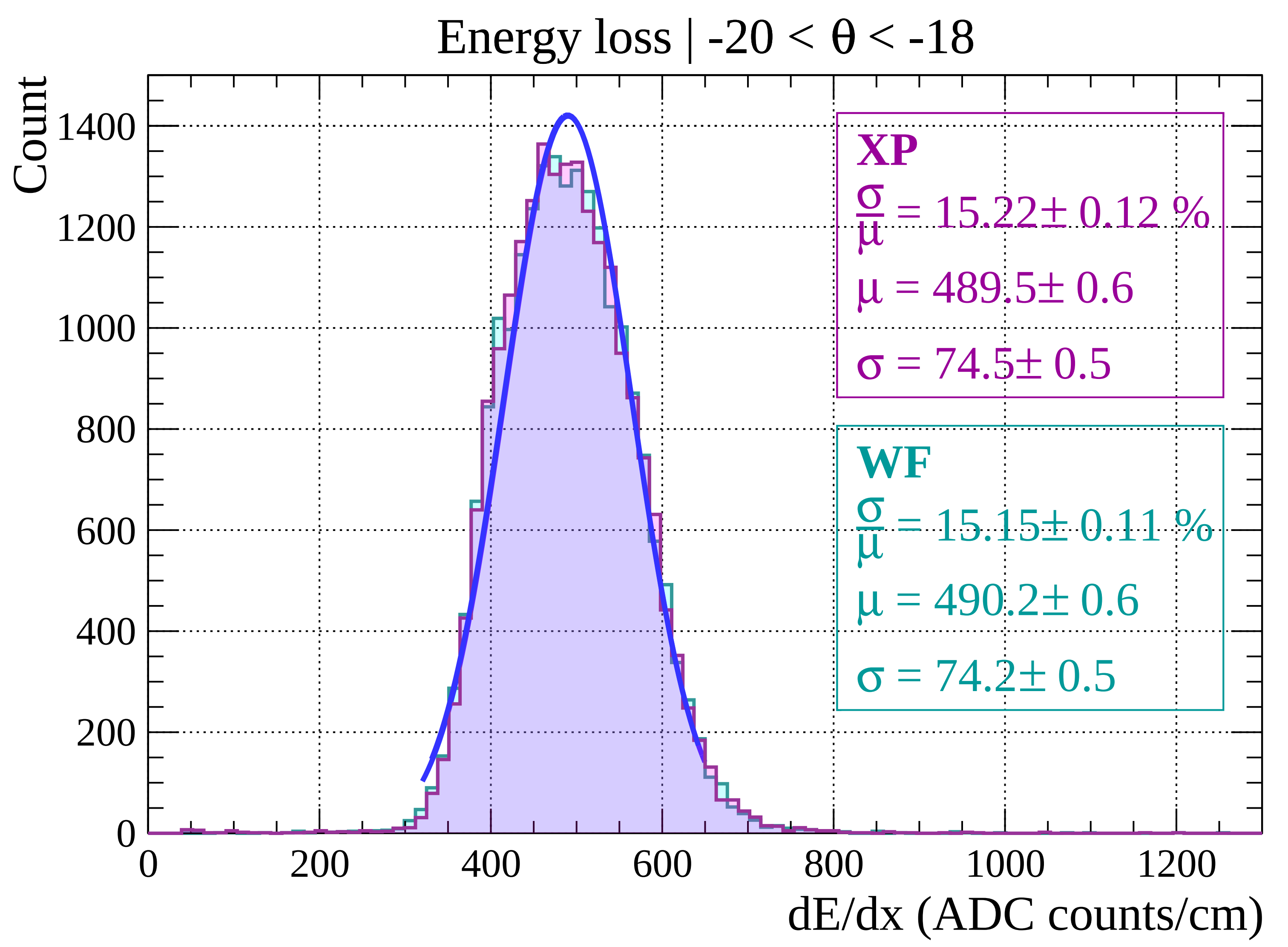
Energy loss | -26 < θ < -24
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 15.01 ± 0.14 %
μ = 496.3 ± 0.7
σ = 74.5 ± 0.6
WF
σ/μ = 15.84 ± 0.17 %
μ = 494.4 ± 0.8
σ = 78.3 ± 0.7



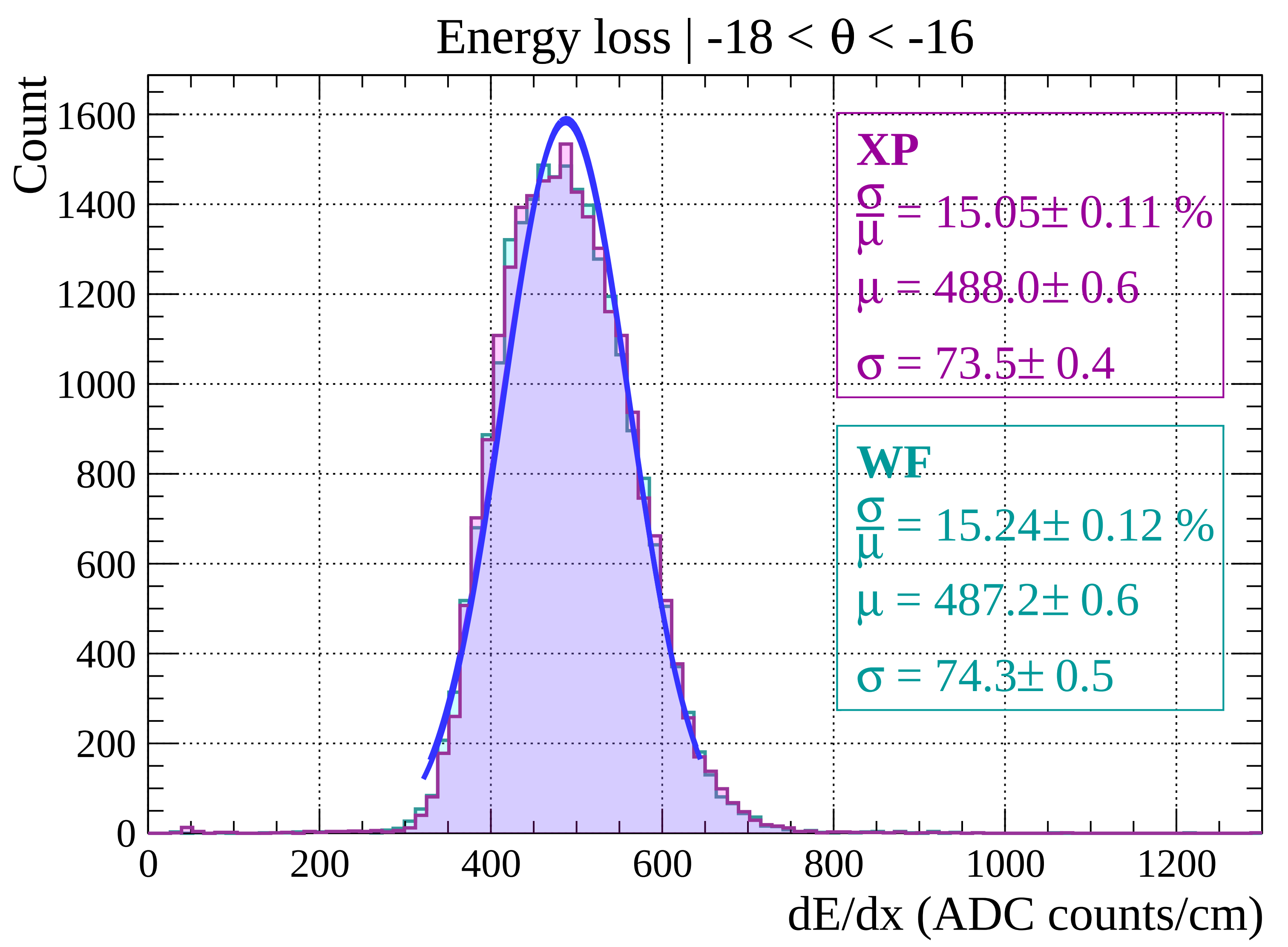
Energy loss | -24 < θ < -22
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 15.13± 0.13 %
μ = 494.2± 0.7
σ = 74.8± 0.6
WF
σ/μ = 15.40± 0.14 %
μ = 493.0± 0.7
σ = 75.9± 0.6



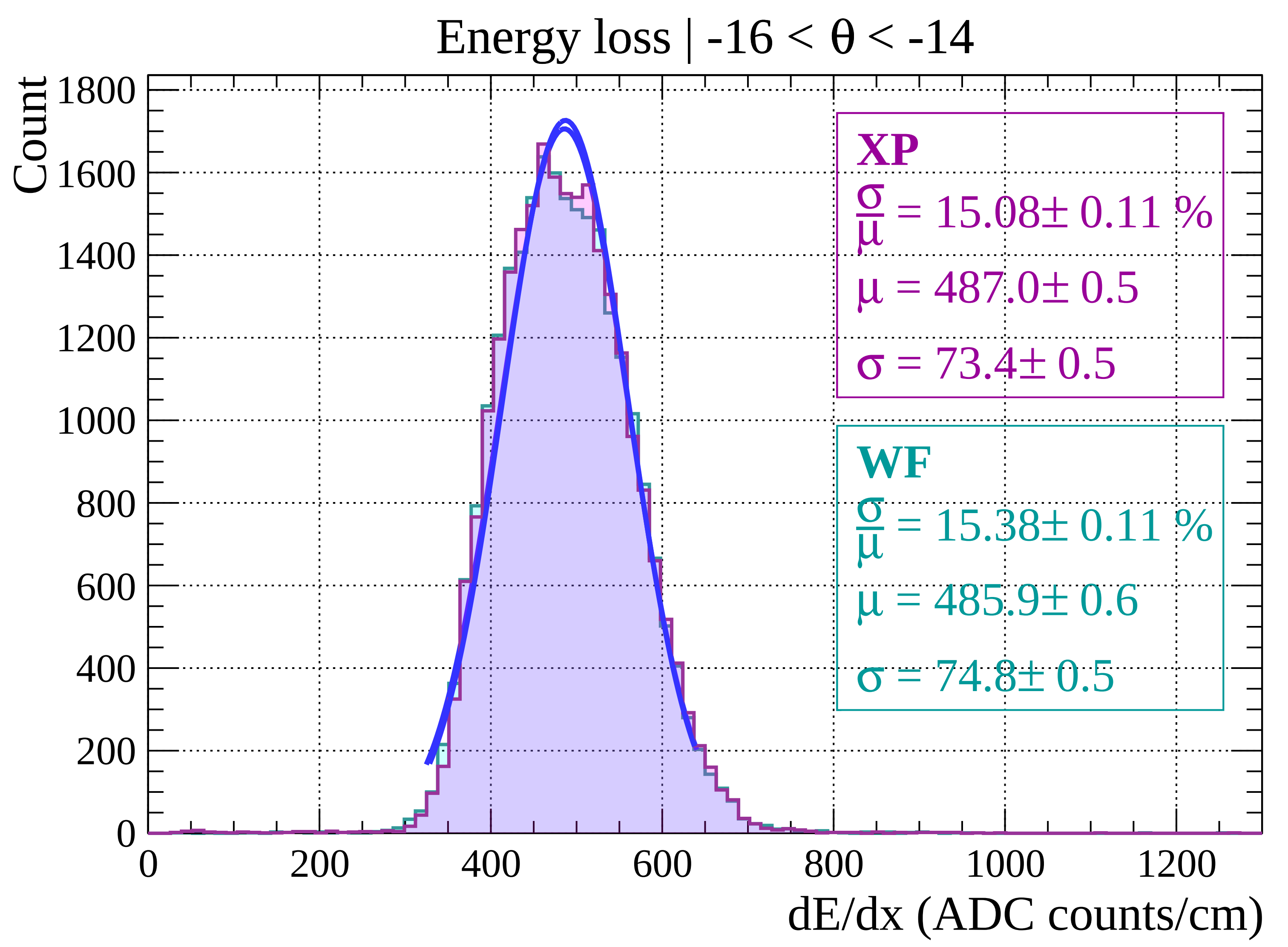
Energy loss | -22 < θ < -20
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 14.96± 0.12 %
μ = 492.0± 0.6
σ = 73.6± 0.5
WF
σ/μ = 14.98± 0.12 %
μ = 491.8± 0.6
σ = 73.7± 0.5



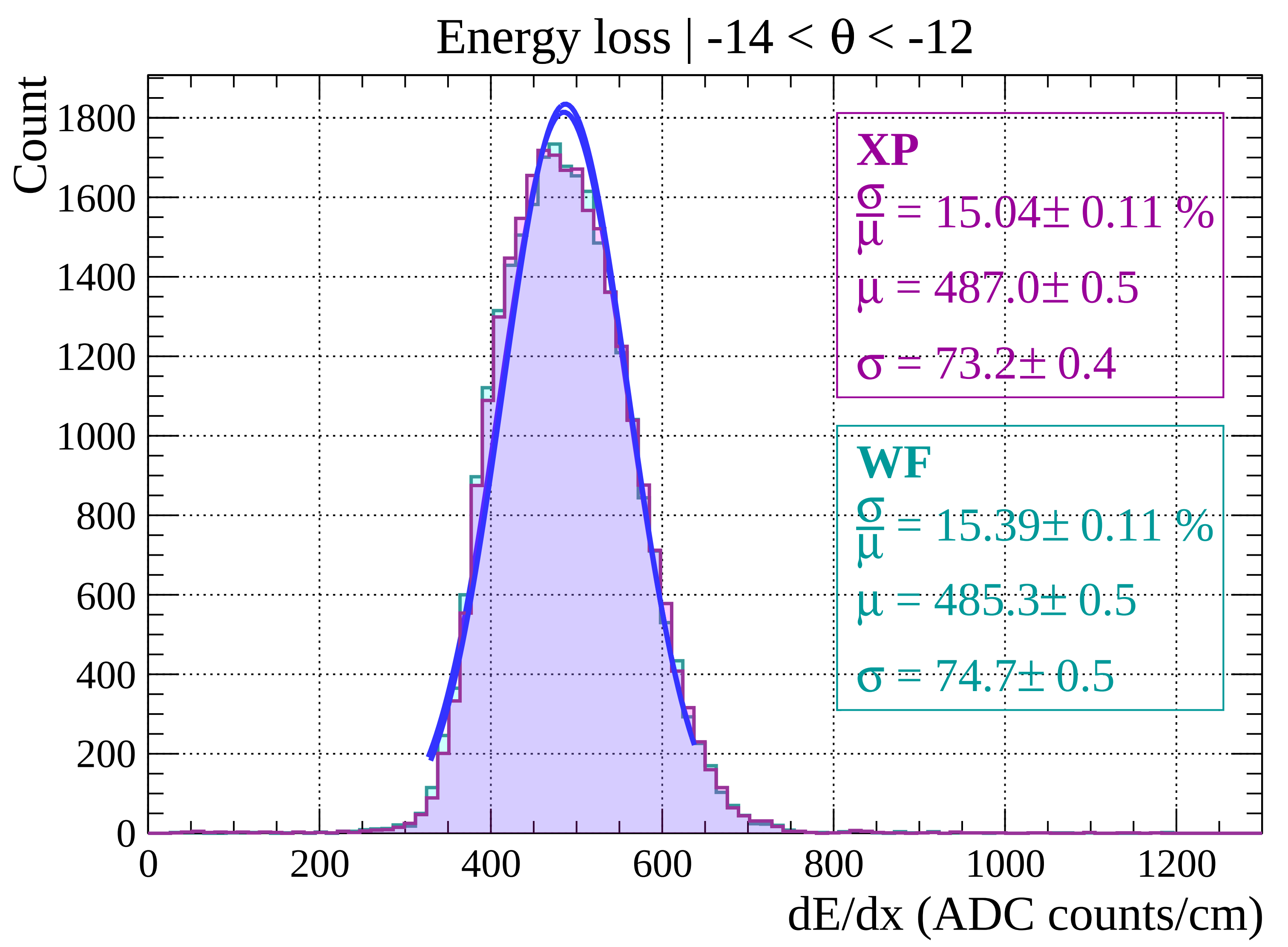
Energy loss | -20 < θ < -18
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.22± 0.12 %
μ = 489.5± 0.6
σ = 74.5± 0.5
WF
σ/μ = 15.15± 0.11 %
μ = 490.2± 0.6
σ = 74.2± 0.5
0
200
400
600
800
1000
1200
1400



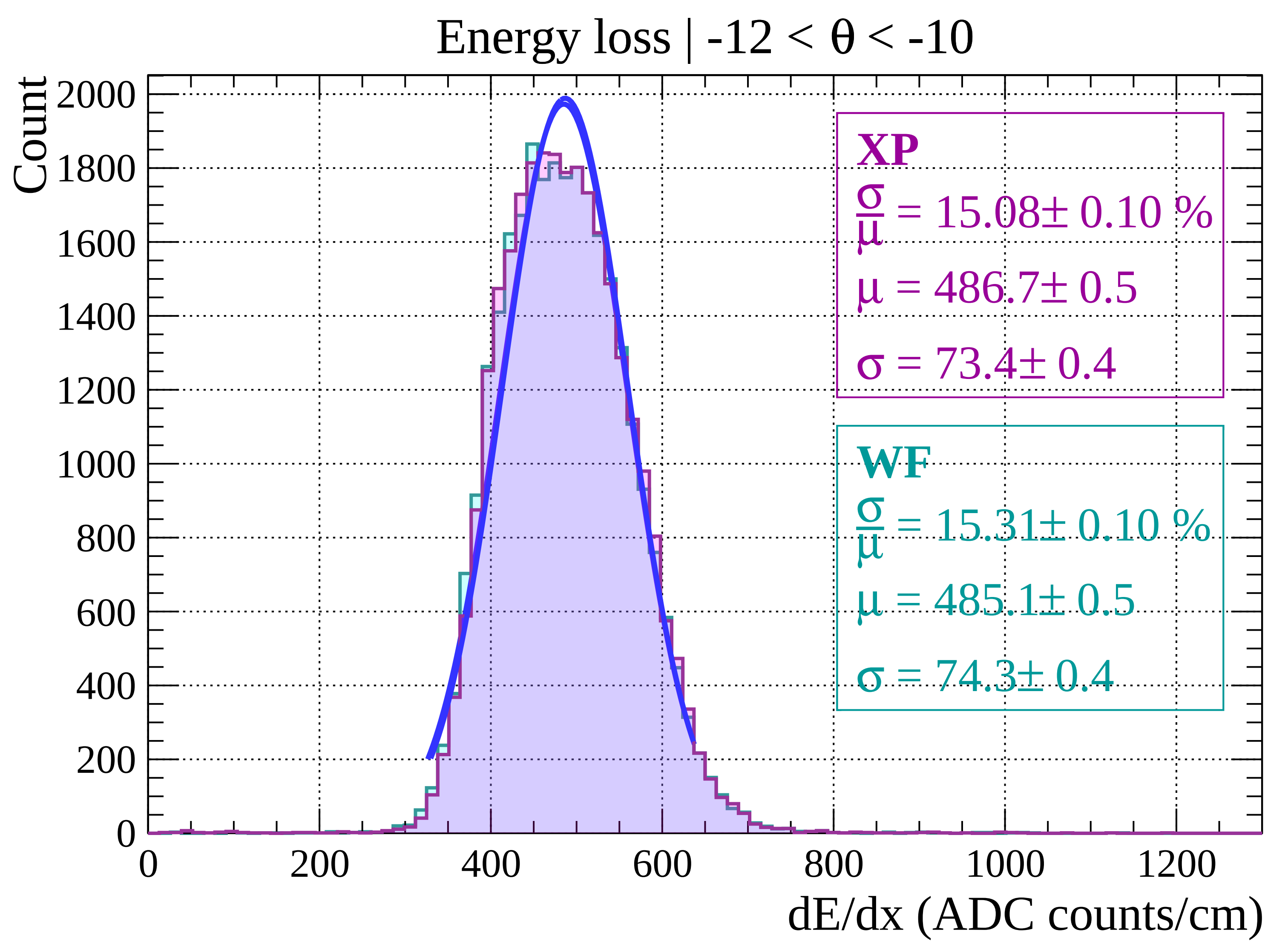
Energy loss | -18 < θ < -16
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.05± 0.11 %
μ = 488.0± 0.6
σ = 73.5± 0.4
WF
σ/μ = 15.24± 0.12 %
μ = 487.2± 0.6
σ = 74.3± 0.5



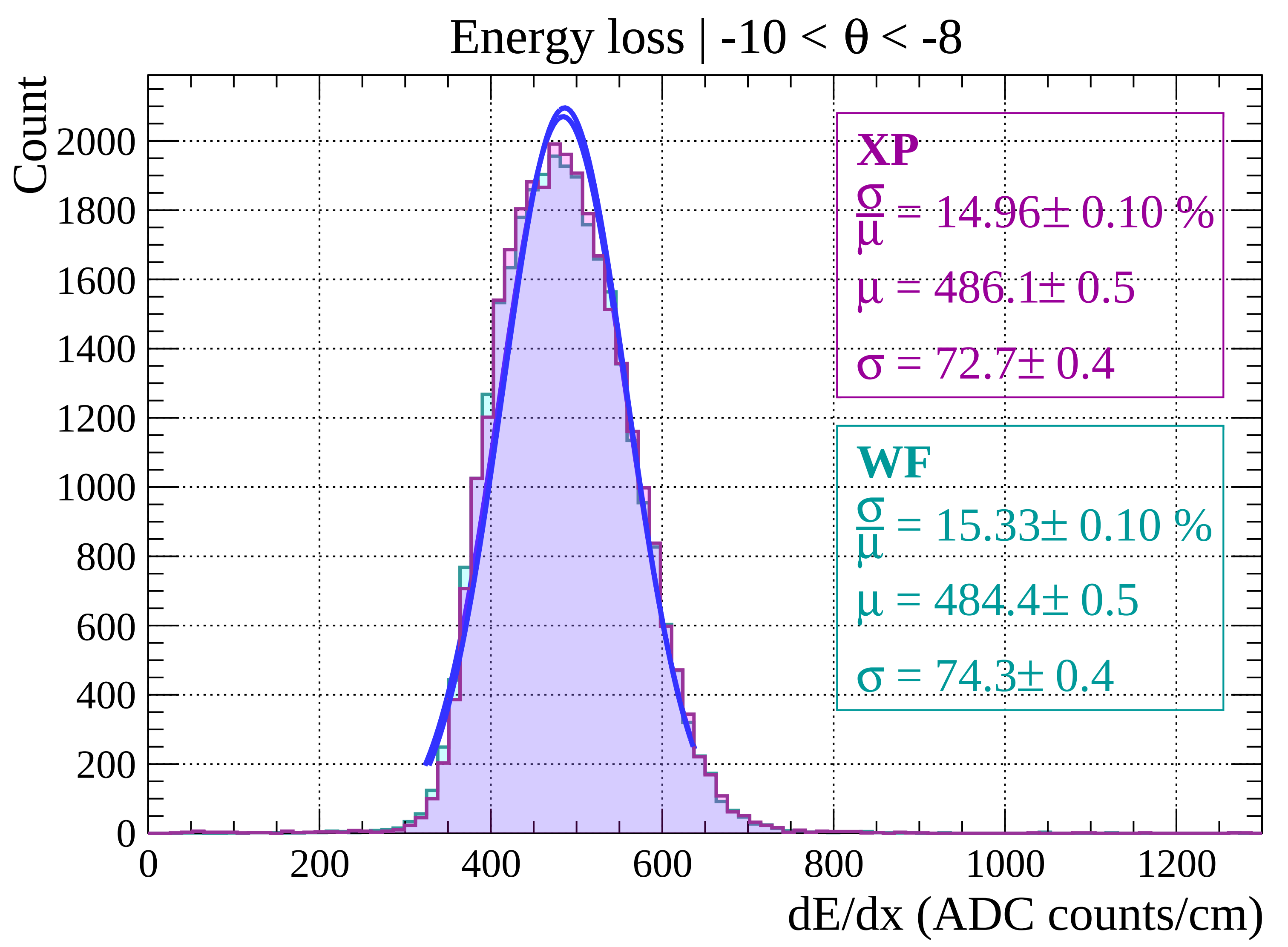
Energy loss | -16 < θ < -14
Count
1800
1600
1400
1200
1000
800
600
400
200
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.08± 0.11 %
μ = 487.0± 0.5
σ = 73.4± 0.5
WF
σ/μ = 15.38± 0.11 %
μ = 485.9± 0.6
σ = 74.8± 0.5



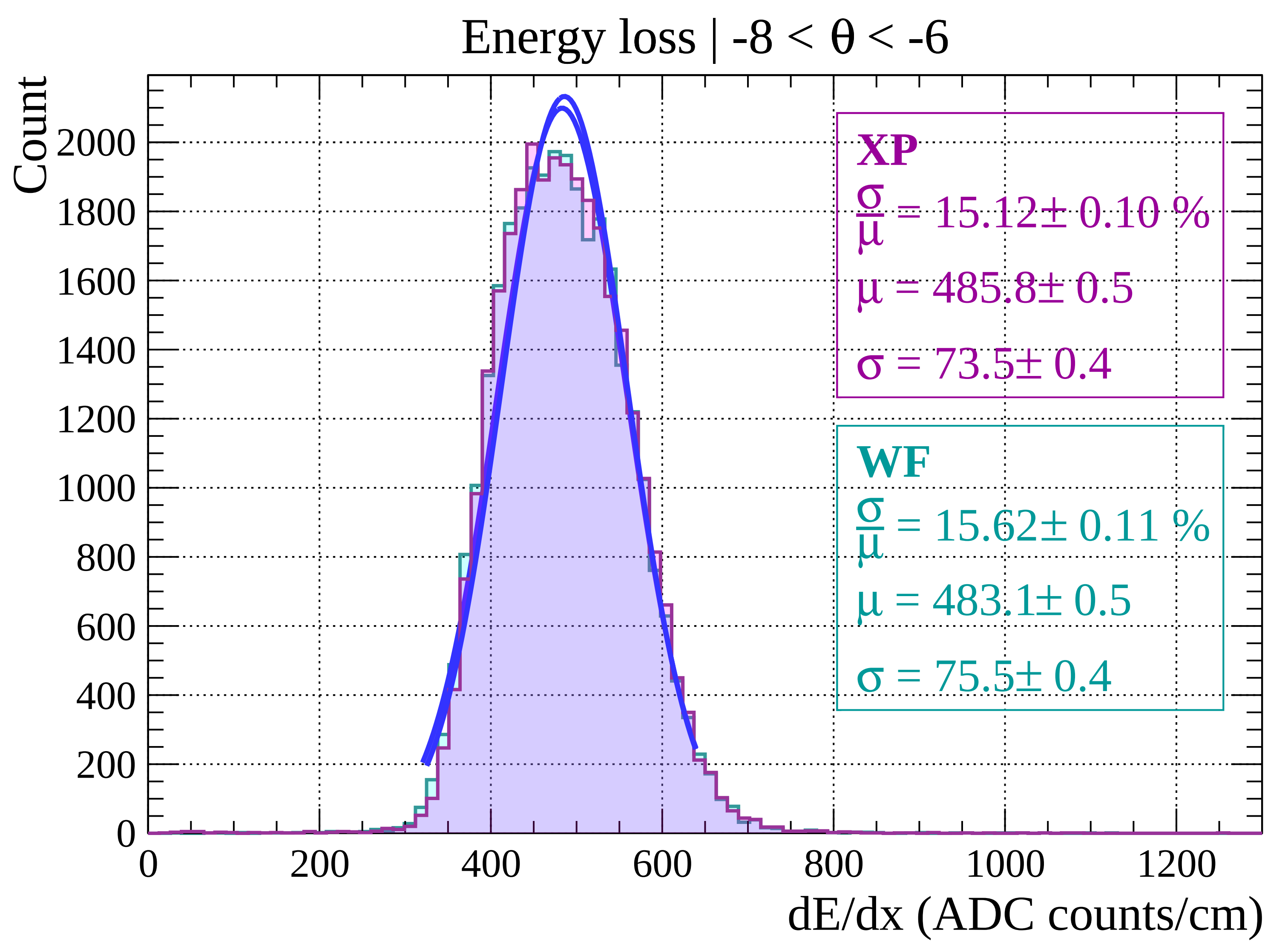
Energy loss | -14 < θ < -12
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.04± 0.11 %
μ = 487.0± 0.5
σ = 73.2± 0.4
WF
σ/μ = 15.39± 0.11 %
μ = 485.3± 0.5
σ = 74.7± 0.5



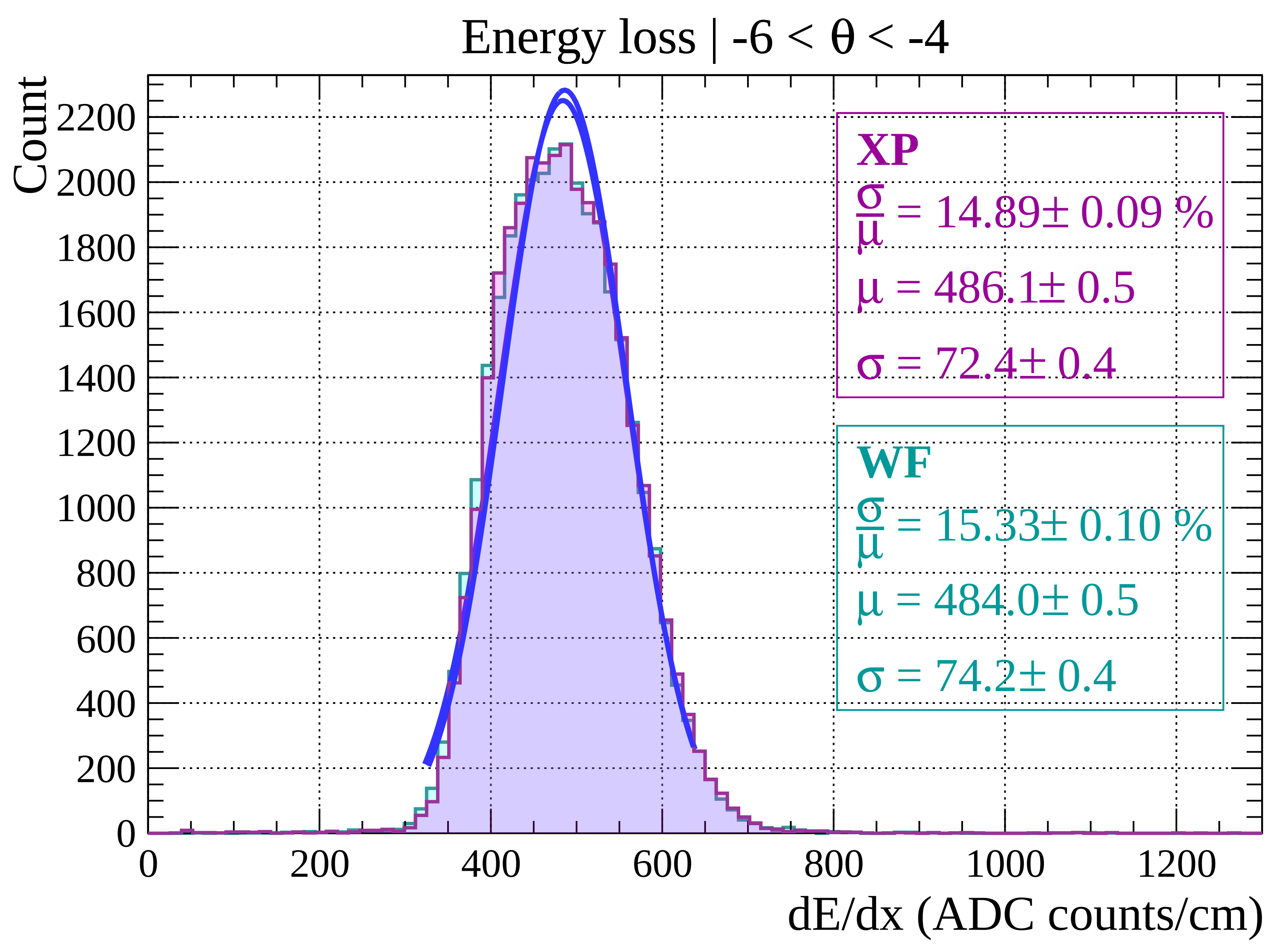
Energy loss | -12 < θ < -10
Count
XP
σ/μ = 15.08± 0.10 %
μ = 486.7± 0.5
σ = 73.4± 0.4
WF
σ/μ = 15.31± 0.10 %
μ = 485.1± 0.5
σ = 74.3± 0.4
dE/dx (ADC counts/cm)

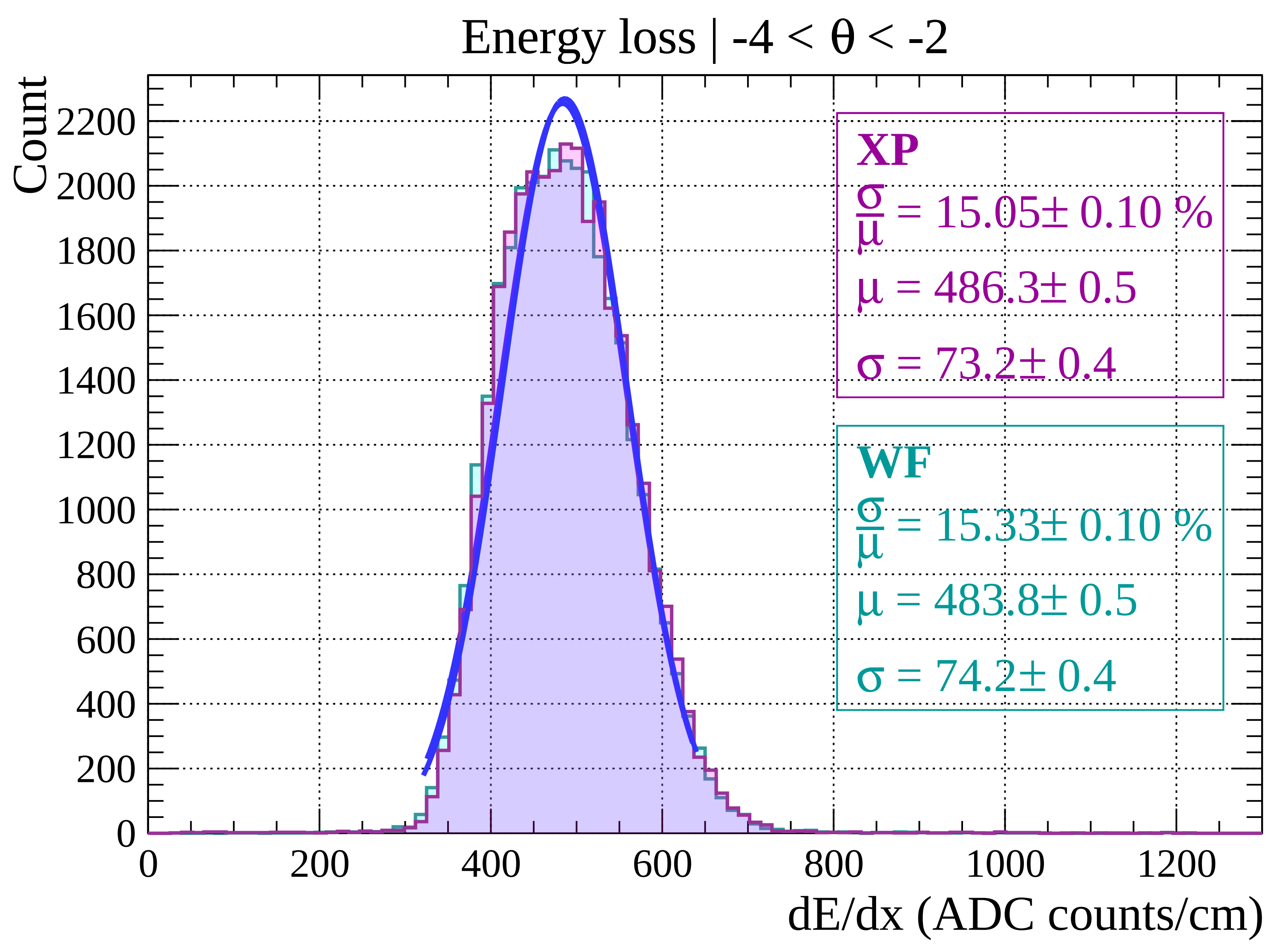


Energy loss | -10 < θ < -8
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 14.96± 0.10 %
μ = 486.1± 0.5
σ = 72.7± 0.4
WF
σ/μ = 15.33± 0.10 %
μ = 484.4± 0.5
σ = 74.3± 0.4

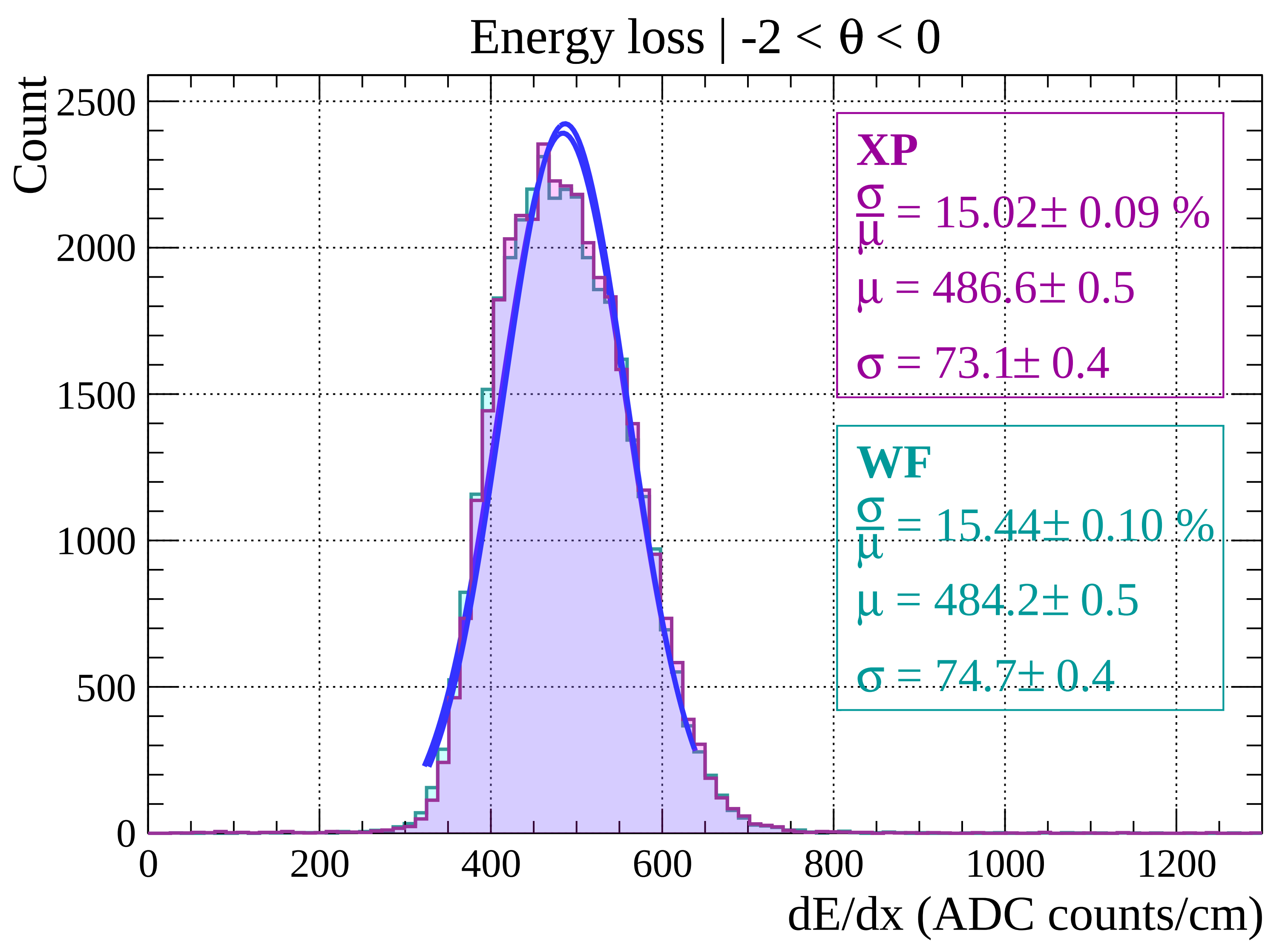


Energy loss | -8 < θ < -6
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
0
200
400
600
800
1000
1200
XP
σ/μ = 15.12 ± 0.10 %
μ = 485.8 ± 0.5
σ = 73.5 ± 0.4
WF
σ/μ = 15.62 ± 0.11 %
μ = 483.1 ± 0.5
σ = 75.5 ± 0.4

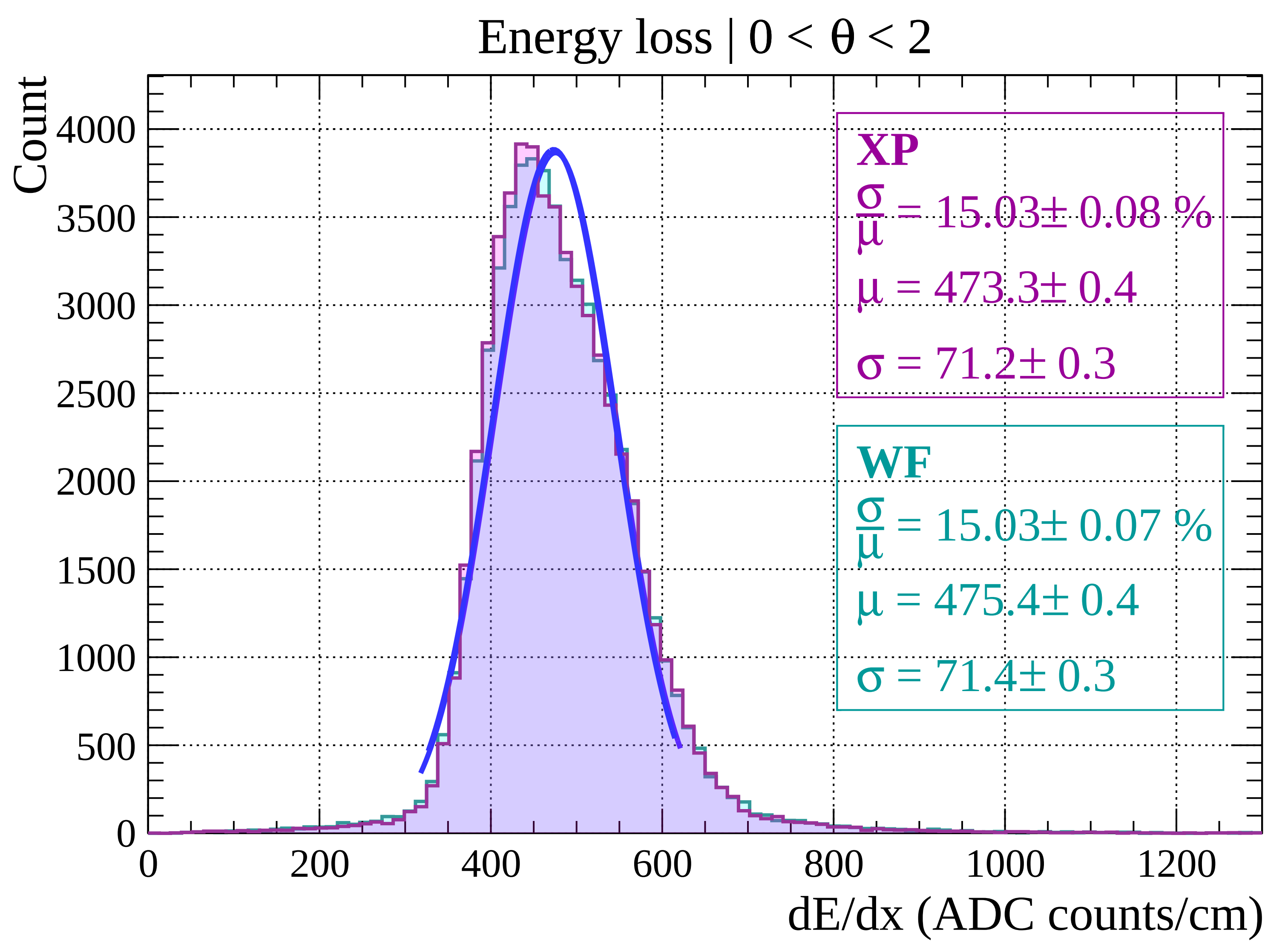

Energy loss | -6 < θ < -4
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 14.89± 0.09 %
μ = 486.1± 0.5
σ = 72.4± 0.4
WF
$\frac{\sigma}{\mu}$ = 15.33± 0.10 %
μ = 484.0± 0.5
σ = 74.2± 0.4



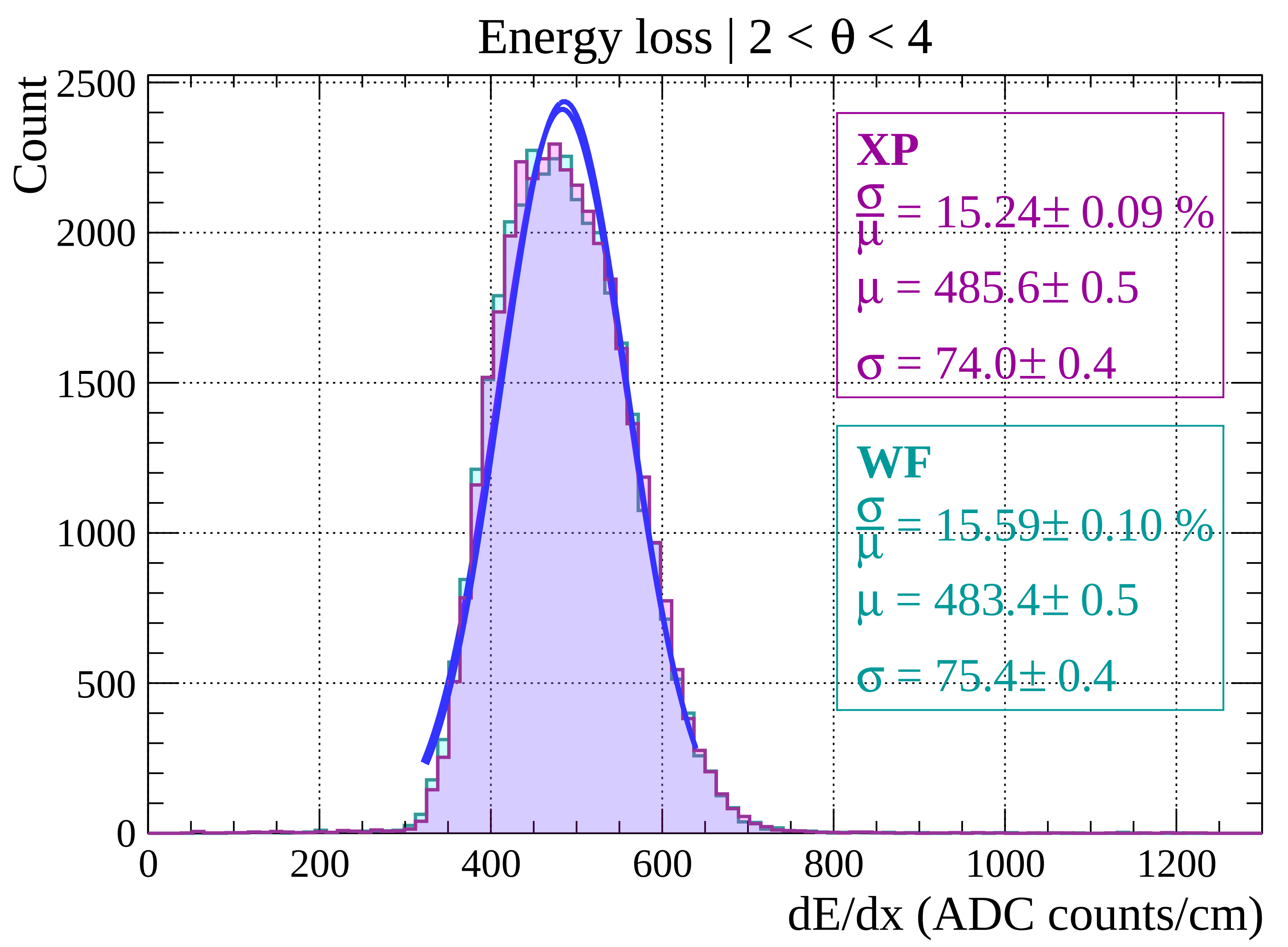
Energy loss | -4 < θ < -2
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
0
200
400
600
800
1000
1200
XP
σ/μ = 15.05± 0.10 %
μ = 486.3± 0.5
σ = 73.2± 0.4
WF
σ/μ = 15.33± 0.10 %
μ = 483.8± 0.5
σ = 74.2± 0.4



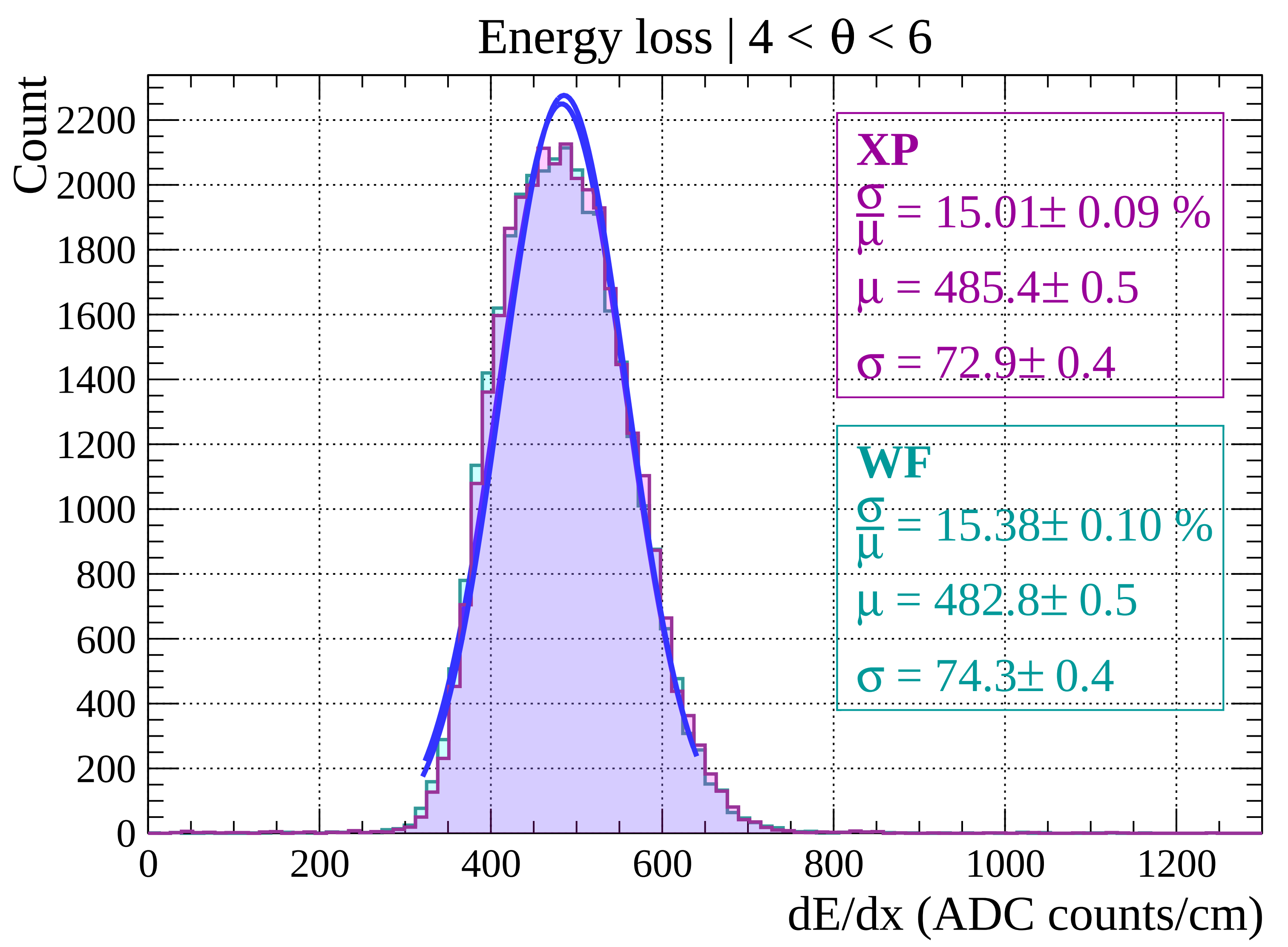
Energy loss | -2 < θ < 0
Count
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.02± 0.09 %
μ = 486.6± 0.5
σ = 73.1± 0.4
WF
σ/μ = 15.44± 0.10 %
μ = 484.2± 0.5
σ = 74.7± 0.4



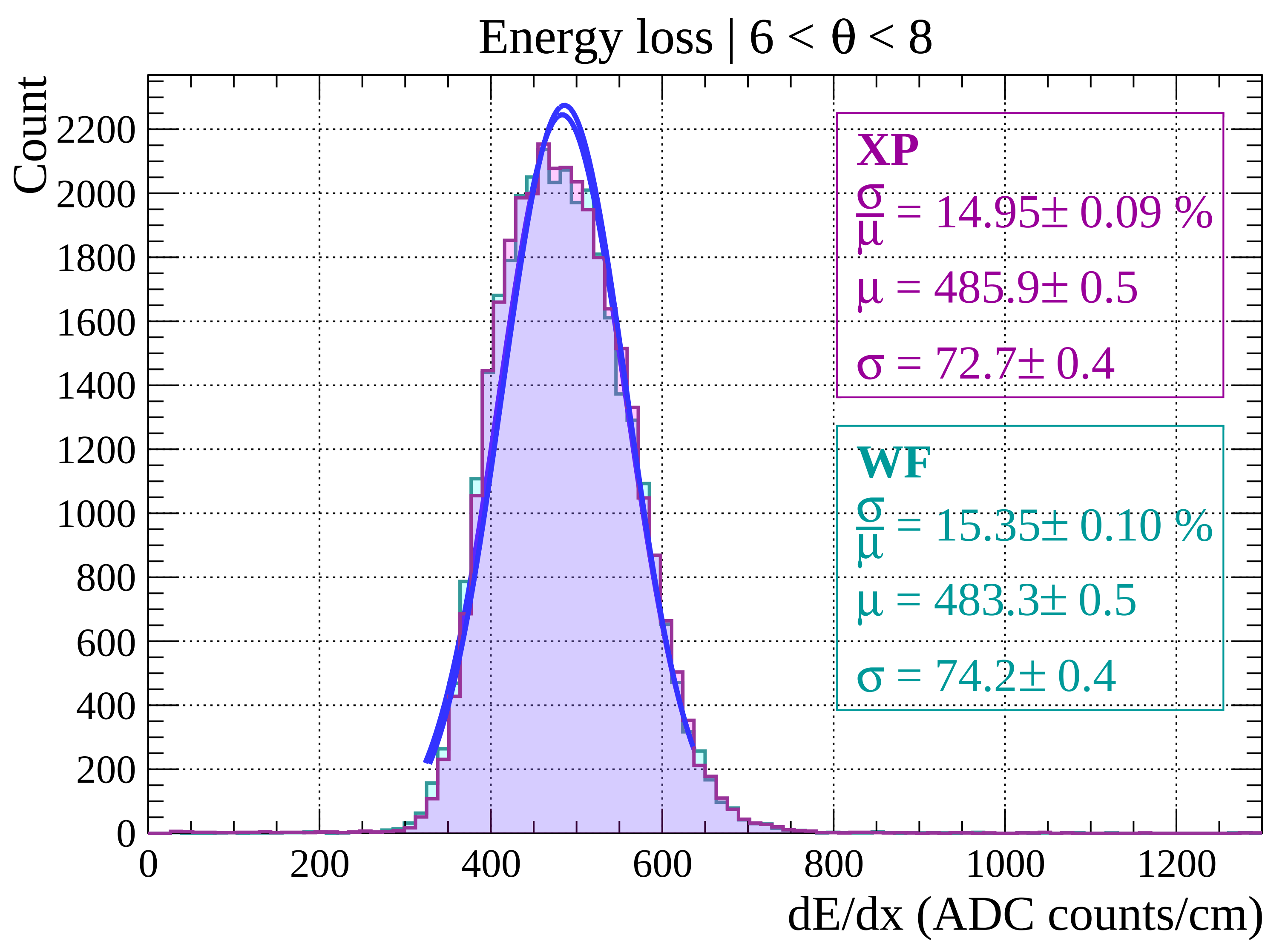
Energy loss | 0 < θ < 2
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.03± 0.08 %
μ = 473.3± 0.4
σ = 71.2± 0.3
WF
σ/μ = 15.03± 0.07 %
μ = 475.4± 0.4
σ = 71.4± 0.3
0
500
1000
1500
2000
2500
3000
3500
4000
0
200
400
600
800
1000
1200



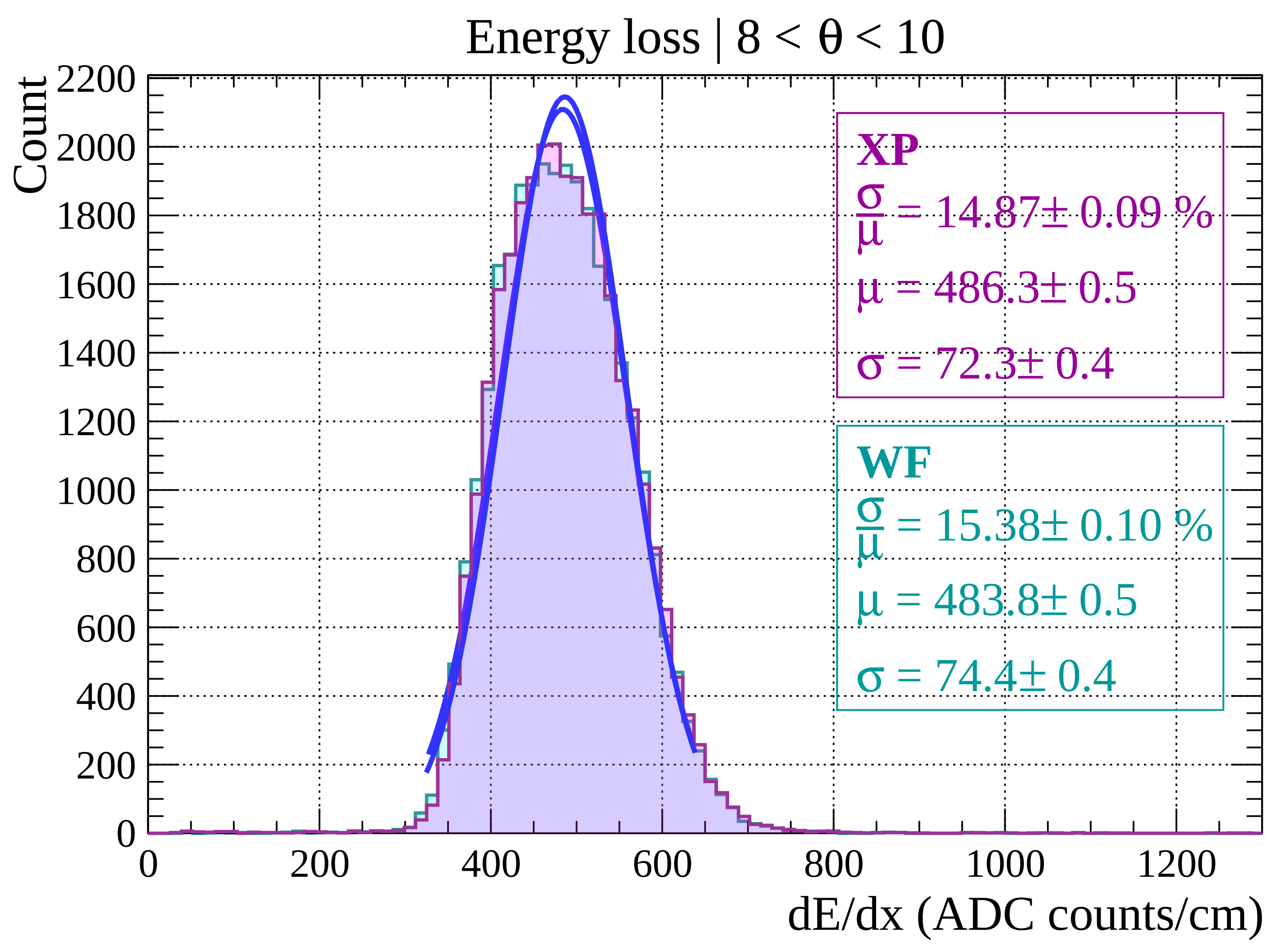
Energy loss | 2 < θ < 4
Count
2500
2000
1500
1000
500
0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 15.24± 0.09 %
μ = 485.6± 0.5
σ = 74.0± 0.4
WF
σ/μ = 15.59± 0.10 %
μ = 483.4± 0.5
σ = 75.4± 0.4

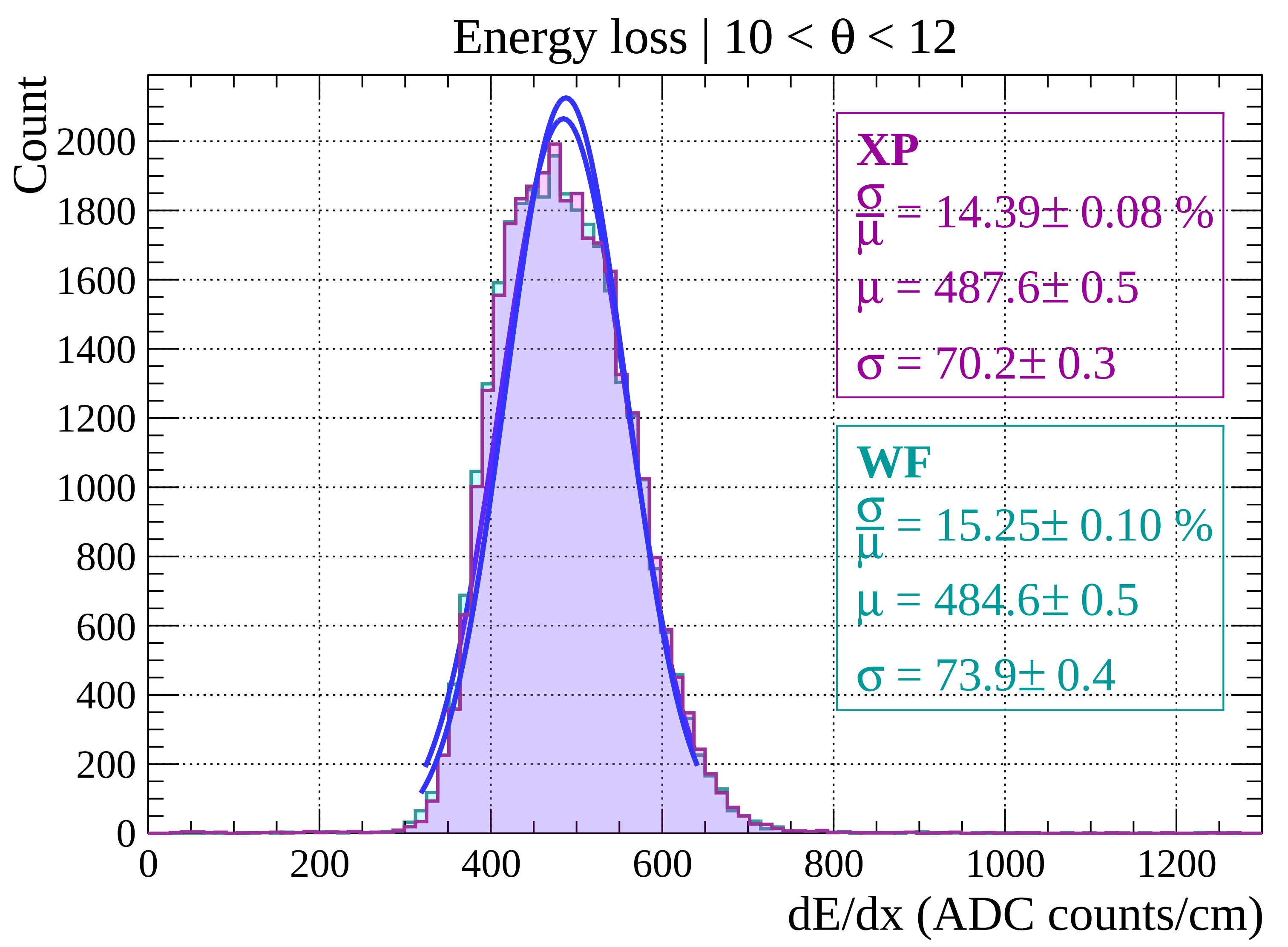


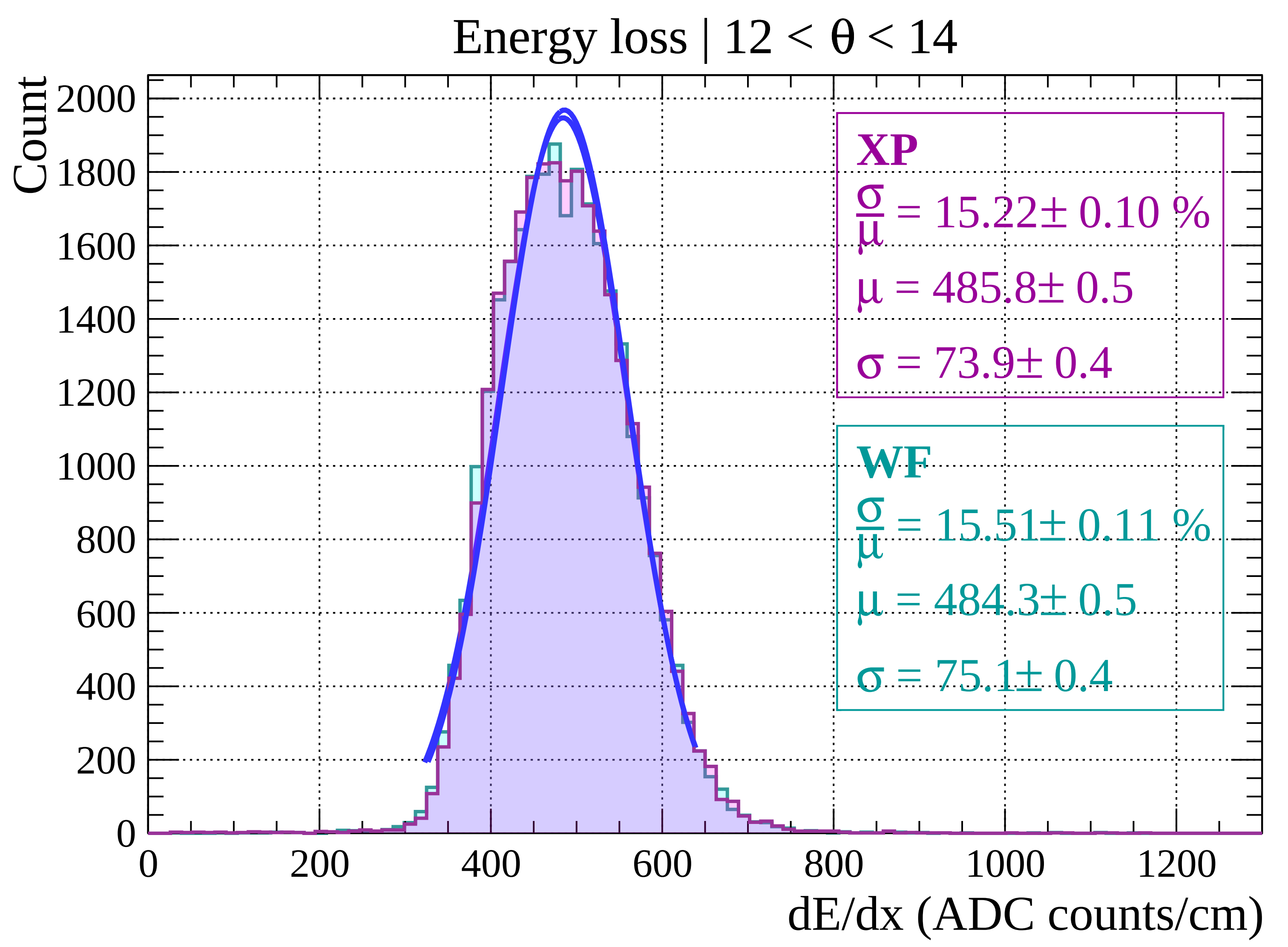
Energy loss | 4 < θ < 6
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.01± 0.09 %
μ = 485.4± 0.5
σ = 72.9± 0.4
WF
σ/μ = 15.38± 0.10 %
μ = 482.8± 0.5
σ = 74.3± 0.4



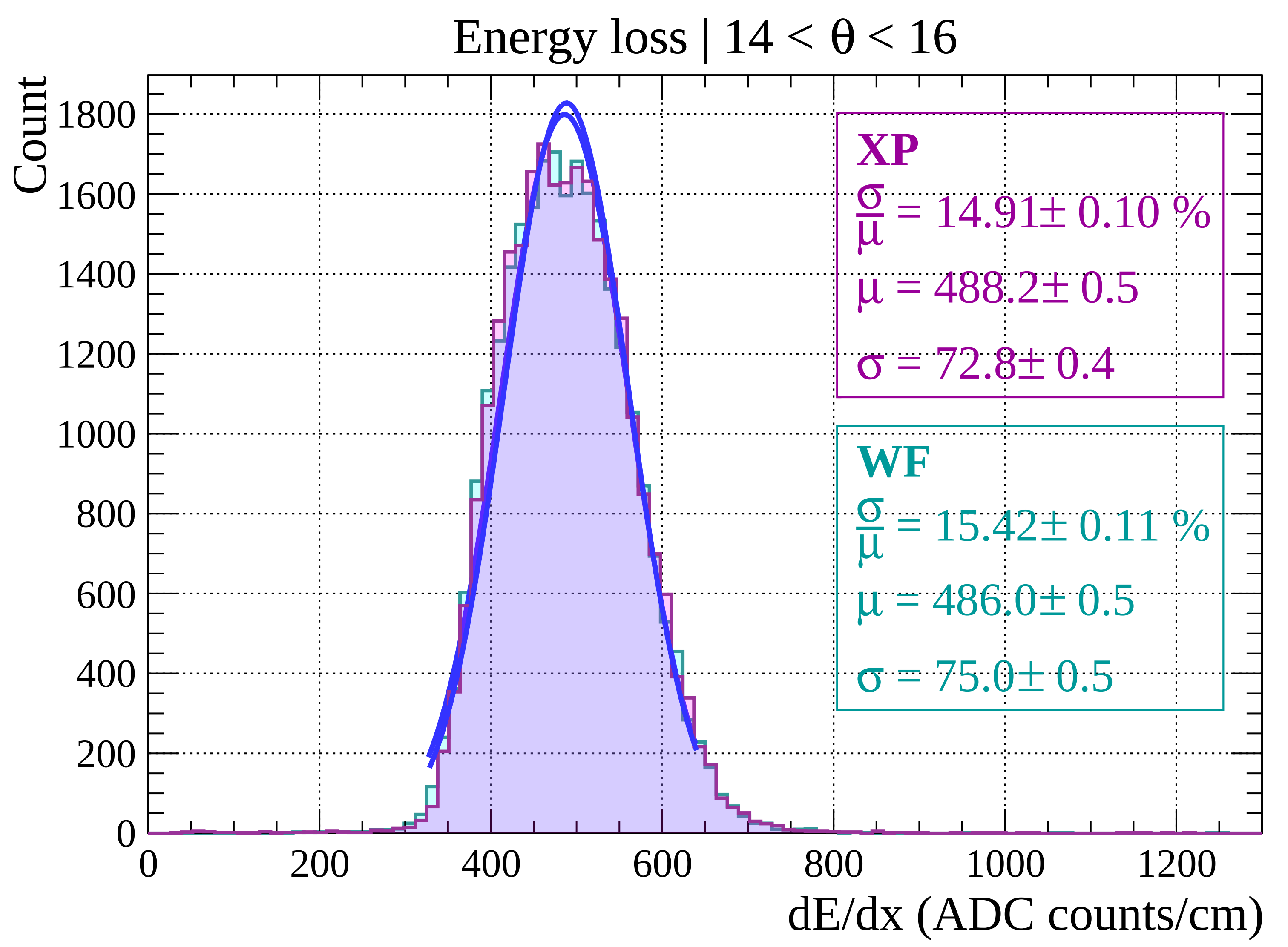
Energy loss | 6 < θ < 8
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200
0
200
400
600
800
1000
1200
XP
σ/μ = 14.95± 0.09 %
μ = 485.9± 0.5
σ = 72.7± 0.4
WF
σ/μ = 15.35± 0.10 %
μ = 483.3± 0.5
σ = 74.2± 0.4


Energy loss | 8 < θ < 10
Count
XP
σ/μ = 14.87± 0.09 %
μ = 486.3± 0.5
σ = 72.3± 0.4
WF
σ/μ = 15.38± 0.10 %
μ = 483.8± 0.5
σ = 74.4± 0.4
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
2200

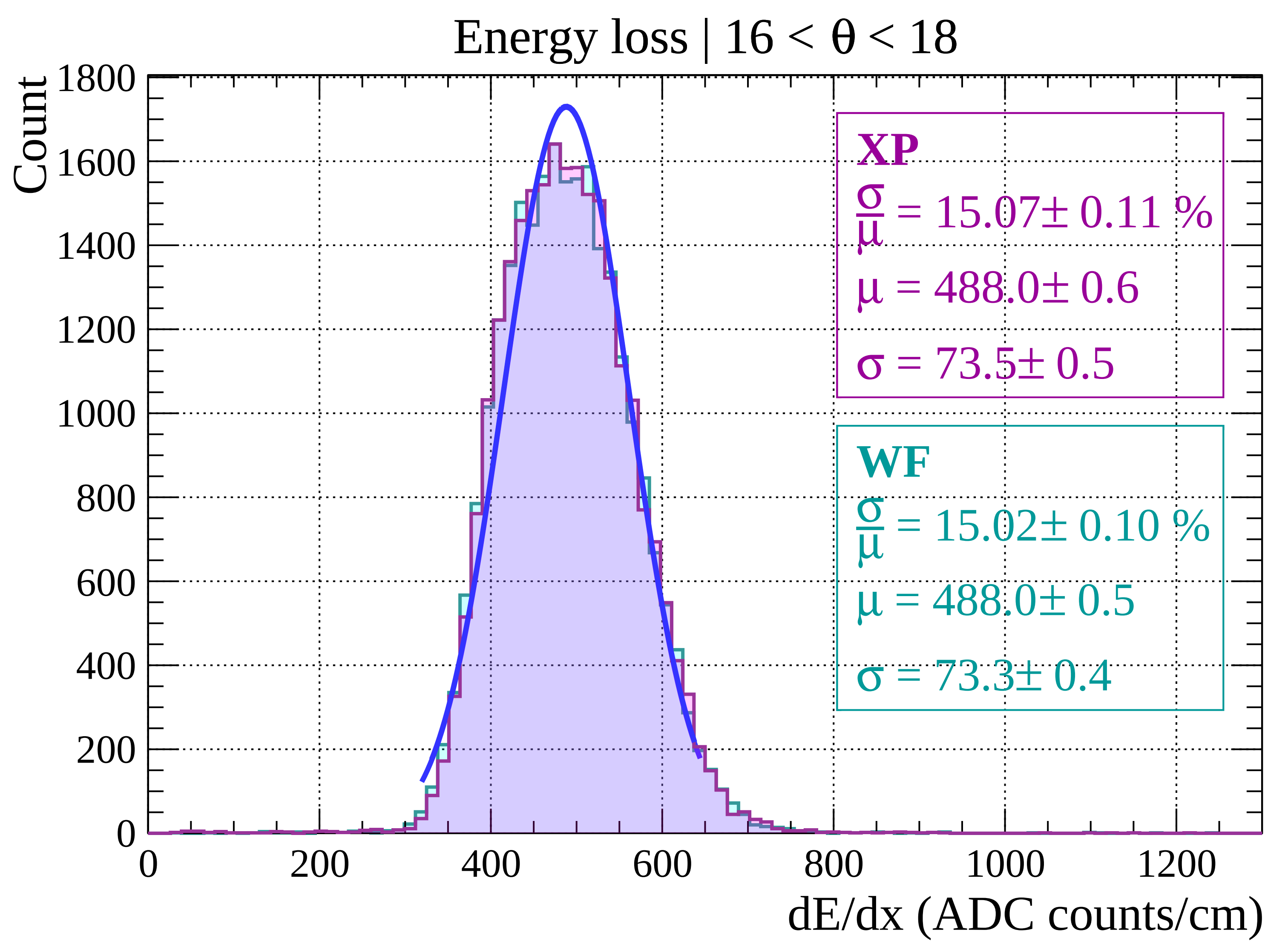

Energy loss | 10 < θ < 12
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
1800
2000
XP
σ/μ = 14.39± 0.08 %
μ = 487.6± 0.5
σ = 70.2± 0.3
WF
σ/μ = 15.25± 0.10 %
μ = 484.6± 0.5
σ = 73.9± 0.4

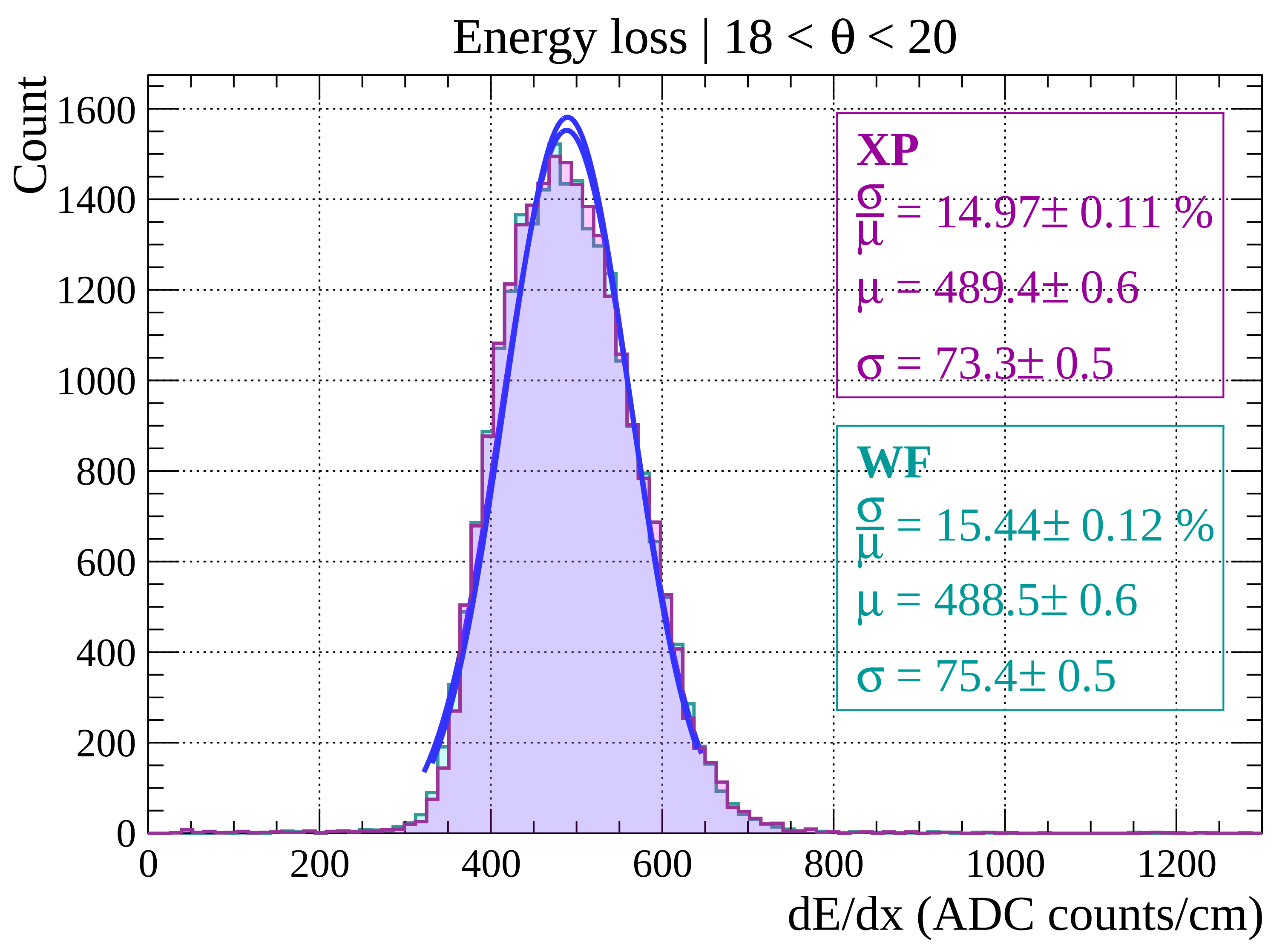


Energy loss | 12 < θ < 14
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 15.22± 0.10 %
μ = 485.8± 0.5
σ = 73.9± 0.4
WF
$\frac{\sigma}{\mu}$ = 15.51± 0.11 %
μ = 484.3± 0.5
σ = 75.1± 0.4

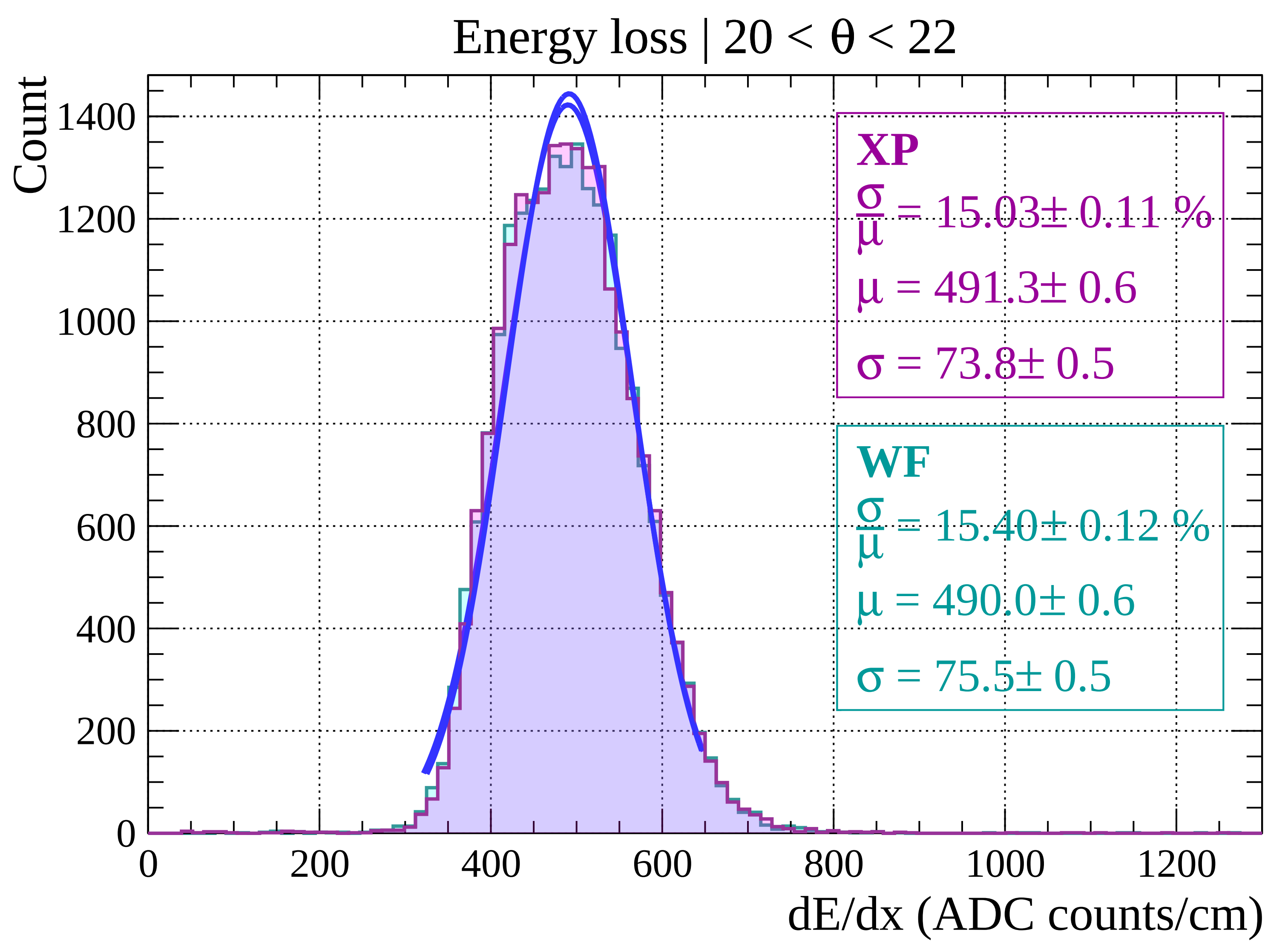


Energy loss | 14 < θ < 16
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 14.91 ± 0.10 %
μ = 488.2 ± 0.5
σ = 72.8 ± 0.4
WF
σ/μ = 15.42 ± 0.11 %
μ = 486.0 ± 0.5
σ = 75.0 ± 0.5

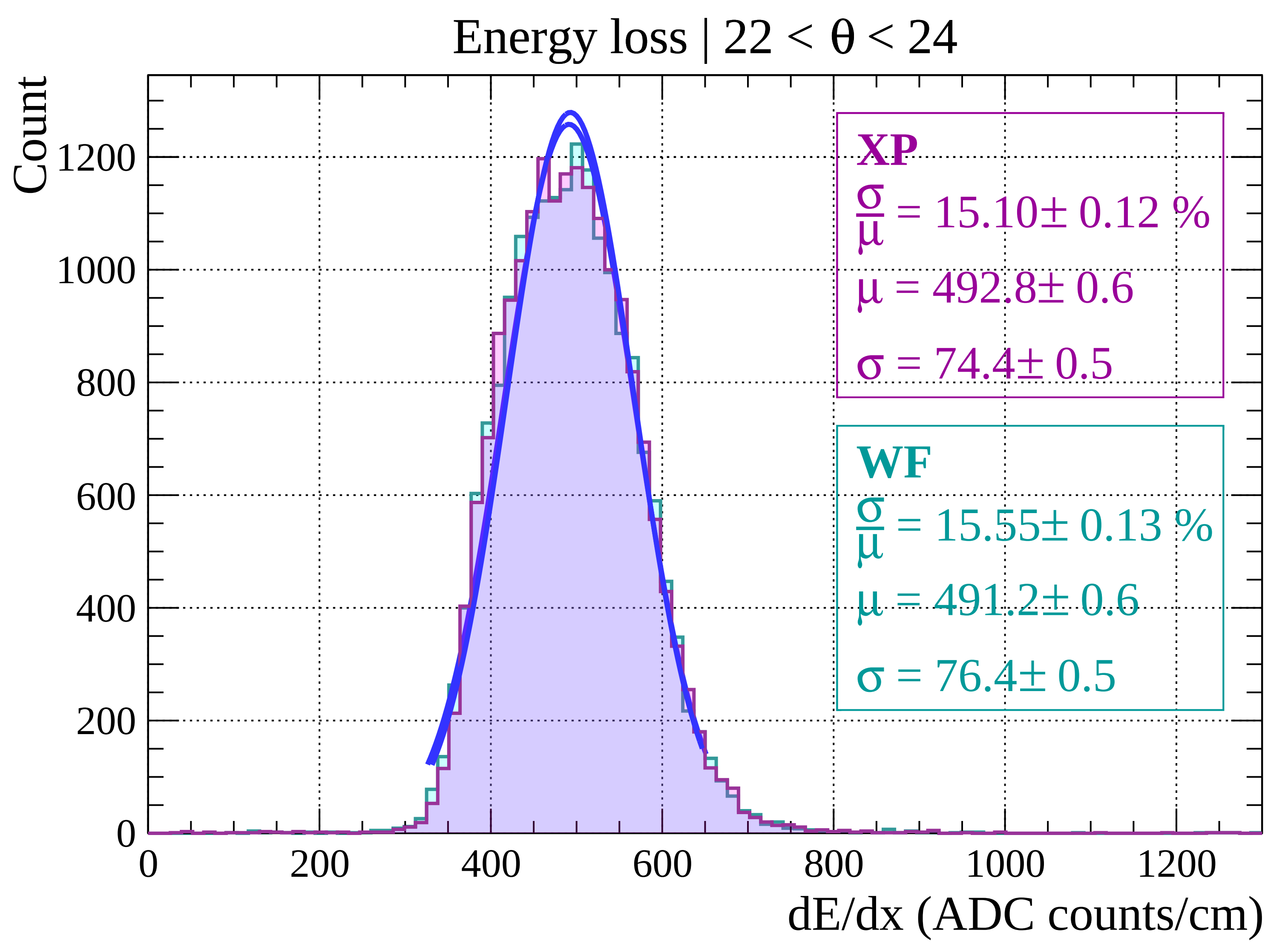

Energy loss | 16 < θ < 18
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.07± 0.11 %
μ = 488.0± 0.6
σ = 73.5± 0.5
WF
σ/μ = 15.02± 0.10 %
μ = 488.0± 0.5
σ = 73.3± 0.4
0
200
400
600
800
1000
1200
1400
1600
1800



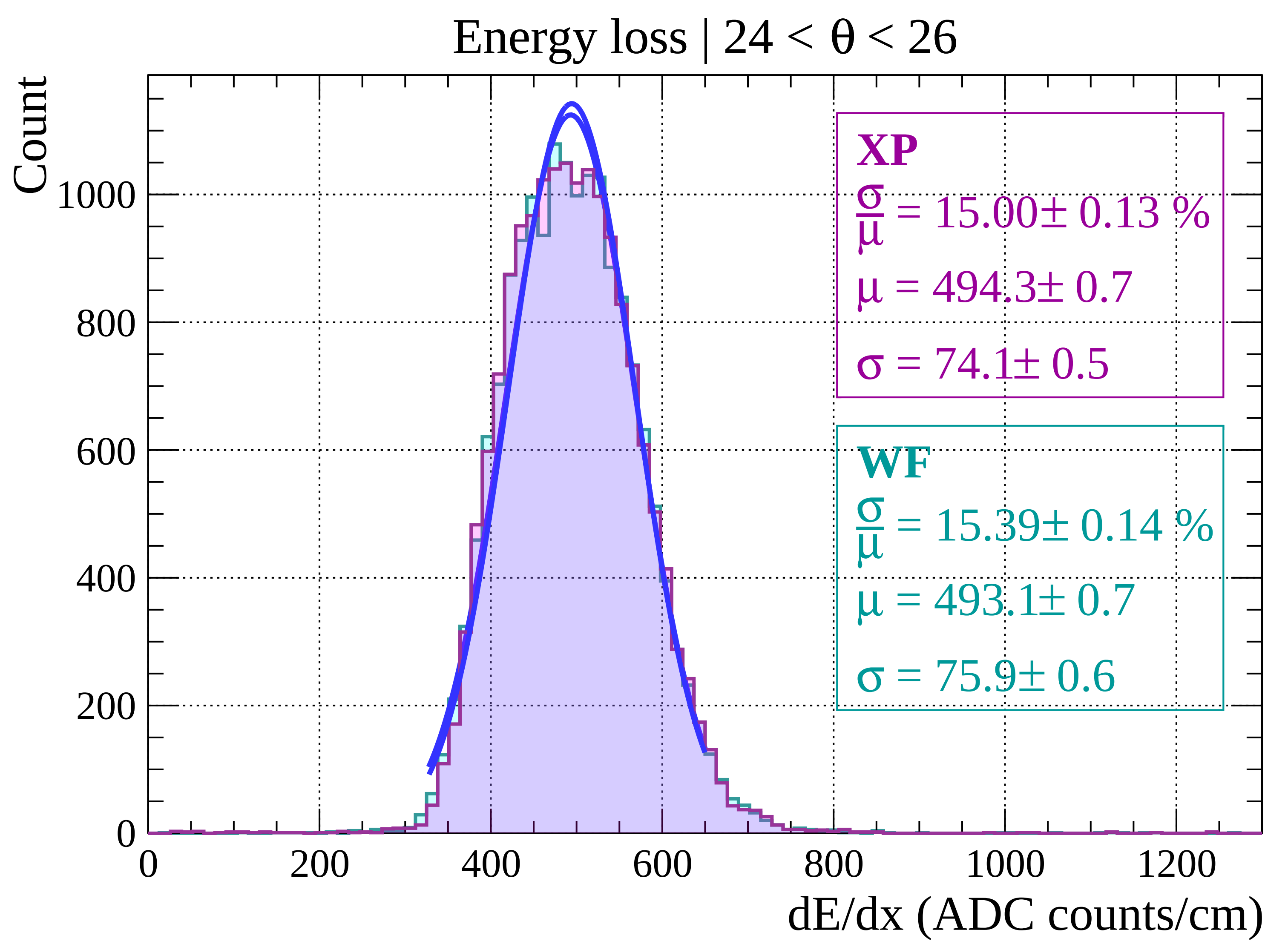
Energy loss | 18 < θ < 20
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1400
1600
0
200
400
600
800
1000
1200
XP
σ/μ = 14.97± 0.11 %
μ = 489.4± 0.6
σ = 73.3± 0.5
WF
σ/μ = 15.44± 0.12 %
μ = 488.5± 0.6
σ = 75.4± 0.5



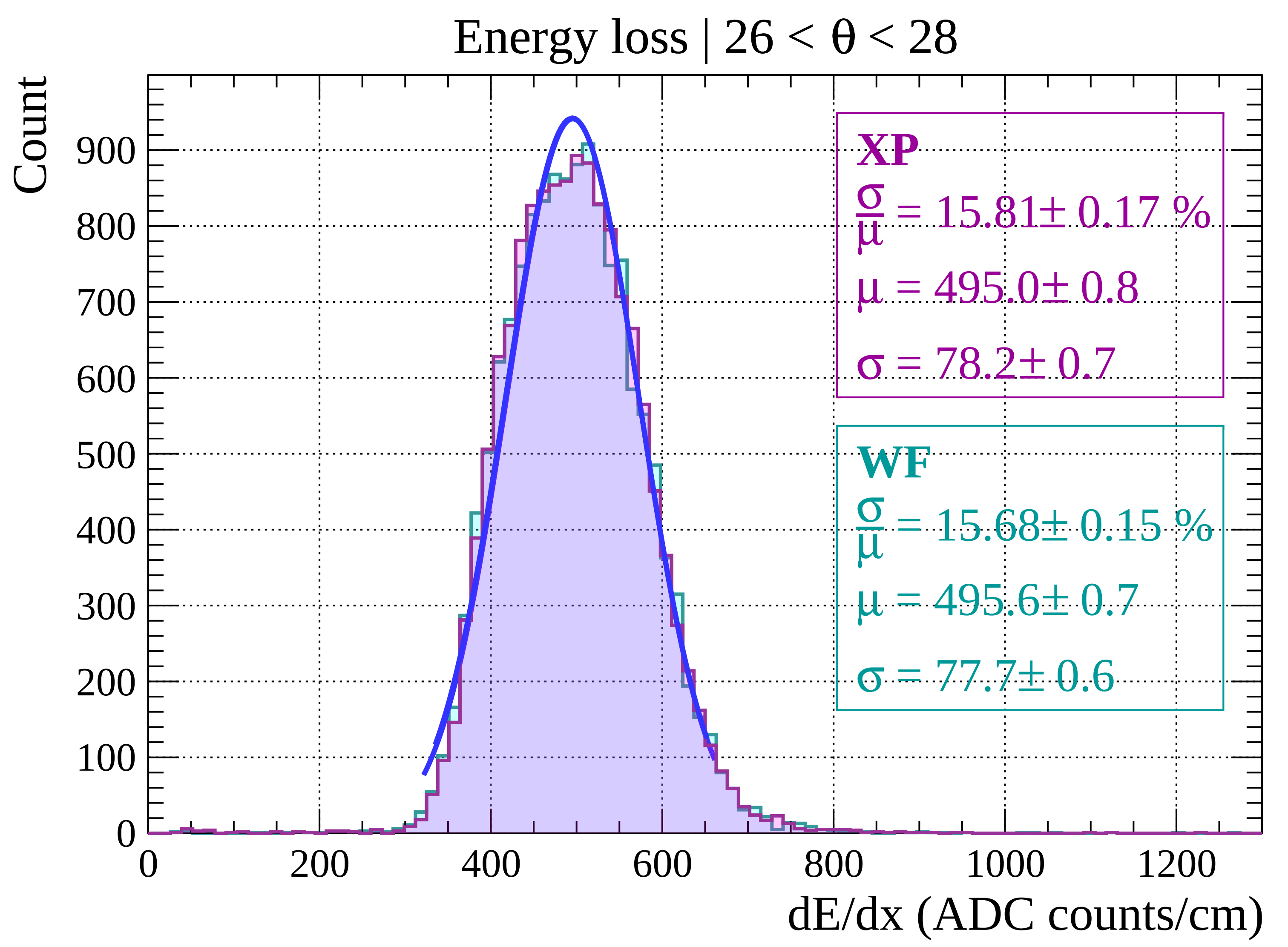
Energy loss | 20 < θ < 22
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.03± 0.11 %
μ = 491.3± 0.6
σ = 73.8± 0.5
WF
σ/μ = 15.40± 0.12 %
μ = 490.0± 0.6
σ = 75.5± 0.5
0
200
400
600
800
1000
1200
1400

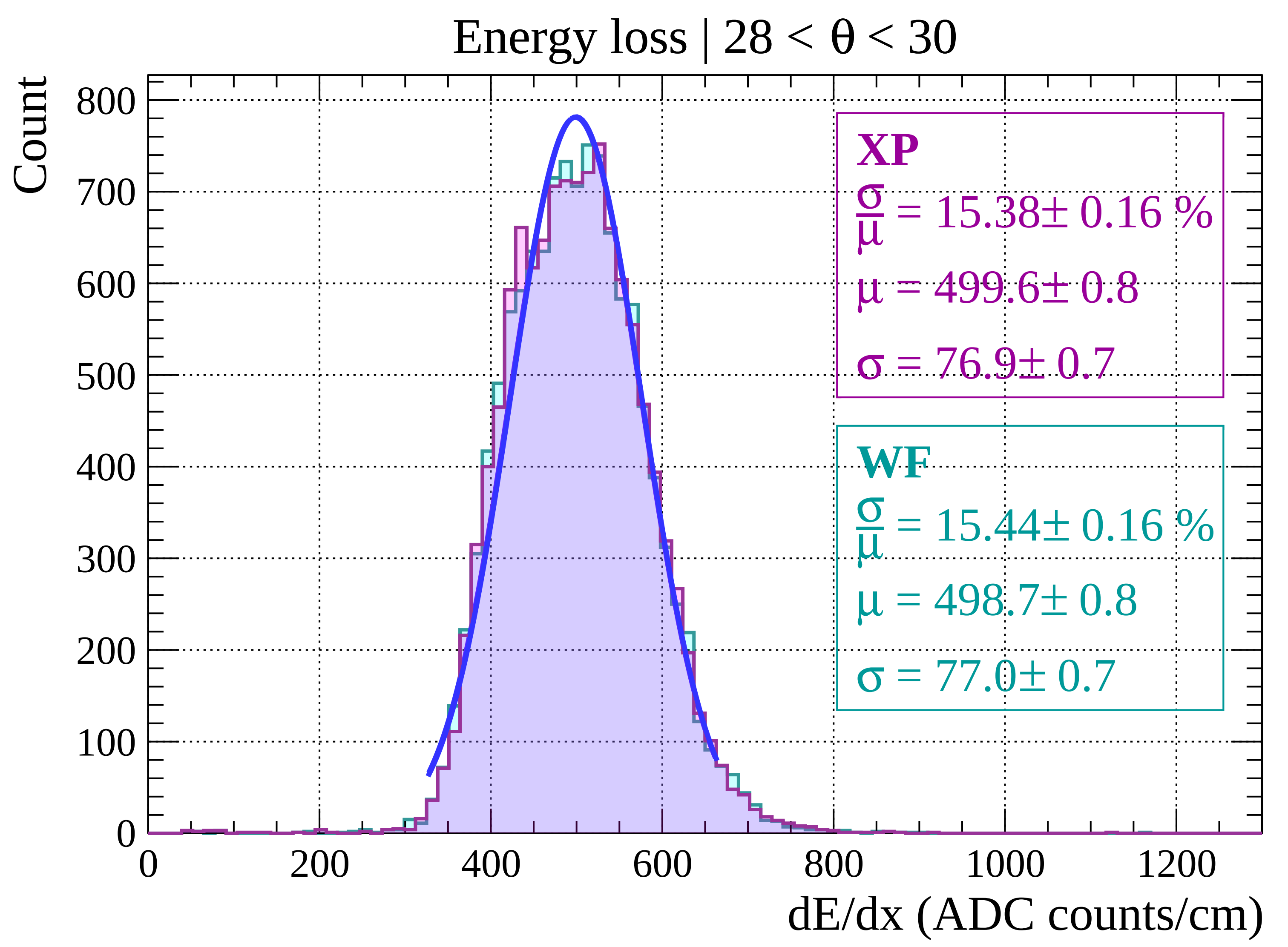


Energy loss | 22 < θ < 24
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.10± 0.12 %
μ = 492.8± 0.6
σ = 74.4± 0.5
WF
σ/μ = 15.55± 0.13 %
μ = 491.2± 0.6
σ = 76.4± 0.5

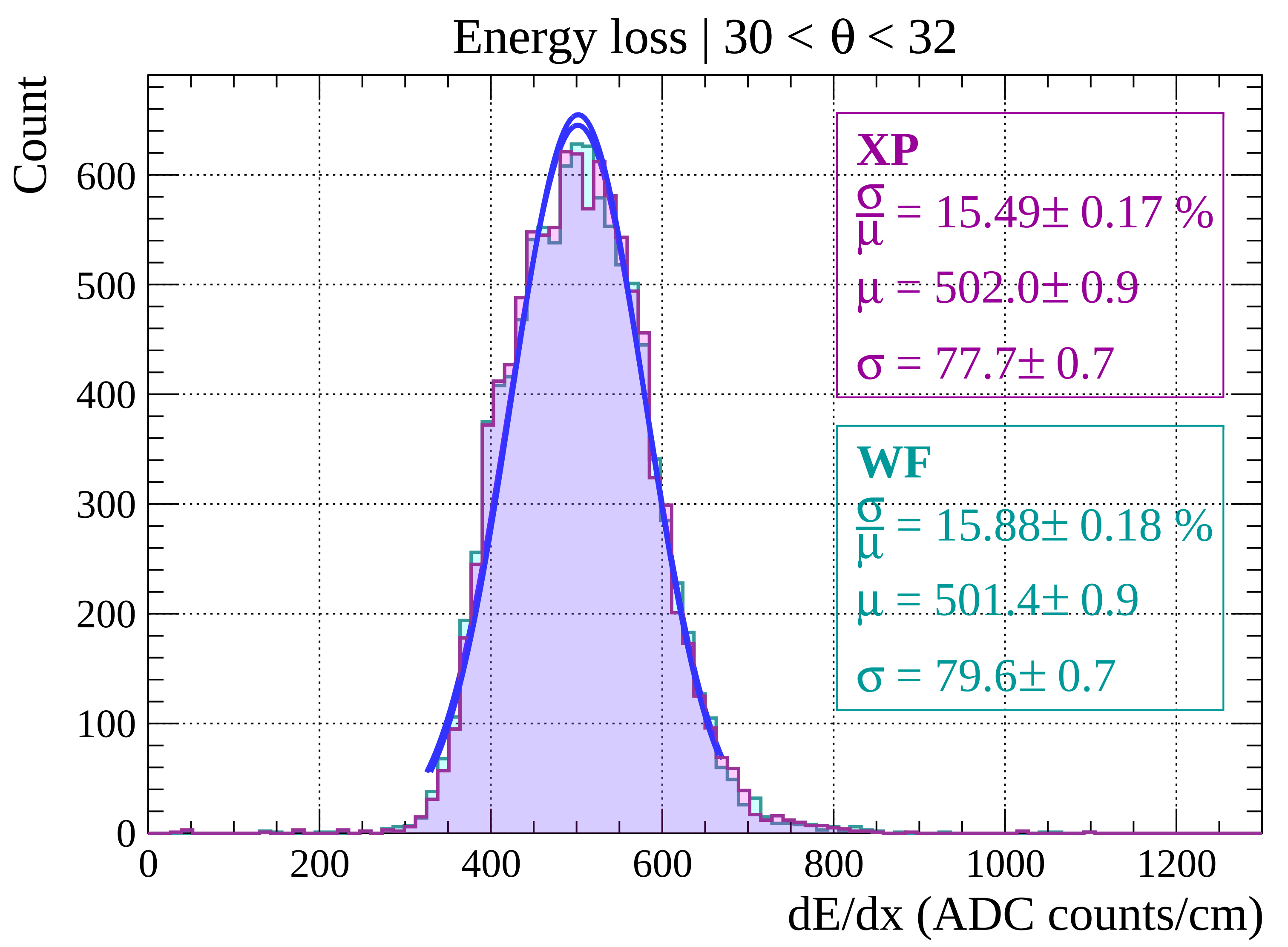


Energy loss | 24 < θ < 26
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
XP
σ/μ = 15.00± 0.13 %
μ = 494.3± 0.7
σ = 74.1± 0.5
WF
σ/μ = 15.39± 0.14 %
μ = 493.1± 0.7
σ = 75.9± 0.6

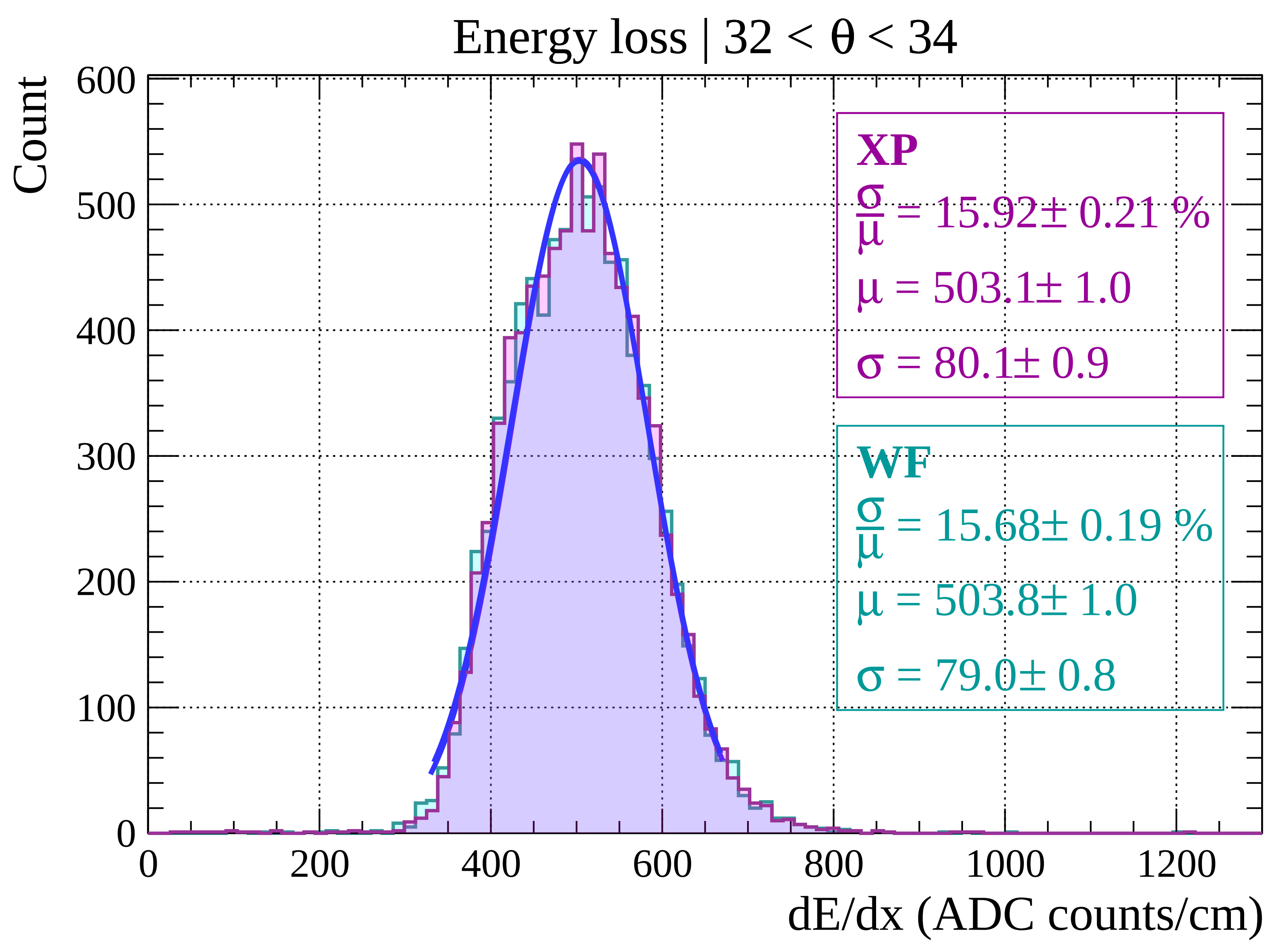

Energy loss | 26 < θ < 28
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
700
800
900
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 15.81± 0.17 %
μ = 495.0± 0.8
σ = 78.2± 0.7
WF
$\frac{\sigma}{\mu}$ = 15.68± 0.15 %
μ = 495.6± 0.7
σ = 77.7± 0.6



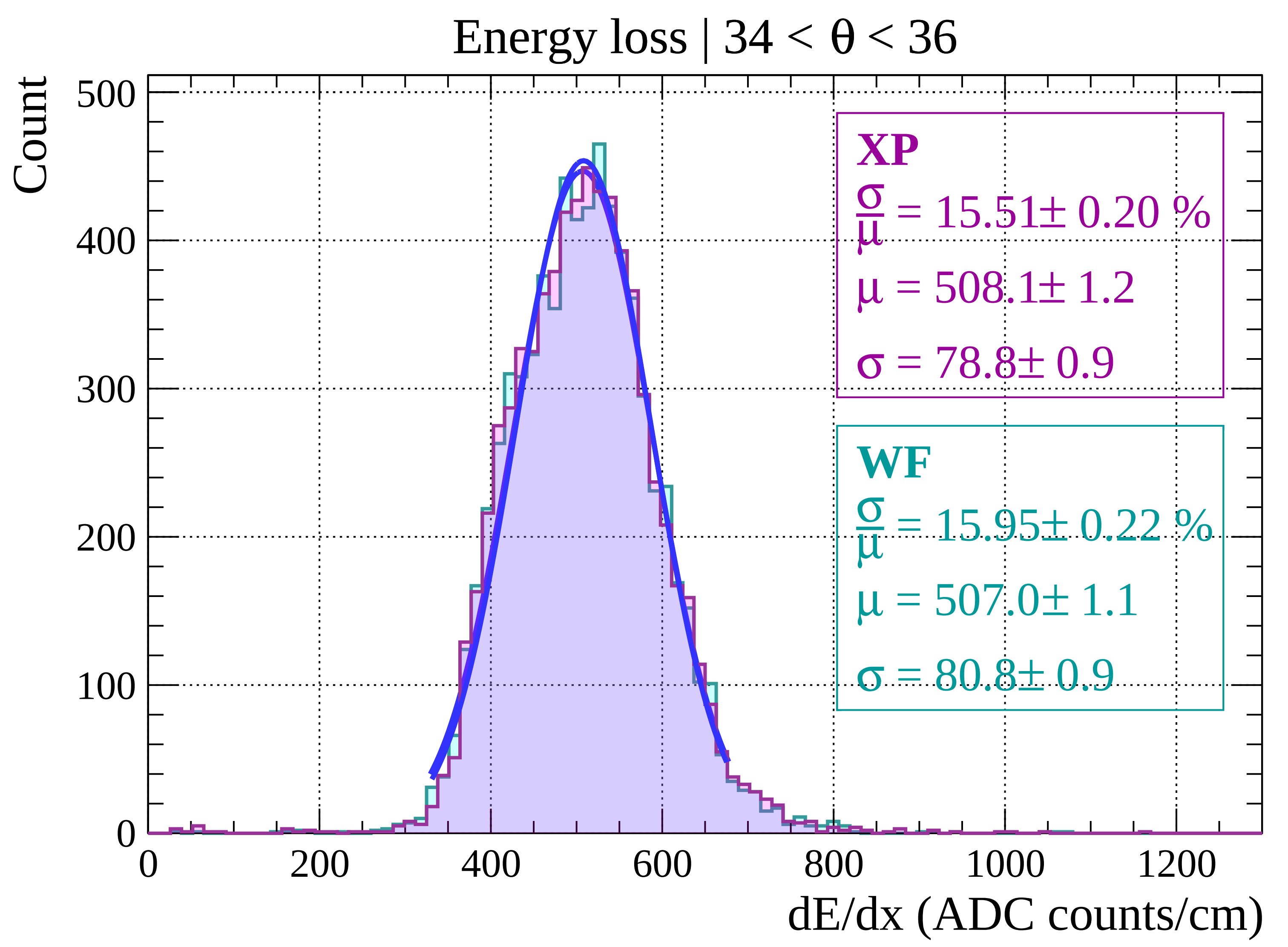
Energy loss | 28 < θ < 30
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 15.38± 0.16 %
μ = 499.6± 0.8
σ = 76.9± 0.7
WF
σ/μ = 15.44± 0.16 %
μ = 498.7± 0.8
σ = 77.0± 0.7



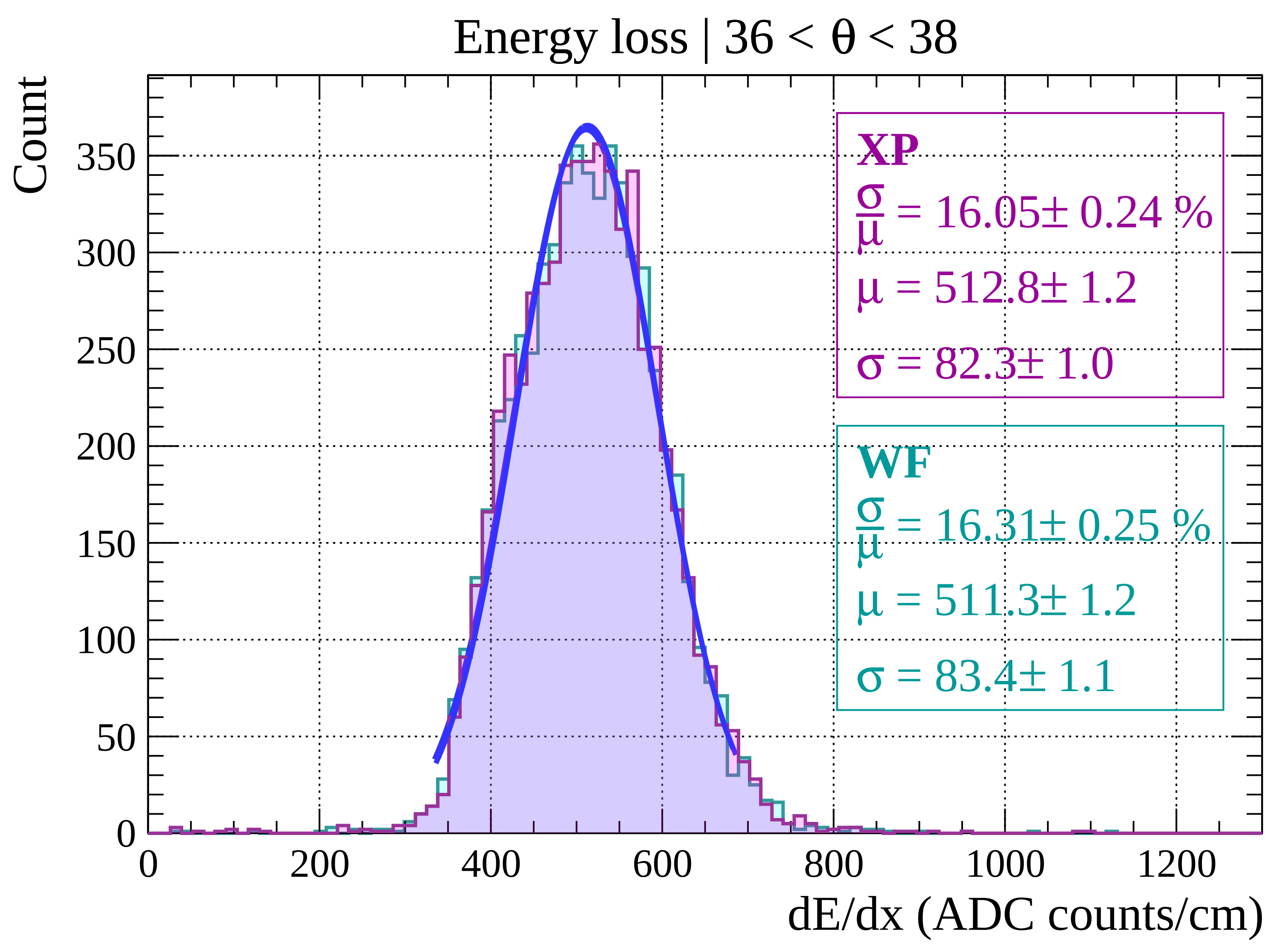
Energy loss | 30 < θ < 32
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 15.49± 0.17 %
μ = 502.0± 0.9
σ = 77.7± 0.7
WF
σ/μ = 15.88± 0.18 %
μ = 501.4± 0.9
σ = 79.6± 0.7



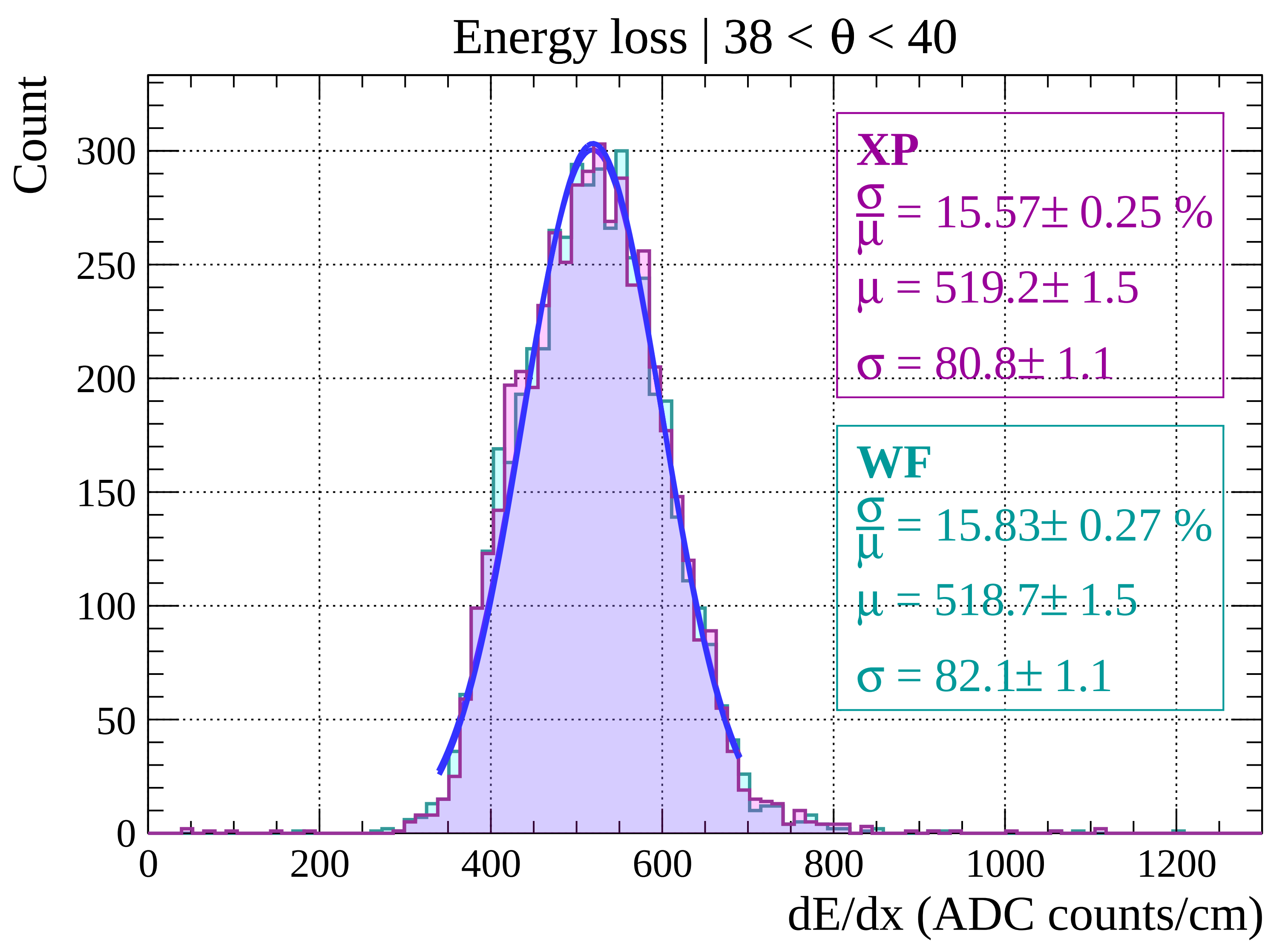
Energy loss | 32 < θ < 34
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
600
0
200
400
600
800
1000
1200
XP
σ/μ = 15.92± 0.21 %
μ = 503.1± 1.0
σ = 80.1± 0.9
WF
σ/μ = 15.68± 0.19 %
μ = 503.8± 1.0
σ = 79.0± 0.8



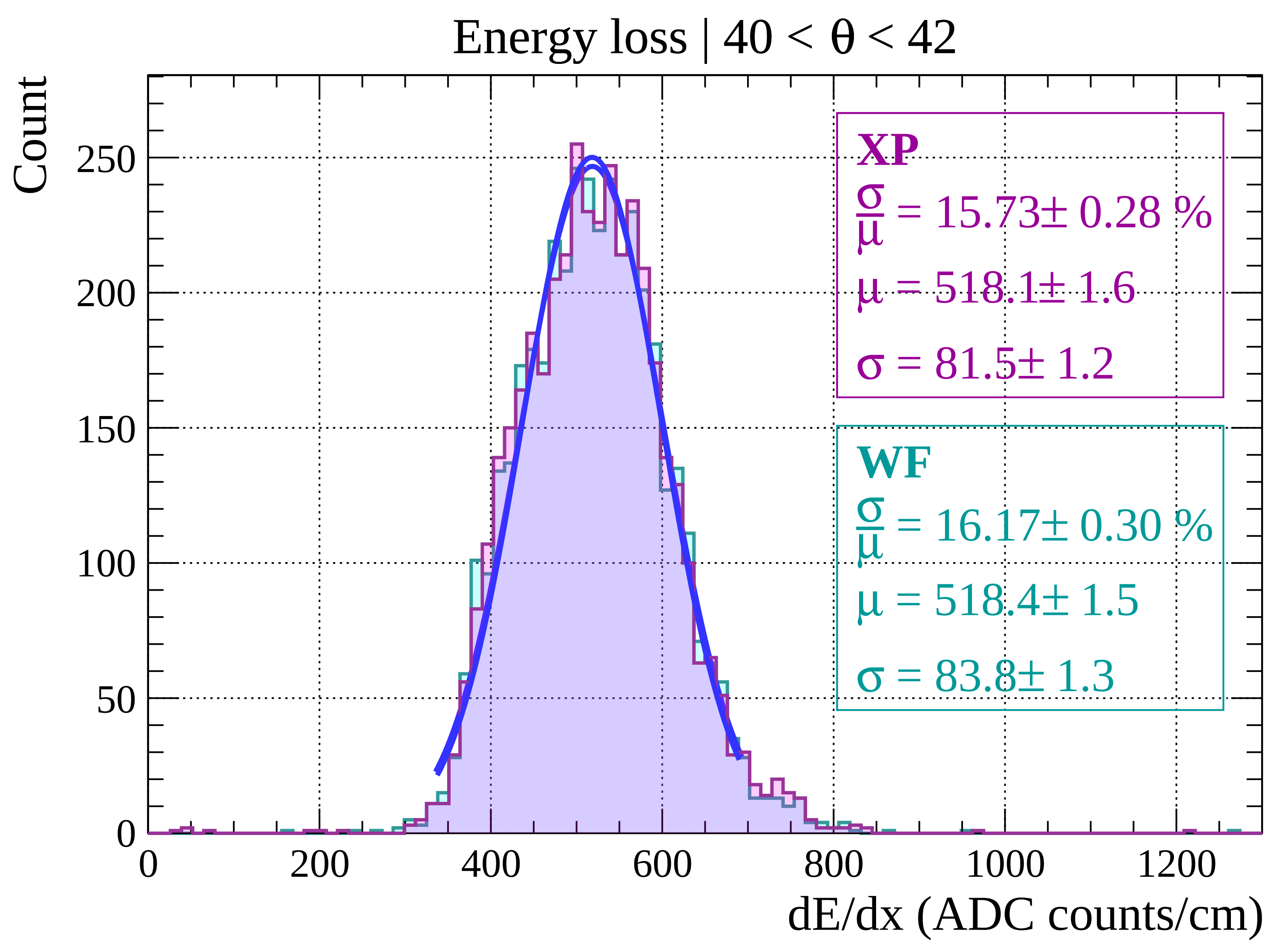
Energy loss | 34 < θ < 36
Count
dE/dx (ADC counts/cm)
0
100
200
300
400
500
0
200
400
600
800
1000
1200
XP
σ/μ = 15.51± 0.20 %
μ = 508.1± 1.2
σ = 78.8± 0.9
WF
σ/μ = 15.95± 0.22 %
μ = 507.0± 1.1
σ = 80.8± 0.9



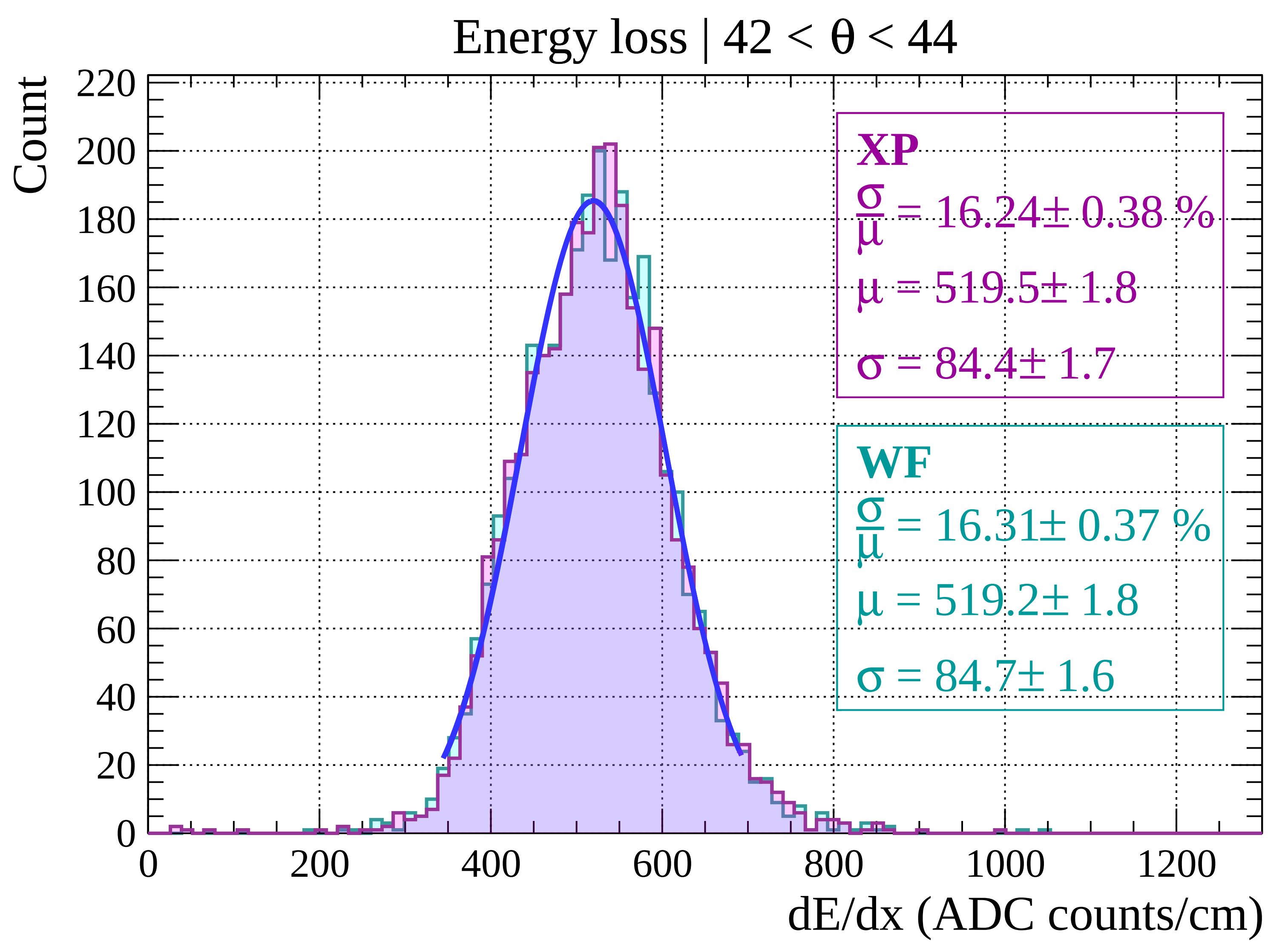
Energy loss | 36 < θ < 38
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.05± 0.24 %
μ = 512.8± 1.2
σ = 82.3± 1.0
WF
σ/μ = 16.31± 0.25 %
μ = 511.3± 1.2
σ = 83.4± 1.1



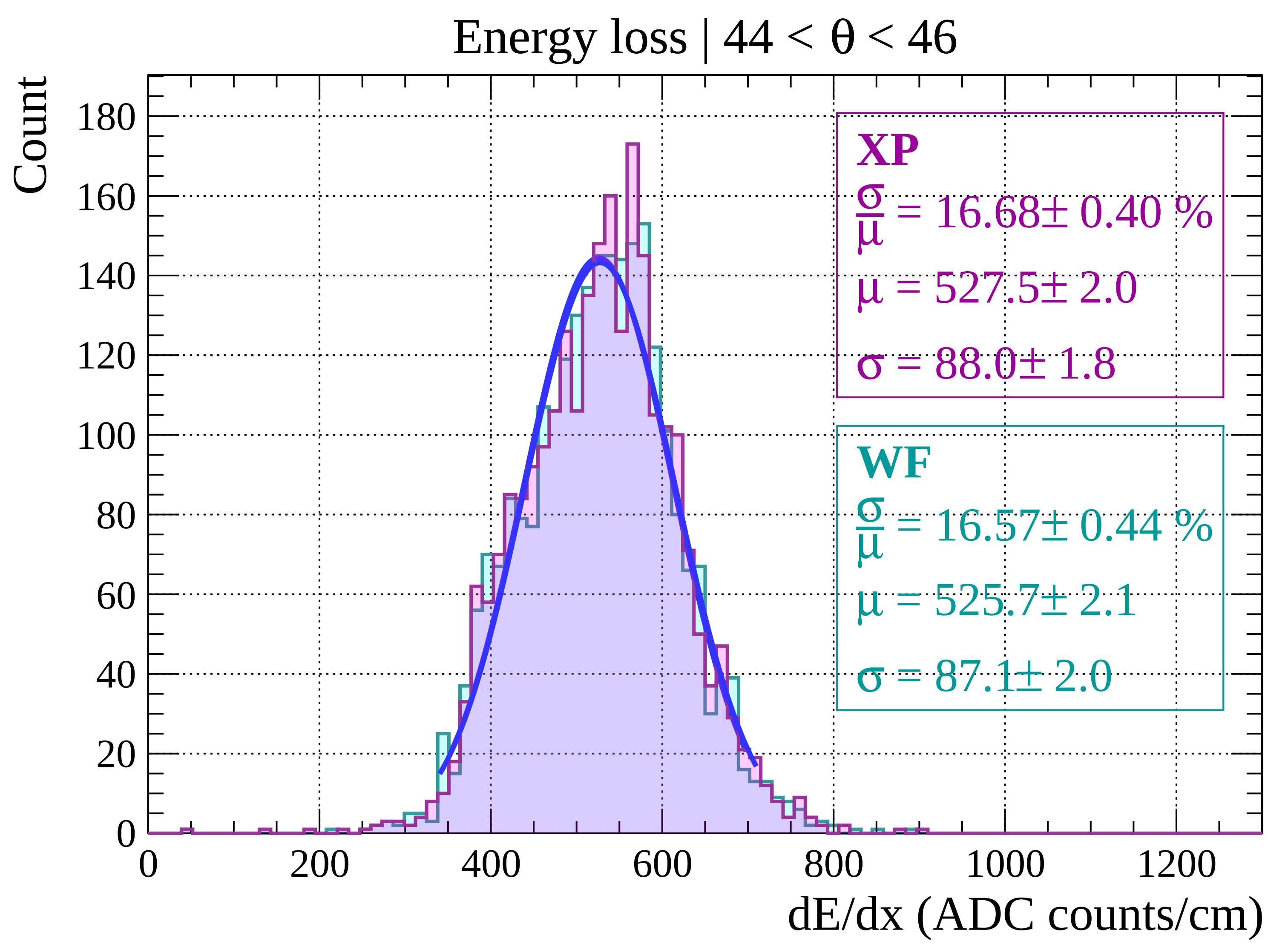
Energy loss | 38 < θ < 40
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
300
0
200
400
600
800
1000
1200
XP
σ/μ = 15.57± 0.25 %
μ = 519.2± 1.5
σ = 80.8± 1.1
WF
σ/μ = 15.83± 0.27 %
μ = 518.7± 1.5
σ = 82.1± 1.1



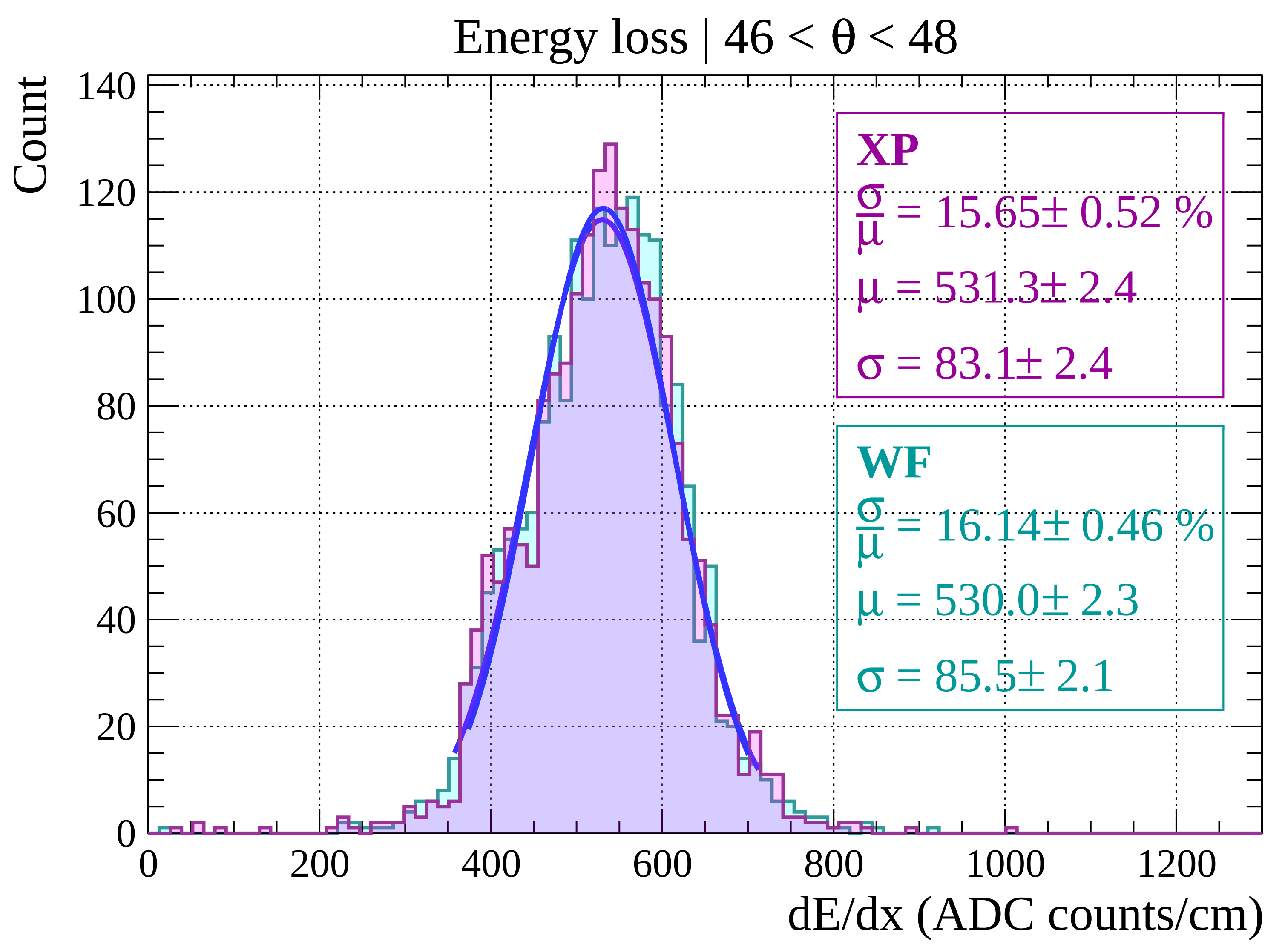
Energy loss | 40 < θ < 42
Count
dE/dx (ADC counts/cm)
0
50
100
150
200
250
0
200
400
600
800
1000
1200
XP
σ/μ = 15.73± 0.28 %
μ = 518.1± 1.6
σ = 81.5± 1.2
WF
σ/μ = 16.17± 0.30 %
μ = 518.4± 1.5
σ = 83.8± 1.3

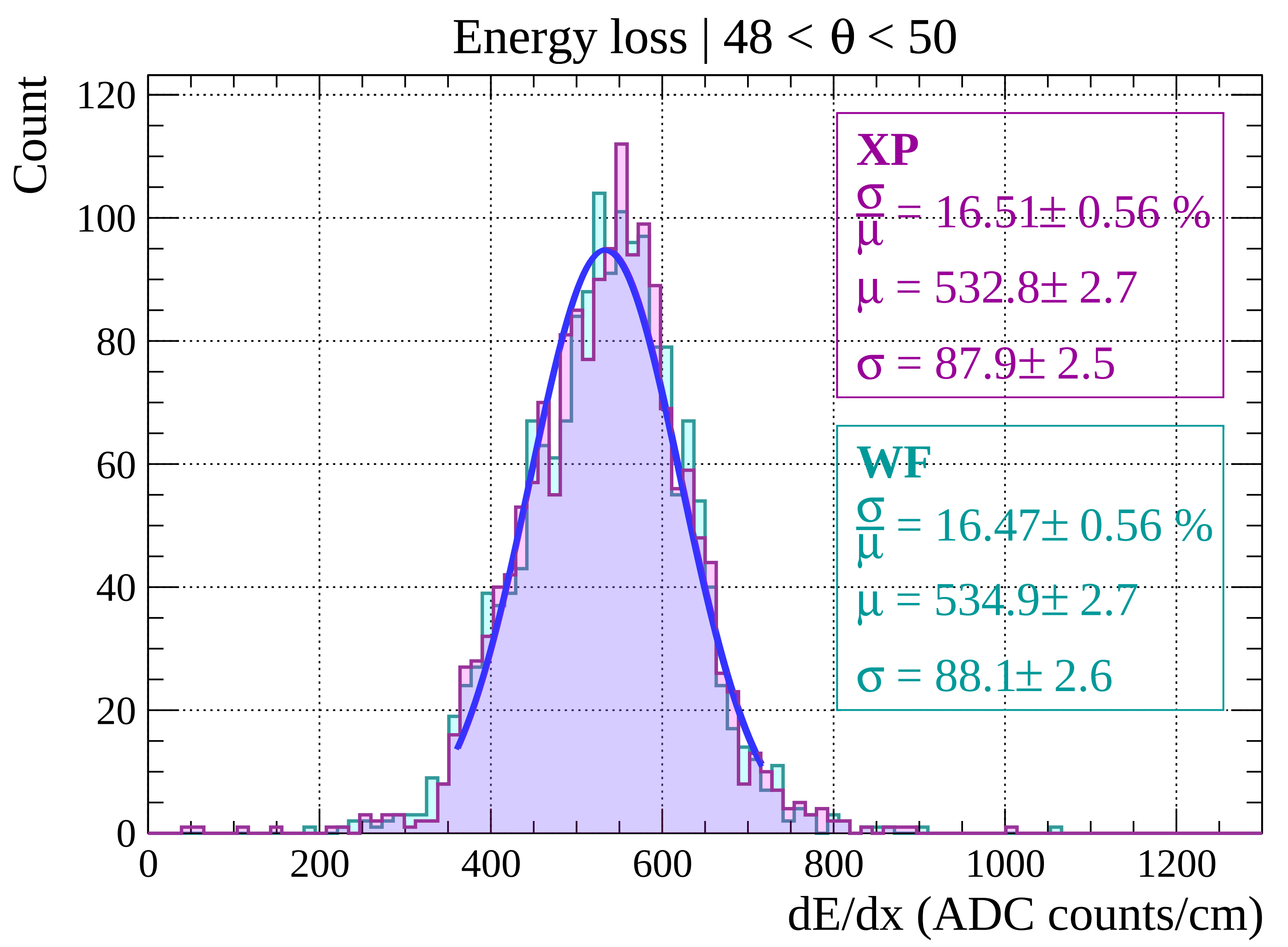


Energy loss | 42 < θ < 44
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
140
160
180
200
220
0
200
400
600
800
1000
1200
XP
σ/μ = 16.24± 0.38 %
μ = 519.5± 1.8
σ = 84.4± 1.7
WF
σ/μ = 16.31± 0.37 %
μ = 519.2± 1.8
σ = 84.7± 1.6

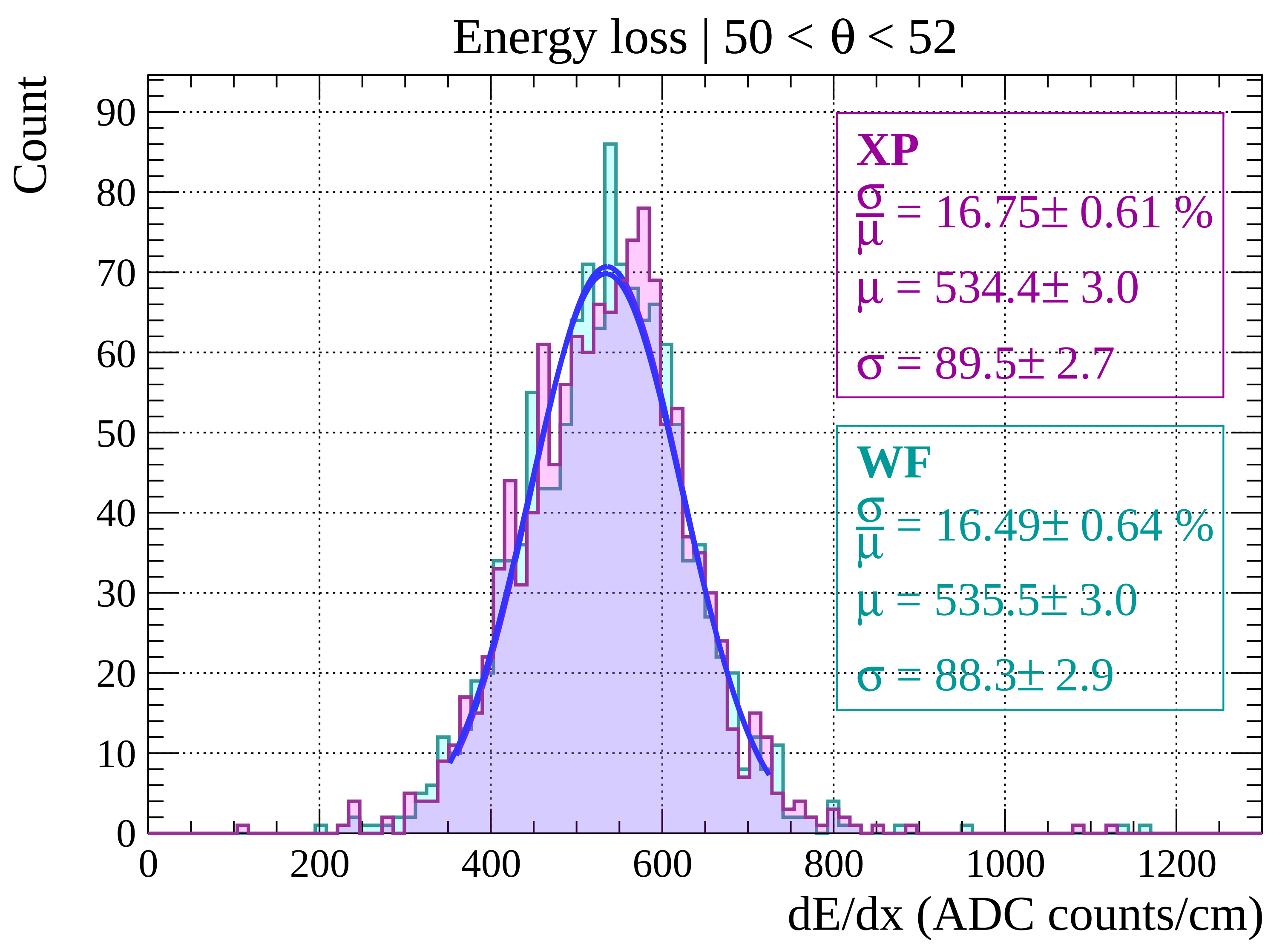


Energy loss | 44 < θ < 46
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
20
40
60
80
100
120
140
160
180
XP
$\frac{\sigma}{\mu}$ = 16.68± 0.40 %
μ = 527.5± 2.0
σ = 88.0± 1.8
WF
$\frac{\sigma}{\mu}$ = 16.57± 0.44 %
μ = 525.7± 2.1
σ = 87.1± 2.0

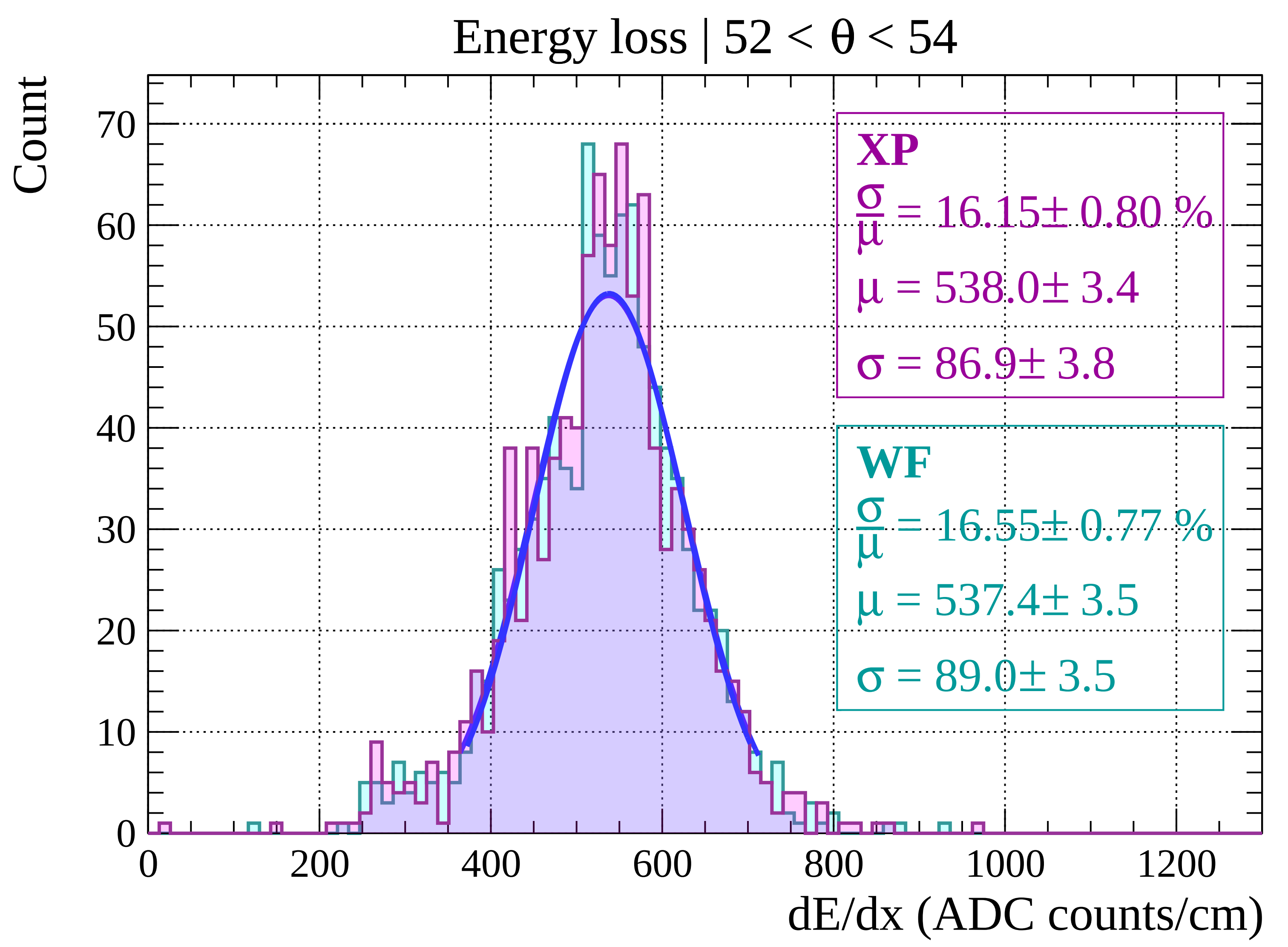

Energy loss | 46 < θ < 48
Count
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 15.65± 0.52 %
μ = 531.3± 2.4
σ = 83.1± 2.4
WF
$\frac{\sigma}{\mu}$ = 16.14± 0.46 %
μ = 530.0± 2.3
σ = 85.5± 2.1
0
20
40
60
80
100
120
140
0
200
400
600
800
1000
1200



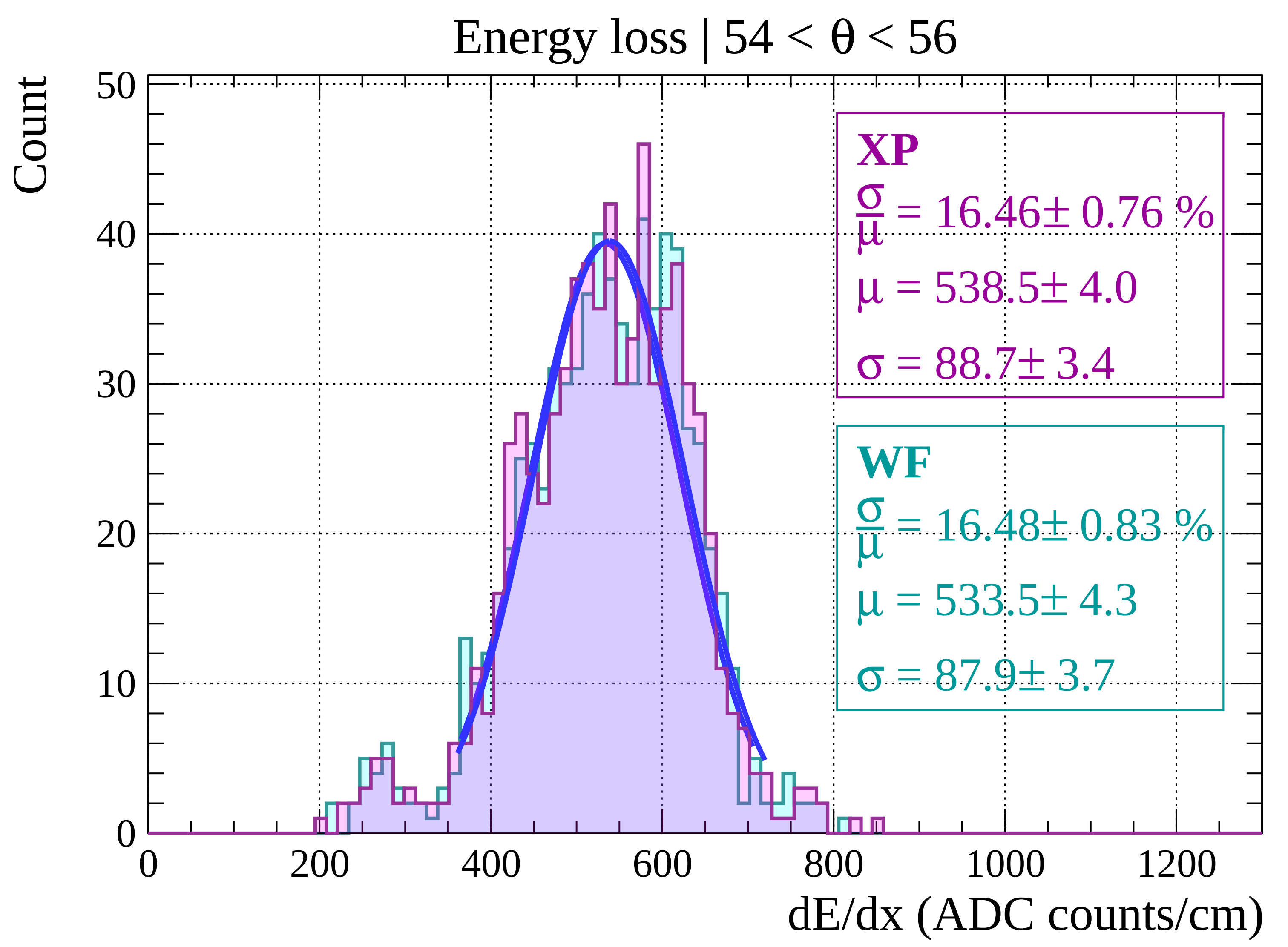
Energy loss | 48 < θ < 50
Count
dE/dx (ADC counts/cm)
0
20
40
60
80
100
120
0
200
400
600
800
1000
1200
XP
σ/μ = 16.51± 0.56 %
μ = 532.8± 2.7
σ = 87.9± 2.5
WF
σ/μ = 16.47± 0.56 %
μ = 534.9± 2.7
σ = 88.1± 2.6



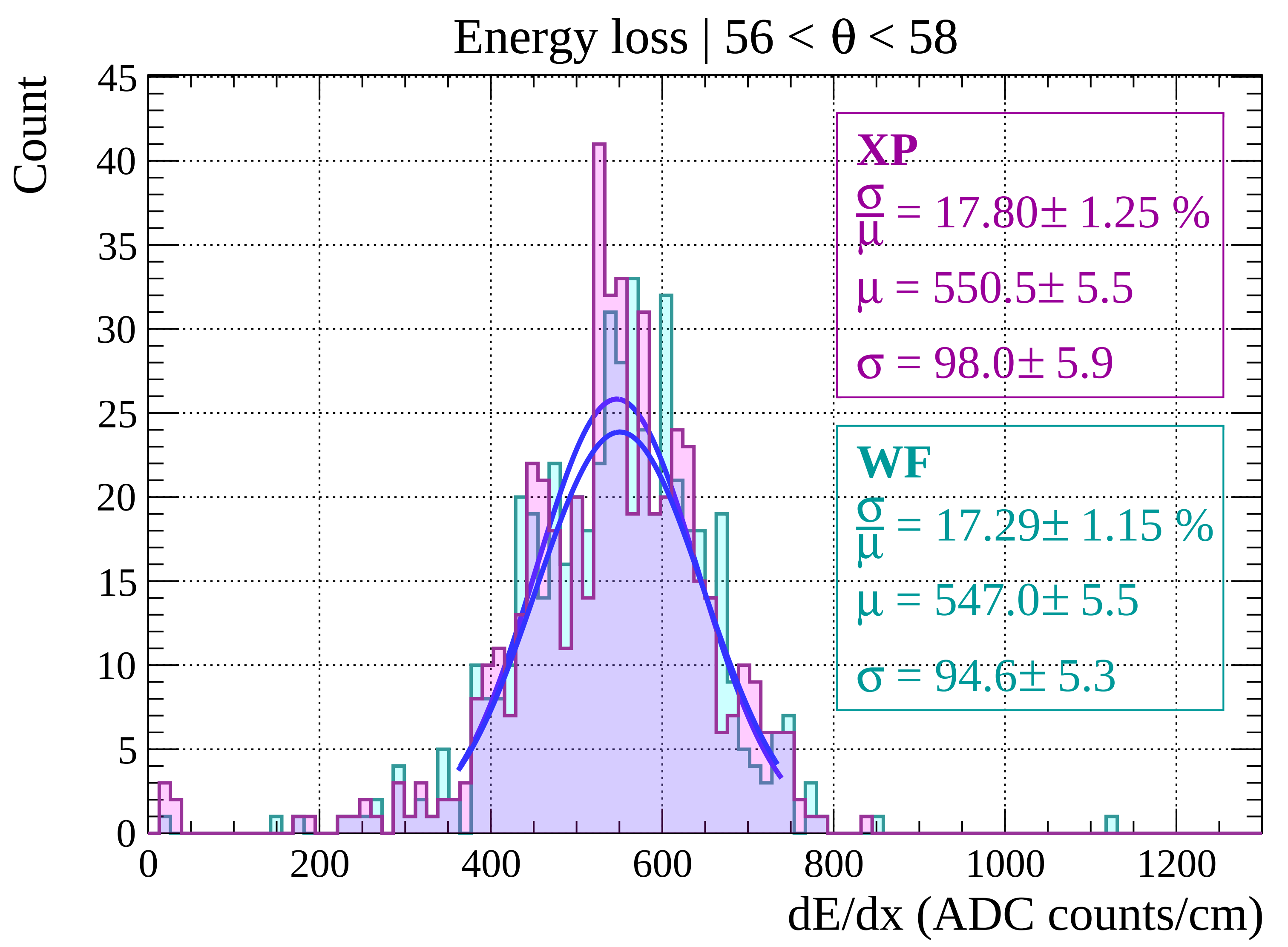
Energy loss | 50 < θ < 52
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
60
70
80
90
0
200
400
600
800
1000
1200
XP
σ/μ = 16.75± 0.61 %
μ = 534.4± 3.0
σ = 89.5± 2.7
WF
σ/μ = 16.49± 0.64 %
μ = 535.5± 3.0
σ = 88.3± 2.9



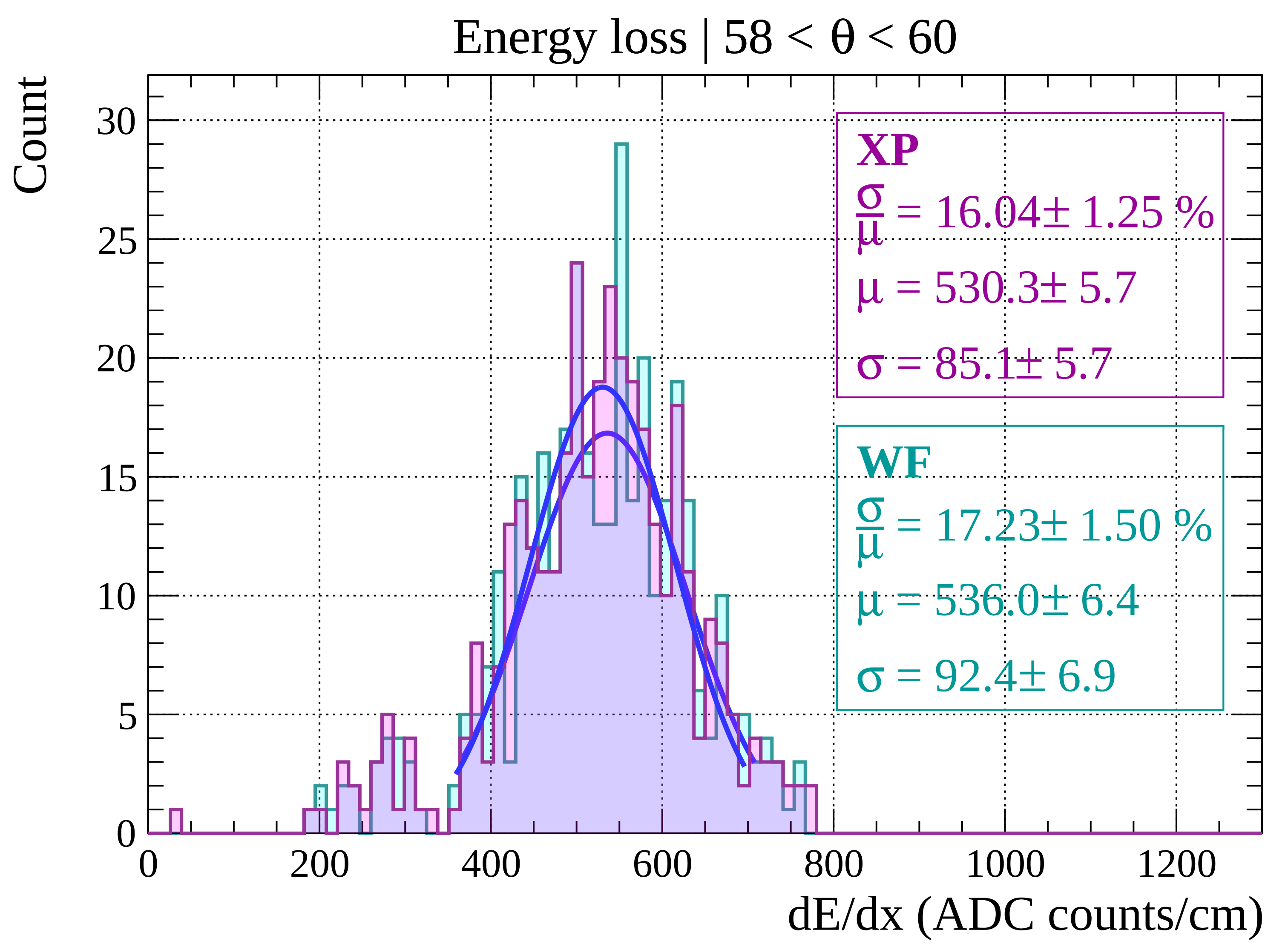
Energy loss | 52 < θ < 54
Count
dE/dx (ADC counts/cm)
XP
σ/μ = 16.15± 0.80 %
μ = 538.0± 3.4
σ = 86.9± 3.8
WF
σ/μ = 16.55± 0.77 %
μ = 537.4± 3.5
σ = 89.0± 3.5



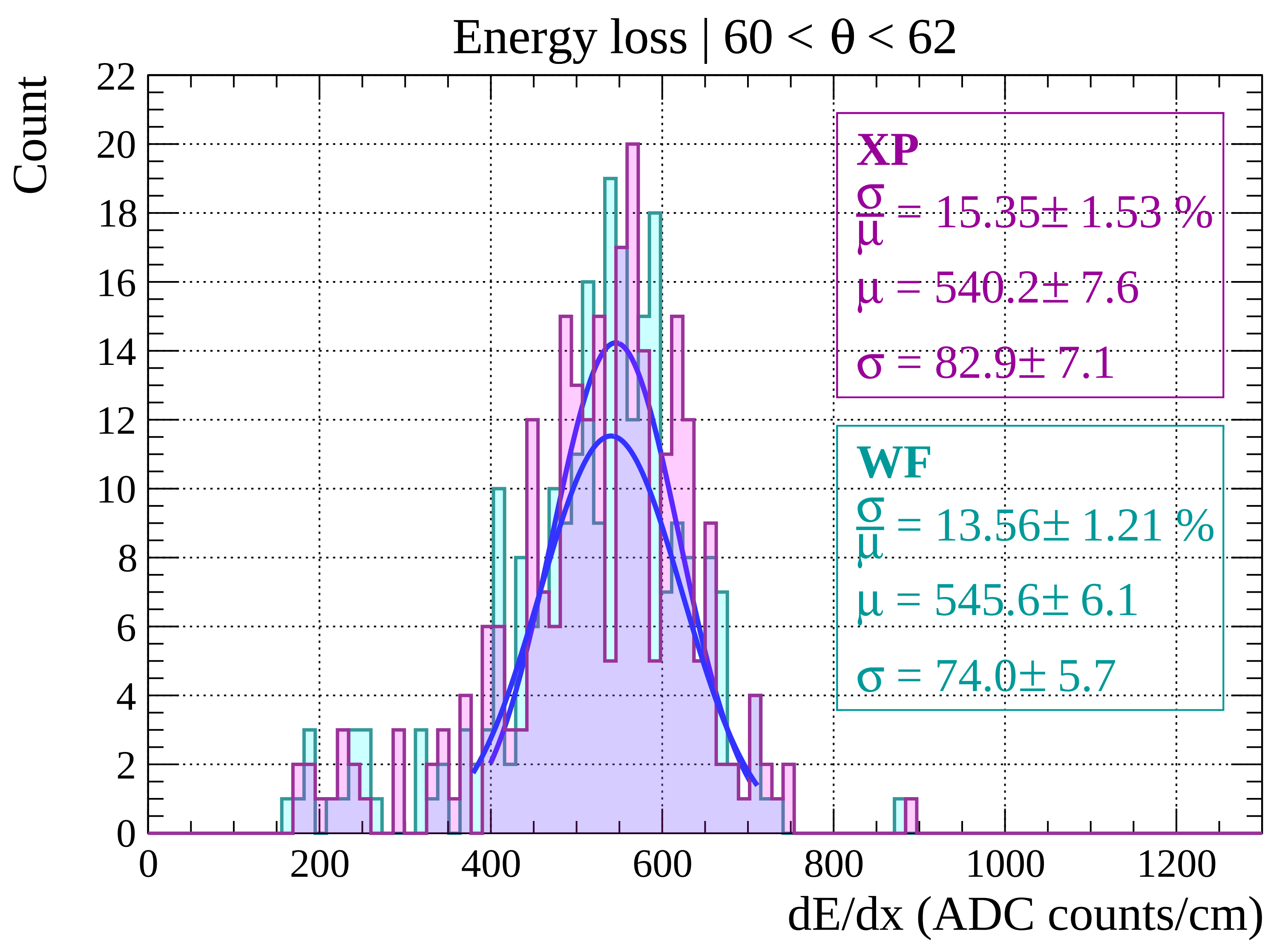
Energy loss | 54 < θ < 56
Count
dE/dx (ADC counts/cm)
0
10
20
30
40
50
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 16.46± 0.76 %
μ = 538.5± 4.0
σ = 88.7± 3.4
WF
$\frac{\sigma}{\mu}$ = 16.48± 0.83 %
μ = 533.5± 4.3
σ = 87.9± 3.7



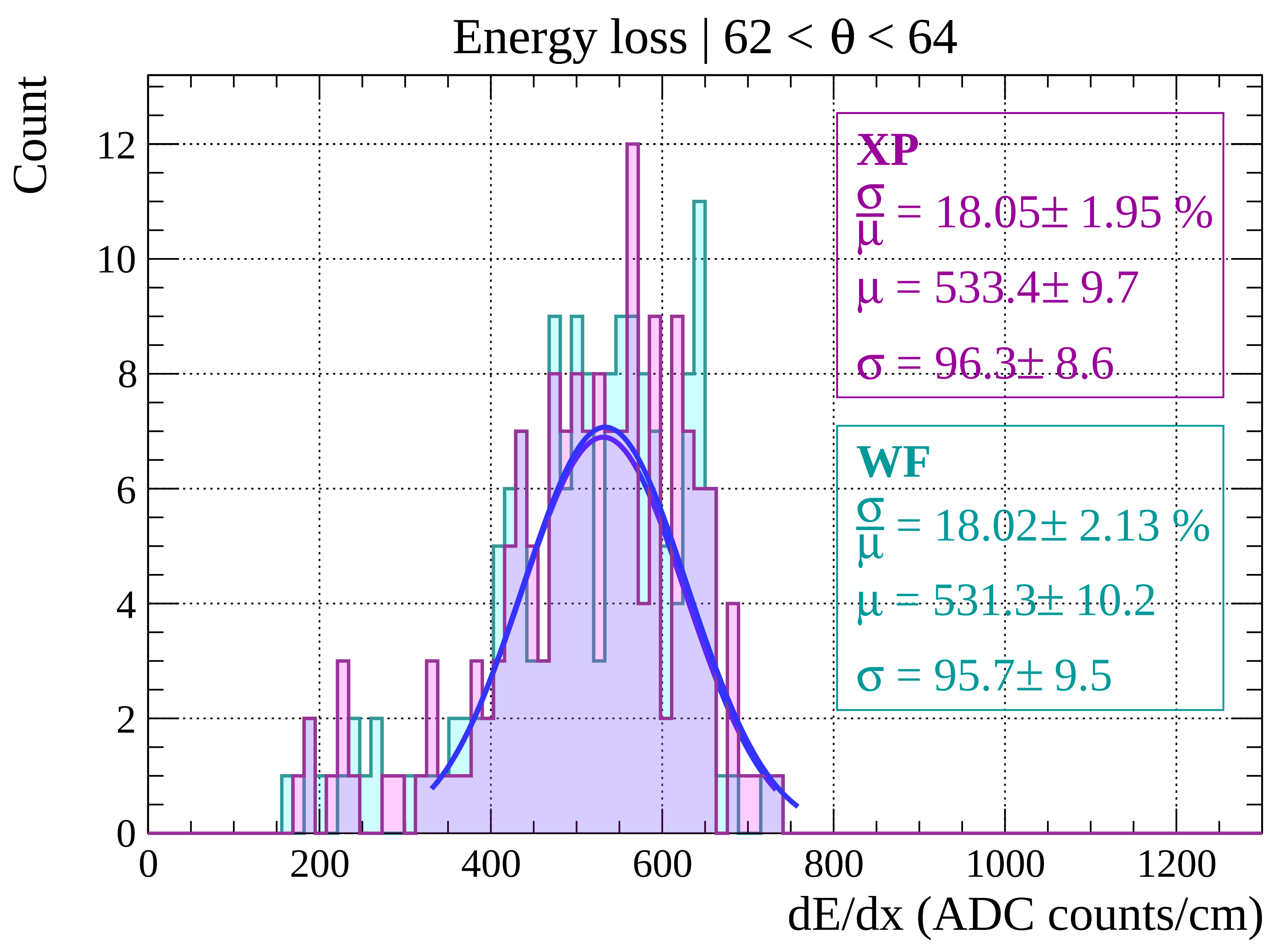
Energy loss | 56 < θ < 58
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
35
40
45
XP
σ/μ = 17.80± 1.25 %
μ = 550.5± 5.5
σ = 98.0± 5.9
WF
σ/μ = 17.29± 1.15 %
μ = 547.0± 5.5
σ = 94.6± 5.3



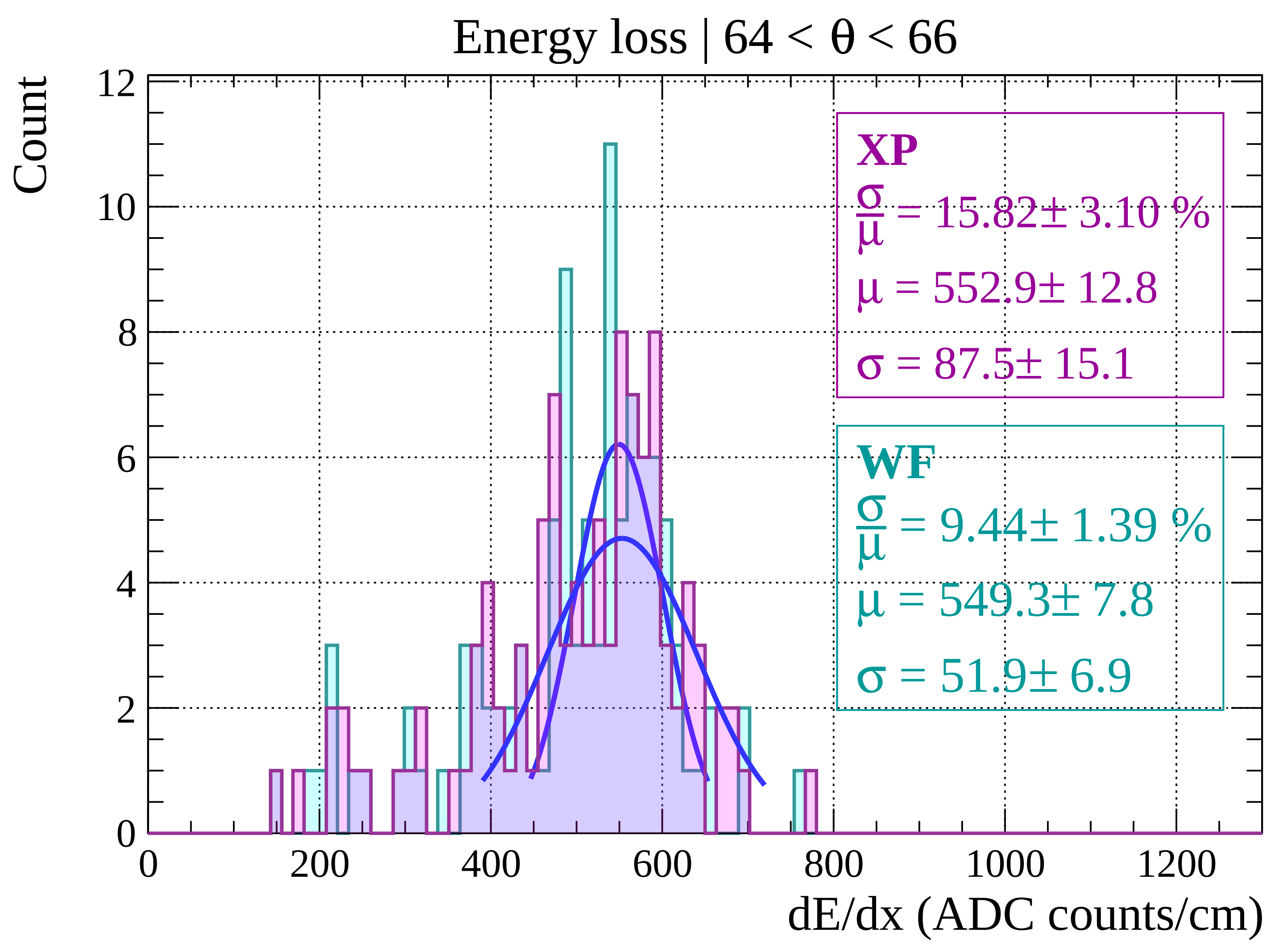
Energy loss | 58 < θ < 60
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
0
5
10
15
20
25
30
XP
σ/μ = 16.04± 1.25 %
μ = 530.3± 5.7
σ = 85.1± 5.7
WF
σ/μ = 17.23± 1.50 %
μ = 536.0± 6.4
σ = 92.4± 6.9



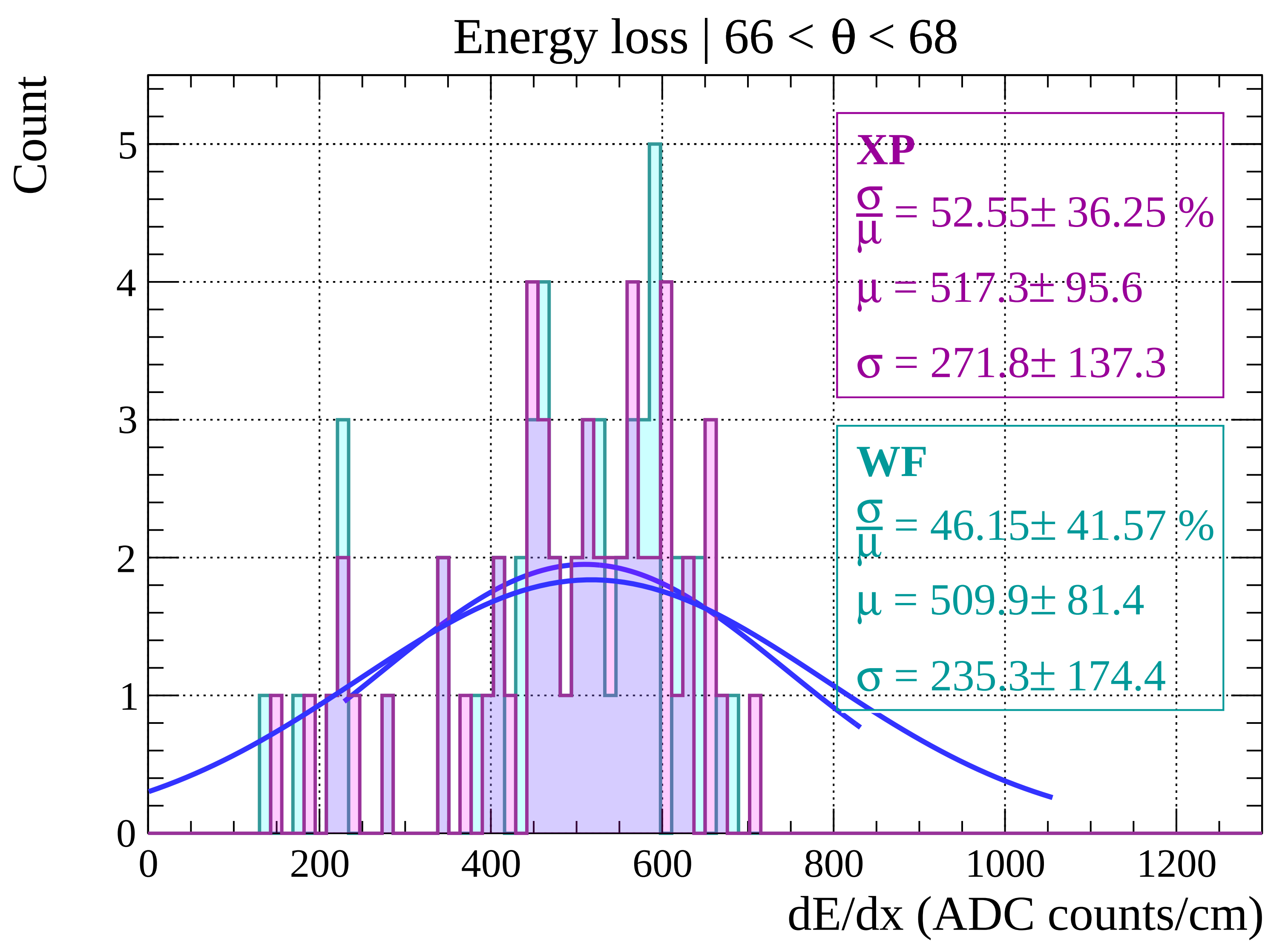
Energy loss | 60 < θ < 62
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
14
16
18
20
22
0
200
400
600
800
1000
1200
XP
σ/μ = 15.35± 1.53 %
μ = 540.2± 7.6
σ = 82.9± 7.1
WF
σ/μ = 13.56± 1.21 %
μ = 545.6± 6.1
σ = 74.0± 5.7



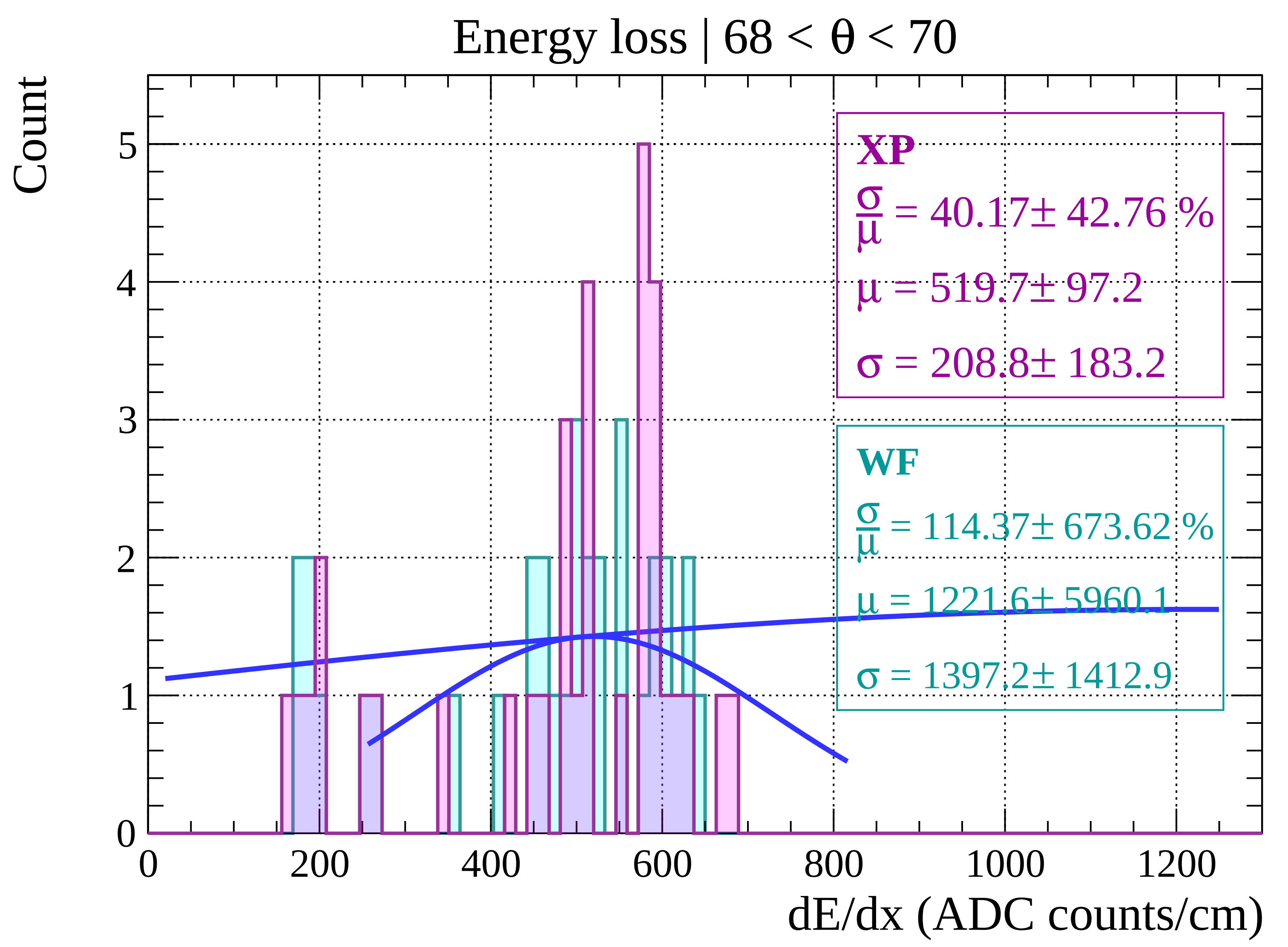
Energy loss | 62 < θ < 64
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
XP
σ/μ = 18.05± 1.95 %
μ = 533.4± 9.7
σ = 96.3± 8.6
WF
σ/μ = 18.02± 2.13 %
μ = 531.3± 10.2
σ = 95.7± 9.5

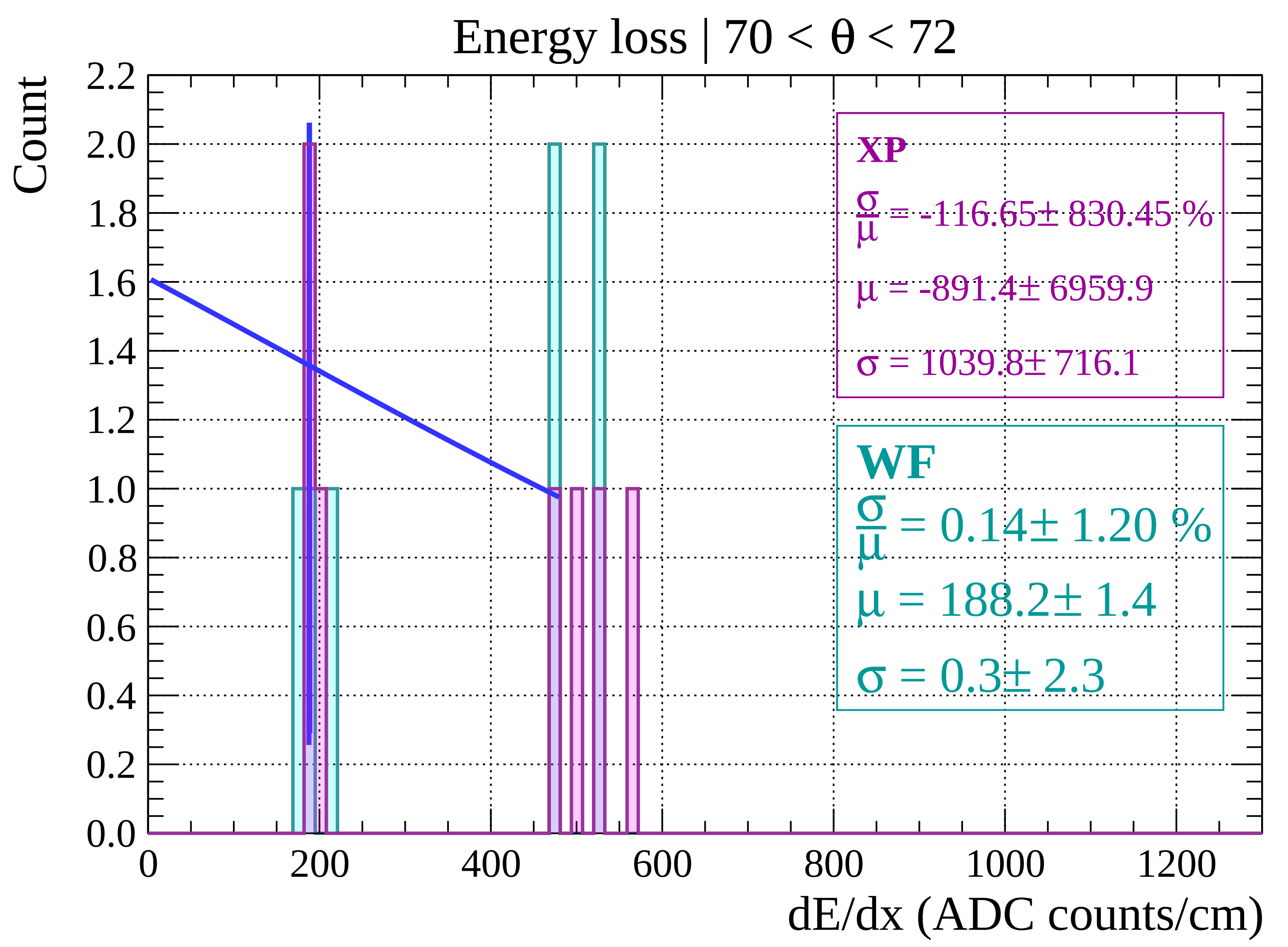


Energy loss | 64 < θ < 66
Count
dE/dx (ADC counts/cm)
0
2
4
6
8
10
12
0
200
400
600
800
1000
1200
XP
$\frac{\sigma}{\mu}$ = 15.82± 3.10 %
μ = 552.9± 12.8
σ = 87.5± 15.1
WF
$\frac{\sigma}{\mu}$ = 9.44± 1.39 %
μ = 549.3± 7.8
σ = 51.9± 6.9

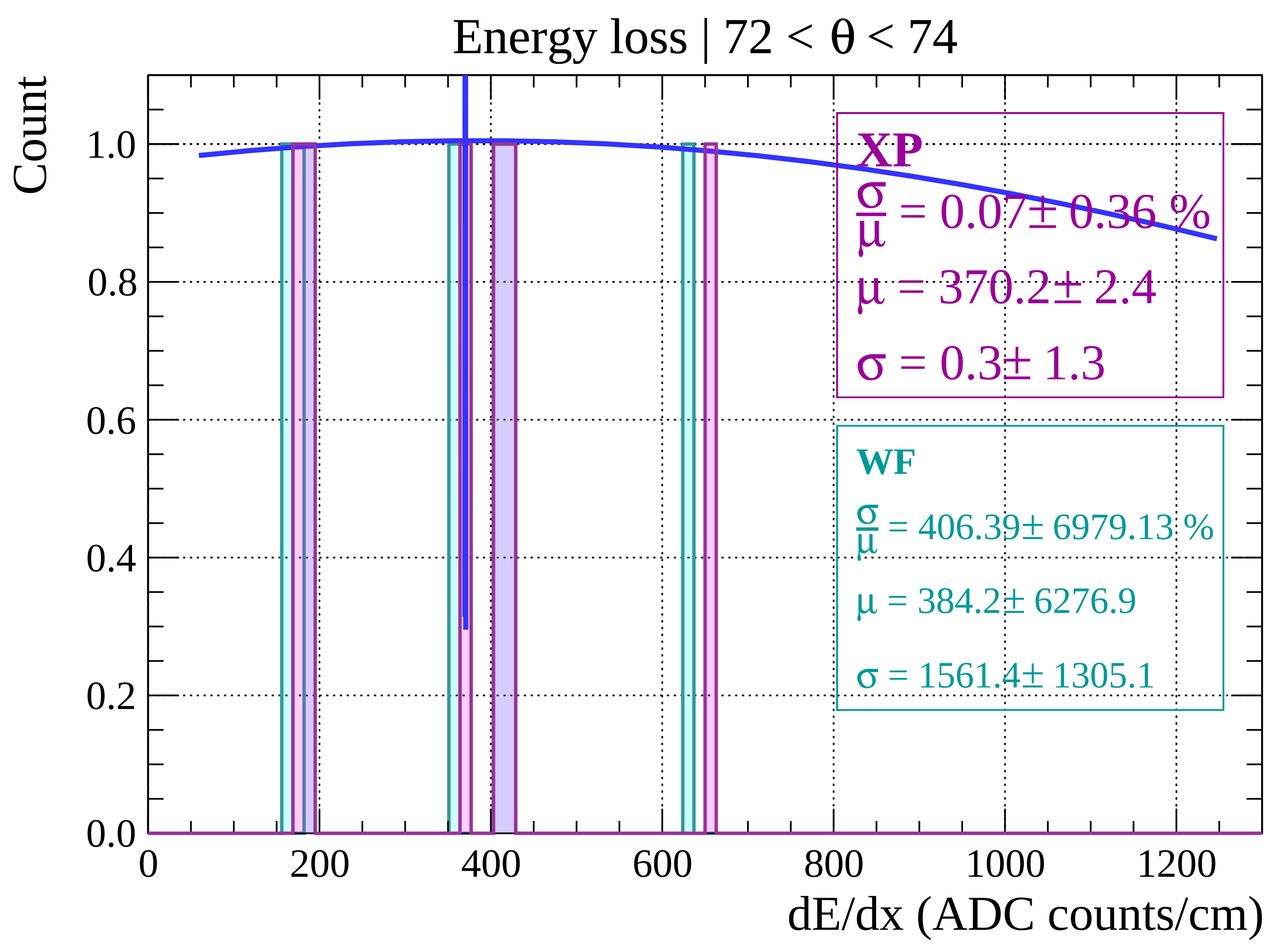


Energy loss | 66 < θ < 68
Count
dE/dx (ADC counts/cm)
0
1
2
3
4
5
0
200
400
600
800
1000
1200
XP
σ/μ = 52.55± 36.25 %
μ = 517.3± 95.6
σ = 271.8± 137.3
WF
σ/μ = 46.15± 41.57 %
μ = 509.9± 81.4
σ = 235.3± 174.4

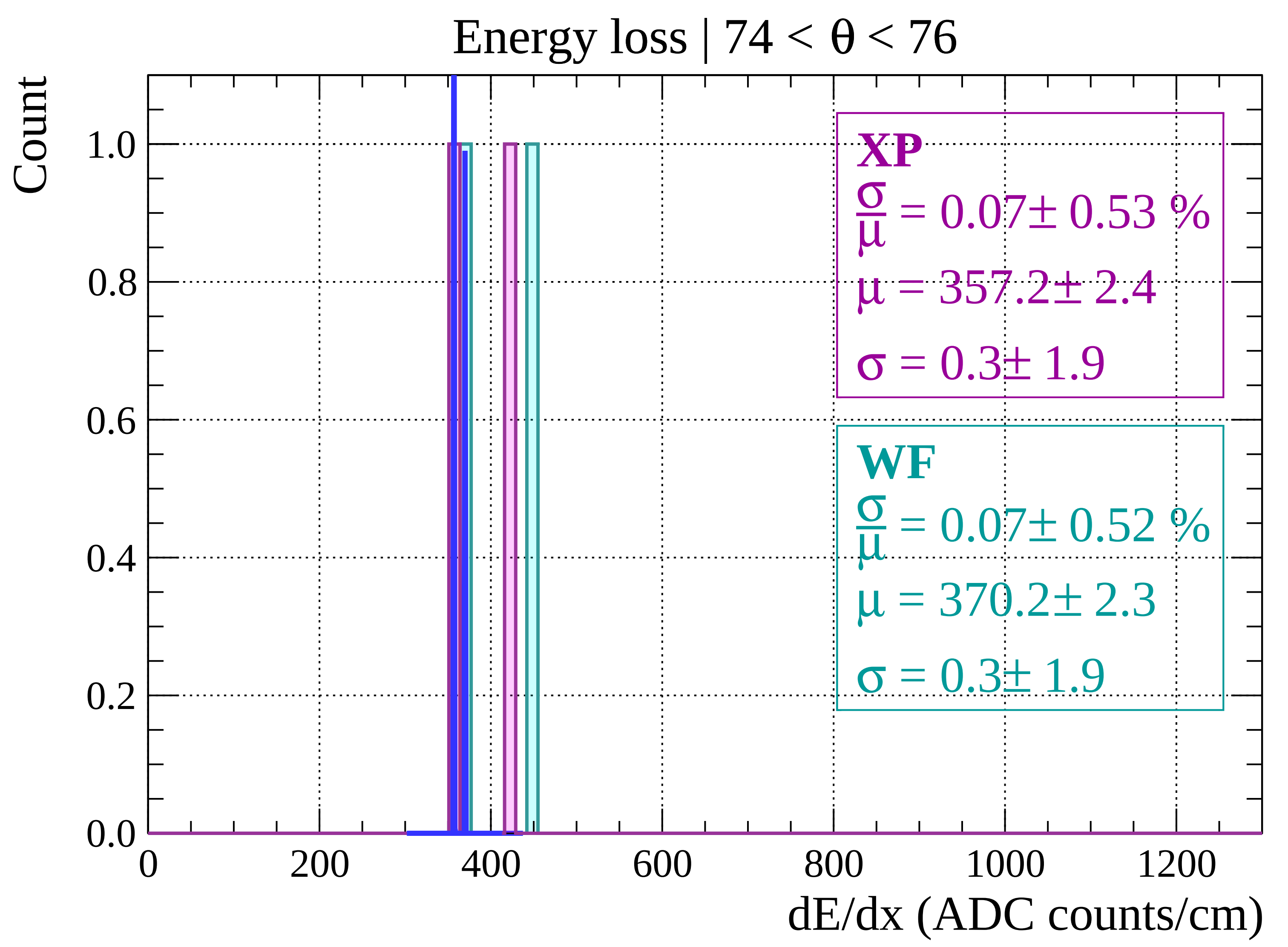

Energy loss | 68 < θ < 70
Count
dE/dx (ADC counts/cm)
0
200
400
600
800
1000
1200
1
2
3
4
5
XP
σ/μ = 40.17± 42.76 %
μ = 519.7± 97.2
σ = 208.8± 183.2
WF
σ/μ = 114.37± 673.62 %
μ = 1221.6± 5960.1
σ = 1397.2± 1412.9



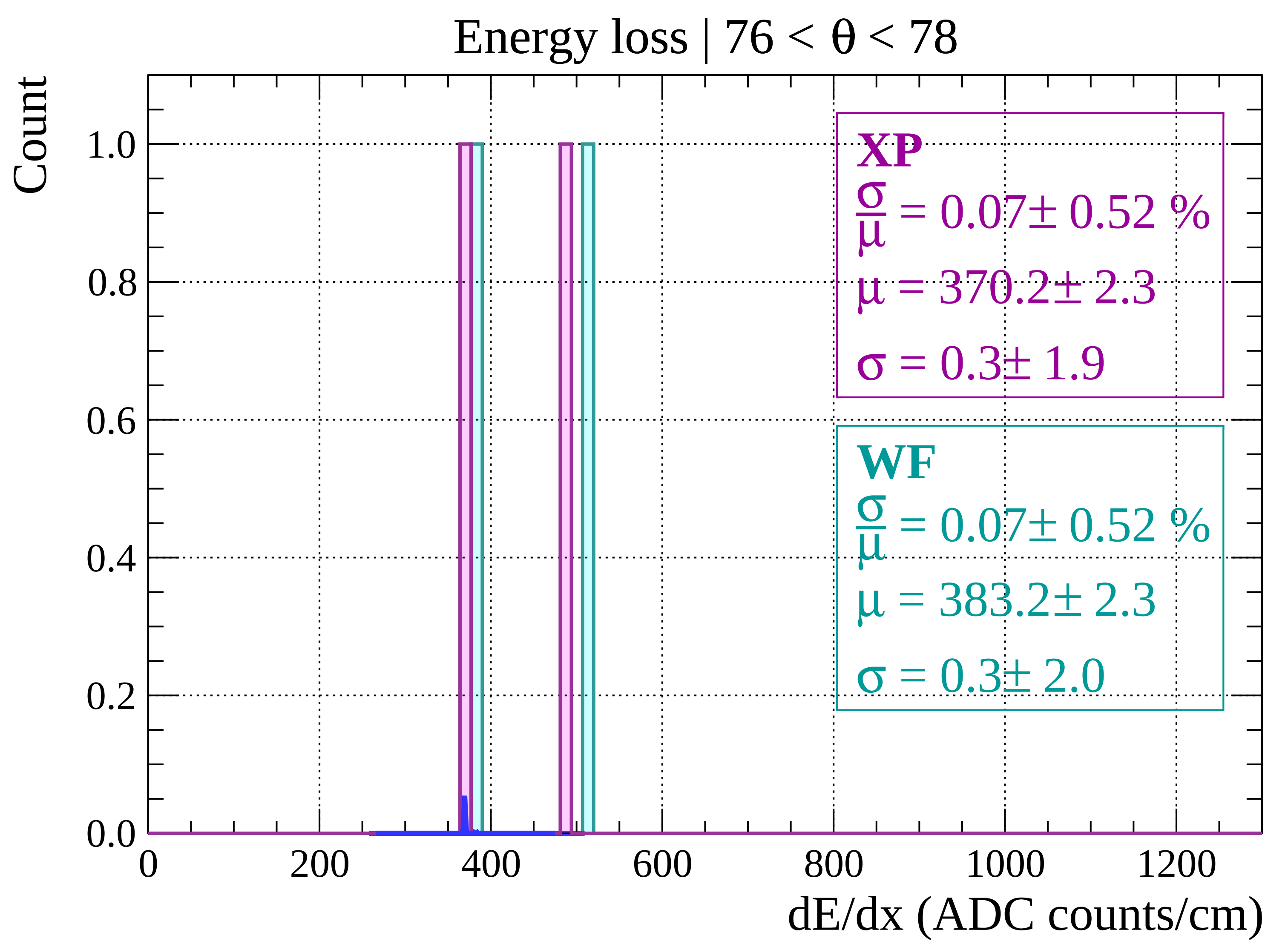
Energy loss | 70 < θ < 72
Count
dE/dx (ADC counts/cm)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
2.0
2.2
0
200
400
600
800
1000
1200
XP
σ/μ = -116.65± 830.45 %
μ = -891.4± 6959.9
σ = 1039.8± 716.1
WF
σ/μ = 0.14± 1.20 %
μ = 188.2± 1.4
σ = 0.3± 2.3



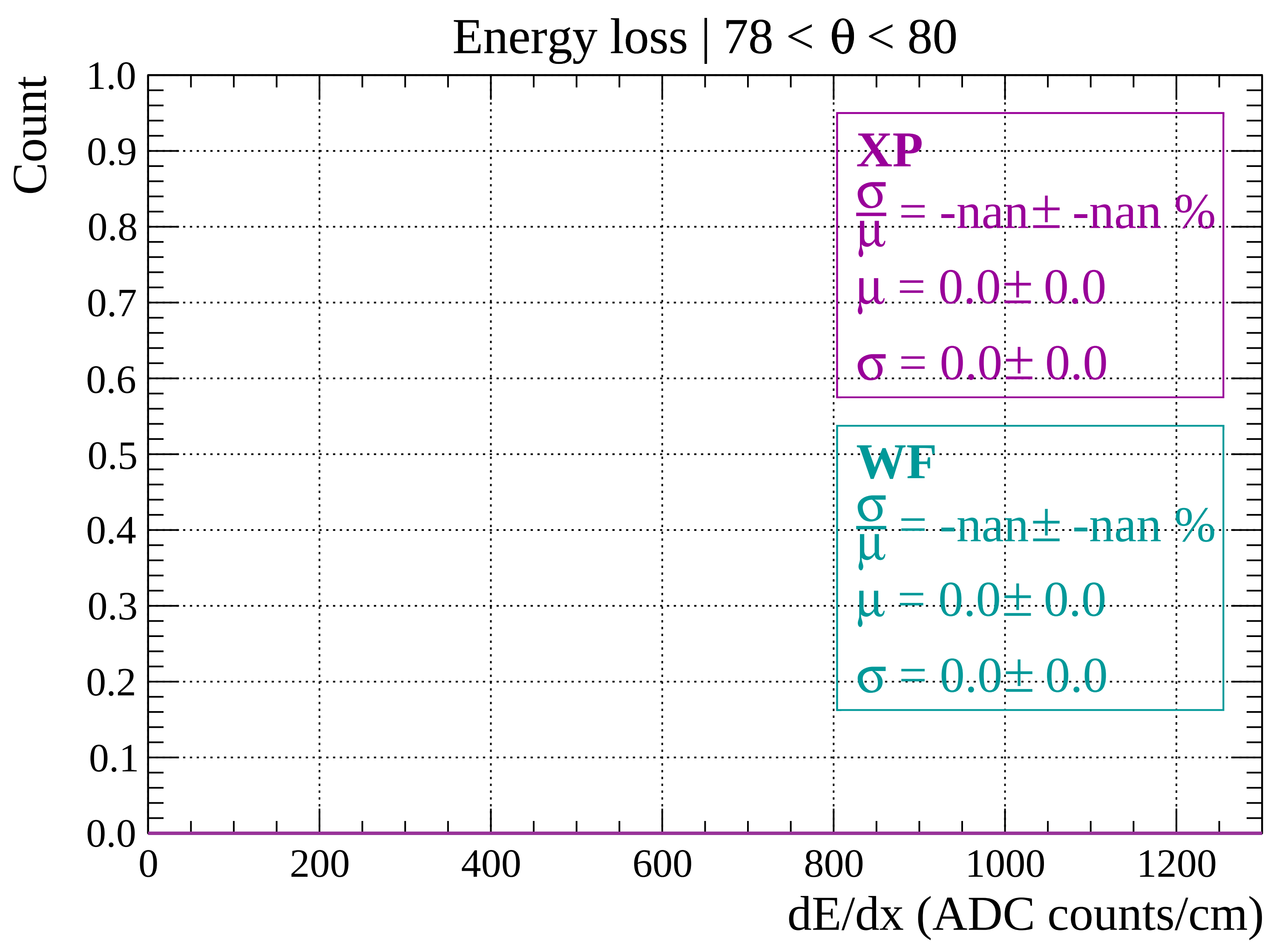
Energy loss | 72 < θ < 74
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 0.07± 0.36 %
μ = 370.2± 2.4
σ = 0.3± 1.3
WF
$\frac{\sigma}{\mu}$ = 406.39± 6979.13 %
μ = 384.2± 6276.9
σ = 1561.4± 1305.1

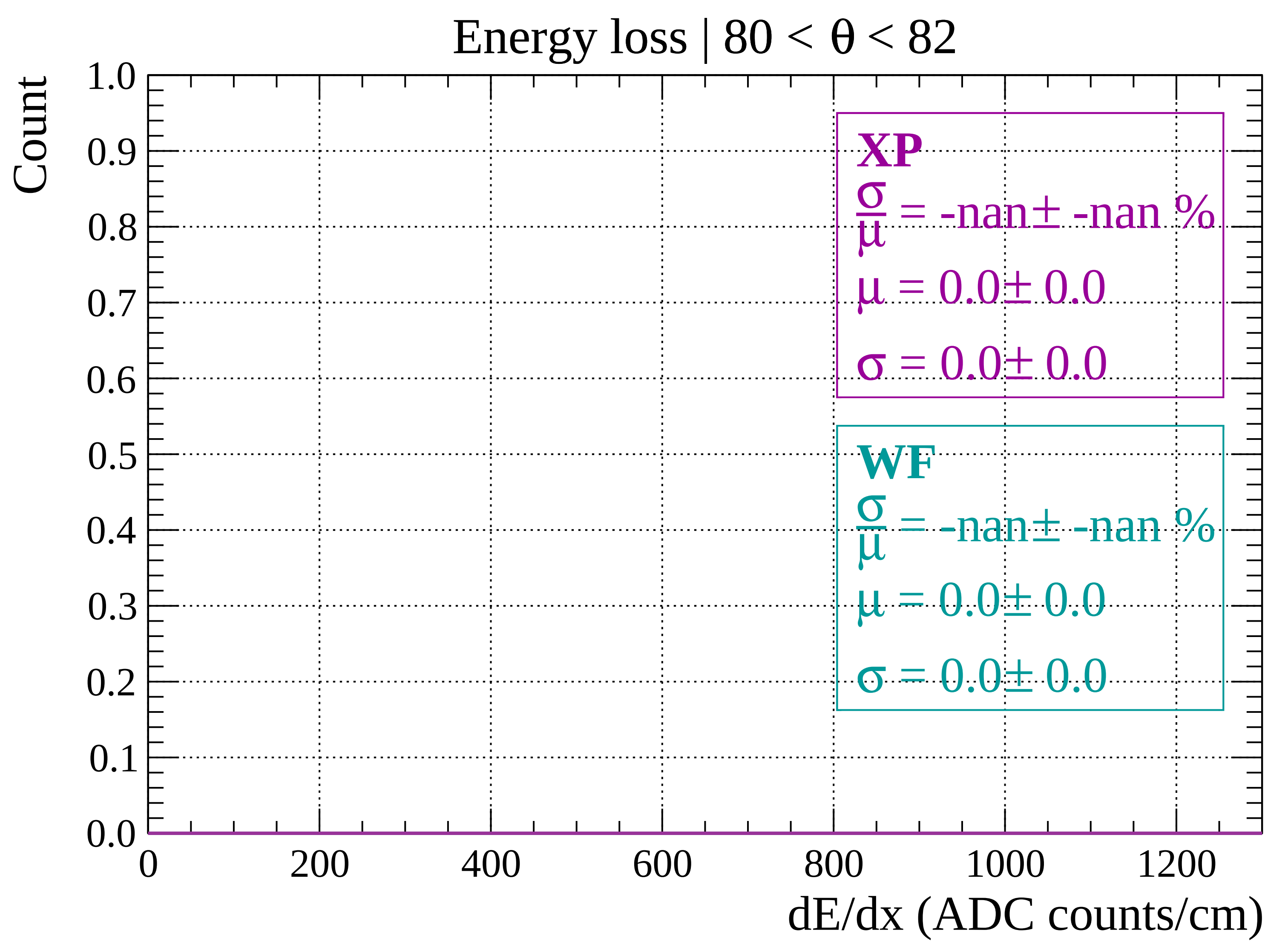


Energy loss | 74 < θ < 76
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = 0.07± 0.53 %
μ = 357.2± 2.4
σ = 0.3± 1.9
WF
σ/μ = 0.07± 0.52 %
μ = 370.2± 2.3
σ = 0.3± 1.9

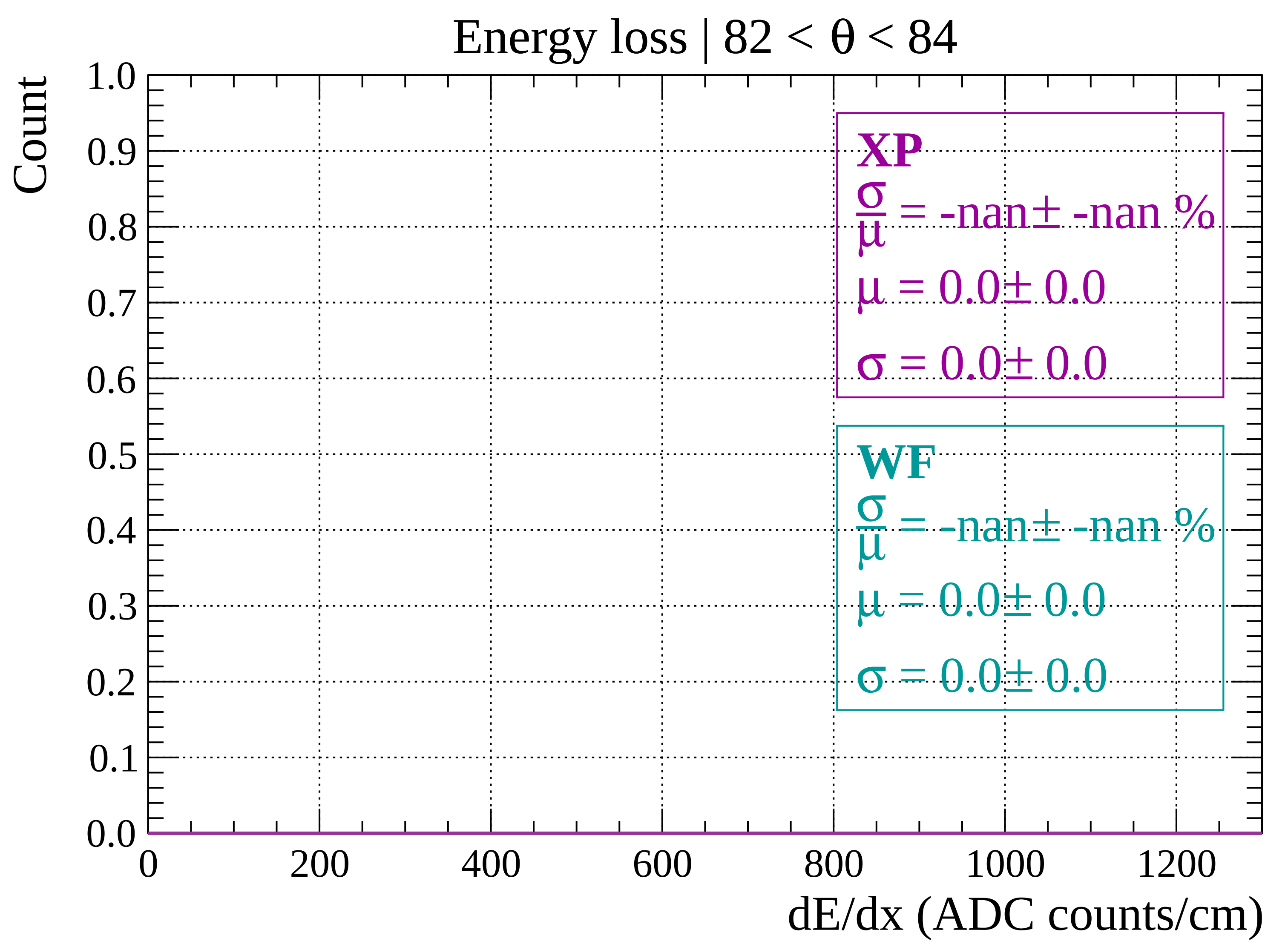


Energy loss | 76 < θ < 78
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
$\frac{\sigma}{\mu}$ = 0.07± 0.52 %
μ = 370.2± 2.3
σ = 0.3± 1.9
WF
$\frac{\sigma}{\mu}$ = 0.07± 0.52 %
μ = 383.2± 2.3
σ = 0.3± 2.0

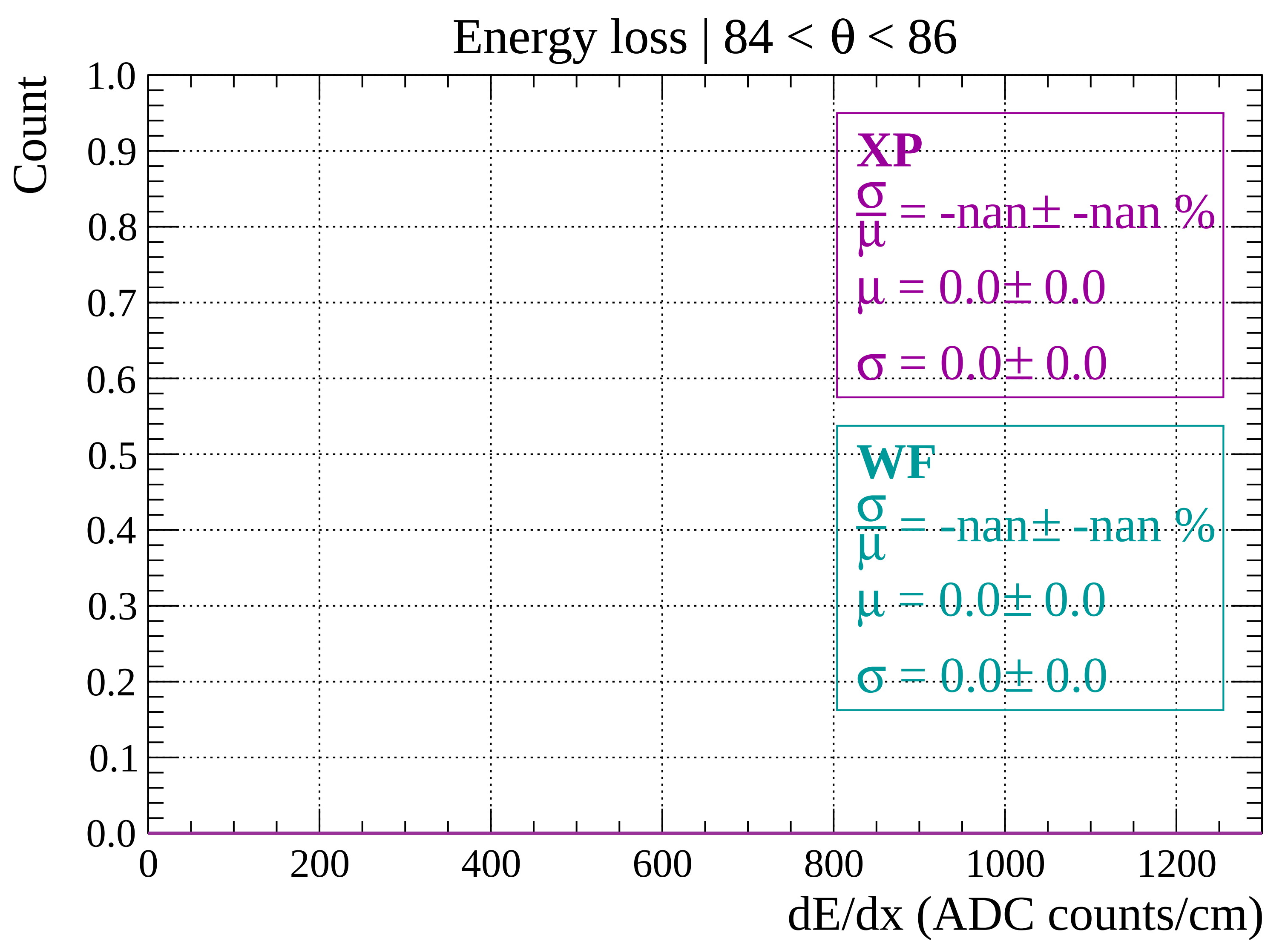

Energy loss | 78 < θ < 80
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

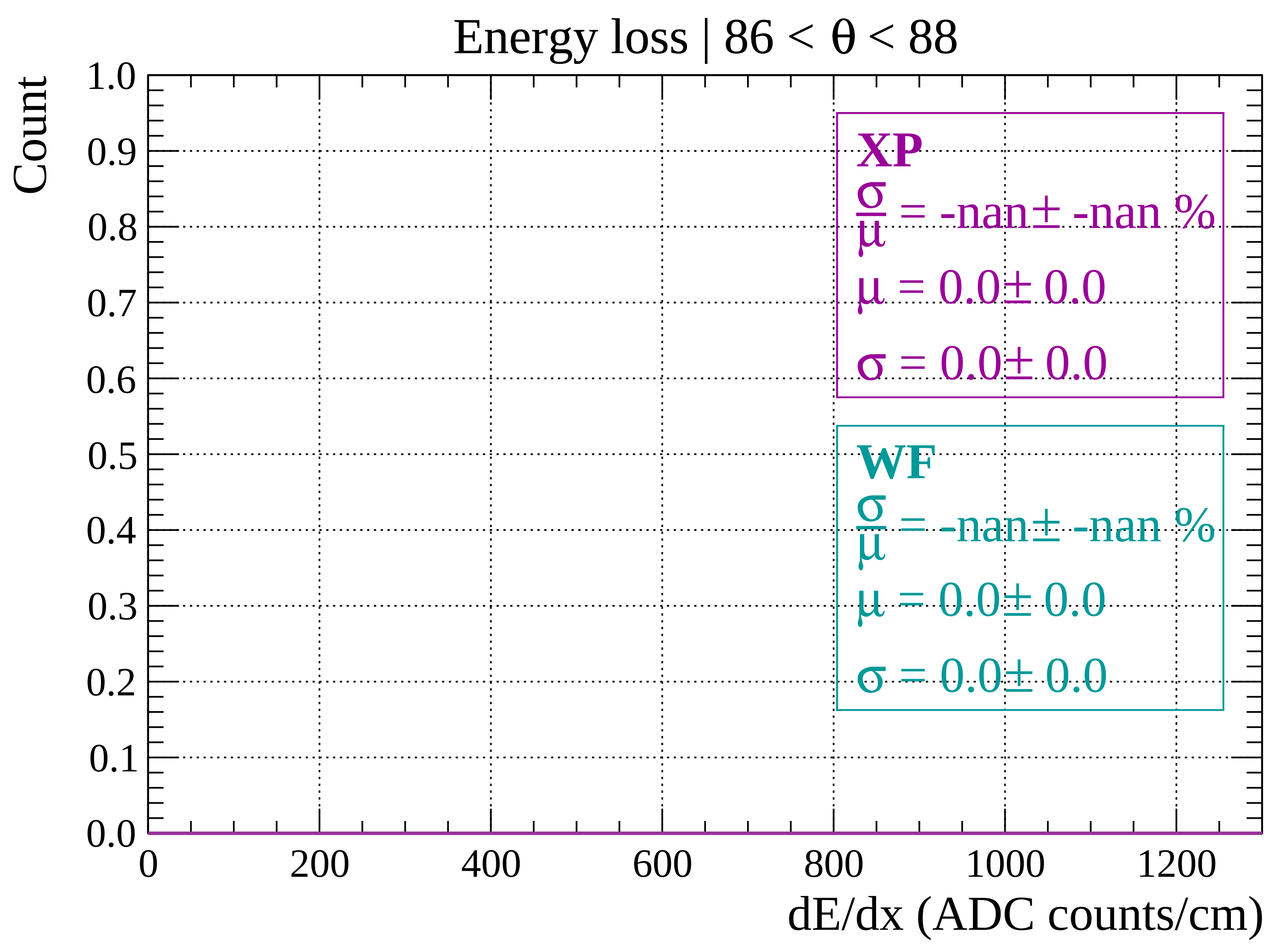


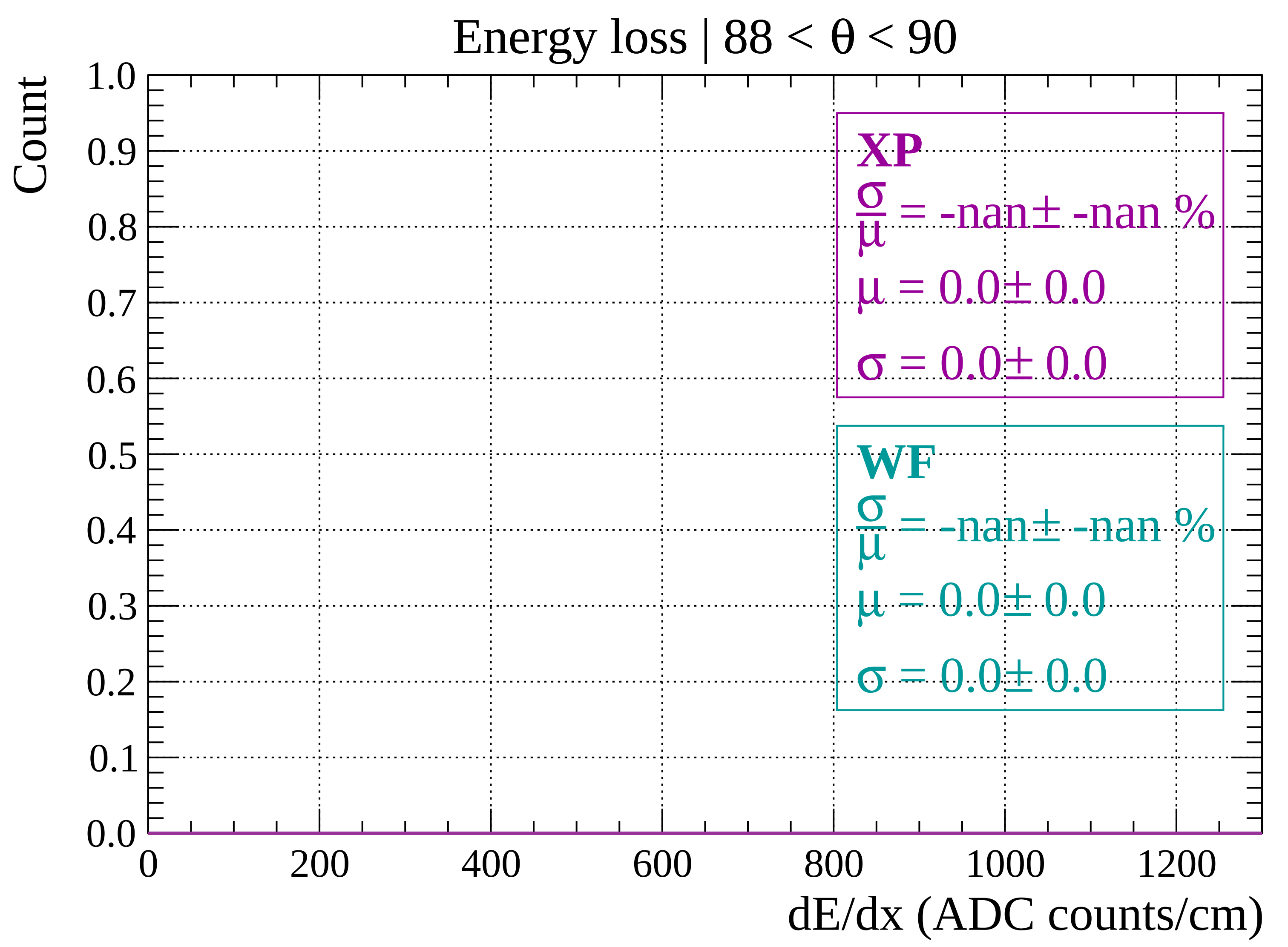
Energy loss | 80 < θ < 82
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



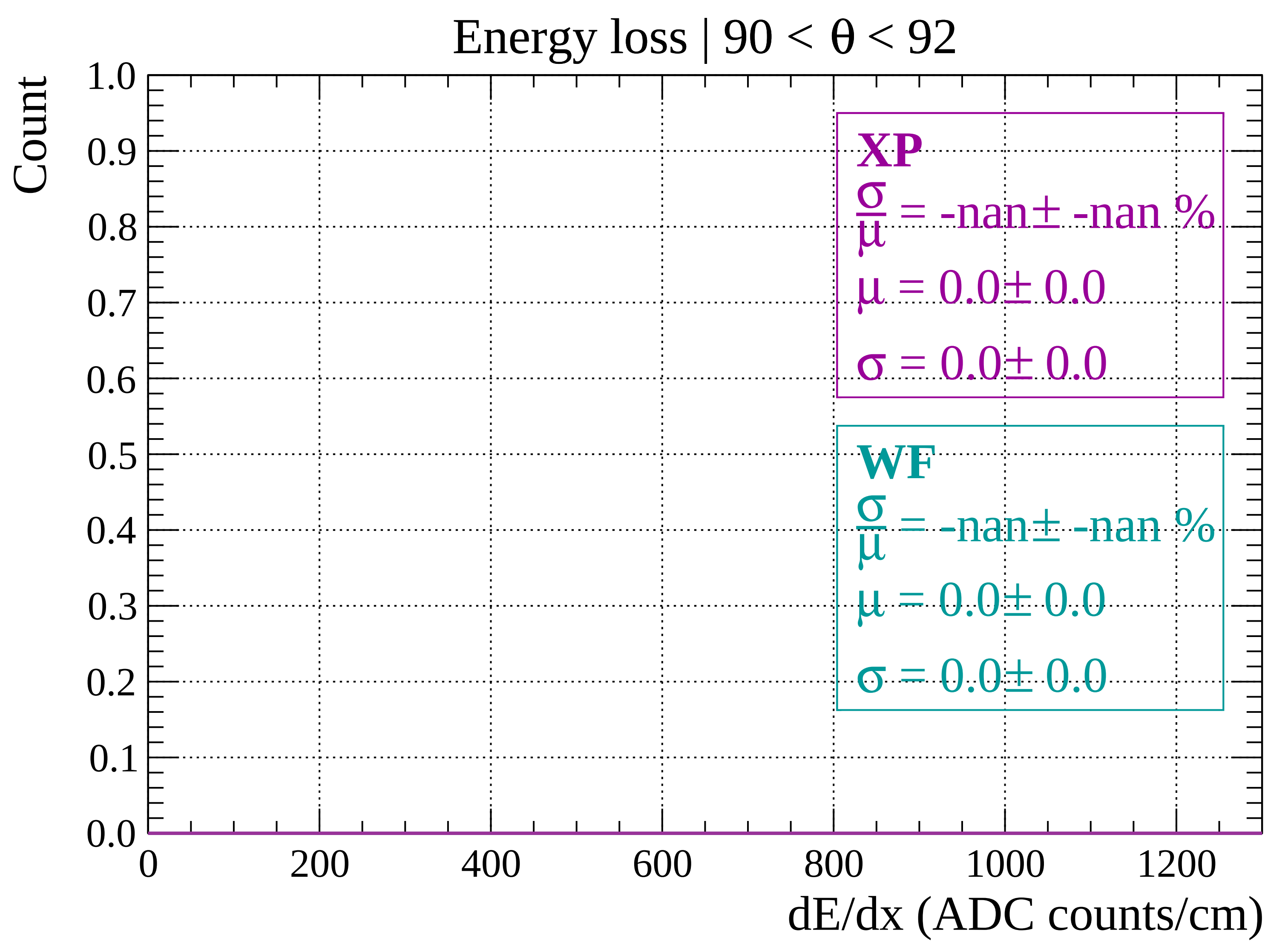
Energy loss | 82 < θ < 84
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

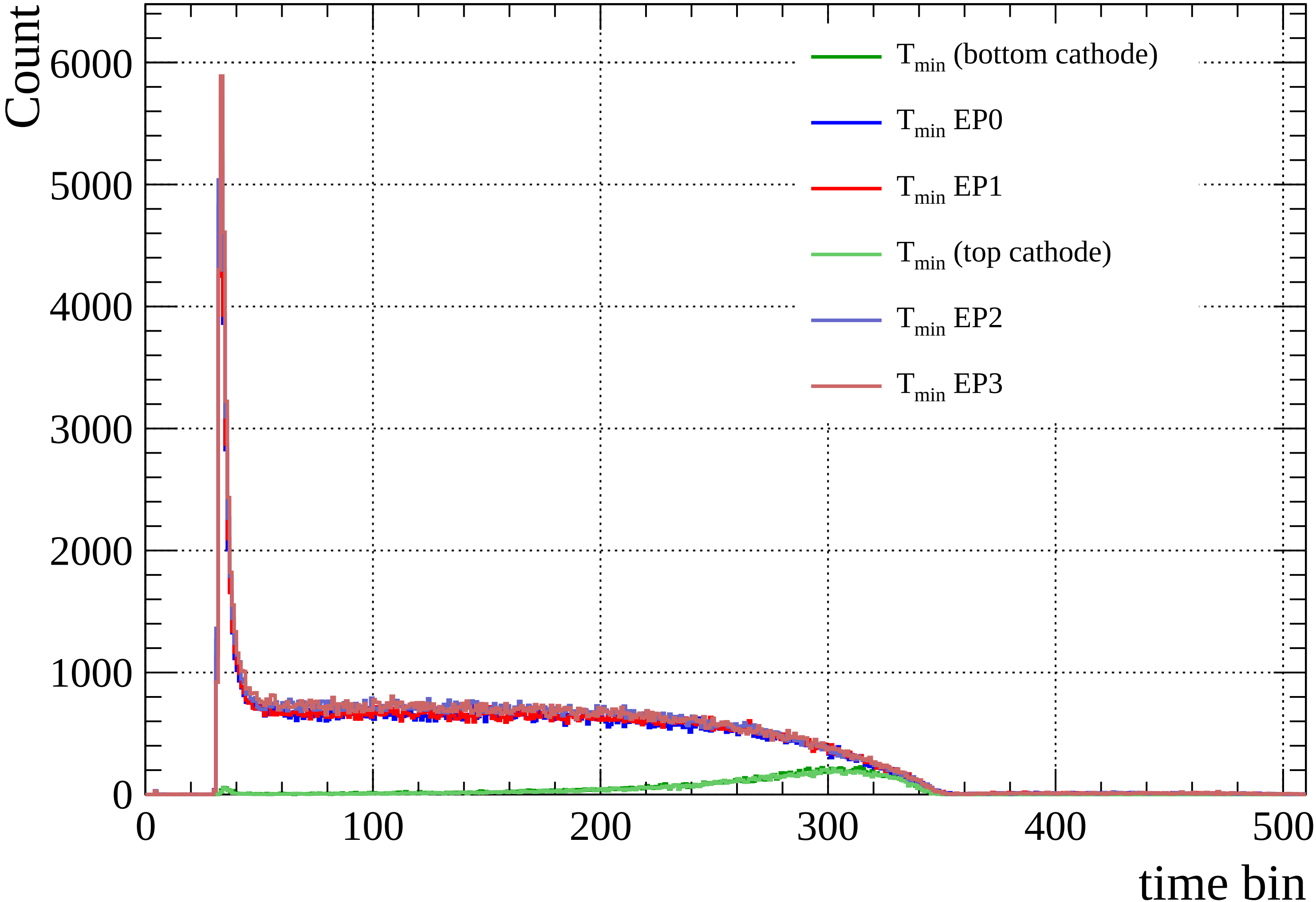

Energy loss | 84 < θ < 86
Count
dE/dx (ADC counts/cm)
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0
200
400
600
800
1000
1200
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

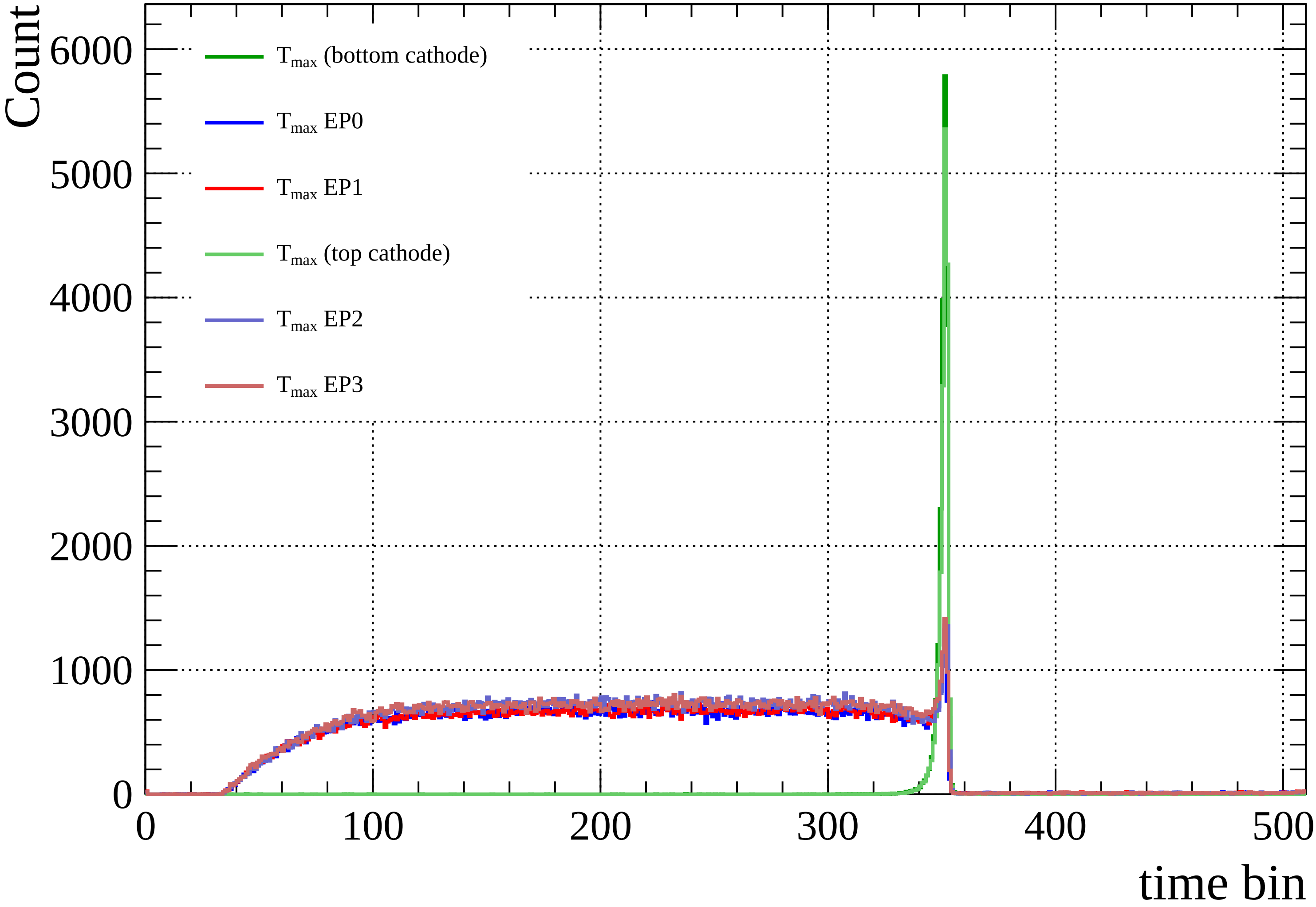

Energy loss | 86 < θ < 88
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Energy loss | 88 < θ < 90
Count
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Energy loss | 90 < θ < 92
Count
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
0
200
400
600
800
1000
1200
dE/dx (ADC counts/cm)
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0


Count
6000
5000
4000
3000
2000
1000
0
0
100
200
300
400
500
time bin
T_min (bottom cathode)
T_min EP0
T_min EP1
T_min (top cathode)
T_min EP2
T_min EP3



Count
time bin
T_{max} (bottom cathode)
T_{max} EP0
T_{max} EP1
T_{max} (top cathode)
T_{max} EP2
T_{max} EP3
0
1000
2000
3000
4000
5000
6000
0
100
200
300
400
500